XXV International Symposium on Solid State Dosimetry

Local Venue: Centro de Desenvolvimento da Tecnologia Nuclear (CDTN)
September 22-26, 2025 — Belo Horizonte, Brazil

Book of Proceedings: Volume 1

Editors
Telma C. F. Fonseca
Bruno M. Mendes
Lucas F. Araújo
Ester M. R. Andrade

BOOK OF PROCEEDINGS

Volume 1

XXV International Symposium on Solid State Dosimetry

Edited by

Telma C. F. Fonseca
Bruno M. Mendes
Lucas F. Araújo
Ester M. R. Andrade

XXV ISSSD

XXV International Symposium on Solid State Dosimetry

Contents

CHAPTER 1

Preface

It is with great pleasure that we welcome all participants to the XXV International Symposium on Solid State Dosimetry (ISSSD) in September of 2025, for the first time, at Centro de Desenvolvimento da Tecnologia Nuclear (CDTN/CNEN) and Universidade Federal de Minas Gerais (UFMG) in Belo Horizonte - Minas Gerais - Brazil. This milestone edition marks over two decades of continuous commitment to the advancement of radiation dosimetry through solid-state materials, techniques, and applications.

Since its inception, the ISSSD has served as a vital platform for scientists, engineers, and professionals to share knowledge, discuss new developments, and foster collaborations in the field of radiation detection and measurement. This 25^{th} edition continues this proud tradition, bringing together leading experts and emerging researchers from across the globe to present their latest findings and insights.

The program of this symposium reflects the diversity and depth of the field, covering topics ranging from fundamental studies of dosimetric materials to applied research in medical physics, environmental monitoring, and radiation protection. Special attention has been given to emerging technologies and interdisciplinary approaches that are shaping the future of dosimetry.

We are especially grateful to all contributors, authors, reviewers, session chairs, and members of the organizing, scientific, and support committees, whose dedication and effort have made this event possible. We also extend our sincere appreciation to our institutional partners and sponsors for their continued support.

We hope that this symposium will not only be an opportunity for scientific enrichment but also a space for meaningful exchanges, new collaborations, and the strengthening of our global community.

Welcome to the XXV ISSSD, and may this be a rewarding and inspiring experience for all.

Organizing Committee
XXV International Symposium on Solid State Dosimetry

Telma C. F. Fonseca
Chair at XXV ISSSD

Bruno M. Mendes
Chair at XXV ISSSD

Juan Azorin Nieto
President of the Sociédad Mexicana de Irradiación y Dosimetria (SMID)

CHAPTER 2

Summary and Perspectives

The XXV International Symposium on Solid State Dosimetry (ISSSD-2025) brought together researchers, professionals, and students from across the globe, reinforcing the vitality and relevance of solid state dosimetry in advancing radiation science and applications.

The 250 submitted works for the event were distributed across eight thematic areas, reflecting a wide range of research interests. Dosimetry (G2) had the highest representation with 90 works (36% of total), followed by Medical Physics (G4) with 36 contributions (14.4%) and Applications of Thermoluminescence (G1) with 35 works (14%). Other areas included Luminescent Materials (G8, 24 works, 9.6%), Radiation Protection (G5, 21 works, 8.4%), Radiobiology (G7, 19 works, 7.6%), Radiation Sources (G6, 15 works, 6%), and Ionizing and Non-Ionizing Radiation (G3, 10 works, 4%).

Regarding the presentation format, a total of 99 oral communications were delivered, with 54 in person and 45 online, while 151 poster presentations were held, including 96 in person and 55 online. Dosimetry (G2) again showed strong engagement with 23 in-person oral presentations, 10 online oral presentations, 38 posters in person, and 19 online posters. Applications of Thermoluminescence (G1) accounted for 8 oral in person, 7 online, 12 posters in person, and 8 online, while Medical Physics (G4) registered 7 oral in person, 9 online, 11 posters in person, and 9 online. Other areas contributed more modest numbers, such as Luminescent Materials (G8) with 19 works, mainly presented as posters, and Radiobiology (G7) and Radiation Protection (G5) with 19 and 21 contributions, respectively, shared between oral and poster sessions. Radiation Sources (G6) and Ionizing and Non-Ionizing Radiation (G3) had smaller representations but maintained coverage of fundamental aspects of the field. Overall, the data demonstrate diverse scientific participation, with a healthy balance between oral and poster formats, and a substantial proportion of online presentations, highlighting the hybrid and inclusive nature of the event.

Reflecting the truly international character of the symposium, participants represented 18 countries, totaling 248 presentations, ensuring broad participation and accessibility despite geographical distances. An analysis of the data reveals a notable predominance from Brazil, with 175 presentations. Mexico followed with a significant participation of 37 presentations. Other Latin American countries also had a notable presence, including Colombia (7), Chile (4), Peru (4), and Argentina (3). The international representation was broadened by Saudi Arabia and Kazakhstan, both with 3 presentations, and Indonesia with 2. Finally, contributing to the event's diversity, Bolivia, Canada, Ecuador, Spain, Guatemala, Malaysia, the United Kingdom, Switzerland, and Uruguay each participated with a single presentation.

CHAPTER 3

Organization, Committees, and Reviewers

3.1 CHAIR

Telma Cristina Ferreira Fonseca (DEN/UFMG-Brazil)
Silvia S. Hidalgo Tobón (UAM/México)

3.2 CO-CHAIR

Bruno Melo Mendes (CDTN/CNEN-Brazil)
Pedro Ramón González Martínez (México)

3.3 ORGANIZING COMMITTEE

Telma Cristina Ferreira Fonseca (DEN/UFMG-Brazil)
Bruno Melo Mendes (CDTN/CNEN-Brazil)
Lucas Fabrício de Araújo (DEN/UFMG-Brazil)
Ester Maria Rodrigues de Andrade (CDTN/CNEN-Brazil)
Ana Carolina Costa da Silva (WiN-Brazil)
Teógenes Augusto da Silva (CDTN/CNEN-Brazil)
Juan Azorín Nieto (UAMI-México)

3.4 SCIENTIFIC COMMITTEE

Andrea Mantuano[1] (UERJ-Brazil)
Ester Maria Rodrigues de Andrade[1] (CDTN/CNEN-Brazil)
Lucas Paixão Reis (IMA/UFMG-Brazil)
Luiz Cláudio Meira-Belo (CDTN/CNEN-Brazil)
Rita de Cássia de Oliveira Sebastião (Chemical/ICEX/UFMG-Brazil)
Sonia Seger Mercedes (DEN/UFMG-Brazil)

[1] President

3.5 PUBLICITY COMMITTEE

Davi Silveira Azevedo[1] (CDTN/CNEN-Brazil)
Bruna Lima Figueiredo[1] (DEN/UFMG-Brazil)
Jaime Torres Juárez (UAMI-México)
Fernando Soares Santos (CDTN/CNEN-Brazil)
Sara Suely Ventura da Silva Lima (CDTN/CNEN-Brazil)
Beatriz Diniz de Oliveira Guedes (CDTN/CNEN-Brazil)
Fernanda Rocha Cavalcante (CDTN/CNEN-Brazil)

3.6 SOCIAL COMMITTEE

Camila Engler[1] (CDTN/CNEN-Brazil)
Elydio Soares[1] (CDTN/CNEN-Brazil)
Laura Takahashi (CDTN/CNEN-Brazil)
Thayná Felizardo (CDTN/CNEN-Brazil)

3.7 SPONSORSHIP COMMITTEE

Bruno Gallotti (DEN/UFMG-Brazil)
Pâmera Paola Silva (Brazil)

3.8 WEBSITE COMMITTEE

Gabriel Góes Braga (COLTEC-Brazil)
Maria Luisa de Paula Ferreira da Silva (UFMG-Brazil)
Norberto Ferreira da Silva (Brazil)

3.9 CONFERENCE VOLUNTEERS

Anna Clara Moreira Da Silva (IMA/UFMG-Brazil)
Asif Amin (Farmácia/UFMG-Brazil)
Bianca De Oliveira Gomes (IB/UNESP-Brazil)
Bruna Lima Figueiredo (DEN/UFMG-Brazil)
Carolina Santos Lamounier (CDTN/CNEN-Brazil)
Danielly Tamires Da Silva Costa (IPEN/CNEN-Brazil)
Daniele Gonçalves Mesquita (UFPel-Brazil)
Davi Silveira Azevedo (CDTN/CNEN-Brazil)
Fernanda Rocha Cavalcante (CDTN/CNEN-Brazil)
Fernando Soares Santos (CDTN/CNEN-Brazil)
João Victor Ramos Oliveira (ICEx/UFMG-Brazil)
José Eliseu de Souza Junior (DEN/UFMG-Brazil)
Ketrim Regina Souza (INB-Brazil)
Lucas Beloti Silva (ICEx/UFMG-Brazil)
Maria Luisa de Paula Ferreira da Silva (ICEx/UFMG-Brazil)
Mateus de Souza Resende (IMA/UFMG-Brazil)
Sâmella de Andrade Brum (DEN/UFMG-Brazil)
Thamyra Cybelle Vieira Dos Santos (INMETRO-Brazil)

[1] President

3.10 REVIEWERS

Abner Duarte Soares (CNEN-Brazil)
Adriana de Souza Medeiros Batista (UFMG-Brazil)
Alex Cristóvão Holanda de Oliveira (FNE-Brazil)
Ana Letícia Almeida Dantas (IRD/CNEN-Brazil)
Andrea Mantuano (UERJ-Brazil)
Angel Ramírez (IGEF/UNAM-Mexico)
Ângela Marques (UFMG-Brazil)
Bernardo Dantas (IRD/CNEN-Brazil)
Bruno Cordeiro Silva (CEFET/MG-Brazil)
Bruno Gallotti (UFMG-Brazil)
Bruno Mendes (CDTN/CNEN-Brazil)
Camila Engler (CDTN/CNEN-Brazil)
Carlos Velasquez (UFMG-Brazil)
Carmen C. Bueno (IPEN/CNEN-Brazil)
Carolina Viloria Barragán (ITM-Colômbia)
Cássio Miri Oliveira (UNIFESP-Brazil)
Crissia Carem Paiva Fontainha (UFMG-Brazil)
Daniel Calheiro (UFMG-Brazil)
Daniel Campolina (CDTN/CNEN-Brazil)
Danilo Junot (UERJ-Brazil)
Denison Santos (IRD/CNEN-Brazil)
Elydio José Soares (CDTN/CNEN-Brazil)
Ester Maria Rodrigues de Andrade (CDTN/CNEN-Brazil)
Estefania Reyes Soto (UAM-México)
Evelise Gomes Lara (RMB-Brazil)
Evelyn Huerta (UAM-México)
Fernando Leyton (UTA-Chile)
Francisco Torres Hoyos (Universidad de Córdoba-Colômbia)
Helio Yoriyaz (IPEN/CNEN-Brazil)
Ivonne Berenice Lozano Rojas (IPN-México)
Jaime Torres Juárez (UAM-México)
Janaína Mendes (INCA-Brazil)
Jhon Brandon Felix Ferreira (UERJ-Brazil)
Jhonny Antonio Benavente Castillo (IPEN/CNEN-Brazil)
Jony Marques (UFMG-Brazil)
José Marques Lopes (IME-Brazil)
José Wilson Vieira (IFPE and UPE-Brazil)
Juan Azorín (UAM-México)
Kelly Cristina Martins Faêda (PUC/MG-Brazil)
Laura Cardoso Takahashi (CDTN/CNEN-Brazil)
Leanderson Cordeiro (UFPE-Brazil)
Leonardo Silva (UFMG-Brazil)
Lucas Daniel Cunha de Paula (CDTN/CNEN-Brazil)
Lucas Fabrício de Araújo (UFMG-Brazil)
Lucas Paixão (UFMG-Brazil)
Luciana Moreira (CEFET/MG-Brazil)
Luciana Nogueira (UFMG-Brazil)
Luciana Penedo (UERJ-Brazil)

Luciene Mota (UFMG-Brazil)
Luís Alexandre Gonçalves Magalhães (LCR/UERJ)
Luiz Claudio Meira-Belo (CDTN/CNEN-Brazil)
Marco Antonio Polo Labarrios (UAM Iztapalapa)
Marco Aurelio de Sousa Lacerda (CDTN/CNEN-Brazil)
Marcia Regina Pereira Attie (UFS-Brazil)
Maria Da Penha Albuquerque Potiens (IPEN/CNEN-Brazil)
Maria do Socorro Nogueira (CDTN/CNEN-Brazil)
Maria Gabrielle de Lima Rocha (FAFAR/UFMG-Brazil)
Maria José Neves (CDTN/CNEN-Brazil)
Marissa Anabel Rivera Cardona (CNEN-Brazil)
Mauricio Moacir da Silva Borges (UERJ-Brazil)
Modesto Sosa (UG-México)
Natalia Regina Souza Araujo (UFMG-Brazil)
Natasha Briggs e Silva (UERJ-Brazil)
Paulo Márcio de Oliveira (UFMG-Brazil)
Pedro R. González (ININ-México)
Priscila Santana (UFMG-Brazil)
Rita Sebastião (UFMG-Brazil)
Rodrigo Gontijo (UFMG-Brazil)
Samara Cristina Ferreira Machado (UERJ-Brazil)
Silvia Hidalgo (UAM-Mexico)
Sonia Mercedes (UFMG-Brazil)
Talita De Oliveira Santos (UFMG-Brazil)
Tarcisio Passos Ribeiro de Campos (CDTN/CNEN and UFMG-Brazil)
Telma Fonseca (UFMG-Brazil)
Teógenes da Silva (CDTN/CNEN-Brazil)
Tiago Jornada (HC/UFPE-Brazil)

CHAPTER 4

Event Organization, Supporting Institutions, and Sponsors

4.1 ORGANIZATION

4.2 SUPPORTING INSTITUTIONS

4.3 SPONSORS

CHAPTER 5

Schedule

XXV International Symposium on Solid State Dosimetry

22-26 SEPTEMBER 2025 - BELO HORIZONTE, BRAZIL

XXV ISSSD

General Schedule

Pre-Symposium - Monday, 22 Sep

Hour			
8h	***Registration***	Development of PET Radiopharmaceuticals - **Dr. Marina Bicalho Silveira** **Dr. Isabel Dregely** **Dr. Emerson Bernardes Soares** (Spectrum)	Introduction to Radiation Detectors - **Dr. Marco A Polo Labarrios** (Lattice)
9h			
9h30			
10h			
10h30	***COFFE BREAK***		
11h			
11h30			
12h	***LUNCH***		
13h		Importancia de la física en una imagen por resonancia magnética en el diagnóstico clínico - **Dr. Silvia S Hidalgo Tobón** (Spectrum)	Solid State Dosimetry: Principles and Applications & OSL Advanced Materials - **Dr. Ivonne Berenice Lozano Rojas** **Dr. Alida Tamar-Gabor** (Lattice)
13h30			
14h			
14h30			
15h			
15h30	***COFFE BREAK***		
16h			
16h30			
17h			

Tuesday, 23 Sep

Registration		
Opening Ceremony (Lumina)		
COFFE BREAK		
Invited Speaker: Wilson Calvo (Lumina)		
Invited Speaker: Yvone Maria Mascarenhas (Lumina)		
LUNCH		
Oral Sessions (Lumina and Spectrum)	*Poster Session:* ***14h30 to 16h30*** *- Referees for evaluation (Hall)*	Roundtable: Radon in Environment: Tracer Applications and Public Health Risks (Lattice)
		Roundtable: AI's challenges and Opportunities: Connecting research, industry and society (Lumina)
COFFE BREAK		
	Poster Session: ***14h30 to 16h30*** *- Referees for evaluation (Hall)*	*Invited Speaker*: Silvia Tobón (Lumina)
Welcome & Honors Night (Lumina)		

Wednesday, 24 Sep

Oral Sessions (Spectrum and Lattice)	*Poster Session:* ***10h00 to 12h30*** *- Referees for evaluation (Hall, Virtual Room)*
COFFE BREAK	
Oral Sessions (Spectrum and Lattice)	*Poster Session:* ***10h00 to 12h30*** *- Referees for evaluation (Hall, Virtual Room)*
LUNCH	
Technical Visit	Oral Sessions (Virtual Room)

Thursday, 25 Sep

Invited Speaker: Luciana Carvalheira (Lumina)		
Invited Speaker: Bernardo Maranhão Dantas Maria Jose Neves (Lumina)		
COFFE BREAK		
Invited Speaker: Ademar Lugão (Lumina)		
Invited Speaker: Richard Hugtenburg (Lumina)		
LUNCH		
Invited Speaker: Helio Yoriyaz (Lumina)	*Poster Session:* ***14h30 to 17h*** *- Referees for evaluation - (Hall)*	Oral Sessions - (Spectrum and Lattice)
Invited Speaker: Georgia Joana (Lumina)		
COFFE BREAK		
Helen Khoury Mentorship WIN-Brasil (Spectrum)	*Poster Session:* ***14h30 to 17h*** *- Referees for evaluation - (Hall)*	Roundtable: Leadership in Scientific and Technological Research (Lumina)
	Oral Sessions (Lumina and Lattice)	

Friday, 26 Sep

	Hour
Invited Speaker: Rolf Behrens (Lumina)	8h
Invited Speaker: Susana Lalic (Lumina)	8h40
	9h
Invited Speaker: Thiago Viana Miranda Lima (Lumina)	9h20
COFFE BREAK	10h
Oral Sessions (Lumina and Spectrum)	10h30
	11h
Invited Speaker: Tarcísio Campos and Alberto Campos (Lumina)	11h30
	12h
LUNCH	12h30
Oral Sessions (Lumina, Spectrum and Lattice)	13h30
	14h
Invited Speaker: Carmen Bueno (Lumina)	14h30
	15h
COFFE BREAK	15h30
Closing Session: Honor award & Helen Khoury award for best works (Lumina)	16h
	16h30
	17h
	17h30
	18h00
	20h00

Monday, 09/22/2025

08:00 *Registration*

	Courses Pre-Symposium	
	Spectrum Auditorium Chair: Marina Bicalho	**Lattice Auditorium Chair: Álvaro M. L. Gómez**
10:30 COFFE BREAK	**08:00** Dr. Marina Bicalho (CDTN/CNEN-Brazil), Dr. Isabel Dregely (Fachhochschule Technikum Wien-Austria) and Dr. Emerson Bernardes (IPEN/CNEN-Brazil) *Development of PET Radiopharmaceuticals*	**08:00** Dr. Marco Antonio Polo Labarrios (UAM-Mexico) and Dr. Álvaro M. L. Gómez (DEN/UFMG-Brazil) *Introduction to Radiation Detectors*

12:00 LUNCH

	Courses Pre-Symposium	
	Spectrum Auditorium Chair: Angel Ramirez	**Lattice Auditorium Chair: Carlos E. Velasquez**
15:00 COFFE BREAK	**13:00** Dr. Silvia S. Hidalgo Tobón (UAM/Mexico) *Importancia de la física en una imagen por resonancia magnética en el diagnóstico clínico*	**13:00** Dr. Ivonne Berenice Lozano Rojas (CICATA-Mexico) *Solid State Dosimetry: Principles and Applications & OSL Advanced Materials*

Tuesday, 09/23/2025

08:00 *Registration*

Opening Cerimony - Lumina Auditorium

09:00 Opening Ceremony – **Amenônia Maria Ferreira Pinto** – Centro do Desenvolvimento da Tecnologia Nuclear (CDTN/CNEN), **Bruno Melo Mendes (ISSSD 2025 – Chair)**, **Henrique Resende Martins** – Vice-director of Escola de Engenharia of Universidade Federal de Minas Gerais (UFMG), **Silvia S. Hidalgo Tobón (ISSSD 2025 – Chair)** – Universidad Autónoma Metropolitana (UAM), **Telma Cristina Ferreira Fonseca (ISSSD 2025 – Chair)** – Universidade Federal de Minas Gerais (UFMG), **Wilson Aparecido Parejo Calvo** – Comissão Nacional de Energia Nuclear (CNEN).

10:00 COFFEE BREAK

Invited Speaker - Lumina Auditorium **Chair: Silvia Hidalgo**

10:30 *Invited Speaker* – Wilson Aparecido Parejo Calvo (CNEN-Brazil),
Current Status of Strategic Projects at CNEN.

11:20 *Invited Speaker* – Yvone Maria Mascarenhas (SAPRA Landauer-Brazil),
Personal Dosimetry: The Importance of Regulation, Technology, and Culture. Evaluation of Dosimetry Services in Latin America and the Caribbean.

12:30 LUNCH

Roundtable Session (Lattice Auditorium)

13:30 Radon in Environment: Tracer Applications and Public Health Risks
Chair: Talita Santos (CDTN/CNEN-Brazil),
Panelists: Elydio Soares, Laura Takahashi, and Nathan da Silva

Radiobiology Spectrum Auditorium Chair: Tarcísio Campos	Luminescent Materials/ Applications of Thermoluminescence Lumina Auditorium Chair: Susana Lalic

Radiobiology

13:30 Jhon Brandon Felix Ferreira
G07-002 - *Low-Dose Neonatal Ionizing Radiation Induces Modest, Sex-Specific Changes in Adult Mouse Behavior and Hippocampal Oxidative Stress*

13:45 Lucas Siqueira de Lima Neto
G07-013 - *Radiometric Analysis by gamma spectrometry of copaiba oil using a Hyper*

14:00 Iris Bomfim Martins Ferreira
G07-015 - *Radiometric analysis of the bivalve mollusk Perna*

Luminescent Materials/ Applications of Thermoluminescence

13:30 João Pedro
VanellusRad - *The First Brazilian Real-Time IoT Electronic Dosimeter for Radiation Monitoring*

14:00 Ángel Ramírez Luna
G01-002 - *Afinando la Cronología Relativa de Cerámicas de la Zona de Playa Vicente, Veracruz, México Aplicando el Método de Datación Absoluta por Termoluminiscencia*

Roundtable Session (Lumina Auditorium)

14:30 AI's Challenges and Opportunities: Connecting Research, Industry, and Society
Chair: Telma Cristina Ferreira Fonseca (DEN/UFMG-Brazil),
Panelists: Ana Carolina Costa, Andre Henrique Alves Carneiro and Rita Sebastião

13:30 to 16:30 - Poster Session (Main Hall)

Note: 14:30 to 16:30 - Referees for evaluation
In-Person

G1 - Applications of thermoluminescence (dosimetry, dating, industrial, etc.)

G01-001 - Olania Herrera González, *Involvement of SMARCB1 subunit of the mSWI/SNF and GLI-1 epigenetic complex and their radiolabeling as a strategy for ionizing therapy in lung cancer*
G01-011 - Samira Batista da Silva, *Enzymatic biocatalysis of alginate spheres and iodinated amino acids for applications in embolization guided by SPECT*
G01-026 - Roseli Künzel, *A low-cost integrated system for thermoluminescence and optically stimulated luminescence measurements*
G01-029 - Yuri Avelino Costa, *Analysis of the average geometric parameters of the closed pores of the fine aggregate matrix using X-ray micro-computed tomography and linear regression*
G01-032 - Danilo O. Junot, *Crab shell biowaste composites as luminescent radiation detectors*
G01-035 - Luiz Matheus de Souza Aguiar Leite, *Development of a radiochromic film using a polyacetylene compound*
G01-036 - Thalles Oliveira Campagnani, *Development of a test bench using a CsI/SiPM detector for mass flow measurement in industrial processes*
G01-038 - Ketrim Regina Souza, *Empowering the Future of Nuclear Energy: A Mentorship Program for the Development of Female Leadership*
G01-040 - Raissa Alexia Camargo Guassu, *Estimation of absorbed dose in critical organs during interventional procedures: experimental and computational approach*
G01-041 - Danilo O. Junot, *Influence of precursor purity on the synthesis and TL/OSL response of $CaSO_4$-based radiation detectors*
G01-044 - Vinicius Costa Silva, *Radiological characterization of the waters feeding the Camorim Reservoir by alpha–beta – Pedra Branca State Park*
G01-045 - Margarete Cristina Guimarães, Thessa Cristina Alonso, *Standardization of gamma irradiation procedures for the preservation of bibliographic heritage*

G5 - Radiological Protection

G05-003 - Adriana de Souza Medeiros Batista, *Assessment of volcanic rock adsorbents for the remediation of acid mine drainage in uranium mining*
G05-006 - Daniel de Castro Pacheco, *Development of a distributed monitoring system for active radiological protection*
G05-007 - Cássio Miri Oliveira, *Dose evaluation and suggestion of typical values for head CT headache protocols: a preliminary study*
G05-015 - Thamyra Cybelle Vieira dos Santos, *Quality assurance in ^{90}Y radioembolization: an intercomparison protocol for activity optimization*
G05-017 - Fábio Lima Guimarães, *Radiometric survey at the Stone House Lookout, Saquarema-RJ*
G05-019 - Marcela Morais Freitas, *Structural and compositional changes in zeolitic materials exposed to uranium mine effluents: implications for environmental radioprotection*
G05-022 - Bianca Rossini Marques, *Validation of analytical methods using gamma spectrometry applied to the radiometric characterization of monazite sand*

G8 - Luminescent materials

G08-004 - Divanizia do Nascimento Souza, *Development of high-sensitivity $CaSO_4$:Tm,Mn detectors for thermoluminescent and optically stimulated luminescent dosimetry*
G08-007 - Roseli Künzel, *Effect of silver doping concentrations on the structural and luminescence properties of $CaMoO_4$ powders*
G08-008 - Paula Thays Santos Malachias, *Investigation of the luminescent properties of zirconia-enriched dental resins*
G08-009 - Bruno Araújo, *Optically stimulated luminescence (OSL) characteristics of seven different color types of quartz samples under 405 nm stimulation*
G08-010 - Rafael dos Santos Viana, *Promotion of the Compton effect of ^{137}Cs photons to enhance the effi-*

ciency of radioluminescent batteries
G08-011 - Anderson Bezerra, *Properties of $CaSO_4$:Eu,Mn for Luminescent Dosimetry*
G08-012 - Patrícia de Lara Antonio, *PTTL and PTOSL of TLD-100 exposed to ^{60}Co and lighting with LEDs*
G08-013 - Daniele Gonçalves Mesquita, *Radiation Detection of Lead-Free $BaZrO_3$ Perovskite Scintillators*
G08-014 - Claudete Roberta Evangelista, *Feasibility of Retrospective Dosimetry Using Ceramic Material from Spark Plugs*
G08-015 - Anderson Manoel Bezerra da Silva, *Structural, Morphological, Luminescent, and Dosimetric Properties of $CaSO_4$:Yb,Mn Phosphors*
G08-016 - André Felipe de Oliveira, *Tailored theranostic nanocomposites: mesoporous silica, gold nanoparticles, and gadolinium complexes for cancer diagnosis and treatment*
G08-020 - Jéssica Pauline Nunes Marinho, *Potential theranostic nanoparticles of HAp-Gd^{3+} with curcumin and folic acid for cancer therapy*

15:30 COFFEE BREAK

Invited Speaker - Lumina Auditorium **Chair: Ivonne B. L. Rojas**

16:00 *Invited Speaker* – Silvia S. Hidalgo Tobón (UAM-Mexico),
State of the art of magnetic resonance imaging: from fundamental physics to clinical applications.

17:00 - 20:00 Welcome & Honors Night - Lumina Auditorium

- For the Ceremony: Bruno M. Mendes, Marco Aurélio de Sousa Lacerda and Telma Cristina Ferreira Fonseca
- Award Ceremony for Honored Researchers: Dr. Tarcísio P. R. de Campos and Dr. Teógenes Augusto da Silva
- Cultural Performance by Aruanda Group
- Welcome Cocktail Gathering

Wednesday, 09/24/2025

Applications of Thermoluminescence
Spectrum Auditorium
Chair: Andrea Mantuano

08:00 Rafael Cogollo
G01-004 - *Dosimetric characteristics and kinetic analysis of pure and cerium-doped alumina (α-Al_2O_3) for low-dose thermoluminescent dosimetry applications*

08:15 Héctor Maya
G01-006 - *Kinetic analysis method for general order TL glow curves*

08:30 Natalia Barbosa Gonzaga Mendes
G01-024 - *Using gamma spectrometry as a tool for assessing residual dose in irradiated topaz*

08:45 Alberto Mizrahy Campos
G01-027 - *Random walk of neutrons in a rotational environment*

09:00 Enrique Alejandro Chávez Camacho
G01-030 - *Analysis of the effects of gamma radiation on amaranth seeds and sprouts: a design of experiments (DoE) approach*

Dosimetry
Lattice Auditorium
Chair: Ana Carolina Costa

08:00 Berenice Guadalupe Olmos Grimaldo
G02-006 - *Medición de radón en la zona del Peñón Blanco, Salinas de Hidalgo, San Luis Potosí, México*

08:15 Renato Monaco Nunes
G02-018 - *A Monte Carlo-based software for photon transport in water using the CUDA platform*

08:30 Iasmin Vieira Nishibayaski
G02-019 - *A new formulation of Fricke gel dosimeter development using salicylic acid*

08:45 Camilla da Silva Sampaio
G02-026 - *Assessing the applicability of TDCR counting for nonpure beta emitters: the case of ^{210}Pb in liquid scintillation cocktails*

09:00 Laura Cardoso Takahashi
G02-040 - *Development of an integrated national database for effective dose calculation from environmental radiation exposure in Brazil*

09:15 Laisa de Almeida Neves
G05-023 - *Analysis of Natural Radioactivity in Beach Sands Along the Rio de Janeiro State Coast by Gamma Spectrometry*

10:00 COFFEE BREAK

Radiation Sources
Spectrum Auditorium
Chair: Denison Santos

10:30 Nayelly Nikole Díaz Gutiérrez
G06-002 - *Estimación del término fuente de un irradiador gamma de ^{60}Co utilizando el código MCNP*

10:45 Ismael Teixeira Vicente
G06-004 - *Alpha particle spectrometry and Monte Carlo evaluation of ^{241}Am sources for use in Elastic Recoil Detection Analysis (ERDA)*

11:00 Estela Maria de Oliveira
G06-006 - *Comparative study of gamma efficiency curves in HPGe detector with different calibration standards*

Medical Physics
Lattice Auditorium
Chair: Lucas Paixão

10:30 Sara Suely Ventura da Silva Lima
G04-004 - *Comparative Monitoring of Lens, Thyroid, and Whole Body in Interventional Cardiology Procedures*

10:45 Prycylla Gomes Creazolla
G02-089 - *Volumetric dose evaluation using 6 MV linear accelerators for blood irradiation*

11:00 Juraci Passos dos Reis Junior
G04-011 - *An AI revolution in radiotherapy in SUS: The example of Mário Kroeff Hospital*

11:15 Natasha Briggs e Silva
G04-020 - *Determination of the mass attenuation coefficient of thermoset and thermoplastic resins at energies used in conventional mammography*

11:15 Roos S. de Freitas Dam
G06-012 - *Neutron Source Applied for Eccentric Scale Prediction in Oil Pipelines using Deep Learning*

11:30 Roger Ferreira da Silva
G04-033 - *Mammography quality assurance based on phantom image with deep learning*

08:00 to 12:30 - Poster Session (Main Hall)

Note: 10:00 to 12:30 - Referees for evaluation
In-Person

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

G02-003 - Emilia Suárez, *Development and Assessment of Perovskite-based X-Ray Detectors for In Vivo Dosimetry Applications*
G02-017 - Tales Noujaim Silveira, *^{68}Ga Production with the GE PETtrace-8 Cyclotron: a Monte Carlo study*
G02-020 - Leanderson Pereira Cordeiro, *AI-based segmentation performance and implications for organ-level dosimetry in radionuclide therapy*
G02-021 - José Marques Lopes, *An Environmental Radiometric Assessment Based on Airborne Gamma Spectrometry Data in the State of Rio de Janeiro*
G02-023 - Hugo Leonardo Lemos Silva, *Analytical and simulation methods in determining radiation exposure time of healthcare waste*
G02-027 - Alessandra Vaz dos Anjos Santos, *Automated System for Visual Detection of Personal Dosimeters Using Convolutional Neural Networks*
G02-028 - Tiago Macedo da Cunha e Silva, *Bayesian-Optimized Monte Carlo Modeling of CyberKnife Collimators and Source Parameters for Small-Field Dosimetry*
G02-030 - Heloísa Milena Caldeira Rhodes, *Comparison of Methods for Determining the Half-Value Layer (Hvl) in Computed Tomography Equipment*
G02-031 - Nicolle Marques Cardoso Ignacio, *Comparison of the Dosimetric Properties of EBT-3 and EBT-4 Radiochromic Films for Application in Hemocomponent Dosimetry*
G02-032 - Fernanda Rocha Cavalcante, *Comparison of X-ray spectral correction software for a CdTe detector*
G02-033 - Guilherme Gazolla Santana, *Computational Modeling of an Anthropomorphic Phantom for Applications in Aerospace Dosimetry*
G02-034 - José Marques Lopes, *Computational Simulation of a NaI(Tl) Detector and Definition of Its Main Parameters for Application in a Neutron Activation Analysis Laboratory*
G02-035 - Lucas Beloti Silva, *Computational Tools for Automatic Conversion of Microscopy Images into Voxelized Cellular Phantoms*
G02-043 - João Vinícius Batista Valença, *Dose estimation in organs of a pediatric patient undergoing brain radiotherapy via Monte Carlo simulation*
G02-045 - Guilherme Cavalcante de Albuquerque Souza, *Dosimetric comparison of methods for obtaining dose profiles in head CT scans using 120 kV X-ray beams*
G02-046 - Frederico Possato Tagliaferro, *Dosimetry Of Electret Microphones Irradiated By X-Rays*
G02-047 - Josemary A. C. Gonçalves, *Energy Dependence of an Epitaxial Diode for Radioprotection Dosimetry*
G02-048 - Enzo Cortez, *Estimating the dose rate constant for Iodine-131 via Monte Carlo simulation: a first step towards pediatric radioiodine therapy shielding evaluation*
G02-050 - João Vinícius Batista Valença, *Evaluating Radiation Dose in Veterinary Professionals: MCNPX-Based Simulation of Equine Radiography Exposure*
G02-051 - Camila Nunes Carvalho Souza, *Evaluating the Performance of Bayesian Optimization in High-Dose-Rate Brachytherapy Planning*
G02-052 - Maria Rita Pascoal de Souza Barros, *Evaluation of Absorbed Dose Uniformity in Irradiated Foods with Different Packaging Geometries Using PMMA Dosimetry*
G02-054 - Gabriel Gomes do Nascimento, *Evaluation of the reproducibility of a new lens dosimeter holder*
G02-061 - João Victor Ramos Oliveira, *Influence of detector geometry on small-field radiotherapy simulations*
G02-066 - Daniel de Castro Pacheco, *Modeling and validation of a Whole Body Counter for in vivo measurements using CAD and MCNP6*

G02-070 - Greiciane de Jesus Cesário, *Monte Carlo Simulation of Depth Dose in Mammography Using Geant4 and 3D-Printed ABS and PETG Phantoms*
G02-071 - Kyssylla Monnyelle Araujo Silva, *Multielectrode Array Devices for dosimetry and dose enhancement factor analysis in nanoparticle radiotherapy*
G02-072 - Mateus de Souza Resende, *^{99m}Tc Radiolabeled Papain Nanoparticle: in silico Dosimetry Evaluation using DM BRA phantom and the experimental biodistribution data*
G02-074 - Henrique Chaves Gulino, *New Low Cost Process of PIN Diode Fabrication for Radiation Detection*
G02-077 - Carmen C. Bueno, *Online response of a homemade diode dosimetry system in a mobile 500-700 keV electron beam facility*
G02-078 - Anderson Manoel Bezerra da Silva, *Optimization of Mn^{2+} and Tb^{3+} Concentrations in $CaSO_4$:Mn,Tb and Advanced Investigation of its TL/OSL Dosimetric Properties*
G02-079 - Rômulo Monteiro Lopes, *Parametrization and Validation of a HPGe Detector by Monte Carlo Simulation*
G02-080 - Hugo Leonardo Lemos Silva, *Assessing Photon and Electron Ranges in Healthcare Waste for Radiation Sterilization*

Note: 10:00 to 12:30 - Referees for evaluation
On-line

G1 - Applications of thermoluminescence (dosimetry, dating, industrial, etc.)

G01-007 - Lenin Estuardo Cevallos Robalino, *Monte Carlo characterization of neutron personal dosimeter based on TLD-600 and TLD-700 irradiated with ^{252}Cf neutron source on the ISO-phantom at LPN-CIEMAT*
G01-010 - Abel Fernando Martínez Cruz, *Thermoluminescence of MgO Chemically Modified with Nd and Li Ions, Obtained by Glycine-based Solution Combustion Synthesis*
G01-013 - Hassan Salah, *Assessing Patient Exposure in PET/CT scans: A Head and Neck Dose Parameter Study*
G01-019 - Mohammed, *Evaluation of Radiation Dose Reduction Strategies in Whole-Body PET/CT Imaging*
G01-021 - Hassan Salah, *Optimizing Radiation Dose Parameters in PSMA PET/CT for Prostate Cancer: Balancing Diagnostic Efficacy and Patient Safety*
G01-022 - Abdelmoneim Sulieman, *Patient-Specific Dose Optimization in Breast PET/CT: Towards ALARA-Compliant Protocols*
G01-033 - Edwar Alonzzo Canaza Mamani, *Datación OSL-SAR y caracterización tecnológica de las cerámicas arqueológicas de los grupos de tradición ceramista Tupiguarani: Pirajuí I y Arataca II, Pernambuco, Brasil*
G01-034 - David Alexander Urbina Leal, *Design and Analysis of FDM Prototypes for Gold-198 Nanoparticles*

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

G02-001 - Alejandro Ferreira Tapia, *Caracterización Física y Computacional de Dispositivos Modeladores de Radiación Electromagnética mediante Simulación Monte Carlo*
G02-008 - Whoody Alem Wanderley Araripe Farias, *A platform for creating computational exposure models: personalized dosimetry with Holmium-166*
G02-009 - Whoody Alem Wanderley Araripe Farias, *All about MARTIN: a BREP phantom with blood and lymphatic vessels applied to Yttrium-90 radioembolization*
G02-012 - Aline Fabiane Gonçalves de Oliveira, *Geospatial assessment of radionuclide dispersion and associated effective dose modeled with ARTM View: a case study at CDTN's radiopharmaceutical unit*
G02-015 - Jardel Lemos Thalhofer, *Radiometric Profile Analysis of Soils from the São José de Itaboraí Paleontological Natural Park*
G02-016 - Marcelo Franklin da Silva Santos, *Synthesis and Luminescent Characterization of $CaSO_4$:Tm,Li via Slow Evaporation, Co-precipitation, and Solid-State Diffusion Routes*
G02-022 - Akemi Yagui, *Organ Dose Estimation in Pediatric Abdomen and Pelvis Radiographs Using Caldose_X Monte Carlo Simulations*
G02-025 - Ranulfo da Silva Dias, *Assembly and Geant4-based computational simulation of a Helium-3 de-*

tector for ground-level cosmic radiation monitoring
G02-029 - Daniel Vieira Lamas, *Characterization of the Ubatuba Formation: Integration of Gamma Radiometry, Total Organic Carbon, and Well Logs from the Upper Cretaceous to Lower Paleogene Intervals of the Campos Basin*
G02-037 - Paulo Sérgio de Abreu Junior, *Cytogenetic assessment of ionizing radiation exposure in Allium cepa using micronucleus frequency: a comparative study between Antarctic and control conditions*
G02-038 - Ana Laura Burin, *Development of a Computational Phantom of the Vertebral Column for Dosimetric Investigations*
G02-039 - Leonardo da Silva Boia, *Development of a Dedicated System for Generating Input Files for the MCNP and Adaptation of a Phantom with Edema for Simulation of Prostate Brachytherapy with* ^{125}I
G02-044 - Wallacy Viana, *Dose Rate Estimation by Extracting and Analyzing Colors from an Image Generated by a CMOS Image Sensor*
G02-059 - Lucas Fabricio de Araujo, *In Silico Investigation of the Impact of Gold Nanoparticles on the Radiosensitivity of 3D HeLa Cells Using the Monte Carlo Method*
G02-060 - Gabriela Rodé de Assis da Silva, *In Situ Gamma-Ray Spectrometric Mapping of Beach Sand Radioactivity in Niterói, Brazil*
G02-067 - Pedro Henrique Avelino de Andrade, *Monte Carlo Modeling of the BPW34 Photodiode for X-ray Dosimetry*
G02-073 - Nadia Rodrigues dos Santos, *Natural Radionuclide Activity and Radiological Risk Assessment in Brazilian Coffee Samples*
G02-086 - Marcelo Luís dos Santos Filho, *Study of* $CaSO_4$*:Tm,Li Based Luminescent Composites Applied to TL/OSL Dosimetry in Mammography*
G02-087 - Leonardo da Silva Boia, *MCNP simulation of the effects of prostatic edema in LDR brachytherapy with* ^{125}I *using the adapted ICRP 110 voxel male phantom*
G02-093 - Daniel V. Lamas, *Quantification of Gamma Radiation on Drilling Cuttings Through HPGe Detector: Comparative Study With Well Logs Spectrometry Data*

G3 - Ionizing and non-ionizing radiation

G03-005 - Samara Mendes Matos, *Effect of Neutron Activation in the Biodegradable Chitosan polymer associated with Gold Nanoparticles.*
G03-010 - Angel Garcia-Duran, *Simulador de detector de Germanio de alta pureza con fuentes radiactivas, embebido en FPGA*

G4 - Medical Physics

G04-001 - Nicolás Eugenio Martín, *Adaptation and performance of CdTe-based XFCT detection system using Ag-tacers for Biomedical Applications*
G04-007 - Priscila Santos Amorim, *Implementation of free software for managing imaging protocols*
G04-008 - Angelica Viridiana Roman Martinez, *Magnetic Resonance Imaging Assessment of Corpus Callosum Volume in Pediatric Epilepsy Patients*
G04-009 - Jaime Torres Juárez, *Optimization of the Pulse Sequence for Visualization of Myocardial Fiber Architecture Using Diffusion Tensor Imaging in the Study of Cardiomyopathies*
G04-012 - Gisell Ruiz Boiset, *Assessment of radiologically tissue-equivalent materials for breast imaging phantoms using X-ray spectrometry*
G04-024 - Jesús Emmanuel Mares Ponce, *Evaluation of elemental Iron content in pharmaceutical formulations using magnetic susceptibility technique*
G04-026 - Camila Hitomi Murata, *Evaluation of the affect on performance of a 3T MRI by the proximity of a subway station and an adjacent MRI scanner*
G04-027 - Carolina Salinas Domján, *Experimental characterization of magnetic fields for dosimetric modelling with Monte Carlo methods in MR-Linac radiotherapy*
G04-029 - Ana Carolina Costa da Silva, *Global Usability of Mobile CT Scanners in Rapid Stroke Diagnosis: A Literature Review*
G04-030 - Raissa Alexia Camargo Guassu, *Impact of Body Mass Index on Radiation Doses in Interventional Radiology*

G5 - Radiological Protection

G05-002 - Wallacy Viana, *Analysis of the characteristics of the Ingeo™ 3D850 polymer in the context of its use in visual inspections in the TRIGA IPR-R1 nuclear reactor*
G05-004 - Hericka O. Kenup-Hernandes, *Computational Modeling for Radiological Risk Assessment: Simulation of Plume Dispersion in Radioactive Waste Fires Using HotSpot*
G05-008 - Akemi Yagui, *Evaluation of radiation exposure in pediatric interventional cardiology procedures*
G05-009 - Hericka O. Kenup-Hernandes, *Gadolinium Adsorption by Functionalized Mesoporous Silicas (MCM- 41/SBA-15): A Proposal for Volume and Dose Reduction in Radioactive Waste*
G05-018 - Watila Lins Silva, *Reference Measurements of Thermal Neutron Flux Using a Cadmium-Encapsulated ^{6}LiI (Eu) Detector at the Neutron Metrology Laboratory*
G05-020 - Fábio Sabará Dias, *Survey of radiation exposure from portable X-ray fluorescence*

G6 - Radiation Sources

G06-005 - Gláucia de Oliveira Barreto, *Axial neutron flux mapping in the Argonauta Research Reactor at IEN-CNEN using neutron activation and γ-ray spectrometry techniques*
G06-008 - João Pedro Carvalho Andrade, *Determination of the mass attenuation coefficient of polymers using Monte Carlo N-Particle simulation*
G06-011 - Gabriel Tura Echeverria, *Identification and quantification of K, Na and Mn in textured soy protein sold in the city of Rio de Janeiro using the Argonauta research reactor of IEN/CNEN and the techniques of instrumental neutron activation analysis and γ-ray spectrometry*

G7 - Radiobiology

G07-004 - Ivana Mara Gomes Andersen Cavalcanti, *Determination of mechanical and physical properties of pig ear cartilage through modeling and simulation using DFT*
G07-014 - Yasmym Sarmento Pereira, *Radiometric analysis of sediments and water from the Lagoons of East Fluminense, Rio de Janeiro.*
G07-016 - Gabriela Rodé de Assis da Silva, *Radiometric characterization of Erythrina verna Vell.: A study of natural radioactivity in plant tissues*

G8 - Luminescent materials

G08-003 - Marcelo Luis dos Santos Filho, *Development of $CaSO_4$:Li Composites by Different Synthesis Routes for Applications in TL and OSL Dosimetry*
G08-005 - Brunno Florencia de Oliveira, *Dosimetric Potential of Cowrie Shell Biomass under High Radiation Doses using TL and OSL Techniques*
G08-018 - Caroline Pereira dos Santos, *Repuesta Termoluminiscente ante rayos gamma de la HAp pura y dopada con tierras raras (Dy3+ y Eu3+) a dos concentraciones 0.1 M y 0.5 M*
G08-019 - André Luiz Tavares e Silva, *Unsupervised learning for pattern recognition in K-feldspar using cathodoluminescence emission and EDX composition mapping*
G08-021 - Joel Arturo Rivera-Garcia, *Estudio de las propiedades luminiscentes del aluminosilicato de estroncio dopado con Ce ($SrAl_2Si_2O_8$:Ce) sintetizado por el método de reacción en estado sólido*

12:30 LUNCH

Applications of Thermoluminescence / Luminescent Materials Virtual Room 1 Chair: Daniel Calheiro/Rodrigo Gadelha	Luminescent Materials / Dosimetry Virtual Room 2 Chair: Marco A. Labarrios/ Álvaro M. L. Gómez	Medical Physics / Ionizing and non-Ionizing / Radiobiology Virtual Room 3 Chair: Jaime T. Juárez/Bruno Gallotti

13:30 Katherine G. Delgado
G01-005 - *Effective dose received by Dentistry students in intraoral dental radiology using TLD dosimetry*

13:45 Nayely D. M. Ramirez
G01-008 - *Synthesis of* Y_2O_3 *nanophosphors and evaluation of their thermoluminescent response for use in radiation dosimetry*

14:00 Abel Fernando Martínez Cruz
G01-009 - *Thermally Stimulated Luminescence of Chemically Modified Magnesium Oxide with* Gd^{3+} *and* Li^{+}*, Obtained by a Glycine-based Combustion Process*

14:15 Radmila Sabitova
G01-016 - *Characterization of additional irradiation positions in the IVG.1M research reactor*

14:30 E. Cruz-Zaragoza
G01-020 - *Optically -and thermally stimulated luminescence detection of irradiated oregano (Lippia graveolens) polymineral*

14:45 Umme Muslima
G01-023 - *Thermoluminescence Characterization of Natural Flake Graphite Under 6 MeV Electron Beam Exposure from Medical LINAC for Dosimetric Applications*

15:00 André Luiz Tavares e Silva
G01-025 - *A Data Science Approach on the Study of* Al_2O_3 *Thermoluminescent Glow Curves Under Varying Heating Rates*

15:15 Gabriela Pontes Cardoso
G08-001 - *Cathodoluminescence behavior of PVDF films doped with* $CuSO_4$ *under high-dose gamma irradiation*

13:30 Jessica Paola C. Fraga
G02-005 - *La termoluminiscencia del Grafito para su posible aplicación dosimétrica*

13:45 José Edgardo Arellano Hernández
G02-004 - *Estudio de la respuesta termoluminiscente de NA2O-Y2O3-P2O5-SIO2 irradiados con Electrones*

14:00 Marcelo F. da Silva Santos
G02-014 - *New Approaches in the Synthesis of* $CaSO_4$*:Tm,Li Phosphors with Sugars as Chelating Agents for Dosimetry Applications*

14:15 Vinícius R. A. A. Martins
G02-055 - *Experimental Evaluation of the Mechanical and Dosimetric Properties of the Red Perspex 4034 Dosimeter After 15 Years of Aging*

14:30 Rayline Leite da Silva
G02-084 - *Standardized Protocol for the Analysis of Cutting Samples and Lithological Materials by Gamma Spectrometry*

14:45 Nadia Rodrigues dos Santos
G02-049 - *Estimation of the soil-to-plant transfer factor of natural radionuclides in organic crop from Pedra Branca State Park*

15:00 Leonardo da Silva Boia
G02-036 - *Construction of the Glioblastoma scenario in the right optic nerve region using DIP techniques for simulation in the MCNP code with conversion to lattice structure by SCAN2MCNP*

15:15 Leonardo Santiago Melgaço Silva
G02-041 - *Development of an optical sensor prototype for dose readings in Fricke gel dosimeters*

13:30 Erika Patricia Azorín Vega
G04-005 - *Dosimetric and Biological Characterization of a 3D Cell Culture Model for the Evaluation of Radiopharmaceutical Induced Bystander Effect*

13:45 Leticia González Zamora
G04-010 - *Study of Prefractal Antennas for Multiband Radiofrequency Applications*

14:00 Maria L. Miranda Maciel
G04-014 - *Calibration of radiochromic films EBT2, EBT3, EBT4 and RTQA2 and assessment of uncertainties for radiotherapy doses*

14:15 Elsa Bifano Pimenta
G04-022 - *Evaluating Lung CT Protocols Using the Detectability Index: An Anthropomorphic Phantom-Based Evaluation of Photon-Counting and Conventional CT*

14:30 Estefania Reyes Soto
G04-002 - *Automated Classification and Manual Segmentation of Pediatric Brain Tumors Using Deep Learning on MRI Data*

14:45 Giulia Rita de Souza Faés
G04-035 - *Radiotherapy Planning with VMAT in Halcyon 2.0: Impacts of Isocenter Configuration on Dosimetric Parameters and Treatment Efficiency*

15:00 Natalia Matos Orozco
G04-003 - *Diseño y cálculo de blindajes para garantizar la protección radiológica en una radiofarmacia productora de radiofármacos*

15:15 Kuanysh Samarkhanov
G05-016 - *Radiological Characterization and Dosimetric Assessment of Liquid Radioactive*

15:30 Júlia Carina Orfão Costa
G08-002 - *Characterizations of $SrMoO_4:Dy^{3+}$ powders synthesized by hydrothermal method*

15:45 Daniele da Silva Machado
G08-006 - *Dosimetric Properties of nanoDot OSL Detectors for Radiotherapy Quality Control: An Initial Characterization and Uncertainty Analysis*

16:00 Carina Oliva Torres Cortés
G08-023 - *Repuesta Termoluminiscente ante rayos gamma de la HAp pura y dopada con tierras raras (Dy3+ y Eu3+) a dos concentraciones 0.1 M y 0.5 M*

16:15 Caroline Pereira dos Santos
G08-017 - *Uncovering the intrinsic nature of the ternary metallic alloys CuNiPd and Cu_2PPd_2: A DFT study*

16:30 Rodrigo Martínez Baltezar
G08-022 - *Point defects study on Aluminum nitride for applications in luminescence dosimetry*

16:45 Pedro R. González
G08-024 - *Thermoluminescent properties of MgO:Sm,Tb+PTFE dosimeters*

17:00 Sandrith M. Pérez Ramos
G01-003 - *Análisis cinético de la curva de brillo termoluminiscente del Óxido de Berilio (BeO) irradiado con Rayos X en un rango de baja dosis*

15:30 Carolina Salinas Domján
G02-090 - *Assessing Water-Equivalence in Gram-Negative Bacteria-Hydrogel Radiation Dosimeters: A Comparative Analysis with Gram-Positive Bacterial Systems*

15:45 Rayline Leite da Silva
G07-017 - *Study of Drinking Water Quality in Rio de Janeiro: Assessment of Gamma Radiation Levels and Physicochemical Parameters*

16:00 Teodoro Córdova Fraga
G04-019 - *COVID largo y fibrosis pulmonar: clasificación basada en tomografía en población mexicana*

16:15 Teodoro Córdova Fraga
G04-013 - *Automated detection and characterization of renal calculi using U-Net and Hounsfield unit analysis in computed tomography scans*

16:30 Priscila Santos Rodrigues
G04-036 - *Three-dimensional dose mapping of gold-198 nanoparticles: A comparative study of 3D-printed matrices using Fricke xylenol gel and Monte Carlo simulations*

16:45 Arthur Reis Martins
G06-015 - *Production of Radioisotopes in Nuclear Microreactors for Medical, Industrial, and Scientific Applications*

17:00 Jardel Lemos Thalhofer
G02-010 - *Assessment of the level of natural radioactivity in the region of the future Municipal Environmental Protection Area (APA) Ilha Verde, Muriqui, Mangaratiba - RJ*

Waste from BN-350 Reactor

15:30 Stanislav Svetachev
G06-007 - *Determination of detection limits of elements in sediment and rock standards for INAA using the IGR reactor*

15:45 Jonathan O. dos Santos
G06-017 - *Response Function Validation of HPGe Detector Through GADRAS and MCNP Simulation*

16:00 Roberto Méndez Villafañe
G06-003 - *Neutron measurements around the research reactor RA-6 at Centro Atómico Bariloche with a Bonner Sphere Spectrometer*

16:15 Roberto Méndez Villafañe
G06-001 - *Development of a detailed model with MCNP of the Neutron Standards Laboratory at CIEMAT and validation with a Bonner Sphere Spectrometer System*

16:30 Fatima Castañeda Ortiz
G07-001 - *Pulsed X-ray-Induced Nuclear Fragmentation in a Radioresistant Ovarian Cancer Cell Line*

16:45 Daniele G. Mesquita
G07-006 - *Gamma Radiation Modulates Genomic Maintenance Pathways in T-Lymphocytes: An Analysis of Clinical Doses*

17:00 Emilio Romero Muñoz
G07-007 - *Germination response of 'Micro-Tom' to controlled magnetization by using a Rodin coil*

17:15 Raúl Ken Delgado Velázquez
G07-010 - *Magnetic stimulation as a promoting factor for germination and development in Guazuma ulmifolia L. seeds*

13:30 to 16:00 TECHNICAL VISIT **Chair: Camila Engler**

Places to visit at CDTN:

- Gamma Irradiation Laboratory - More information, acess here
- Dosimeter Calibration Laboratory - More information, acess here
- TRIGA IPR-R1 Reactor - More information, acess here

Thursday, 09/25/2025

Invited Speaker - Lumina Auditorium **Chair: Bernardo Dantas**

08:00 *Invited Speaker* – Luciana Carvalheira (IEN/CNEN-Brazil),
Neutrons, Knowledge, and New Generations: How Brazil's Argonauta Reactor is Shaping the Future of Radiochemistry and Its Impact on Global Dosimetry Challenges.

09:00 *Invited Speaker* – Bernardo Maranhão Dantas (IRD/CNEN-Brazil),
Diseminación de técnicas de monitoreo in vivo ocupacional en una Red de Clínicas de Medicina Nuclear.

09:25 *Invited Speaker* – Maria Jose Neves (CDTN/CNEN-Brazil),
Range Radiation Exposure Indicator.

10:00 COFFEE BREAK

Invited Speaker - Lumina Auditorium **Chair: Telma Cristina Ferreira Fonseca**

10:30 *Invited Speaker* – Ademar Lugão (IPEN/CNEN-Brazil),
Use of ionizing radiation for the production of nanoparticles and advanced curatives.

11:30 *Invited Speaker* – Richard Hugtenburg (Swansea University-United Kingdom),
New Technologies as a Route to Particle Therapy: Laser-Hybrid Acceleration for Radiobiology Applications – The LhARA Project.

12:30 LUNCH

Invited Speaker - Lumina Auditorium **Chair: Richard Hugtenburg**

13:30 *Invited Speaker* – Hélio Yoriyaz (IPEN/CNEN-Brazil),
Proton Beam Therapy: Main Characteristics and Challenges.

Dosimetry
Spectrum Auditorium
Chair: Peterson Squair

13:30 Divanizia do Nascimento Souza
G02-042 - *Dose Distribution in an Anthropomorphic Cervix Phantom Using Two Computer Modeling Approaches*

13:45 Francisco Harlley Dantas Hauradou Xavier
G02-057 - *Generator of Health Optimized Simulation Templates – GHOST a 3D Slicer plugin to voxelized medical images for MCNP*

14:00 Francisco H. D. Hauradou Xavier
G02-094 - *Optimization of dose rate map acquisition time for internal dosimetry in nuclear medicine using 3D U-Net convolutional neural networks*

14:15 José Elizeu de Souza Junior
G02-062 - *Influence of Photon Emission Spectra on Air Kerma Rate Constants for Radionuclides: A Monte Carlo EGSnrc Study*

Medical Physics/Dosimetry
Lattice Auditorium
Chair: Lucas Paixão

13:30 Ana Beatriz Caldeira de Andrade
G04-025 - *Evaluation of spatial resolution in PET equipment with a Jaszczak-analog phantom built by additive manufacturing*

13:45 Crislane de Jesus Cesário
G04-028 - *Experimental Evaluation of 3D-Printed Materials (ABS and PETG) for Mammographic Phantom Applications Using $CaSO_4$:Dy Dosimetry*

14:00 Nathalia Luzia Aidar Alves
G04-032 - *Machine Learning in PSQA: Performance of a Random Forest Model on Radiotherapy Metrics*

14:15 André Luiz Espindola Fidelis
G04-037 - *Validation of a 3D-Printed Anthropomorphic Eye Phantom for Quality Assurance in External Beam Radiotherapy Using TLDs and Radiochromic Film*

14:30 Leonardo Gualberto Silva Macedo
G02-095 - *Mass Analysis of Computational Mouse Models Developed at CDTN: A Comparative Study of Male and Female Versions*

14:45 Beatriz Diniz de Oliveira Guedes
G02-082 - *Spectrophotometric Analysis of a 3D-Printed Photopolymerizable Resin Exposed to High-Dose Gamma Radiation*

15:00 Hirys Sales
G02-053 - *Evaluation of Polymeric Materials as Muscle Tissue Equivalents for Physical Phantoms Used in the Calibration of In Vivo Monitoring Systems*

13:30 - 17:00 - Poster Session (Main Hall)

Note: 14:30 to 17:00 - Referees for evaluation
In-Person

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

G02-081 - Felipe Mamed de Souza, *SMS Notification System for Potential Radiological Risk Scenarios*
G02-085 - Alcilene Cristina da Silva, *Study and Characterization of the EPR Response of New Materials for High Dose Dosimetry*
G02-088 - Ana Letícia Almeida Dantas, *The IRD In Vivo Monitoring Laboratory - LABMIV: Instrumental Capabilities applied in routine and emergency situations*
G02-091 - Maryanna Regina de Souza Roberto, *OSL Tandem Dosimeter for ICRU 95 Operational Quantities*
G02-092 - Thalys Matheus Gama de Oliveira, *Photons Response of an Albedo Monitor*

G3 - Ionizing and non-ionizing radiation

G03-001 - Paula Regina Rodriguez Coelho, *"Analysis of spatial resolution in two different phantoms as a quality parameter in MRI pediatric pathology"*
G03-002 - José Marques Lopes, *Adjustment Factor for Gamma Spectra Due to the Influence of Sample Density on the Shielding Background Radiation*
G03-003 - Davyd Soares Dunda, *Application of Steam Methodology in Science Teaching: Prototyping a Cloud Chamber*
G03-004 - José Marques Lopes, *Comparison of environmental sampling results using portable and benchtop scintillation detectors*
G03-006 - Adriana de Souza Medeiros Batista, *Effect of repolymerization and gamma irradiation on the morphology of styrene-divinylbenzene composites with magnetite*
G03-007 - Paulo Cezar Rocha Silveira, *Gamma Irradiation Effects on the Mechanical and Ballistic Performance of 3D-Printed Polylactic Acid*
G03-008 - Luciana Carvalheira, *Gold nanoparticles with potential for use in multimodal combat of head and neck cancer*

G4 - Medical Physics

G04-006 - Otávio Gabriel Moreira Silva, *SNR and CTDIvol study as a function of the Noise Index in Computed Tomography exams*
G04-016 - Rute Cristina de Oliveira Soares, *Chest Computed Tomography in Oncology: Clinical Contributions and Dosimetric Assessment*

G04-017 - Gustavo Freire Pereira da Silva, *Comparative Validation of the DEEDZ Methodology and a Photon-Interaction-Based Model Implemented in TEMPy for Estimating Effective Atomic Number and Electron Density*
G04-018 - Rute Cristina de Oliveira Soares, *Contributions of Computed Tomography to Breast Cancer Staging and Follow-Up: Justification of Practice and Dosimetric Analysis*
G04-021 - Iasmin Vieira Nishibayaski, *Dosimetric evaluation of EBT3 and EBT4 in superficial HDR brachytherapy for keloids*
G04-023 - Camila Engler, *Evaluation of Breast Positioning Quality and Its Association with Technical Parameters in Screening Mammography*
G04-031 - Douglas Philipe Martinho Dos Santos, *Investigation of Diagnostic Reference Levels (DRL) in medical radiography: dose-area product (DAP)*
G04-034 - Kellen Adriana Curci Daros, *Radiation dose from pediatric coronary artery calcium scoring CT scan to evaluate chronic kidney disease-mineral and bone disorder: Observational study*
G04-038 - Arícia Ravane Pereira da Cruz, *Validation of an alanine-based postal dosimetry system for quality control in radiotherapy*

G6 - Radiation Sources

G06-013 - Kathleen Martins, *Neutronic Analysis of an Accelerator-Driven Subcritical Reactor Utilizing Thorium Dioxide and Reprocessed Nuclear Fuels*

G7 - Radiobiology

G07-003 - Christiana da Silva Leite, *Comparative Study of Organ Mass in Healthy and Tumor-Bearing Mice for Preclinical Dosimetry Purposes*
G07-005 - Camila Ramos Silva, *Development and validation of a body-shielding device for radiotherapy of breast cancer in mice using a panoramic gamma source*
G07-008 - Luciana Penedo de Melo Teixeira, *Impact of Physical Exercise on Irisin Expression in the Brain of Irradiated Mice: Effects on Behavior and Memory*
G07-009 - Olania Herrera González, *Involvement of SMARCB1 subunit of the mSWI/SNF and GLI-1 epigenetic complex and their radiolabeling as a strategy for ionizing therapy in lung cancer*
G07-011 - Camila de Almeida Salvego, *Nitric Oxide-Donor Nanoparticles Potentiate Radiotherapy of TripleNegative Breast Cancer*
G07-018 - Bruno Gallotti, *Growth Inhibition of S. cerevisiae BY4741 Exposed to Gold Nanoparticles and Low-Energy Radiation*
G07-019 - Maurício Moacir da Silva Borges, *The Role of Caffeine in Reducing TNF-α and Improving Cognitive in Irradiated Mice in the Brain*
G07-020 - Luis Martín Garcia Ortiz, *Cellular response in human leukocytes (in vitro), due to exposure to UVB radiation using apoptotic biomarkers*

Invited Speaker - Lumina Auditorium — Chair: Teógenes da Silva

14:30 *Invited Speaker* – Georgia Joana (CNEN-Brazil) and Ketrim Souza (INB-Brazil),
WiN Brasil - Diversity & Inclusion Management.

15:30 COFFEE BREAK

16:00 - Mentorship Helen Khoury WiN Brasil (Spectrum Auditorium)

Empowering the Future of Nuclear Energy: A Mentorship Program for the Development of Female Leadership
Moderator: Telma Cristina Ferreira Fonseca (DEN/UFMG-Brazil),
Panelists: Ketrim Souza and Winners (WiN Brasil)

Roundtable Session (Lumina Auditorium)

16:00 Leadership in Scientific and Technological Research
Moderator: Luciana Carvalheira (IEN/CNEN-Brazil),
Panelists: Adriana Marques and Anna Letícia Barbosa de Sousa

Radiation Protection
Lumina Auditorium
Chair: Teógenes da Silva

17:00 Mirella Maturo da Cruz
G05-021 - *Transfer Factors of Natural Radionuclides in Organic Foods from Rio de Janeiro, Brazil*

17:15 Elydio J. D. Soares
G05-010 - *Investigation of radon exhalation from mortars made from niobium tailings and the mitigating role of coating materials*

17:30 Raphael Francisco Gomes dos Santos
G05-005 - *Determination of Radiation Protection Procedures for an Offshore Plant Inspection in Brazil Using Radiotracer Techniques*

Radiation Sources / Radiation Protection
Lattice Auditorium
Chair: Jhonny A. Benavente Castillo

17:00 Gabriel do Nascimento Freitas
G06-014 - *Nuclear simulation workflow to correct gamma ray well logs for radioactive contaminated drilling fluids*

17:15 José Alípio dos Santos Filho
G06-016 - *Proposal for the construction of a neutron generator for application in research and industry*

17:30 Welen Nunes de Lima
G05-014 - *Proposal for the use of thorium as fuel in TRIGA® research reactors*

Friday, 09/26/2025

Invited Speaker - Lumina Auditorium **Chair: Richard Hugtenburg**

08:00 *Invited Speaker* – Rolf Behrens (PTB-Germany),
From dosemeter development to routine use – Standards and Uncertainties.

08:40 *Invited Speaker* – Susana Lalic (DFI/UFS-Brazil),
Plant Stem Cells as an Ethical Alternative for Biodosimetry.

09:20 *Invited Speaker* – Thiago Viana Miranda Lima (University of Lucerne-Switzerland),
Current and Future Trends in Nuclear Medicine Dosimetry.

10:00 COFFEE BREAK

Radiological Sources
Spectrum Auditorium
Chair: Carmen C. Bueno

10:30 Matheus R. do Nascimento
G03-009 - *Speknife: open-source software for X-ray spectra correction in solid state dosimetry*

10:45 Eduardo B. de Paula
G05-011 - *Metrological Enhancement of Radiopharmaceuticals Half-Lives Determination via γ-ray Spectrometry and Algorithmic Data Processing*

11:00 Igor Andrade Machado
G02-083 - *SpecUnPy code: updates and what is coming*

Dosimetry
Lumina Auditorium
Chair: Peterson Squair

10:30 Jhonny A. Benavente Castillo
G02-007 - *Simulación del blindaje del búnker de un acelerador ciclotrón utilizando el código MCNP*

10:45 Natasha Briggs e Silva
G02-011 - *Preliminary Evaluation of a Positioning Phantom for Dosimetric Audit in Ir-192 HDR Brachytherapy: Results from an IAEA Collaborative Project*

Invited Speaker - Lumina Auditorium **Chair: Bruno M. Mendes**

11:30 *Invited Speaker* – Tarcísio P. R. de Campos (DEN/UFMG and CDTN/CNEN-Brazil) and Alberto Mizrahy Campos (Augsburg-Germany),
Challenges in Radioisotope Decontamination and Production through Neutron Spectral Selection.

12:30 LUNCH

Dosimetry

Spectrum Auditorium
Chair: Maria da Penha A. Potiens

13:30 Isabela N. S. Ferreira
G02-064 - *Mapping of natural radionuclides U, Th and K in the city of Belo Horizonte/Minas Gerais*

13:45 Claudio A. Federico
G02-065 - *Measurements of Ionizing Radiation Dose Onboard*

Dosimetry

Lattice Auditorium
Chair: Bruno M. Mendes

13:30 Celso G. de Paulo
G02-069 - *Monte Carlo simulation of a HPGe detector*

13:45 Ana G. Leão Ferro
G02-075 - *Occupational exposure assessment in ROLL procedures with Tc-99m: A Monte Carlo simulation approach*

Applications of Thermoluminescence
Lumina Auditorium
Chair: Teógenes da Silva

13:30 Tarcísio P. R. Campos
G01-046 - *The reduction of the radiological impact of nuclear fuel waste*

14:00 Júlia B. Severo
G01-039 - *Environmental Gamma Radiation and Radon Exposure in Areas Surrounding Urban*

an Aircraft from Southeastern Brazil to the Antarctic Continent

14:00 Paulo J. Iack Silva
G02-068 - *Monte Carlo N-Particle (MCNP) Simulations for Scintillation Emission in Quantum Dot-Doped Systems*

14:00 Rafael S. Oliveira
G02-076 - *On the adaptation of a commercial Bonner Sphere Spectrometer to be used with thermoluminescent detectors*

Quarries in Belo Horizonte, Brazil

Invited Speaker - Lumina Auditorium **Chair: Telma Cristina Ferreira Fonseca**

14:30 *Invited Speaker* – Carmen Bueno (IPEN/CNEN-Brazil),
Radiation processing with electron beams: On the feasibility of using electronic dosimeters for real-time dosimetry.

15:30 COFFEE BREAK

16:00 CLOSING SESSION

- Honorary Award in Memory of Helen Jamil Khoury: Carmen Bueno
- Helen Khoury Award for Best Works: Andrea Mantuano & Ester Andrade
- Final Session: Summary and Perspectives: Telma Cristina Ferreira Fonseca & Bruno M. Mendes

CHAPTER 6

Pre-Symposium Courses

6.1 DEVELOPMENT OF PET RADIOPHARMACEUTICALS

Pre-symposium course speakers

- Dr. Marina Bicalho (CDTN/CNEN-Brazil)
- Special Guest: Dr. Isabel Dregely (Fachhochschule Technikum Wien-Austria)
- Special Guest: Dr Emerson Bernardes (IPEN/CNEN-Brazil)

Dr. Marina Bicalho Silveira is an Industrial Pharmacist (UFMG, 2007) with an M.Sc. (2010) and Ph.D. (2016) in Radiation Science & Technology (CNEN). Specialized in Radiopharmacy (SBRAFH, 2023), she works as a Senior Technologist at CDTN/CNEN, leading Quality Assurance for PET radiopharmaceutical production. Her expertise covers GMP, regulatory compliance, and R&D of novel radiopharmaceuticals, including preclinical studies. She also serves as a Professor in CDTN's Graduate Program, bridging industry and academia.

Dr. Isabel Dregely is Head of the Competence Center for Artificial Intelligence and Data Analytics at Fachhochschule Technikum Wien. She previously served as Senior Lecturer at King's College London and was a UKRI Future Leaders Fellow. Her research background includes advanced MRI techniques, particularly in prostate and cardiovascular imaging, with postdoctoral work at UCLA and experience at the Technical University of Munich. She holds a PhD in Physics from the University of New Hampshire, where she focused on hyperpolarized xenon MRI and RF coil development. Dr. Dregely combines expertise in biomedical imaging with AI and data analytics applications.

Dr. Emerson Soares Bernardes is a pharmacist and researcher with over two decades of experience in immunology, molecular biology, and nuclear medicine. He served as Manager of the Radiopharmacy Center at the Nuclear and Energy Research Institute (IPEN), where he led pioneering projects in the development of radiopharmaceuticals. He currently holds a position as a Visiting Researcher at Erasmus Medical Center in the Netherlands, contributing to innovation initiatives and patent development. His career is distinguished by the integration of scientific research, strategic management, and entrepreneurship.

Description: This course covers the basics of PET radiopharmaceuticals-production, clinical applications, and their impact on nuclear medicine-while showcasing CDTN'S cutting-edge technologies and their role in enhancing diagnostics and patient care in Brazil.
Dr. Marina Bicalho - Theoretical Aspects and State of the Art in PET Radiopharmaceuticals.
Dr. Emerson Bernardes - Advances in new PET Radiopharmaceuticals
Dr. Isabel Dregely - Artificial Intelligence Applied to PET Imaging

6.2 INTRODUCTION TO RADIATION DETECTORS

Pre-symposium course speakers

- Dr. Marco Antonio Polo Labarrios (UAM-Mexico)
- Dr. Álvaro M. L. Gómez (DEN/UFMG-Brazil)

Dr. Marco Antonio Polo Labarrios holds a Ph.D. and Master's degree in Energy Engineering from UNAM and a Bachelor's in Energy Engineering from UAM Iztapalapa. He teaches part-time at Universidad Iberoamericana and UNAM, focusing on Transport Phenomena and Renewable Energy. His research involves the simulation of energy processes based on mass, momentum, and energy transport. He has published 21 scientific articles, participated in multiple national and international conferences, and supervised four undergraduate theses. Dr. Polo Labarrios was part of the National Commission on Nuclear Safety and has been a Level 1 member of the National System of Researchers since 2021.

Dr. Álvaro M. L. Gómez holds a Ph.D. and M.Sc. in Nuclear Science and Technology from the Federal University of Minas Gerais – UFMG (2017; 2021). Holds a Bachelor's degree in Physics (diploma validated by UFMG) and two postgraduate specializations: Bioengineering – Universidad Distrital Francisco José de Caldas (2008; 2014) (Colombia), and University Teaching – Universidad Cooperativa de Colombia (2013). Has research experience in ionizing radiation dosimetry, development of tissue-equivalent materials for computed tomography phantoms, material characterization, and teaching experience at the Department of Nuclear Engineering (DEN) at UFMG, lecturing on radiation protection, radiation detection and nuclear instrumentation, radioisotope applications, and introduction to nuclear energy.

Description: Throughout the course, we will explore different types of radiation detectors, their operating principles, and criteria for their proper selection and use. The goal is to provide students with a solid foundation that enables them to identify the most suitable equipment for each situation and use it efficiently, maximizing both measurement accuracy and radiological protection.

6.3 IMPORTANCIA DE LA FÍSICA EN UNA IMAGEN POR RESONANCIA MAGNÉTICA EN EL DIAGNÓSTICO CLÍNICO.

Pre-symposium course speakers

- Dr. Silvia S Hidalgo Tobón (UAM-Mexico)
- Msc. Jaime Torres Juárez (UAM-Mexico)

Dr. Silvia Hidalgo earned her Bachelor's degree in Physics in 1998 from the University of the Americas-Puebla, Mexico. In 2000, she earned her Master of Science in Medical Physics from the Institute of Physics at the National Autonomous University of Mexico (the first Master of Science in Medical Physics from the UNAM program). In 2005, she earned her PhD in Physics, specializing in Magnetic Resonance Imaging from the Sir Peter Mansfield Magnetic Resonance Center at the University of Nottingham, England. She completed a postdoctoral fellowship in Magnetic Resonance Imaging at the Robarts Research Institute, University of Western Ontario, Canada. She also completed a postdoctoral fellowship at the Autonomous Metropolitan University, Iztapalapa. Her scientific interests are related to advanced functional magnetic resonance imaging, neural networks applied to neuroscience, and magnetic resonance safety in daily clinical practice. She has taught courses and supervised theses at the undergraduate, master's, and doctoral levels in magnetic resonance imaging at the UAM, UAEM, and UNAM. She has given lectures at various national and international forums on nuclear magnetic resonance imaging. She is a member of the International Society of Magnetic Resonance in Medicine, RSNA, the Mexican Physical Society, IEEE, and the Neuroscience Society, among others.

Descripción: Se presentarán las bases de la física de una imagen por resonancia magnética y su relación con el radio diagnóstico. La seguridad en resonancia magnética que evita complicaciones y accidentes que con la continua capacitación disminuyen notablemente. Así como las técnicas nuevas que están incluyendo inteligencia artificial para un mejor diagnóstico por imagen.

6.4 SOLID STATE DOSIMETRY: PRINCIPLES AND APPLICATIONS & OSL ADVANCED MATERIALS.

Pre-symposium course speakers

- Dr. Ivonne Berenice Lozano Rojas (IPN-Mexico)

Dr. Ivonne, a distinguished Mexican scientist holding a Ph.D. from Centro de Investigación en Ciencia Aplicada y Tecnología Avanzada, Unidad Legaria, conducts pioneering research in TL/OSL dosimetry, radiation detectors, and food irradiation detection, significantly advancing medical physics and safety.

Description: This minicourse focuses on quartz as a natural radiation dosimeter, exploring its defects and their role in optically stimulated luminescence (OSL). Advanced OSL materials, including synthetic and natural quartz, are analyzed for radiation dose assessment in dating and dosimetry applications.

CHAPTER 7

Symposium Speakers

7.1 DR. ADEMAR BENÉVOLO LUGÃO: USE OF IONIZING RADIATION FOR THE PRODUCTION OF NANOPARTICLES AND ADVANCED CURATIVES.

Dr. Ademar Lugão holds Bachelor’s in Chemical Engineering from the Polytechnic School of the University of São Paulo – USP (1980), earned a Master’s degree in Nuclear Technology from IPEN-USP (1985), and a Ph.D. in Nuclear Materials Technology, also from IPEN-USP (2004). With more than 30 years of experience as a research leader in polymer irradiation, he is currently the Manager of the Center for Chemistry and the Environment at the Nuclear and Energy Research Institute – IPEN. His main research focuses on the synthesis, processing, and characterization of polymers; structural modification of polymers through energetic processes, including crosslinking, grafting, branching, and chain scission; development of hydrogels for health and environmental applications; clean technologies for producing polyolefins with controlled rheology; starch foams for sustainable packaging; advanced low-cost dressings tailored to the needs of the Brazilian Public Health System (SUS); albumin and papain nanoparticles for encapsulating radiotherapeutics and chemotherapeutics; and radioactive gold nanoparticles (Au-198) for applications in radiodiagnosis and radiotherapy (theranostics). He currently serves as the Coordinator of the Nanomaterials and Nanocomposites Coordination Hub of SISNANO (National System of Nanotechnology Laboratories) of the Ministry of Science, Technology, and Innovation (MCTIC).

Description: Use of ionizing radiation for the production of nanoparticles and advanced curatives.

7.2 DR. BERNARDO DANTAS: DISEMINACIÓN DE TÉCNICAS DE MONITOREO IN VIVO OCUPACIONAL EN UNA RED DE CLINICAS DE MEDICINA NUCLEAR

Dr. Bernardo Maranhão Dantas holds a PhD in Sciences (Biology – Nuclear Biosciences) from the State University of Rio de Janeiro (1998). He is a Senior Technologist at the Brazilian National Nuclear Energy Commission (CNEN) and a faculty member of the Graduate Program at the Institute of Radiation Protection and Dosimetry, focusing on Radiation Biophysics. He has published 87 peer-reviewed articles and has supervised numerous students across undergraduate, master’s, and doctoral levels. He has led five research and development projects and currently coordinates the CNPq-certified research group “Occupational Internal Dosimetry.”

Description: Professionals who handle radiopharmaceuticals for diagnostic and therapeutic purposes in nuclear medicine are subject to the intake of radionuclides through inhalation and ingestion. The radiological protection aspects of this practice are regulated in Brazil by the National Nuclear Energy Commission (CNEN). The International Atomic Energy Agency (IAEA) recommends the implementation of occupational monitoring programs when there is a risk of annual effective doses exceeding 1 mSv. The control of radionuclide intake can be carried out using internal dosimetry techniques. Currently, in Brazil, there are not enough laboratories capable of providing internal monitoring services to meet the entire demand if this requirement were to be enforced by CNEN. This paper presents an overview of the current situation in Brazil and economically viable technical alternatives, aiming to make the implementation of routine internal monitoring programs for workers occupationally exposed to significant risks of radionuclide exposure in nuclear medicine feasible.

7.3 DR. CARMEN BUENO: RADIATION PROCESSING WITH ELECTRON BEAMS: ON THE FEASIBILITY OF USING ELECTRONIC DOSIMETERS FOR REAL-TIME DOSIMETRY

Dr. Carmen holds Bachelor's and Licentiate degrees in Physics from PUC/SP, as well as Master's and Doctorate degrees in Nuclear Instrumentation from the same university. She began her academic career at PUC/SP, where she was appointed as a Full Professor of Nuclear Physics. In 1995, she started working as a researcher at the Nuclear and Energy Research Institute (IPEN-CNEN/SP) and later completed her postdoctoral work at the Instrumentation and Experimental Particle Physics Laboratory (LIP) at the University of Coimbra. She also holds a second Doctorate in nuclear technology from IPEN-USP. She has published nearly 150 full-length articles in international journals and conference proceedings and has supervised numerous master's, doctoral, and postdoctoral students. Currently, she is a senior researcher at IPEN with over 30 years of experience in nuclear instrumentation for radiation measurement and dosimetry. Most of her research focuses on developing semiconductor-based dosimetry systems for medical and radiation processing fields. She currently serves on the Steering Committee of the National Institute of Science and Technology (INCT) project, "Nuclear Instrumentation and Applications in Industry and Health" (INAIS). Additionally, she is a member of the Advisory Boards of the Brazilian Society of Solid-State Dosimetry (SSD) and the Brazilian Association of Nuclear Energy (ABEN).

Description: In this lecture, we will briefly introduce the radiation processing of products, a relatively new industry with wide applications and significant commercial success, primly involving gamma rays and electron beams (EB). The main features of the irradiation facilities, the doses delivered, and the regulatory requirements for the processed products will be discussed to highlight the role of dosimetry in this field. Several passive solid-state dosimeters, which are well-established in radiation processing dosimetry, will be introduced to emphasize the need for online dosimeters, whether as routine tools or for monitoring the operating conditions of EB facilities. An overview of efforts to develop online dosimeters, with a focus on electronic devices, will be provided. Subsequently, the lecture will explore the operational characteristics and the physics behind the dose and dose rate response of diodes, explaining how our group has addressed issues related to their use as real-time dosimeters in fixed and mobile EB facilities. The latest data obtained with these diodes, both as routine dosimeters and as electron beam monitors, will be presented to assess the feasibility of their application. The current status of online high-dose dosimeters and the challenges of meeting the metrological standards required in radiation processing dosimetry will also be discussed.

7.4 DR. GEÓRGIA SANTOS JOANA: WIN BRASIL - DIVERSITY & INCLUSION MANAGEMENT

Bachelor's degree in Physics from the Federal University of Minas Gerais - UFMG (2003), master's degree in Radiation, Mineral, and Materials Science and Technology from the Nuclear Technology Development Center of the National Nuclear Energy Commission - CDTN/CNEN (2005), and a Ph.D. in Nuclear Science and

Technology from the Postgraduate Program of the Department of Nuclear Engineering at UFMG (2018). She is currently a technologist at the National Nuclear Energy Commission. She previously worked in the field of Medical Physics at the Superintendence of Sanitary Surveillance of the Minas Gerais State Health Secretariat, where she coordinated the State Quality Control Program in Mammography - PECQMamo. She has experience in Medical Physics and Health Surveillance, with an emphasis on Radiological Protection, Regulation, Safety, and Quality in health services. Her research focuses on Radiological Protection, Quality Control, Radiation Dosimetry and Metrology, and Risk Analysis and Management in the field of medical applications of radiation.

Description: WiN Brasil - Diversity & Inclusion Management

7.5 DR. HELIO YORIYAZ: PROTON BEAM THERAPY: MAIN CHARACTERISTICS AND CHALLENGES

Bachelor in Physics (University of São Paulo, 1979–1982), Master's (1984–1986) and PhD(1998–2000) in Nuclear Technology from the University of São Paulo. Former RHAE fellow atthe Paul Scherrer Institute (Switzerland, 1993–1994) and IAEA Fellow at Oak Ridge AssociatedUniversities (USA, 1996–1997). Researcher at the Nuclear and Energy Research Institute(IPEN), working in the field of Nuclear Engineering with an emphasis on Medical Physics, particularly in Monte Carlo simulation, dosimetry and nuclear medicine. Currently a faculty member of the Graduate Program in Nuclear Technology at USP and the Professional Master's Program in Radiation Technology in Health at IPEN. Bachelor's in Physics (University of São Paulo, 1979–1982), Master's (1984–1986) and PhD(1998–2000) in Nuclear Technology from the University of São Paulo. Former RHAE fellow atthe Paul Scherrer Institute (Switzerland, 1993–1994) and IAEA Fellow at Oak Ridge AssociatedUniversities (USA, 1996–1997). Researcher at the Nuclear and Energy Research Institute(IPEN), working in the field of Nuclear Engineering with an emphasis on Medical Physics, particularly in Monte Carlo simulation, dosimetry and nuclear medicine. Currently a faculty member of the Graduate Program in Nuclear Technology at USP and the Professional Master's Program in Radiation Technology in Health at IPEN.

Description: Proton therapy is an increasingly relevant form of radiotherapy because it enables areduction in the total dose compared to photons, preserving critical structures whilesimultaneously delivering higher and better-conformed doses to the tumor volume. Thiseffect is achieved through the superposition of several weighted Bragg curves with differentincident energies, resulting in a broadened Bragg peak, known as the Spread-Out BraggPeak (SOBP). Although already considered technologically mature, proton therapy still presents importantchallenges that need to be investigated. Among these, the precise calculation of the relativebiological effectiveness (RBE) stands out, defined as the ratio between the doses requiredfor two different types of radiation to produce the same biological effect. This aspect isdirectly linked to the physical interaction processes between the proton and matter. When a proton with kinetic energy passes through a medium, it interacts with electrons andnuclei of the target, causing its deceleration. These interactions are known as electronicstopping and nuclear stopping, respectively. The stopping power describes the force themedium exerts on the ion, corresponding to the average energy loss per unit path length. Inthe case of proton-matter interaction, electronic stopping is dominant. This energy loss dynamic is fundamental to understanding the dose deposition profile. Asthe proton slows down, the energy transferred to biological tissue per unit path lengthincreases, reaching a maximum at a specific depth. This region, known as the Bragg peak,coincides with the maximum stopping power and represents the area of greatest interest forclinical applications. Therefore, its precise positioning is critical during treatment planning.To study these processes in greater detail, advanced approaches have been employed, suchas real-time Time-Dependent Density Functional Theory (TDDFT). This is a non-perturbative method based on modern quantum simulations, which has proven particularlyuseful for investigating electronic stopping in complex systems like water and DNA,bringing computational modeling closer to the real conditions found in biological tissues.

7.6 DR. LUCIANA CARVALHEIRA: NEUTRONS, KNOWLEDGE, AND NEW GENERATIONS: HOW BRAZIL'S ARGONAUTA REACTOR IS SHAPING THE FUTURE OF RADIOCHEMISTRY AND ITS IMPACT ON GLOBAL DOSIMETRY CHALLENGES.

Dr. Luciana Carvalheira is a Black Brazilian scientist and singer from Rio de Janeiro, leading pioneering projects at the Institute of Nuclear Engineering and serving as a professor and vice-coordinator in postgraduate programs for Nuclear Sciences. She co-founded startups like Magtech Brasil and Radíon, driving innovation in radiochemistry and nanomaterials. Her research spans neutron activation, medical polymers, and environmental monitoring, alongside mentoring future scientists through WiN Brasil's Helen Khoury Program. As a member of MunaN, she advocates for the inclusion of young Black women in nuclear sciences. Honored by ALERJ for empowering women and recognized as a leader by ENAP's LideraGOV 4.0, she embodies audacity, joy, and a relentless pursuit of scientific freedom and innovation.

Description: For over 60 years, Brazil's Argonauta research reactor has addressed hands-on education in physics and engineering of reactors, and radiochemistry. As a hub for RD&I, Argonauta drives partnerships with institutions like IME, UFRJ and startups in radiotracer technologies. Modernization efforts, including fuel improvements and new radioisotopes production, exemplify its commitment to advancing radiation science nuclear applications worldwide.

7.7 DR. MARIA NEVES: RANGE RADIATION EXPOSURE INDICATOR

Dr. Maria José Neves is a Senior Researcher at the Brazilian National Nuclear Energy Commission (CNEN), based at the Centre for the Development of Nuclear Technology (CDTN), where she leads the Radiobiology Laboratory. She holds a Bachelor's degree in Medical Biological Sciences from the Faculty of Medicine of Ribeirão Preto, University of São Paulo (USP-RP), as well as a Master's and Ph.D. in Physiology from the same institution, with a focus on metabolism. She completed her postdoctoral training in Biochemistry at the Katholieke Universiteit Leuven (KUL), Belgium. Her main research interests include cancer cell metabolism and the biological effects of ionizing radiation.

Description: A small plastic or glass tube (with a final volume of 1.5mL or smaller) containing a solution, composed of sodium selenite and glycose, can be fixed to the surface of a material and will indicate whether it was itself exposed to moderate doses of range radiation (5 – 100 kGy) in a range irradiator contained in the source of ^{60}Co. This type of irradiator is widely used in the irradiation of foods such as dried grains, fruits, tubers, etc., sterilization of surgical material, among other activities. Given its low cost and ease of preparation, it can be frozen and kept in stock at the time of use, which is why we consider an alternative option to confirm exposure to high-range radiation.

7.8 DR RICHARD HUGTENBURG: NEW TECHNOLOGIES AS A ROUTE TO PARTICLE THERAPY: LASER-HYBRID ACCELERATION FOR RADIOBIOLOGY APPLICATIONS - THE LHARA PROJECT.

Dr Richard Hugtenburg is a medical physics expert and research academic at Swansea University, UK, specialising in radiotherapy physics. Dr Hugtenburg began his career in Christchurch, New Zealand, working as a medical physicist while studying for a PhD on the use of computational methods in radiation oncology. He moved to the UK in 1997 and continued to practice in radiotherapy physics; first at the Queen Elizabeth Medical Centre in Birmingham, then at Singleton Hospital in Swansea. His research examines dosimetry methods associated with emergent radiotherapy technologies, utilising Monte Carlo modelling for a diverse range of radiation physics problems, including the analysis of energy deposition in biological systems, the optimisation of dosimeters and detectors, nanoparticulates, shielding and therapy systems.

Description: Particle therapy is a form of external beam radiotherapy that utilises ions to achieve precise (sub-mm) physical distributions of physical dose with the benefits of high LET to combat radioresistant tumours. Particle therapies are also expected to offer better normal-tissue sparing and reduce long-term side-effects in patients, where the current state-of-the-art in radiotherapy has contributed to significant improvements in the survivability of many cancers in recent years. The LhARA project is a consortium of UK universities that considers three new technologies as a means to reduce the cost, physical footprint and flexibility of particle therapies, where currently treatments, such as carbon ion therapy, are restricted to a few laboratories world-wide. The laser-hybrid method uses high-powered lasers to generate ions with kinetic energies of 10s of MeV and at high dose-rates suitable to examine high dose-rate sparing effects such as FLASH radiotherapy.

7.9 DR. ROLF BEHRENS: FROM DOSEMETER DEVELOPMENT TO ROUTINE USE - STANDARDS AND UNCERTAINTIES

Dr. Rolf Behrens holds a diploma at Technical University of Braunschweig (Germany) and a doctorate (Dr. rer. nat.) in physics from the University of Jena (Germany). He is a senior scientist at the Physikalisch-Technische Bundesanstalt (PTB), Germany's national metrology institute, where he leads the Working Group on Dosimetry for Brachytherapy and Beta Radiation Protection. Dr. Behrens is responsible for quality assurance in the Ionizing Radiation division, serves as Radiation Protection Officer for brachytherapy and beta radiation, and acts as an auditor for the German Accreditation Service (DAkkS). He heads and participates in several national and international standardization committees, including the IEC and ISO working groups on dosimetry and uncertainties, and chairs the German Committee on Radiation Protection Dosimetry. His research focuses on dosimetry of beta and photon radiation for radiation protection and brachytherapy, Monte Carlo simulations, spectrometry, Bayesian data evaluation, measurement uncertainty, and dosimetry for the lens of the eye.

Description: Radiation protection uses protection quantities (not directly measurable) and operational quantities (measured to estimate protection quantities). Standards ensure the quality of dose measurements, established by IEC and ISO, and adopted by regional organizations (CEN, CENELEC, GCC, SARSO, ARSO). Type tests for photon dosemeters use ISO 4037 reference fields; beta and neutron fields follow ISO 6980 and ISO 8529. Standards like IEC 62387 and IEC 61526 address other influencing factors such as angle, dose rate, temperature, and electromagnetic immunity. Uncertainties are calculated by combining the effects of these influences, as described in IEC TS 62461 (updated 2025). Performance is regularly checked using comparison measurements following ISO 14146 (updated 2024, includes EURADOS). Click here to view the presentation slides.

7.10 DR. SILVIA HIDALGO TOBÓN: STATE OF THE ART OF MAGNETIC RESONANCE IMAGING: FROM FUNDAMENTAL PHYSICS TO CLINICAL APPLICATIONS.

Dr. Silvia Hidalgo earned her Bachelor's degree in Physics in 1998 from the University of the Americas-Puebla, Mexico. In 2000, she earned her Master of Science in Medical Physics from the Institute of Physics at the National Autonomous University of Mexico (the first Master of Science in Medical Physics from the UNAM program). In 2005, she earned her PhD in Physics, specializing in Magnetic Resonance Imaging from the Sir Peter Mansfield Magnetic Resonance Center at the University of Nottingham, England. She completed a postdoctoral fellowship in Magnetic Resonance Imaging at the Robarts Research Institute, University of Western Ontario, Canada. She also completed a postdoctoral fellowship at the Autonomous Metropolitan University, Iztapalapa. Her scientific interests are related to advanced functional magnetic resonance imaging, neural networks applied to neuroscience, and magnetic resonance safety in daily clinical practice. She has taught courses and supervised theses at the undergraduate, master's, and doctoral levels in magnetic resonance imaging at the UAM, UAEM, and UNAM. She has given lectures at various national and international forums on nuclear magnetic resonance imaging. She is a member of the International Society of Magnetic Resonance in Medicine, RSNA, the Mexican Physical Society, IEEE, and the Neuroscience Society, among others.

Description: State of the art of magnetic resonance imaging: from fundamental physics to clinical applications.

7.11 DR. SUSANA LALIC: PLANT STEM CELLS AS AN ETHICAL ALTERNATIVE FOR BIODOSIMETRY

Dr. Susana de Souza Lalic is Full Professor of Physics at the Federal University of Sergipe. She holds a B.Sc. (1997) and Ph.D. in Physics (2002) from the University of São Paulo and a postdoctoral fellowship at the University of Pisa (2011). She is a faculty member of the Ph.D. Program in Industrial Engineering (Nuclear Engineering and Industrial Safety) at the University of Pisa and on the Board of Directors of the Master Course in CBRNe Protection at Tor Vergata University of Rome. She is Counselor of the Brazilian Physical Society (SBF), Chair of its Medical Physics Committee, and Vice-chair of the International Solid State Dosimetry Organization (ISSDO). She previously served as Secretary-General of SBF and Coordinator of the Graduate Program in Physics at UFS. Her research focuses on medical physics, condensed matter physics, and applied nuclear physics, with emphasis on radiation detectors, material characterization, radiobiology, and radiation metrology.

Description: This study investigates an ethical approach to biodosimetry, utilizing plants as a sustainable alternative to traditional animal models. Plants, known for their sensitivity to environmental stressors such as radiation, provide a valuable platform for identifying genotoxic agents while reducing reliance on animal testing. The research highlights the utility of the onion (Allium cepa) as an in vivo model for assessing radiation-induced genetic and chromosomal damage. While Allium cepa is a widely recognized standard in environmental contamination studies, it remains underexplored as a tool for radiation detection. A comprehensive evaluation of radiation effects and their temporal progression is made possible by analyzing key variables, including linear energy transfer, dose, and dose rate. The study also examines the potential of photobiomodulation to enhance biological recovery in tissues damaged by radiation, offering a novel intervention strategy. To ensure robustness, findings from the Allium cepa model are cross-validated against other in vivo, in vitro, and in silico systems, establishing its efficacy as an ethical and effective alternative for radiation biodosimetry.

7.12 DR. TARCÍSIO CAMPOS AND DR. ALBERTO CAMPOS: CHALLENGES IN RADIOISOTOPE DECONTAMINATION AND PRODUCTION THROUGH NEUTRON SPECTRAL SELECTION

Dr. Tarcisio P. R. de Campos is retired full professor from the Federal University of Minas Gerais (UFMG), currently collaborating with both UFMG and the Nuclear Technology Development Center (CDTN). He holds a degree in Engineering from UFMG, an MSc in Nuclear Engineering from UFRJ, a PhD from the University of Illinois at Urbana-Champaign, and a postdoctoral fellowship at MIT as a Fulbright Fellow. His expertise spans nuclear engineering, radiobiology, radiochemistry, radiotherapy, and the development of radioisotopes, simulators, and medical software. He has published extensively, holds 14 patents, and has supervised over 75 graduate theses. He has coordinated over 28 research projects and actively collaborates with national and international research institutions.

Dr. Alberto M. Campos is Postdoctoral researcher in Augsburg, Germany (2024–present), with a PhD in Mathematics from the Institute for Pure and Applied Mathematics (IMPA, 2024). He completed a research stay at Instituto Superior Técnico (IST), Lisbon (2022), holds a Master's and Bachelor's degree in Mathematics from UFMG, and a technical degree in Electrotechnics from CEFET-MG. He has served as a teaching assistant in various probability and analysis courses at IMPA and UFMG, and has presented talks at the Erdős Institute (Budapest) and the Department of Nuclear Engineering at UFMG. His research interests lie broadly in Probability Theory, with a focus on random covering problems, percolation, and random walks.

Description: We present investigations into the production and decontamination of radioisotopes using a selective Spectral Resonant Selector (RSS), focusing on neutron radiative capture reactions in precursor nuclides and transmutation reactions of long-lived radioactive waste. The physical and mathematical aspects of biased Brownian motion are addressed, through which a tailored neutron spectrum can be generated to target specific nuclides. Reaction rates for both precursor nuclides and decontamination processes are presented. Two distinct neutron spectra are considered—one at low energy and another at resonance energy—allowing interaction with a broad range of target nuclides. The production of isotopes such as Sm-153, Lu-177, Re-188, Ho-166, and Yb-169 serves as representative examples of potential radionuclide generation. Additionally, the decontamination of long-lived actinides, including transuranic radionuclides and fission products, through fission or radiative capture reactions, illustrates a promising approach to nuclear waste mitigation. The specific activities and benchmark parameters of various radioisotopes under defined conditions demonstrate the effectiveness of the RSS in both radioisotope production and nuclear decontamination.

7.13 DR. THIAGO LIMA: CURRENT AND FUTURE TRENDS IN NUCLEAR MEDICINE DOSIMETRY

Dr. Thiago holds a BSc in Physics with Medical Physics from UFRJ and a PhD in Medical Physics from University College London. He is the chief diagnostic medical physicist at Lucerne University Hospitals (LUKS) and research physicist at Genolier Hospital. He also lectures at the University of Lucerne. He chairs the education committee of the Swiss Society of Radiobiology and Medical Physics and represents the society in EFOMP working groups. His work focuses on dosimetry, protocol optimization, radioprotection, and research in nuclear medicine and medical imaging.

Description: This lecture delves into the evolution of nuclear medicine dosimetry. It will explore the progression from early rudimentary methods, which relied on basic measurements and empirical estimates, to the sophisticated imaging techniques and computational models employed today. These advancements have significantly enhanced the accuracy of absorbed dose quantification, improving both safety and efficacy in treatments. The lecture will also discuss the future of dosimetry, highlighting the integration of artificial intelligence and personalized medicine. This forward-thinking approach promises even more precise dose calculations tailored to individual patient characteristics, reflecting a continuous effort to optimize therapeutic protocols and improve patient outcomes in nuclear medicine.

7.14 DR. YVONE MASCARENHAS: PERSONAL DOSIMETRY - THE IMPORTANCE OF REGULATION, TECHNOLOGY, AND CULTURE. EVALUATION OF DOSIMETRY SERVICES IN LATIN AMERICA AND THE CARIBBEAN

Dr. Yvone Maria Mascarenhas has a degree in Physics from the University of São Paulo (USP), and a master's, doctorate, and post-doctorate from the USP's Institute of Physics and Chemistry of São Carlos (IFSC), with research in basic physics as well as radiology and dosimetry. She has worked as a coordinator for research projects funded by agencies such as CNPq and FAPESP and has extensive experience in the field of Physics, working primarily in radiological protection, x-ray production, radiodiagnosis, and dosimetry.

Description: Personal dosimetry is a fundamental tool in radiation protection, ensuring that bestpractices in facilities using ionizing radiation effectively minimize risks to exposed workers.The use of radiation is highly regulated worldwide, and to make personal dosimetry widelyaccessible, technology plays a key role—not only as a scientific instrument but also as areliable and reproducible measurement technique. Accurate analysis of population-level dosimetry data is only possible with the correct andappropriate use of personal dosimeters. This can be achieved only when a strongradiation protection culture is promoted and adopted by all stakeholders involved inpersonal dosimetry. In Latin America and the Caribbean, the evaluation of Individual Monitoring Services(SMIE) varies by country, reflecting differences in regulations, technological infrastructure,and safety culture. This study presents an overview of the current state of personaldosimetry in the region, with emphasis on the 2022 intercomparison exercise organized byREPROLAM. A total of 26 services—both public and private—were evaluated usingdifferent irradiation beams and dose levels. The results show that 69.2% of the laboratories complied with ISO 14146:2018. However,some exhibited calibration deviations, particularly with X-ray beams. Notably, OSL(Optically Stimulated Luminescence) dosimetry has seen significant growth,accounting for 34% of the systems evaluated in 2022. In Brazil, the development of Personal Dosimetry Services has been shaped by twodecades of intercomparison exercises and over 30 years of a structured certificationprocess. Despite this progress, challenges remain in standardizing methodologies andenhancing laboratory training to improve dosimetry accuracy across the region.This analysis underscores the critical importance of regulation, technologicaladvancement, and the consolidation of a radiological safety culture as pillars forstrengthening occupational dosimetry in Latin America and the Caribbean.

7.15 DR. WILSON CALVO: CURRENT STATUS OF STRATEGIC PROJECTS AT CNEN

Dr. Wilson Calvo holds a diploma in Metallic and Ceramic Materials Engineering from the Federal University of São Carlos - UFSCar (1987) and holds a Master's (1997) and a PhD (2005) degrees in Nuclear Technology - Applications, from the University of São Paulo - USP. He began working at the Nuclear and Energy Research Institute (IPEN) in 1988 and went on to pursue a successful career in both administration and technology. He is a Senior Technologist, professor, and postgraduate advisor in the area of Nuclear Technology at IPEN/USP. He published nearly 100 full-length articles in international conferences and indexed magazines, and supervised more than 15 master's, doctoral, and postdoctoral students. He is a leading expert in Nuclear Engineering, with a focus on the applications of Nuclear Techniques in Industry, Health, Agriculture, and the environment. His emphasis is on radioisotope technology (radiotracers and sealed radioactive sources) and ionizing radiation (electron beams, X-rays, and gamma rays). Owing to his great contribution in the nuclear applications field, he was awarded the Nuclear Merit Award from ABDAN (2020). Regarding his administrative career, he served as a Manager of the Radiation Technology Center (2001-2013), Director of Administration and Infrastructure (2014-2016), and Superintendent of IPEN (2017-2023). He was also Vice President of the Board of Directors of the São Paulo Technology-Based Incubator at USP/IPEN-Cietec (2017-2023). He is a Member of the Deliberative Committee of CNEN (2023-), the Advisory Board of ABENDI, and the International Irradiation Association (IIA). Currently, he is the Director of Research and Development at the National Nuclear Energy Commission - CNEN.

Description: The National Nuclear Energy Commission's guidelines for technological innovation focus on three main areas: nuclear technologies—including reactors, instrumentation, materials, and the fuel cycle—applications of ionizing radiation in health, industry, agriculture, and the environment, as well as radioprotection and dosimetry. This lecture will highlight several strategic projects, including CENTENA, GraNioTer, the tracer technique for offshore oil and gas platforms and processing plants, mobile units equipped with electron accelerators, and a multipurpose reactor. The main features of each project, their contributions to addressing major national challenges in developing nuclear technology, and the current progress of each initiative will be discussed. In short, CENTENA aims to design, build, and commission the first nuclear and environmental technology facility in South America for disposing of low-level radioactive waste, thereby establishing a foundation of safety and sustainability for Brazil. GraNioTer's goal is to establish itself as a national technological hub, promoting research, development, and innovation based on graphene, niobium, rare earths, and other materials and minerals with increasing global demand. Another key project is the application of the radiotracer technique to offshore oil and gas platforms and processing plants, which seeks to develop a prototype and method for measuring gas flow in pipelines. Equally important is the Mobile Unit, equipped with an electron beam accelerator, which aims to expand the nation's capacity to treat industrial effluents and improve water quality in urban areas, addressing one of the primary challenges faced by modern society.

7.16 DR. BRUNO MELO MENDES

Dr. Bruno Melo Mendes holds a Bachelor's degree in Biological Sciences and a Master's and PhD in Nuclear Science and Technology from the Federal University of Minas Gerais (UFMG). He is currently a researcher at the Nuclear Technology Development Center (CDTN/CNEN), responsible for the Internal Dosimetry Laboratory, and a professor in the CDTN postgraduate program. He also coordinates the Monte Carlo Modelling Expert Group (MCMEG). His expertise lies in the nuclear field, with a focus on radiopharmacy, computational modeling (Monte Carlo), and internal dosimetry.

7.17 DR. TELMA CRISTINA FERREIRA FONSECA

Dr. Telma C. F. Fonseca is professor at the DEN/UFMG, faculty member of the PCTN/UFMG and member of the Diversity & Inclusion Management Committee (2024–2027) of WinBrasil – Women in Nuclear, and coordinates the Monte Carlo Modelling Expert Group (MCMEG). She holds degrees in the area of computational sciences, a Master's in Radiation Sciences from UFMG, and a PhD in Biomedical Sciences from KU Leuven (Belgium), where her research focused on internal dosimetry. She was a researcher at the Radiation Protection Dosimetry and Calibrations group (RDC) at SCK-CEN in Belgium. Additionally, she completed a postdoctoral fellowship at the Centre for Development of Nuclear Technology (CDTN/CNEN), where she also led the Internal Dosimetry Laboratory. Her expertise and research interests include dose calculation in radiotherapy, radiobiology, nuclear instrumentation, nanodosimetry, computational modeling, and the use of Monte Carlo methods in radiation protection and radiobiology. Website: www.telmafonseca.com.br

CHAPTER 8

Round Tables

8.1 AI'S CHALLENGES AND OPPORTUNITIES: CONNECTING RESEARCH, INDUSTRY, AND SOCIETY

Moderator

- Dr. Telma Fonseca (DEN/UFMG)

Panelists

- Dr. Rita Sebastião (ICEX/UFMG)
- Dr. Ana Carolina (Siemens Healthineers)
- Dr. Andre Henrique Alves Carneiro (USP)

Description: The roundtable is part of the Tech Area on Artificial Intelligence, an innovative space within ISSSD 2025 dedicated to the convergence of technology, healthcare, and innovation. Its main goal is to foster dialogue among startups, researchers, healthcare professionals, and investors, encouraging the use of AI in clinical practice. During the session, strategic topics will be discussed, such as: Challenges and opportunities of AI in clinical radiology Explainability and ethics of diagnostic algorithms Automation and integration of hospital workflows Success stories from innovative startups Trends for the next 5 years in medical AI The roundtable will feature multidisciplinary experts — including radiologists, data scientists, entrepreneurs, and regulators — and will be moderated by professionals with experience in healthcare innovation. There will also be room for audience interaction, with guided questions designed to stimulate debate.

Dr. Telma C. F. Fonseca is professor at the DEN/UFMG, faculty member of the PCTN/UFMG and member of the Diversity & Inclusion Management Committee (2024–2027) of WinBrasil – Women in Nuclear, and coordinates the Monte Carlo Modelling Expert Group (MCMEG). She holds degrees in the area of computational sciences, a Master's in Radiation Sciences from UFMG, and a PhD in Biomedical Sciences from KU Leuven (Belgium), where her research focused on internal dosimetry. She was a researcher at the Radiation Protection Dosimetry and Calibrations group (RDC) at SCK-CEN in Belgium. Additionally, she completed a postdoctoral fellowship at the Centre for Development of Nuclear Technology (CDTN/CNEN), where she also led the Internal Dosimetry Laboratory. Her expertise and research interests include dose calculation in radiotherapy, radiobiology, nuclear instrumentation, nanodosimetry, computational modeling, and the use of Monte Carlo methods in radiation protection and radiobiology. Website: www.telmafonseca.com.br

Dr. Rita C. O. Sebastião is a full professor at Universidade Federal de Minas Gerais (UFMG) with experience in Chemistry, working mainly on the topics: inverse problems, artificial neural networks, nuclear magnetic resonance, thermal decomposition, chemical kinetics, combustion processes, and thermodynamics. Extensive experience in education, research, and extension, especially in projects aiming for the popularization of science, educational innovation, and entrepreneurship, connecting the university to basic education. Experience in developing and managing projects on socio-environmental sustainability and education. She is currently a member of the Postgraduate Program in Chemistry (PPGQUI), Postgraduate Program in Technological Innovation (PPGIT) and Professional Master's in Technological Innovation and Intellectual Property (MPITPI) at UFMG.

Dr. André Carneiro is a Senior Data Scientist and Management Consultant with over 6 years of experience working with leading Brazilian companies. Skilled in developing mathematical, statistical, and machine learning models to deliver actionable insights and optimize business decisions. Currently, I am a Senior Data Scientist at Ambev Tech, one of Brazil's largest companies, where I focus on recommendation systems, clustering, predictive modeling, causal inference, and advanced techniques such as double machine learning (e.g., causal forests). In addition to my corporate experience, I have a strong academic background with over 5 years of research experience. I hold a PhD in Production Engineering from Escola Politécnica of the University of São Paulo (Poli-USP), as well as a double degree in Production Engineering from Poli-USP and a Master's degree in Management and Industrial Engineering from the Instituto Superior Técnico of the University of Lisbon. My research focuses on machine learning, statistical control, and project monitoring. I am also a collaborating researcher at LABDAPS (Laboratório de Big Data e Análise Preditiva em Saúde) of the Public Health School of the University of São Paulo, where I pursue postdoctoral research on fairness and social justice in machine learning models, aiming to enhance predictive performance for underrepresented populations.

8.2 RADON IN ENVIRONMENT: TRACER APPLICATIONS AND PUBLIC HEALTH RISKS

Moderator

- Dr. Talita Santos (CDTN/CNEN)

Panelists

- MSc. Elydio Soares (CDTN/CNEN)
- MSc. Laura Takahashi (CDTN/CNEN)
- Dr. Nathan da Silva (CDTN/CNEN)

Description: The two main aspects related to radon will be presented and discussed: the concern regarding the impacts on public health and the applications of radon isotopes as tracers of environmental phenomena.

Dr. Talita de Oliveira Santos holds a degree in Radiology Technology from the Federal Center for Technological Education of Minas Gerais (2007), a master's degree in Nuclear Sciences and Techniques from the Federal University of Minas Gerais (2010) and a PhD in Nuclear Sciences and Techniques from the Federal University of Minas Gerais (2015). She is currently an adjunct professor at the Federal University of Minas Gerais and a collaborator of the National Commission for Nuclear Energy. She has experience in Physics, with an emphasis on Radiation Physics, working mainly on the following topics: natural radioactivity, application of radiation to the environment, nuclear instrumentation and radioprotection.

8.3 LEADERSHIP IN SCIENTIFIC AND TECHNOLOGICAL RESEARCH

Moderator

- Dr. Luciana Carvalheira (IEN/CNEN)

Panelists

- Adriana Marques (Agência Nacional de Água - ANA)
- Anna Letícia Barbosa de Sousa (CNEN)

Description: This roundtable will convene leading experts to explore the pivotal role of leadership in advancing nuclear scientific and technological research. The discussion will address key challenges, including regulatory frameworks, ethical considerations, nuclear applications, diversity and inclusion, effective communication, and the integration of the next generation into the global nuclear sector. By examining how leadership drives innovation, navigates complex challenges, and shapes the future of nuclear science and technology, this session aims to foster meaningful dialogue. Participants will gain insights into emerging technologies, explore strategic leadership approaches, and strengthen collaboration among researchers, policymakers, and industry leaders.

Dr. Luciana Carvalheira is a Black Brazilian scientist and singer from Rio de Janeiro, leading pioneering projects at the Institute of Nuclear Engineering and serving as a professor and vice-coordinator in postgraduate programs for Nuclear Sciences. She co-founded startups like Magtech Brazil and Radíon, driving innovation in radiochemistry and nanomaterials. Her research spans neutron activation, medical polymers, and environmental monitoring, alongside mentoring future scientists through WiN Brazil's Helen Khoury Program. As a member of MunaN, she advocates for the inclusion of young Black women in nuclear sciences. Honored by ALERJ for empowering women and recognized as a leader by ENAP's LideraGOV 4.0, she embodies audacity, joy, and a relentless pursuit of scientific freedom and innovation.

CHAPTER 9

Mentorship Helen Khoury WIN-Brasil

The Helen Khoury Mentorship WinBrasil2025
Program: Advancing Diversity, Inclusion, and Female Leadership in the Nuclear Sector

The Helen Khoury Mentorship Program, promoted by Women in Nuclear Brazil (WiN Brasil), was established to foster the professional development of women in the nuclear sector, integrating diversity in age, educational background, and technical expertise. Over seven months, the program combined one-on-one mentoring sessions with collective workshops, encouraging both academic and professional growth.

Mentors and mentees co-developed final projects that reflected the program's multidisciplinary and inclusive approach. Outcomes included: the design of a critical meaningful learning model for future mentorship cycles; the creation of Radioarretadas, a local initiative to engage women from Pernambuco's countryside in nuclear discussions; research on gender and racial equity within Eletronuclear; studies on hybridization of nuclear and renewable energy; structured career development strategies in nuclear physics; participation in scientific proposal submissions (PPSUS/Innovation 2025); scientific dissemination projects on radiation and radioactivity; and professional milestones such as article submissions, approvals in competitive examinations, and postgraduate advancements.

The results demonstrated progress in strengthening both technical and behavioral competencies, expanding professional networks, and integrating diverse regional and cultural perspectives. By connecting institutions such as CNEN, IPEN, IEN, Eletronuclear, INMETRO, and universities across Brazil, the initiative reinforced the importance of knowledge sharing and inter-institutional collaboration. It is concluded that the Helen Khoury Mentorship has been consolidated as a model for professional development aligned with sectoral demands and individual expectations. The program not only contributes to participants' growth but also reinforces the presence and leadership of women in the nuclear sector, leaving a legacy of diversity, inclusion, and innovation.

Keywords: Mentorship, Diversity, Inclusion, Female Leadership, Nuclear Sector.

CHAPTER 10
Honors Awards

The International Symposium on Solid State Dosimetry (ISSSD) is proud to honor outstanding contributions in the field of radiation science and its applications in medicine, industry, and research. The 2025 Awards recognize excellence, innovation, and leadership that have advanced knowledge, safety, and education in nuclear science.

10.1 PROF. DR. TARCISIO PASSOS RIBEIRO DE CAMPOS

We are honored to recognize the exceptional contributions of Prof. Tarcísio Passos Ribeiro de Campos, a distinguished figure in nuclear engineering and radiation science in Brazil. With a career spanning over four decades, Prof. Campos has significantly advanced the fields of medical dosimetry, radiation protection, and nuclear technology.

As a retired full professor at the Federal University of Minas Gerais (UFMG), he continues to collaborate with the university and the Nuclear Technology Development Center (CDTN). Prof. Campos earned his undergraduate degree in Engineering from UFMG (1983), a master's in Nuclear Engineering from the Federal University of Rio de Janeiro (1985) with a focus on Reactor Physics and Nuclear Resonances, and a Ph.D. in Nuclear Engineering from the University of Illinois at Urbana-Champaign (UIUC, USA, 1993). At UIUC, he conducted research at the Beckman Institute in neuroscience, robotics, and visuo-motor coordination. He completed a postdoctoral fellowship at the Massachusetts Institute of Technology (MIT, USA, 1998) under the MIT-Harvard research program as a Fulbright Fellow, focusing on Boron Neutron Capture Therapy (BNCT) for brain tumors.

Prof. Campos has extensive experience across multiple areas of Engineering and Nuclear Energy, with current emphasis on materials, radiation science, radioisotopes, radiochemistry, and radiobiology. His research spans radioactive biomaterials, ionizing radiation effects, radiotherapy, brachytherapy, experimental and computational dosimetry, radiopharmaceutical production, development of physical radiology simulators, radiotherapy software, neutron capture therapy, radiobiology, radiochemistry, oncology, nuclear instrumentation, automation, and the design of particle accelerators, isotope transmuters, radioisotope generators, and nuclear reactor cores.

He holds 14 patents and 1 software registered with INPI, with 2 additional invention patents submitted to WIPO, and has developed 4 computational systems. Prof. Campos has published 148 articles in indexed journals, 340 full conference papers, and 34 abstracts, and has supervised 57 master's theses, 21 doctoral dissertations, 2 co-supervisions, and currently has 2 ongoing doctoral supervisions. He has also mentored 13 postdoctoral researchers, including 1 senior postdoc.

Prof. Campos has collaborated with numerous institutions, including departments of Surgery, Biology, Pharmacy, Veterinary, and Medicine at UFMG, as well as radiotherapy centers at Santa Casa de Misericórdia, Hospital Mário Pena, Hospital São Francisco, and CDTN. He has coordinated over 28 research projects funded by government agencies.

He is a member of the editorial boards of the Journal of Radiology (Austin Publishing Group) and the Revista Mineira de Radioterapia. Over 30 years of teaching, he has delivered undergraduate and graduate courses in applied radiobiology, applied radiochemistry, medical imaging concepts, biomedical radiation applications, particle accelerators applied to biomedicine, and radiation protection, within the Radiation Sciences research line focused on engineering and health applications.

Prof. Campos's dedication to education, research, and innovation has left a lasting legacy, inspiring generations of scientists and advancing nuclear science and technology both in Brazil and internationally.

10.2 PROF. DR. TEÓGENES AUGUSTO DA SILVA

We are honored to recognize the outstanding contributions of Prof. Teógenes Augusto da Silva, a distinguished researcher and educator in nuclear physics and medical physics in Brazil. With a career spanning over four decades, Prof. Silva has made significant advances in radiation metrology, radiological protection, and dosimetry.

He graduated in Physics from the Federal University of Rio de Janeiro (1976) and earned his Master's (1981) and Ph.D. in Nuclear Engineering from the Programa de Engenharia Nuclear – COPPE/UF (1996). During his doctoral studies, he was a CNPq sandwich fellow at the National Institute of Standards and Technology (NIST, Gaithersburg, USA) from December 1992 to June 1995.

Prof. Silva served as a senior researcher at the Comissão Nacional de Energia Nuclear (CNEN) from July 1977 until his retirement in September 2023. He was a faculty member in the graduate program at the Nuclear Technology Development Center (CDTN, Belo Horizonte) from March 2015 to September 2023 and served as an adjunct professor at the Postgraduate Program in Nuclear Science and Techniques, DEN/UFMG from March 2000 to June 2017.

His research has focused on Medical Physics, emphasizing radiation metrology, radiological protection, and dosimetry, particularly in the calibration of dosimeters, medical and dental radiodiagnostics, quality control, and radiation safety. He contributed as a Mineiro Researcher (2011–2019) and as a CNPq Productivity Fellowship (PQ 2) awardee (2009–2022). He also served as Vice-Coordinator of the CNPq INCT Radiation Metrology in Medicine project (2008–2017).

Prof. Silva held leadership positions at CDTN, including Chief of the Nuclear and Radiological Safety Division (1988–2010) and Chief of the Reactor and Radiation Division (2014–2018). After his retirement, he continues as a collaborator at CDTN (from October 2024) and currently serves as Executive Secretary and Administrative Director of the INCT for Nuclear Instrumentation and Applications in Industry and Health (INAIS, 2023–present). Through his extensive career, Prof. Silva has left a profound impact on radiation safety, dosimetry, and medical physics in Brazil, mentoring generations of scientists and promoting excellence in research and education.

10.3 PROF. DR. HELEN JAMIL KHOURY - IN MEMORIAM

Prof. Helen Khoury was a leading figure in radiological protection and scientific outreach in Brazil, dedicating over four decades to teaching, research, and development in the nuclear sector. She served as a full professor at the Department of Nuclear Energy (DEN) at the Federal University of Pernambuco (UFPE) and held a Ph.D. in Nuclear Physics from the Pontifical Catholic University of São Paulo.

Her work focused on nuclear dosimetry and instrumentation, with applications in medical imaging, radiation therapy, and radiation metrology. She led the Occupational Radiological Protection Network for Latin America and the Caribbean (Reprolam), fostering professional training and knowledge exchange, and founded the Nuclear Science Museum at DEN/UFPE, a pioneering initiative for popularizing nuclear science in Brazil.

Prof. Khoury's legacy lives on in the countless professionals she trained, the initiatives she helped establish, and the lives she inspired. Her dedication and vision continue to guide the field of radiological protection and nuclear science.

CHAPTER 11

Helen Khoury Award for Best Works

In recognition of excellence in research and innovation, the XXV International Symposium on Solid State Dosimetry is proud to present the Helen Khoury Award. This award honors the legacy of Professor Helen Khoury, whose significant contributions to the field of dosimetry continue to inspire generations of scientists worldwide.

The award is granted in two categories — Panel and Communication — highlighting outstanding contributions in each format. In addition to the winners, the Scientific Committee has selected the five best-ranked works in each category to receive an Honorable Mention, acknowledging the high quality and impact of their research.

11.1 BEST WORK IN COMMUNICATION

ID work code: G02-011
Title of Work: **Preliminary Evaluation of a Positioning Phantom for Dosimetric Audit in Ir-192 HDR Brachytherapy: Results from an IAEA Collaborative Project**
Authors: *Natasha Briggs e Silva, Alexander Camargo Firmino da Silva, Aneli Silva, Sandro Passos Leite, Alexandre Bernardino Pinto Jorge, Camila Salata, Maria Helena Marechal, Ademir Xavier da Silva, Andrea Mantuano Coelho da Silva.*
Presenters: *Natasha Briggs e Silva*

Mini-bio: Master in Nuclear Engineering, graduated from the Military Institute of Engineering (IME). Bachelor's degree in Physics with an emphasis in Medical Physics from Souza Marques College. Occupational Safety Technician from the Celso Suckow da Fonseca Federal Center for Technological Education (CEFET/RJ). Has experience as a teacher in the field of Occupational Safety Technology. Currently working as a Researcher in Dosimetry and Radioprotection at the Laboratory of Radiological Sciences (LCR/UERJ), and pursuing a Ph.D. in Nuclear Engineering at COPPE/UFRJ, focusing efforts on research and development in the field of Dosimetry for HDR brachytherapy.
More info:
`https://www.linkedin.com/in/natasha-briggs-2b722bb3/`
`http://lattes.cnpq.br/2115738852789471`

11.1.1 Honorable Mentions in Communication

ID work code: G05-023
Title of Work: **Analysis of Natural Radioactivity in Beach Sands Along the Rio de Janeiro State Coast by Gamma Spectrometry**
Authors: *Laisa de Almeida Neves, Wander Nascimento de Araújo Júnior, Jonathan Oliveira, Leandro Barbosa*

da Silva, Ademir Xavier da Silva.
Presenters: *Laisa de Almeida Neves*

ID work code: G02-057
Title of Work: **Generator of Health Optimized Simulation Templates – GHOST a 3D Slicer plugin to voxelized medical images for MCNP**
Authors: *Francisco Harlley Dantas Hauradou Xavier, Paula Selvatice Pereira Teles, Ademir Xavier da Silva, Mirta Bárbara Torres Berdeguez*.
Presenters: *Francisco Harlley Dantas Hauradou Xavier*

ID work code: G07-002
Title of Work: **Low-Dose Neonatal Ionizing Radiation Induces Modest, Sex-Specific Changes in Adult Mouse Behavior and Hippocampal Oxidative Stress**
Authors: *Jhon Brandon Felix Ferreira, Luciana Penedo de Melo Teixeira, Maria Eduarda Gomes Moreira da Silva, Heitor gabriel de Souza Saraiva, Mauricio Moacir da Silva Borges, Bianca Torres Ciambarella, Carlos Eduardo Veloso de Almeida, Samara Cristina Ferreira Machado*.
Presenters: *Jhon Brandon Felix Ferreira*

ID work code: G02-019
Title of Work: **A new formulation of Fricke gel dosimeter development using salicylic acid**
Authors: *Iasmin Vieira Nishibayaski, Anna Luiza Fraga Silveira, Luiz Cláudio Meira Belo*.
Presenters: *Iasmin Vieira Nishibayaski*

11.2 BEST WORK IN PANEL

ID work code: G02-066
Title of Work: **Modeling and validation of a Whole Body Counter for in vivo measurements using CAD and MCNP6**
Authors: *Daniel de Castro Pacheco, Bruno Melo Mendes, Luiz Cláudio Meira Belo*.
Presenters: *Daniel de Castro Pacheco*

Mini-bio: Graduated in Mechanical Engineering (2020), with over 5 years of experience in the development and coordination of industrial engineering projects for refineries, mining, and steel industries, participating in the preparation of technical and managerial documents, as well as in the coordination and management of people and projects. Holds a Master's degree in Science and Radiation Technology (2023), focused on the development of a computational model for shielding studies in in vivo monitoring, using a CAD tool associated with a Monte Carlo code. Currently a Ph.D. candidate in Science and Radiation Technology at the Nuclear Technology Development Center (CDTN) / National Nuclear Energy Commission (CNEN), working on the development of the dosimetry monitoring and data collection system at CDTN, with an approach based on the Internet of Things (IoT), Internet of Everything, and Digital Twins. Also worked in supporting and developing projects in the field of Information and Communication Technology (ICT), focused on smart cities.
More info:
`https://www.linkedin.com/in/daniel-de-castro-pacheco`
`http://lattes.cnpq.br/0207136434699068`

11.2.1 Honorable Mentions in Panel

ID work code: G02-070
Title of Work: **Monte Carlo Simulation of Depth Dose in Mammography Using Geant4 and 3D-Printed ABS and PETG Phantoms**
Authors: *Greiciane de Jesus Cesário, Crislane de Jesus Cesário, Linda Viola Ehlin Caldas, Edilaine Honório da*

Silva.
Presenters: *Greiciane de Jesus Cesário*

ID work code: G02-035
Title of Work: **Computational Tools for Automatic Conversion of Microscopy Images into Voxelized Cellular Phantoms**
Authors: *Lucas Beloti Silva, Lucas Fabricio de Araujo, Telma Cristina Ferreira Fonseca*.
Presenters: *Lucas Beloti Silva*

ID work code: G04-038
Title of Work: **Validation of an alanine-based postal dosimetry system for quality control in radiotherapy**
Authors: *Aricia Ravane Pereira da Cruz, Lucas D. Albino, Ernesto Roesler, Claudio C. B. Viegas, Josemary Angélica Corrêa Gonçalves, Carmen Cecília Bueno, Vinicius Saito Monteiro de Barros, Viviane Khoury Asfora*.
Presenters: *Aricia Ravane Pereira da Cruz*

ID work code: G02-081
Title of Work: **SMS Notification System for Potential Radiological Risk Scenarios**
Authors: *Felipe Mamed de Souza, Alessandra Vaz dos Anjos Santos, Paulo Cézar Rocha Silveira, Wallace Vallory Nunes*.
Presenters: *Felipe Mamed de Souza*

CHAPTER 12

Symposium Papers

Effective dose received by Dentistry students in intraoral dental radiology using TLD dosimetry

Katherine Gutierrez Delgado[1]*, Enrique Gaona[1], Nelly Molina-Frechero[1], Guadalupe Robles Pinto[1], Lorena López González[1], Raúl Enrique Castañeda Castaneira[1].

[1] UniversidadAutónoma Metropolitana, Unidad Xochimilco, Maestría en Ciencias Odontológicas, Calz. del Hueso 1100, 04960 Ciudad de México, México

[*] k.thdelgado1@gmail.com

Abstract

INTRODUCTION. The purpose of the study was to determine the annual effective dose received by dental students during their clinical practice with intraoral dental radiology in two groups, one using radiological safety procedures and the other group without radiological safety procedures. Students are classified as public personnel and not as personnel occupationally exposed to radiation. MATERIALS AND METHODS. The study was conducted at the university's Dental Clinic and involved 32 students. Four-crystal TLD dosimeters were used: two of lithium borate $Li_2B_4O_7$ and two of calcium sulfate $CaSO_4$ for the chest and one of lithium borate for the hands. Monitoring of the effective dose to the chest and absorbed dose to the hands in the students was performed over a period of 20 working days during clinical care with intraoral dental radiology. RESULTS. The overall results of the estimated annual effective doses to the Thorax received by Group A (radiographic examinations without radiation safety procedures) and Group B (radiographic examinations with radiation safety procedures using distance and using a film-holder/digital sensor device inside the patient's mouth) were 1.42 ± 0.20 mSv/year and 0.32 ± 0.23 mSv/year, respectively. The difference in the mean values between the two groups was statistically significant at $p < 0.05$, with the mean annual effective dose for Group A exceeding the annual effective dose limit recommended by ICRP Publication 103 of 1 mSv/year for the public. The mean annual absorbed dose to the hands was 3.01 ± 0.38 mGy without radiation safety procedures and 0.61 ± 0.24 mGy with radiation safety procedures. DISCUSSION. Several studies have found effective doses in dental radiology students and workers greater than 1 mSv/year. Exceeding the recommended effective dose limits for the thorax for the general public is due to the lack of a radiation safety culture in dental school curricula. CONCLUSIONS. Dental students are considered members of the public in relation to their clinical practice with intraoral dental radiology, but they receive unjustified effective doses to the thorax that exceed the public limit of 1 mSv/year recommended by ICRP publication 103. They also receive unjustified absorbed doses to the hands, eye, thyroid and breast because the dental curriculum does not cover topics such as radiation protection, radiation biology, image quality and legislation. One obstacle to a culture of radiation safety in dental radiology is self-referral in dentistry, where the dentist decides whether an x-ray is needed, what type of X-ray image use, what exposure parameters should be used, what clinical findings the radiological image has and what radiation protection procedures should be used without referral from another specialist. Finally, we can say that what they have learned in dentistry/stomatology courses continues to be applied in their professional practice, such as performing intraoral dental radiological examinations without radiation protection procedures.

Keywords: Dentistry students; intraoral dental radiology; effective dose; TLD.

Introduction

The objective of this study was to determine the annual effective dose received by dental students during their clinical practice with intraoral dental radiology. Radiographs in dentistry are very important because they provide information for dentists in the diagnosis, follow-up, and treatment planning of lesions in the oral cavity, as well as surrounding tissues, dental support structures, and the upper and lower jaw bones [Ubeda 2018; Barba 2022; Wlches-Visbal 2021].

Dentistry/Stomatology programs in Mexico do not include radiation safety education (theory and practice) in their curriculum and do not cover topics such as radiation protection, radiation biology, image quality and regulations. Students learn to perform radiographic examinations with minimal, non-specialized guidance from professors who do not have a specialty in dental radiology.

Let us review other studies in the practice of dental radiology: a) Valenzuela et al., [2021] evaluated the radiation safety knowledge in dental students and found that only 66.7% had a basic understanding of radiation protection regulations. b) Kaviani et al., [2007] reports that students use distance as a measure of radiation protection and it is the most used, however only 38% know the correct angle from which to shoot in relation to the X-ray tube in intraoral radiology. c) Mupparupu et al., [2022] in a study with 139 first-year dental students, the results showed that 56% of the students believed that routine dental x-rays should be performed for all patients without justification. Tirado-Amador et al., [2015] found that having biosafety protocols in infection control and waste management as well as radiation protection measures (justification, optimization, dose limit, shielding, distance, dosimeter monitoring, continuous training and limiting exposure times) are ignored by many professionals and students.

In the study by Loya et al., [2016] they reported effective doses in Dentistry students of 2.59 mSv/year. Álvarez et al., [2024] reported effective doses to the thorax and hands of 1.58 mSv/year and 3.46 mGy/year respectively and concludes that exceeding the dose limit of 1 mSv/year for the public is due to a lack of training in radiological safety of students.

Figure 1 illustrates students taking an intraoral radiograph in the traditional manner as it has always been done, a student holds the device containing the film/digital sensor inside the patient's mouth and with other hand holds the collimator of the X-ray tube while another student presses the shutter button to generate X-rays. Under these conditions, we performed measurements of the effective dose to the chest and the measurement of the absorbed dose to the hands by placing thermoluminescent dosimeters (TLD) without radiological protection procedures.

Figure 1. Students taking intraoral radiographs in a traditional manner.

Figure 2 illustrates students taking an intraoral radiograph using film holders/digital sensor in the patient's mouth to avoid hand-holding the image receptor and the X-ray exposure (shot) is performed at a distance of 2 meters from the patient. Under these conditions, TLD dosimeters are placed to measure the effective doses to the chest and the absorbed dose to the hands to determine if the implementation of film /digital sensor holders and a distance of 2 meters from the patient is sufficient for the annual effective doses received to be below 1 mSv for the public recommended by the ICRP (International Commission on Radiological Protection) in its publication 103 [ICRP 2007].

Figure 2. Students taking an intraoral x-ray using film holders and at a distance of 2 meters from the patient.

Materials and methods

The study was conducted at the Nezahualcóyotl Dental Clinic of the Autonomous Metropolitan University, Xochimilco Unit, Mexico. 32 students were selected from the sixth and twelfth trimester Dentistry program, divided into two groups A and B, Group A obtained radiographic images in a traditional way without radiological safety procedures (Figure 1) and Group B used film holders/digital sensor inside the patient's mouth to avoid placing the image receptor with their hands inside the patient's mouth, in addition the student made the X-ray exposure (shot) 2 meters from the patient (Figure 2). Each student was given two thermoluminescent dosimeters (TLD), one to measure effective dose to the thorax and another to measure absorbed dose to the hands, the thorax dosimeter had 4 crystals, two of Lithium Borate Li2B4O7 and two of Calcium Sulfate CaSO4 and the hand dosimeter had a Lithium Borate crystal, the dosimeter reader was Panasonic, model Ud-716AGL from the company Asesoría Integral en Dosimetría Termoluminiscente, SA de CV (AIDTSA) with dosimetry services authorized by the regulatory agency of Mexico (National Commission for Nuclear Safety and Safeguards). A Corix 70-USV X-ray equipment was used in the radiographic examinations. The dose received by the students was monitored over a 20-day period during clinical care. The number of radiographic images during that period was also recorded. The annual dose was determined based on the estimated number of radiographs per year. The database was processed using the SPSS 22 statistical package.

Results

Table 1 describes the overall results of the estimated annual effective doses to the thorax received by students in Group A (radiographic examinations without radiation protection procedures) and Group B (radiographic examinations with radiation safety procedures). The average annual effective dose for Group A exceeds the 1 mSv/annual public dose limit recommended by ICRP Publication 103 [ICRP 2007]. However, for Group B, the average annual effective dose received is below the dose limit of ICRP Publication 103 [ICRP 2007].

Table 1. General results of the estimated annual effective doses to the thorax

Student No.	Without Radiation Safety Procedures, Group A (mSv)	With Radiation Safety procedures, Group B (mSv)
1	0.54	0.36
2	1.21	0.28
3	0.79	0.18
4	2.06	0.23
5	1.99	0.45
6	0.55	0.21
7	0.54	0.46
8	1.32	0.28
9	1.63	0.30
10	1.12	0.33
11	1.03	0.17
12	2.23	0.47
13	3.75	0.36
14	1.50	0.37
15	1.27	0.36
16	1.27	0.33
Mean ± Standard Error	1.42 ± 0.20	0.32 ± 0.23

Figure 3 illustrates the comparison of the 95% confidence intervals of the estimated annual effective doses to the thorax between the two groups of students with reference to the dose limits of ICRP publication 103 [ICRP 2007], the difference in annual effective doses between Groups A and B is statistically significant for $p < 0.05$.

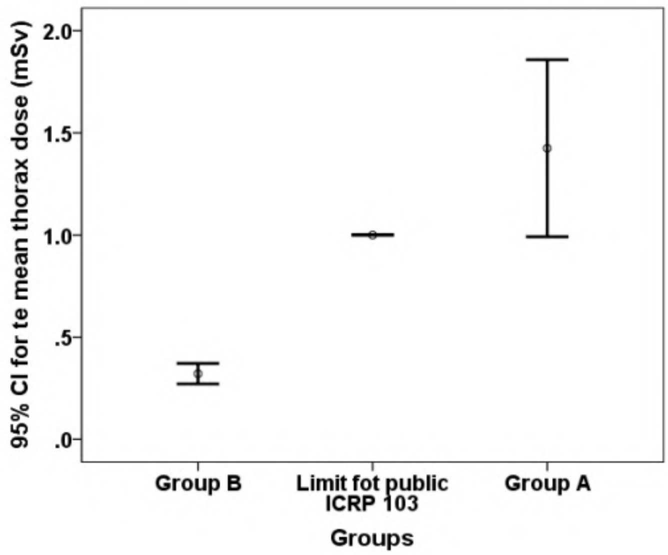


Figure 3. The graph shows the 95% confidence intervals in annual effective doses to the thorax between groups and compares them with the public dose limit of 1 mSv/annual from ICRP publication 103 [ICRP 2007].

Table 2 describes the overall results of the estimated annual absorbed doses received by students in Group C (radiographic examinations without radiation safety procedures) and Group D (radiographic examinations with radiation safety procedures). The average annual absorbed dose received by Group C is higher than the average dose received by Group D.

Table 2. Describes the general results of the estimated annual absorbed doses at the hands of Dentistry students.

Student No.	Without Radiation Safety Procedures, Group C (mGy)	With Radiation Safety procedures, Group D (mGy)
1	1.10	0.72
2	2.08	0.63
3	1.97	0.63
4	4.71	0.64
5	7.55	0.63
6	1.93	0.56
7	1.03	0.55
8	3.19	0.69
9	3.35	0.35
10	3.02	0.48
11	3.25	0.51
12	3.78	0.66
13	3.25	0.68
14	3.00	0.67
15	2.95	0.67
16	2.75	0.69
Mean ± Standard Error	3.01 ± 0.38	0.61 ± 0.24

Figure 4 shows the comparison of the 95% confidence intervals in the annual absorbed doses received by hands between groups C and D. We found that there is a statistically significant difference between groups for $p < 0.05$ in performing radiographic examinations with and without radiological safety procedures

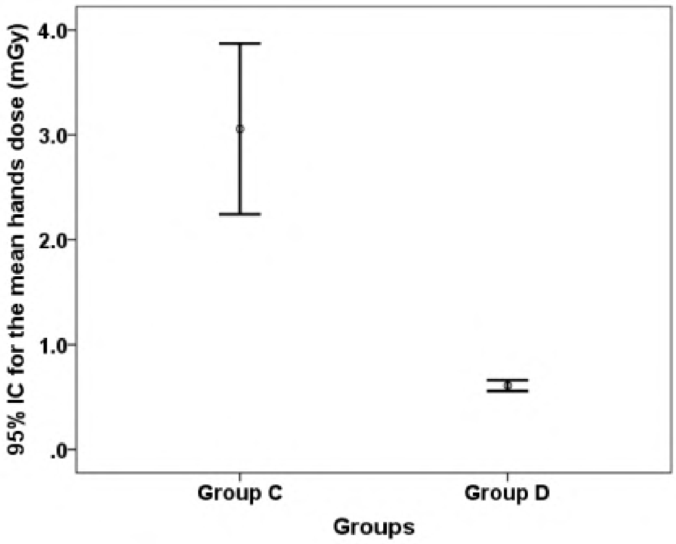


Figure 4. Illustrates the comparison of 95% confidence intervals in annual absorbed dose to hands between groups.

Discussion

We begin this section with some statistics that show the magnitude of the annual increase in the number of dental radiological studies. A sample of patients in the United States during the 2014-2015 period estimated approximately 500 million intraoral radiographs were taken annually, and at least 1.5 billion dental radiographs were taken globally, however, the doses to dentists are small, but the collective doses cannot be ignored [IAEA 2022].

According to Table 1, second column, and Figure 3, we determined an annual effective dose to students of 0.32 mSv. Similar reports have been published by other authors, Kim et al., [2016] found effective doses of 0.18 to 0.13 mSv/year in dentists. Basheer et al., [2021] reported effective doses of 0.55 mSv/year in dental students. Cheng et al., [2022] carried out a sampling from 2013 to 2020 of the effective doses in dental radiology personnel in the range of 0.15-1.11 mSv/year. These dose values were obtained by applying radiological safety procedures and generally do not exceed the public dose limits of 1 mSv/annual recommended by ICRP in its publication 103 [ICRP 2007].

According to our experience with dental students nationwide, most intraoral radiographic examinations are performed as shown in Figure 1 without radiological safety procedures. Part of this study was conducted under these conditions and the estimated average effective dose to the thorax was 1.42 mSv (Table 1, first column), which exceeds the annual effective dose limit of 1 mSv recommended for the public by ICRP publication 103 [ICRP 2007]. In other studies, the annual effective doses were similar. Alashban et al. [2020] found an average effective dose of 0.72 mSv/year in dental radiology workers, although 2% exceeded 2 mSv/year. Loya et al. [2016] estimated effective doses of up to 2.59 mSv/year in dental students. Reddy et al., [2015] reported effective doses in dentists no greater than 1.5 mSv/year. In the studies conducted by Reddy et al., [2015] and Cheng et al., [2022] they comment that the annual effective doses in dental radiology workers and students can exceed the effective dose limits recommended by ICRP publication 103 of 1 mSv/year [ICRP 2007] when preventive radiation protection strategies are not implemented.

In this study we only determined the estimated effective doses annually to the thorax and absorbed doses to the hands, but students also receive unjustified doses to the lens, thyroid and breast tissue, where the thyroid and breast are very sensitive to the effects of ionizing radiation that increase the risk factors of suffering a stochastic biological effect (cancer) in the future or other non-cancer diseases [Preston et al., 2003].

Conclusions

Dental students are considered members of the public in relation to their clinical practice with intraoral dental radiology, but they receive unjustified effective doses to the thorax that exceed the public limit of 1 mSv/year recommended by ICRP publication 103 [ICRP 2007]. They also receive unjustified absorbed doses to the hands, eye, thyroid and breast because the dental curriculum does not cover topics such as radiation protection, radiation biology, image quality and legislation.

The remaining work is to continue measuring the effective doses to dental radiology students and workers from the lens, thyroid, and breast tissue. In addition, we are promoting the creation of a standard for intraoral dental radiology and dental tomography to improve radiation protection for students/workers and patients, including image quality.

Acknowledgements

We would like to thank Asesoría Integral en Dosimetría Termoluminiscente, SA de CV (AIDTSA) for their support of the TLD dosimetry service for our study. Their dosimetry service is authorized by Mexico's regulatory agency (National Commission for Nuclear Safety and Safeguards).

We thank the participating students and staff of the Nezahualcóyotl Stomatology Clinic (LDC) of the Autonomous Metropolitan University, Xochimilco Unit, for their support of the study.

References

Alashban Y., Shubayr N., Almalki M., Albeshan S., Aldawood S., Aldosari A. (2021). Assessment of radiation dose for dental workers in Saudi Arabia (2015–2019), Journal of King Saud University-Science, Volume 33, Issue 1,101250, ISSN 1018- 3647,https://doi.org/10.1016/j.jksus.2020.101250.

Alvarez O, D. E., Gaona, E., Molina F, N., Castañeda C, R. E., Lopez G, L. (2024), & Sociedad Mexicana de Irradiación y Dosimetría, Ciudad de México (Mexico). Determination of effective dose to dentistry students in the care of patients with dental radiology using TLD. 24 international symposium on solid state dosimetry; 6. Symposium on clinical medical physics (ISSSD 2024), Ciudad de Mexico (Mexico). Publisher: International Atomic Energy Agency, International Nuclear Information System (INIS Repository): INIS RN: 55098873, Report number: INIS-MX—3769. https://inis.iaea.org/records/f79w1-7h715.

Barba Ramírez L., Ruiz García de Chacón V., Hidalgo Rivas A. (202). El uso de rayos X en odontología y la importancia de la justificación de exámenes radiográficos. Av Odontoestomatol [Internet]. Ago [citado 2025 Jul 22]; 36(3): 131-142. Disponible en:http://scielo.isciii.es/scielo.php?script=sci_arttext&pid=S0213- 12852020000300002&lng=es. Epub 05-Dic-2022. https://dx.doi.org/10.4321/s0213- 12852020000300002.

Basheer BT, Allahim RI, Alarfaj SS, Alkharashi TA, Fallatah AA, Alqahtani AS, Aljarallah SS. (2021). Evaluation of Radiation Exposure in Open Dental Clinics Using Thermoluminescence Dosimeters and Questionnaires. Contemp Clin Dent. Jul-Sep; 12(3):266-275. doi: 10.4103/ccd.ccd_598_20. Epub 2021 Sep 21. PMID: 34759684; PMCID: PMC8525804.

Cheng FC, Lee MC, Chen MH, Ting CY, Su CP, Liu SY, Chiang CP. (2022). Occupational radiation exposure for various medical radiation workers, especially the dental radiation workers, in Taiwan from 2013 to 2020. J Dent Sci. Oct; 17(4):1544-1552. doi: 10.1016/j.jds.2022.06.011. Epub 2022 Jun 28. PMID: 36299339; PMCID: PMC9588832.

ICRP, 2007. The 2007 recommendations of the International Commission on Radiological Protection. Ann. ICRP. 103.

International Commision on Radiological Protection. (1991). 1990 Recommendations of the International Commission on Radiological Protection. Vol 21, Publication 60. (1-3).

International Atomic Energy Agency (2022). Safety Reports Serie Vol.108, Radiation Protection in Dental Radiology. Vienna. Disponible en: https://wwwpub.iaea.org/MTCD/Publications/PDF/PUB1972_Web.pdf

Kaviani F, Esmaeili F, Balayi E, Pourfattollah N. (2007). Evaluation of X-ray Protection Methods Used in Dental Offices in Tabriz in 2005-2006. J Dent Res Dent Clin Dent Prospects. Spring; 1(1):49-52. doi: 10.5681/joddd.2007.008. Epub 2007 Jun 10. PMID: 23277833; PMCID: PMC3522909.

Kim YJ, Cha ES, Lee WJ. (2016). Occupational radiation procedures and doses in South Korean dentists. Community Dent Oral Epidemiol. 2016 Oct; 44(5):476-84. doi: 10.1111/cdoe.12237. Epub 2016 May 5. PMID: 27146959

Loya M, Sanín LH, González PR, Ávila O, Duarte R, Ojeda SL, Montero-Cabrera ME (2016). Measurements of radiation exposure of dentistry students during their radiological training using thermoluminescent dosimetry. Appl Radiat Isot. 2016 Jan; 107:234-238. doi: 10.1016/j.apradiso.2015.10.031. Epub 2015 Oct 28. PMID: 26562447

Mupparapu M, Kulkarni R. (2022). Assessment of radiology baseline knowledge of freshman North American dental students prior to any formal instruction. Eur J Dent Educ. Aug; 26(3):563-568. doi: 10.1111/eje.12732. Epub 2021 Dec 21. PMID: 34862838.

Preston DL, Shimizu Y, Pierce DA, Suyama A, Mabuchi K. (2003). Studies of mortality of atomic bomb survivors. Report 13: Solid cancer and noncancer disease mortality: 1950-1997. Radiat Res. Oct; 160(4):381-407. doi: 10.1667/rr3049. PMID: 12968934.

Reddy SS, Rakesh N, Chauhan P, Clint JB, Sharma S. (2015). Is dosimetry still a necessity in current dental practice? J Radiol Prot. Dec; 35(4):911-6. doi: 10.1088/0952- 4746/35/4/911. Epub 2015 Nov 30. PMID: 26619179.

Tirado-Amador Lesbia Rosa, González-Martínez Farith Damián, Sir-Mendoza Francisco Javier. (2015). Uso controlado de los rayos X en la práctica odontológica. Rev. Cienc. Salud [Internet]. Jan [cited 2025 July 22]; 13(1): 99-112. Available from: http://www.scielo.org.co/scielo.php?script=sci_arttext&pid=S1692- 72732015000100009&lng=en. https://doi.org/10.12804/revsalud13.01.2015.08.

Ubeda Carlos, Nocetti Diego, Aragón Marco. (2018). Seguridad y Protección Radiológica en Procedimientos Imagenológicos Dentales. Int. J. Odontostomat. [Internet]. Sep [citado 2025 Jul 22]; 12(3): 246-251. Disponible en: http://www.scielo.cl/scielo.php?script=sci_arttext&pid=S0718-381X2018000300246&lng=es. http://dx.doi.org/10.4067/S0718- 381X2018000300246.

Wilches-Visbal Jorge Homero, Castillo Pedraza Midian Clara, Jamil Khoury Helen. (2021). Protección Radiológica en Radiología Dental. CES odontol. [Internet]. June [cited 2025 July 22]; 34(1): 52-67. Available from: http://www.scielo.org.co/scielo.php?script=sci_arttext&pid=S0120-971X2021000100052&lng=en. Epub Feb 16, 2022. https://doi.org/10.21615/cesodon.34.1.6.

Valenzuela Medina C, Hidalgo Rivas A. (2021). Evaluación del conocimiento en protección radiológica en odontología. Revisión narrativa. Av Odontoestomatol [Internet]. Dic [citado 2025 Jul 22]; 37(4):77-182. Disponible en: http://scielo.isciii.es/scielo.php?script=sci_arttext&pid=S0213-12852021000400005&lng=es. Epub 05-Dic-2022. https://dx.doi.org/10.4321/s0213-12852021000400005

XXV ISSSD International Symposium on Solid State Dosimetry
September 22 – 26, 2025 Belo Horizonte, Minas Gerais - Brazil
CDTN UFMG

Monte Carlo characterization of neutron personal dosimeter based on TLD-600 and TLD-700 irradiated with ^{252}Cf neutron source on the ISO-phantom at LPN-CIEMAT

Lenin E. Cevallos-Robalino[1,2]*, Roberto Méndez Villafañe[2], Hector Rene Vega-Carrillo[3], Marco Aurélio de Sousa Lacerda[4], Víctor David Larco-Torres[1], Orlando Barcia-Ayala[1]

[1] Universidad Politécnica Salesiana, NANOTECH research group, 090109, Guayaquil, Ecuador

[2] Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas CIEMAT, Laboratorio de Patrones Neutrónicos, 28040, Madrid, Spain

[3] Universidad Autónoma de Zacatecas, CAR-UaEN, 98060, Zacatecas, México

[4] Centro de Desenvolvimento da Tecnologia Nuclear (CDTN/CNEN), 31270-901, Belo Horizonte, MG, Brazil

* lcevallos@ups.edu.ec

Abstract

This work evaluates the neutron response of neutron personal dosimeter, based on a commercial model, constituted by four thermoluminescence dosimeter (TLD), two TLD-600 and two TLD-700 with and without a Cd foil inside a plastic badge. A group of six dosimeters, modeled with the MCNP6.2 Monte Carlo code (Kulesza et al., 2022), was placed on a water-equivalent ISO-phantom and simultaneously exposed to a ^{252}Cf neutron source.

A current irradiation in the Neutron Standards Laboratory at CIEMAT (LPN-CIEMAT), (Spain) has been simulated (Méndez-Villafañe et al., 2019). For this purpose, a detailed simulation of the LPN including the irradiation bench and other details related to neutron source capsule has been developed. The six dosimeters were positioned on the front surface of the ISO-phantom at varying source-to-detector distances: 50, 75, 100, and 150 cm (Kim et al., 2010).

The neutron response of the dosimeters and the effect of simultaneous irradiation of six monitors were evaluated. The contribution of backscattered neutrons on the response, as a function of the distance for the real neutron field associated with the LPN, was evaluated through two kinds of simulations: (a) in vacuum, and (b) considering all the structures of the LPN (walls, bench, pool, etc). The results clearly show a decrease in neutron response with increasing distance (Le et al., 2024) and a minimum influence of the other dosimeters. This work contributes to improving the dose assignment and the calibration process of neutron personal dosimeters following the recommendations proposed by ISO 8521-2 (ISO 8529-2, 2000).

Keywords: MCNP; TLD600; TLD700; neutron personal dosimeter; thermoluminescence.

1 Introduction

There are different types of neutron personal dosimeters for the purposes of neutron monitoring and safety assessment active and passive (Thermo Fisher Scientific, 2022b, 2022a). In the calibration of neutron personal dosimeters, the quantity of interest is the neutron personal dose equivalent, *Hp*(10**)**, which represents the absorbed dose at a depth of 10 mm in a phantom that simulates human tissue (Kim et al., 2010).

The Neutron Standards Laboratory (LPN: Laboratorio de Patrones Neutrónicos) of the CIEMAT or Centro de Investigaciones Energéticas Medioambientales y Tecnológicas, has a ^{252}Cf calibration neutron source inside a capsule holder and stored under at least 1m of water in a pool where an automatic system allows their remote manipulation and its controlled through a software from the Control Room. This source is recommended by International Organization for Standardization (ISO 8529-1, 2021) for dosimeter calibration purposes.

A detailed model was carried out using the Monte Carlo method with the MCNP6.2 code, aiming to characterize the response of TLD-600 and TLD-700, corresponding to LiF:Mg,Ti enriched in ^{6}Li and in ^{7}Li respectively and provided by ThermoFisher Scientific, under the neutron field emitted by the ^{252}Cf source at LPN-CIEMAT. For this purpose, the geometry of the calibration setup was virtually replicated, including the encapsulated source, the calibration laboratory (LPN), the ISO-conforming water phantom, and the arrangement of the dosimeters on the phantom at different distances.

The neutron fluence at the calibration point was determined from the neutron emission rate of the ^{252}Cf source of LPN-CIEMAT (Fig. 1) in the positions where the personal dosimeters on-phantom were placed to 50, 75, 100 and 125 cm from the source.

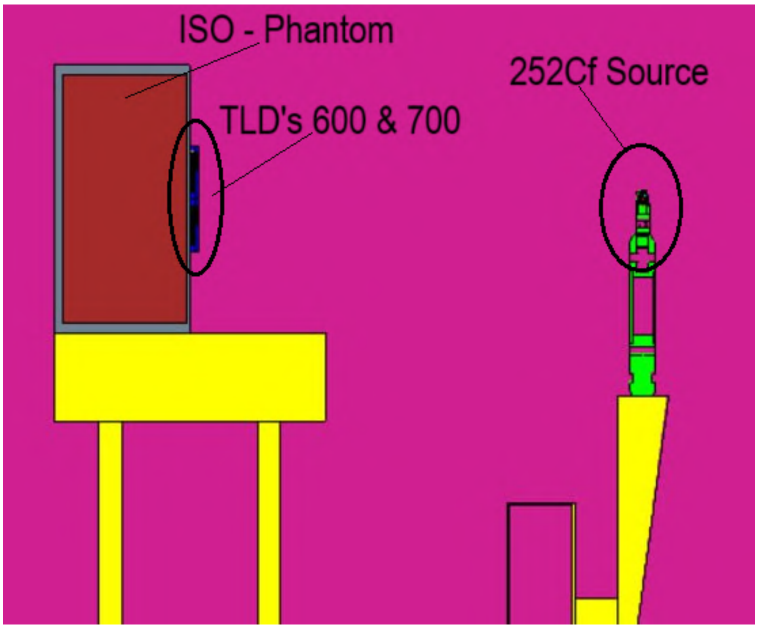


Figure 1: Scheme of the MC model of the irradiation bench at LPN-CIEMAT with ^{252}Cf source and ISO-phantom with TLD’s 600 and 700.

The aim of this work was to characterize neutron personal dosimeters based on TLD’s 600 and 700 irradiated with **^{252}**Cf neutron source on the ISO-phantom at LPN-CIEMAT. The system was simulated by Monte Carlo methods using MCNP6 code (Kulesza et al., 2022). This simulation allowed for an accurate evaluation of the neutron fluence and the personal dose equivalent *Hp*(10) under real irradiation conditions, providing essential information for the experimental interpretation and validation of the results.

2 Material and method

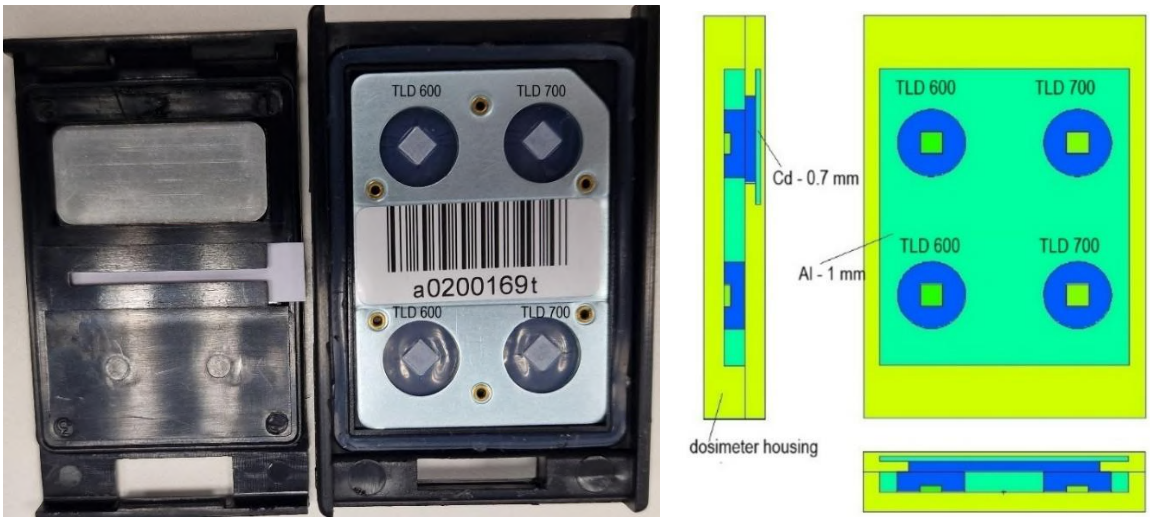
2.1 Neutron source

The source to irradiate the TLD cards consist of 236 µg of ^{252}Cf (5 GBq) with an emission rate $5.471 \cdot 10^{8}$ s^{-1} ± 2.6% (k = 2) on 20th May 2012 (Méndez-Villafañe et al., 2019), traceable to the National Institute of Standards and Technology (USA). This source is inside a stainless-steel double capsule with dimensions 7.8 mm-diameter and 10.0 mm-high (Guzman-Garcia et al., 2015). The Dosimeters card was irradiated on the ISO Phantom sited at LPN-CIEMAT bench. A detailed model of the phantom with a set of TLD cards on its surface was performed using the MCNP6.2 code, and the transport of neutrons from the source to the neutron detectors was simulated. This model has been inserted into a previous virtual model of the laboratory (Guzman-Garcia et al., 2015), used to verify the shielding of the facility (Mendez-Villafane et al., 2014), and where a detailed simulation of the ^{252}Cf neutron source has been employed (Mendez-Villafane et al., 2019-2).

2.1.1 TLD 600 & 700 dosimeters

The pair of TLD-700 and TLD-600 is widely used in mixed neutron-gamma fields to obtain not only gamma dose but also neutron dose due to the presence of the ^{6}Li isotope (D'Avino et al., 2022), beeing one of the most popular dosimeters for thermal neutrons (Takenaga, 1977). The main difference between 600 and 700 TLD chips is that they are enriched by ^{6}Li and ^{7}Li isotopes, respectively (Behmadi et al., 2024). Due to these differences in the neutron capture cross section, it is possible to detect thermal and epithermal neutrons with LiF detectors (Obryk et al., 2014). Alpha particles are produced through the nuclear reaction ^{6}Li(n,α) ^{3}H in the TLD cards (Sokolov et al., 2017).

The TLD card has four thermoluminescent chips (Fig. 2); two of these are TLD-600 and the other two are TLD-700. TLD-600 chip is LiF:Mg,Ti, having 95.6% ^{6}Li and 4.4% ^{7}Li, whereas the TLD-700 is enriched up to 99.99% ^{7}Li. The (n, α) cross section of ^{6}Li for thermal neutrons is several orders of magnitude larger than the ^{7}Li cross section; therefore, the TLD-600 has a larger response to thermal neutrons, while the response of TLD-700 to neutrons is practically negligible by comparison. On the other hand, both types of TLDs have identical density, chemical composition, and effective atomic number; therefore, TLD600 and TLD700 respond to gamma-rays in the same way (Vega-Carrillo et al., 2025).



a) b)

Figure 2: a) Structure of a Thermoluminescence Dosimeter Card; b) Thermoluminescence dosimeter card modeling through MCNP6 code.

2.1.2 ^{252}Cf neutron source in the neutron irradiation room

The neutron personal dose equivalents in the neutron irradiation room were calculated by simulating the LPN-CIEMAT room. The simulations were performed by adapting a realistic geometry that included air, walls, doors, a rail system, a source holder and guide, and the geometry of the neutron source itself.

The neutron spectrum emitted by the ^{252}Cf source was employed using a discretized structure of 89 energy groups, following ISO 8529-1:2021 (ISO 8529-1, 2001). This approach was chosen to more accurately characterize the ^{252}Cf neutron spectrum in simulation processes. The energy range below 35 keV was organized into 20 energy groups, while energies above 35 keV were distributed into 69 energy groups (Méndez et al., 2021). The source has 236 μg ^{252}Cf dispersed in a ceramic matrix encapsulated in 2 mm thick A316 stainless steel to ensure its integrity under all conditions (Guzman-Garcia et al., 2015; Méndez-Villafañe et al., 2019).

2.1.3 ISO water slab Phantom

The calibration and dose assignment of personal neutron dosimeters requires the use of the ISO water slab phantom (30×30×15) cm^3**,** in which the dosimeter is placed in the centre to simulate the human thorax according to ICRU (Otto, 2024) and following the ISO 8529-3 recommendations.

The phantom used in the simulations was composed of polymethylmethacrylate (PMMA) filled with water.

2.2 Methodology

The TLD setup consisted of two pairs of chips: TLD-600, based on ^{6}Li, which is sensitive to both gamma and neutron radiation, and TLD-700, based on ^{7}Li and only responsive to gamma radiation. One pair of TLD-600 and TLD-700 is covered by a sheet of Cd to distinguish between thermal and epithermal, and fast neutrons. The dosimeter consists of a set of four chips and the Cd sheet inside a badge (ABS plastic) to be placed on the thorax of the workers to measure *Hp*(10) magnitude. A group of 6 dosimeters has been simulated on the ISO phantom (Fig.3) to study the variations in the response due to the different positions on the phantom for a given distance to the neutron source.

In the TLD-600 chips, neutrons were detected via the well-known ^{6}Li(n,α)^{3}H reaction, whose cross-section decreases as 1/v with neutron energy. Hence, TLD-600 is considered a detector for thermal neutrons. A Cd sheet is used in these detectors to separate the thermal contribution.

The behavior of this material as a dosimeter comes from the proportionality between the resulting thermosluminescent (TL) signal and the absorbed dose in the chips. The simulations, performed with MCNP6.2, aimed to evaluate the number of ^{6}Li reactions in the TLD chips, related to the TL response, when they are exposed to neutron fields. The tally F4 has been used with a multiplier FM4 to obtain the number of reactions inside the TLD chip as the product of fluence rate and cross section for this material.

Multiple simulations were conducted to estimate the number of tritium (^{3}H) nuclei and alpha particles generated in the TLD-600 chips during the detection period. The TLD cards were exposed to fast neutrons by varying the distance from the centre of the ^{252}Cf source to the dosimeters placed on the ISO slab phantom, at 50, 75, 100, and 125 cm.

This relationship enables analysis of the expected signal ratio between the two types of dosimeters, thereby separating the neutron and gamma contributions.

The response relative to the reference distance, used to compare how the signal decreases when moving away from the source, is defined in equation (1):

$$R_{rel(d)} = \frac{R(d)}{R(d_r)} \qquad (1)$$

Where $R(d)$ is the response at the distance d and $R(d_r)$ is the response to the reference distance, that corresponds to 75, following the ISO recommendations. Indicating the relative signal drop over distance, checking the effect of dispersion.

The figure 3 shows the configuration of the 6 dosimeters and the TLD-600 and 700 chips simulated through MCNP6 code.

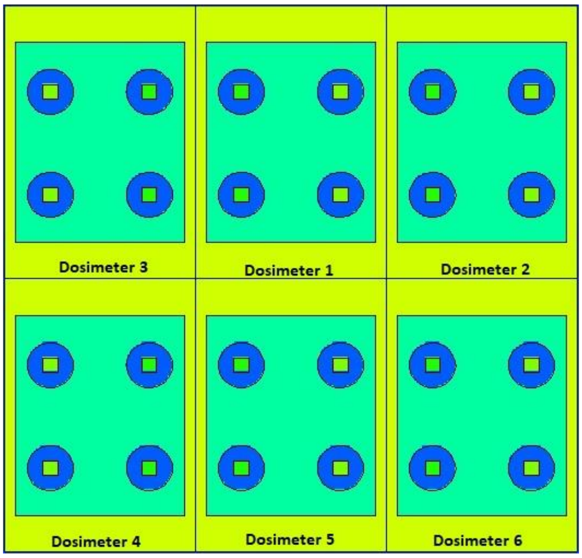


Figure 3: Scheme of the 6 dosimeters with the TLD chips simulated through MCNP6 code.

3 Results

Figure 4 shows the average responses of TLD-600 and TLD-700 chips as a function of the distance from the ^{252}Cf source (with and without Cd foils) in vacuum, and considering all the structures of the LPN (walls, bench, pool, etc).

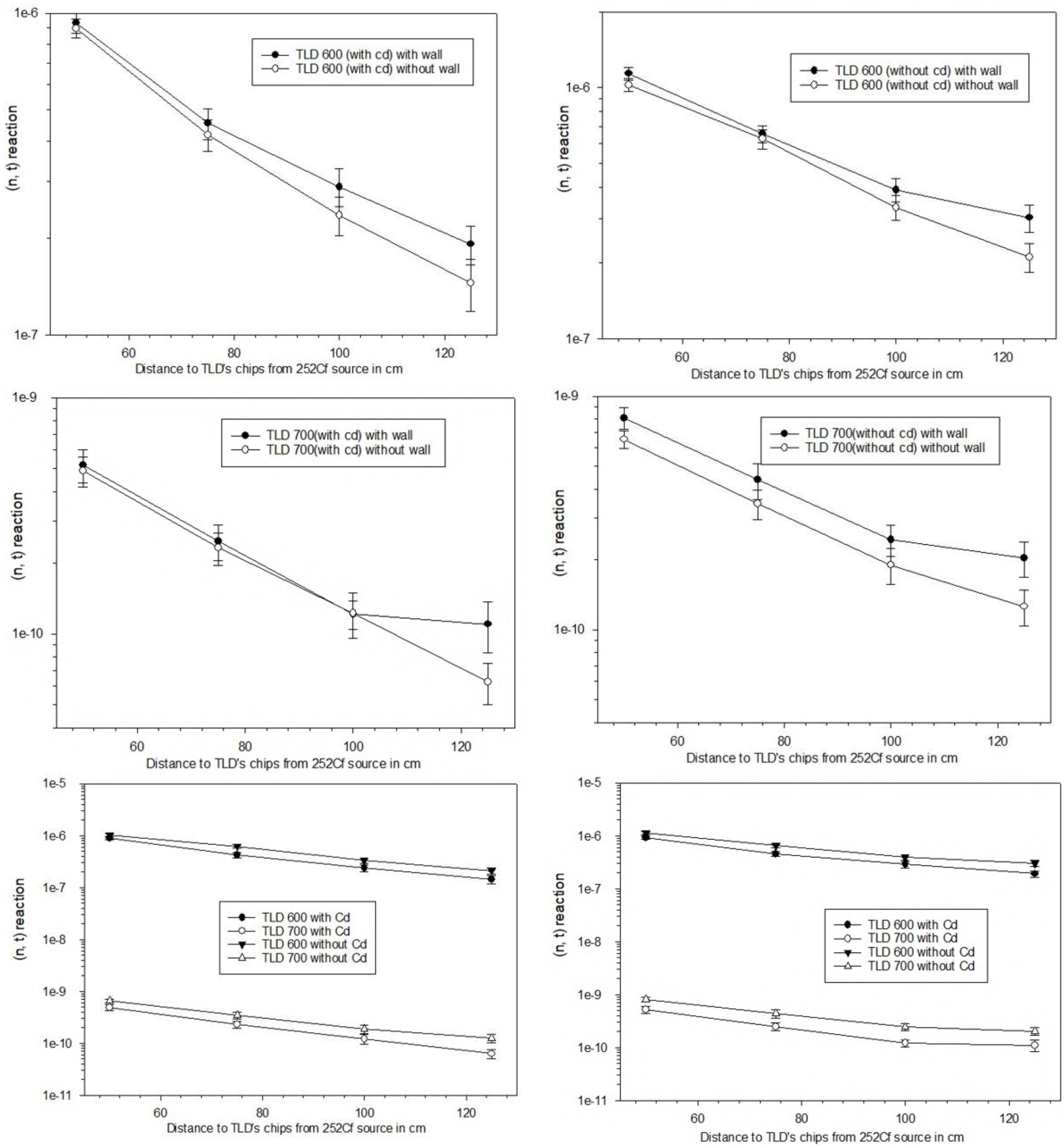


Figure 4: Average responses of TLD-600 and TLD-700 chips as a function of the distance from the ^{252}Cf source (with and without Cd foils) in vacuum, and considering all the structures of the LPN (walls, bench, pool, etc).

4 Conclusions

The dosimetric responses of the TLDs were evaluated at four source-to-phantom distances: 50 cm, 75 cm, 100 cm, and 125 cm, using a ^{252}Cf neutron source. The evaluation was carried out using the MCNP6 code, which enabled a detailed model of the geometry, materials, and neutron interactions.

A 0.46 mm cadmium (Cd) filter was placed over the top detectors to suppress the contribution from thermal neutrons. This shielding significantly influenced the relative responses observed at different distances.

These results confirm the effectiveness of the cadmium filter in isolating the fast neutron component, allowing a clearer distinction of the dosimetric contribution from different energy regions of the neutron spectrum.

The distances were selected in accordance with the recommended measurement setup for calibration fields using isotopic neutron sources, as indicated in ISO 8529-1 (ISO 8529-1, 2021)

The simulations were performed at the LPN-CIEMAT, including the geometry and room size. Therefore, the results are highly dependent on physical configuration, including the backscattered neutrons.

To evaluate the contribution of wall-induced backscatter, comparative simulations were performed with and without the surrounding walls.

This analysis confirms the need to account for laboratory-specific conditions when performing dosimetric characterizations and interpreting neutron field homogeneity and intensity distributions.

Acknowledgements

The first author Lenin E. Cevallos-Robalino, thanks to the Universidad Politécnica Salesiana (UPS) for the support received. In addition, he wishes to express his special gratitude to Roberto Méndez-Villafañe for allowing him to carry out the postdoctoral stay. Marco Aurélio de Sousa Lacerda is grateful for the financial support from FAPEMIG APQ-02622-24 and to CNPq for the PQC 308483/2025-1 fellowship and financial support from INCT/INAIS 406303/2022-3.

References

Behmadi, M., Mohammadi, S., Ravari, M. E., Mohammadi, A., Loushab, M. E., Bahreyni Toossi, M. T., & Ghergherehchi, M. (2024). Neutron dosimetry with a pair of TLDs for the Elekta Precise medical linac and the evaluation of optimum moderator thickness for the conversion of fast to thermal neutrons. *Nuclear Engineering and Technology*, *56*(2), 753–761. https://doi.org/10.1016/j.net.2023.11.014

D'Avino, V., Ambrosino, F., Bedogni, R., Campoy, A. I. C., La Verde, G., Vernetto, S., Vigorito, C. F., & Pugliese, M. (2022). Characterization of Thermoluminescent Dosimeters for Neutron Dosimetry at High Altitudes. *Sensors*, *22*(15), 1–11. https://doi.org/10.3390/s22155721

Guzman-Garcia, K. A., Mendez-Villafañe, R., & Vega-Carrillo, H. R. (2015). Neutron field characteristics of Ciemat's Neutron Standards Laboratory. *Applied Radiation and Isotopes, 100*, 84–90. https://doi.org/10.1016/j.apradiso.2014.10.028

ISO 8529-1. (2001). *International Standard. Reference Neutron Radiations – Part 1: Characteristics and Methods of Production.* (First Edit).

ISO 8529-1. (2021). Neutron reference radiations fields - Part 1: Characteristics and methods of production. In *International Organization for Standardization* (Vols. 8529-1:202). chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://cdn.standards.iteh.ai/samples/80060/97743810cea14c43b572fbd946d4eb20/ISO-8529-1-2021.pdf

ISO 8529-2. (2000). *International Organization for Standardization, Reference neutron radiations – Part 2: Calibration fundamentals of radiation protection devices related to the basic quantities characterizing the radiation field, ISO 8529-2:2000,*.

Kim, J. H., Park, H., & Choi, K.-O. (2010). Calibration of neutron personal dosimeters in a realistically simulated neutron irradiation room. *Radiation Measurements*, *45*(10), 1544–1547.

https://doi.org/10.1016/j.radmeas.2010.07.003

Kulesza, J. A., Adams, T. R., Armstrong, J. C., Bolding, S. R., Brown, F. B., Bull, J. S., Burke, T. P., Clark, A. R., Forster III, R. A. (Art), Giron, J. F., Grieve, T. S., Josey, C. J., Martz, R. L., McKinney, G. W., Pearson, E. J., Rising, M. E., Solomon Jr., C. J. (CJ), Swaminarayan, S., Trahan, T. J., … Zukaitis, A. J. (2022). MCNP Code Version 6.3.0 Theory & User Manual. In *LA-UR-22-30006, Rev. 1* (Vol. 28). https://www.osti.gov/servlets/purl/1889957/

Le, N. T., Hoang, T. P. H., Hui, Z., Bui, D. K., Nguyen, N. Q., Hoang, V. K., Dinh, T. H., Nguyen, M. C., Cao, V. H., & Tran, H. N. (2024). Intercomparison of neutron personal dose equivalent measured by thermoluminescence dosimeters. *Applied Radiation and Isotopes*, *205*(December 2023), 111175. https://doi.org/10.1016/j.apradiso.2024.111175

Mendez-Villafane, R., Guerrero, J. E., Embid, M., Fernandez, R., Grandio, R., Perez-Cejuela, P., Marquez, J. L., Alvarez, F., & Ortego, P. (2014). Design and verification of the shielding around the new Neutron Standards Laboratory (LPN) at CIEMAT. *Radiation Protection Dosimetry*, *161*(1–4), 393–397. https://doi.org/10.1093/rpd/nct348

Méndez-Villafañe, R., Lacerda, M. A. S., Campo, X., Lorente, A., Ibañez, S., Gallego, E., & Vega-Carrillo, H. R. (2019). Neutron spectra from Neutron Standards Laboratory (LPN/CIEMAT) sources with two Bonner sphere spectrometers. *Radiation Physics and Chemistry*, *155*, 97–100. https://doi.org/10.1016/j.radphyschem.2018.06.015.

Méndez-Villafañe, R. et al, " Estimation of the anisotropy emission of LPN/CIEMAT neutron sources: effect of heavy capsule holders", Radiat. Prot. Dosim. 186, 4, 2019, https://doi.org/10.1093/rpd/ncz050

Méndez, R., Gómez-Ros, J. M., Thomas, D. J., Thompson, A. K., & Bedogni, R. (2021). Revision of the ^{252}Cf and D_2O-moderated ^{252}Cf reference neutron fields for use in radiation protection dosimetry. *Radiation Physics and Chemistry*, *184*, 109433. https://doi.org/10.1016/j.radphyschem.2021.109433

Obryk, B., Batistoni, P., Conroy, S., Syme, B. D., Popovichev, S., Stamatelatos, I. E., Vasilopoulou, T., & Bilski, P. (2014). Thermoluminescence measurements of neutron streaming through JET Torus Hall ducts. *Fusion Engineering and Design*, *89*(9–10), 2235–2240. https://doi.org/10.1016/j.fusengdes.2013.12.045

Otto, T. (2024). ICRU Report 95: new operational quantities for external radiation exposure. *Annals of the ICRP*, *53*(1_suppl), 74–79. https://doi.org/10.1177/01466453241283931k

Sokolov, A., Fehrenbacher, G., Kozlova, E., Radon, T., & Zahradnik, I. (2017). Optimization of a neutron dosimeter for the high energy accelerators. *EPJ Web of Conferences*, *153*, 01013. https://doi.org/10.1051/epjconf/201715301013.

Takenaga, M. (1977). Thermoluminescent Response to Thermal Neutrons of Mixture of CaSO4: Tm and Non-Luminous ^{6}LiF. *Journal of Nuclear Science and Technology*, *14*, 292–299.

Thermo Fisher Scientific. (2022a). *Harshaw 4500 dual TLD reader and workstation*. https://tools.thermofisher.com/content/sfs/brochures/D16603.pdf

Thermo Fisher Scientific. (2022b). *Thermo scientific harshaw TLD materials and dosimeters.* https://assets.thermofisher.com/TFS-Assets/LSG/Catalogs/Dosimetry-Materials-Brochure.pdf

Vega-Carrillo, H.R., Baltazar-Raigosa. A., Hernandez-Davila, V.M., Gallego, E., Garcia-Fernandez, G., & Bedenko, S.V. (2025). Monte Carlo design of a novel passive Regular Parallelepiped Neutron Spectrometer (TLD600/RPS). *Radiation Physics and Chemistry, 236*, 112861. https://doi.org/10.1016/j.radphyschem.2025.112861

Using Gamma Spectrometry as a Tool for Assessing Residual Dose in Irradiated Topaz

B. G. M. Natália[1]*, A. M. C. Daniel[1]

[1] Centro de Desenvolvimento da Tecnologia Nuclear, CDTN/CNEN, UFMG Campus Pampulha, 31.270-901, Belo Horizonte, Brazil

* nbg@cdtn.br, campolina@cdtn.br

Abstract

The irradiation of topaz in nuclear reactors is a widely recognized technique for modifying its optical properties, especially for producing blue hues such as “Swiss Blue” and “London Blue”, which are highly valued in the gemstone market. However, during exposure to the neutron flux, certain trace elements present in the crystal matrix undergo neutron activation, resulting in the formation of radioactive isotopes that make the material temporarily radioactive. Ensuring the radiological safety of these gemstones before commercialization requires careful monitoring of this residual activity. In this context, gamma spectrometry—especially using high-purity germanium (HPGe) detectors—plays a fundamental role in the identification and quantification of the radionuclides formed, as well as in estimating the decay period necessary to achieve the release levels defined by national and international regulatory standards. This study presents the application of gamma spectrometry in monitoring residual activity in irradiated topaz, discusses the main activated radionuclides, and highlights the importance of this technique as a tool for quality control and radiological safety in the post-irradiation phase.

Keywords: Topaz; Gamma Spectroscopy; Reactor; Neutron; Residual Dose.

1 Introduction

Topaz is a natural mineral composed of aluminum fluorsilicate which, when exposed to ionizing radiation, can undergo structural and electronic alterations that affect its coloration. This process has been widely adopted by the gemological industry to commercially enhance gemstones, especially in the production of intense blue varieties. One of the most effective ways to induce these changes is the exposure of colorless topaz to a fast neutron flux in nuclear reactors. Although effective from a gemological point of view, this procedure results in the activation of trace elements in the crystal, generating radionuclides.

Among the most predominant radionuclides observed in irradiated topaz are ^{46}Sc, ^{182}Ta, ^{65}Zn, ^{134}Cs, and ^{54}Mn, all with long half-lives and primarily gamma emitters. These isotopes pose radiological risks if the gemstones are handled or commercialized before proper decay. Regulatory bodies such as the International Atomic Energy Agency (IAEA) and the National Nuclear Energy Commission (CNEN) in Brazil establish exemption and clearance limits to ensure public safety.

The correct identification of radionuclides and the precise estimation of the time required for them to reach safe levels are fundamental. For this purpose, gamma spectrometry is a well-established analytical technique, even recognized by international standards such as the IAEA, allowing high-resolution detection of the gamma photons emitted during radioactive decay. The use of high-purity germanium (HPGe) detectors provides precise quantification of the radionuclides, ensuring compliance with radiological protection standards and offering a solid basis for quality control protocols in the gemstone sector ORTEC, a subsidiary of AMETEK, Inc. (2015); Gilmore (2008); Crouthamel (1970); Krambrock et al. (2007); Leal et al. (2007).

2 Metodology

Topaz samples used in this work originate from two distinct locations in the state of Minas Gerais: Marambaia (Region 6) and Teófilo Otoni (Region 1). Following the criteria established by the International Atomic Energy Agency (IAEA) standards, the samples were cleaned with nitric acid and subjected to ultrasonication to avoid further activation of impurities. Subsequently, they were irradiated in the TRIGA IPR-R1 nuclear reactor, a TRIGA Mark I type reactor, located at the Center for Nuclear Technology Development (CDTN/CNEN) in Belo Horizonte. After a waiting period of several months to reduce the activity of the samples, gamma spectrometry analysis was performed.

The TRIGA IPR-R1 reactor has several irradiation positions, with the central tube being the region with the highest flux of fast and epithermal neutrons. The thermal, fast, and epithermal neutron fluxes recorded in this region are, respectively: $(3{,}14 \pm 0{,}02) \times 10^{12}$, $(2{,}48 \pm 0{,}01) \times 10^{12}$ e $(0{,}24 \pm 0{,}01) \times 10^{12}$ nêutrons/($cm^2 \cdot s$) Menezes et al. (2023). Since it is undesirable to irradiate these samples with thermal neutrons, a cadmium holder coated with aluminum was developed to irradiate the samples in the central tube. This acts as a filter for thermal neutrons, given the high absorption cross-section for this part of the neutron spectrum. This holder, a cylinder closed at the bottom and open at the top, with a height of 10 cm and a diameter of 2 cm, was designed with a lid of the same material, ensuring the filtering of the spectrum from all directions and reducing unwanted activation. The gemstones were placed inside, respecting the limit of up to two-thirds of the tube's total volume, and irradiated for 14 hours at a power of 80 kWt. After irradiation, the samples were stored for one week in a lead castle located in the controlled area of the CDTN, being continuously monitored by a Geiger-Müller detector (RADOS RDS-80A, serial no. 260019), until the radiation levels reached values close to the *background*, allowing for safe handling.

The gamma spectrometry analyses presented in this article were performed four months after the irradiation, following the methodology previously described. The objective of this analysis is to identify the remaining activated radionuclides and determine the residual activity of each. The measurements were carried out with a high-purity germanium (HPGe) detector, ORTEC GMX50-83, coupled to a MultiChannel Buffer (MCB), model ORTEC DSPEC-LF, and a computer running GammaVision V8 software, also from ORTEC. Each sample was positioned at a fixed distance of 5 cm from the detector's face, with an acquisition time of 24 hours per spectrum.

The energy and efficiency calibration of the system was performed using multigamma standard sources, namely: Co-60, Cs-137, Eu-152, Zn-65, Co-57, and Cd-109. The energy calibration was based on the characteristic peaks of Co-60, Cs-137, and Eu-152, while the efficiency calibration employed an extended combination of these radionuclides, with energies ranging from 14.41 to 1408.01 keV.

The calibration data indicated agreement between the actual values and those adjusted by polynomial regression, with relative deviations below 3% for most of the analyzed energies. The efficiency curve was fitted using a Tcc polynomial, with specific coefficients available in the calibration report.

The specific activities of the radionuclides were calculated based on the net areas of the gamma peaks, corrected for system efficiency and the respective emission probabilities. The calibration also included an evaluation of the resolution (FWHM), with adjusted values for each energy channel, ensuring the reliability of spectral identification and activity quantification.

The radiological compliance assessment of the samples considered the following main regulatory parameters:

- **Annual dose limit for the public:** 1/year of effective dose and 50/year of equivalent dose to the skin.
- **Exemption limit from radiological control:** 10, as established by CNEN and IAEA.
- **Average annual dose limit for planned public exposure:** 10/year, according to CNEN standard NN 3.01.
- **Dose limit for commercial release of irradiated gemstones:** between 0.01 and 0.02/year, as per the *IAEA Safety Standards Series SSG-36*.

To evaluate regulatory compliance, the criterion of the sum of maximum concentration fractions established by CNEN NN 3.01 was used (Equation 1), in accordance with both national and international standards de Energia Nuclear (2005); IAEA (2014); de Energia Nuclear (2014); Comissão Nacional de Energia Nuclear (2021).

$$X_m = \left(\sum \frac{f(i)}{X(i)} \right)^{-1} \tag{1}$$

where:

- $f(i)$ is the fraction of the activity concentration of radionuclide i in the mixture;
- $X(i)$ is the applicable level for radionuclide i, as presented in Tables C-I and C-III;
- n is the number of radionuclides present.

To estimate the equivalent and effective dose for gemstones T6_1, T1_6, and T2_6, simulations were performed using the Visual Monte Carlo (VMC) software VMC Software (2025). The position of the gemstone was modeled as if it were a necklace, located at 7.8 cm on the X-axis and 139.2 cm on the Z-axis of the mathematical phantom. A total count of 100,000 events was simulated for 300 days of continuous use of the topaz.

The resulting values were compared with reference limits in order to meet the requirements established by CNEN NN 3.01, as well as international standards such as IAEA GSR Part 3, thereby supporting safe and evidence-based decisions regarding the release of these materials.

3 Results and Discussion

Gamma spectrometry analysis enabled the identification and quantification of several activated radionuclides in irradiated topaz samples originating from two distinct regions of Minas Gerais, Brazil: Teófilo Otoni (Region 1) and Marambaia (Region 6). The main radionuclides detected were: Mn-54, Sc-46, Cs-134, Ta-182, and Zn-65 — all gamma emitters with half-lives greater than 40 days.

Specific Activity

Sample T4_1, from Teófilo Otoni, exhibited the highest total specific activity, with Mn-54 contributing 14.78 Bq/g, which accounts for 81% of the total exemption ratio. The estimated time for this sample to reach safe release levels is 176 days, considering exclusively the decay of Mn-54.

In the Marambaia samples, T1_6 and T2_6, Mn-54 was also the predominant radionuclide, with 10.68 Bq/g and 10.37 Bq/g, respectively. These values correspond to 95% and 97.6% of the total exemption ratio in the respective samples. Consequently, the estimated exemption times were 29.76 days (T1_6) and 16.39 days (T2_6), significantly shorter compared to the sample from Region 1.

The other radionuclides — Sc-46, Cs-134, Ta-182, and Zn-65 — showed contributions well below 10 Bq/g and individual ratios below 0.01 in most cases, being considered negligible in the exemption time calculation.

The Figure 1 presents a comparative graph of the three irradiated topaz samples analyzed in this study.

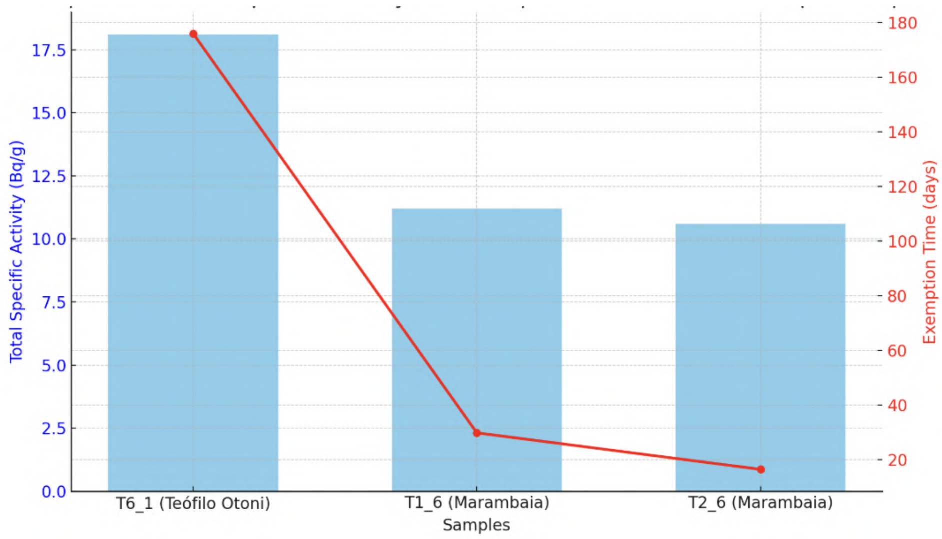


Figure 1: Comparison of Specific Activity and Exemption Time in Irradiated Topaz Samples

The blue bars represent the total specific activity values (in Bq/g) determined by gamma spectrometry for each sample, while the red line indicates the estimated radiological exemption time (in days), calculated based on the half-life of the predominant radionuclide, Manganese-54 (Mn-54).

This graphical approach allows for a comprehensive visualization of the relationship between the residual radioactive load of the samples and the time required for them to reach safe release levels according to established regulatory criteria.

It is evident that sample T6_1 (Teófilo Otoni) requires the longest exemption time (176 days), due to its higher specific activity, while the Marambaia samples demand significantly shorter periods for release.

According to national and international standards, the specific activity limit of 10 Bq/g is still exceeded, after four months of decay, only for the radionuclide ^{54}Mn.

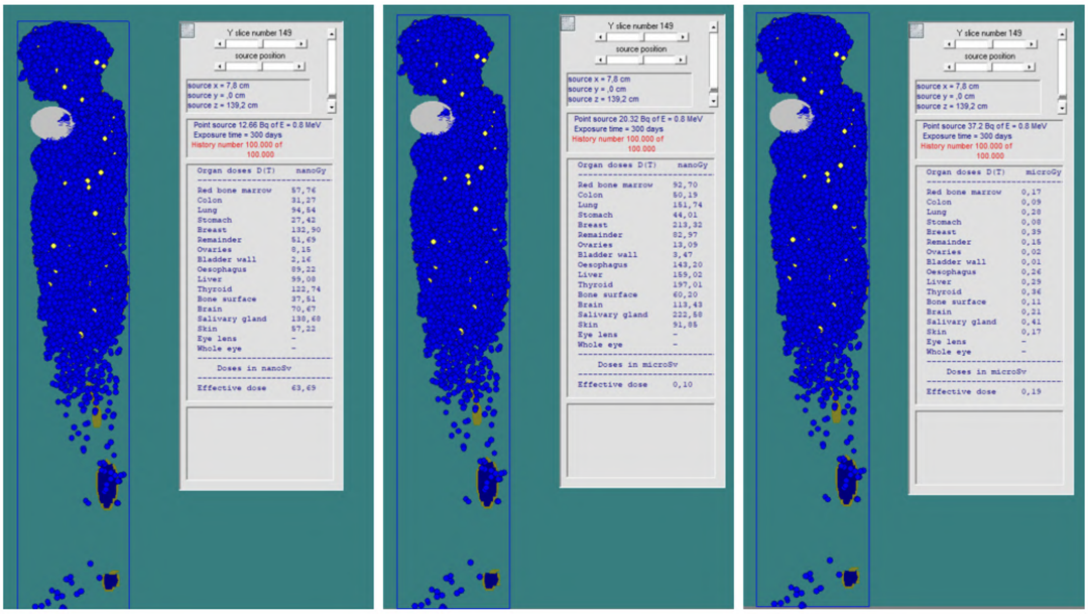
Effective Dose

Considering that ^{54}Mn (0.836 MeV) contributed between 81% and 98% of the total effective dose, dose control was primarily based on this radionuclide and calculated using the Visual Monte Carlo (VMC) software.

The results for gemstone T6, which exhibited the lowest residual dose, indicated an effective dose of 0.064 μSv due to ^{54}Mn. Gemstones T1 and T2 showed doses of 0.10 μSv and 0.19 μSv, respectively. Although these values exceed the exemption level recommended by the IAEA for commercially exempt gemstones (0.01–0.02 μSv), they remain significantly below the annual effective dose limit for the general public (1 mSv/year), and below the 10 μSv per-event threshold for planned exposure situations.

The Figure 2 shows the calculations of absorbed doses (Gy) and effective doses (Sv) for ^{54}Mn in the three gemstones.

Figure 2: Absorbed dose calculation (Gy) and effective dose (Sv) for ^{54}Mn in gemstones T6, T1, and T2



The highest absorbed dose values were recorded in the thyroid, breasts, and salivary glands, as expected due to the anatomical proximity of these tissues to the position of the gemstone and their high radiosensitivity.

For the thyroid, the absorbed doses were 0.123 μGy, 0.197 μGy, and 0.360 μGy. For breast tissue, the doses were 0.133 μGy, 0.213 μGy, and 0.390 μGy, while for the salivary glands, the values reached 0.139 μGy, 0.222 μGy, and 0.410 μGy.

Considering the radiation weighting factor is 1, the absorbed dose in this case is numerically equal to the equivalent dose, with the only difference being the unit used — Sievert (Sv).

Although there are no specific equivalent dose limits for the public regarding these radiosensitive organs, comparisons can be made using the reference value established for the skin, as defined by CNEN.

The annual dose limit for the skin is 50 mSv/year. The equivalent doses estimated by the program for the skin in gemstones T6, T1, and T2 were, respectively, 0.057 μGy, 0.092 μGy, and 0.17 μGy — values significantly lower than the dose limits set by CNEN.

4 Conclusions

The analysis based on the presented results demonstrates that, although the irradiated topaz samples from the regions of Teófilo Otoni (T6_1) and Marambaia (T1_6 and T2_6) contained multiple activated radionuclides, ^{54}Mn predominated, being responsible for the majority of the specific activity and the radiological exemption time.

Sample T6_1 from Teófilo Otoni stood out due to its higher specific activity and, consequently, the longest time required to reach safe release levels (176 days). In contrast, the Marambaia samples required significantly shorter decay periods for exemption (29.76 days and 16.39 days, respectively).

Regarding the effective dose, the calculated values were below the annual limits established for both occasional and planned public exposures. For a more detailed analysis, it would be worthwhile to also calculate the doses associated with the other radionuclides, even if present in smaller amounts.

Additionally, the estimated equivalent dose values for the thyroid, breast, salivary glands, and skin are significantly lower than national regulatory limits, taking as a reference the annual dose limit for the skin, which is 50 mSv/year.

Therefore, gamma spectrometry proved to be an effective technique for the identification and quantification of activation radionuclides present in the topaz samples, enabling a reliable assessment of compliance with the radiological safety criteria established by IAEA and CNEN regulations for radiological release.

Furthermore, the *VMC* program demonstrated efficiency in estimating the effective dose associated with the analyzed samples.

Acknowledgements

This work was supported by CDTN/CNEN and CNPq. Special thanks go to Elton Faria Araújo and Walter Lopes Ferreira who contributed by providing the samples for this research.

References

Comissão Nacional de Energia Nuclear (2021) Norma CNEN NN 5.01: Regulamento para o Transporte Seguro de Materiais Radioativos. aprovada pela Comissão Deliberativa da CNEN em 05 de março de 2021

Crouthamel C. (1970) *Applied Gamma-Ray Spectrometry*. Elsevier, Amsterdam

de Energia Nuclear C. N. (2005) *Diretrizes Básicas de Proteção Radiológica*. CNEN. Brasil

de Energia Nuclear C. N. (2014) Norma cnen nn 8.01: Licenciamento de instalações radiativas. Technical report. Comissão Nacional de Energia Nuclear (CNEN). Rio de Janeiro, Brasil

Gilmore G. (2008) *Practical Gamma-ray Spectrometry*. John Wiley & Sons, Chichester. 2 edition

IAEA (2014) *Radiation Protection and Safety of Radiation Sources: International Basic Safety Standards*. number GSR Part 3 in IAEA Safety Standards Series. International Atomic Energy Agency, Vienna, Austria

Krambrock K., Ribeiro L. G. M., Pinheiro M. B. V., Leal A. S., Menezes M. A. B. C., and Spaeth J.-M. (2007) Color centers in topaz: comparison between neutron and gamma irradiation. *Physics and Chemistry of Minerals* 34:437–444

Leal A., Krambrock K., Ribeiro L., Menezes M., Vermaercke P., and Sneyers L. (2007) Study of neutron irradiation-induced colors in brazilian topaz. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 580:423–426

Menezes M., Sabino C., and Jaćimović R. (2023) k_0-Neutron Activation Analysis at CDTN, Brazil: 27 years of history, development and main achievements. *Journal of Radioanalytical and Nuclear Chemistry* 332:3457–3468

ORTEC, a subsidiary of AMETEK, Inc. (2015) *GammaVision V8 User's Manual: Gamma-Ray Spectrum Analysis and MCA Emulator for Microsoft Windows*. United States. version 8 edition. oRTEC Part No. 783620 0915, Manual Revision K

VMC Software (2025) Vmc dose calculation. acesso em: 29 abr. 2025

A Data Science Approach on the Study of Al2O3 Thermoluminescent Glow Curves Under Varying Heating Rates

A. L. T. e Silva[1]*, L.C. Meira-Belo[1]

[1] Centro de Desenvolvimento da Tecnologia Nuclear, Radiation Dosimety Section, 30161-970, Belo Horizonte, Brazil

[*] aluiztasil@yahoo.com.br

Abstract

A data-driven classification and enrichment framework was developed to improve the extraction of kinetic parameters from thermoluminescent (TL) glow curves and to enhance their interpretability. The study analyzed 81 TL measurements of aluminium oxide (TLD-500 and ALOX-520) obtained under varying heating rates. Each glow curve was visually inspected and annotated with boolean morphological attributes—such as single peak, multi-peak, noisy signal, or background-only—derived from a reference dataset of over 6,000 standardized curves. These annotations enabled a conditional processing pipeline, allowing tailored treatments including smoothing, deconvolution, or exclusion, depending on curve morphology. Kinetic parameters were extracted using the Chen Peak Shape Method, Initial Rise, and Area methods. Dimensionality reduction via Principal Component Analysis (PCA) revealed structured variance associated with both heating rate and kinetic order, suggesting that these factors significantly influence TL curve behavior. The proposed workflow demonstrates the utility of structured dataset enrichment in TL analysis and supports future applications in automated classification and machine learning-based interpretation.

Keywords: *Machine Learning*; Thermoluminescence; Data Mining.

1 Introduction

Thermoluminescence (TL) is a luminescent phenomenon observed in insulating and semiconducting materials when exposed to thermal stimulation after prior exposure to ionizing radiation. This process involves charge carriers—electrons and holes—becoming trapped at defect sites within the material's crystal lattice. Upon heating, these trapped electrons gain sufficient energy to escape their traps and recombine with holes, releasing stored energy in the form of light. The intensity of this emitted light, recorded as a glow curve (GC), provides valuable information about the material's radiation exposure history and its structural characteristics. This makes TL an essential tool for studying defect states, charge recombination mechanisms, and kinetic properties in various materials. Once the glow curves are acquired, interpreting their shape and extracting meaningful physical parameters requires modeling based on TL kinetic theory. These glow curves are interpreted according TL kinetic models (BOS, 2006).

Traditional TL analysis relies heavily on expert-driven inspection and manual fitting of glow curves using predefined kinetic models (Yukihara, 2023). These conventional methods, while grounded in solid physical theory, are often labor-intensive and subject to operator bias—especially when dealing with noisy, multi-peak, or background-dominated signals. Moreover, manual classification lacks scalability and reproducibility when applied to large datasets. In contrast, a data-driven approach leverages structured datasets, machine learning classifiers, and conditional processing pipelines to automate and standardize the analysis. By annotating glow curves with morphological attributes and training models to generalize these classifications, the data-driven methodology allows for scalable processing of thousands of curves with minimal human intervention. This approach also enables the discovery of latent patterns and non-intuitive correlations that may not be evident in manual analysis, thereby enhancing the interpretability and reliability of kinetic parameter extraction. The objective of this paper is to propose and discuss a data-driven approach to analyze TL data, focusing on Al2O3 GCs heated under different rates.

2 Materials and Methods

2.1 Dataset Selection

The thermoluminescent (TL) glow curve dataset used as a standard to enrich the data processing study was obtained from a historical archive of measurements performed with a Risø TL/OSL reader at the Luminescent Dosimetry Laboratory (LCD/CDTN). Each record comprises two primary components: (i) a glow curve vector representing light intensity as a function of temperature, and (ii) metadata fields including heating rate, irradiation time, sample position, number of data points (n_points), pre-heat time, and pre-heat temperature.

To standardize the dataset for machine learning applications, a multi-stage filtering procedure (Figure 1) was applied. First, only glow curves recorded in TL mode were retained, representing approximately 75% of the initial dataset. Second, curves were restricted to those with a fixed resolution of 350 channels, ensuring compatibility across models and preserving numerical consistency in feature representation. Finally, the irradiation time was fixed at 1 second to eliminate variability due to dose history. This sequential filtering reduced the initial pool of ~60,000 glow curves to a curated training dataset of 6,152 TL records.

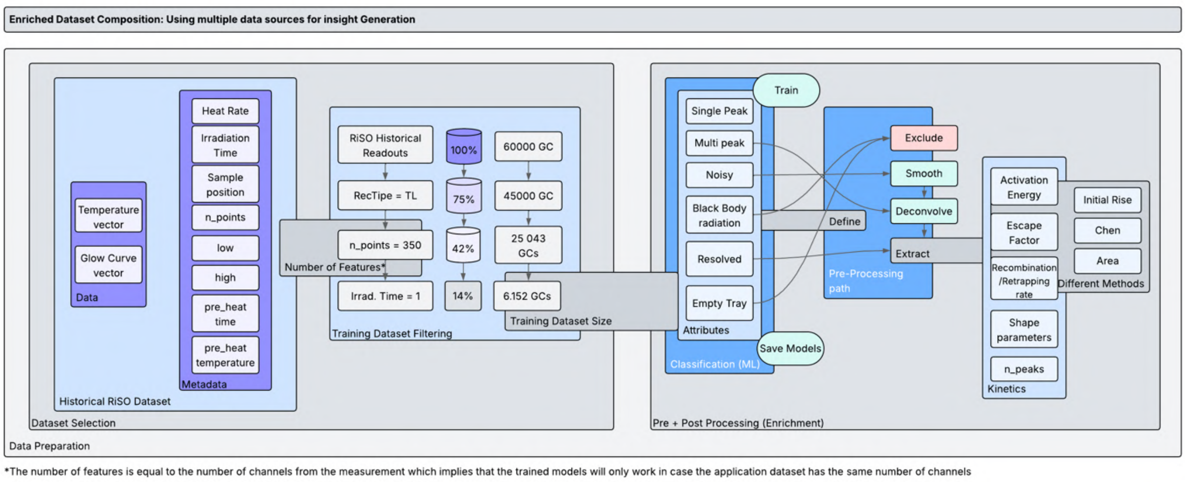


Figure 1 - Composition of an Eriched dataset, the use o multiple Datasources for insight generation

2.2 Morphological Attribute Classification

Each glow curve was manually or algorithmically labeled with one or more boolean morphological attributes to characterize its shape and quality (Figure 2). The attributes include: single peak, multi-peak, noisy, black body radiation, resolved peaks, and empty tray. These annotations were used to train supervised machine learning classifiers for downstream classification tasks and to define conditional pre-processing pipelines.

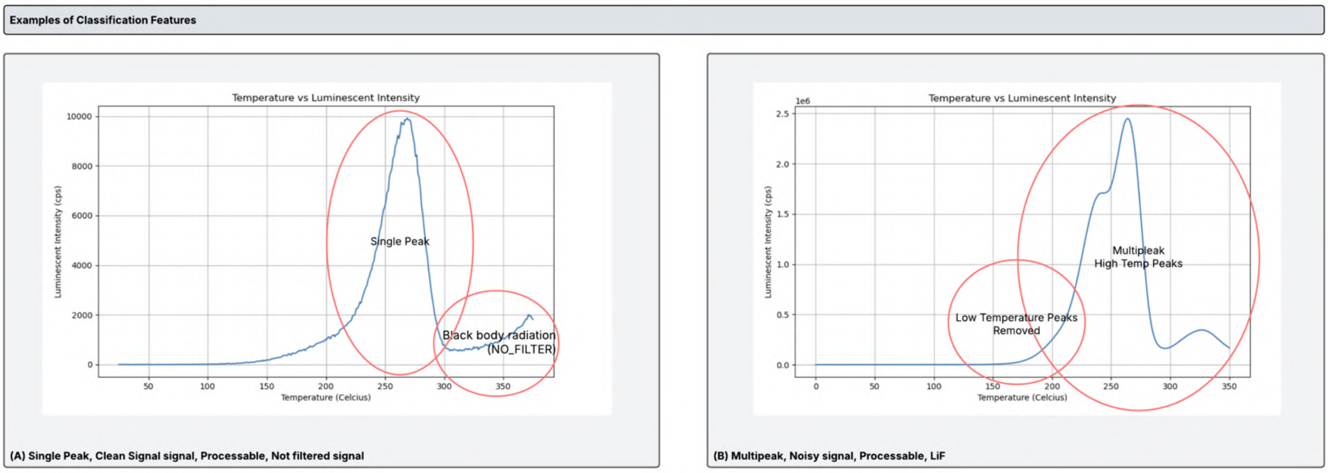


Figure 2 - Example of Glow Curve Morphological Characteristics

2.3 Conditional Pre-Processing Pipeline

Based on their classification, glow curves were directed into distinct processing paths:

- Exclude: Curves marked as invalid or empty were removed from further analysis.
- Smooth: Curves labeled as noisy underwent smoothing using a moving average or Savitzky–Golay filter to reduce high-frequency noise.
- Deconvolve: Multi-peak or complex curves were processed using Gaussian or general-order kinetic deconvolution to isolate overlapping components.
- These pre-processing steps ensured that only high-quality or correctly corrected data were subjected to kinetic analysis.

2.4 Kinetic Parameter Extraction

Considering that the heat rate is provided as GC metadata, following pre-processing, kinetic parameters were extracted from each glow curve using three established methods (Yukihara, 2023):

- Initial Rise Method –provides information regarding the activation energy (E);
- Chen Peak Shape Method - provides information regarding peak symmetry, E, escape factor s and Kinetic Order (b);
- Area Under the Curve Method - provides information regarding E, s and b;

The parameters computed include activation energy, escape factor, retrapping or recombination rate, peak shape descriptors, and the number of resolved peaks. These metrics offer insights into the physical trapping processes underlying each glow curve.

2.5 Model Training and Deployment

The enriched and pre-processed dataset was used to train a classification model capable of predicting morphological attributes directly from raw glow curves. A Support Vector Machine (SVM) model, as proposed by Amit and Datz (2019) for detecting anomalies in in TL GC, was selected due to its capacity to operate effectively in high-dimensional spaces and to identify linear boundaries within large, fixed-size feature sets. The model was trained using standardized glow curve inputs with a fixed dimensionality of 350 channels and was subsequently saved and integrated into the enrichment pipeline. This integration enables automated prediction of morphological attributes in future datasets, ensuring consistency and scalability of the classification workflow.Results

An experimental setup was programmed on the Risø DA-20 TL/OSL reader, where four ALOX-520 samples were irradiated using a Sr/Y beta source with a fixed exposure time of 1 second. The samples were subsequently read under variable heating rates ranging from 0.5 K/s to 10 K/s. A total of 162 glow curves (GCs) were recorded.

In the initial preprocessing stage, the glow curves were morphologically classified, and GCs exhibiting characteristics of blank signals or dominant black body radiation were excluded. This filtering step resulted in a curated dataset of 81 valid glow curves.

These 81 GCs were then subjected to a feature engineering pipeline, which involved the extraction of kinetic parameters, peak descriptors, and integrated intensities. The final dataset comprised 25 engineered features derived from physical and mathematical analysis of each glow curve (see Table 1).

Table 1 - Features from Enriched TL dataset

Feature Name	Theoretical Description	Origin / Method
tau_chen(eV)	Symmetry factor (τ)	Chen's method (eV)
omega_chen(eV)	Half-width (ω)	Chen's method (eV)
delta_chen(eV)	Peak width (Δ)	Chen's method (eV)
delta_error	Δ error	Propagation from Chen fit
ordem_cinetica	Kinetic order	Manual or automated classification
tmax	Peak temperature	Experimental value (°C)
t1	Temperature at 10% max	Glow curve point
t2	Temperature at 90% max	Glow curve point
IM	Index of maximum	Glow curve index
shape_factor	Shape factor (μ)	t2-t1 / ω
Sf_error	Shape factor error	Area Method error
delta	Peak width (Δ)	Empirical extraction
tau	Symmetry factor (τ)	Empirical extraction
omega	Half-width (ω)	Empirical extraction
Ea_ir	Activation energy	Initial Rise method (eV)
Ea_error	Ea error	Fitting residual
Ea_rsquared	Ea R^2	Goodness-of-fit
heat_rate	Heating rate	Experimental setting (°C/s)
Tmax	Max temperature	Experimental setting (°C)
irr_time	Irradiation time	Experimental setting (s)
trial	Trial ID	Experiment identifier
sample_number	Sample ID	Specimen identifier
high_temp	High temp limit	Readout parameter
low_temp	Low temp limit	Readout parameter
bg	Background index	Background position
i_max	Maximum intensity	Peak amplitude
integral	Glow curve integral	Integrated luminescence

To investigate the interdependencies among extracted kinetic, morphological, and signal-based parameters, a scatter matrix was constructed using 14 representative features from the thermoluminescence (TL) dataset (Figure 3). This multidimensional visualization enables pairwise assessment of linear relationships, redundancy, and potential clustering patterns within the dataset.

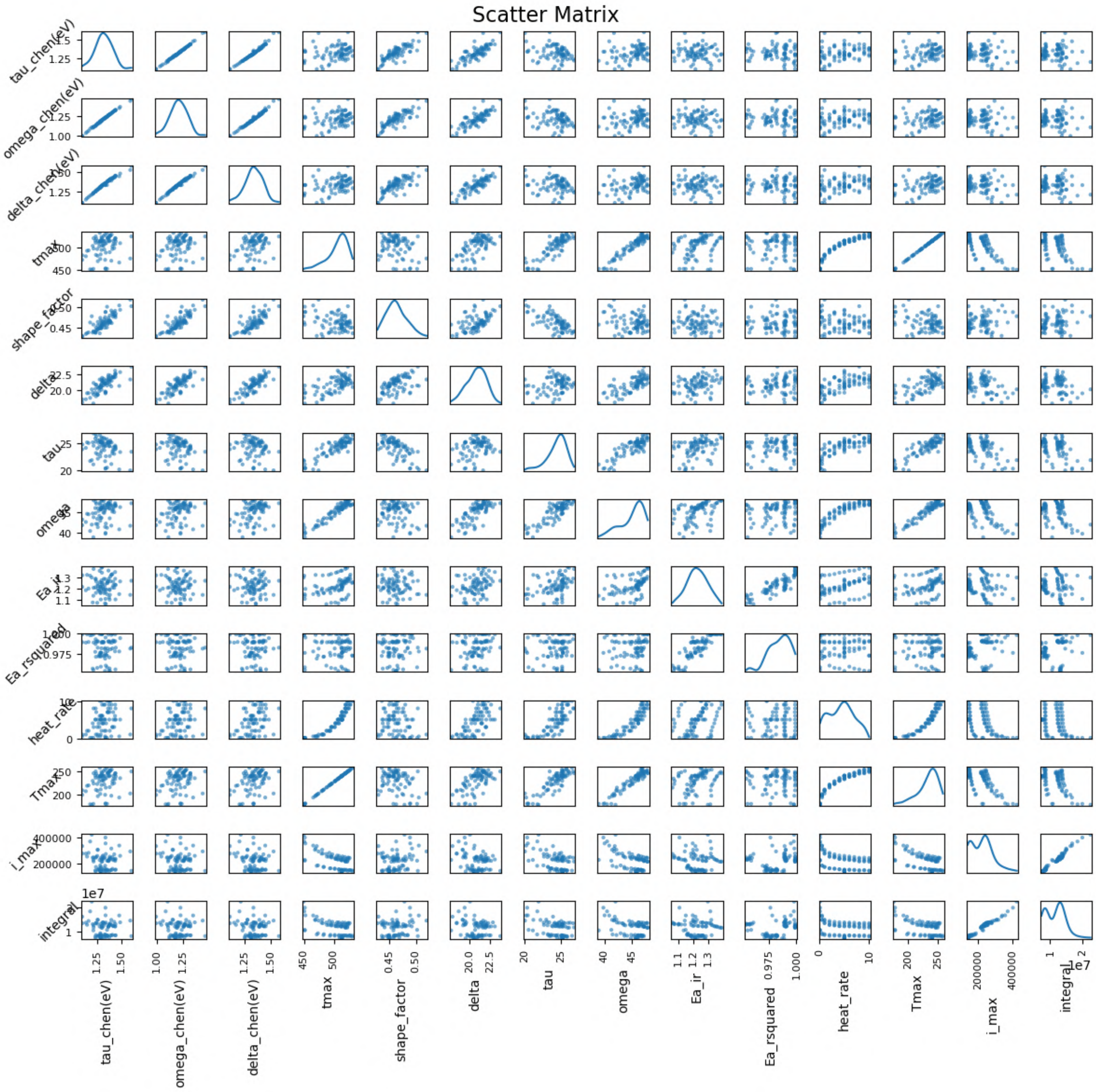


Figure 3 - Scatter Matrix of Important Features from Enriched Dataset

Notably, a strong linear correlation was observed among the Chen-derived kinetic parameters: tau_chen(eV), omega_chen(eV), and delta_chen(eV). These variables, derived from peak shape metrics, are inherently interrelated, reflecting their shared mathematical formulation. The near-perfect collinearity suggests that their simultaneous use in multivariate modeling may introduce multicollinearity, and dimensionality reduction or feature selection is recommended.

The integrated TL signal (integral) and maximum intensity (i_max) also showed high positive correlation, consistent with theoretical expectations where greater peak amplitude corresponds to increased total photon emission. This validates the reliability of both descriptors for quantifying TL emission strength.

Moderate correlations were identified between classical peak shape descriptors (shape_factor, delta) and kinetic parameters (tau, omega), supporting their utility as complementary indicators of recombination dynamics and trap population behavior. The parameters tmax and Tmax also demonstrated strong association, suggesting potential redundancy or variation in nomenclature for peak position.

In contrast, parameters derived from the initial rise method, such as Ea_ir and Ea_rsquared, exhibited low correlation with both the shape-based and intensity-based variables. This indicates that the initial rise analysis may capture distinct aspects of trap behavior or be influenced by factors not reflected in the main peak morphology, such as low-temperature noise or recombination pathway variation.

The heat_rate variable, used as a controlled experimental parameter, showed negligible correlation with the kinetic descriptors. This confirms that heating rate was not confounded with trap properties, consistent with proper quenching correction.

Overall, this correlation analysis reveals key relationships and redundancies in the dataset, guiding informed variable selection for downstream modeling. Specifically, kinetic parameters extracted via Chen's method are highly redundant, while signal-based descriptors provide robust intensity metrics. Initial rise results remain orthogonal and may warrant separate treatment in classification or kinetic modeling tasks.

3.1 Applying PCA

To reduce the dimensionality of the dataset and identify latent structures within the glow curve descriptors, Principal Component Analysis (PCA) was performed on a standardized feature matrix comprising 16 numerical variables (see Table 1). These variables represent kinetic parameters, morphological features, and intensity-based descriptors extracted from the curated population of 81 glow curves. PCA was selected as a suitable unsupervised technique to capture major variance directions while mitigating multicollinearity between highly correlated variables, particularly those derived from Chen's method (Pagonis et al., 2006)

Prior to applying PCA, all features were standardized using StandardScaler from Scikit Learn(Pedregosa et al., 2011) to ensure zero mean and unit variance, thereby eliminating scale-related bias in the eigenvalue decomposition.

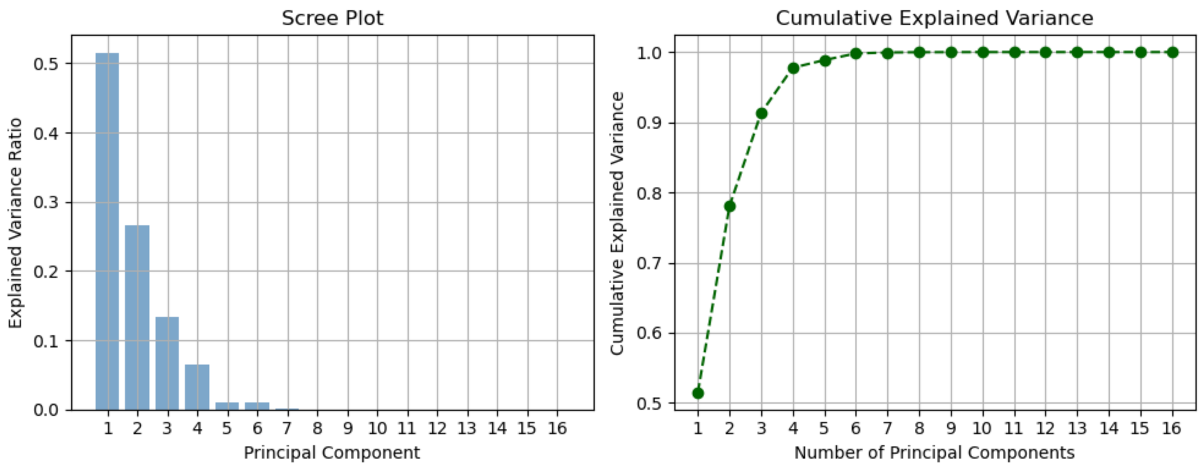


Figure 4 - Scree Plot and Cumulative Variance from dataset PCA

Figure 4 shows the distribution of variance across the principal components. The scree plot (left) reveals that the first three components account for the majority of the dataset's variance, with PC1, PC2, and PC3 explaining approximately 52%, 27%, and 13%, respectively. From PC4 onward, the explained variance sharply declines, indicating that additional components contribute minimally.

The cumulative explained variance plot (right) confirms that the first three components capture over 90% of the total variance, while the first five components exceed 97%. This supports the use of a reduced PCA space with three to five components for downstream analysis without significant information loss.

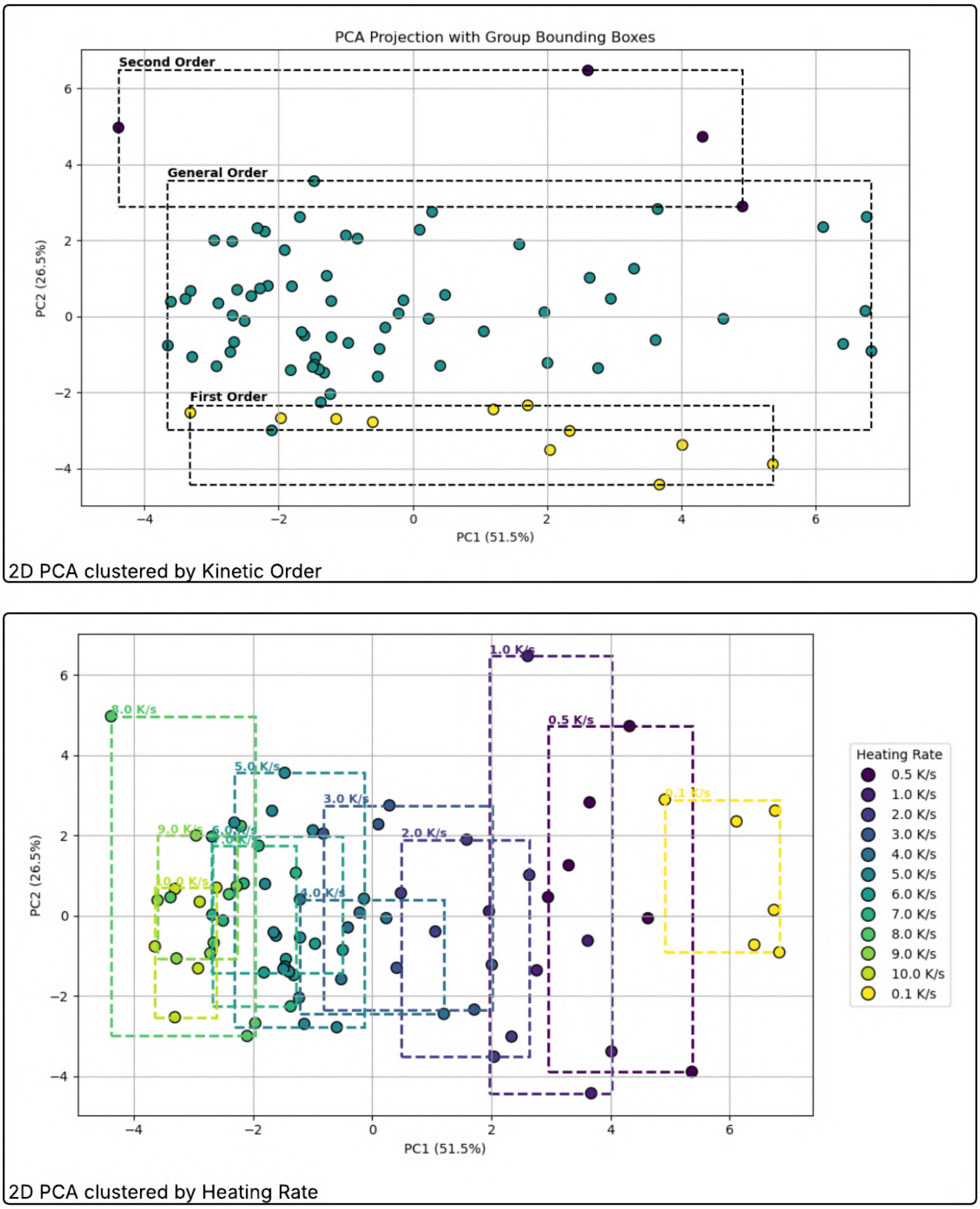


Figure 5 - PCA Applied with clustering for Kinetic Order (upper) and Heat Rate (lower)

Figure 5 shows two PCA projections based on the engineered glow curve dataset, capturing 51.3% and 17.5% of the total variance in the first and second principal components, respectively. In the top panel, the samples are grouped according to their kinetic order classification (*First Order*, *Second Order*, and

General Order). Each group is enclosed by a dashed bounding box, revealing partial overlap but some separation—particularly for the First Order group, which is more concentrated in the lower-left quadrant of the PCA space.

In the bottom panel, the same PCA projection is used, but the samples are colored and grouped by heating rate (ranging from 0.1 K/s to 10 K/s). Each heating rate is enclosed by a color-matched bounding box. The distribution reveals a gradient along PC1, where lower heating rates (e.g., 0.1–1.0 K/s) are clustered toward negative PC1 values, while higher heating rates (e.g., 5–10 K/s) are more dispersed or shifted toward positive PC1 values.

These observations indicate that both kinetic order and heating rate contribute meaningfully to the variance in the dataset. The PCA projection effectively captures this structure, supporting its use as a tool for visualizing relationships and preparing the data for downstream classification or clustering tasks.

4 Conclusions

This study demonstrated the effectiveness of a structured, data-driven approach to Thermoluminescent (TL) glow curve analysis, combining manual classification, feature enrichment, and conditional preprocessing. By leveraging boolean morphological attributes and tailored signal treatments, it was possible to improve the robustness of kinetic parameter extraction across a diverse set of aluminium oxide glow curves. The integration of Principal Component Analysis revealed that both heating rate and kinetic order introduce meaningful variance into the dataset, reinforcing the importance of standardized acquisition protocols in TL experiments. Overall, the proposed methodology enhances the interpretability and processing accuracy of TL data and provides a scalable foundation for future automation and machine learning applications in luminescence dosimetry and materials characterization.

Acknowledgements

FAPEMIG, CNPQ and CAPES and CDTN/CNEN for sponsoring this work.

References

Amit, G.; Datz, H. (2019). Radiat. Meas. 125: 1–6.

BOS, A. J. J. Theory of thermoluminescence. Radiation measurements, v. 41, p. S45-S56, 2006.

Ferreira, H.R.; Santos, A. (2015). Radiat. Prot. Dosim. 163(2): 166–172.

Pagonis, V.; Kitis, G.; Furetta, C. Numerical and Practical Exercises in Thermoluminescence, 1st ed., Roma, Italia: 208p., (2006).

Pedregosa, F.; Varoquaux, G.; Gramfort, A.; Michel, V.; Thirion, B.; Grisel, O.; Blondel, M.; Prettenhofer, P.; Weiss, R.; Dubourg, V.; Vanderplas, J.; Passos, A.; Cournapeau, D.; Brucher, M.; Perrot, M.; Duchesnay, É. (2011). J. Mach. Learn. Res. 12: 2825–2830.

Yukihara, E.G. (2023). Ceram. Int. 49: 24356–24369.

XXV ISSSD
International Symposium on Solid State Dosimetry
September 22 – 26, 2025
Belo Horizonte, Minas Gerais - Brazil
CDTN UFMG

Random walk of neutrons in a rotational environment

Alberto M. Campos[1], **Tarcísio P.R. Campos**[1,2]

[1] Universität Augsburg, Department of Mathematics, D-86135 Augsburg, Germany;

[2] Centro de Desenvolvimento de Tecnologia Nuclear, Universidade Federal de Minas Gerais, Av. Pres. Antônio Carlos, 6627 - Pampulha, 31270-901, Belo Horizonte, Brazil

Abstract

This paper investigates the motion of a free particle, specifically a neutron, interacting with a rotating medium through random collisions. Between interactions, the particle moves in a straight line, but at random times it collides with the environment, which rotates at constant angular velocity. We show that, due to energy loss and the geometric structure of the space, the neutron's motion diverges from the classical Brownian behavior. Instead, the particle exhibits a drift induced by a virtual centripetal-like effect, which emerges from the asymmetry of the collisions. A probabilistic-geometric analysis confirms that the particle asymptotically follows the rotation of the medium while drifting away from the origin. These findings suggest new applications for neutron spectral selection in nuclear technology, including radioisotope production and long-lived radionuclide transmutation in spent fuel.

Key words: Biased Brownian motion; neutron energy spectral segregation; reaction rate optimization.

1 Introduction

The first mention of the Brownian motion in the literature is related to the movement of pollen particles, where in still water the pollen perform a movement that approximates the Brownian motion; the original paper was not published, but one can follow the same experiment in Ford (1991). Interesting, if the water moves, the trace of the pollen particle is almost deterministic following the flow of the water. This raises intriguing open questions: How does the movement of the medium affect the random motion and is it possible to explain the deterministic component of its movement induced by the environment using its randomness? If the particles are not glue to the medium but suffer random punctual interactions alternating to independent linear travels, it diverges from their initial position?

The paper focuses on understanding how the medium affects random motion in a specific case where the neutron particle is moving in a rotational environment. In the problem formally stated, the particle is floating in the medium, moving along a straight line without interference from the space in which it is traveling until a certain point in time. At that moment, the particle with a certain relative velocity will collide with the space, altering its direction and velocity, repeating such behavior over and over.

Without solving mathematically due to the high number of collisions, one can stipulate a solution by a physical approximation. The particle as the process evolves will rotate following the velocity of the space, be dragged away by a centripetal force, and also have a random perturbation due to the random nature of the problem; see Löwen (2019) for some experiments and simulations of such movement.

In this article, we will analyze the impact of multiple collisions over the movement, showing that the physics stipulation needs further analysis. Spoiling the result, the particle will follow the rotation movement of the space, but a like-centripetal force will not appear with the same intensity as expected. The particle is walking in a straight line except in collisions, and thus this force that drags the particle always from the origin is not a classic centripetal force, existing only on the average results of these random collisions.

Such distinct neutron behavior may transform a distribution of neutrons into a biased environment to a suitable energetic condition, bringing a novel application in nuclear energy being a moderator neutron's energy. The present article will also describe some issues on those applications.

2 Methodology: the model of random walk with rotation bias

Consider a neutron particle at time $t > 0$ as a triplet (X_t, V_t, t), where $X_t \in \mathbb{R}^3$ is the position, $V_t \in \mathbb{R}^3$ is the velocity, and t is the time, also set the initial values $X_0 = x_0$, $V_0 = v_0$. The neutron is not bound or glued to the space and will interact with it in the form of collisions at random times $(T_k)_{k\in\mathbb{N}}$. As an abuse of notation, with the times $(T_k)_{k\in\mathbb{N}}$ fixed, denote for every $k \in \mathbb{N}$ the values $X_k = X_{T_k}$ and $V_k = V_{T_k}$ as the position and velocity in the k-th interaction of the particle. In particular, for $t \in [T_k, T_{k+1}]$, the position of the particle is $X_t = X_k + (t - T_k)V_k$, satisfying the classical laws of mechanics.

The space in which the neutron moves also performs an independent movement— a rotation with constant angular speed ω. For any point $x = (x_1, x_2, x_3) \in \mathbb{R}^3$ and time $t > 0$, its position at time $t > 0$ is given by

$$\mathcal{P}^{\omega}_{\text{rot}}(x,t) = \big(x_1 \cos(t\omega) - x_2 \sin(t\omega),\ x_1 \sin(t\omega) + x_2 \cos(t\omega),\ x_3\big).$$

This motion induces a constant velocity field such that, at any time, the velocity of the space at the point $y = (y_1, y_2, y_3)$ is given by

$$F_\omega(y) = (-\omega y_2,\ \omega y_1,\ 0).$$

The linear motion of the neutron in $\mathbb{R}^3$ is compounded with the motion of the space, resulting in a new trajectory in the environment (not necessarily a straight line), denoted by $\gamma \subset \mathbb{R}^3$. Let $(\gamma_k)_k$ be the sequence of trajectories followed by the particle between successive interaction times $(T_k)_k$, defined by

$$\gamma_k(t) = \mathcal{P}(X_k + (T_k - t)V_k,\ t), \quad t \in [T_k, T_{k+1}).$$

We denote by $|\gamma|$ the length of any fixed trajectory γ.

The target atoms of the medium involved in the collisions present the velocity of the space together with a fluctuation of the thermal energy. Specifically, let $(\Delta_k)_k$ the thermal energy of the target atoms, modeled as an i.i.d. sequence of random vectors in $\mathbb{R}^3$, where each Δ_k is distributed according to the normal distribution $\mathcal{N}(0, \sigma^2)$ for some small $\sigma > 0$.

Basically speaking, the outcome of a collision depends on the angle between the velocities of the target and the incident particle. However, because of the fluctuations caused by the thermal energy, this quantity exhibits random behavior. In addition, to avoid nonphysical energy growth, it is important to consider energy loss in collisions.

A natural way to incorporate all these effects is to introduce a sequence $(\eta_k)_k$ of random outcomes corresponding to a reference collision: that of a neutron with velocity $(1,0,0)$ striking a particle in the stopped environment. Then, by rescaling and rotating the system, we can relate these vectors to the outcomes of general collisions.

Therefore, fix any sequence $(\eta_k)_k$ of random vectors i.i.d. supported on the open ball $B(0,1) = \{x \in \mathbb{R}^3 : \|x\| < 1\}$. Given an incident particle with velocity v colliding with a target atom moving at velocity $u + \Delta$, we describe the particle velocity output v_{new} of the collision as:

$$v_{new} = |v-(u+\Delta)|R[v-(u+\Delta)]\,\eta + (u+\Delta), \tag{1}$$

where, for any vector w, $R[w]$ denotes the 3×3 rotation matrix that maps the vector $(1,0,0)$ to $w/|w|$, preserving both angles and orientation.

The total distance a particle travels in the space between collisions is given by a sequence of exponential random variables of i.i.d. $(\xi_k)_{k\in\mathbb{N}}$ with rate $\lambda > 0$. That is, each ξ_k satisfies $\mathbb{P}(\xi_k > s) = e^{-s\lambda}$, and the particle presents a mean free path of size $\Sigma = \frac{1}{\lambda}$. This value of distance, together with some calculations, indirectly informs the new time in which the particle interacts with space. To illustrate, consider (X_k, V_k, T_k) a particle in collision $k-th$, define the distance walk by the particle to time $t > 0$ as a function:

$$D^{\omega}_{X_k,V_k}(t) = \int_0^t \|V_k - F_{\omega}(X_k + sV_k)\|\, ds, \tag{2}$$

and with that define $T_{k+1} = T_k + [D^{\omega}_{X_k,V_k}]^{-1}(\xi_k)$. Notice that by the definition of the problem $|\gamma_k| = \xi_k$.

3 Results and discursion

3.1 Main theorem and proof idea

By an argument from Einstein (1905), later confirmed experimentally by Perrin (1909), a neutron colliding with a still environment behaves like a Brownian motion. Our main result is to expose that this behavior is far from trivial in the rotating case: due to the collisions, the particle now rotates with the space and experiences a centripetal-type effect that tends to push it away from the origin.

We emphasize that, by construction, the particle moves in a straight line between interactions, so this centripetal effect is not a mechanical force in the classical sense, but rather a virtual force induced by the random collisions.

To state the result, denote for every $x = (x_1, x_2, x_3) \in \mathbb{R}^3$ the norm $\|x\| = \sqrt{x_1^2 + x_2^2 + x_3^2}$. Also, define the radial (horizontal) distance by $d_r(x) = \sqrt{x_1^2 + x_2^2}$.

For a sequence of events or sets $(A_n)_n$, define

$$\liminf_{n\to\infty} A_n := \bigcup_{n=1}^{\infty} \bigcap_{m>n} A_m.$$

In other words, if this event occurs, then there exists some value $m_0 > 0$ such that for all $n > m_0$, the events A_n also occur.

With this notation, it is possible to prove the following theorem that govern the particle motion in the rotating environment. Such theorem will not be proved at the present short paper, but its implications will be discussed later.

Theorem 1. *For every $\omega > 0$ the particle (X_t, V_t, t) in a rotational environment satisfies:*

$$d_r(X_t) \to \infty \text{ and } \frac{V_t}{F_{\omega}(X_t)} \to 1 \quad \text{a.s. and in probability.}$$

Moreover, there exists an value $\beta > 0$ such that $\mathbb{P}\left(\liminf_{n\to\infty} \{d_r(X_n) > \beta n\}\right) = 1$.

The proof of Theorem 1 is long, but to clarify its core idea, we outline the main steps below.

The first step is to show that, due to energy loss, the difference between the particle's velocity and the velocity of the rotating space, namely $V_t - F_\omega(X_t)$, is a random variable with uniformly bounded mean. This implies that when the particle is in high-energy positions, $d_r(X_t)$ large, this difference becomes comparatively small with high probability.

The second step is to convert this metastable condition into a recursive effect. Geometrically, when the particle's velocity is close to that of the space, any attempt to move toward the origin is with high probability canceled: the small relative velocity with respect to the environment, together with the convexity of the circular domain, implies that the next collision is, with high probability, in a region of larger radius. That is, even when trying to move inward, the particle tends to collide in an outer region.

Therefore, when the particle is far from the origin, one can assume, with high probability, that its velocity is nearly aligned with the velocity of the space. At the scale of a single collision, the system appears locally uniform, and the particle has approximately equal probabilities of moving inward or outward. However, there remains a positive probability that, upon entering the inner region, the particle takes extra time to collide and ends up in a higher-energy state. This bias, repeated over time, causes the particle to drift toward infinity. From this observation, Theorem 1 follows as a consequence of the Borel–Cantelli lemma and Cramér's theorem on large deviations for sums of independent random variables; see Durrett (1996) for details.

Since, one can proof that the particle cannot remain trapped in a finite domain, it will eventually return to high-energy regions. Thus, with probability one, the particle diverges, and we obtain:

$$d_r(X_t) \to \infty$$

almost surely and in probability. Furthermore, we can also show that

$$\frac{V_t}{\mathbf{F}_r(X_t)} \to 1$$

almost surely and in probability.

Even more, the exponential decay provided by Cramér's theorem, combined with the Borel–Cantelli argument, shows that for every $\varepsilon > 0$, we eventually have $d_r(X_n) > n(\mathbb{E}[Y] - \varepsilon)$ for all sufficiently large n. Choosing $\varepsilon > 0$ appropriately, we conclude that the neutron reaches, at time t, the average velocity corresponding to the space velocity at position X_t.

3.2 Repercussions in nuclear energy

An environment set up as a spinning hollow cylinder may operate as a possible neutron spectral selector (RSS). It works by adjusting the angular velocity and peripheral radium to transform an inner central neutron source to a specific output neutron distribution, with average energy E_m and standard deviation ε, spelled in the periphery of the modulator. According to Theorem 1 the input neutrons diverge to the periphery and its velocity asymptotically approaches the medium velocity. Based on the operation medium parameters, this outer distribution may be enclosed in a cold neutron energy at 1 meV up to 25 *meV* or it may be subject to a match with a resonance energy E_o of a target nuclide.

There are many applications for a neutron distribution enclosed in a suitable spectral energy range. An possibility involves the production of radioisotopes whose studies have already been published Campos and de Campos (2025). The reaction rates of precursor nuclides and impurities can be selectively enhanced by tailoring the neutron energy spectrum Campos and de Campos (2025). Two different configurations, RSS MB and RSS MR, can be used as part of the problem-solving strategy. In the RSS MB setup, using a low emision neutron source of $10^8 n/s$ and an irradiation time of 3 days, specific activities (in $MBqg^{-1}$) were achieved as follows: approximately 896 for ^{153}Sm, 306 for ^{177}Lu, 804 for ^{188}Re, and 1190 for ^{166}Ho. In contrast, the RSS MR configuration yields, for example, 36 for ^{177}Lu, 10 for ^{169}Yb, and 135 for ^{192}Ir, in $MBqg^{-1}$.

To evaluate RSS performance, the efficiency of radioisotope production was compared with the conventional nuclear reactor (NR) activation method using a defined activation potential parameter η, taken as the ratio of the integrals in the energy domain of the reaction rates for RSS and NR. The non-dimensional parameter η does not take into account the intensity of emission of the nuclear system. The RSS MR system demonstrated significantly enhanced values of η, reaching 3.2 million for ^{153}Sm, 1.1 thousand for ^{177}Lu, and 620 for ^{188}Re, which substantially outperforms traditional nuclear reactors.

In particular, based on the low neutron emission, the production of ^{188}Re via the ^{187}Re(n,γ)^{188}Re reaction is impractical in standard reactors due to the high level of contamination of ^{186}Re. However, the RSS MR configuration enables the generation of ^{188}Re with an activity approximately 60 times greater than that of the main contaminant. These results underscore the effectiveness of RSS configurations in producing high-purity radioisotopes with enhanced specific activities and favorable benchmark metrics.

An other possible application will involve the removal of radionuclides by exposing a set of spent fuel rods to a neutron spectrum that covers a well-resolved resonance or a $1/v$ cold region at 1 up to 25 meV. Effective removal constants and correspondent effective half-lives based on the operational protocols with 10^{12} $n.s^{-1}$ can be performed on a cold neutron at 5 meV, as an example. Early evaluations, not demonstrated in this short paper, provide an effective half-life for ^{235}U, ^{239}Pu, ^{242}Pu, ^{241}Am and ^{244}Cm of 13.38 y, 12.27 y, 16.00 y, 14.64 y, 12.17 y, respectively; while ^{240}Pu and ^{241}Pu can reduce to 2.79 y and 5.09 y. In contrast, ^{237}Np, ^{238}U and ^{129}I exhibit a longer persistence with 63.45 y, 164.87 y and 314.88 y, respectively. ^{137}Cs and ^{90}Sr were more difficult to transmute in the cold spectrum. The radiotoxicity of ^{235}U, ^{236}U decreased over 5×10^9 s timescale; Pu isotopes exhibited lower levels below 1 $\mu Sv.cm^{-3}$ after the same period; ^{244}Cm and ^{241}Am showed a reduction of 50 mSv/cm^{-3} to 0.5 $mSv.cm^{-3}$ over ten decades; ^{237}Np, ^{99}Tc and ^{129}I remained persistent but below 1 $\mu Sv.cm^{-3}$, while ^{151}Sm became negligible after a few years . These findings underscore the potential of using neutron spectral selection as an effective strategy for managing long-lived radionuclide in spent nuclear fuel over practicable timescales.

Removal of radionuclides from spent fuel rods can be achieved by exposing a set of rods to customized neutron spectra, particularly within well-defined resonance regions or the $1/v$ cold neutron range, typically spanning from 1 to 25 meV. For example, using cold neutrons at 5 meV, effective transmutation rates and corresponding half-lives can be established on the basis of operational protocols. Preliminary evaluations indicate that effective half-lives for key radionuclides under such conditions are significantly reduced: approximately 13.38 y for ^{235}U, 12.27 y for ^{239}Pu, 16.00 y for ^{242}Pu, 14.64 y for ^{241}Am, and 12.17 y for ^{244}Cm. Shorter effective half-lives are also observed for ^{240}Pu and ^{241}Pu, at 2.79 and 5.09 years, respectively. In contrast, nuclides such as ^{237}Np, ^{238}U, and ^{129}I exhibit much greater persistence, with effective half-lives of 63.45, 164.87, and 314.88 years, respectively.

The transmutation of ^{137}Cs and ^{90}Sr under cold neutron conditions proved to be less effective. Radiotoxicity assessments show a significant decrease in ^{235}U and ^{236}U levels on a time scale of approximately 5×10^9 seconds. Plutonium isotopes dropped to levels below 1 $\mu Sv.cm^{-3}$ during the same period. In particular, ^{244}Cm and ^{241}Am were reduced from 50 $mSv.cm^{-3}$ to 0.5 $mSv.cm^{-3}$ over ten decades. Although ^{237}Np, ^{99}Tc, and ^{129}I remained present, their radiotoxicity fell below $1 \mu Sv.cm^{-3}$. Meanwhile, ^{151}Sm became negligible after only a few years.

There is also a feasibility of decontaminating long half$-$live radioisotopes by burning them by fission or neutron radiative capture reactions in a selective energetic neutron domain, in the cold region of 1 to 25 meV in which the cross section (XS) has $1/v$ behavior. The depletion of the masses of 24 actinide radionuclides can be demonstrated as a function of a spectral selector operating in the cold spectrum, with an energy value of 1 ± 0.5 meV as a reference. In a realistic case, a correction factor should be applied to the exact value of the E_m average energy of the neutron distribution growing in the spectral selector. A higher value of central neutron emission S_n can generate a rapid mass depletion, while a higher temperature between 1 and 25 meV reduces mass depletion.

4 Conclusions

Consider a rotating environment and a particle walking on it. The particle travels in a straight line and only changes direction in collisions with the environment that occur at random times. Through simple assumptions, such as the energy loss in collisions and the fact that the collisions satisfy the laws of Newton, it can be demonstrated by simulations and straightforward calculations that the particle behaves differently from the classical behavior of a Brownian motion. In the rotational model, the particle's position satisfies a strong law described as a spiral movement. In contrast to classical mechanics, the force pushing the particle away is not the centripetal force, but a random force due to collisions, since the particle is almost always traveling in a straight line. This behavior was presented by two distinct theorems in which a geometrical-probabilistic argument shows how this new mechanics takes place, and how the Brownian motion approximation shifts to the biased random walk regime. This new approach opens up a possible application explored in the recent literature, such as Campos and de Campos (2025) on the movement of neutrons in rotational environment and the study of the equations that describe the diffusion of such particles.

These applications highlight the potential of spectral neutron selection as a practical and effective strategy for the long-term management of nuclear waste, for the mitigation of radiotoxicity in spent nuclear fuel, and for radioisotope production.

Acknowledgements

This research developed in 2022 was an independent work of both authors without grant. At the moment, the authors acknowledge CNPq — Conselho Nacional de Desenvolvimento Científico, CAPES — Coordenação de Aperfeiçoamento de Pessoal de Nível Superior, CDTN — Centro de Desenvolvimento da Tecnologia Nuclear, INCT.INAIS—Instituto Nacional de Ciência e Tecnologia, UFMG — Federal University of Minas Gerais, and Mathematical Department at University of Augsburg, Germany; Mathematical Department at Federal University of Minas Gerais, Brazil; and IMPA — Instituto de Mathematica Pura e Aplicada. Alberto M. Campos thanks the project 446293/2024-5 of CNPq — CNPq/MCTI/FNDCT Nº 21/2024 - Programa Conhecimento Brasil - Atração e Fixação de Talentos, and thanks the project PROBRAL-CAPES through Project 88887.979525/2024-00..

References

Campos A. M. and de Campos T. P. R. (2025) Feasibility of radioisotope production by neutron spectral selection: Mathematical and computational approaches. *Applied Rad and Isot,* page 111782

Durrett R. (1996) *Probability: theory and examples*. Duxbury Press, Belmont, CA. second edition

Einstein A. (1905) Über die von der molekularkinetischen theorie der wärme geforderte bewegung von in ruhenden flüssigkeiten suspendierten teilchen. *Annalen der Physik* 322:549–560

Ford B. J. (1991) Robert brown, brownian movement, and teethmarks on the hatbrim. *The Microscope 39: 161–171-*

Löwen H. (2019) Active particles in noninertial frames: How to self-propel on a carousel. *Phys Rev E* 99:062608

Perrin J. (1909) Mouvement brownien et réalité moléculaire. *Annales de chimie et de Physique* 18:5–114

Analysis of the average geometric parameters of the closed pores of the fine aggregate matrix using x-ray micro-computed tomography and linear regression.

R.F. Costa[1,2]*, Y A. Costa[1], L. S. Santos[1], N. B. F. Pessoa[3], L. R. Taveira[1], I. C. B. Lima[4], R. F. G. Santos[5]

[1]Instituto Federal de Educação, Ciência e Tecnologia de Goiás, Goiânia, GO, Brazil.
[2]Universidade Estadual de Goiás, Morrinhos, GO, Brazil
[3]Universidade Federal de Goiás, Goiânia, GO, Brazil.
[4]Universidade Federal do Rio de Janeiro, PEN - COPPE - Rio de Janeiro, RJ, Brazil
[5]Universidade Estácio de Sá, Rio de Janeiro, RJ, Brazil
* rogerio.costa@coppe.ufrj.br

Abstract

The Fine Aggregate Matrix is an important part of the Hot Mix Asphalt Cement and is constituted of asphalt Binder, fine aggregates, filling material and pores. There are little published researches about the pore's distribution inside the Fine Aggregate Matrix; however, the asphaltic concrete projects have to consider pores structures, that are related to moisture damage. Traditional methods of porosity assessment, as Mercury's porosimetry and Arquimedes method, are well documented, but only provide overall average results for the entire sample, in this context, the X-ray microtomography stands out, because it allows us to know parameter series related to the object internal structure. The target of this research was to estimate the closed pore's geometric parameters of the FAM2 using linear regression and compare it with the experimental results obtained from the X-ray microtomography. Through the results it was possible to verify that the theoretical model of the volume and the closed pore's area underestimated the experimental values in 13.68% and 7.92%, respectively. The calculated sphericity from the volume and the average area was 0.9556, as the experimental model was 0.9706. By the hypothesis test it was concluded that the model's results cannot be differentiated. The analysis of the theoretical model for the calculated rays from the volume and the closed pores' area had a lower average than the experimental model as well, with relative differences of 6.70% and 5.33%, respectively. The theoretical sphericity underestimated the experimental sphericity in 2.7974%, furthermore, it was verified that the model's results are differentiable, because the p-value was lower than 5%. Thus, it was possible to use the linear regression with the sample's volumetric density as the predictor variable to predict the geometric parameters of closed pores.

Keywords: Fine Aggregate Matrix; Closed pore; X-ray microtomography; Linear regression.

1 Introduction

The pores in any structure can or not be in contact with the material's surface. When the pores are connected to the sample's external part, they are material conductors, therefore, interfere in the structure's mechanical resistance, because a lot of fractures start in surface openings, as pores (Richerson, 1992). The pores that are not connected to the surface are called closed pores, although they can be internally connected inside the material (Klobes et al., 2006). Closed pores can be generated by the compression process that can happen in the opened pores, due to the gas's adjustments form the solid phase and consequently the trapped gasses cannot escape from the structure (Richerson, 1992). The more porous the structure, the less fatigue resistance it will have (Mehta and Monteiro, 2008). The pores are the empty spaces in a material and the opened pore and the closed pore volume's sum is equal to the total pore's volume (Vp) of the sample (Libardi, 2018). Still, the porosity, which is a physical propriety of the matter, is the relative fraction of the volume (V) of any solid structure filled by pores and is related to the capacity of this material of absorbing fluids (Ahmed, 2010). The porosity (P) can be expressed as a percentage defined as

$$P = 100\%.\frac{V_p}{V} \quad (1)$$

The geometry description of the opened pore is carried out by models of a small number of shapes, such as cylinders, prisms, cavities and windows, slits or spheres (Rouqueirol et al., 1994). It is difficult to determine and to analyze the pores geometry in a structure, due to its great variability. In some researches, the pores are filled with a fluid that solidifies and then the rock is dissolved by acidification. The final product is the pore molds. One application of this methodology is the use of molten lead in sandstones or epoxy resin in carbonate rocks (Selley and Sonnenberg, 2015). However, to determine the closed pore geometry, it is necessary to use the x-ray microtomography (Costa and Lima, 2022; Osmari et al., 2020).

The geometric parameter used to determine the pore's spherical shape is called sphericity (ϕ), which measures how much their shape approaches to a sphere. A perfect sphere has a sphericity of 1, while the particles with more irregular shapes have lower values, approaching 0. The f can be defined as the relation between the superficial area and the pore's volume, numerically; such measure indicates the similarity of the pore with a sphere (Mendes, 1972) or the ratio between the circle's diameter with the area equal to the pore's projection and the diameter of the smallest circle circumscribed around the pore, with values ranging from 0 to 1 (Rittenhouse, 1943). In short, those values vary approximately from 0.45 to elongated pores to 0.97 to very spherical pores (Rittenhouse, 1943).

The Hot Mix Asphalt (HMA) is a material that contains large pores with irregular shapes, small with spherical shapes and cylindrical (Costa and Lima, 2022). Those are incorporated to the structure during mixing operations (Richerson, 1992). It is used to coat pavement and constituted basically of a combination of aggregates and asphalt binders. The combination of aggregates in the mixture includes various gradations, which, usually, present an aggregate's maximum nominal size of 12.5 or 19.0 mm for coarse aggregates and sizes lower than 0.075 mm for fine aggregates (Cooley and Hurley, 2004). Lots of researchers have been investigating the relation between the combination of smaller aggregates in the asphalt mix and the asphalt binder, as known as the fine aggregate matrix (FAM), because they believe that some wear initiate where the aggregates have maximum dimensions of 2.5 mm (Kim et al., 2003; Aragão et al., 2017).

The FAM consists mainly in asphalt binder, fine aggregates, filling and pores as one of the main constituents of mixtures (Kim et al., 2003). Until the moment, the current methods of HMA projects need to consider pore's structures, what are highly related to moisture damage (Masad et al., 2006). On the other side, there is a necessity of understanding the permeability and the moisture damages based on the pore's size distribution (Al-Omari et al., 2002). Traditional methods of porosity evaluation, such as mercury porosimetry and the Archimedes method, are well documented and provide good results for various materials. However, they are only average global results for the entire sample (Pessoa et al., 2014); besides, they do not determine the closed porosity nor describe the sample's geometry of the pores. In this context, X-ray microtomography stands out because, beyond the information about a number of additional parameters related to the internal structure of the object (Gopalakrishnan et al., 2006), it can be estimated by mathematical calculations the geometry of the closed and opened pores (Costa and Lima, 2022).

2 Experimental program

2.1 Determination of the number of samples

The analysis of all population elements allows a better understanding of the study object; however, the time and cost restrictions and the advantages of using inferential techniques justify the utilization of sampling strategies (Martins and Domingues, 2017). The calculation of the sample number (n), from a population with N elements, depends on the bearable sampling error (e), on the trust level that is related to the parameter (z) of the normal distribution, and on the population's variability (s). In this research, the FAM specimens' population was considered finite and given by the equation (2):

$$n = \frac{z^2.s^2.N}{e^2(N-1)+z^2.s^2} \qquad (2)$$

The size of the population (N) was calculated by dividing the FAM's total volume (Vt) by the sample's volume (Vs), as described by Pessoa et al. (2025) The standard deviation was obtained from specimens located at the outer radius of the Fine Aggregate Matrix with an average pore volume of 11.88% of the total sample volume (Osmari et al., 2020), because this was the higher standard deviation observed, and, in that condition, the number of samples for our study was maximized (Pessoa et al., 2025). The maximum sampling error allowed (d) was equal to 0.59%, which is equivalent to 5% of the pores' medium volume (Osmari et al., 2020; Martins and Domingues, 2017). Table 1 shows the necessary quantity to determine the sample size for the Fine Aggregate Matrix 1 (FAM1), Fine Aggregate Matrix 2 (FAM2), and Fine Aggregate Matrix 3 (FAM3).

Table 1. Quantities needed to determine the random sample size for FAMl

Quantity	V_t (mm^3)	V_s (mm^3)	N	s (%)	d (%)	n
Values	494550	3165.12	157	0.9	0.59	8.43

Thereby, the results showed that 8.43 FAM1, FAM2, and FAM3 samples should be used in the research. However, nine samples were utilized for each FAM in order to reduce errors and increase the statistics.

2.2 FAM design method

In this study, the FAM samples were designed according to the method proposed by Amelian et al. (2019) at the Jacques de Medina Geotechnics and Paving Laboratory, located at the Department of Civil Engineering at the Federal University of Rio de Janeiro (COPPE/UFRJ). This method defines the volumetric characteristics of the FAM based on the main parameters of the HMA mixing design. The binder content determined for FAM is 8.49%, while the binder content for the corresponding HMA is 4.90% (Amelian et al., 2019). The specimens (FAM1, FAM2, and FAM3) were compacted in the Superpave Gyratory Compactor (SGC) at a pressure of 600 ± 18 kPa and a rotation angle of 1.25 ± 0.02° with three different densities. Next, nine samples were extracted from the following positions: X, A1, A2, B1, B2, C1, C2, D1, and D2, using a hole saw with a diameter of 12 ± 1 mm. The samples and their respective positions are shown in Fig. 1a, and Fig. 1b shows the matrix with the holes made by the drills.

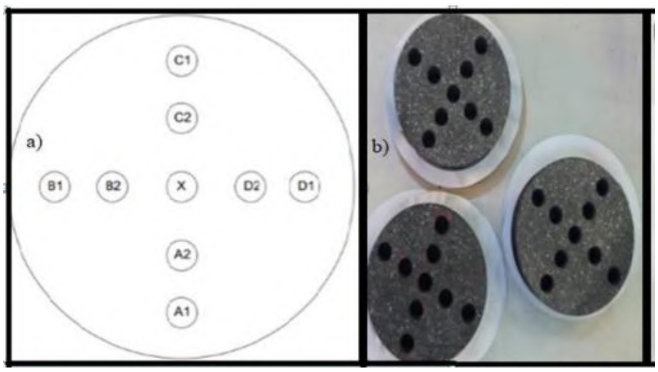


Figure 1. (a) Position of the samples within the FAM specimen and (b) Specimen with the holes made by the drills.

The samples were then sawn to obtain a final length of approximately 48 ± 1 mm. The samples FAM1, FAM2, and FAM3 were compacted in the Superpave rotary compactor (SGC). Then, they were sawn to a final length of nearly 48 mm. Nine specimens of each FAM1, FAM2, and FAM3 were extracted with a cylindrical drill bit connected to a drill with a diameter of 12 mm in different positions, as shown in Figure 1b. Four samples were extracted at a distance of 6 cm from the center of the compacted samples by the SGC, the external ring, and were called specimens A1, B1, C1, and D1. The samples of the inner ring were extracted at a distance of 3 cm from the center and were called specimens A2, B2, C2, and D2. Finally, the center-extracted samples were called X. Figure 2b shows the bunker after the sample extraction.

2.3 Image generation and microstructural analysis

The image analysis of the specimens of FAMs was performed on the SkyScan model 1173 bench microtomograph, installed at the X-rays Applied Laboratory at the State University of Londrina. A 120 kV voltage, 66 µA electrical current, 0.3-degree angular step, 1 mm thick aluminum filter, and 8 µm resolution were used to perform the procedure. At this resolution it was possible to radiograph the entire sample by running 3 scans for each sample at different heights and then stitching them together. After the acquisition process, the radiographic images were reconstructed with the NRecon program (NRecon, 2011), whose algorithm is based on the works of Feldkamp et al. (Feldkamp et al., 1984). In this study, grade 4 Gaussian smoothing filters, level 40 artifact reduction, and 60% grade beam-hardening artifact correction were used. SkyScan software, CTAn (CTAnalyser, 2012), was used for image processing and analysis. The quantified parameters were the total volume of objects converted to binary within the volume of interest (VOI). Based on this, the total porosity volume of the FAMs was calculated. All objects within the region of interest (ROI) were analyzed.

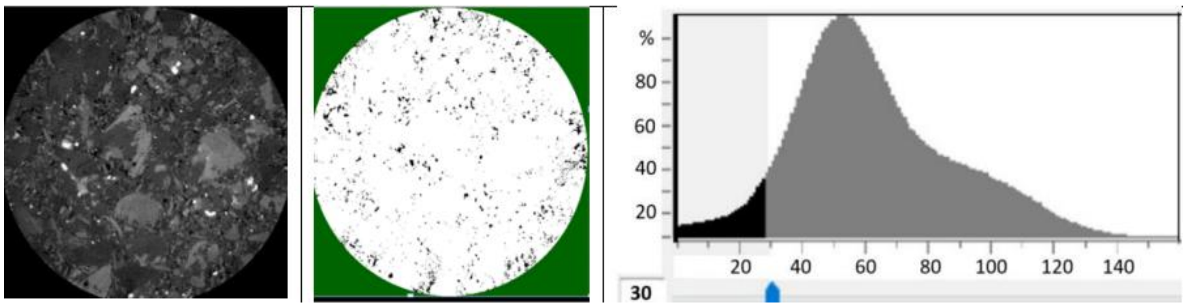


Figure 2. Cross-sectional image and binarized grayscale histogram image.

It is important to highlight that a threshold value (TH) which separates the two objects included in the ROI must be chosen for the quantification process. There is no default method for determining the value of TH. In this study, a global TH value of 30, out of a range from 0 to 255. Thus, the pores, represented by the black color, were separated from the FAM solid phase indicated by the white color. After segmentation by global thresholding, the porosity values of the 9 specimens of each FAM1, FAM2, and FAM3 were determined by µCT at 8 µm resolution.

In 3D morphometric analysis, the primary object surface index, which measures the surface area of all solid objects within the VOI, is calculated according to the Marching Cubes method which triangularizes the surface of the sample structure (Muller et al., 1994) and constructs tetrahedrons on the triangularized surface (Guilak, 1995). Furthermore, the empty spaces in 3D of the structure can be measured through the diameter of the largest sphere that fulfills two conditions: the sphere surrounds the point and the sphere is entirely limited within the empty surface (Hidelbrand and Ruegsegger, 1997).

2.4 Linear regression between the average volume and area of closed pores and the density

In this research, a linear regression (Bussab and Morettin 2010; Crespo, 2002) was used to relate the sample's volumetric density to the closed pores' average volume (Pessoa et al., 2025) and the sample's volumetric density to the closed pores' average area. Thereunto, we used FAM1's and FAM3's samples, nine for each. Therefore, it was possible to build two first-degree equations between the average volume of closed pores and density (Equation 3) and the closed pores' average area and density (Equation 4):

$$V_{UT} = \rho\beta_{1V} + \beta_{0V} \qquad (3)$$

$$and$$

$$A_{UT} = \rho\beta_{1A} + \beta_{0A} \qquad (4)$$

where: β_{0V} is the measured linear coefficient in mm^3, β_{1V} is the measured line angular coefficient in mm^6/g, β_{0A} is the measured linear coefficient in mm^2, and β_{1A} is the measured line angular coefficient in mm^5/g).

2.5 Sphericity

From the obtained results by equations (2) and (3), it was possible to determine the sphericity (Costa and Lima, 2022; Wadel, 1932; Pinto et al., 2009) of each sample and the average sphericity of FAM2 through equation (4). Next, the result was compared with the experimental values (Table 5). Following, the hypothesis test was applied to verify if the observed differences were statistically significant:

$$\phi = \frac{\sqrt[3]{\pi}(6.\mathrm{V})^{2/3}}{A} \qquad (5)$$

where: V and A are the volume of the contained sphere within the pore and the area of the closed pore, respectively.

3 Results and Discuss

The sample's densities of FAM specimens were calculated by diameter measures and cylinder length, as well as their masses. In Table 2 there are the values of the volumetrical density for each FAM sample.

Table 2. Density of samples of FAM1, FAM2, and FAM3 measured in g/mm^3.

FAM1 Sample	X	A2	A1	B2	B1	C2	C1	D2	D1
$\rho(10^{-3}g/mm^3)$	2.30	2.26	2.15	2.30	2.28	2.31	2.17	2.22	2.24
FAM2 Sample	AX	A2	A1	B2	B1	C2	C1	D2	D1
$\rho(10^{-3}g/mm^3)$	2.34	2.39	2.40	2.35	2.33	2.36	2.37	2.35	2.34
FAM3 Sample	AX	A2	A1	B2	B1	C2	C1	D2	D1
$\rho(10^{-3}g/mm^3)$	2.42	2.48	2.44	2.46	2.45	2.43	2.50	2.49	2.43

Through the Table 2 it was possible to verify that the FAM densities increase with the compaction. Thus, FAM3 is the densest, and FAM1 has the lowest density. Furthermore, within the same FAM, the values undergo small variations in density, with FAM1 having the greatest spread, i.e., the least compacted.

The volume of closed pores (V), the area of closed pores (A) and the number of closed pores (N) for each sample of FAM1 and FAM3 were obtained by X-ray microtomography, from 3D analyses using the CTan software, as can be seen in Table 3. Through these data, the average pore volume (V_{UE}) and the average pore area (A_{UE}) were determined for each sample and the results are presented in Tables 3 and 4 for FAM1 and FAM3, respectively.

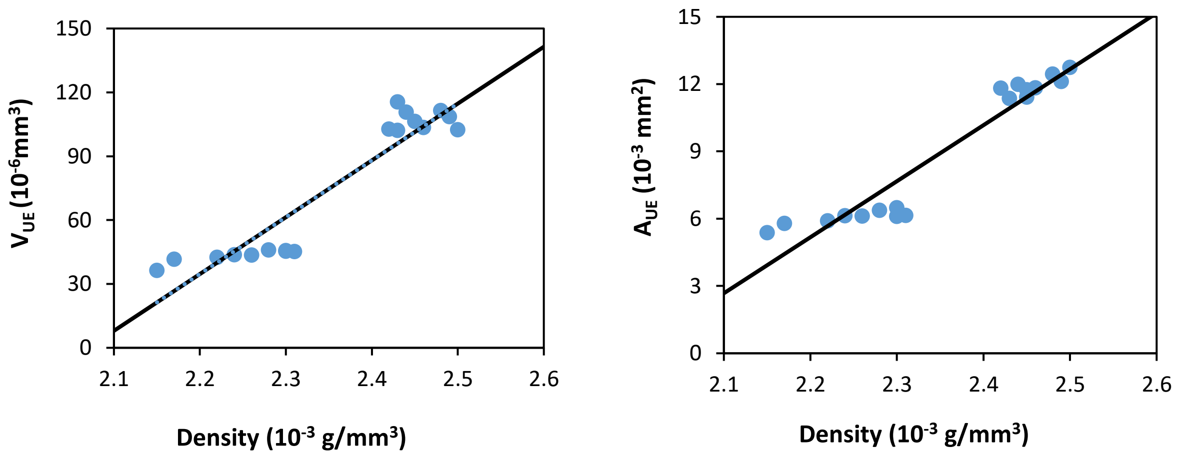
Table 3. Geometric parameters of the closed pores of FAM1 obtained by 3D using CTan software.

Sample	N	V (mm^3)	A (mm^2)	V_{UE} ($10^{-6}mm^3$)	A_{UE} ($10^{-3}mm^2$)
X	578244	26.39972	3748.25581	45.65498	6.48214
A1	490993	17.83492	2635.91939	36.32418	5.36855
A2	524995	22.86132	3210.05323	43.54578	6.1144
B1	594639	27.28794	3789.20753	45.88993	6.37228
B2	503961	22.804200	3073.34408	45.24993	6.09838
C1	460217	19.15096	2661.16591	41.61289	5.78242
C2	503137	22.73770	3087.92939	45.19187	6.13735
D1	538291	23.48593	3294.83919	43.63055	6.12093
D2	539108	22.877100	3183.71573	42.43509	5.90552

Table 4. Geometric parameters of the closed pores of FAM3 obtained by 3D using CTan software.

Sample	N	V (mm^3)	A (mm^2)	V_{UE} ($10^{-6}mm^3$)	A_{UE} ($10^{-3}mm^2$)
X	655844	67.31236	7751.46710	102.63471	11.81907
A1	717509	79.40303	8604.12587	110.66485	11.99166
A2	659919	73.54005	8209.57553	111.43799	12.44028
B1	747803	79.47373	8780.48654	106.27629	11.74171
B2	683256	70.69840	8080.38734	103.47278	11.82630
C1	716674	73.40605	8183.09189	102.42599	11.41815
C2	637456	73.60676	8118.98162	115.46955	12.73654
D1	720430	73.57872	8186.01475	102.13168	11.36268
D2	672740	73.03392	8150.59910	108.561810	12.11553

Tables 3 and 4 show that increasing density increases total volume, total area of closed pores, and number of closed pores in the sample. Conversely, the average closed pore volume and average closed pore area decrease with compaction. From these results, first-degree equations were determined to estimate the porosity of FAM2 using its volumetric density. Figures 3a and 3b show the distribution of closed porosity for each sample based on density.



(a) (b)

Figure 3. Linear regression between density and (a) mean closed pore volume and (b) mean closed pore area.

Table 5 shows equations of linear regression, obtained by the pore's volume and the mean area of the samples (Table 3 and 4) and of their respective densities (Table 2). Also, it contains the determination coefficient (R^2), correlation coefficient (R) and the degree of freedom (df) required to find the $T_{critical}$ and the $T_{calculated}$ by the hypothesis test.

Table 5. Hypothesis test for the existence of a linear correlation

Equation	$V_{UT} = 0.27086\rho - 0.00056182$	$A_{UT} = 25.156\rho - 0.050168$
R^2	0.902	0.907
R	0.950	0.953
df	17	17
P-value	$6.72.10^{-24}$	$4.67.10^{-24}$
α	0.05	0.05

The linear regression equations (Table 5) presented a positive angular coefficient, since the geometric parameters increase with increasing of FAM density. It was verified, through R^2, that density explains more than 90% of the variations in the volume and area of closed pores, as there is a strong correlation between the variables, as R is higher than 0.9 (Crespo, 2002). The hypothesis test confirmed the existence of the model (P-value < 0.05) (Martins, Domingues, 2017). Using the values obtained by the regression equations (Table 5), we determined the theoretical average of volume (V_{UT}) and area (A_{UT}) of FAM2 closed pores from the sample density (Table 2), following, they were compared with the experimental average pore volume (V_{UE}) and the experimental average pore area (A_{UE}), respectively, obtained by x-ray microtomography. The results are presented in Tables 6.

Table 6. Comparison between geometric parameters obtained by linear regression and X-ray microtomography and their respective relative differences (Δ), means (μ), and standard deviations (σ).

Sample	V_{UT} (10^{-6} mm^3)	V_{UE} (10^{-6} mm^3)	Δ(%)	A_{UT} (10^{-3} mm^2)	A_{UE} (10^{-3} mm^2)	Δ(%)
X	71.99240	81.49429	11.65958	8.69704	9.46536	8.11718
A1	88.24400	80.70033	9.34776	10.20640	9.40005	8.57815
A2	85.53540	89.98200	4.94165	9.95484	10.03601	0.80879
B1	69.28380	84.97519	18.46585	8.44548	9.72790	13.18291
B2	74.70100	85.17547	12.29752	8.94860	9.86294	9.27046
C1	80.11820	81.06518	1.16817	9.45172	9.21946	2.51924
C2	77.40960	86.90225	10.92337	9.20016	9.90255	7.09302
D1	71.99240	98.98602	27.27013	8.69704	10.70872	18.78544
D2	74.70100	111.33022	32.90142	8.94860	11.03669	18.91953
μ	77.10864	88.95677	13.31897	9.17221	9.92885	7.62066
σ	6.41607	10.14098	-	0.59589	0.60105	-

The theoretical model for the average volume and average area underestimated the experimental model by 13.31897% and 7.62066% (Table 6). Furthermore, the largest difference between the volumes and areas is in sample D2. The values of the geometric parameters obtained by linear regression are more concentrated around the mean than the values obtained by X-ray microtomography for FAM2. This is proven by the standard deviation of the theoretical model, which is equal to $6.461607.10^{-6}$ mm^3 and $0.59589.10^{-3}$mm^2, while to the experimental measure was equal to $10.14098.10^{-6}$ mm^3 and $0.60105.10^{-3}$ mm^2, for the volume and area, respectively.

The theoretical results of the mean sphericity (ϕ_T) of closed pores for each FAM2 sample obtained by linear regression were compared with the experimental sphericity (ϕ_E) of FAM2 and their respective relative differences (Δ), means (μ), and standard deviations (σ). The results are presented in Tables 7.

Table 7. Comparison between sphericity obtained by linear regression and X-ray microtomography

Sample	ϕ_T	ϕ_E	Δ(%)
X	0.962	0.960	0.238
A1	0.939	0.961	2.274
A2	0.943	0.967	2.474
B1	0.966	0.961	0.513
B2	0.959	0.949	1.006
C1	0.951	0.982	3.169
C2	0.955	0.958	0.340
D1	0.962	0.966	0.385
D2	0.959	1.014	5.469
μ	0.955	0.969	1.428
σ	0.009	0.019	-

Through Table 7 it can be observed that the theoretical model for the average sphericity underestimated the experimental model in 1.428 %, furthermore, the values of the theoretical sphericity obtained by the equation 4, are more concentrated around the mean than the experimental sphericity values for FAM2, this is proven by the standard deviation of the theoretical model which is equal to 0.009, while for the experimental measurement it was 0.019. The largest differences observed were less than 5.470%, demonstrating that the theoretical model produced results very similar to the experimental ones. By the hypothesis test, it was concluded that the specimens cannot be differentiated because the p-value was equal to 8.914%.

4 Conclusions

A study was conducted to determine the geometric parameters of the FAM2 closed pore using linear regression, based on sample density. The model used the mean volume and mean area of the closed pore in the linear regression with the sample density, and showed mean relative differences compared to the experimental model of 13.319% and 7.621%, respectively. Hypothesis testing confirmed the existence of the models. Furthermore, the sphericity showed a mean relative difference of 1.428% between the results obtained by linear regression and the experimental results, with the largest difference observed for sample D2 at 5.469%. The statistical differences were not considered significant. Thus, it was concluded that using the statistical model to determine mean volume, mean area, and sphericity of pores based on sample density proved appropriate and can help reduce research costs, given that the use of X-ray computed microtomography techniques is expensive, and many researchers lack the resources to finance testing. Determining these geometric parameters is essential, as it is associated with the material's rheological properties.

Acknowledgements

The authors thank the Brazilian agencies CNPq (National Council for Scientific and Technological Development) for financial support and the Laboratory of Applied Nuclear Physics, Department of Physics, State University of Londrina, and Laboratory of Applied Physics to Biomed and Environmental Science, State University of Rio de Janeiro, for help during image acquisition and reconstruction.

References

Ahmed T. (2010). Chapter 4 - fundamentals of rock properties. In: Ahmed, T. (Ed.), Reservoir Engineering Handbook, 4◦ Edition. Gulf Professional Publishing. https://doi.org/10.1016/B978-1-85617-803-7.50012-2.

Al-Omari A.; Tashman L., Masad E., Cooley A., and Har-Man, T. (2002) Proposed Methodology for Predicting HMA Perme- ability. Journal of the Association of the Asphalt Paving Tech- nologists. 71:30-58. http://worldcat.org/issn/02702932

Amelian S., Y. R. Kim P. H., Osmari F. T. S., Aragão F. T. S., Braz D., and Nogueira L. P. (2019) Development of a Volumetric Mix Design Approach for Fine Aggregate Matrix (FAM) and Validation with Micro-CT Method. in *98th Conference: Transportation Research Board, Washington, USA, January.*

Aragão F. T. S., Vargas G. A. B., Hartmann D.A., Oliveira, A.D., and Kim, Y. R. (2017) Characterization of Temperature- and Rate-Dependent Fracture Properties of Fine Aggregate Bituminous Mixtures Using an Integrated Numerical-Experimental Approach. Engineering Fracture Mechanics, 180: 195 – 212. https://doi.org/10.1016/j.engfracmech.2017.05.036

Bussab W. O., and Morettin P. A. (2010) Estatística Básica 6 ed, São Paulo, Saraiva.

Cooley L. A.; Hurley G. C. (2004) Potential of using stone matrix asphalt (sma) in Mississippi. National Center for Asphalt Technology. /view/dot/62099/dot_62099_DS1.pdf

Costa R. F. and Lima I. C. B. (2022) Analysis of pore geometry in the compacted fine aggregate matrix by x-ray microtomography. Brazilian Journal of Radiation Sciences, 10:1-13. https://doi.org/10.15392/2319-0612.2022.1862

Crespo A. A. (2009) Estatística Fácil 19 ed, São Paulo, Saraiva.

CTAnalyser (2012) The User's Guide, SkyScan/Bruker micro-CT, Kartuizerweg 3B 2550 Kontich, Belgium.

Feldkamp L. A.; Davis L. C., and Kress J. W. (1984) Practical Cone Beam Algorithm. Journal of Optical Society of America, 1: 612–619. https://doi.org/10.1364/JOSAA.1.000612

Gopalakrishnan K., H. Ceylan F. Inanc J. Gray, and Heitzman M. A. (2006) Characterization of Asphalt Materials Using X-Ray High-Resolution Computed Tomography Imaging Techniques. Proc., Airfield and Highway Pavements Speci alty Conference, Atlanta, Ga.

Guilak, F., 1995. Compression-induced changes in shape and volume of the chondrocytenucleaus. J. biomechanics 28 (2), 1529–1541. https://doi.org/10.1016/0021-9290(95)00100-X

Hidelbrand, T., Ruegsegger, P., 1997. A new method for the model independent assessment of thickness in three dimensional images. J. Microscocy 185, 67–75. https://doi.org/10.1046/j.1365-2818.1997.1340694.x.

Kim Y. R., Little D. N. and Song I. (2003) Effect of Mineral Fillers on Fatigue Resistance and Fundamental Material Characteristics: Mechanistic Evaluation. Transportation Research Record: Journal of the Transportation Research Board, 1832: 1-8. https://doi.org/10.3141/1832-0

Klobes P., Meyer K., and Munro R. G. (2006) Porosity and specific surface area measurements for solid materials. National Institute of Standards and Technology. Washington, DC.

Libardi P.L. (2018). Dinâmica da água no solo 1º edição São Paulo, Editora: Edusp.

Martins G. A., Domingues O. (2017) Estatística Geral e Aplicada. 6º ed., São Paulo, Editora Atlas S.A.

Masad E., Castelblanco A., and Birgisson B. (2006) Effects of air void size distribution, pore pressure, and Bond Energy on Moisture Damage. Journal of Testing and Evaluation 34: 15-23. https://doi.org/10.1520/JTE13112

Mehta P. K. and Monteiro P. J. M. (2008) Concreto: estruturas, propriedades e materiais. 3ª ed. São Paulo, IBRACON.

Mendes J. C. (1972) Stratigraphy and sedimentology: structural geology and aerophotogeology. National Book Institute, Brasília.

Muller, R., Hidelbrand, T., Ruegsseger, P., 1994. Non-invasive bone biopsy: a new method to analyse and display the threed-dimensional structure of trabecular bone.Phys. Med. Biol. 39 (1), 145–164. https://doi.org/10.1088/0031-9155/39/1/009.

NRecon (2011) SkyScan/Bruker Micro-CT, Kartuizerweg 3B 2550 Kontich, Belgium.

Osmari P. H., Costa R. F., Aragão F. T. S., Braz D.; Barroso R. C., Nogueira L. P., Ng, A. K (2020) Determination of volumetric characteristics of FAM mixtures using X-Ray Micro-Computed Tomography and their effects on the rheological behavior of the material. Transportation Research Record: Journal of the Transportation Research Board, 2674:.97-107. https://doi.org/10.1177/0361198120914607

Pessoa J. R. C., Domiguez, J. S., Carvalho, G. and Assis, J. T. (2014). Concrete porosity determined by X ray microtomography and image processing. J. Chem. Chem. Eng. 8: 371-377. https://doi.org/10.17265/1934-7375/2014.04.007

Pessoa N. B. F., Costa R.F., Santos L. S., Barbosa C. M., Raitz C., and Lima I. C. B. (2025) Comparison between pore size distributions of the fine aggregate matrix obtained by micro-computed X-ray tomography and a set of linear regressions, Applied Radiation and Isotopes, 221: 1-11.https://doi.org/10.1016/j.apradiso.2025.111819

Pinto T. C. S., Lima O. A., and Filho, L. S. L. (2009) Sphericity of apatite particles determined by gas permeability through packed beds. Mining, Metallurgy & Exploration, 26: 105 -108.

Richerson D. W. (1992) Modern Ceramic Engineering. Properties, Processing and Use in Design. 2 ed. New York, Marcel Dekker.

Rittenhouse G. (1943) A visual method of estimating two-dimensional sphericity. J Sedim Petrol. 13: 79-81.

Rouqueirol. J., Avnir D., Fairbridge, C. W., Everett, D. H., Haynes J. H., Pernicone, N., Ramsay, J. D. F., Sing K. S. W., K. K., and Unger K. K. (1994) Recommendations for the characterization of porous solids, Pure & Appl. Chern. 66: 1739-1758. https://doi.org/10.1351/pac199466081739

Selley R. C. and Sonnenberg S. A. (2015) Geology of Petroleum, 3º ed Elsevier. Wadel, H., Volume, shape and roundness scale for sedimentary rock particles, Journal of Geology, 40: 443-451.

Wadel, H. (1932) Volume, shape and roundness scale for sedimentary rock particles, Journal of Geology, 40: 443-451.

Design and Analysis of FDM Prototypes for Gold-198 Nanoparticles.

D. A. U. Leal[1]*, P. S. Rodrigues[1], L. V. Angelocci[1], M. E. Z. Rigo[1], P. V. S. Tavares[1], M. M. F. Gesserame[1], W. A. R. Arcos[1], G. G. do Nascimento[1], M. P. A. Potiens[1], I. M. M. A. Medeiros[1], A. C. K. de Souza Silva[1], O. Rodrigues Jr. [1], M. E. C. M. Rostelato[1], C. A. Zeituni[1]

[1] Institute for Energy and Nuclear Research (IPEN), Universidade de São Paulo (USP), 05508-000, São Paulo, SP, Brazil
*Corresponding author: davidurbinaleal@usp.br

Abstract

Additive manufacturing is a main technology of Industry 4.0, also known as 3D printing, which allows the manufacture of customized three-dimensional structures with geometric versatility, high precision and efficiency through layer-by-layer deposition of materials. Among 3D printing techniques, fused deposition modeling (FDM) stands out for its low operating cost, ability in processing diverse thermoplastic filaments and reduces waste to develop a prototype compared to other 3D printing technologies. This approach has gained particular relevance in the field of medical physics, where it is applied in the development of experimental systems for dosimetry studies and support of radioactive sources.

In this study, 3D models were designed using Autodesk Fusion 360 software and hollow cylinders were fabricated by FDM 3D printing using four different materials: polylactic acid (PLA), acrylonitrile butadiene styrene (ABS), polyethylene terephthalate ethylene glycol (PETG) and acrylonitrile styrene acrylate (ASA). The prototypes were printed with dimensions of 23 mm diameter x 40mm high x 1 mm wall thickness and 0.08 mm layer resolution, which required high printing quality to ensure dimensional accuracy and structural stability, and were placed inside Fricke Xylenol Gel (FXG) molds and 1 mL solution containing AuNPs for 24hs under controlled conditions; a dosimetric medium that allows 3D mapping of the dose distribution of liquid radioactive sources. The physicochemical properties of each of the materials were evaluated by Fourier Transform Infrared Spectroscopy (FTIR), comparing the spectrum before and after exposure to gold-198 nanoparticles (^{198}AuNPs), identifying the functional groups present in each material analyzed and determining which 3D material performed best for this application.

Keywords: Additive Manufacturing; 3D printed polymers, Nanoparticle dosimetry.

Introduction

Cancer is one of the leading causes of death worldwide, with millions of new cases and deaths reported annually. In response to its global impact, technological advancements in medicine have played a pivotal role in transforming cancer diagnosis and treatment. Cutting-edge innovations—such as precision imaging, molecular diagnostics, targeted therapies, and minimally invasive procedures—have significantly improved early detection and increased patient survival rates. These breakthroughs have enabled more personalized and effective treatment plans. However, cancer treatment remains a complex challenge, as it requires the selective elimination of tumor cells while minimizing harm to surrounding healthy tissues, a balance that continues to drive research and development in medical technology.

In recent years, the integration of nanotechnology into medical applications has led to groundbreaking advancements, particularly in the development of theragnostic systems. Among these, radioactive gold nanoparticles (^{198}AuNPs) have emerged as highly promising agents capable of combining diagnostic imaging and targeted radiotherapy within a single system. Their unique physicochemical characteristics, including favorable biodistribution, biocompatibility, and the emission of therapeutic beta particles, make them ideal candidates for precision oncology (Daems et al., 2021). These multifunctional nanoparticles have the potential to transform clinical strategies by enabling more accurate tumor localization and effective treatment with minimal damage to healthy tissues. Nevertheless, as these technologies advance toward clinical implementation, there is a pressing need to establish dedicated dosimetric protocols that ensure safe, reproducible, and effective therapeutic outcomes.

Dosimetry is the fundamental science concerned with determining radiation dose through simulation or measurement techniques. Dosimetry is fundamental for quantifying the known dose-effect relationship, which is characterized by the incidence of various biological changes as a function of the dose received. Dose, in turn, is defined as the quantity expressing the energy absorbed per unit mass in a specific target tissue (Attix, 2008).

The study of dosimetry is essential to ensure the effectiveness of radiation treatment delivery, which depends on the interaction of ionizing radiation with matter and on the realistic acceptance of dosimetric precision. Dose is a scalar function within a four-dimensional (4D) space, representing the magnitude of energy deposition at a specific point in three-dimensional space over the course of irradiation time (Van Dyk, Battista, & Bauman, 2017). Although various types of dosimeters exist, the complex dose distributions surrounding a ^{198}Au source demand more advanced techniques. In this context, gel dosimetry becomes necessary (Ibbott, 2004), as volumetric chemical gel dosimeters offer an innovative method for generating true 3D dose data from a single irradiation procedure.

Chemical dosimeters are based on detectors composed of substances that are sensitive to ionizing radiation. Upon exposure, these detectors undergo fundamental changes in their chemical properties, which are directly related to the absorbed radiation dose. They are particularly suitable for higher dose ranges commonly encountered in radiotherapy (Nezhad and Geraily, 2022). Gel-based chemical dosimeters are generally categorized into three main groups according to their chemical characteristics: Fricke gel dosimeters, polymer gel dosimeters, and radiochromic plastic dosimeters (Oldham et al., 2001).

In this context, Fricke-Xylenol Gel (FXG) stands out as a relevant dosimetric medium, offering a valuable tool for three-dimensional (3D) dosimetry to monitor and estimate the radiation dose delivered during theragnostic procedures. The FXG was the dosimeter used in the development of this study (Schreiner, 2015).

Additive manufacturing (AM) is a rapidly evolving technology that enables the production of physical parts directly from three-dimensional (3D) digital models, primarily through layer-by-layer material deposition. The AM process typically involves two main stages: a digital rendering phase, during which the part geometry is designed and processed using computer-aided design (CAD) tools; and a fabrication phase, in which the physical component is built (Acierno et al., 2023)

According to the ISO/ASTM 52900 standard, AM is defined as a “process of joining materials to make parts from 3D model data, usually layer upon layer, as opposed to subtractive manufacturing and formative manufacturing methodologies”. Earlier terminology used to describe AM includes *additive fabrication*, *additive processes*, *layer manufacturing*, *solid freeform fabrication*, and *freeform fabrication*(Acierno et al., 2023).

This technology is based on slicing parameters such as layer thickness, flow rate, infill density, raster angle, raster pattern, air gap, nozzle diameter, and the thickness of top and bottom layers play a central role in determining the final geometry and mechanical performance of printed components. In addition, part orientation within the build platform—defined by its alignment relative to the X, Y, and Z axes (horizontal, vertical, or inclined)—is a key factor influencing surface quality, dimensional accuracy, and interlayer adhesion. These variables, along with thermal processing conditions, are widely recognized as critical parameters in fused deposition modeling (FDM), as they directly affect the reproducibility, structural integrity, and overall performance of the fabricated parts(Mallikarjuna et al, 2023).

Most commercially available FDM systems or FDM 3D printers operate at maximum extrusion temperatures in the range of 280–350 °C. Consequently, thermoplastic materials with high melting points are generally unsuitable for processing via standard FDM, limiting the material selection to polymers and composites that can be extruded at moderate temperatures (Mallikarjuna et al, 2023).

In the FDM process, raw materials are used in the form of filaments. These are polymeric materials such as acrylonitrile butadiene styrene (ABS), polylactic acid (PLA), a variation of polyethylene terephthalate (PET) called PETG, acrylonitrile styrene acrylate (ASA), polyamide (PA), polycarbonate (PC), polymethyl methacrylate (PMMA), polyethylene (PE) and polypropylene (PP). Research is also ongoing to discover new materials for this process, such as ceramic, composite, and metallic materials.

Firstly, polylactic acid (PLA) is a biodegradable thermoplastic derived from renewable biomass sources such as corn starch and vegetable oils. It is environmentally friendly and biocompatible, making it suitable for a wide range of applications. PLA exhibits a glass transition temperature between 50 °C and 70 °C and a melting temperature typically ranging from 180 °C to 220 °C. Due to its low energy requirements and cost-effectiveness, it is widely used in fused deposition modeling (FDM) (Santana et al, 2018).

Acrylonitrile butadiene styrene (ABS) is a widely used thermoplastic in additive manufacturing due to its high impact resistance, dimensional stability, good chemical resistance and low price. It also exhibits low water absorption, high color fastness, and improved oxidative stability when formulated with antioxidant. These characteristics make ABS particularly suitable for fused deposition modeling (FDM) applications. Its typical melting temperature ranging is from 220°C to 240 °C (Wady et al, 2019).

Nowadays, a material that's getting more popular among 3D printing users and filament producers is PETG, especially when you need to make flexible and durable parts. In general, PETG consists of a polymer with a glass transition temperature around 80ºC, with mechanical properties similar to those of PET, offering the advantages of remarkable toughness, flexibility, and high processing capacity. Its melting temperature ranging is from 220°C to 250 °C(Santana et al, 2018).

Polymers containing aromatic structures, such as acrylonitrile styrene acrylate (ASA), exhibit greater radiation resistance than those with predominantly saturated aliphatic backbones [3,10,28,29]. ASA is widely recognized in additive manufacturing for its superior weatherability, offering enhanced resistance to ultraviolet (UV) radiation, moisture, and thermal degradation compared to PLA, ABS, and PETG. In addition, ASA demonstrates improved printability over ABS, with reduced warping and cracking during large-scale prints. However, due to the emission of styrene vapors during processing, printing with ASA requires adequate ventilation or the use of a fume extraction system. Its melting temperature ranging is from 240°C to 260 °C (Wady et al, 2019).

In summary, fused deposition modeling (FDM) stands out as one of the most accessible and versatile additive manufacturing techniques, particularly within the biomedical field. Its ability to fabricate customized models, prototypes, and end-use components in a cost-effective and straightforward manner makes it highly suitable for applications such as anatomical models, orthopedic implants, scaffolds for tissue engineering, and dosimetry experimental systems. FDM continues to expand its relevance across personalized medicine and biomedical engineering, driven by ongoing advancements in material development and process optimization (McMahon et al, 2011).

This study aims to evaluate the chemical compatibility of 3D-printed cylindrical supports with radioactive gold nanoparticles (^{198}AuNPs), employing a dual-response protocol based on Fricke-Xylenol Gel (FXG) dosimetry, which functions simultaneously as a dose detector and a pH indicator. The supports were fabricated via fused deposition modeling (FDM) using four different thermoplastic filaments—ABS, ASA, PETG, and PLA—with standardized dimensions of 23 mm in diameter, 40 mm in height, and 1 mm in wall thickness. Each structure was filled with 1 mL of AuNP-containing solution and subsequently immersed in FXG for 24 hours under controlled conditions (refrigerated and light-protected). This approach provides a systematic framework to assess potential material-induced interactions that may compromise the physicochemical integrity of nanoradiopharmaceutical systems.

Materials and Methods.

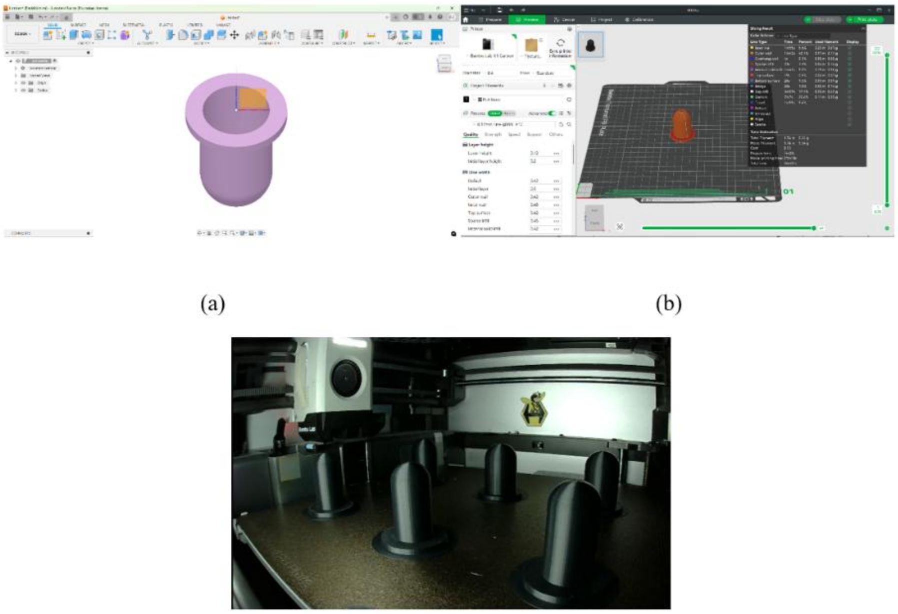
The methodology adopted in this study is organized into several stages:

First, gold nanoparticles (AuNPs) were synthesized as described by Kannan et al. (2006) and Katti et al. (2006). In this methodology, metallic gold is dissolved in an $HCl:HNO_3$ acid solution, purified by controlled heating and sequential additions of HCl, and subsequently combined with gum arabic (0.2 mM, 1:1 ratio) at 100°C. Reduction with sodium citrate (39 mM) and NaOH (13 mM) produced AuNPs evidenced by their characteristic wine-red color. To obtain these radioactive gold nanoparticles, stable gold-197 was neutron-activated in a flux of approximately 2×10^{13} $n{\cdot}cm^{-2}{\cdot}s^{-1}$, yielding ^{198}Au with a half-life of 2.7 days. The irradiated gold was subsequently dissolved and stabilized with gum arabic, producing colloidal ^{198}Au nanoparticles. The final suspension used in this study had an activity of 16.65 MBq (0.45 mCi).

Afterwards, the FXG dosimeter was prepared by dissolving 270 Bloom bovine gelatin (5%) in 75% ultrapure water at 70°C, cooling to 35°C, and adding the other components (50 mM H_2SO_4, 1 mM NaCl, 1 mM ferrous ammonium sulfate, and 0.01 mM xylenol orange) dissolved in the remaining 25% water. After homogenization, the solution was stored in specific bottles for the Vista 16 tomograph and refrigerated for 12 hours (Tavares et al., 2024).

Then, the prototype was modeled using Autodesk Fusion 360 (see Figure 1 (a)). The prototype has a cylindrical geometry and the following dimensions: height of 40 mm, diameter of 23 mm with a through hole to be hollow internally, and wall thickness of 1 mm.

The 3D printing of the cylindrical supports was carried out using the Bambu Lab X1 Carbon printer, configured through the Bambu Studio slicing software (see Figure 1 (b) e Figure 1 (c)). The slicing settings were optimized to ensure dimensional accuracy and print consistency, employing a 0.4 mm nozzle and a high-resolution layer height of 0.08 mm (High Quality preset). Specific extrusion temperatures were defined for each filament based on their thermal processing requirements: 220 °C for PLA, 270 °C for ABS, 250 °C for PETG, and 260 °C for ASA. These temperature settings were selected to ensure adequate layer adhesion and minimize printing defects for each material.



(a) (b)

(c)

Figure 1:Build 3D prototypes. (a) modelling the prototype in Fusion 360 software. (b) Configuration 3D printing parameters in Bambu Studio Software. (c) Printing the prototypes in Bambu Lab X1 Carbon.

Based on the previously described procedures, the experimental stage of this study was conducted as follows: the Fricke-Xylenol Gel (FXG) dosimeter was prepared in plastic containers, in which the 3D-printed cylindrical supports were fully immersed. The internal cavity of each support was filled with 1 mL of a gold nanoparticles (^{198}AuNPs) solution, and the entire system was maintained under controlled conditions—specifically, refrigerated and protected from light—for a period of 24 hours. This procedure was independently performed for each support material: ABS, ASA, PETG, and PLA.

In addition, dose distribution analysis is performed by optical computed tomography (OCT) to quantify changes in the optical density of the gel (Oldham et al., 2001).

Fourier-transform infrared (FTIR) spectroscopy was employed to identify the principal molecular components of the 3D printing filaments and to compare the spectra with similar materials reported in the literature. The analyses were conducted using a Bruker Invenio S FTIR spectrometer, equipped with a diamond ATR accessory. Spectra were recorded in the range of 4000 to 550 cm^{-1}, with 32 scans per sample. The chemical characterization was performed both before and after exposure of the printed supports to the gold nanoparticles, including conditions with and without direct contact with the AuNP solution.

Results and Discussion

This approach allowed simultaneous monitoring of (1) potential corrosive processes, inferred from visual changes in the support-gel and support-gold nanoparticle interfaces, as illustrated in Figures 2(a) and 2(c); and (2) pH variations resulting from the atypical basic behavior of ^{198}AuNPs. The latter was quantified by optical density analysis using the Vista 16 spectrophotometric system. For this purpose, the same precursors and reagents used in the synthesis of ^{198}AuNPs were used, excluding the radioactive components. Figure 2(c) highlights the formation of a distinct stain at the interface after 24 hours of contact with the nanoparticles, indicating a possible chemical interaction between the AuNP solution and the surface of the support.

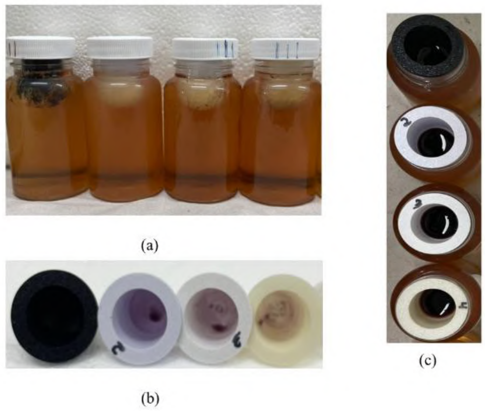


Figure 2: The experimental dosimetry FXG, AuNP solution and 3D supports. (a) FXG plastic containers with 3D Prototype Supports filled with AuNPs (b)The 3D Prototype supports after 24hs (c) The AuNPs in the different 3D Prototype supports.

The three-dimensional reconstruction of the optical computed tomography (OCT) images allowed the evaluation of the dose distribution along the X-axis in the FXG dosimeters attached to the supports containing ^{198}AuNPs. After 24 hours of exposure, no visual changes were observed in the FXG containers (see Figure 2(b)), indicating that there was no apparent degradation of the dosimetric material.

Analysis of the reconstructed images confirmed the absence of direct contact between the FXG gel and the AuNPs, as evidenced by attenuation values approximately ten times lower than those measured in the control sample, in which the FXG was in direct contact with the nanoparticles. While the control group exhibited a characteristic radial diffusion pattern with high attenuation values, the FXG samples incorporated into the supports maintained a homogeneous profile throughout the analyzed region, demonstrating the effectiveness of the physical barrier provided by the printed structures (See Figure 3).

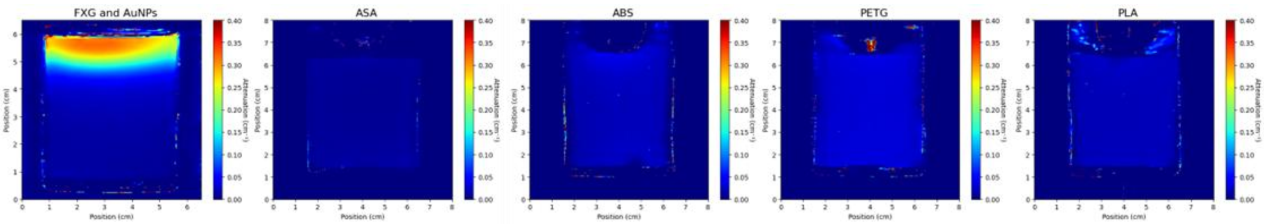


Figure 3: Optical attenuation maps comparing FXG exposed to AuNPs without supports and with ABS, ASA, PETG, and PLA 3D prototypes supports with a diameter of 23 mm.

The FTIR spectra are presented in Figure 4. Figure 4(a) shows the analysis of the 3D-printed support prototypes in their as-printed condition, while Figure 4(b) displays the spectra of the same prototypes (denoted as ABS', ASA', PETG', and PLA') after 24 hours of contact with the FXG dosimeter and gold nanoparticles. Overall, both spectra exhibit a consistent behavior, with similar absorption profiles observed across the different materials before and after exposure.

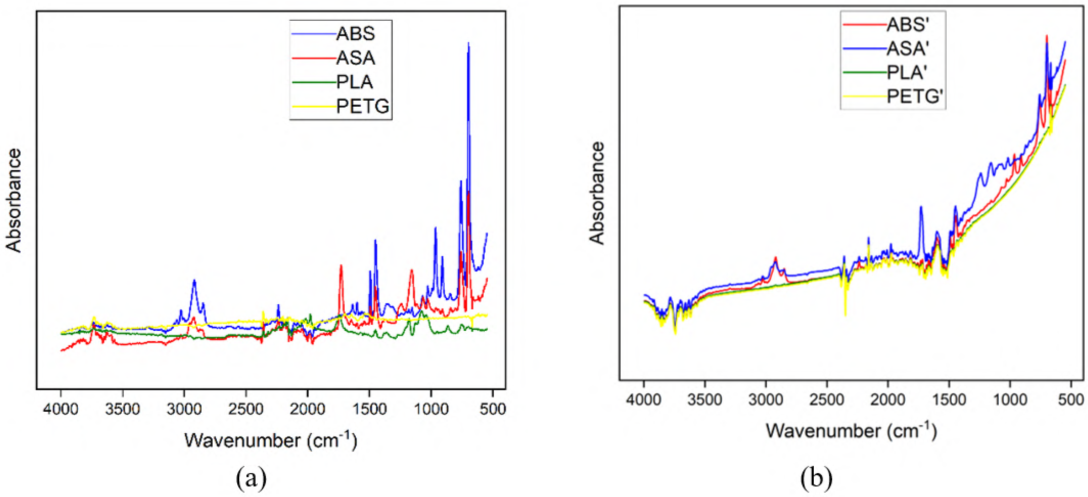


Figure 4: FTIR Spectra for 3D prototypes (a)FTIR spectra for prototypes in their as-printed condition (b)FTIR spectra for 3D prototypes support after 24hs of contact with FXG dosimeter and AuNP.

The FTIR spectra before exposure (Figure 4(a)) show well-defined chemical features for each polymer. ABS and ASA displayed typical nitrile absorption bands (~2230 cm^{-1}) and aromatic ring vibrations below 1000 cm^{-1}. PLA exhibited a strong ester carbonyl peak near 1745 cm^{-1} and C–O–C stretching between 1200–1000 cm^{-1}, while PETG revealed similar C=O characteristics around 1725 cm^{-1}.

After 24 hours of contact with FXG containing AuNPs (Figure 4(b)), significant changes were observed. The spectra of all polymers became more uniform, suggesting a possible superficial coating or diffusion barrier effect by the FXG gel. The distinctive nitrile peak of ABS was no longer visible, and carbonyl bands of PLA, PETG, and ASA were markedly reduced or obscured. These changes may reflect surface adsorption or coverage rather than chemical modification, since no new absorption bands indicative of covalent bonding or degradation products emerged.

The global increase in baseline absorbance below 1000 cm^{-1} for all materials could result from residual FXG or gel components adhering to the surface, altering the scattering or absorbance behavior. The spectral flattening across polymers suggests that after gel contact, surface-specific features were masked, but no chemical degradation or polymer backbone cleavage is evident.

Conclusion

In summary, the integration of 3D-printed supports (ABS, PETG, PLA, and ASA) with ^{198}Au nanoparticles in FXG dosimetry systems was successfully demonstrated. FTIR analysis confirmed that the polymers retained their chemical integrity after 24 hours of exposure, with spectral variations attributable to surface adsorption of the gel rather than bulk chemical reactions. OCT and dose distribution analyses further indicate that the high-energy β^- emission of ^{198}Au results in localized superficial dose deposition, effectively contained by the polymeric supports. These findings validate the proposed methodology for radiological protection studies, reinforcing the suitability of additive manufacturing for producing stable and functional components in gel dosimetry applications.

Acknowledgements

Acknowledgment to the agencies FAPESP 2017/50332-0, CAPES PROEX, CNEN 01342,006411/2023-03 for project funding and scholarship support.

References

- Acierno, D.; Patti, A. Fused Deposition Modelling (FDM) of Thermoplastic-Based Filaments: Process and Rheological Properties—An Overview. Materials, 2023, 16, 7664.
- Attix, F. H. Introduction to Radiological Physics and Radiation Dosimetry. John Wiley & Sons, 2008.
- Daems, N. et al. Gold nanoparticles meet medical radionuclides. Nuclear Medicine and Biology, 2021.
- Ibbott, G. S. Applications of gel dosimetry. In: Journal of Physics: Conference Series. IOP Publishing, p. 007, 2004.
- Kannan, R., Rahing, V., Cutler, C., Pandrapragada, R., Katti, K. K., Kattumuri, V., Robertson, J. D., Casteel, S. J., Jurisson, S., Smith, C., and Katti, K. V. (2006). Nanocompatible chemistry toward fabrication of target-specific gold nanoparticles. Journal of the American Chemical Society, 128(35):11342–11343.
- Katti, K. V., Kannan, R., Katti, K. K., Kattumori, V., Pandrapraganda, R., Rahing, V., Cutler, C., Boote, E., Casteel, S. J., Robertson, J. D., and Jurisson, S. S. (2006). Hybrid gold nanoparticles in molecular imaging and radiotherapy. Czechoslovak Journal of Physics, 56(1):D23–D34.
- Mallikarjuna, et al. A review on the melt extrusion based fused deposition modeling (FDM) background, materials, process parameters and military applications. International Journal on Interactive Design and Manufacturing (IJIDeM), 2023.
- McMahon, S. J., et al. (2011). Radiother. Oncol., 100(3): 412–416.
- Nezhad, Z. A.; Geraily, G. A review study on application of gel dosimeters in low energy radiation dosimetry. Applied Radiation and Isotopes, 179, p. 110015, 2022.
- Olding, T., Holmes, O., and Schreiner, L. J. (2010). Cone beam optical computed tomography for gel dosimetry I: scanner characterization. Physics in Medicine & Biology, 55(10):2819.
- Santana, et al. Estudo comparativo entre PETG e PLA para impressão 3D através de caraterização térmica, química e mecânica. Revista Materia, 2018.
- Schreiner, L. J. True 3D chemical dosimetry (gels, plastics): Development and clinical role. In Journal of Physics: Conference Series. IOP Publishing, p. 012003, 2015.
- Tavares, P. V. S., Diniz, R. E., Rodrigues Jr, O., and Campos, L. L. (2024). Attenuation images of optical CT using Fricke xylenol orange gel for dose mapping in radiotherapy. Radiation Physics and Chemistry, 224:112016.
- Van Dyk, J.; Battista, J. J.; Bauman, G. S. Accuracy requirements for 3D dosimetry in contemporary radiation therapy. In: Clinical 3D Dosimetry in Modern Radiation Therapy. CRC Press, p. 15–62, 2017.
- Wady, P., et al. (2019). Additive Manufacturing, 28:100907.

XXV ISSSD September 22 – 26, 2025 Belo Horizonte, Minas Gerais - Brazil
International Symposium on Solid State Dosimetry

Development of a test bench using a CsI/SiPM detector for mass flow measurement in industrial processes

S. C. P. Oliveira, T. O. Campagnani, L. S. Thein, J. Q. C. Duarte, D. A. M. Campolina*

Centro de Desenvolvimento da Tecnologia Nuclear, Serviço de Tecnologia de Reatores, Belo Horizonte, Brazil

* campolina@cdtn.com

Abstract

Monitoring mass flow in industrial processes such as cement manufacturing, mining, and petrochemical plants presents significant challenges, especially for non-intrusive solutions operating in harsh environments. This work reports the development and computational testing of a laboratory test bench for mass-flow measuring gauges based on γ-ray transmission using a sealed ^{60}Co source and a compact detection head coupling CsI(Na) scintillators to Silicon Photomultipliers (SiPMs). The proposed system aims to overcome the limitations of traditional NaI(Tl)/PMT detectors by improving robustness, reducing temperature dependence, and enabling modular source and target geometries. The test bench was conceptually designed to replicate the main geometrical and radiometric conditions found in conveyor-based flow-meters and to support interchangeable radioactive sources and bulk-material targets. To guide the system design and optimization, a Monte Carlo simulation model was developed using OpenMC. This allowed investigation of parameters such as source geometry, crystal dimensions, shielding configuration, and bulk-material composition. Parallel studies quantified the temperature sensitivity of both the crystal and the SiPM, reinforcing the need for robust temperature compensation via software-based "soft sensor" algorithms. The adopted compensation strategy is based on the nonlinear autoregressive with exogenous input (NARX) model. Preliminary simulation results demonstrate that maintaining sample heights below 15.7 cm ensures linear detector response, while the optimal crystal geometry for efficient signal acquisition is a $4.0\,\text{cm} \times 4.0\,\text{cm} \times 5.0\,\text{cm}$ CsI(Na). These findings, obtained exclusively from computational studies, are being used to finalize the design of the physical test bench. Future experimental work will focus on validating these results and implementing real-time temperature compensation for stable operation in industrial conditions.

Keywords: Soft-Sensor; Gammametry; CsI(Na)/SiPM detectors; Monte Carlo.

1 Introduction

Industrial process monitoring in sectors such as cement manufacturing, mining, and petrochemical processing requires precise, real-time measurement of material flow rates on conveyor systems and pipelines. Among the various techniques for non-intrusive flow measurement, γ-ray transmission methods are particularly valuable due to their ability to penetrate dense materials and provide continuous monitoring without interfering with production (Mayet et al., 2023).

Traditional γ-ray mass flow gauges employ sealed radioactive sources, typically ^{60}Co, paired with scintillation detectors to monitor material streams in industrial environments. The ^{60}Co isotope, with its characteristic gamma energies and long half-life, provides a stable and penetrating radiation source suitable for dense material analysis (Roshani et al., 2019). Previous work at the TRIGA Reactor Unit Lab (URT) has developed a rod-type ^{60}Co radioactive source specifically tailored for level measurement in continuous casting processes and has successfully conducted sealing and qualification tests to ensure reliable encapsulation of radioactive sources (Gonçalves et al., 2024; Caldeira et al., 2025).

The predominant detector technology in industrial γ-ray flow measurement systems has traditionally relied on sodium iodide doped with thallium [NaI(Tl)] scintillators coupled to photomultiplier tubes (PMTs). While NaI(Tl) crystals offer excellent light output and energy resolution, they present significant operational challenges in industrial environments, such as hygroscopicity and substantial temperature sensitivity, which can compromise measurement accuracy (Mayet et al., 2024).

Recent advances in semiconductor photodetectors have introduced silicon photomultipliers (SiPMs) as promising alternatives to traditional PMTs. SiPMs offer advantages including compact size, low operating voltage, insensitivity to magnetic fields, and robust mechanical construction (Acerbi and Gundacker, 2021). Parallel developments in scintillator technology have identified cesium iodide doped with sodium [CsI(Na)] as an attractive alternative to NaI(Tl) for industrial applications, due to its lower hygroscopicity and emission spectrum well-matched to SiPMs (Gridin et al., 2015).

The transition from NaI(Tl)/PMT to CsI(Na)/SiPM detector systems introduces several technical challenges such as the complex temperature dependence of the CsI/SiPM combination: while the CsI light yield decreases by approximately $0.3\,\%\,°C^{-1}$, the SiPM gain increases by 2 to $3\,\%\,°C^{-1}$, resulting in nontrivial effects on measurement stability (Kim et al., 2024). Additionally, SiPMs are susceptible to intrinsic noise sources such as dark counts, after-pulsing, and temperature-induced gain variations (Zappalà et al., 2018). Optimizing the signal-to-noise ratio and achieving a linear detector response across the full measurement range further requires careful control of source geometry and activity distribution, as non-uniform profiles can introduce significant non-linearities (Roshani et al., 2022).

To address these challenges, this research is structured along two complementary fronts:

Soft-Sensor Algorithm Development: The first front focuses on the design and validation of a soft-sensor compensation algorithm. This algorithm dynamically corrects for temperature-induced variations in real time, using sensor data and predictive modeling to ensure stable and accurate mass flow measurements under varying industrial conditions. Characterizing and compensating for temperature sensitivity is thus a central objective, as robust compensation is essential for reliable detector operation.

Simulation-Based Test Bench Design: The second front involves the development of a simulation model of a laboratory test bench using Monte Carlo methods (Romano et al., 2015). This virtual bench reproduces the main geometrical and radiometric conditions of industrial flow meters, allowing systematic evaluation of different detector configurations and source geometries prior to physical implementation incorporating a radioactive source fabricated at the IPR-R1 reactor.

2 Temperature Sensitivity and Compensation

According to the crystal datasheet, the photon emission curve varies with temperature (Amcrys, 2009), and similarly, the solid-state photomultiplier (SiPM) sensor exhibits a temperature-dependent response curve (Onsemi, 2022). Both these curves are non-linear, and their interaction results in a strongly non-linear overall sensor response, which leads to distortions in the detected pulse peak depending on the system temperature.

The manufacturer's approximations indicate a decrease in photon emission from the crystal of approximately $0.3\,\%\,°C^{-1}$ and a variation in the photomultiplier response between $2\,\%\,°C^{-1}$ and $3\,\%\,°C^{-1}$, but these values represent only rough estimates valid within a limited temperature range. Since the mass-flow gauge is intended for use in diverse industrial environments, it is essential to implement a robust temperature compensation mechanism to correct the output signal accordingly.

To address this, the present work will develop and evaluate a software-based correction approach using soft sensors, which are mathematical models designed to process and adjust signals from physical sensors (Abeykoon, 2019). Soft sensors are particularly useful when direct measurement is challenging due to harsh environments or complex signal interactions, and they can also serve as backup systems in case of sensor failure (Stebner et al., 2021; Vaitkus et al., 2020).

Several modeling techniques exist for soft sensors, including neural networks and deep learning (Yan et al., 2017; Lin and Yan, 2015), geometric methods such as k-nearest neighbors or support vector machines (Vaitkus et al., 2020), and simpler approaches like polynomial models from the ARX family (Mohler et al., 2018). Given the high-speed nature of the sensor response and the short duration of the pulses, computational efficiency is a key concern in this application. Therefore, this study will focus on the polynomial NARX (Nonlinear Auto-Regressive with eXogenous inputs) model, which offers a good balance of nonlinear modeling capability and low prediction cost (Aguirre, 2007).

The NARX polynomial model will be employed in the form described by equation 1, where past output samples (y) and exogenous inputs (u_1, u_2) are used as regressors to predict the current output signal, with Ξ representing the prediction error. The model can capture highly nonlinear dynamic behavior through its polynomial function F^ℓ of degree ℓ:

$$y(k) = F^\ell\left[y(k-1),\ldots,y(k-n_y),u_1(k-1),\ldots,u_1(k-n_u),u_2(k-1),\ldots,u_2(k-n_u)\right] + \Xi(k). \tag{1}$$

Sampling frequency will be selected according to the Nyquist-Shannon theorem, ensuring it is at least twice the highest frequency component present in the sensor signals. In practice, a sampling frequency at least three times greater than the maximum signal frequency will be used to preserve signal integrity and facilitate noise filtering (Aguirre, 2007).

Regarding model delays, the maximum output lag n_y and input lag n_u will correspond to the largest pulse width detectable by the sensor. Since pulses are generally independent except during relaxation periods where overlaps might occur, this delay parameter is straightforward to determine.

For training the model, a reference temperature will be chosen, and the sensor pulses collected under this condition will form the baseline signal. This baseline signal will serve as both the exogenous input u_1 and the output y at the reference temperature input u_2. Signals recorded at other temperatures will be aligned to this reference by adjusting pulse peak positions and flattening effects, effectively constructing the output y for those temperatures. This procedure allows the model to learn how pulse shapes change with temperature variations, enabling accurate temperature compensation in real-time operation.

This strategy outlines the proposed methodology for temperature compensation using a polynomial NARX model; however, it should be noted that results from the implementation and validation of this approach are not yet available and will be presented in future work.

3 Monte Carlo Modeling and Simulation

The methodology employed in this study utilized the OpenMC Monte Carlo code, a community-developed neutron and photon transport simulation framework (Lund and Romano, 2018). OpenMC implements the main physical processes required for photon transport simulation, including coherent (Rayleigh) scattering, incoherent (Compton) scattering, photoelectric absorption with fluorescent emission, pair production, and annihilation radiation emission, similarly to what is done in MCNP (Pelowitz and et al., 2023). However, OpenMC relies exclusively on preprocessed nuclear data libraries, and does not include explicit physical models for continuous processes such as bremsstrahlung emission.

The folloring Simplificatrions were adopted for the modeling:

(i)The simulation does not account for cosmic rays or ambient background radiation, focusing solely on particles emitted by the modeled source.

(ii)The interaction of gamma radiation with the photoluminescent crystal, including scintillation and light emission effects, was not simulated. This simplification is due to the fact that OpenMC models particle transport but does not include optical photon processes. Therefore, all gamma rays reaching the crystal volume are counted regardless of their energy or the distance traveled within the crystal.

All simulations were performed using continuous-energy cross-section data from the ENDF/B-VIII.0 library. Statistical uncertainties were maintained below 2 % through adequate particle history sampling (200000 histories per case). The simulation geometry incorporated material compositions and densities representative of the target materials (coal and hematite) and scintilator crystal under investigation.

The materials used in the simulations were hematite (Fe_2O_3, iron(III) oxide) and bituminous coal, both commonly applied in mass flow measurement systems. The isotopic compositions of these materials are listed in Table 1.

Table 1: Elemental composition of *Iron Ore* and *Coal (Bituminous)* materials used in the OpenMC simulation.

Element	**Symbol** **Iron Ore**	**Mass Fraction**	**Element**	**Symbol** **Coal (Bituminous)**	**Mass Fraction**
Iron	Fe	0.60	Oxygen	O	0.13
Oxygen	O	0.30	Carbon	C	0.78
Silicon	Si	0.05	Hydrogen	H	0.05
Aluminum	Al	0.04	Nitrogen	N	0.02
Manganese	Mn	0.01	Sulfur	S	0.02
Density		3.5 g/cm^3	**Density**		0.793 g/cm^3

Figure rod source located beneath the conveyor emits gamma radiation, which interacts with the conveyor structure and the bulk material. The scattered and attenuated photons are then detected by the detector positioned above the entire system.

3.1 Parameter Studies

Three parameter variation were conducted to evaluate geometric effects on gamma ray detection: Bulk Material Geometry Variation, Material Position Variation and Detector Crystal Geometry Variation.

The first investigation examined the influence of bulk material geometry on detector response while maintaining constant volume. The bulk material depth was fixed at 10 cm, while height and width parameters were varied systematically to maintain a constant volume of 2500 cm. This approach isolates geometric effects from material quantity effects, enabling assessment of how radiation path length and angular distribution influence detection efficiency.

The second study investigated proximity effects by varying the position of bulk materials relative to the detector system. Material samples (coal and hematite) with fixed dimensions (height = 15.81 cm, width = 15.81 cm) were positioned at locations ranging from -10 cm to 10 cm relative to the detector centerline. This configuration enables evaluation of measurement artifacts caused by material approach and departure from the detection zone, simulating portal monitoring scenarios.

The third investigation examined detector crystal geometry effects on measurement. Crystal dimensions were varied while maintaining constant active volume (80 cm^3), with crystal side dimensions ranging from 2.6 to 4.0 cm and corresponding length adjustments from 11.834 to 5.000 cm. This study assesses the impact of detector aspect ratio on photon detection efficiency and energy resolution.

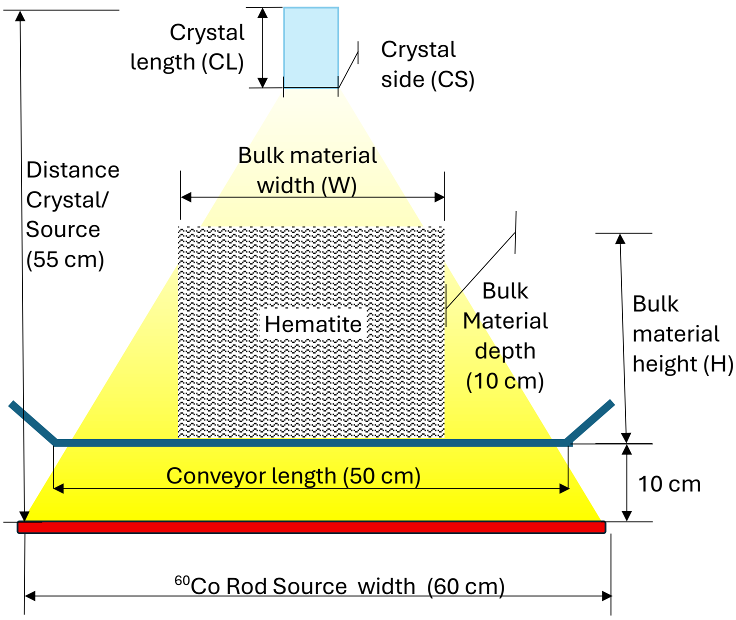


Figure 1: Reference geometry used for simulation of the radiometric bulk flow measurement system.

4 Results

4.1 Parameter Studies

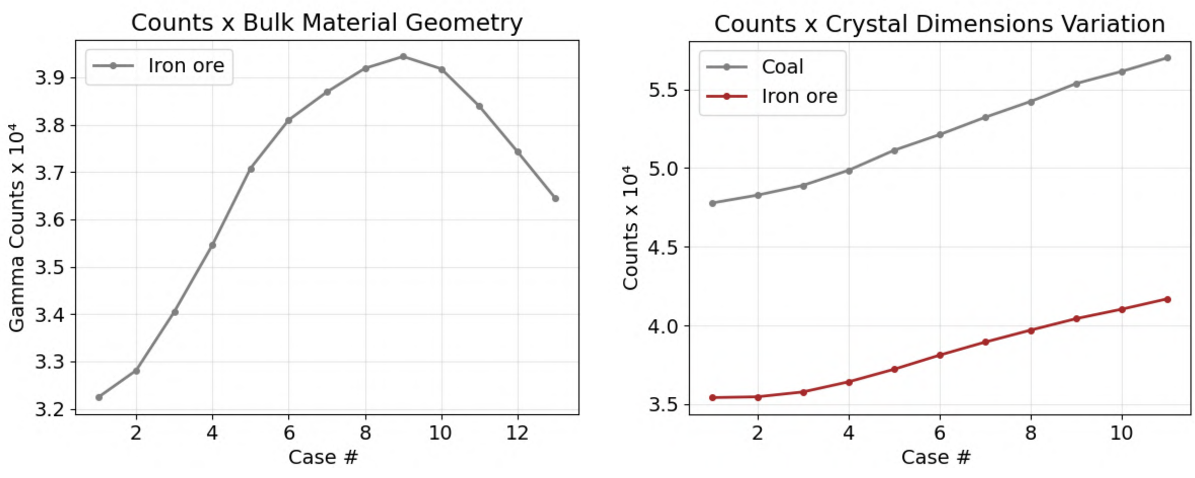
Figure 2(a) plots the detector response as a function of the material height and width while the total volume was kept constant. For heights up to 15.71 cm (Case 6), the counting curve remains linear, indicating that self-attenuation is uniform and does not distort the flux that reaches the detector. When the height is increased to 17.86 cm the slope of the curve starts to deviate. This loss of linearity confirms that a tall, narrow target shields the γ-particles differently, altering the counting.

Figure 2(b) shows that starting from case 3, the count rate increases linearly with the crystal side length, indicating that any crystal with a side length equal to or greater than 3.6 cm is suitable for the application. Considering the minimum crystal height and the requirement for the ADC board (analog-to-digital converter, housing eight 3 mm SiPMs) to fit within the crystal base area, the crystal with a 4.0 cm side length (case 6) was adopted as the reference configuration.

The response profile shown in Figure 3(a) was obtained by translating the target (bulk material) from $z = -5$ to 15 cm with respect to the detector centre $z = 5$. A small pre-rise and post-fall of the counting rate is observed for coal. The presence of material enhances the counts registered close to the overlap region (z approximately 0 cm and 10 cm). Figure 3(b) presents the geometric setup implemented in the OpenMC simulation environment in the YZ plane, illustrating the relative movement of the bulk material sample with respect to the detector position.

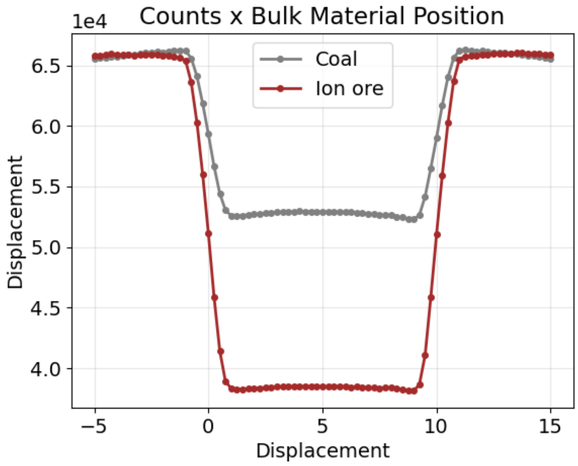
Table 2: Geometrical configurations investigated in Studies 1 and 3. Study 1: bulk materials with fixed depth of 10 cm and volume of 2,500 cm^3. Study 3: detector crystals with fixed active volume of 80 cm^3.

Study 1 – Bulk Material Geometry			**Study 3 – Detector Crystal Geometry**	
Case #	Height (cm)	Width (cm)	Crystal side CS (cm)	Crystal length CL (cm)
1	5.00	50.00	3	8.889
2	7.14	35.00	3.2	7.813
3	9.29	26.91	3.4	6.922
4	11.43	21.88	3.6	6.173
5	13.57	18.42	3.8	5.542
6	15.71	15.91	4	5
7	17.86	14.00	4.2	4.535
8	20.00	12.50	4.4	4.132
9	22.14	11.30	4.6	3.781
10	24.29	10.30	4.8	3.472
11	26.43	9.46	5	3.2
12	28.57	8.75		
13	30.00	8.33		

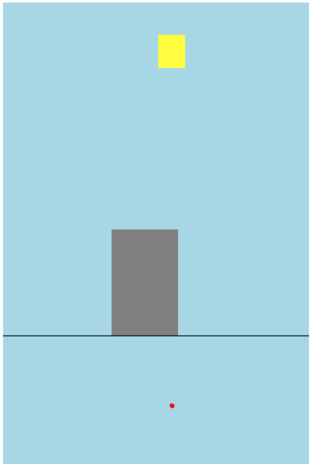


(a) (b)

Figure 2: Detector response as a function of (a) bulk material geometry and (b) crystal side and length.



(a)



(b)

Figure 3: (a) Detector response profile as the bulk material sample is translated; (b) Geometric setup in the YZ plane (OpenMC).

4.2 Physical Structure of the Test Bench

The proposed laboratory test bench is designed to provide a controlled environment for the evaluation and calibration of radiation-based mass flow measurement systems. The physical setup features a carousel-type conveyor, which facilitates the easy placement and movement of test material samples through the detection zone. The configuration allows for the systematic positioning of different types of bulk material (e.g., coal, hematite), simulating real industrial conveyor operations.

The rod-type ^{60}Co radioactive source is mounted beneath the conveyor for γ-ray irradiation of the material as it passes over the source. Directly above the passage, the detection system, consisting of a CsI(Na) scintillator crystal coupled to a Silicon Photomultiplier (SiPM), is positioned to collect transmitted gamma photons. The geometric arrangement—including the fixed distance between the source and detector, adjustable height and width of the test material path, and a modular conveyor that allows quick swapping of samples and fine adjustments to material dimensions—supports systematic studies on self-attenuation, geometrical effects, and detector positioning.

Figure 4 represents the laboratory mass flow test bench, highlighting the carousel-type conveyor for easy placement of test materials as they pass through the rod-type radioactive source (positioned below) and the detector system.

4.3 Data Acquisition and Control Electronics

The electronic architecture of the test bench leverages a modular approach for high-performance signal acquisition and flexible system integration. The core components include an Intel Cyclone 10 FPGA and a Texas Instruments LAUNCHXL-F28379D DSP, interconnected via an SPI (Serial Peripheral Interface) bus.

In this configuration, the FPGA operates as the SPI master, managing the timing and control of data transfers to the DSP (configured as the SPI slave). The communication supports 8, 16, or 32-bit word lengths, with 16-bit mode currently in operation. The SPI logic on the FPGA implements a finite state machine (FSM) with three distinct states: IDLE (awaiting new data), LOAD (preparing the data for transmission), and SEND

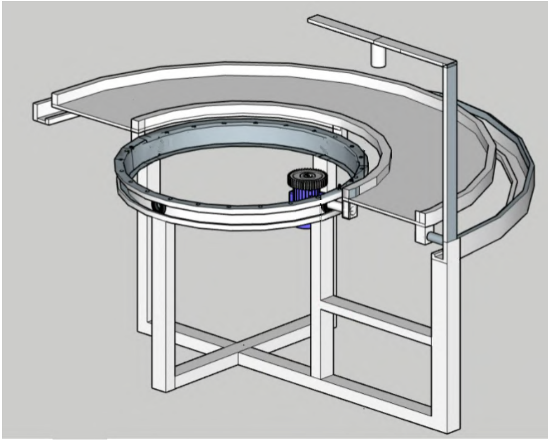


Figure 4: Mass flow test bench: carousel-type conveyor.

(serially transmitting data while controlling the chip select signal CS), ensuring robust and deterministic data flow.

For hardware integration, pin assignments on both the FPGA and DSP are configured to match the SPI signaling requirements. The FPGA reads digitized signals from the analog-to-digital converter (ADC) associated with the detector, packages the data, and transmits it to the DSP for further processing and analysis. Additional microcontrollers (such as an Arduino Nano) have been successfully employed for initial test data reception and validation, confirming the operational integrity of the SPI link.

This electronic subsystem is engineered to provide low-noise, high-fidelity acquisition of the detector's output pulses, supporting real-time analysis and calibration routines. Combined with precision timing and configurable logic, the system will enable flexible implementation of signal processing algorithms, real-time compensation strategies, and diagnostic routines essential for accurate and stable mass flow measurement in laboratory deployments.

5 Conclusions

The Monte Carlo results demonstrate that a sample height of 15.71 cm is enough to break the linear relation between activity and count rate. For conveyor-belt applications, the working height should therefore be kept below 39 % of the full 40 cm range to preserve calibration linearity.

The position-sweep study confirms that neighboring samples can perturb each other's reading when their leading edges are within a few centimeters of the detector centre.Follow-up work will not only investigate whether increasing belt speed (i.e., reducing residence time in the overlap zone) mitigates this effect to an acceptable level, but will also examine if gamma effects or other secondary reactions occur in the coal samples, since such phenomena were not observed in the ore material.

In summary, although the current results are based solely on preliminary Monte Carlo simulations, ongoing computational studies—including simulations of the photoluminescent response of the crystal using Geant4, which supports detailed modeling of optical photon generation, scintillation, and light transport phenomena—are being conducted to guide the design and construction of the fixed laboratory test bench.

Acknowledgements

The authors are grateful to Brazilian research funding agencies, CNEN (Comissão Nacional de Energia Nuclear – Brazil), CNPq (Conselho Nacional de Desenvolvimento Científico e Tecnológico – Brazil), CAPES (Coordenação de Aperfeiçoamento de Pessoal de Nível Superior – Brazil) and Fapemig (Fundação de Amparo à Pesquisa do Estado de Minas Gerais (MG – Brazil) for the support.

References

Abeykoon C. (2019) Design and applications of soft sensors in polymer processing: A review. *IEEE Sens J* 19:2801–2813

Acerbi F. and Gundacker S. (2021) Characterization of VUV4 SiPM for liquid argon detector. *Journal of Instrumentation* 16:P07021

Aguirre L. A. (2007) *Introdução à Identificação de Sistemas–Técnicas Lineares e Não-Lineares Aplicadas a Sistemas Reais*. Editora UFMG

Amcrys (2009) Scintillation material data sheet. Datasheet. available at: `http://www.amcrys.com/pdf/4281_.pdf` (Accessed: 2025)

Caldeira S., Ribeiro E., Vilela J. J., da Silva M. A., Peressinotto J., Campagnole dos Santos A. A., Pimenta Mourão R., and Campolina D. (2025) Tests for qualification of encapsulations for sealed radioactive sources. *Journal of Radioanalytical and Nuclear Chemistry* 334:565–573

Gonçalves D., Silva M., and Campolina D. (2024) Performance check bench of cobalt-60 radioactive source for steel plant. in *Proceedings of the INAC 2024: International Nuclear Atlantic Conference. Nuclear Energy: assuring energy, health and food*. pages 4–4

Gridin S., Belsky A., Vasil'ev A. et al. (2015) Luminescence and scintillation properties of CsI: A potential cryogenic scintillator. *Physica Status Solidi (b)* 252:804–810

Kim D. H., Kim J. H., Park H. S. et al. (2024) Investigation of scintillation properties of a CsI(Tl) crystal at low temperature for dark matter search. *Journal of Instrumentation* 19:P07011

Lin Y. and Yan W. (2015) Study of soft sensor modeling based on deep learning. in *2015 American Control Conference (ACC)*. IEEE

Lund A. L. and Romano P. K. (2018) Implementation and validation of photon transport in openmc. Technical report. Argonne National Laboratory (ANL), Argonne, IL (United States)

Mayet A. M., Alizadeh S. M., Kakarash Z. A. et al. (2023) Evaluation of liquid-gas flow in pipeline using gamma-ray absorption technique and advanced signal processing. *Nukleonika* 68:87–96

Mayet A. M., Alizadeh S. M., Nurgalieva K. S. et al. (2024) Identification of the structure of liquid–gas flow in a horizontal pipeline using the gamma-ray absorption and a convolutional neural network. *Applied Sciences* 14:4854

Mohler I., Ujević Andrijić Ž., and Bolf N. (2018) Soft sensors model optimization and application for the refinery real-time prediction of toluene content. *Chem Eng Commun* 205:411–421

Onsemi (2022) Silicon photomultipliers (sipm), low-noise, blue-sensitive. Datasheet. available at: `https://www.onsemi.com/pdf/datasheet/microc-series-d.pdf` (Accessed: 2025)

Pelowitz D. B. and et al. (2023) *MCNP® User's Manual - Code Version 6.3*. Los Alamos National Laboratory. Los Alamos, NM. lA-UR-23-20045

Romano P. K., Horelik N. E., Herman B. R. et al. (2015) OpenMC: A state-of-the-art Monte Carlo code for research and development. *Annals of Nuclear Energy* 82:90–97

Roshani G. H., Nazemi E., Feghhi S. A. H. et al. (2019) Precise volume fraction measurement for three-phase flow meter using ^{137}Cs gamma source and one detector. *Radiochimica Acta* 107:459–469

Roshani M., Phan G., Roshani G. H. et al. (2022) Enhanced multiphase flow measurement using dual non-intrusive techniques and ANN model for void fraction determination. *Processes* 10:2371

Stebner S. C., Maier D., Ismail A., Balyan S., Dölz M., Lohmann B., Volk W., and Münstermann S. (2021) A system identification and implementation of a soft sensor for freeform bending. *Materials* 14:4549

Vaitkus V., Brazauskas K., and Repšytė J. (2020) Soft-sensors based on black-box models for bioreactors monitoring and state estimation. in *Proceedings of the 2020 12th International Conference on Bioinformatics and Biomedical Technology*. ACM, New York, NY, USA

Yan W., Tang D., and Lin Y. (2017) A data-driven soft sensor modeling method based on deep learning and its application. *IEEE Trans Ind Electron* 64:4237–4245

Zappalà G., Auricchio N., Andersen M. I. et al. (2018) Spatially resolved dark count rate of SiPMs. *European Physical Journal C* 78:567

Environmental Gamma Radiation and Radon Exposure in Areas Surrounding Urban Quarries in Belo Horizonte, Brazil

J. B. Severo[1]*, I. N. S. Ferreira[1], L. C. Takahashi[1], B.B.G. Correa[1], W. Brito[1], Z. Rocha[1] and R. G. Passos[1]**

[1] Centro de Desenvolvimento da Tecnologia Nuclear (CDTN), Serviço de Análise e Meio Ambiente (SEAMA), 31270-901, Belo Horizonte, Brazil

* julia.severo@cdtn.br
***In memoriam*

Abstract

The city of Belo Horizonte, Brazil, was historically developed over a geologically active basement rich in granites and gneisses, with widespread use of materials extracted from urban quarries in civil construction and street paving. This geological setting, combined with the anthropogenic use of naturally radioactive rocks, has raised concerns about increased environmental radiation levels, particularly gamma radiation and radon gas concentrations. This study investigates the relationship between disused quarries located within the urban perimeter of Belo Horizonte, the environmental gamma radiation levels in their surroundings, and indoor radon concentrations. To evaluate these relationships, gamma radiation data were collected using a portable RS-230 BGO gamma spectrometer, georeferenced with GPS, and processed using ArcGIS® software through inverse distance weighting (IDW) interpolation. A heat map of gamma radiation distribution (in counts per second – CPS) across the city was generated. Circular buffer zones (100 m, 300 m, 500 m, and 1000 m) were established around four abandoned quarries: Engenho Nogueira, Prado Lopes, Lagoinha, and Pompeia. The gamma radiation means within each radius were calculated and classified into four categories: low (<150 CPS), medium (150–250 CPS), high (250–350 CPS), and very high (>350 CPS). Based on CPS values, it is also possible to estimate human exposure through dose calculations, contributing to the assessment of potential health risks associated with gamma radiation. The results revealed elevated gamma radiation levels near the quarries, especially within 100 meters, with the highest means observed at Lagoinha (411 CPS) and Prado Lopes (369 CPS). A general decreasing trend in radiation levels with distance from the quarries was observed, suggesting a direct influence of these rock formations on the surrounding gamma radiation levels. Deviations were noted, particularly due to the proximity of certain quarries, such as Prado Lopes and Lagoinha, which may have resulted in overlapping influences. Additionally, radon concentration data from the “Indoor Radon Monitoring Campaign in Belo Horizonte” (CMRAI-BH) were analyzed for residences located within a 1000-meter radius of the quarries. The mean radon levels (in Bq/m³) found near the quarries were as follows: Engenho Nogueira – 39.5, Prado Lopes – 94.0, Lagoinha – 35.2, and Pompeia – 69.1. These values reflect multiple influencing factors, including building type, floor level, ventilation, and construction materials. The study highlights the potential health risks posed by residual radioactivity in urban areas due to both natural geological features and the legacy of construction practices involving quarry materials. The findings emphasize the importance of integrating geological and environmental data into urban planning and public health assessments, especially in regions with high natural radioactivity.

Keywords: Natural Radioactivity; Gamma Radiation; Radon; Urban Quarries.

1 Introduction

The presence of outcropping rocks and intense quarrying in the city of Belo Horizonte, Minas Gerais, are key factors in understanding the population's exposure to natural radioactivity. Since its foundation, the city has used materials extracted from urban quarries, such as granites, quartzites and gneisses, both in civil construction and in street paving. Many of these quarries have been deactivated over time, but there are still dozens of abandoned cavities on the urban perimeter of the capital of Minas Gerais. Because they are located in a naturally radioactive geological context, these rock formations may have contributed to the spread of materials with a high potential for emitting radon and gamma radiation in various areas of the city (CORREA, 2019).

Natural radioactivity refers to the radiation emitted by materials that have existed on Earth since its formation, and is a constant feature of the environment. This radiation has two main origins: cosmic rays and the natural radionuclides present in the Earth's crust. Among the most important elements in this process are uranium-238, thorium-232 and potassium-40. From the decay of uranium-238, radon-222 is formed, a radioactive noble gas that seeps into the environment from soil and rocks (UNEP, 2016). Colorless, odorless and tasteless, radon poses a risk mainly through inhalation: its short half-life decay products, such as polonium-218 and polonium-214, can deposit in the lung epithelium, emitting alpha particles that cause damage to respiratory tissue (ICRP, 2010).

Natural sources are responsible for more than 80% of the radiation exposure of the world's population. The global average annual effective dose per person from these sources is approximately 2.4 mSv, which can vary from 1 mSv to more than 10 mSv depending on geographical location. Radon and its decay products represent a significant portion of this exposure, contributing around 1.2 to 1.3 mSv per year (UNSCEAR, 2019). According to the World Health Organization (WHO, 2009), radon is responsible for around 42% of the total annual dose of natural radiation received by individuals, and is considered the second leading cause of lung cancer in the world, behind smoking.

The Metropolitan Region of Belo Horizonte (RMBH), located on a geological foundation rich in granites and gneisses, presents particularly favorable conditions for the emission of natural radiation. The region's geological substrate, belonging to the Iron Quadrangle, is known to contain high concentrations of uranium, thorium and potassium (TAKAHASHI, 2022).

Duarte et. al (2017) points out that previous studies in the RMBH have detected radioactive anomalies in residential areas, with radon concentrations higher than international reference levels. In particular, high concentrations were found in the center of Belo Horizonte. Although the local geology favors the presence of radionuclides, the study pointed out that part of these anomalies can be attributed to the use of quarry materials in paving the streets, demonstrating a relevant anthropogenic impact.

High levels of radon were identified within Avenida do Contorno, one of the oldest and most urbanized areas of the city. The use of materials from quarries in construction may have intensified the levels of radiation detected, highlighting the importance of considering the origin of construction materials as a risk factor for urban environmental radioactivity (LARA, 2017).

Given this context, this study aims to investigate the relationship between quarries in the urban area of Belo Horizonte, ambient gamma radiation rates and the presence of radon indoors.

2 Materials and Methods

To assess the influence of quarries and outcropping rocks on natural radioactivity in the city of Belo Horizonte, gamma radiation count data obtained by the Natural Radioactivity Laboratory (LRN) of the Nuclear Technology Development Center (CDTN) were used, using scans made with the RS-230 BGO portable spectrometer (Radiation Solutions, Canada), coupled to a global positioning system (GPS).

The measurement campaigns were carried out by researchers from the LRN/CDTN, who traveled to various points in the city, simultaneously recording the gamma radiation values and the respective geographical coordinates. The data obtained was processed with ArcGIS® software, using geoprocessing tools for spatial interpolation. Spatial interpolation was carried out using the inverse distance weighting (IDW) method, which estimates values for a continuous surface (raster). This made it possible to generate a heat map representing the distribution of total gamma radiation in the city, expressed in counts per second (CPS).

Based on the map drawn up, the average gamma counts in areas close to the outcropping rocks studied by Correa (2019) were determined. In this study, the author assessed the presence and distribution of natural radionuclides (238U, 232Th and 40K) in rock formations in the Metropolitan Region of Belo Horizonte (RMBH). The research was conducted with rock samples collected from four deactivated quarries (Engenho Nogueira, Prado Lopes, Lagoinha and Pompeia) which were analyzed using gamma spectrometry and neutron activation.

To investigate the possible influence of these rocks on environmental radioactivity, circular zones with radii of 100 m, 300 m, 500 m and 1,000 m were delineated from the center of the old quarries. For each radius, CPS averages were calculated in ArcGIS® software based on the interpolated data.

In addition, data on radon concentrations in indoor environments obtained by Takahashi et al. (2022), who investigated the levels of this gas in 520 environments in Belo Horizonte, including homes, schools and commercial establishments, were analyzed as part of the "Campaign for Monitoring Radon in Indoor Environments in Belo Horizonte (CMRAI-BH)". The measurements were carried out using CR-39 trace detectors. The measurement points were defined based on voluntary adherence to a spatial reference grid built with data from the Belo Horizonte City Hall, divided into 1.2 x 1.2 km quadrants. For this analysis, homes within a radius of 1,000 meters of the quarries studied were selected in order to investigate possible correlations between proximity to rock outcrops, environmental gamma radiation levels and radon concentrations inside the properties.

3 Results and Discussion

Based on the data obtained, a map was drawn up using ArcGIS® software, showing the locations of the quarries evaluated (names of the quarries from left to right: Engenho Nogueira, Prado Lopes, Lagoinha and Pompéia), the gamma radiation rates (in counts per second - CPS), the boundaries of the concentric circles around each quarry (with radii of 100 m, 300 m, 500 m and 1000 m), as well as the homes where the radon concentrations were analyzed (in Bq/m³). These elements are shown in Figure 1.

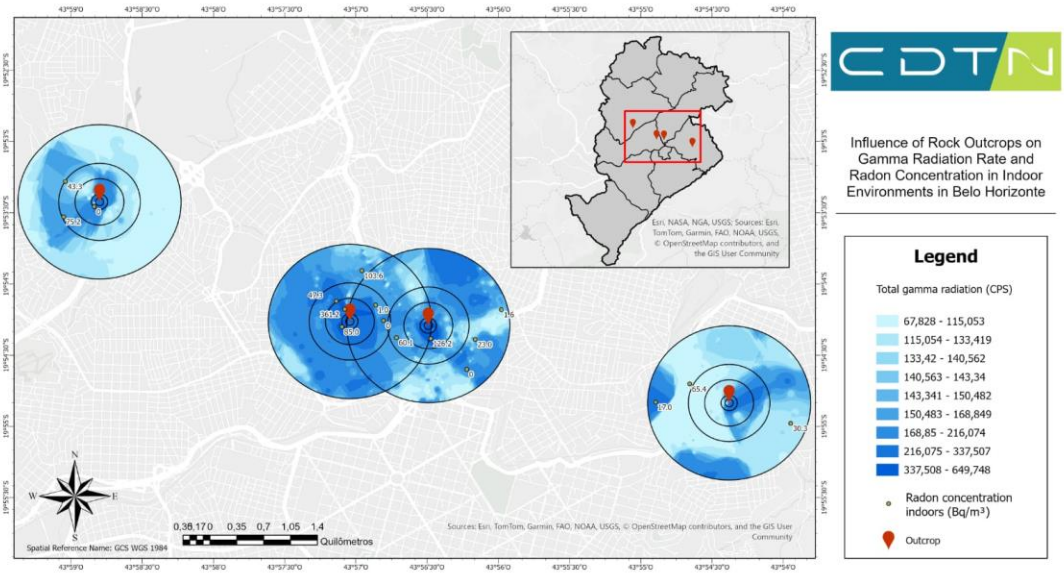


Figure 1: Influence of Rock Outcrops on Gamma Radiation and Radon Concentration in Indoor Environments in Belo Horizonte

The averages of the gamma radiation rates as a function of distance for each quarry are shown in Table 1.

Table 1: Mean gamma radiation count rates (CPS) at different distances from the quarries

Quarry	50 m	100 m	300 m	500 m	1000 m
Engenho Nogueira	292,6	253,3	170,4	150,0	127,7
Prado Lopes	360,1	368,7	305,3	235,5	187,4
Lagoinha	410,7	338,8	215,2	177,9	178,3
Pompéia	211,9	210,4	156,1	137,5	136,2

For classification purposes, the following reference intervals were used for gamma radiation rates: low (<150 CPS), medium (150-250 CPS), high (250-350 CPS) and very high (>350 CPS) (DUARTE, 2017). Based on these ranges, it was observed that, at a distance of 50 meters from the quarries, the average gamma radiation rates were classified as very high in the Lagoinha and Prado Lopes regions, high in Engenho Nogueira and medium in the Pompeia region.

A preliminary analysis indicates a general tendency for average gamma radiation rates to decrease with increasing distance from the quarries, a pattern especially evident in the Engenho Nogueira, Pompeia and Lagoinha quarries. This behavior suggests a direct influence of the proximity of the quarries on the levels of radiation detected in the surrounding area.

In the specific case of the Lagoinha quarry, there was a slight increase in the average CPS between the radii of 500 m (177.9 CPS) and 1000 m (178.3 CPS). However, this variation is small and does not represent a significant change. The Prado Lopes quarry also showed atypical behavior, with an increase in the average CPS between distances of 50 m (360.1 CPS) and 100 m (368.7 CPS). Although this growth deviates from the expected pattern of a reduction with distance, it is also a minor variation. A relevant factor to consider is the proximity of the Prado Lopes and Lagoinha quarries. This proximity may be influencing the values recorded in both areas, making it difficult to distinguish between the individual contributions of each quarry to the radiation levels detected in the surrounding area.

In addition to monitoring gamma radiation, it is essential to assess the concentration of radon indoors. In this context, the homes measured by Takahashi (2022) within 1000 meters of the quarries were surveyed, with the following average results: Engenho Nogueira - 39.5 Bq/m³; Prado Lopes - 94.0 Bq/m³; Lagoinha - 35.2 Bq/m³; and Pompeia - 69.1 Bq/m³. These values reflect an initial analysis, but already indicate the importance of considering multiple variables that directly influence the concentration of radon indoors, such as the type of construction, ventilation, proximity to the ground and the materials used in the buildings. The presence of single-storey houses, for example, can favor the accumulation of radon due to greater interaction with the ground, while well-ventilated environments tend to have lower concentrations (LARA, 2017). In this case, among the residences evaluated, one was on the first floor in the Engenho Nogueira quarry region (n = 3); three were on the first floor in the Prado Lopes quarry region (n = 7); none of those analyzed in the Lagoinha quarry region were on the first floor (n = 6); and, in the Pompeia quarry region, one residence analyzed was also on the first floor (n = 3).

Although the available data is limited, it is possible to observe that the Prado Lopes quarry, which had a high rate of gamma radiation in its surroundings (360.1 CPS at 50 m), was associated with the highest average radon concentration in nearby homes. These results suggest a possible correlation between the radiological potential of outcropping rock formations and the presence of radon indoors, but reinforce the need for more detailed studies, taking into account the environmental and structural variables of each residence.

The results obtained in this study are in line with the findings of Correa (2019). In general, the author confirmed that the rocks from these quarries, which are widely used in construction and paving in Belo Horizonte, have naturally high levels of radioactive elements. Detailed analysis revealed variations in the concentrations of uranium, thorium and potassium, both between the different quarries and between collection points in the same quarry (Prado Lopes). Particularly high levels of uranium were found in the Engenho Nogueira and Rua Pedro Lessa quarries (Prado Lopes), and thorium in the Carlos Góis Municipal School (Prado Lopes), regions where high rates of gamma radiation were also recorded. The Pompeia quarry, on the other hand, had lower concentrations of these elements, which is consistent with the lower gamma radiation rates observed in this study. In the Lagoinha quarry, the main results indicate that the gneiss rock sampled has high concentrations of elements from the thorium and potassium-40 series, and this is the region where the highest gamma radiation rates were found.

The results obtained in this study suggest the need to expand the investigation by taking gamma radiation and radon concentration measurements indoors in other areas of Belo Horizonte. This comparison is important to better understand the real influence of urban quarries on the exposure levels found, distinguishing between what can be attributed to the natural background of the region and what is enhanced by these outcropping rocks. Continuing this type of analysis will contribute to a better understanding of the spatial distribution of natural radioactivity in urban environments and the factors that increase the population's exposure.

4 Conclusions

The data obtained is in line with previous studies on radiometric anomalies in Belo Horizonte, as well as specific studies of quarries in the region. The initial analysis made it possible to establish a possible correlation between total gamma radiation rates and the presence of quarries in Belo Horizonte. In addition, the areas analyzed also showed a relationship between high levels of gamma radiation and significant radon concentrations in homes. To better understand the influence of quarries on radon concentration, it is necessary to consider other variables, such as the floor on which the radon concentration was measured, and the ventilation conditions of the rooms.

Acknowledgements

The authors would like to thank Fundação de Amparo à Pesquisa do Estado de Minas Gerais (Fapemig) and Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) for their financial support and research grants. Our thanks also go to Centro de Desenvolvimento da Tecnologia Nuclear (CDTN/CNEN) for the infrastructure and laboratories provided, which were essential for carrying out this study. Finally, we would like to express our gratitude to the volunteers who contributed to the radiometric campaign, and to the LRN/CDTN researchers who helped acquire the data used in this study

References

CORRÊA, B. B. G. Radioatividade natural nas rochas aflorantes da Região Metropolitana de Belo Horizonte. Trabalho de Conclusão de Curso (Graduação em Tecnologia em Radiologia) – Faculdade de Medicina, Universidade Federal de Minas Gerais, Belo Horizonte, 2019.

DUARTE, M. P.; TAVEIRA, N. F.; SANTOS, T. O.; LARA, E. G.; BARRETO, A. A.; SILVA, J. A. R.; ROCHA, Z.; ASSOCIAÇÃO BRASILEIRA DE ENERGIA NUCLEAR (ABEN). Interpretation of the natural radioactive anomalies in the central area of Belo Horizonte/Brazil. International Nuclear Atlantic Conference (INAC 2017), Belo Horizonte, MG (Brazil), 2017.

INTERNATIONAL COMMISSION ON RADIOLOGICAL PROTECTION (ICRP). ICRP Publication 115: Lung Cancer Risk from Radon and Progeny and Statement on Radon. Annals of the ICRP, v. 40, n. 1, 2010.

LARA, E. G. Potencial de radônio no ar de ambientes interiores residenciais: aspectos geológicos e construtivos da Região Metropolitana de Belo Horizonte. 2017. Trabalho de Conclusão de Curso (Graduação em Geologia) – Universidade Federal de Minas Gerais, Belo Horizonte, 2017.

LARA, Evelise Gomes. Radônio nos gases do solo: distribuições e correlações com litologias e pedologias da RMBH – Região Metropolitana de Belo Horizonte. 2013.

TAKAHASHI, L. C. et al. Região Metropolitana de Belo Horizonte: potencial de estudo da radioatividade natural. In: SEMANA NACIONAL DE ENGENHARIA NUCLEAR E DA ENERGIA E CIÊNCIAS DAS RADIAÇÕES, 2022, Belo Horizonte. Anais eletrônicos. Belo Horizonte: SENCIR, 2022.

UNEP – UNITED NATIONS ENVIRONMENT PROGRAMME. Radiation Effects and Sources. What is radiation. 2016.

UNSCEAR – UNITED NATIONS SCIENTIFIC COMMITTEE ON THE EFFECTS OF ATOMIC RADIATION. Sources, Effects and Risks of Ionizing Radiation. New York: United Nations, 2000.

UNSCEAR – UNITED NATIONS SCIENTIFIC COMMITTEE ON THE EFFECTS OF ATOMIC RADIATION. UNSCEAR 2019 Report: Sources, effects and risks of ionizing radiation. New York: United Nations, 2020.

WHO – WORLD HEALTH ORGANIZATION. Who Handbook on Indoor Radon: A Public Health Perspective. WHO Library: Cataloguing-in-Publication Data, France, 2009.

Estimation of Absorbed Dose in Critical Organs During Interventional Procedures: Experimental and Computational Approach

Raissa Alexia Camargo Guassu[1*], Nayara da Costa Oliveira Sousa[1], Daniel Molena Seraphim[1], Túlio Guilherme Soares Marques[1], Matheus Alvarez[2], Diana Rodrigues de Pina[3]

[1] Sao Paulo State University Julio de Mesquita Filho, Institute of Bioscience, R. Prof. Dr. Antônio Celso Wagner Zanin, 250 - Distrito de Rubião Junior - Botucatu/SP - CEP 18618-689.

[2] Sao Paulo State University Julio de Mesquita Filho, Medical school, Av. Prof. Mário Rubens Guimarães Montenegro, s/n - UNESP - Campus de Botucatu - Botucatu/SP - CEP 18618687.

[3] Department of Tropical Diseases and Imaging Diagnosis, São Paulo State University Julio de Mesquita Filho, Medical school, Av. Prof. Mário Rubens Guimarães Montenegro, s/n - UNESP - Campus de Botucatu - Botucatu/SP - CEP 18618-687,

[*]raissa.guassu@unesp.br

Abstract

Interventional procedures performed in hemodynamic units involve significant exposure to ionizing radiation, with potential dose absorption by radiosensitive organs. Despite ongoing efforts to optimize fluoroscopy protocols, there is still limited objective data regarding internal dose distribution, which hinders the accurate assessment of biological risks. This study aimed to estimate the absorbed dose in critical organs during a coronary angioplasty procedure using an experimental and computational approach. A coronary angioplasty procedure was simulated using a GE Innova IGS 320 system and an Alderson-RANDO anthropomorphic phantom, in which 250 thermoluminescent dosimeters (TLD-100) were distributed in anatomical regions corresponding to the reference organs defined in ICRP Publication 103. The protocol included three common clinical projections (90°, LAO30, CAU30), simulating real exposure scenarios. Monte Carlo simulations were also performed using the PCXMC 2.0 software, based on the same parameters, to enable cross-validation and explore organ dose variation under different anatomical and technical conditions. The experimental results showed higher doses in organs near the primary beam, such as lungs, heart, and esophagus, especially in oblique projections. Overall, PCXMC provided consistent estimates for centrally located organs, though it significantly underestimated doses in peripheral regions such as thyroid and breasts. The agreement between TLD and PCXMC results supports the complementary use of both methods, reinforcing the validity of combining physical dosimetry with computational modeling to improve dose estimation. This hybrid approach contributes to optimizing radiation protection strategies in interventional cardiology, promoting adherence to the ALARA principle and enhancing patient and occupational safety.

Keywords: Dosimetry; Cardiovascular diseases; PCXMC; fluoroscopy; Monte Carlo simulation; Organ dose.

1 Introduction

Cardiovascular diseases (CVDs) stand out as one of the leading causes of death worldwide, affecting millions of people each year. It is estimated that these conditions account for over 30% of global deaths, with an increasing trend over recent decades driven by factors such as poor diet, sedentary lifestyles, and population aging (World Health Organization, 2023; Faria et al., 2023). The obstruction of blood flow in the coronary arteries often results in myocardial infarction, one of the most common manifestations of these diseases, leading to serious complications such as heart failure, arrhythmia, and irreversible damage to the heart (Faria et al., 2023). These conditions impair cardiovascular function, significantly reducing patients' quality of life, limiting their physical capabilities, and requiring continuous medical care. The high morbidity rate associated with CVDs places enormous pressure on healthcare systems, which face a growing demand for specialized treatments (Ferretti et al., 2014; Morais et al., 2011)

In this context, coronary angioplasty emerges as a fundamental therapeutic solution, correcting blockages in the coronary arteries and restoring blood flow to the heart. This treatment is performed using fluoroscopy-based coronary angiography systems, which allow real-time visualization of the arteries, facilitating the removal of obstructions and the placement of stents (Miller et al., 2010; Shah et al., 2015). These systems provide accurate and detailed images essential to ensuring the procedure's effectiveness and minimizing risks. However, despite their high efficacy, coronary angioplasty involves exposure to ionizing radiation, making rigorous evaluation of radiation levels essential to ensure patient safety (Miller et al., 2010; Tunçman et al., 2025).

During these procedures, critical organs, such as the thyroid, lungs, breasts, and liver, are particularly vulnerable, receiving significant radiation doses. Prolonged exposure, especially in repeated or lengthy procedures, increases the risk of long-term adverse effects, such as cancer. In this context, the International Commission on Radiological Protection (ICRP), through its Publication 103, recommends the ALARA principle (As Low As Reasonably Achievable), which aims to minimize radiation dose without compromising treatment quality (ICRP, 2007). Applying this principle is fundamental to reducing health risks and ensuring that radiation exposure remains within safe limits (Sanchez et al., 2015).

To ensure compliance with these limits, thermoluminescent dosimeters (TLDs) have been widely used in dosimetry studies due to their high accuracy and reproducibility. This technique has been adopted for decades and is crucial for measuring the radiation dose absorbed by tissues, serving as an essential tool for validating computational simulations (Barbosa et al., 2023; Kim et al., 2012). TLDs provide empirical data used to validate simulation estimates and monitor exposure during medical procedures, being employed in both clinical and research contexts. Their continuous use and the regular validation of simulations are critical to ensuring radiological safety and improving clinical practices (Tunçman et al., 2025).

The Monte Carlo method, widely used in PCXMC 2.0 software, simulates the interaction of radiation with tissues by creating 3D models that consider parameters such as projection type, focus-to-patient distance, kV, and mA. The use of PCXMC provides a broader view of radiation distribution, which cannot be obtained through TLDs alone. Although PCXMC offers detailed estimates, they must be validated to ensure accuracy. In this context, the experimental data obtained with TLDs play a crucial role in validating the simulations, ensuring that computational estimates are reliable (Fujii et al., 2017; Iriuchijima et al., 2018).

Despite advances in dosimetry techniques, there are still significant gaps in the literature regarding the validation and application of computational simulations in interventional radiology, particularly in the context of coronary angioplasty. Although there are studies using TLDs and simulations, few address the direct comparison between both methods or explore the implications of this combination for improving radiation protection practices (Barbosa et al., 2023; Vañó et al., 2008). Additionally, variability in radiation doses depending on the parameters used during procedures is still not fully understood. This research aims to fill this gap by providing a detailed analysis of dose distribution and a robust comparison between the two methods (Sanchez et al., 2015).

Despite advances in dosimetry techniques, there are still significant gaps in the literature regarding the validation and application of computational simulations in interventional radiology, particularly in the context of coronary angioplasty (Barbosa et al., 2023; Vañó et al., 2008). Although there are studies using TLDs and simulations, few address the direct comparison between both methods or explore the implications of this combination for improving radiation protection practices. Additionally, variability in radiation doses depending on the parameters used during procedures is still not fully understood. This research aims to fill this gap by providing a detailed analysis of dose distribution and a robust comparison between the two methods (Barbosa et al., 2023; Vañó et al., 2008; Sanchez et al., 2015).

The objective of this study is to compare the dose estimates obtained through Monte Carlo simulations using the PCXMC software with the experimental measurements obtained with TLDs during the coronary angioplasty procedure. Through this comparison, the study seeks to validate the simulations and provide reliable data to improve radiation protection practices, minimizing the risks of radiation exposure. The work also aims to explore how combining these two techniques can contribute to advances in clinical practice and patient safety, providing a more accurate and effective dosimetry model.

2 Materials and Methods

2.1 Equipment and Exam Configuration

The coronary angioplasty procedure was simulated in a controlled environment using the GE Innova IGS 320 fluoroscopy system (GE Healthcare), installed in the hemodynamics unit of a large regional reference hospital. This equipment is widely used in diagnostic and therapeutic procedures and complies with all current operational and regulatory standards, undergoing regular quality assurance testing.

2.2 Anthropomorphic Phantom

The Alderson-RANDO anthropomorphic phantom (model ART-200, RSD Phantoms, California, USA) was used widely in dosimetry studies for its ability to simulate the interaction of ionizing radiation with the human body. This model represents an adult male individual, with a height of 175 cm and a weight of 73.5 kg, consistent with a standard patient profile.

The phantom is composed of 34 transverse slices, each 2.5 cm thick, containing uniformly spaced (1.5 to 2.5 cm) cylindrical holes with a diameter of 1.0 mm. These allow for the precise placement of detectors in specific anatomical locations. Internally, it incorporates tissue-equivalent materials such as muscle, bone, lung, and cavities, optimized to simulate radiation attenuation and scattering across different densities and compositions.

Although it does not include a head or upper and lower limbs, its detailed structure enables accurate dosimetric measurements, especially in interventional procedures such as coronary angioplasty. Figure 1 illustrates the phantom model used.

Figure 1: Alderson-RANDO anthropomorphic phantom

2.3 Thermoluminescent Dosimeters (TLDs)

A total of 250 thermoluminescent dosimeters (TLDs) of type TLD-100 were used, manufactured with lithium fluoride crystals doping with magnesium and titanium (LiF:Mg,Ti). As shown in Figure 2, the dosimeters have a flat chip shape with dimensions of 3.2 × 3.2 × 0.89 mm and were placed in the holes of the phantom, covering various anatomical regions.

Figure 2: Numbered TLD dosimeters.

The TLDs underwent standardized thermal treatments. The pre-irradiation annealing was performed in two steps, totaling 4 hours: 1 hour at 400 °C followed by 2 hours at 100 °C. This procedure aims to reestablish the thermodynamic equilibrium of the crystal material and prevent interference from previous exposures.
After exposure, the dosimeters were subjected to a pre-readout annealing at 100 °C for 10 minutes to

eliminate charges from shallow traps, which are unstable at room temperature. This step ensures higher accuracy and reproducibility in reading.

The readout was performed using an automatic reader, the RadPro Freiberg Instruments TLDcube. Based on previous studies and experimental tests, a heating rate of 10 °C/s was adopted, with thermal count recording between 60 °C and 300 °C, the range in which the most stable emission peaks of the material occur. This configuration helps minimize the effects of ambient temperature and the time between irradiation and readout, ensuring greater reliability in the results.

2.4 Simulated Protocol

The simulation was conducted to realistically represent a clinical coronary angioplasty procedure, with a total exposure time of 15 minutes. Three different projections were used, with variations in technical parameters (kV, mA, and time). The projections included: 90°, LAO30, and CAU30. The source-to-skin and source-to-detector distances were adjusted according to tube positioning in each projection, and accumulated dose values were recorded directly on the equipment monitor in terms of air kerma at the reference point (Ka,r, in mGy) and dose-area product (DAP, in Gy·cm^2).

2.5 Monte Carlo Simulation in PCXMC

For comparison purposes, an additional simulation was performed using the PCXMC 2.0 software (STUK, Finland), based on the Monte Carlo method. The three main projections used in the experimental setup (90°, LAO30, and CAU30) were individually simulated, each with its corresponding technical parameters manually entering the software, including kV, mAs, incidence angles, source-to-skin, and source-to-detector distances. This approach preserved the specific characteristics of each projection adopted in the practical simulation.

The phantom selected in PCXMC represented a standard patient (adult male, 175 cm, 73 kg), matching the physical model of the Alderson-RANDO phantom used in the experimental part.

Absorbed organ doses were individually extracted from the reports generated for each projection (in mGy) and later summed by anatomical structure to estimate the total accumulated dose over the course of the procedure. The list of organs considered was the same as those evaluated experimentally using TLDs, enabling a direct comparison between the two methods, as exemplified in Figure 3, which shows a visual representation of one of the simulated projections in PCXMC.

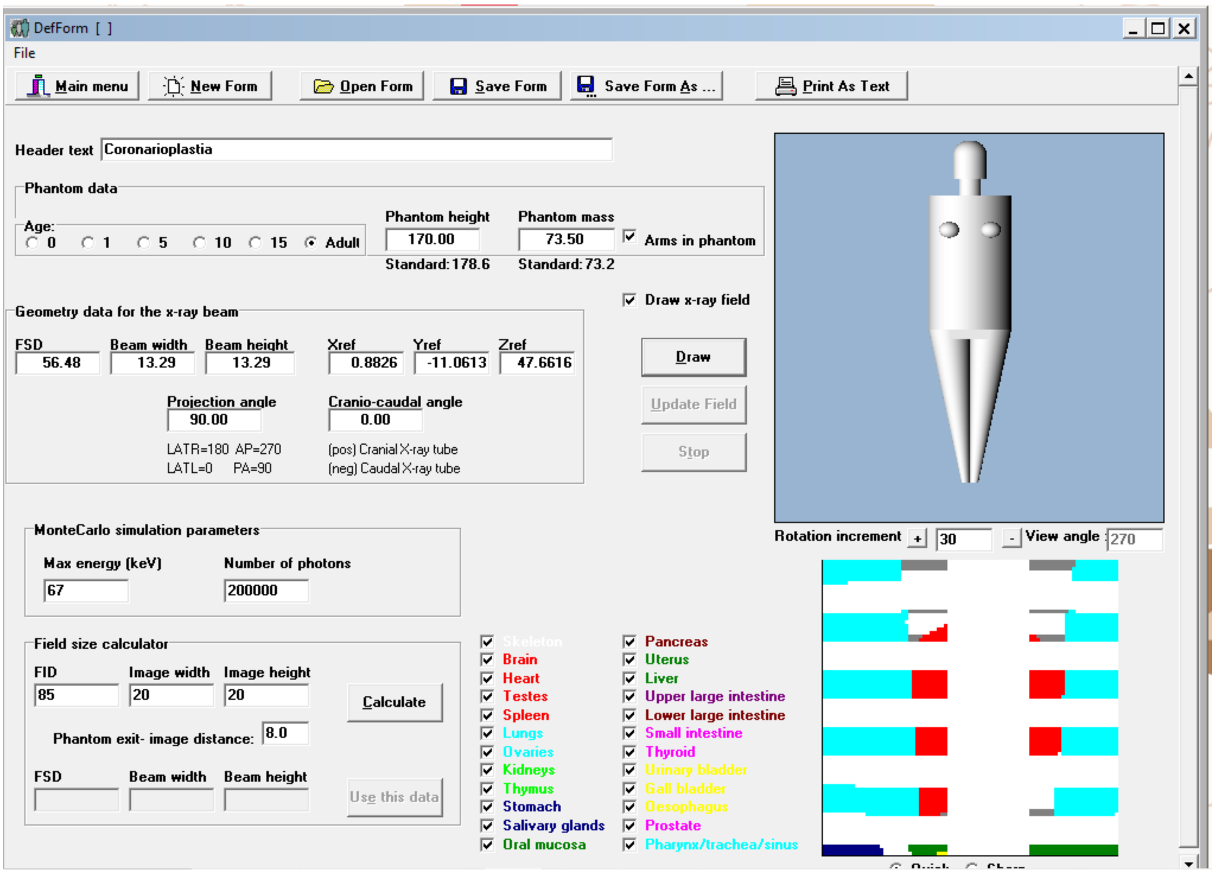


Figure 3: Visual representation of one of the simulated projections in PCXMC

2.6 Effective Dose Calculation and Comparative Analysis

The effective dose was estimated from the mean absorbed doses per organ (in mGy), obtained experimentally using TLDs, based on the tissue weighting factors (W_T) recommended by ICRP Publication 103. The individual contribution of each organ to the effective dose was calculated according to the following equation:

$$E = \sum (D_T x w_T) \quad (1)$$

Where D_T represents the mean absorbed dose in tissue or organ T, and W_T is the corresponding tissue weighting factor

The mean organ doses obtained using TLDs were also compared with those estimated through simulation in PCXMC. For each anatomical structure, the relative percentage difference between the methods was calculated:

$$difference(\%) = \left(\frac{D_{TLD} - D_{PCXMC}}{D_{PCXMC}}\right) x\,100 \quad (2)$$

3 Results

The coronary angioplasty procedure was simulated using an Alderson Rando-type anthropomorphic phantom, with the distribution of 250 TL dosimeters across different anatomical regions. The simulation included multiple C-arm projections to replicate the clinical behavior of the procedure.

Table 1 presents the technical parameters used for each projection during the simulated procedure, including kV, mA, exposure time, as well as the dose values recorded by the equipment monitor in terms of Ka,r (mGy) and DAP (Gy·cm²).

Table 1: Technical parameters by projection during the coronary angioplasty simulation

Projection	**kV**	**mA**	**Duracion (min)**	**Ka,r (mGy)**	**DAP (Gy·cm²)**
90° (initial)	66	9,7	7	79	12,17
LAO30	61	14,7	3	98	15,05
CAU30	71	19,3	3	117	17,70
90° (final)	67	10,5	2	184	27,14
Total	—	—	15	**206**	**30,97**

Table 2: presents the values of mean dose, total dose, and the number of dosimeters allocated per organ

Organ	Number of Records	Mean Dose (mGy)	Dose total (mGy)
Right Lung	61	28.602279	1744.739
Left Lung	60	28.729467	1723.768
Heart	27	19.929407	538.094
Esophagus	4	29.190000	116.760
Stomach	8	5.793500	46.348
Left Breast	4	10.949000	43.796
Liver	10	4.097200	40.972
Right Breast	4	9.476000	37.904
Outside Left Breast	1	20.744000	20.744
Outside Right Breast	1	15.035000	15.035
Left Kidney	8	1.737875	13.903
Right Kidney	8	1.247625	9.981
Right Thyroid	2	4.831000	9.662
Left Thyroid	2	4.588500	9.177
Central Thyroid	1	6.736000	6.736
Pancreas	4	1.607000	6.428
Central Thyroid	1	6.322000	6.322
Spleen	2	2.834500	5.669

Outside Thorax	1	5.263000	5.263
Outside Thyroid	1	1.536000	1.536
Right Ovary	4	0.349000	1.396
Right Bone Marrow	7	0.179714	1.258
Outside Abdomen	1	1.097000	1.097
Left Bone Marrow	7	0.140571	0.984
Bladder	6	0.115333	0.692
Left Ovary	4	0.165750	0.663
Testicle	8	0.072875	0.583
Prostate	2	0.027000	0.054

Figure 4 shows the distribution of mean absorbed doses per organ, obtained from the TLD readings, organized in decreasing horizontal scale.

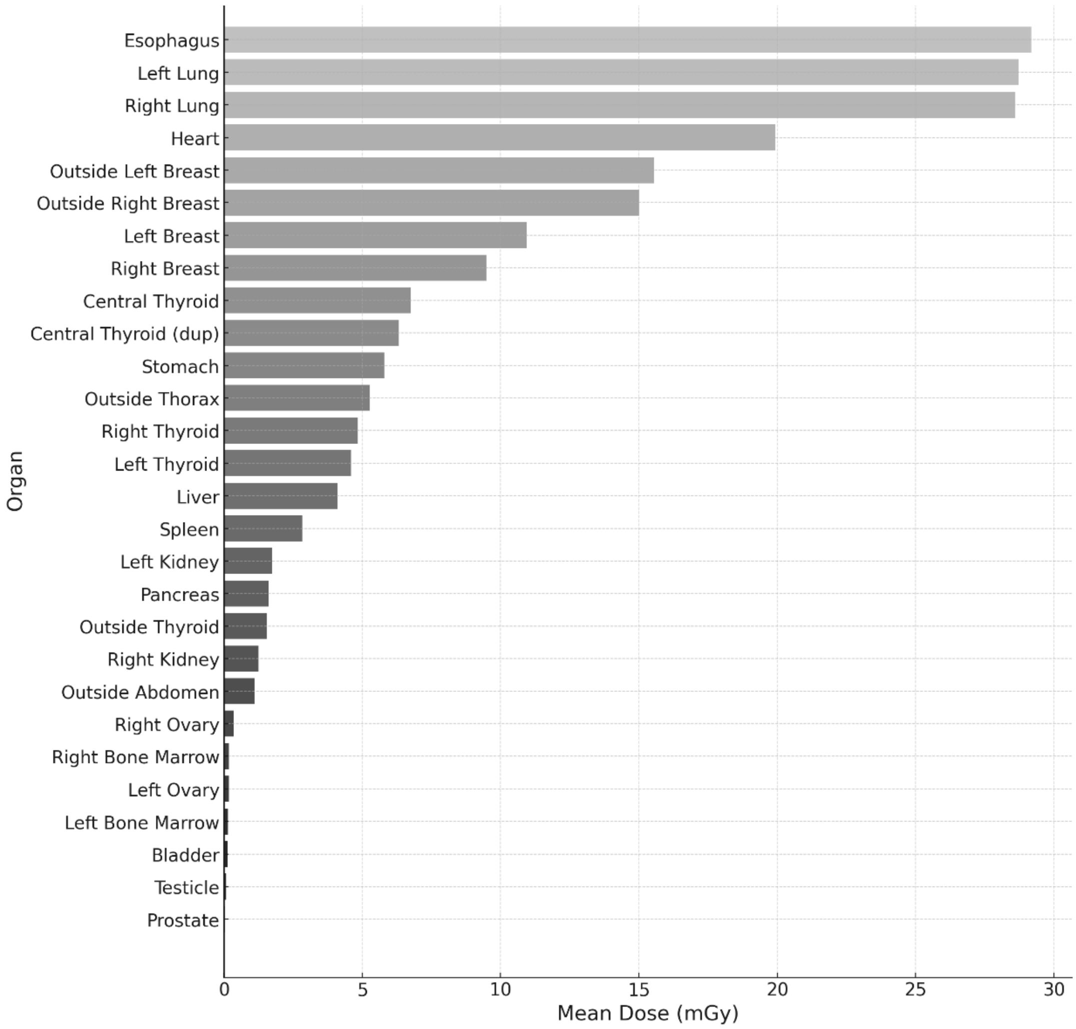


Figure 4: Distribution of Mean Dose (mGy) by Organ

Table 3 presents the individual contribution of the main organs to the effective dose, based on the mean doses measured experimentally and the tissue weighting factors established by ICRP Publication 103. The values are ordered according to their relative contribution to the total effective dose.

Table 3: Estimated Contribution to the Effective Dose

Organ	Mean Absorbed Dose (mGy)	Tissue Weighting Factor (wT)	Contribution (mSv)
Esophagus	29.19	0.04	0.00117
Lungs	28.67	0.12	0.00344
Heart	19.93	0.12	0.00239
Breast	10.21	0.12	0.00123
Stomach	5.79	0.12	0.00070
Thyroid	5.32	0.04	0.00021
Liver	4.10	0.04	0.00016
Spleen	2.83	0.01	0.00003
Pancreas	1.61	0.01	0.00002
Kidney	1.49	0.05	0.00007
Ovary	0.26	0.08	0.00002
Bone Marrow	0.16	0.12	0.00002
Testicle	0.07	0.08	0.00001

Prostate	0.03	0.01	<0.000001

Table 4 presents a comparison between the organ-absorbed doses obtained experimentally using TLDs and those estimated by Monte Carlo simulation in the PCXMC software, considering the sum of the three simulated projections (90°, LAO30, and CRA30).

Table 4: Percentage Difference Between Organ Doses Estimated by TLD and PCXMC

Organ	PCXMC (mGy)	TLD Mean (mGy)	Difference (%)
Esophagus	25.44	29.19	+14.74%
Lungs	28.23	28.67	+1.55%
Heart	18.53	19.93	+7.55%
Breast	4.02	10.21	+154.00%
Stomach	14.05	16.62	+18.30%
Thyroid	1.41	5.32	+277.30%
Liver	22.25	24.92	+12.00%
Spleen	2.59	5.10	+96.90%
Pancreas	8.35	11.49	+37.60%
Kidney	7.59	7.59	0%
Ovary	2.06	2.72	+32.03%
Bone Marrow	19.62	21.85	+11.36%

Testicle	0.46	0.90	+95.65%
Prostate	4.88	5.53	+13.31%

4 Discussion

In this study, 250 TLDs were positioned in the Alderson-Rando anthropomorphic phantom with the aim of experimentally quantifying the distribution of radiation in different organs during a simulated coronary angioplasty procedure. The projections used (90°, LAO30, CAU30) reflect those most applied in clinical practice, particularly for optimal visualization of coronary bifurcations. Kočka et al. (2020) reported that angulations such as LAO ~37° and CAU up to ~49° are frequently employed in coronary procedures, reinforcing the clinical relevance of the simulation conducted in this study.

The highest mean doses were observed in the lungs (28.7 mGy), esophagus (29.2 mGy), and heart (≈19.9 mGy), reflecting the proximity of these structures to the primary radiation field and areas of greatest exposure during fluoroscopy. In contrast, more distant organs, such as the bladder, testicles, and prostate, received significantly lower doses, as expected due to their anatomical position and distance from the primary area of interest in the procedure.

The accumulated dose at the reference point (Ka,r) of 206 mGy recorded during the simulation was lower than the overall typical value (1112 mGy) and the third quartile (1706 mGy), previously established in clinical surveys conducted by our group, which analyzed the influence of body mass index (BMI) on the definition of reference levels. However, this value is consistent with the typical dose observed for patients with normal BMI (≈268 mGy), reinforcing the adequacy of the anthropomorphic phantom used in terms of both anatomical structure and body composition.

The comparison between the experimental (TLDs) and computational (PCXMC) methods showed considerable differences in organ-absorbed dose estimates. Overall, PCXMC underestimated the doses compared to the TLD measurements. The greatest discrepancies were found in the thyroid (+277%), breasts (+154%), and testicles (+95.7%). These differences may be attributed to geometric simplifications in the mathematical models used by PCXMC, which do not accurately represent the morphology and anatomical distribution of peripheral tissues. Additionally, the fixed beam positioning and lack of real-time variation in the simulation limit the accurate reproduction of secondary radiation scatter.

Vañó et al. (2008) demonstrated that although PCXMC is effective for estimating effective dose and doses to central organs, simulations tend to show larger deviations in regions with complex geometry or where radiation reaches predominantly through scatter. These findings are in line with the results of the present study, highlighting the role of TLDs as the gold standard for validating computational models.

On the other hand, the agreement observed for dose values in central organs, such as lungs, heart, and liver (with differences below 15%) confirms that PCXMC is a valid tool for initial dose estimates in regions of high direct exposure. However, when greater accuracy is required, particularly in regulatory contexts, clinical research, or estimates involving radiosensitive tissues, calibration of simulated models based on experimental data is strongly recommended.

Thus, the integration of computational simulations and physical measurements offers a robust and complementary approach for dose assessment in interventional radiology. TLDs provide empirical measurements that capture the nuances of radiation interaction with the human body, while PCXMC enables quick and practical estimates, especially in clinical scenarios where direct dosimetry is not feasible.

5 Conclusions

The coronary angioplasty simulation using TLDs and PCXMC revealed relevant differences between the methods, with PCXMC underestimating doses in peripheral organs. The anthropomorphic phantom proved to be compatible with normal BMI patients, validating its use. The combination of experimental dosimetry and computational simulation proved effective for assessing and optimizing radiation exposure in interventional radiology.

References

AAPM. (2016). Dosimetry in interventional radiology: A comprehensive review. Journal of Applied Clinical Medical Physics, 17(3), 5926. https://doi.org/10.1120/jacmp.v17i3.5926

Barbosa, M. H., Moreira, T. M., Tavares, J. L., Andrade, É. V. de, Bitencourt, M. N., de Freitas, K. B. C., & Cardoso, G. L. (2013). Complicações em pacientes submetidos à Angioplastia Coronariana Transluminal Percutânea. Enfermería Global, 31(3), 24–33.

Faria D., Hennessey B., Shabbir A., Mejía-Rentería H., Wang L., Lee J. M., Matsuo H., Biscaglia S., Koo B. K., Xu B., Baptista S., Gonzalo N., and Escaned J. (2023). Functional coronary angiography for the assessment of the epicardial vessels and microcirculation. EuroIntervention, 19(3): 203–221. https://doi.org/10.4244/EIJ-D-22-00969

Fujii K., Nomura K., Muramatsu Y., Obara S., Akahane K., and Kusumoto M. (2017). Organ dose evaluations based on Monte Carlo simulation for CT examinations using tube current modulation. Radiation Protection Dosimetry, 174(3): 387–394. https://doi.org/10.1093/rpd/ncw144

ICRP. (2007). The 2007 Recommendations of the International Commission on Radiological Protection. ICRP Publication 103. Ann. ICRP, 37(2–4).

Iriuchijima A., Fukushima Y., and Ogura A. (2018). Comparison of organ dose calculation using Monte Carlo simulation and in-phantom dosimetry in CT examination. Nippon Hoshasen Gijutsu Gakkai Zasshi, 74(2): 166–171. https://doi.org/10.6009/jjrt.2018_JSRT_74.2.166

Kočka, V., et al. (2020). Optimizing fluoroscopic projections in coronary angiography: minimizing radiation while preserving image quality. European Heart Journal, 41(3), 270-278.

Miller, D. L., et al. (2010). Clinical radiation management for fluoroscopically guided interventional procedures. Radiology, 257(2), 321–332. https://doi.org/10.1148/radiol.10091269

Nogueira, M. S., et al. (2023). Organ dose estimation in coronary angiography using TLDs in anthropomorphic phantom: validation and comparison with PCXMC simulation. Radiation Physics and Chemistry, 212, 110084. https://doi.org/10.1016/j.radphyschem.2023.110084

Sánchez R. M., Vano E., Fernandez J. M., and Escanedo J. (2015). Avaliação de um display em tempo real para mapa de dose cutânea em procedimentos de cateterismo cardíaco. Radiation Protection Dosimetry, 165(1–4): 240–243. https://doi.org/10.1093/rpd/ncv033

Shah A., Das P., Subkovas E., Buch A. N., Rees M., and Bellamy C. (2015) Radiation Dose During Coronary Angiogram: Relation to Body Mass Index. Heart Lung and Circulation 24(1): 21–25.

Smith-Bindman, R., et al. (2012). Radiation dose associated with common computed tomography examinations and the associated lifetime attributable risk of cancer. Archives of Internal Medicine, 169(22), 2078–2086. https://doi.org/10.1001/archinternmed.2009.427

Vañó, E., Fernández, J. M., & Ten, J. I. (2008). Real-time measurement and estimation of radiation doses in interventional cardiology. Radiation Protection Dosimetry, 131(3), 374–377. https://doi.org/10.1093/rpd/ncn046

Tunçman Kayaokay D., Kesmezacar F. F., Yeyin N., Çavdar Karaçam S., Günay O., Karabağ T., Demir M., Issa B., Tekin H. O., and ALMisned G. (2025). Comprehensive organ-specific radiation dose mapping in coronary angiography using the Alderson Rando phantom: An experimental approach to optimize patient safety. Radiation Physics and Chemistry, 233, 112739. https://doi.org/10.1016/j.radphyschem.2025.112739

WHO. (2024). Cardiovascular diseases (CVDs): Key facts. World Health Organization. https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)

World Heart Federation. (2023). Deaths from cardiovascular disease surged 60% globally over the last 30 years. https://world-heart-federation.org/news/deaths-from-cardiovascular-disease-surged-60-globally-over-the-last-30-years-report

Reducción del impacto radiológico de los residuos nucleares

A. M. Campos[1,2], T. P. R. Campos[2,3]*

[1] Universitat Augsburg, Department of Mathematics, D-86135 Augsburg, Germany.

[2] Universidade Federal de Minas Gerais, 30270901, Belo Horizonte, Brazil.

[3] Centro de Desenvolvimento da Tecnologia Nuclear, 30270901, Belo Horizonte, Brazil.

* tprcampos@yahoo.com.br

Abstracto

El objetivo principal de este artículo es evaluar la viabilidad de disminuir el impacto radiológico de los radioisótopos de larga vida presentes en los residuos nucleares, sometiéndolos a reacciones de fisión o captura de neutrones dirigidas dentro de un espectro de energía definido. El estudio emplea modelos matemáticos y simulaciones computacionales para investigar las velocidades de reacción de radionúclidos de larga vida dentro de un espectro de neutrones especialmente diseñado, utilizando un selector espectral resonante (RSS). En condiciones operativas controladas de un RSS de baja energía, el análisis incluye las tasas de transmutación de radionúclidos y la correspondiente reducción de masa. Se cuantifica el agotamiento normalizado de los isótopos radiactivos durante los períodos de funcionamiento en el manejo de actínidos de larga vida.

Palabras clave: Selector espectral resonante; actinios de vida media larga; secciones transversales 1/v; reacciones de fisión; reacción de captura radiactiva;

1. Introducción

Según el Organismo Internacional de Energía Atómica (OIEA, 2009), los residuos nucleares se clasifican de acuerdo con su peligrosidad potencial, nivel de radiactividad y vida media. Los residuos de alta actividad, que comprenden principalmente el combustible nuclear gastado y los subproductos de la reprocesamiento, presentan una elevada radiactividad y contienen radioisótopos de larga vida. Dentro de esta categoría, los residuos transuránicos (TRU) abarcan materiales contaminados con elementos como el plutonio y el americio. En general, los residuos de alta actividad (RAA) contienen actínidos como uranio, neptunio, americio y curio, responsables de su elevada radiotoxicidad a largo plazo (OIEA, 2009).

Las prácticas actuales de tratamiento de desechos, documentadas por el OIEA, se basan en el almacenamiento temporal seguido de la eliminación geológica permanente. (TECDOC, OIEA, 2016) Aunque se ha propuesto la transmutación en reactores de alto flujo, presenta riesgos debido a la contaminación del reactor y a las limitaciones operativas. (Sun et al., 2022)

Los reactores térmicos y rápidos generan residuos que se acumulan a lo largo de múltiples ciclos de combustible, almacenados inicialmente en piscinas de agua y posteriormente en contenedores dentro de la planta de la industria nuclear. Incidentes como “Three Mile Island”, Chernóbil y Fukushima han subrayado las preocupaciones de seguridad a largo plazo del combustible gastado, que sigue siendo peligroso durante décadas o milenios. (Marguet, 2022) La composición de RAA varía según el diseño del reactor, el tipo de combustible, el enriquecimiento, el quemado y el período de enfriamiento. Los actínidos y las TRU, formados durante el funcionamiento del reactor, son particularmente desafiantes debido a su vida media prolongada y su impacto ambiental. Su eliminación segura es una de las principales preocupaciones de la industria nuclear.

Este estudio propone una nueva estrategia de tratamiento utilizando un selector espectral resonante (RSS), que modula la velocidad y la dirección de los neutrones para crear un espectro selectivo. Esto permite la transmutación dirigida de radionúclidos de larga vida con una contaminación mínima, utilizando equipos compactos y eficientes.

Campos y Campos (M.C. Alberto, P.R.C. Tarcisio, 2023 y 2025) introdujeron el marco teórico del comportamiento de los neutrones en sistemas de dispersión giratoria. Su modelo demostró la formación de espectros de neutrones direccionales alrededor de un cilindro giratorio, promoviendo la captura selectiva de neutrones por isótopos en la periferia radial. El diseño RSS emplea un espectro de energía de neutrones modulado adaptado a las secciones transversales de absorción de isótopos específicos de RAA. Las energías de neutrones fríos (1-25 meV) y los picos de resonancia permiten mejorar las reacciones (n,γ) y (n,f) en los isótopos objetivo, al tiempo que minimizan las reacciones en materiales no objetivo.

Las configuraciones RSS, MB o MR, integran pre moderadores y moduladores giratorios para dar forma al espectro de neutrones a través de la dispersión controlada. El sistema consta de componentes cilíndricos concéntricos: una estructura de soporte, un núcleo central, un reflector periférico, anillos de modulación giratorios (cilindros huecos) y un soporte de objetivos. Los subsistemas adicionales incluyen unidades de calefacción/refrigeración, fuente de neutrones, motores magnéticos de alta velocidad, sensores y controles de presión. El RSS permite la transmutación selectiva o fisión de radionúclidos, convirtiéndolos por reacción de captura de neutrones o por fisión, respectivamente. Esta técnica se dirige a nucleídos con secciones transversales significativas en regiones de neutrones fríos o regiones de resonancia (10^{-3} a 103 eV), lo que reduce las absorciones no deseadas.

Este artículo tiene como objetivo demostrar específicamente la viabilidad teórica de la descontaminación de radionúclidos de larga duración basados en actínidos mediante reacciones optimizadas de fisión y captura radiactiva en los dominios de energía neutrónica fría, 1 meV hasta 25 meV.

2 Materiales y métodos

Las configuraciones de RSS, basadas en la disposición específica de sus módulos giratorios y estáticos, junto con los compartimentos blancos, se pueden aplicar en procesos de transmutación radiactiva o fisión inducida. Los núclidos contenidos en los soportes de materiales blanco se bombardean en el RSS, que utiliza un espectro modulado derivado de una fuente de neutrones primarios. Este espectro está diseñado para lograr una distribución normal $D_n(E_m,\varepsilon)$ con una media E_m y una desviación estándar ε, lo más cercana posible a la energía de resonancia resuelta específica E_o, o en la región de baja energía, donde se maximizan las tasas de captura de neutrones o fisión.

La configuración de RSS se presenta en la literatura (M.C. Alberto, P.R.C. Tarcisio, 2023 y 2025[a,b]). Está sujeto a una patente internacional, cuya descripción completa se encuentra en WIPO PCT/BR2023/050225 (2023), ya divulgada con detalles geométricos y operativos y dibujos. (PCT/BR2023/050225, 2023).

El sistema RSS comprende varias configuraciones definidas por la composición del material, el diámetro y la velocidad de rotación. El Moderador Primario (PM) está hecho de polietileno deuterado con un diámetro de 0,075 metros y permanece estacionario. El pre-moderador de baja energía (PMB), construido a partir de un compuesto de PTFE-grafito, tiene un espesor de 0,05 metros y también es estacionario. En la configuración RSS-MR, el modulador resonante (MR) está compuesto de grafito, con un diámetro externo de 0,40 metros y funciona a una velocidad de rotación de 5.10^5 rpm. En la configuración RSS-MB, el modulador de baja energía (MB) también está hecho de grafito, tiene un diámetro externo de 0,20 metros y gira a 1.10^5 rpm.

El compartimento dos materiales blancos (CP) está construido en acero inoxidable, con un diámetro de 0,015 metros, y permanece estacionario. El reflector (RF) también es estacionario, hecho de PTFE-grafito, y tiene un diámetro de 0,10 metros. La fuente de neutrones (FT) está configurada para diferentes reacciones nucleares o de fusión. Esta fuente tiene un diámetro de 0,012 metros y es estacionaria.

Las distintas reacciones nucleón-nucleón generan fuentes de neutrones, a saber, fuentes f_1, f_2 y f_3. En primer lugar, la fuente f_1 representa las reacciones de fusión de D-D y la f_2 por D-T, donde los objetivos son D y T en Ti. Un acelerador de deuterio para reacciones (d,n) proporciona los iones incidentes D^+ (o D^-). Además, la fuente f_3 representa H+ (o H^-) para (p,n), proporcionada por un acelerador de iones con suficiente energía para bombardear materiales de elementos químicos ligeros como Li, Be o B, ubicados en el objetivo central en PM. La reacción Li^7(p,n) fue la elección para la fuente f_3. Los iones se mueven directamente hacia el tubo en el centro de RSS, golpeando su objetivo metálico TiD ou TiT. Los neutrones de las reacciones (*,n) tienen espectros distintos, específicos para cada reacción nucleón-nucleón.

El análisis de la operación RSS, basado en la física y las matemáticas, conduce a un comportamiento asintótico donde el producto escalar vectorial $\mathbf{v}_a.\mathbf{v}_n \rightarrow 1$ y la relación $E_n/E_a \rightarrow 1$ como modulador, con velocidad angular ω_n y radio r, gira. En este contexto, v_a denota la velocidad lineal del medio en rotación, v_n representa la velocidad lineal de la partícula que colisiona con el medio, E_n es la energía cinética de la partícula y E_a es la energía cinética del medio, como se mostró anteriormente en (Campos A.M. et al, 2023).

Las simulaciones de transporte de neutrones demostraron el comportamiento colectivo de las partículas en la salida de los moderadores y moduladores dinámicos. En estas simulaciones se emplearon bibliotecas de datos nucleares: JEFF-3.3, EAF-2000, TENDL-2019, ENDFB-V/VIII.0 y los códigos nucleares estocásticos MCNP6 (MCNP, 2013; Durkee et al, 2016)

Además, las simulaciones se acoplan a subrutinas desarrolladas por los autores, considerando evaluaciones probabilísticas y estadísticas, así como fenómenos físicos de dispersión nuclear e interacciones de absorción que ocurren dentro de los materiales en el RSS. Las rutinas tratan las colisiones bidimensionales y tridimensionales de dos cuerpos con densidad atómica y masas del núcleo distinto de los moduladores, incluidas las velocidades iniciales incidentes y angulares. Las simulaciones sostuvieron el tratamiento multigrupo sosteniendo discretización espacial, angular y de grupos de energía.

En nuestros cálculos, los datos de entrada fueron datos de parámetros de resonancia de radioisótopos de nucleídos precursores, datos de sección transversal nuclear y sección transversal de fuente de interacciones nucleón-nucleón. Se realizaron evaluaciones previas sobre la generación de neutrones en fuentes f_1, f_2 y f_3 , así como sobre la difusión de neutrones en moderador estático soportado por el código MCNP (MCNP, 2013; Durkee et al, 2016). El cálculo proporcionó la corriente J en la superficie de entrada de los moduladores dinámicos. Más tarde, las rutinas realizaron análisis de dispersión en colisiones bidimensionales y tridimensionales en los átomos de los moduladores. El balance de energía y momento en colisiones validó cada interacción en los datos de salida. Además, se produjo la conversión de las distribuciones de partículas de neutrones en la superficie de salida de los moduladores a las distribuciones de neutrones normales equivalentes $D_n(E_m,\varepsilon)$. Estos datos fueron útiles para las evaluaciones de la velocidad de reacción.

La discretización espacial radial realizó hasta 10^2 grupos, y lo mismo en distribuciones de energía y angulares. Las simulaciones en MCNP ejecutan 10^8 partículas que generan J^{+}_{en} corriente de entrada; mientras que las rutinas internas se realizan con 3000 partículas, lo que permite 800 colisiones por partícula, en cada uno de los 10^2 grupos radial, energético y angular.

Las simulaciones también consideraron la vibración térmica y las variaciones de temperatura del medio. Los datos del MCNP tuvieron en cuenta el enfoque S(α,β) en los datos transversales (MCNP, 2013). En las rutinas internas, la energía térmica convertida en vectores aleatorios se sumó a los movimientos atómicos dos materiales en los moduladores dinámicos. La física y las matemáticas adicionales involucradas en el modelado se pueden encontrar en la literatura (Duderstadt y Hamilton, 1976)

La integral de la velocidad de reacción IRR en el dominio del espectro de energía de la distribución de salida de neutrones $D_n(E_m,\varepsilon)$ y la sección transversal microscópica de la captura radiactiva (n,γ) en la región de resonancia con energía E_o tuvo la siguiente expresión:

$$\mathrm{IRR} = \int_{\mathrm{Em}-2\varepsilon}^{\mathrm{Em}+2\varepsilon} dE \;\; S\gamma\,(E)\;\mathrm{PDF}\,(D_n(E_m,\varepsilon),E). \tag{1}$$

en el que $E_m > 2\varepsilon$. La fórmula de Wigner expresa las secciones transversales resonantes de captura microscópica en el dominio de la energía de una resonancia, de la siguiente manera:

$$S\gamma = (g\frac{\Gamma_{\mathrm{n}}}{\Gamma_T}(2.608 * 10^6/E_o)\;\;((A+1)/A)^2)\frac{\Gamma_\gamma}{\Gamma_T}\sqrt{E_o/E}\,(1+(2\,(E-E_o)/\Gamma_T)^2 \tag{2}$$

En este caso, la energía de resonancia es E_o, A es la masa atómica del núclido, los parámetros g, Γn, $\Gamma\gamma$ $\Gamma_{,\ T}$ y E_o son parámetros de resonancia que se encuentran en la literatura de datos nucleares. (Mughabghab, S.F, 2018; Landolt-Börnstein, 2009)

Por otro lado, la integral de la velocidad de reacción IRC en el dominio del espectro de la distribución de *neutrones* $D_n(E_m,\ \varepsilon)$ multiplicada por la sección transversal microscópica de la captura radiactiva (n γ), en la región de baja energía con un comportamiento 1/v expresa:

$$\mathrm{IRC} = S\gamma\int_{\mathrm{Em}-2\varepsilon}^{\mathrm{Em}+2\varepsilon} dE\;\;\mathrm{PDF}\,(D_n(E_m\,,\varepsilon)\,,E\,)\,.\sqrt{1/E} \qquad , \tag{4}$$

donde la sección transversal microscópica en la región espectral E_T de los neutrones térmicos proporcionó:

$$S\gamma = So\sqrt{E_m\,/E_T} \qquad , \tag{5}$$

en la región del espectro frío y XS con comportamiento 1/v, en el que E_m es la energía de referencia de la sección transversal S_o tomada, y E_T es la energía en temperatura real en el experimento. El límite inferior de integración en las ecuaciones 1 y 4 debe ser mayor que cero, o cero.

En un escenario donde la distribución de neutrones sigue una distribución normal equivalente, la evaluación de la integral considera la función de densidad de probabilidad (PDF) de la función de distribución normal equivalente. Tenga en cuenta que la notación PDF representa la normalización de la distribución $D(E_m,\ \varepsilon)$, proporcionando una distribución normal equivalente a la distribución normalizada.

Hay muchas formas de quemar nucleídos RAA en RSS. Una propuesta simple en este estudio fue exponer nucleídos RAA a un espectro de neutrones fríos evaluado de 1 a 25 meV. Las velocidades de reacción se consideraron XS (n, γ) más (n, f) en este dominio de energía. Se investigó la masa de los radionúclidos y sus más adicionales en las transformaciones $X_a(n,\gamma)X_b$ en este dominio de energía. Los parámetros de resonancia Γ_f, Γ_g, Γ_T, $g\Gamma_n$, están identificados en la literatura. Esos valores se tomaron en bibliotecas de datos nucleares: JEFF-3.3, EAF-2000, TENDL-2019, ENDFB-V/VIII.0, así como en Atlas de resonancia neutrónica (Mughabghab, S.F, 2018; Landolt-Börnstein, 2009).

3 Resultados

Figura 1: Valores de la velocidad de reacción para radionúclidos de actínidos de RAA en exposición al espectro de neutrones de la configuración RSS MB.

La Figura 1 muestra la velocidad de reacción de los radionúclidos actínidos, tomada de la sección transversal máxima S_{fo} de (n,f) y S_{go} (n,γ) en el neutrón frío, en las energías de 1±0,5 meV y 25±1 meV como referencia, multiplicada por integrales IRC a la energía de referencia. El protocolo frío "C" para ejecutar RSS en el modulador MB fue una elección basada en el S_gC1 y el S_gC25. Los valores de S_gC25 se evaluaron aplicando el factor de corrección sobre S_gC1.$1/\sqrt{25}$

La Figura 2 presenta el agotamiento de las masas de los 24 radionúclidos actínidos en función del tiempo de operación del RSS en el espectro frío. Se tomó como referencia el valor de energía de 1±0,5 meV. En un caso realista, se debe aplicar un factor de corrección al valor exacto de la energía media de E_m. Las inclinaciones de las curvas exponenciales dependen de la constante de remoción del núclido "a", es decir, que están relacionadas con la intensidad de neutrones de la fuente, tomada a partir da constante de remisión λ_r^a y S_n de 10^9 n..s^{-1}. Un valor más alto de emisión puede generar un rápido agotamiento de la masa, mientras que una temperatura más alta entre 1 y 25 meV reduce el agotamiento de la masa.S_n.

La masa de los residuos de radionúclidos de larga vida puede reducirse en el funcionamiento del RSS teniendo en cuenta los datos de entrada especificados. En este cálculo se incluyó el agotamiento de la masa aditiva por la auto transmutación de algunos RAA en otros radioisótopos de RAA. Algunos radionúclidos como U-236, U-238, U-234, Ra-226, son más persistentes en la reducción de masa en la exposición de neutrones fríos. Sin embargo, 34 de los 39 radionúclidos estudiados redujeron su masa a valores inferiores al 1% al final del tiempo de operación. La masa incremental de la auto transmutación se produce a 10 radionúclidos; sin embargo, el crecimiento en masa se limitó al 1% para Am-242, Np-236, U-235, Pu-241, Cm-247, Cm-243, Cm-245. Solo el nucleído Cf-251 alcanzó el 10% de la masa original en el 10% del tiempo de operación, siendo consumido posteriormente.

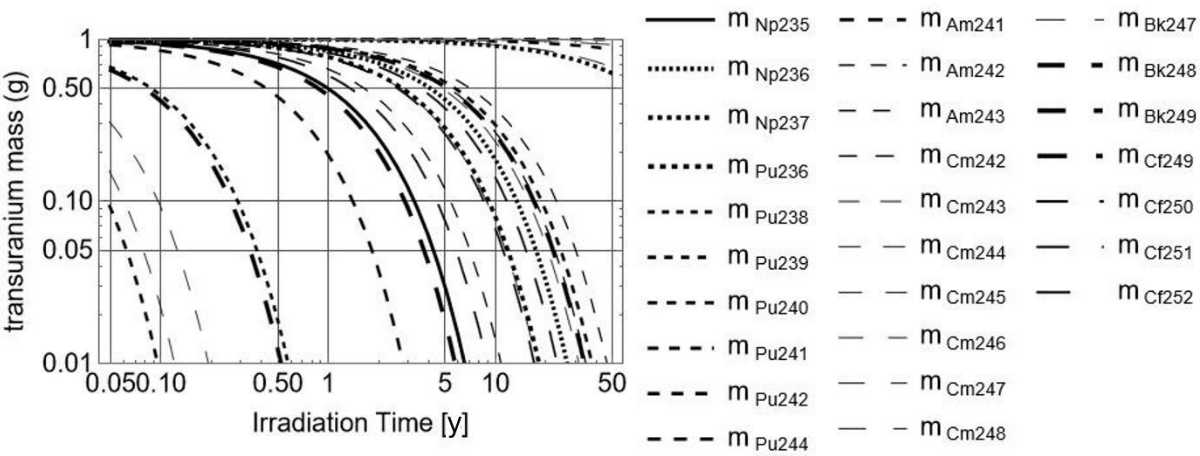


Figura 2**:** Reducción de masa del actínido RAA por fisión y captura radiactiva en el dominio de energía fría.

La principal vía de combustión fue la fisión de los radioisótopos RAA. Las reacciones de fisión impulsan el crecimiento de los productos de fisión también con radioisótopos de larga vida, como I-129, Tc-99, por ejemplo. Sin embargo, estos productos tienen una vida media inferior a la de los núclidos progenitores y se producen en una masa mucho menor. Otro protocolo puede ser la propuesta de quemar dichos productos de fisión en un futuro próximo.

Durante el tiempo de operación del RSS, algunos núclidos como U-238, U-235, U-233 se pueden separar para ser utilizados en otros procesos nucleares. Además, Cf-252 y Am-241 se pueden utilizar como fuentes de neutrones. El proceso de combustión libera calor por fisión y reacciones de captura radiactiva que puede ser recuperado por el intercambiador de calor y utilizado para calefacción o producción de electricidad, reduciendo los costos de consumo eléctrico en la máquina RSS.

Los moduladores resonantes pueden diseñarse basándose en la expresión E_m igual a $K_f.(w_n. R)^2$, en el que w_n es la velocidad angular y R es el radio externo. En un futuro próximo, se puede proponer un protocolo resonante para la descontaminación de los radionúclidos Th-230, Ra-226, Th-229, U-236, U-238, Pu-244 y Pu-242 con actividad más persistente.

4 Conclusión

Existe una posibilidad teórica de descontaminar radioisótopos de vida media larga mediante su transmutación por reacciones de fisión o de captura radiactiva de neutrones, en un dominio energético de neutrones selectivo, concretamente en la región fría de 1 a 25 meV, donde la sección eficaz sigue una dependencia 1/v.

Agradecimientos

Esta investigación desarrollada en 2022 fue un trabajo independiente sin subvención. Los autores reconocen CNPq – Conselho Nacional de Desenvolvimento Científico, CAPES – Coordenação de Aperfeiçoamento de Pessoal de Nível Superior, CDTN – Centro de Desenvolvimento da Tecnologia Nuclear, INCT. INAIS – Instituto Nacional de Ciencia y Tecnología, UFMG – Universidad Federal de Minas Gerais, y Departamento de Matemáticas en la Universidad de Augsburgo, Alemania, y Departamento de Matemáticas de Minas Gerais, Brasil.

Referencias

Alberto M.C., Tarcisio P.R.C. (2025) Feasibility of radioisotope production by neutron spectral selection: Mathematical and computational approaches, *Applied Radiation and Isotopes*, 220:111782, ISSN 0969-8043, 10.1016/j.apradiso.2025.111782.

Campos, A M ; Campos, T.P.R. (2025) Feasibility of a Novel Nuclear Core for ^{232}Th Cycle. *Nuclear Science and Engineering*, v. 199, p. 1-20. 10.1080/00295639.2025.2508556

Campos, A.M., Campos, T.P.R. (2023), Random walk in a rotational environment, in arXiv:2312.02727 [math.PR], https://doi.org/10.48550/arXiv.2312.02727.

Duderstadt J.J., Hamilton L.J. (1976), *Nuclear Reactor Analysis*, John Wiley Sons, Inc, Michigan.

Durkee J.W., Russel C.J., Lauries S.W. (2016), MCNP 6 moving Objects part I: Theory, *Prog. Nucl. Energy*.87:104-121

Durkee J.W., Russel C.J., Lauries S.W. (2016), MCNP 6 moving objects. Part II: Simulations. *Prog. Nucl. Energy*. 87: 122–143

EAF-2000 (2000) Neutron-induced cross section library, The European Activation File: EAF-99-JEFF-3 activation file.

ENDF-B/VIII.0 (2003) Evaluated Nuclear Data Library National Nuclear Data, https://www.nndc.bnl.gov/endf-b8.0/index.html, Accessed: may, 2003.

IAEA (2009), No. NW-T-2.20, Disposal approaches for long lived low and intermediate level radioactive waste, in *Nuclear Energy Series*, Vienna Internacional Center, 1400, Austria. http://www.iaea.org/Publications/index.html.

JEFF-3.3; Joint Evaluated Fission and Fusion File (JEFF), NEA Data Bank, https://www.oecd-nea.org/dbdata/jeff/jeff33/index.html, Accessed 2023.

Koning A.J., Rochman D., Sublet J., Dzysiuk, N., Fleming M. and Van der Marck S. (2019), TENDL: Complete Nuclear Data Library for Innovative Nuclear Science and Technology, in *Nuclear Data Sheets* 155, TALYS-based evaluated nuclear data library, TENDL-2019.

Landolt-Börnstein (2009), Numerical Data and Functional Relationships in Science and Technology, New Series / Editor in Chief: W. Martienssen. Group I: Elementary Particles, Nuclei and Atoms. V.24. Neutron Resonance Parameters, S.I. Sukhoruchkin, Z.N. Soroko and F. Gunsing, V.G. Pronyaev, Edited by H. Schopper.

Marguet S. (2022*), A Brief History of Nuclear Reactor Accidents, From Leipzig to Fukushima,* Springer Praxis Books (PRAXIS), Springer Cham, ISBN 978-3-031-10499-2 , https://doi.org/10.1007/978-3-031-10500-5

MCNP. (2013) *Monte Carlo N–Particle Transport Code System* MCNP 6.1. RSICC Code Package CCC-810, Los Alamos national Laboratory.

Mughabghab, S.F (2018*), Atlas of Neutron Resonances, Resonances Parameters and Thermal Cross Sections Z=1-100,* 6th Editon, National Nuclear Data Center, Brookhaven National Laboratory, USA, Copyright 2018, Elsevier, ISBN: 978-0-444052035-7

PCT/BR2023/050225, Instituto Nacional da Propriedade Industrial (Brasil) - RO/BR, WIPO, World Intellectual Property Organization, 34 chemin des Colombettes, CH-1211 Geneva, Switzerland | www.wipo.int.

Sun X.Y., Luo W., Lan H.Y., Song Y.M., Gao Q.Y., Zhu Z.C., Chen J.G., Cai X..Z. (2022), Transmutation of long-lived fission products in an advanced nuclear energy system, in *Scientific Reports* 12: 2240.

TECDOC (2016), Predisposal Management of Radioactive Waste from Nuclear Power Plants and Research Reactors, in IAEA *Safety Standards Series* No. SSG-40, Specific Safety Guides STI/PUB/1719 | 978-92-0-109815-3, 83 pages.

MEDICIÓN DE RADÓN EN LA ZONA DEL PEÑÓN BLANCO, SALINAS DE HIDALGO, SAN LUIS POTOSÍ, MÉXICO.

B. G. Olmos Grimaldo[1], V. Yutsis[2], M. A. Rojas Beltrán[3], A. Méndez Gaona[2], C. Letechipía de León[1]

[1] Universidad Autónoma de Zacatecas/Unidad Académica de Ciencias Nucleares/Maestría en Ciencias Nucleares/Ciprés No. 10, Fracc. La Peñuela, C.P. 98060, Zacatecas, Zac. México

[2] Instituto Potosino de Investigación Científica y Tecnológica/Geociencias Aplicadas/Geofísica, Camino a la Presa San José 2055, Col. Lomas 4 sección, CP. 78216, San Luis Potosí, S.L.P., México

[3] Universidad Autónoma de San Luis Potosí/Facultad de Ingeniería/Área Ciencias de la Tierra/Ingeniería En Geología, Dr. Manuel Nava No. 8, Zona Universitaria Poniente, C.P. 78290 San Luis Potosí, S.L.P., México

[*] bereguog@gmail.com

Resumen

El radón es un gas radioactivo que se forma por la descomposición del uranio, torio y radio, en las rocas, suelo y agua. Es el gas noble radiactivo que está dañando nuestro ADN pulmonar, por lo que cobra relevancia ahora más que nunca (Stanley et al., 2019; cancer.gov., 2015). En la zona del Peñón Blanco, en los límites entre los Estados de Zacatecas-San Luis Potosí, se reporta la presencia de una mina de fluorita con valores de uranio (Vélez 1972), por estas condiciones geológico-mineras, es un lugar propicio para la emisión y concentración de radón. Debido que es una zona con pocos estudios enfocados a la radioactividad natural, el objetivo de este trabajo fue cuantificar las emisiones de radón, y entender su relación con las rocas de la zona de estudio. La metodología seguida para este trabajo fue realizar una línea de sección en campo, y se llevaron a cabo mediciones con un detector electrónico portátil, modelo RAD7, en modo de detección rápida. Las mediciones fueron hechas en un muestreo aleatorio siguiendo el cauce principal del arroyo, que conforma una cuenca unitaria. Con esta técnica se realizaron 6 puntos de medición y se obtuvieron las concentraciones de radón (^{222}Rn) y torón (Tn o ^{220}Rn)). Los valores directos obtenidos de Radón en campo y sin corrección oscilan en un rango entre 340 Bq/m^3 a 41600 Bq/m^3; por lo que se puede observar que son niveles muy altos, al realizar la corrección para los datos, se obtienen valores superiores a los anteriores, siendo el máximo de 54900 Bq/m^3 y el menor de 450 Bq/m^3. Aún con la corrección, los valores son extremadamente altos y exceden a los valores recomendados por la Organización Mundial de la Salud, la cual propone un valor máximo de 100 Bq/m^3; aunque se puede tomar como límite, el que establece un valor aceptable de 300 Bq/m^3. Pero como se observa en los resultados de la zona, aún el valor de campo más pequeño sobrepasa el límite aceptable (Tarakanov, 2022). Actualmente, se están realizando caracterizaciones mineralógicas para obtener las fases de uranio presentes en la zona (Pi-Puig et al., 2023), y correlacionarlas con los valores de radón, los cuales se restringen a la zona de falla donde se emplaza la mina de fluorita, y tratar de establecer un modelo geológico que explique los valores excepcionalmente altos en la zona del Peñón Blanco.

Palabras clave: fluorita; geología; litología; radiación natural; radón; uranio

1 Introducción

La radiación natural forma parte del medio ambiente siendo sus principales componentes la radiación cósmica y la que se presenta en rocas, agua, suelo y aire.

Esta radiación tiene variaciones naturales, por lo que se debe tomar en cuenta diversos factores como es la altitud y latitud, el vivir a 1000 metros sobre el nivel del mar, tiene un aumento del 20% de radiación cósmica en comparación con las personas que viven al nivel del mar (Gómez-Quiñones, 2016).

Otro factor importante es la distribución de los componentes de la corteza terrestre, ya que no se encuentran distribuidos de una manera homogénea, siendo el 98% principalmente 8 elementos, y los de más elementos se encuentran en pequeñas cantidades en las rocas que comprenden la corteza terrestre, que varían por diversos procesos. Cuando las concentraciones de los elementos superan el contenido promedio en las rocas, se considera un yacimiento geológico (Porto & Merino, 2010). Lo cual permite correlacionar los elementos naturales con la radiación natural de la zona, dependiendo de la composición de la roca y minerales del sitio de interés.

Hay cuatro series primordiales de elementos radioactivos las cuales son: uranio-238 (^{238}U), uranio-235 (^{235}U), uranio-234 (^{234}U) y torio-232 (^{232}Th), desprendiéndose múltiples elementos. Sin embargo, las series en la cuales se enfoca este trabajo son: serie del uranio 238 (^{238}U) de donde se desprende el radón 222 (^{222}Rn) (Figura 1) y la serie del torio 232 (^{232}Th), donde se desprende el radón 220 (^{220}Rn) también conocido comúnmente como Torón (Figura 2).

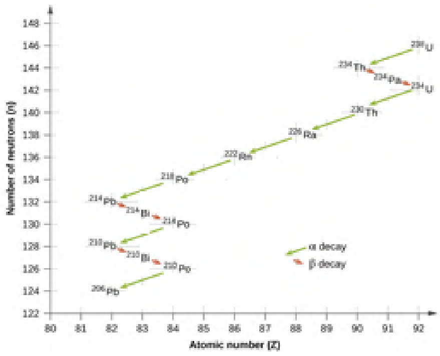


Figura 1. Serie primordial del uranio-238. Puede observar la serie de decaimiento del uranio-238 (^{238}U), donde un núcleo de uranio-238 se 14 transforma para convertirse en plomo-206 (^{206}Pb) un elemento estable, modificada y recuperada de Flowers et al., (2022).

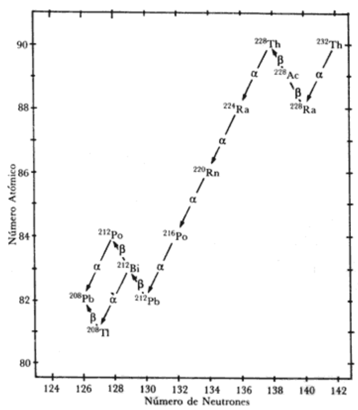


Figura 2. Serie primordial de torio-232. Puede notarse que un núcleo de torio-232 (^{232}Th) experimenta 10 transformaciones para convertirse en plomo-208 (^{208}Pb) estable, recuperada de Bulbulian (1996).

El radón es un gas radiactivo que es incoloro, inoloro e insípido; existen 3 radioisótopos de radón (radón 222, 220 y 219) que se desprenden de las series de decaimiento del uranio y del torio, siendo el más abundante el radón 222 aportando el 80%, del radón total en el medio ambiente (Barbosa et al., 2015), que al emanar de las rocas, se dispersa, migrando hacia lugares cerrados lo cual aumenta su concentración, provocando una gran exposición a las personas que viven cerca de yacimientos de uranio, y en lugares con actividad tectónica (Ruano-Ravina, A. et al. 2014). Este trabajo se llevó a cabo en una zona con un yacimiento de uranio, por lo que los valores de radón-222(^{222}Rn) y radón-220(^{220}Rn o Tn) se podrían encontrar aumentados, por lo que el objetivo de este trabajo es cuantificar el radón y torón de la zona, así como entender su relación con las rocas de la zona de estudio.

2 Materiales y Métodos

2.1 Selección del sitio

En 1955 México se creó la comisión nacional de energía nuclear (CNEN), por el descubrimiento de la importancia del uranio a nivel mundial, siendo a partir de allí la formación de varias instituciones para su estudio (Archivo General de la Nación, 2023). Entre 1956-1959 se realizaron trabajos de búsqueda de material radiactivo en diferentes zonas, siendo hasta 1967-1968 reportados en varias regiones, variando las reservas encontradas, siendo una de ellas San Luis Potosí. (Instituto Nacional de Investigaciones Nucleares, s/f) En trabajos previos realizados principalmente por Uranio mexicano (URAMEX), se reconoció la existencia de un yacimiento de pequeñas dimensiones, de uranio, en la zona del Peñón Blanco, localizada en los límites de los Estados de San Luis Potosí-Zacatecas. El lote minero las Viborillas (Comisión Nacional de Energía Nuclear, 1967), se localiza en el municipio de Salinas de Hidalgo (Figura 4), San Luis Potosí. Esta mina fue muestreada por uranio, en los años sesenta a-ochentas por URAMEX (Uranio Mexicano), encontrando poco potencial de recursos mineros, por lo que fue descartado para seguir con trabajos de exploración (Uranio Mexicano, 1981). Adicionalmente, hasta la fecha no se han realizado trabajos sobre las concentraciones de radón en esta zona, a pesar de la estrecha relación que existe entre estos dos elementos y su importancia en términos de medio ambiente y salud.

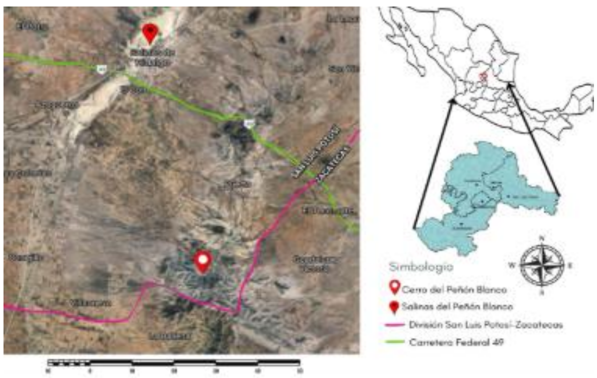


Figura 3. Localización del cerro del Peñón Blanco. Este sitio se localiza en los límites de Zacatecas-San Luis Potosí. El acceso es por la carretera federal No. 54. en el kilómetro 94.4 se encuentra el entronque que lleva a la comunidad El Alegre, continuando por la terracería que comunica El Alegre y San Juan Sin Agua, hasta llegar a la mina de fluorita “La Víbora”, figura recuperada de Morales Flores et al., (2024).

Para este trabajo solamente se seleccionó una cuenca unitaria, sobre el cual se seleccionaron 6 sitios potenciales y toman las mediciones de radón-222 y radón-220 (torón) (Figura 3), como lo es en la zona de falla y de la mina de fluorita.

Figura 3. Localización de los puntos de muestreo de radón. Figura de elaboración propia a partir de datos de Google Earth Pro y datos de campo.

2.2 Medición de Radón

El sitio del muestreo debe de tener una zona con suelo ya que para la colocación del instrumento y realizar la medición, se debe realizar una excavación de aproximadamente 40 cm para introducir la sonda, en la cual pasa el gas radón hacia un detector de estado sólido de partículas alfas.

Se utilizó el detector de radón modeloRAD7, marca Durridge (Figura 5), que tiene como principio cuantificar la radiación alfa (en un rango de 6 a 9 MeV), de forma indirecta del radón y torón, cuando los descendientes de estos, depositadas en la superficie del detector se desintegran y emiten partículas alfa, con una energía característica, directamente dentro del detector de estado sólido. La acumulación de varias señales genera un espectro (Durridge, 2014).

La calibración del equipo se realizó como lo sugiere el fabricante, con un intervalo de confianza de 95% o una incertidumbre 2σ.

Se llevaron a cabo 4 mediciones por punto de muestreo, con 5 minutos de diferencia, y el valor obtenido es el promedio de las mediciones en cada punto. Los datos que se obtienen directamente en campo se les debe realizar una corrección automatizada por medio del programa CAPTURE el cual toma en cuenta tanto la temperatura como la humedad (Durridge, 2014).

Figura 5. Instrumento de medición de gas radón, modelo Durridge RAD7. Imagen propia tomada en los trabajos realizados en campo.

3 Resultados

Los valores obtenidos mediante esta técnica fueron las concentraciones de radón (Rn) y torón (Tn) en 6 puntos de medición, realizadas sobre la cuenca unitaria. Los valores directos obtenidos de radón y torón en campo y sin corrección oscilan en un rango entre 340 Bq/m^3 a 41600 Bq/m^3; por lo que se puede observar que son niveles muy altos. Al realizar la corrección se obtienen valores superiores a los anteriores, siendo el máximo de 54900 Bq/m^3 y el menor de 450 Bq/m^3, tabla 1.

Tabla 1. Valores de radón en cada uno de los puntos de muestreo. Además se realizaron mediciones de Bq/m^3, Temperatura y porcentaje de humedad relativa.

Punto	**Elevación (mnsm)**	**Rn Bq/m^3**	**Tn Bq/m^3**	**T °C**	**RH %**
001	2371	22000	5300	20.4	44
002	2377	54900	7700	22.5	47
003	2342	11100	4600	23.1	48
004	2305	690	5100	28.9	48
005	2288	2450	1800	27.7	53
006	2207	450	1110	24.6	43

Por lo que estos valores que se obtuvieron en la zona del Peñón Blanco mediante el muestreo realizado con el RAD 7, se realizó en una zona donde se localiza el yacimiento de uranio y fallas inactivas, donde el radón está migrando de capas profundas a la superficie, en esta tabla 1 podemos observar que los tres primeros puntos de muestreo (001, 002, 003), que se encuentran cerca del contacto del intrusivo con la roca encajonante por falla, tienen los valores más altos (22000 Bq/m3, 54900 Bq/m3, 11100 Bq/m3), siendo también superiores.

Bená et al., (2022) menciona que hay múltiples estudios de la relación entre la alta concentración de radón y fallas activas, pero que se tienen pocos estudios realizados en fallas inactivas, en su trabajo destaca que la concentración más alta es de 800 Bq/m^3.

En la tabla 1 se puede observar que la elevación mayor (2377 msnm) tiene los valores más altos de radón (54900 Bq/m^3), donde la mina de fluorita está ubicada en un contacto por falla, entre las rocas intrusivas del peñón, y las rocas encajonantes de las formaciones sedimentarias (calizas, lutitas, areniscas), permitiendo que el gas emane con mayor facilidad.

4 Conclusiones

Todas las concentraciones obtenidas mediante esta medición, exceden a los valores recomendados por la Organización Mundial de la Salud, la cual propone un valor máximo de 100 Bq/m^3-300 Bq/m^3. También se pueden observar que el resultado con valor mínimo de radón-222 (450 Bq/m^3), sobrepasa el límite aceptable.

Podemos observar que los valores mayores están relacionados a la zona del yacimiento de uranio, sin embargo la concentración de radón, también puede estar relacionado como un indicador de la actividad tectónica de las fallas del sitio, Por lo que es un dato significativo en relación a la exposición de la radiación.

Los valores de radón son mayores en la zona del Peñón Blanco, a los que se han reportados en el volcán Popocatépetl en suelo en el estudio reportado por Peña et al. (2004), siendo que este volcán se encuentra en una zona tectónicamente activa, y la emanación de los gases en esta zona es constante y fluctuante dependiendo de las erupciones que estén presente, habiendo que las mediciones en suelo que se realizaron en las zonas cercanas al Popocatépetl fueron en dos sitios diferentes y en dos campañas, mostrando un promedio de 1.5kBq/m^3 y 2.5 kBq/m^{3,} mientras que los obtenidos en el cerro del Peñón Blanco también en suelo en un promedio de 11.54 kBq/m^3.

La finalidad de este trabajo no fue realizar un análisis de riesgo ambiental, ni monitoreo ambiental, si no que pretende aportar información relevante como antecedente de la concentración de radón en esta zona. Debido a la ubicación de la zona de estudio donde existen comunidades cercanas al mismo. Estas zonas urbanas probablemente estén expuestas a las concentraciones del radón, y puede ser un posible riesgo de salud pública en la misma población, como lo establecen otras publicaciones de riesgo ambiental.

Agradecimientos

Dr. Vsevolod Yutsis (IPICYT), M.C. Marco Antonio Rojas Beltran (UASLP), M..C. Alejandro Méndez Gaona (IPICYT), Ing. Carlos Eduardo Olmos Grimaldo (Capston Copper), Ing. María Guadalupe Grimaldo Teniente, Dra. Consuelo Letechipía de León por todo el apoyo recibido en este trabajo.

Referencias

Archivo General de la Nación. (15 de marzo 2023. El uranio, un mineral estratégico para la industria energética en México. gob.mx. Recuperado el 13 de febrero de 2025, de https://www.gob.mx/agn/articulos/el-uranio-un-mineral-estrategico-para-la-industria-energetica-en-mexico

Barbosa-Lorenzo R., Ruano-Ravina A., Cerdeira Caramés S., & Barros-Dios J. M.(2015), «Radón residencial y cáncer de pulmón. Un estudio ecológico en Galicia», *Medicina Clínica*, vol. 144, n.º 7, pp. 304-308, doi: 10.1016/j.medcli.2013.11.034.

Barros Dios, J. M. (2011). El radón: ¿riesgo para la salud?. Revista De Salud Ambiental, 11(1-2), 27–31. Recuperado a partir de https://ojs.diffundit.com/index.php/rsa/article/view/171

Benà, E., Ciotoli, G., Ruggiero, L., Coletti, C., Bossew, P., Massironi, M., Mazzoli, C., Mair, V., Morelli, C., Galgaro, A., Morozzi, P., Tositti, L., & Sassi, R. (2022). Evaluation of tectonically enhanced radon in fault zones by quantification of the radon activity index. Scientific Reports, 12(1), 21586. https://doi.org/10.1038/s41598-022-26124-y

Bulbulian, S. (1996). La Radiactividad. Fondo de cultura económica.

Cancer.gov. (2015) Radón https://www.cancer.gov/espanol/cancer/causas-prevencion/riesgo/sustancias/radon

Comisión Nacional de Energía Nuclear. (17 de enero de 1967). Reporte Parcial L-009.

Durridge, R. I. (2014). detector de radón RAD7, manual de usuario. https://durridge.com/documentation/RAD7%20Manual%20Spanish.pdf

Flowers, P., Theopold, K., Langley, R., & Robinson, W. R. (2022, junio 2). 21.3 Decaimiento radiactivo. Química 2ed; OpenStax. https://openstax.org/books/qu%C3%ADmica-2ed/pages/21-3-decaimiento-radiactivo

Gómez Quiñones, I. S. (2016). Flujo de radón en suelos para uso habitacional en la ciudad de Fresnillo, Zacatecas. Universidad Autónoma de Zacatecas, Unidad Académica de Estudios Nucleares Instituto Nacional de Investigaciones Nucleares. (s/f). Breve retrospectiva del Uranio en México. gob.mx. Recuperado el 16 de julio de 2025, de https://www.gob.mx/inin/articulos/breve-retrospectiva-del-uranio-en-mexico?idiom=es

Peña, P., Segovia, N., López M, B. E., Cisniega, G., Valdés, C., Armienta, M. A., Mena, M., & Universidad Autónoma Metropolitana, Unidad Iztapalapa, México D.F. (México). (2004). Determination of concentration of radon, volatile organic compounds (VOC) and water chemistry in springs near to Popocatepetl volcano. 7. International Conference; 17. National Congress on Solid State Dosimetry, Puebla (México).

Morales Flores, F. J., Castellanos Gutiérrez, Y., Moreno Nava, I., Cadena íñiguez, J., Olivera Méndez, A., & Balvanera Levy, P. (2024). Historiografía de un paisaje: servicios ecosistémicos del cerro del Peñón Blanco de Salinas, San Luis Potosí, México. Acta universitaria, 34, 1–20. https://doi.org/10.15174/au.2024.4237

Pi-Puig, T.; Solé, J.; Alba-Aldave, L (2023). Uranium in Fluorite, a Case Study: The La Azul Fluorspar Deposit, Taxco, Guerrero, México. Minerals. doi;10.3390/min13121495.

Porto, J. P., & Merino, M. (2010, mayo 20). Yacimiento. Definición.de; Definicion.de. https://definicion.de/yacimiento/

Ruano-Ravina, A., Quindós-Poncela, L., Sainz Fernández, C., & Barros-Dios, J. M. (2014). Radón interior y salud pública en España: tiempo para la acción. Gaceta Sanitaria, 28, 439-441.

Stanley, F.K.T., Irvine, J.L., Jacques W.R. et al., 2019. Radon exposure is rising steadily within the modern north American residential environment, and is increasingly uniform across seasons. Scientific Reports, 9, 18472, https://doi.org/10.1038/s41598-019-54891-8

Tarakanov V. (2022) ¿Qué es el radón y en qué lugares estamos expuestos a este gas?

Uranio Mexicano (1981). Distribución de reservas uraníferas.

Vélez-Ocón C. (1972), Solicitud de asignación de terrenos para la exploración y explotación de Uranio.

Preliminary Evaluation of a Positioning Phantom for Dosimetric Audit in Ir-192 HDR Brachytherapy: Results from an IAEA Collaborative Project

N. Briggs[1,2], A. C. F. Silva[1], S. P. Leite[2], A. Silva[3], A. Bernardino[3], M. H. Marechal[2], C. Salata[4], A. X. Silva[1], A. Mantuano[2,5]

[1] Universidade Federal do Rio de Janeiro/Coordenação dos Programas de Pós Graduação e Pesquisa de Engenharia da Universidade Federal do Rio de Janeiro, 21941-616, Cidade Universitária, Rio de Janeiro, Brasil

[2] Universidade do Estado do Rio de Janeiro/Laboratório de Ciências Radiológicas, Departamento de Ciências Radiológicas, R. São Francisco Xavier, 524, CEP: 20550-900, Maracanã, Rio de Janeiro, Brasil

[3] Hospital do Câncer/Instituto Nacional de Câncer III /Física Médica, Rua Visconde de Isabel, 274, 20560-121, Centro, Rio de Janeiro, Brasil

[4] Comissão Nacional de Energia Nuclear, 22290-040, Botafogo, Rio de Janeiro, Brasil

[5] Universidade do Estado do Rio de Janeiro, Instituto de Física/Departamento de Física Aplicada, R. São Francisco Xavier, 524, 20550-013, Rio de Janeiro, Brasil

natashabriggs.silva@gmail.com

Abstract

Dosimetric audits are essential to ensure quality and safety in brachytherapy, reducing risks to patients (Izewska, 2018). These audits verify whether the clinical dosimetry is adequate to ensure tumor control, minimize toxicities, and preserve the patient's quality of life, in addition to ensuring compliance with international standards (IAEA, 2016). This work presents the development and validation of a methodology for external auditing in high-dose-rate (HDR) brachytherapy using an Ir-192 source, conducted within the scope of a research project coordinated by the International Atomic Energy Agency (IAEA), with the participation of nine countries. A positioning phantom, manufactured through 3D printing, was designed to accommodate various models of intracavitary applicators, allowing adaptable and reproducible assembly conditions for dosimetric evaluation. Optically Stimulated Luminescence (OSL) dosimeters (nanoDot model) were used for point dose measurements around the applicator, in configurations previously standardized based on the results obtained from earlier tests leading to the final prototype. The coordinates of these points were registered in the Treatment Planning System (TPS) and used as a reference for experimental comparison, respecting distances and depths equivalent to those defined in the planning. This methodology enabled direct analysis of the agreement between the doses calculated by the TPS and the absorbed doses measured experimentally at the same evaluation points. The data obtained from the initial experiments were extremely important for assessing the functionality of the phantom for dosimetry in HDR brachytherapy with applicators, thus reproducing a scenario as close as possible to a clinical treatment (end-to-end). The dosimeter values positioned in parallel showed a percentage difference of 3.36% up to the current phantom configuration, with an average absorbed dose to water (Dw) of (7.79 ± 0.24) Gy. A significant percentage variation was observed in the responses of dosimeters placed at the same point in different experiments, reinforcing the need for the use of radiopaque markers in the phantom to more accurately match the TPS-defined positions with the actual

location of the OSLs. The project was funded by project E24023, International Atomic Energy Agency (IAEA), Contract number 24544.

Keywords: *HDR Brachytherapy; OSL Dosimetry; Phantom Design; TPS Validation; Monte Carlo MCNP; Audit Protocol.*

1 Introduction

Hight dose rate (HDR) brachytherapy plays a crucial role in modern radiotherapy, offering high precision in dose delivery to the tumor volume and a rapid dose fall-off in adjacent tissues (Viswanathan et al., 2012). To ensure that this technique achieves its maximum effectiveness with safety, rigorous dosimetry is required, as deviations in dose values may compromise local disease control and increase the incidence of late adverse effects (IAEA, 2019) (Pötter et al., 2021).

In the specific case of cervical cancer, the third most common type among women in Brazil, with approximately 17,000 new cases estimated for 2023 according to INCA (2021), the combination of external beam radiotherapy and HDR brachytherapy is considered the gold standard for patients with locally advanced stages (Viswanathan et al., 2012) (Pötter et al., 2021). The high geometric complexity involved in treatments using gynecological applicators requires special attention to independent dose verification, especially in protocols aimed at ensuring inter-institutional quality control.

Despite the existence of international protocols established by organizations such as the IAEA and GEC-ESTRO, there is still a lack of standardized, reproducible, and low-cost devices that can be udes in external audits, particularly in resource-limited countries. The use of technologies such as 3D printing enable the creation of customized positioning phantoms, adaptable to different applicator geometries and with the potential for distribution within dosimetric audit networks (Izewska et al., 2018) (Franco et al., 2021).

Beyond accessibility and reproducibility, the development of these phantoms must adhere to metrological principles applied to clinical dosimetry, including material stability, radiological equivalence, and traceability of measurements. These considerations are fundamental to ensuring international comparability and reliability in experimental evaluations, especially in audits involving high-activity radioactive sources such as Ir-192 in HDR brachytherapy (Andreo et al., 2005).

The objective of this work is to present the progress in the development of a 3D-printed phantom, designed to act as a positioning device for applicators in external audits of high dose rate (HDR) brachytherapy using an Ir-192 source. The phantom was conceived to allow the insertion of optically stimulated luminescence (OSL) dosimeters, aiming at the experimental evaluation of dose distribution.

2 Materials and Methods

The dosimetric kit for HDR brachytherapy audit includes a positioning phantom, a frontal slot for dosimeter placement, OSL dosimeters (nanoDots), and an acrylic box designed to be filled with water. The phantom features a compartment for positioning various applicators, such as tandem and ring types, and its front face contains five dedicated slots for inserting OSLs (nanoDots). The fully assembled kit for dosimetric purposes is shown below in Figure 1.

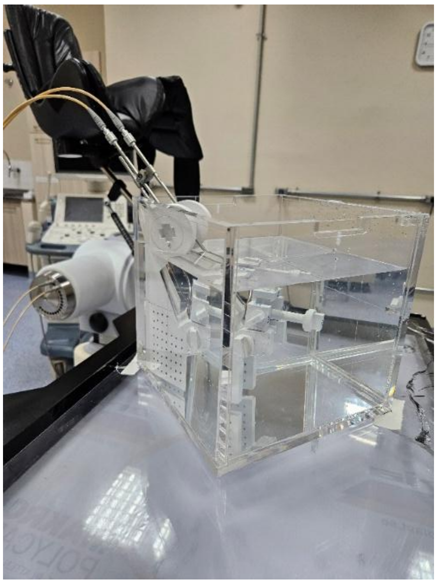


Figure 1: dosimetric kit for HDR bravchytherapy

The configuration of this phantom model for dosimetry allows for the inclusion of variables introduced by the applicator's geometry combined with the path followed by the source during the test, which simulates a real patient treatment. This setup makes it possible to reproduce a scenario closer to actual clinical conditions in terms of dose delivery and distribution, resulting in a more accurate comparison between the points marked in the TPS and the experimental data obtained from the placement of the OSLs at these same locations.

2.1 OSL (nanoDot) Dosimeters Positioning

In the case of 2D brachytherapy, immediately after acquiring the orthogonal radiographic images (lateral and anteroposterior), the dosimeters are positioned exactly 20.0 mm from the center on the front support of the phantom, as shown in Figure 2. Once this front part is properly fitted and locked in place, the acrylic box must be filled with water up to the level specified in the protocol, and only then can the dosimetry procedure begin.

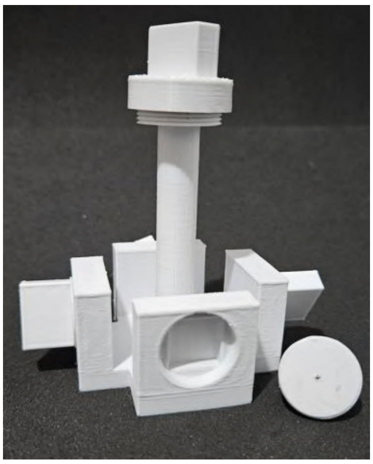


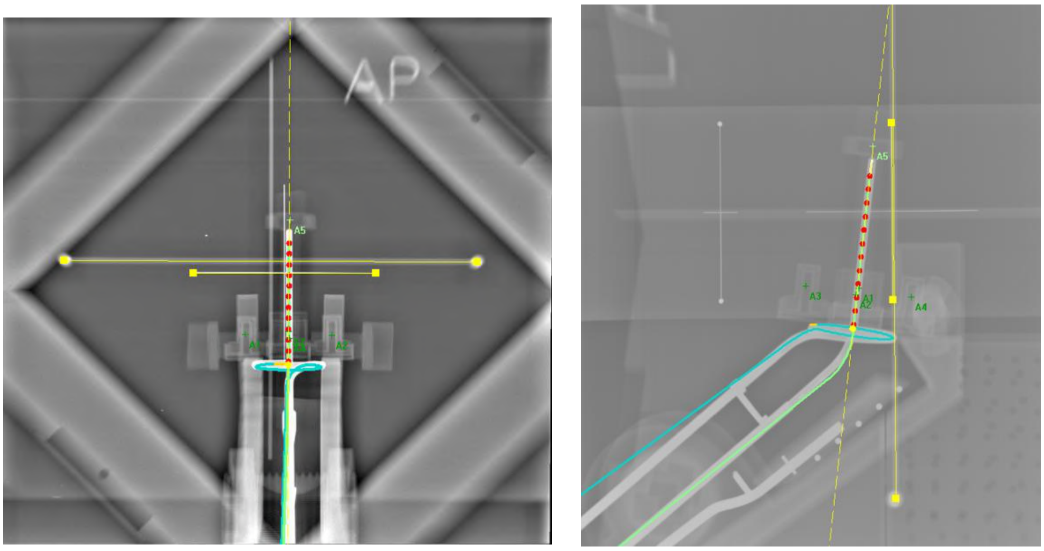
Figure 2: Front part with spaces for OSLs (nanoDots)

Proper positioning of the OSL dosimeters is a critical factor to ensure the reliability of dosimetric data, especially in situations where there is variation in the direction of radiation incidence (Kerns et al., 2011). These dosimeters exhibit angular dependence, meaning their response varies according to the angle of radiation incidence relative to the sensitive plane of the detector (Yukihara and McKeever, 2008). Therefore, the exposure geometry of the dosimeters must be as perpendicular as possible to the predominant direction of the incident radiation, thus minimizing angular effects.

2.2 Positioning of Geometric Points in the TPS for Dosimetric Assessment

The prescribed dose in the experimental phase was 800 cGy, and the dose distribution was calculated according to the AAPM TG-43U1 formalism (Rivard et al., 2004), assuming a homogeneous aqueous medium and an encapsulated Ir-192 source.

The reference points are defined directly on the radiographic images in the lateral and apical projections (Figure 3), previously imported into the treatment planning system, allowing their visualization and spatial localization during plan reconstruction.



(a) (b)

Figure 3: Lateral (profile) and apical (frontal) radiographic views imported into the TPS. (a) Anteroposterior (AP) view. (b) Lateral (LAT) view.

This model was modified after initial testing demonstrated the need to include radiopaque markers, aiming to improve the identification and reproducibility of the reference points in the images. This modification highlights the importance of greater detail in the phantom design, in order to support more accurate and traceable dosimetric point marking (VIM, 2012).

The points defined in the TPS represent point dose values used for direct evaluation of the calculated distribution at specific coordinates, without considering volumetric averages (Rivard et al., 2004).

2.3 Processing of Dosimetric Data: Comparison Between TPS and Experimental Dw Values

To formulate statistically consistent data, the irradiations were performed in triplicate, with repetitions on different days, allowing the evaluation of variability among the experimental tests.

The OSL readings generate a report of absorbed dose to water (cGy) for each dosimeter. These values are compared to the dose values at the reference points defined in the treatment planning system (TPS).

Two dosimeters, A1 and A2, were positioned in parallel orientation, at the same height, and separated by 40.0 mm—each placed 20.0 mm from the central axis where the applicator probe is inserted. As these are symmetrical regions relative to the applicator, they correspond to areas of highest dose agreement (Dw).

Figure 4 illustrates the correlation between the Dw values (Gy) obtained from the OSL dosimeters and the calculated values from the TPS reference points. The corresponding TPS values at the evaluated points were 7.87 Gy (at all locations), representing 98.43% of the prescribed dose. The mean value from the irradiations (OSL data) was 7.79 ± 0.24 Gy (tipe A uncertainty).

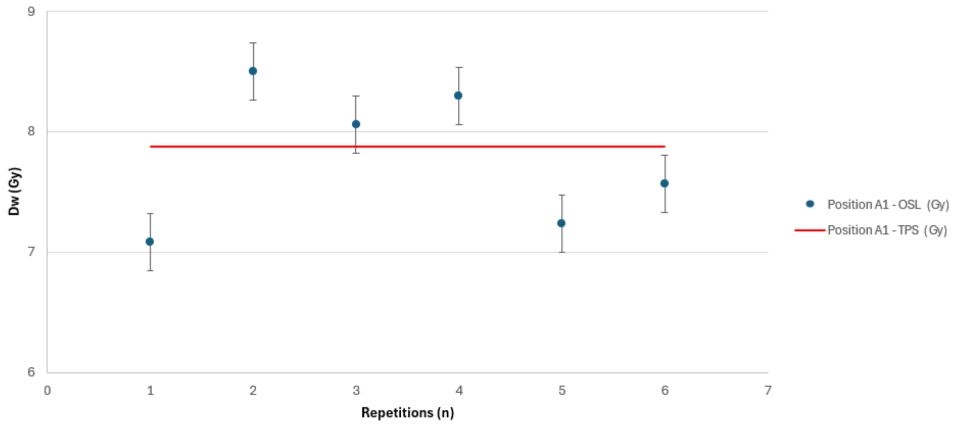


Figure 4: Comparison of measured and planned dose at A1: OSL vs. TPS

In Figure 5, the illustration corresponds to the Dw (Gy) values from the OSL dosimeters compared to the TPS reference point values for position A2, which was placed at a distance of 40.0 mm from the dosimeter at position A1. The values presented in the graph for the TPS points were 7.82 Gy (at all locations), corresponding to 97.84% of the prescribed dose. The mean value from the irradiations (OSL data) was 7.54 ± 0.08 Gy (uncertainty).

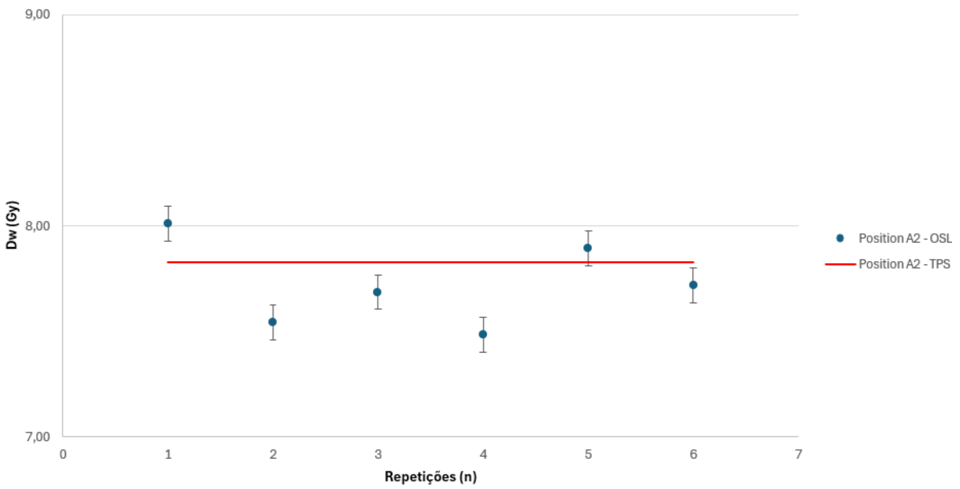


Figure 5: Comparison of measured and planned dose at A2: OSL vs. TPS

2.5 Computer Simulation MCNP

For the purpose of validating the experimental method and the model simulated by the TPS, Monte Carlo simulations were performed using the MCNP 6.3 software (Kulesza et al., 2022). This method is widely used in the literature for the validation of dosimetric parameters in brachytherapy, as described by Rivard et al. (2004) and Sahoo et al. (2007).

The software was run on a desktop computer with an AMD Ryzen 7 5800X 4.6 GHz processor and 32 GB of memory. The geometry was modeled using the same dimensions as the experimental phantom. The dose deposited in each detector was calculated using the F6 tally in MCNP, which provides results in MeV/g per particle. An input file was created for each dwell position of the iridium seed, totaling 13 input files.

All simulations were performed using 1 and 7 histories to minimize the statistical uncertainty of the simulation results to below 1%. Twenty processors were employed, and the simulation time for each input file was approximately 1 minute.

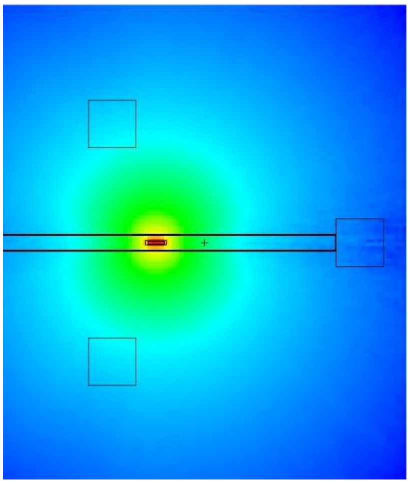


Figure 6: Dose map made with MCNP

The iridium source was modeled according to the parameters described in the official calibration document (Moura, 2015) and configured to emit an isotropic beam. The emission spectrum used was obtained from the work of Rostami A (2019) and includes all characteristic energies from Ir-192 decay.

In the computational simulation, it was possible to determine correction factors for the material used in the phantom fabrication, taking into account the thickness variations within the model. Additionally, dose map data were generated based on the Ir-192 source positioning, as illustrated in Figure 6. Figure 7 shows the interaction of particles with the walls of the phantom's frontal part, where the OSL dosimeters (nanoDots) are internally positioned.

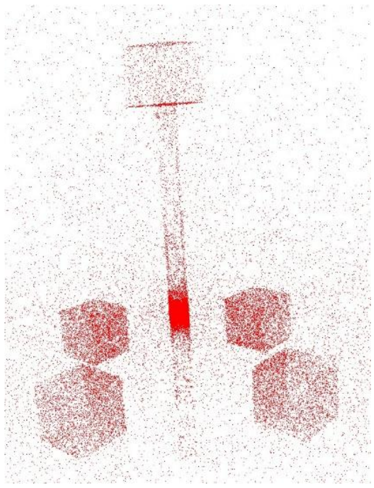


Figure 7: Interaction of particles with the phantom and dosimeter.

The computational simulation enables controlled modification of irradiation parameters, allowing for the evaluation of correction factors needed in cases of substitution or modification of the materials used in the prototype's 3D printing. In this way, it is possible to estimate the impact of different materials on attenuation properties and absorbed dose distribution without the need for immediate physical testing.

3 Results and Discussions

The verification of the experimental data using the prototype positioning phantom, developed for HDR brachytherapy dosimetry with an Ir-192 source, proved to be effective in fulfilling its proposed function, as it accurately replicates the clinical conditions observed in the treatment environment.

The dosimeters were positioned at locations equivalent to those defined in the TPS plan, enabling a direct comparison with the prescribed and calculated doses according to the TG-43 formalism. Additionally, the model considers the trajectory of the source during its movement through the applicator channels—a critical variable that can impact dose distribution values. The proposed model incorporates the main variables that influence the integral absorbed dose (ICRU Report 50, 1993) within the volume of interest under real conditions, validating its application as a reliable tool for dosimetric verification studies in HDR brachytherapy.

A percentage variation of 0.61% was observed between points A1 and A2 in the TPS, while the experimental data obtained using OSL dosimeters showed a variation of 0.87% between these same points. When comparing the TPS-calculated values with the experimental data, the observed percentage variations were 1.08% for position A1 and 1.34% for position A2. These results indicate good agreement between the experimental model and the treatment planning, with minor discrepancies consistent with the uncertainties associated with both methods.

The use of orthogonal radiographs (LA and AP views) in 2D brachytherapy planning presents inherent limitations for the precise localization of reference points, as spatial reconstruction depends on the bidimensional interpretation of complex anatomical structures. This limitation has already been addressed in documents such as ICRU Report 38 (1985) and reinforced by Rivard et al. (2004), which emphasize the need for more accurate imaging techniques to ensure the accuracy of prescription and control points.

The two points presented in this work are part of a dataset with five points that were statistically analyzed for the verification of possible improvements in the dosimetric set. The OSLs (nanoDots) are acquired by a company that provides the statistical data and these values are added to the uncertainty values associated with the phantom set with dosimeters. An acceptance criterion of ±5% was adopted for the dosimetric comparisons (TPS vs. experimental OSL data), in line with the IAEA code of practice for brachytherapy

(TRS-492, 2022) that supports acceptance limits of 5% in audits, and with classical guidelines (ICRU 38, ESTRO Booklet No. 8, TG-43) for quality control and clinical uncertainties in HDR brachytherapy (ICRU 38, 1985) (ESTRO, 2004) (Rivard et al., 2004).

4 Conclusions

This model was presented as part of a coordinated research project (CRP E24023) of the International Atomic Energy Agency (IAEA), initially involving researchers from ten countries (Brazil, China, Croatia, Greece, India, Iran, Mexico, Russian Federation, South Africa, and Spain).

The execution of experimental tests led to improvements in the phantom; in this final phase, data analysis from the prototype model enabled further enhancements, such as the insertion of radiopaque markers that assist in more precise localization of points of interest in the images imported into the TPS.

The results obtained in this study indicate that the proposed model shows good agreement with the TPS data and the experimental results. Additional studies are being conducted with variations in geometry, 3D printing materials, and irradiation configurations, in order to expand the applicability and robustness of the developed model.

Acknowledgements

The authors would like to thank the Cancer Control University Center (CUCC) – Pedro Ernesto University Hospital, at the State University of Rio de Janeiro (UERJ), for their institutional support and for enabling the initial experimental tests of this study. The authors gratefully acknowledge the International Atomic Energy Agency (IAEA) for providing financial support and promoting this project through international meetings and discussions. We also personally thank Professor Dr. Carlos Eduardo de Almeida for his support and dedication to the development of this project.

References

Andreo, P., Burns, D. T., Huq, M. S., et al. 2005. Fundamentals of absorbed dose dosimetry in radiotherapy. IAEA Training Course Series No. 33, Vienna, 2005.

ESTRO. 2004. Bookle No. 8 – Quality Assurance in Brachyterapy. Brussels: European Society for Radiotherapy and Oncology, 2004.

IAEA. 2016. International Atomic Energy Agency. Accuracy requirements and uncertainties in radiotherapy. Vienna: IAEA, 2016. (IAEA Human Helth Series, No. 31).

IAEA. 2019. Accuracy requirements and uncertainty analysis in high dose rate brachytherapy. IAEA Human Health Reports No. 45, 2019.

IAEA. 2022. Dosimetry in Brachytherapy: Na International Code of Pratice for Dosimetry in Brachytherapy (TRS-492). Vienna: International Atomic Energy Agency, 2022.

INCA. Instituto Nacional de Câncer. 2022. *Estimativa 2023: incidência de câncer no Brasil*. Ministério da Saúde, Rio de Janeiro, 2022.

International Commission on Radioations Units and Mesurements – ICRU. *ICRU Report 38:* Dose and Volume Specifications for Reporting Intracavitary Therapy in Gynecology. Bethesda, MD: ICRU Publications, 1985.

International Commission on Radioations Units and Mesurements – ICRU. ICRU Report 50: Prescribing Recording, and Reporting Photon Beam Therapy. Bethesda, MD: ICRU Publications, 1993.

International Vocabulary os Metrology – Basic and General Concepts and Associated Terms (VIM). 2012. 3rd edition. Join Committe for Guides in Metrology (JCGM 200:2012). Disponível em: https://www.bipm.org/en/publications/guides/vim.html. Acesso em: 07 de julho de 2025.

Izewska, J., Lechner, W., Wesolowska, P., Glick, S. J., Ikejimba, L. C. 2018. Global availability of dosimetry audits in radiotherapy: The IAEA dosimetry audit networks database. *Physics and Imaging in Radiation Oncology*, v. 5, p. 1-4, 2018.

Izewska, J., Lechner, W., Wesolowska, P. 2018. Global availability of dosimetry audit networks for radiotherapy: IAEA survey. Physica Medica, 2018.

Kerns, J. R., Kry, S. F., Sahoo, N., Followill, D. S. and Ibbott, G. S. 2011. Angular dependence of the nano dot OSL dosimeter. Med. Phys. 38(7), 3955–3962 (2011).

Kulesza J. A., Adams T. R., Armstrong J. C., Bolding S. R., Brown F. B., Bull J. S., Burke T. P., Clark A. R., Forster III R. A. A., Giron J. F. et al. (2022) Mcnp® code version 6.3. 0 theory & user manual. Technical report. Los Alamos National Laboratory (LANL), Los Alamos, NM (United States).

Franco, F. et al. Low-cost 3D printed phantom for end-to-end quality assurance in HDR brachytherapy. Physica Medica, 2021; 89:56–64.

Moura, E. S. 2015. *Desenvolvimento de um objetivo simulador para investigação de heterogeneidade em Braquiterapia de alta taxa de dose.* 2015 (Doctoral dissertation, *Dissertação (Doutorado)* IPEN/SP).

Pötter, Richerd Chargari, Cyrus et al. *Braquiterapia adaptativa guiada por ressonância magnética em câncer cervical localmente avançado (EMBRACE – I): um estudo de corte prospectivo multicêntrico.* The Lancet Oncology, Volume 22, Edição 4, 538-547.

Rivard, M. J., Coursey, B. M., DeWerd, L. A., Hanson, W. F., Huq, M. S., Ibbott, G. S., Mitch, M. G., Nath, R., Williamson, J. F. 2004. Update of AAPM Task Group No. 43 Report: Arevised AAPM protocol for brachyterapy dose calculations. Medical Physics, 31(3) 633-674. https://doi.org/10.1118/1.1646040

Rostami A, Hoseini M, Ghorbani M, Knaup C. Dosimetric investigation of a new high dose rate 192Ir brachytherapy source, IRAsource, by Monte Carlo method. Rep Pract Oncol Radiother. 2020 Jan-Feb;25(1):139-145. doi: 10.1016/j.rpor.2019.12.022. Epub 2019 Dec 26. PMID: 32051681; PMCID: PMC7002884.

Sahoo, S. K., Selvam, T. P., Vishwakarma, R. S. 2007. Dosimetric study of 192Ir HDR brachytherapy source using Monte Carlo technique. Applied Radiation and Isotopes, 65(3), 317-324. https://doi.org/10.1016/j.apradiso.2006.09.00

Viswanathan, Akila & Beriwal, Sushil & de Los Santos, Jennifer & Demanes, David & Gaffney, David & Hansen, Jorgen & Jones, Ellen & Kirisits, Christian & Thomadsen, Bruce & Erickson, Beth. (2012). American Brachytherapy Society consensus guidelines for locally advanced carcinoma of the cervix. Part II: High-dose-rate brachytherapy. Brachytherapy. 11. 47-52. 10.1016/j.brachy.2011.07.002.

A Monte Carlo-Based software for photon transport in water using the CUDA platform

R. M. Nunes[1,2*], **H. Yoriyaz**[1]

[1] Universidade de São Paulo, Instituto de Pesquisas Energéticas e Nucleares, 05508-000, São Paulo, Brazil

[2] Instituto de Pesquisas Tecnológicas do Estado de São Paulo, Tecnologias Digitais, 05508-901, São Paulo, Brazil

[*] renato.nunes@usp.br

Abstract

The programs commonly used to simulate dose estimation in radiotherapy are based on the Monte Carlo method (MCM) and are executed on computer CPUs (Andreo, 2018). This code structure demands considerable computational power and can result in lengthy simulations, often lasting several hours, since achieving statistically valid results using MCM may require the simulation of around 10^9 particles (Blanchet, 2017). Therefore, this work aimed to develop a photon transport and dose calculation software using the CUDA platform, executed on computer GPUs. This approach takes advantage of the parallel processing potential of GPUs to perform faster simulations. The PENELOPE (NEA, 2019) software was used as a reference for implementing the simulated physical models and tabulated physical quantities, while the MCNP6 software (Pelowitz, 2013) acted as a validator for the obtained results. The developed software is currently limited to simulating photon transport, not accounting for the simulation of other particles, and the simulated targets are composed entirely of water. Additionally, all targets were structured using voxels, which allows greater modeling flexibility and better discretization of the calculated dose values. Simulations were performed using sources with three different energy values: 6 MeV, 10 MeV, and 15 MeV, focusing on a voxelized water cube. The simulations were replicated in MCNP6, and the results obtained by both programs showed good agreement, with average doses in the targets differing by no more than 5.40%. Moreover, analyzing the individual doses in each voxel, the largest discrepancy observed across all simulations was 10.70%. Regarding execution times, a significant advantage was observed when using GPUs for processing the simulations, as in all analyzed cases the execution time on the GPU was reduced by more than 90%, which indicates that this approach can bring good solutions to the lengthy simulations problem, and the development of more robust programs running on GPUs can be a valid alternative for radiotherapy simulations.

Keywords: Dosimetry; Photon transport; Parallelism; CUDA; Voxels.

1 Introduction

Radiotherapy stands as a widely adopted technique for the treatment of tumors in patients (Moding et al., 2013), where a precisely directed radiation beam aims to reach and destroy cancerous cells, thereby inducing cancer regression or even complete eradication (Fairbanks and Furnari, 2019). Despite its efficacy, this approach presents inherent challenges. The intricate complexities of human tissues, with different dimensions and interaction cross-sections, coupled with the intrinsic physical properties of radiation, often lead to unintended irradiation and damage of healthy tissues surrounding the tumor during the procedure, resulting in undesirable side effects for the patient (Dilalla et al., 2020).

To mitigate these challenges and enhance treatment precision, computational simulations have emerged as an indispensable tool in modern radiotherapy planning. These simulations enable the digital reproduction of human organs of interest, allowing for the virtual application of radiation to these tissues. This process generates realistic predictions of dose distribution and particle interactions, which are crucial for guiding real patient treatments. Several software tools are available for performing these simulations, with some of the most widely used being MCNP (Pelowitz, 2013), PENELOPE (NEA, 2019), and Geant4 (Agostinelli et al., 2003).

In these simulations, the trajectory and interactions of individual radiation particles are meticulously calculated from their source until they are either absorbed or traverse all tissues of interest. Given the inherently non-deterministic nature of particle behavior, the Monte Carlo Method is widely utilized to provide a statistically robust approach for obtaining results consistent with real-world scenarios, requiring a large number of simulations.

Current programs are structured to be executed on the computer's CPU (Central Processing Unit). Although modern CPUs already have several cores, allowing some degree of parallel task execution, their architectures are designed for linear and individualized task processing, which is not suitable for more specific cases that require a high volume of parallel calculations.

This current scenario results in lengthy execution times, since achieving statistically valid results with MCM often necessitates simulating a vast number of particles, frequently exceeding 10^9 for complex cases (Blanchet, 2017).

An alternative to reduce these prolonged simulation times is the development of a program capable of performing the processing of particle transport and interaction simulations using the computer's GPU (Graphics Processing Unit).

Unlike CPUs, GPUs have an architecture geared towards parallel data processing and are thus capable of performing a large number of calculations simultaneously. Their structure, based on Streaming Multiprocessors (SMs), allows the creation of hundreds of execution blocks containing numerous threads running in parallel. In this way, it becomes possible to simultaneously simulate hundreds of particles, reducing the time required to obtain the necessary results for the patient radiotherapy treatment planning.

In this context, the primary objective of the present article is to detail the development of a software program designed to simulate particle trajectories through a voxelized target and calculate the deposited dose. This program takes advantage of the parallel processing capabilities of computer GPUs, specifically utilizing the NVIDIA CUDA platform.

Currently, the software is limited to the exclusive transport of photons, absorbing any electrons generated in the interactions with matter and computing their energy as local dose. Furthermore, the only material available for voxel composition is liquid water. In future versions, the addition of more materials and radiotherapy-relevant tissues is planned.

2 Methodology and Materials

2.1 Monte Carlo Method

The Monte Carlo Method (MCM) is a statistical computational technique that employs algorithms to solve a diverse range of problems. It is particularly valuable for addressing problems that have an intrinsically stochastic nature or those for which an analytical solution is either difficult to obtain or nonexistent (Yoriyaz, 2009).

There are various algorithms that fall under the MCM category, but in general, all share a common logical structure: knowing the Cumulative Distribution Functions (CDFs) or the Probability Density Functions (PDFs) that describe the phenomenon of interest, a large quantity of pseudo-random numbers is generated. These numbers are then analyzed and compared against the reference functions to yield results consistent with the problem description.

In the developed program, MCM is implemented to perform the probabilistic calculations that accurately represent the stochastic properties of photon interactions with water.

2.2 MCNP and PENELOPE

MCNP (Monte Carlo N-Particle) and PENELOPE (PENetration and Energy Loss of Positrons and Electrons) are well-established Monte Carlo-based programs extensively used for simulating the transport of electrons and photons across various energy ranges, as well as their interactions with user-specified geometries and material compositions.

Given their long-standing consolidation and widespread acceptance in the field, these programs served as references for the development of the new code.

PENELOPE-2018, being an open-source code, served as the foundation for implementing the physical models of radiation interactions with matter and for tabulating physical quantities. MCNP6, on the other hand, served as a validator for the obtained results and as a benchmark for comparing simulation execution times.

2.3 CUDA

CUDA is a parallel computing model developed by NVIDIA (NVIDIA, 2024), focused on the development of applications that use the company's GPU hardware for data processing. Built upon the C/C++ programming languages, CUDA provides a comprehensive set of tools, such as functionalities for managing GPU memory, programming and launching kernels, which are responsible for the executions of the parallelized calculations in the GPU.

2.4 Simulated Model

The computational model implemented for the simulations is visually represented in Figure 1. This model features a flat source, measuring 40 cm x 40 cm, which emits monoenergetic photons exclusively in the Z-direction. The target is defined as a cubic volume of water, with dimensions of 40 cm x 40 cm x 40 cm and is discretized into 8000 individual voxels, each measuring 2 cm in each direction (20 voxels per side).

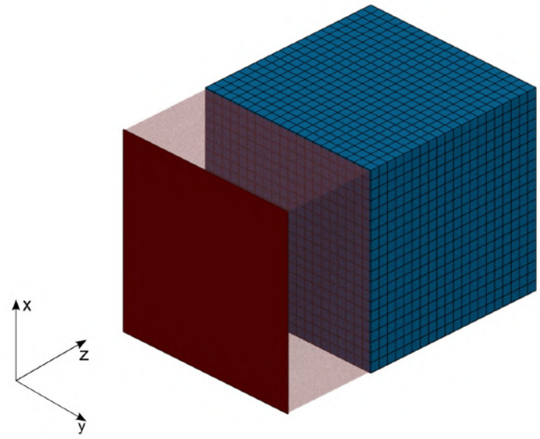


Figure 1: Computational model implemented for the simulations.

The voxelized structure of the target offers significant advantages, enabling a more granular and precise analysis of dose calculations within each specific region of the target. Furthermore, this approach simplifies the creation of models with more complex geometries, which is crucial for realistic simulations.

For the simulations conducted with the developed software, 1 billion monoenergetic particles were simulated at three distinct energy levels: 6 MeV, 10 MeV, and 15 MeV, while 5 million particles were simulated in MCNP. The dose values calculated in each voxel and the total execution times were subsequently compared between the two programs. To ensure consistency in the simulated physical models and allow for a direct comparison, MCNP was configured to suppress the simulation of electrons during its runs. In both programs, any electrons generated as a result of physical interactions between photons and the material medium were immediately absorbed, and their energy was accounted for as deposited dose within the respective voxel where the interaction occurred.

2.5 Equipment

Two computer systems were utilized to perform the simulations. The computer employed for the MCNP simulations featured the following specifications:

- CPU: Intel Core(TM) i7-12700KF (12 Cores, 20 Threads);
- GPU: Nvidia RTX 3050 6GB GDDR6;
- RAM: 2x PNY 16GB, 2667MHz, DDR4;
- Storage: SSD M.2 PCIe X4 NVMe 512GB;

And the developed software was executed on a computer with the following characteristics:

- CPU: Intel Core(TM) i7-12700 (12 Cores, 20 Threads);
- GPU: Nvidia RTX 3060 Ti 8GB GDDR6X;
- RAM: 2x Kingston Fury Beast 16GB, 5600MHz, DDR5;
- Storage: SSD M.2 PCIe X4 NVMe 1TB;

3 Results

3.1 Dose Calculations

Based on the dose values obtained in each voxel from the simulation, the average dose across the entire target was calculated. For the 6 MeV photon beam simulation, MCNP yielded an average dose of $4.9103(1) \times 10^{-5} MeV/g$, whereas the developed program resulted in a value of $5.0033(2) \times 10^{-5} MeV/g$, representing a difference of 1.89% between the two results. At the voxel level, the most significant discrepancy in calculated average dose occurred in the voxel with coordinates (12, 20, 20), where the developed program's dose was 7.57% higher than the value found by MCNP.

For simulations involving 10 MeV energy particles, the overall average dose in the target obtained by MCNP was $7.4724(2) \times 10^{-5} MeV/g$. For the developed program, this value was $7.8762(3) \times 10^{-5} MeV/g$, representing a 5.40% difference. The voxel exhibiting the largest discrepancy was at coordinates (2, 16, 1), with an average dose 10.70% higher than the MCNP value.

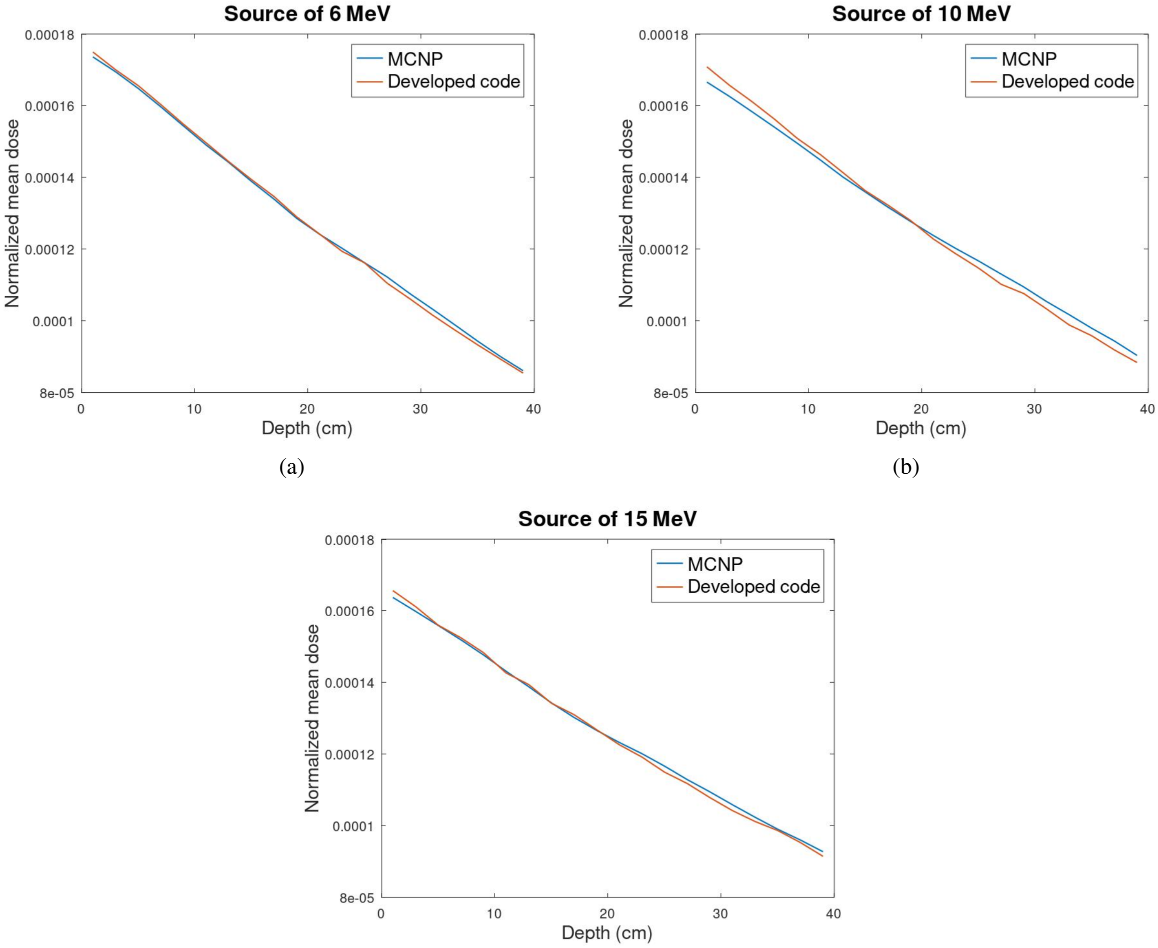
In the case of the 15 MeV monoenergetic beam, the average dose in the entire target was $1.10659(8) \times 10^{-4} MeV/g$ in MCNP and $1.10109(5) \times 10^{-4} MeV/g$ in the developed program, corresponding to a difference of 3.30%. The voxel at coordinates (12, 20, 18) showed the greatest difference in results, with an average dose 7.41% superior to the value reported by MCNP.

These results indicate a slight tendency toward dose overestimation by the developed software.

To analyze the behavior of the average dose in relation to the depth of the target, graphs were generated considering only the central row of voxels in the Z direction. The selected rows correspond to voxels with X and Y coordinates fixed at 10, and the Z coordinate varying from 1 to 20.

To focus on these behaviors, the mean doses from the central values were normalized by the respective average dose in the entire target, facilitating the comparison of the curves' patterns. Figure 2 presents these graphs for the three simulated energies.

In Figure 2(a) is shown the graph for the 6 MeV simulations. The curves follow a very similar pattern, although the results from the developed code are slightly higher than MCNP's in the superficial region and slightly lower in deeper regions. For the 10 MeV source (Figure 2(b)), the discrepancy between the curves is more pronounced, where the one for the developed code shows a pattern of greater dose deposition in the target's superficial regions. Finally, Figure 2(c) also shows a trend of higher dose deposition in the initial voxels for the 15 MeV source, although less significant than for 10 MeV.



(a) (b) (c)

Figure 2: Normalized dose deposition versus depth for three different sources. (a) 6 MeV source. (b) 10 MeV source. (c) 15 MeV source.

These observations indicate that the developed program's tendency to overestimate dose primarily occurs in the superficial region of the target. Nevertheless, the results do not significantly differ from those obtained using MCNP. For the normalized central dose values, the greatest discrepancy occurred in the 10 MeV source simulation at voxel (10, 10, 17), where the developed code's result was only 2.82% below the MCNP value.

3.2 Execution Times

An analysis of the simulation execution times clearly demonstrates the significant time savings achieved by utilizing the developed software. For the 6 MeV monoenergetic source, the execution time of the developed program was 7 minutes and 27 seconds, while in MCNP it was 144 minutes and 36 seconds, representing a reduction of almost 95% in execution time.

Similarly, for 10 MeV photons, the new software completed the simulation in 6 minutes and 17 seconds, representing only 4.2% of the execution time in MCNP, which lasted 150 minutes and 53 seconds.

In the last case, for the 15 MeV source, the execution time in the developed program was 5 minutes and 2 seconds, while in MCNP it was 209 minutes and 39 seconds, which indicates a reduction in execution time of 97.5%.

Table 1 summarizes the execution times of both programs.

Table 1: Execution times and comparisons.

Energy	Developed code	MCNP	Time reduction
6MeV	7min27s	144min36s	94.8%
10MeV	6min17s	150min53s	95.8%
15MeV	5min02s	209min39s	97.5%

4 Conclusions

Although the developed software slightly overestimates the dose values deposited in the target voxels compared to MCNP, both programs showed good agreement in their results. The average doses in the entire target did not differ by more than 5.40%, and voxel-level discrepancies rarely exceeded 10.00%. Nevertheless, further analysis is needed to understand the causes of this overestimation and minimize this issue. In addition, to better validate the developed program, more complex simulations should be performed, using smaller voxels and sources with a broader energy spectrum.

Regarding execution times, the significant advantage that the parallelism provided by GPUs brings to simulations is clear. Despite the simplicity of the proposed simulation model, it was possible to achieve execution time reductions of over 90%. Regardless of the limitations and simplifications of the presented software, the study and development of more robust programs that utilize GPU parallelism to perform radiotherapy treatment simulations prove to be a promising approach, as the time savings provided by these tools can bring significant benefits in developing patient treatment plans.

Acknowledgements

We gratefully acknowledge the Instituto de Pesquisas Tecnológicas do Estado de São Paulo for providing the resources and institutional support for the development and execution of this study.

References

Agostinelli S., Allison J., and Amako K. andApostolakis J. (2003) Geant4 - a simulation toolkit. *Nucl Instrum Meth A* 506:250–303

Andreo P. (2018) Monte carlo simulations in radiotherapy dosimetry. *Radiat Oncol* 13

Blanchet D. (2017) A monte-carlo benchmark of tripoli-4® and mcnp on iter neutronics. *European Physical Journal Web of Conferences* 153

Dilalla V., Chaput G., Williams T., and Sultanem K. (2020) Radiotherapy side effects: integrating a survivorship clinical lens to better serve patients. *Current oncology* 27:107–112

Fairbanks L. and Furnari L. (2019) Radioterapia guiada por imagem: Técnicas e controle de qualidade. *Revista Brasileira de Física Médica* 13:99–108

Moding E. J., B. K. M., and G. K. D. (2013) Strategies for optimizing the response of cancer and normal tissues to radiation. *Nat Rev Drug Discov* 12:526–542

NEA (2019) *PENELOPE-2018: A Code System for Monte Carlo Simulation of Electron and Photon Transport*. OECD Publishing

NVIDIA C. (2024) *CUDA C++ Programming Guide Release 12.6*

Pelowitz D. B. (2013) *MCNP6TM user's manual - Version 1.0*. os Alamos National Laboratory Tech. Rep.

Yoriyaz H. (2009) Método de monte carlo: princípios e aplicações em física médica. *Revista Brasileira De Física Médica,* 3:141–149

XXV ISSSD
International Symposium on Solid State Dosimetry
September 22 – 26, 2025 Belo Horizonte, Minas Gerais - Brazil
CDTN UFMG

A new formulation of Fricke gel dosimeter development using salicylic acid

I. V. Nishibayaski[1]*, A. L. F. Silveira[1], L. C. Meira-Belo[1]

[1] Centro de Desenvolvimento da Tecnologia Nuclear, Seção de Dosimetria das radiações, Av. Presidente Antônio Carlos, 6627, 31270-90, Belo Horizonte, Brazil

* iasmin.nishibayaski@cdtn.br

Abstract

Three-dimensional dosimeters determine radiological parameters such as exposure, kerma, and absorbed dose in a volumetric distribution. Fricke gel, for instance, is a 3D dosimeter based on the conventional Fricke solution, considered a reference standard in chemical dosimetry according to International Organization for Standardization and American Society for Testing and Materials (ISO/ASTM). The conventional Fricke solution, composed of ferrous ammonium sulfate in an aqueous solution of sulfuric acid, has been modified to a three-dimensional dosimeter by adding a gelling agent to the solution. Beyond this addition, another important modification was the addition of ion indicators, such as xylenol orange, which increases dosimeter gel stability and enables the use of optical analyses for dosimetric calculations. This work proposes a new formulation of Fricke gel, with salicylic acid as an ion indicator (FSG). After its exposure to a cobalt-60 source installed at the Laboratory of Gamma Irradiation, in the Nuclear Technology Development Center (LIG/CDTN) in Belo Horizonte, Brazil, the dose response of this dosimeter was evaluated by spectrophotometry. The results showed that FSG demonstrated a linear dose-response relationship in the gamma irradiation field for a range between 5 Gy and 200 Gy. In addition, a comparison between this dosimeter and the well established Fricke xylenol gel was made and the results showed that FSG has a higher sensitivity compared to the conventional Fricke gel based on xylenol orange as an ion indicator (FXG). Although salicylic acid has not previously been applied in Fricke gel dosimetry, it is a promising alternative due to its accessibility, affordability, and non-toxic properties as an ion indicator.

Keywords: Dosimetry; Fricke gel; Salicylic acid.

1 Introduction

Radiation dosimetry is a fundamental process that quantifies parameters such as exposure, kerma, and absorbed dose, based on the interaction between ionizing radiation and matter (Attix, 2004). This evaluation is essential for the safe and optimized use of radiation, particularly in medical and technological applications. Accurate dosimetric measurements not only ensure patient and operator safety but also support the development and improvement of radiation-based technologies.

The choice of an appropriate dosimetric system depends on the intended application. Factors such as radiation type and energy, dose range, cost, availability, sensitivity to environmental conditions, and spatial resolution must be considered. A widely accepted reference system is the Fricke dosimeter, which is highly tissue-equivalent due to its 96.1 % water content by weight and is frequently used in calibration procedures (ICRU, 2008).

The Fricke dosimeter consists of an aqueous solution of ammonium ferrous sulfate [$Fe(NH_4)_2(SO_4)_2 \cdot 6H_2O$] and sulfuric acid ($H_2SO_4$). First described by Fricke and Morse in 1927 (Fricke and Morse, 1927), its mechanism is based on the oxidation of ferrous ions (Fe^{2+}) to ferric ions (Fe^{3+}) upon absorption of ionizing radiation. This transformation results from the radiolysis of water, which leads to the formation of reactive species such as hydrogen atoms, hydroxyl radicals, hydrogen peroxide, hydronium ions, molecular hydrogen, and solvated electrons. These species interact with Fe^{2+}, promoting its oxidation as summarized in Equation 1 (Schreiner, 2004).

$$Fe^{2+} + OH\bullet \rightarrow Fe^{3+} + OH^{-}$$

$$Fe^{2+} + H_2O_2 \rightarrow Fe^{3+} + OH\bullet + OH^{-}$$

$$H\bullet + O_2 \rightarrow HO_2\bullet$$
$$Fe^{2+} + HO_2\bullet \rightarrow Fe^{3+} + HO_2^{-}$$

$$HO_2^{-} + H_3O^{+} \rightarrow H_2O_2 + H_2O \qquad (1)$$

The concentration of Fe^{3+} formed is directly proportional to the energy absorbed by the solution, enabling the calculation of the absorbed dose through Equation 2 (Schreiner, 2004):

$$\Delta\left[Fe^{3+}\right] = \frac{D \cdot G\left(Fe^{3+}\right) \cdot 10\rho}{N_A \cdot e} \qquad (2)$$

Where D represents the absorbed dose, $G(Fe^{3+})$ is the radiochemical yield of ferric ions, ρ is the density of the solution, N_a is Avogadro's number, and e is the elementary charge.

The Fricke dosimeter is known for its simplicity, sensitivity, and reproducibility. Over time, several modifications have been introduced to expand its applicability. Among these, the incorporation of gel matrices and metal ion indicators has enabled spatially resolved dose measurements and facilitated optical detection, enhancing its versatility.

Three-dimensional (3D) dosimetry allows the evaluation of spatial dose distributions within a volume and has relevant applications in radiotherapy, diagnostic imaging, and internal dosimetry. Gel dosimeters, in particular, can qualitatively visualize the absorbed dose without the need for external imaging systems, reducing uncertainties in dose delivery. Additionally, such gels can be applied to dose assessment in computed tomography (Courbon *et al.*, 1999).

To allow spatial resolution, Fricke solutions can be incorporated into a gel matrix that restricts ion diffusion (Gore *et al.*, 1984). In this format, ferric ions generated by radiation remain close to their production site, enabling the mapping of dose distributions.

Traditionally, Fricke gels are prepared using gelatin (animal-derived) or agarose (plant-based). Synthetic polymers, such as polyvinyl alcohol (PVA) (Chu *et al.*, 2000) and polyethylene oxide (PEO) (Araujo *et al.*, 2021), have emerged as promising gelling agents due to their improved chemical stability. PEO, in particular, is biocompatible, highly soluble in water, and presents elevated viscosity (Bailey, 2012). Its thermoplastic properties also allow it to be molded for specific applications.

Another strategy for enhancing Fricke gel dosimeters involves the inclusion of colorimetric indicators (chelating agents) that bind to Fe^{3+}, allowing dose measurement via optical methods. These agents also help reduce ion diffusion, improving spatial accuracy (Rae, 1995). Xylenol orange (XO) is the most widely used indicator and it forms chromophoric groups with Fe^{3+}. These complexes absorb light in the visible spectrum, imparting visible color to the gel (Scotti, 2022).

Although not primarily used as a chelating agent, salicylic acid, it can form complexes with metal ions, especially Fe^{3+}, in aqueous solution, exhibiting a pinkish coloration (Silva e Sá, 2006). This interaction, although not the primary function of salicylic acid, can be exploited in chemical analyses and dosimetric studies using Fricke. This study investigates the use of Salicylic acid as an alternative indicator to XO in Fricke gels. Agarose was used as gelling agent. The dosimetric response of this novel formulation was evaluated and compared to that of traditional Fricke xylenol gel (FXG) when both were exposed to gamma radiation.

2 Materials and methods

2.1. Preparation of Fricke gel dosimeters

Two different Fricke gel dosimeter formulations were developed in this study. The first formulation, referred to as FSG, employed agarose as the gelling agent and salicylic acid as the indicator. The second formulation, FXG, was used as a reference and consisted of porcine gelatin and xylenol orange, following the traditional Fricke gel composition.

The preparation process was conducted in two stages. Initially, the gel matrices were formulated: agarose was used at 1 % (w/v), and porcine gelatin at 5 % (w/v), based on the final volume of the dosimeter. Agarose and gelatin were separately dissolved in triple-distilled water and heated to 150 ± 1 °C under stirring. After complete dissolution, the mixtures were cooled naturally in the environment up to 50 ± 1 °C.

In the second stage, the Fricke solution was prepared with 50 mM sulfuric acid, 1.0 mM ammonium ferrous sulfate, and 1.0 mM sodium chloride. The FSG formulation included 4.0 mM salicylic acid as the indicator, while xylenol orange was used at a concentration of 0.1 mM in the other formulation. The final mixture had a proportion of 75 % (v/v) gelatin matrix and 25 % (v/v) Fricke solution and they were poured into PMMA cuvettes (1.5 mL, 4.5 × 10 × 32 mm, optical path: 10 mm). All samples were stored in the dark at 2.5 ± 1 °C for 24 hours to complete the gelation process and prevent photodegradation.

2.2. Irradiation procedure

The dosimeters were irradiated at ambient temperature (27 ± 1 °C) using a cobalt-60 gamma source (activity: 51598.17 Ci) located at the Gamma Irradiation Laboratory (LIG/CDTN) in Belo Horizonte, Brazil. Irradiation was performed in triplicate for each formulation over a dose range of 5 to 200 Gy. For doses from 5 to 30 Gy, 5 Gy increments were used; for 50 to 200 Gy, increments of 50 Gy were applied. The dose rate was kept constant at 9.24 $Gy \cdot h^{-1}$.

These ranges were selected based on the operational limits of conventional Fricke dosimeters. It is important to note that gelation and indicator incorporation can shift these limits by altering ion mobility and chemical reactivity.

2.3. Optical analysis

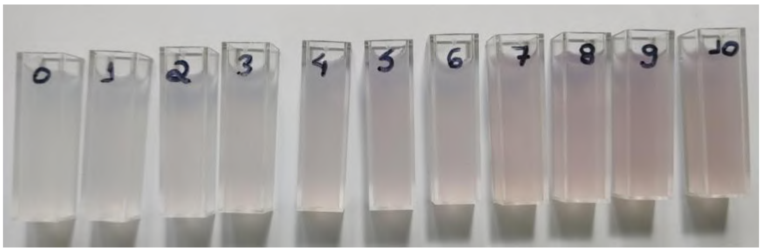
Optical absorption spectra were obtained using a Shimadzu UVmini-1240 spectrophotometer operating in the 300–700 nm range. For each absorbed dose, the absorption spectrum was recorded for three identically irradiated samples, and the results were averaged. Spectra were acquired before and approximately one hour after irradiation.

To isolate the spectral changes induced by radiation, the pre-irradiation spectrum was subtracted from the post-irradiation spectrum. To reduce baseline fluctuations and instrumental noise, spectra were normalized by subtracting the minimum absorbance value. Discontinuities in the data, likely due to lamp variation or mechanical factors, were numerically corrected by replacing peak values with adjacent point averages before reintegration of the curve.

3 Results and discussion

3.1. Absorbed dose response

Figure 1 illustrates the colorimetric changes observed in the gel formulations following gamma exposure.



(a)

(b)

Figure 1: Color change of (a) FSG and (b) FXG following gamma irradiation at doses ranging from 5 Gy to 200 Gy.

All Fricke gel dosimeters exhibited noticeable color changes following gamma irradiation. The formation of salicylic acid-Fe^{3+} complex, which appears pink post-irradiation in Fricke gels containing the chelating agent, is responsible for the progressive coloration. Similarly, FXG changed color due to the chromophore groups in the complex formed between Fe^{3+} and xylenol orange molecules, transitioning from orange to violet.

Among the formulations, FXG displayed the highest sensitivity, with visible color change occurring from the lowest dose of 5 Gy. FSG also showed changes at 5 Gy, but saturation in this formulation did not occur for the absorbed doses provided in this study. Nonetheless, FXG showed color saturation at approximately 30 Gy.

3.2. Spectrophotometric results

Figure 2 shows the average absorption spectra for all two dosimeter types. The FXG spectra were not subtracted by the minimum absorbance value because FXG baseline spectrum is more stable comparing to FSG.

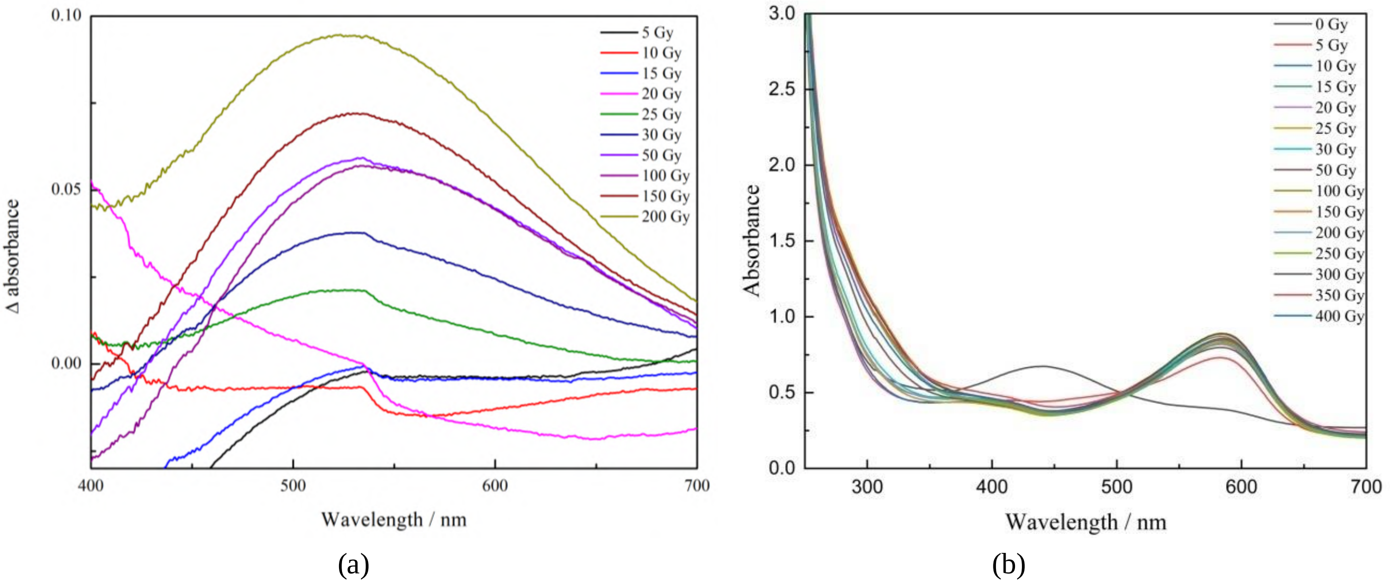


Figure 2: UV-visible spectrum of (a) FSG and (b) FXG after irradiation.

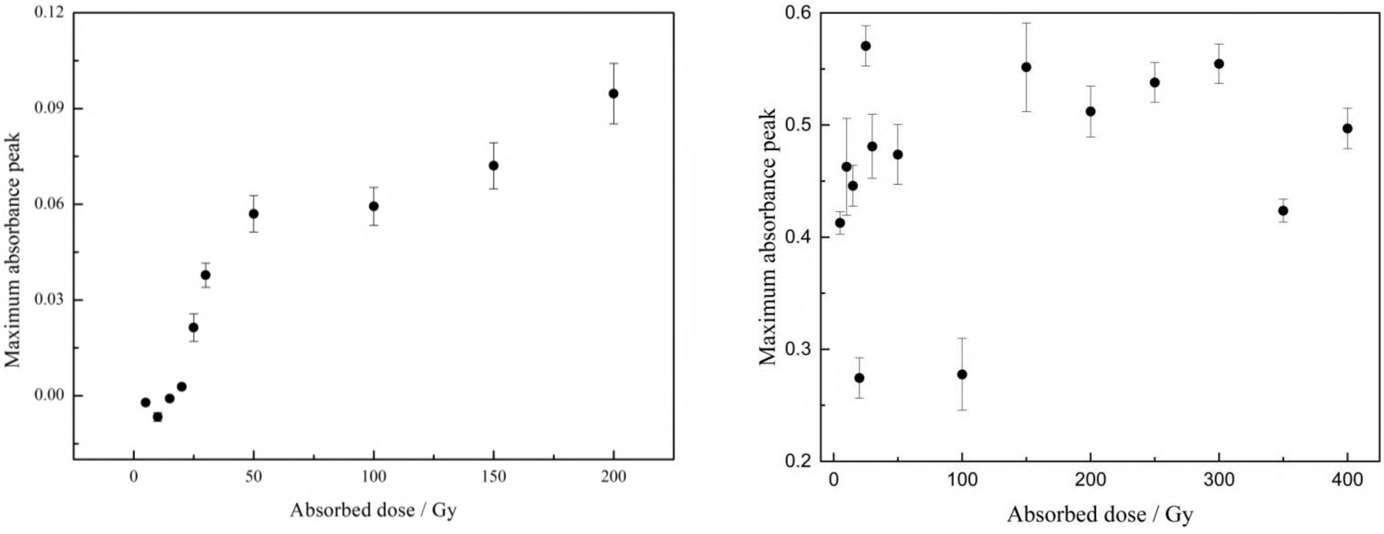
The maximum absorbance band appeared at approximately 530 nm in Fricke gel with salicylic acid, corresponding to the absorption of the salicylic acid-Fe^{3+} complex formed during irradiation. The irradiated FSG absorbed electromagnetic wavelengths within this range and emitted wavelengths corresponding to the remaining visible spectrum, imparting a pinkish tone to the dosimeter. Increased radiation exposure resulted in a higher maximum absorption peak for FSG, which was attributed to producing more Fe^{3+} ions. This led to more chelation reactions between Fe^{3+} ions and salicylic acid molecules, further intensifying the tone of the gel. The probability of forming salicylic acid-Fe^{3+} complexes depended on the concentrations of both Fe^{3+} and this chelating agent.

The FXG spectrum displayed two distinct absorption peaks: one at approximately 442 nm, corresponding to the absorption of XO-Fe^{2+}, and another at 573 nm, associated with XO-Fe^{3+} formed post-irradiation. Hence, the peak at 442 nm decreased as the absorbed dose increased due to oxidation of XO-Fe^{2+}, while the peak at 573 nm increased proportionally with the production of XO-Fe^{3+}. The metallic complexes XO-Fe^{2+} and XO-Fe^{3+} contributed to the orange and violet tones, respectively, observed in the FXG dosimeter.

Figure 3 depicts the dose-response curves for both formulations, using the absorbance at the maximum peak wavelength. All gels showed increased absorbance with increasing absorbed dose, though saturation behavior modified: FXG saturated at lower doses while FSG did not show saturation until absorbed dose of 200 Gy.

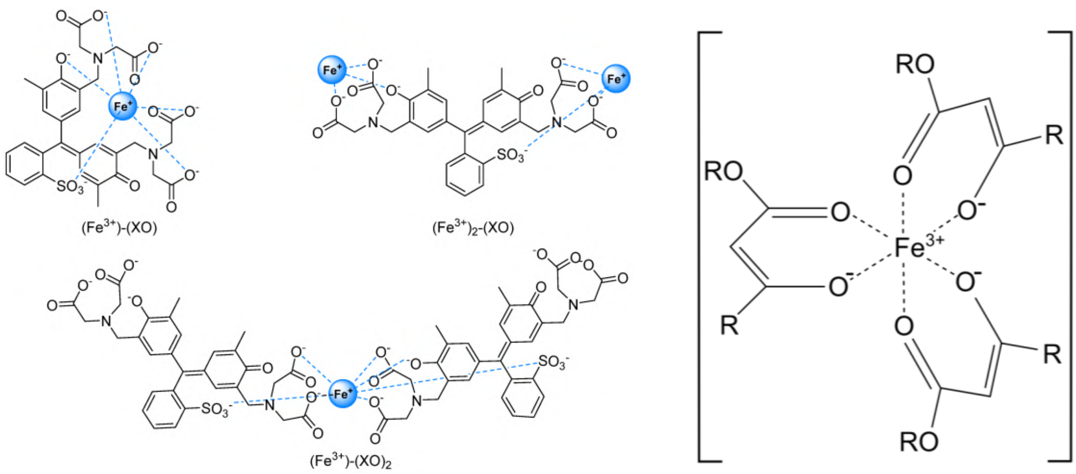
Saturation in the FSG formulation likely resulted from two independent factors. The first involved the depletion of Fe^{2+} in the gel due to irradiation-induced oxidation, which halted Fe^{3+} production and the formation of the salicylic acid-Fe^{3+} complexes. The second factor was the insufficient availability of salicylic acid molecules, which prevented complex formation even with Fe^{3+} production. For FXG, saturation was due to the complete conversion of XO-Fe^{2+} to XO-Fe^{3+}.

Despite identical concentrations of ammonium ferrous sulfate, differences in saturation were influenced by the choice of gelling agent and indicator. Indicators impacted the absorbed dose range of the Fricke dosimeter because the reaction between Fe^{3+} ions and molecular indicators varied depending on the chelating agent. For example, a single Fe^{3+} ion can bind to multiple salicylic acid molecules, unlike xylenol orange, which binds to Fe^{3+} only at its ends (see Figure 4). Hence, formulations with salicylic acid achieved higher saturation points compared to FXG.



(a) (b)

Figure 3. Spectrophotometric dose response of (a) FSG and (b) FXG as a function of absorbed dose.



(a) (b)

Figure 4. Chemical structures of (a) the XO-Fe^{3+} complex and (b) salicylic acid-Fe^{3+} complex (Scotti, 2022).

4 Conclusions

This study developed and evaluated an alternative Fricke gel dosimeter formulation using agarose as the gelling agent and salicylic acid as an ion indicator. These gels were irradiated at various absorbed doses using a cobalt-60 source that emitted a gamma radiation beam. Molecular absorption spectroscopy in the visible region was employed to measure absorbed doses.

The new formulation using salicylic acid responded to gamma irradiation, showing measurable optical changes. Although FXG, the conventional Fricke gel made with porcine gelatin and xylenol orange, showed highest initial sensitivity, it saturated quickly. Salicylic acid formulation offered extended dynamic ranges, remained unsaturated up to the highest tested dose. Despite slightly lower sensitivity, the benefits of salicylic acid (cost-effectiveness, availability, and non-toxicity) make it a viable alternative to traditional indicators.

Acknowledgements

The authors would like to thank the Centro de Desenvolvimento da Tecnologia Nuclear / Comissão Nacional de Energia Nuclear (CDTN/CNEN) for providing the infrastructure that enabled this study. The authors are grateful to the Brazilian funding agencies: Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES), Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq), and Fundação de Amparo à Pesquisa do Estado de Minas Gerais (FAPEMIG) for their financial support. This work was conducted as part of the Brazilian Institute of Science and Technology for Nuclear Instrumentation and Applications to Industry and Health (INCT.INAIS), CNPQ project 406303/2022–3.

References

Araujo B. C. R., Ferreira B. D. L., Virtuoso L. S., Meira-Belo L. C., Fonseca T. C. F., Santos A. M. M., Lula I., and Sebastião R. C. O. (2021) A new formulation for polymer Fricke dosimeters and an innovative application of neural networks to study dose profiles from spin-echo NMR data. *Radiation Physics and Chemistry* 184:1–9.

Attix F. H. (2004) *Introduction to Radiological Physics and Radiation Dosimetry*. Wiley-VCH, Weinheim.

Bailey F. E. J. (2012) *Poly (ethylene oxide)*. Academic Press, New York.

Chu K. C., Jordan K. J., Battista J. J., Van Dyk J., and Rutt B. K. (2000) Polyvinyl alcohol–Fricke hydrogel and cryogel: two new gel dosimetry systems with low Fe^{3+} diffusion. *Physics in Medicine and Biology* 45:955–969.

Courbon F., Ibarolla D., Caselles O., Francois P., Bonnet A., and Berry I. (1999) Internal dosimetry using magnetic resonance imaging of polymer gel irradiated with iodine-131: Preliminary results. *European Journal of Nuclear Medicine* 26:1181–1181.

Éder da Silva e Sá (2006) *Determinação espectrofotométrica de ácido salicílico em produtos dermatológicos*. Universidade Federal de Santa Catarina, Centro de Ciências Físicas e Matemáticas, Departamento de Química.

ICRU REPORT 80 (2008) Dosimetry systems for use in radiation processing. Journal of the ICRU 8(2).

Rae W. I. D., Willemse C. A., Lötter G., Engelbrecht J. S., and Swarts J. C. (1995) Chelator effect on ion diffusion in ferrous-sulfate-doped gelatin gel dosimeters as analyzed by MRI. *Medical Physics* 23:15–23.

Schreiner L. J. (2004) Review of Fricke gel dosimeters. *Journal of Physics: Conference Series* 3:9–21.

Scotti M., Arosio P., Brambilla E., Gallo S., Lenardi C., Locarno S., Orsini F., Pignoli E., Pedicone L., and Veronese L. (2022) How xylenol orange and ferrous ammonium sulphate influence the dosimetric properties of PVA–GTA Fricke gel dosimeters: A spectrophotometric study. *Gels* 8:1–17.

Assembly and computer simulation by Geant4 of a ^{3}He detector for cosmic radiation in the ground

Ranulfo da Silva Dias[1,2], Thiago Augusto do Carmo[1,2], Felipe Lopes Frigi[1,2], Guilherme Gazolla Santana[1,2], Roberto Bedogni[5], Odair Lélis Gonçalez[1,2], Maurício Tizziani Pazianotto[1,2], Edson Ramos de Andrade[2,3,4], Claudio Antonio Federico[1,2,*]

[1] Aeronautics Technological Institute (ITA), Brazilian Air Force, São Paulo, Brazil

[2] Institute for Advanced Studies (IEAv), Brazilian Air Force, São Paulo, Brazil

[3] Defense Engineering Graduate Program, Military Institute of Engineering (IME), Rio de Janeiro, Brazil

[4] University of Western São Paulo, (UNOESTE), São Paulo, Brazil

[5] National Institute of Nuclear Physics - Frascati National Laboratories (LNF), Rome, Italy

[*] claudio.federico40@gmail.com

Abstract

This paper describes the development, experimental characterization, and computer simulation implemented in Geant4 of a neutron detection system based on helium-3 (^{3}He) proportional tubes, with the aim of monitoring cosmic radiation at ground level, with special attention to the South Atlantic Magnetic Anomaly (SAMA) region. Primary cosmic radiation, when interacting with the Earth's atmosphere, generates secondary particles, among which neutrons are of particular interest due to their high penetrating power and significant contribution to the ambient dose at various altitudes. The experimental system consists of an array of 20 ^{3}He detectors operating at a pressure of 4 atm and 6 atm. The tubes are inside a structure made of polyethylene and lead moderators. The assembly was set up at the Ionizing Radiation Laboratory (LRI) of the Institute for Advanced Studies (IEAv), and measurements were taken using a radioactive source of ^{241}Am-Be, known to emit neutrons with an average energy between 3 and 5 MeV. The results showed good agreement between the experimental and simulated data regarding the spatial distribution of the detector response. The model proved to be robust and can be improved for future applications. The final installation of the system will be carried out at the Pico dos Dias Observatory (MG), in Brazópolis, Minas Gerais (MG), Brazil (1840 m altitude), providing continuous monitoring of cosmic radiation in the national territory and data on SAMA.

Keywords: Helium-3 Detector; Cosmic Radiation; Neutron; South Atlantic Magnetic Anomaly.

1 Introduction

Primary cosmic radiation consists mainly of protons, alpha particles, and high-energy heavy nuclei, which interact with the Earth's atmosphere to produce an extensive cascade of secondary particles, including mesons, muons, electrons, photons, and neutrons Heinrich et al. (1999). As they penetrate the innermost layers of the atmosphere, these secondary particles undergo multiple interactions, losing energy mainly through ionization and inelastic scattering processes, resulting in a gradual reduction in average energy and an increase in particle flux at ground level.

Among these particles, neutrons are particularly relevant for radiation monitoring applications, as they have high penetrating power and contribute significantly to the ambient dose equivalent at high altitudes and at ground level. Efficient detection of these neutrons is essential for assessing exposure in aircraft and in specific geographic regions, such as the South Atlantic Magnetic Anomaly (SAMA), where the Earth's magnetic field has different characteristics from the rest of the globe Almeida Filho et al. (2023).

Helium-3 (^{3}He) detectors stand out as one of the most effective solutions for detecting thermal neutrons, due to their high capture cross section (approximately 5330 barns for thermal neutrons), high selectivity, and low sensitivity to photons. Knoll (2010). The fundamental nuclear reaction used is:

$$^{3}\mathrm{He} + \mathrm{n} \rightarrow ^{3}\mathrm{H} + \mathrm{p} \;\; (Q = +0,764\ \mathrm{MeV})$$

in which the neutron is captured by ^{3}He, resulting in the production of tritium (3H) and a proton (p). The energy released is deposited in the gas, generating ionizations that are collected as an electrical signal. This process gives the detector high efficiency for low-energy neutrons, also allowing the detection of fast neutrons when coupled with suitable moderators. The implementation of fixed ground-based neutron flux monitors in Brazil is strategic for providing regional data, supporting estimates of ambient dose equivalent on board aircraft, and characterizing the specific impact of the South Atlantic Magnetic Anomaly (SAMA). However, to date, most measurements are performed globally without detailed coverage of national airspace Federico et al. (2013, 2010).

The study proposes the assembly, characterization, and experimental validation of a neutron detection system with ^{3}He proportional tubes operating at pressures of 4 atm and 6 atm, coupled with different thermalization materials. The experimental measurements were performed at the Ionizing Radiation Laboratory (LRI) of the Institute for Advanced Studies (IEAv), using a ^{241}Am-Be source to verify the response. In addition, a detailed Monte Carlo simulation was developed in the Geant4 code, replicating the geometry of the system and the experimental parameters, allowing for a critical comparison between simulated and experimental data. The final installation of the detector will be carried out in an experiment container maintained by IEAv at the Pico dos Dias Observatory in Brazópolis-MG, Brazil (1840 m altitude), with the objective of continuously monitoring cosmic radiation on Brazilian soil, providing support for aerospace applications and studies related to radiological safety in commercial flights.

2 Methodology

2.1 System with ^{3}He detectors

The neutron detector system consists of two polyethylene moderator layers, one upper and one lower (73.9 × 75.5 × 9.5 cm), with a central lead layer (73.9 × 74.5 × 3 cm), whose function is to promote the multiplication of high-energy neutrons. The base is an iron plate (95.2 × 95.3 × 1.0 cm), supported on a wooden pallet containing slats arranged in three bases, providing stable support. Around the base, there is a hollow polyethylene box, similar to a wall (89.2 × 89.2 × 13 cm; thickness de 4.1 cm) covered by a hollow lead box (80 × 80 × 12.5 cm; thickness 3 cm).

Structure 1 contains holes for the detectors to be 24.25 cm from the edge, with a spacing of 47.42 mm, a radius of 27.51 mm, and a depth of 77 cm on one side of the sides, which allows for the installation of ^{3}He detectors, responsible for capturing neutrons.

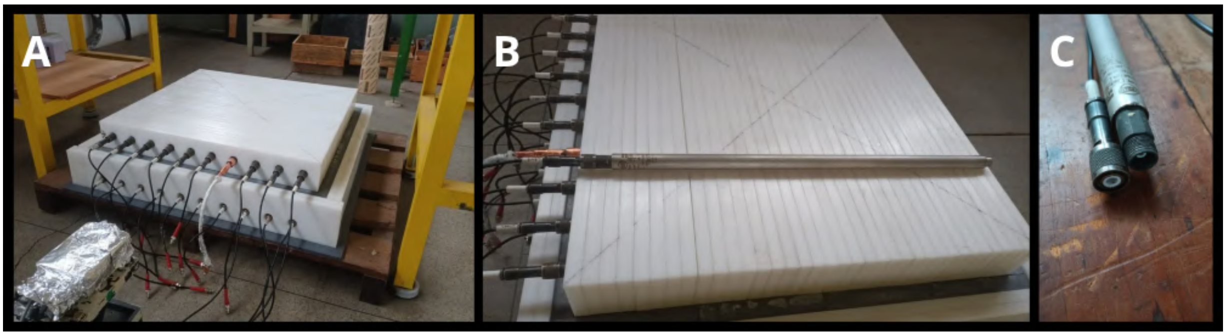


Figure 1: View of the system in the laboratory: (A) Front view of the detector; (B) Side view with the detector at the top as a comparison for the size of the system; (C) Detail of the detector connectors.

The detectors used were manufactured by Harshaw (Allied Chemical Corporation) in 1975 and are 71.8 cm long and 25.55 mm in diameter. The connectors are 4.2 cm long and 1.0 cm in diameter. The active region filled with ^{3}He gas has an approximate length between 64 and 66 cm.

To investigate the internal structure, an X-ray was performed 2. In the image, inside the red square, a white volume can be seen near the base of the connector, suggesting the presence of an internal ring. Considering common manufacturing practices, this ring is likely made of ceramic, used as an insulator in detectors of this type. Its estimated dimensions are approximately 0.3 cm in length and a radius of around 12.05 mm Fares et al. (2021).

Figure 2: Detector X-ray image

In total, the system has 20 proportional counters, with 10 of 4 atm detectors installed in the upper layer (top) and 10 of 6 atm detectors positioned in the lower layer (base). All these geometric and material characteristics of the detectors were implemented in the simulation developed in Geant4.

2.2 Experimental measurements

Figure 3 shows how the detectors are connected in parallel to a standard NIM signal analysis system, whose neutron detection pulses from the detector array are digitized in 512 channels and stored in the form of a pulse amplitude spectrum connected to a multichannel system installed on a computer. The signal integration time in the experimental testing stages was 5 min (300 s). The applied voltages were 1400 V for the 4 atm detector and 1600 V for the 6 atm detector.

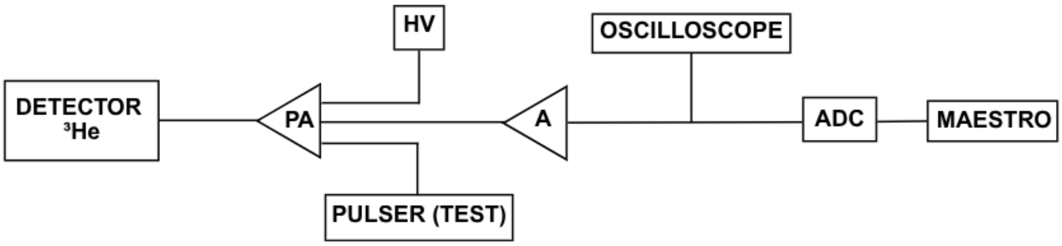


Figure 3: Block diagram of the electronic reading system of the ³He neutron detector. The signal is amplified by a preamplifier (PA) and a linear amplifier (A), observed on an oscilloscope, digitized by an analog-to-digital converter (ADC), and processed by MAESTRO software. A high-voltage (HV) source polarizes the detector, and a pulser is used for system testing.

These tests sought to evaluate the efficiency of the system of 20 ³He detectors and their positions for the neutron spectrum of the ^{241}Am-Be source with average energy between 3 and 5 MeV. The source has an activity of $Q = 3{,}7 \times 10^9$ Bq with a neutron emission rate of Q = $(2{,}45 \times 10^5 \pm 7359)$ n/s.

The relative response of the ³He tubes is obtained to ensure that all detectors have the same pulse amplitude response, with a variation of 3% to 5% in detection efficiency between them. To this end, individual tests were performed on the different detectors under the same test conditions (same hole), and the background of the detectors was measured without the source in order to subtract the noise from the ambient cosmic radiation itself. At the LRI, the ^{241}Am-Be source was positioned 68 cm from the surface of the upper polyethylene layer at its normal incidence. In addition to this experimental configuration, keeping the source at the same height of 68 cm, at a distance of 100 cm from the center of the system, the source was also positioned to irradiate the side and rear of the system.

Detection efficiency is determined experimentally with known emission sources and, when these are unavailable, through computer simulations Pazianotto et al. (2011). The efficiency depends on the energy spectrum of the neutrons. The procedure for measuring efficiency (ε), expressed in cont.cm^2/n, is based on the calculations described below. The count rate recorded by the detector (R in counts/s) is given by:

$$R = \varepsilon\phi \tag{1}$$

Where ϕ (n/cm^{2}s) is the incident neutron flux.

The neutron flux from the source located at a distance d (in meters) from the detector, considering the isotropic point source approximation, is given by:

$$\phi = \frac{Q}{4\pi d^2} \rightarrow \frac{\phi_o}{d^2} \tag{2}$$

Where Q is the neutron emission rate of the source (in n/s) and ϕ_o is the flux at unit distance (d=1m).

Combining equations 1 and 2, the detection efficiency is obtained from the experimental measurement of the count rate R and the value of the unit flux ϕ_o of a previously calibrated source.

$$\varepsilon = \frac{d^2 R}{\phi_o} \tag{3}$$

2.3 Computational Simulation

The simulation was performed via Monte Carlo using Geant4 software (version 11.02). Figure 4 shows different views of the geometric model, highlighting the materials and arrangement of the detectors.

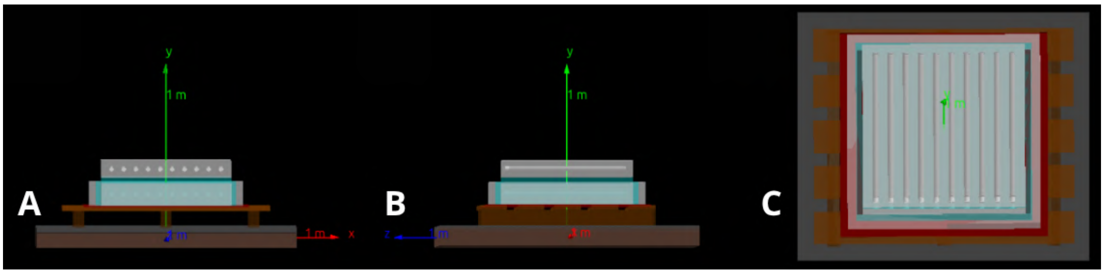


Figure 4: Visualization in Geant4: (A) Front view; (B) Side view; (C) Top view with ³He tubes.

The General Particle Source (GPS) generator was used, with the ^{241}Am-Be source of the ISO $8529-1:2001$ ISO (2001). The physics list is `QGSP_BIC_HPT`, suitable for thermal neutron transport and hadronic interactions. To account for the resulting protons, the *G4PSPopulation* class associated with the *G4SDParticleFilter* filter was used. The scattering matrix $S(\alpha,\beta)$ was applied to correct cross sections in bound and free hydrogen Folger et al. (2004) Boudard et al. (2013) Wright and Kelsey (2015).

Table 1 shows the materials and their respective densities used in the construction of the detector geometry in Geant4, defined from the NIST (National Institute of Standards and Technology) database. CERN (2025) EMBRAPA (2025) PNNL (2021).

Table 1: Materials used and their respective densities in GEANT4.

Material	**Name in Geant4**	**Density (g/cm³)**	**Notes**
Air	G4_AIR	~0.00120479	Default GEANT4 value at 1 atm and 20°C
Polyethylene	G4_POLYETHYLENE	0.94	Default GEANT4 value
Lead	G4_Pb	11.35	Default GEANT4 value
Iron	G4_Fe	7.874	Default GEANT4 value
Concrete	G4_CONCRETE	2.3	Default GEANT4 value
Soil	Soil	1.5	Manually defined in the code
Wood (Cellulose)	Wood	0.7	Manually defined in the code
Stainless steel	G4_STAINLESS-STEEL	8.0	Default GEANT4 value
Ceramic (Alumina)	Ceramic	3.95	Manually defined in the code
Aluminum	G4_Al	2.699	Default GEANT4 value
^{3}He (4 atm)	Helium3_4atm	0.0050	Manually defined in the code
^{3}He (6 atm)	Helium3_6atm	0.0075	Manually defined in the code

Air was defined as the material of the world in Geant4, serving as the medium for constructing the geometry of the system. Concrete and soil were implemented as structural base layers, while wood was used to represent the pallet. The detector body was modeled in stainless steel, the connectors in aluminum, and ceramics as internal insulation. Finally, lead and polyethylene were used in the shielding and moderation regions surrounding the system.

For ^{3}He density was calculated using the ideal gas equation, where $R = 0.082$ L·atm·mol^{-1}·K^{-1}, $T = 296.15$ K, and the volume V was estimated at 0.246 L. The results of the calculations are presented in Table 2.

Table 2: Parameters calculated for ^{3}He gas in detectors.

Pressure (atm)	**n** (mol)	**m** (g)	**V** (L)	**Density** (g/L)
4	0.040	0.12	0.246	0.50
6	0.061	0.18	0.246	0.74

The source was positioned aligned with the central axis of the upper polyethylene plate, 68 cm from the center of the detector, with isotropic emission in a 55° cone angle, covering the entire detector area and a 75° angle for firing at the back and right side. In Geant4, 55° and 75° are the half-angles defined between the central axis of the cone and its wall. The simulation defines these angles so that all particles remain within this cone, but this does not represent the actual physical behavior of the emission, being a resource to optimize computational processing and accelerate the obtaining of results. Due to the angular limit of the source used in the simulations, a correction factor was applied based on the fraction of the solid angle covered in relation to the total space of 4π steradians, given by the following expression:

$$\Omega = 2\pi(1 - cos(\theta)) \tag{4}$$

Where θ is the angle of the cone. The correction factors for 55º and 75º are $\frac{\Omega}{4\pi} \approx 0.213$ and $\frac{\Omega}{4\pi} \approx 0.373$, respectively. These factors were applied by multiplying the simulated results to reflect the total isotropic emission scenario. This procedure ensures the compatibility of the simulated data with real experimental situations and allows direct comparisons between different geometric and angular configurations. In addition, to ensure good statistics, 2.000.000.000 events were simulated Federico et al. (2009).

3 Results and Discussion

The graph in figure 5 shows the average population of protons recorded in the interaction in the 4 atm ^{3}He detectors for both the simulated and experimental data, with the source in the central region at a height of 68 cm at normal incidence. The simulation curve shows a larger population of protons mainly in the central detectors than at the ends because they are lost due to distance or thermalized by the material. In this graph, the simulation curve was above the experimental curve, both in the order of 10^{-3} and 10^{-4}.

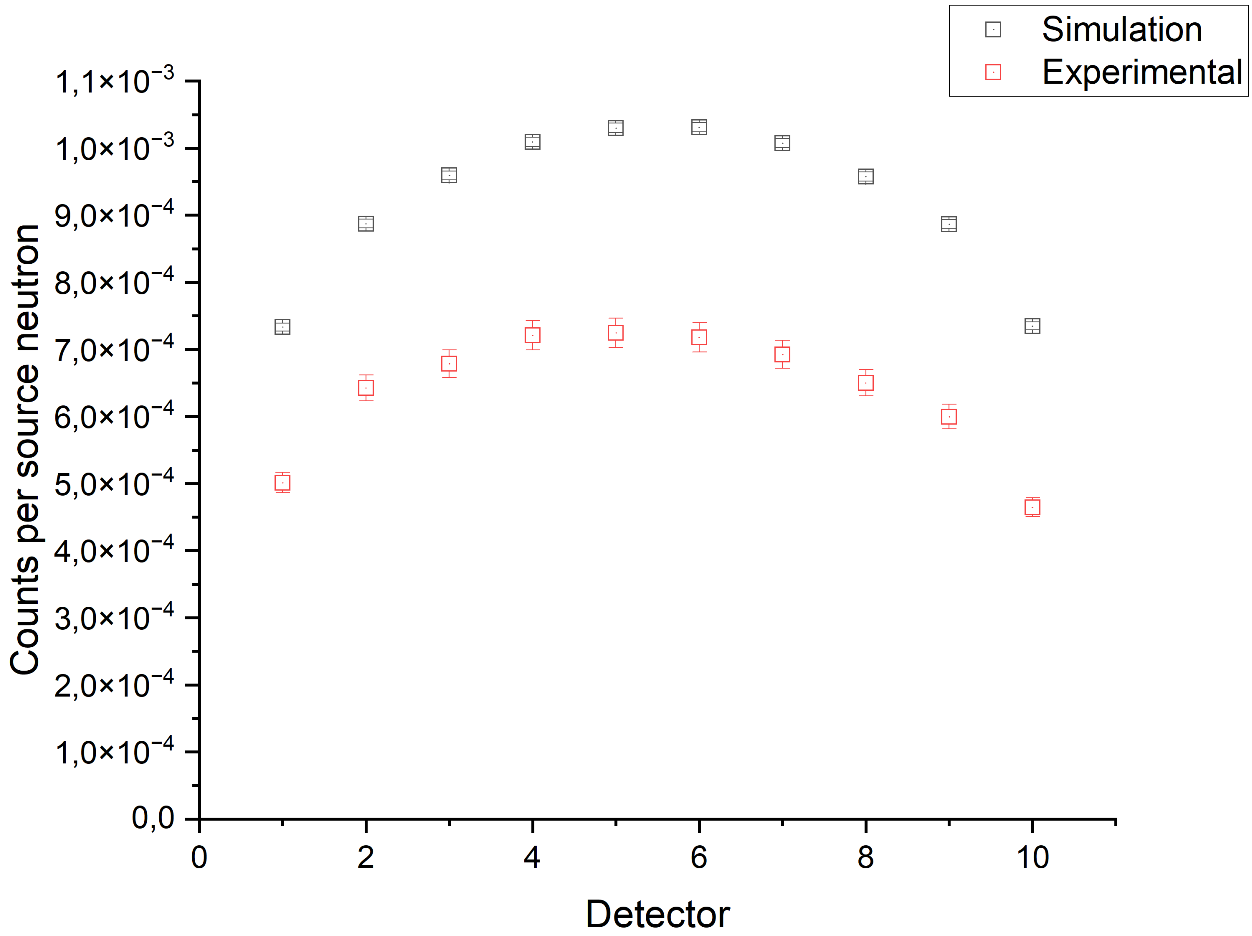


Figure 5: Average population of protons detected for the simulated and experimental central shot for the 4 atm detectors.

The graph in figure 6 shows the average population of protons recorded in the 6 atm ^3He detectors for both the simulated and experimental data in the order of 10^{-3} and 10^{-4}. The behavior of the curve resembles that of figure 5.

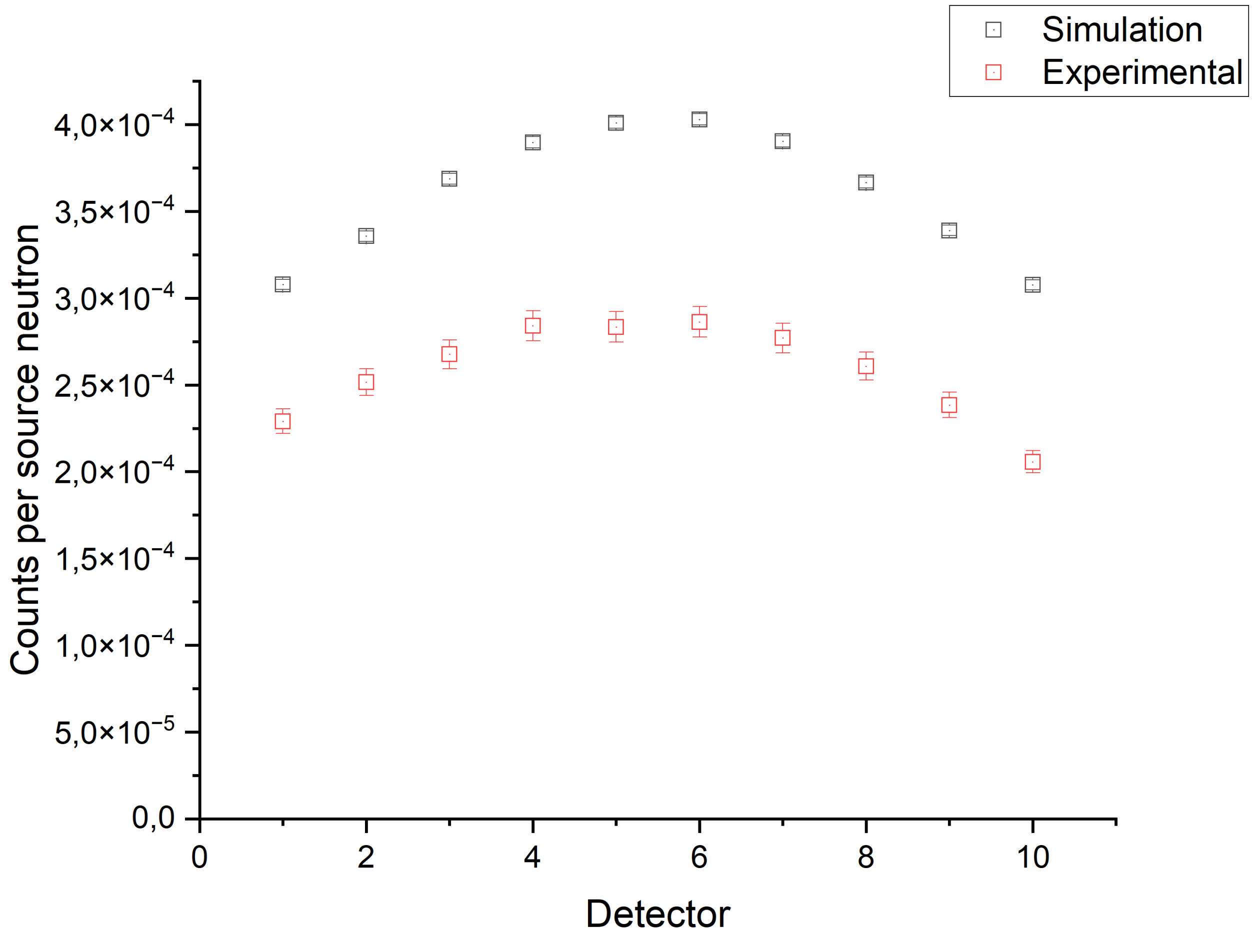


Figure 6: Average population of protons detected for the simulated and experimental central shot for the 6 atm detectors.

Figure 7 shows a comparison of experimental and simulated data for firing in the rear region of the detection system. The simulated data was 4 atm higher than the experimental data. The simulated data curve was distant from the experimental data, but followed the same behavior. All values for 4 atm, both experimental and simulated, were in the order of 10^{-4}.

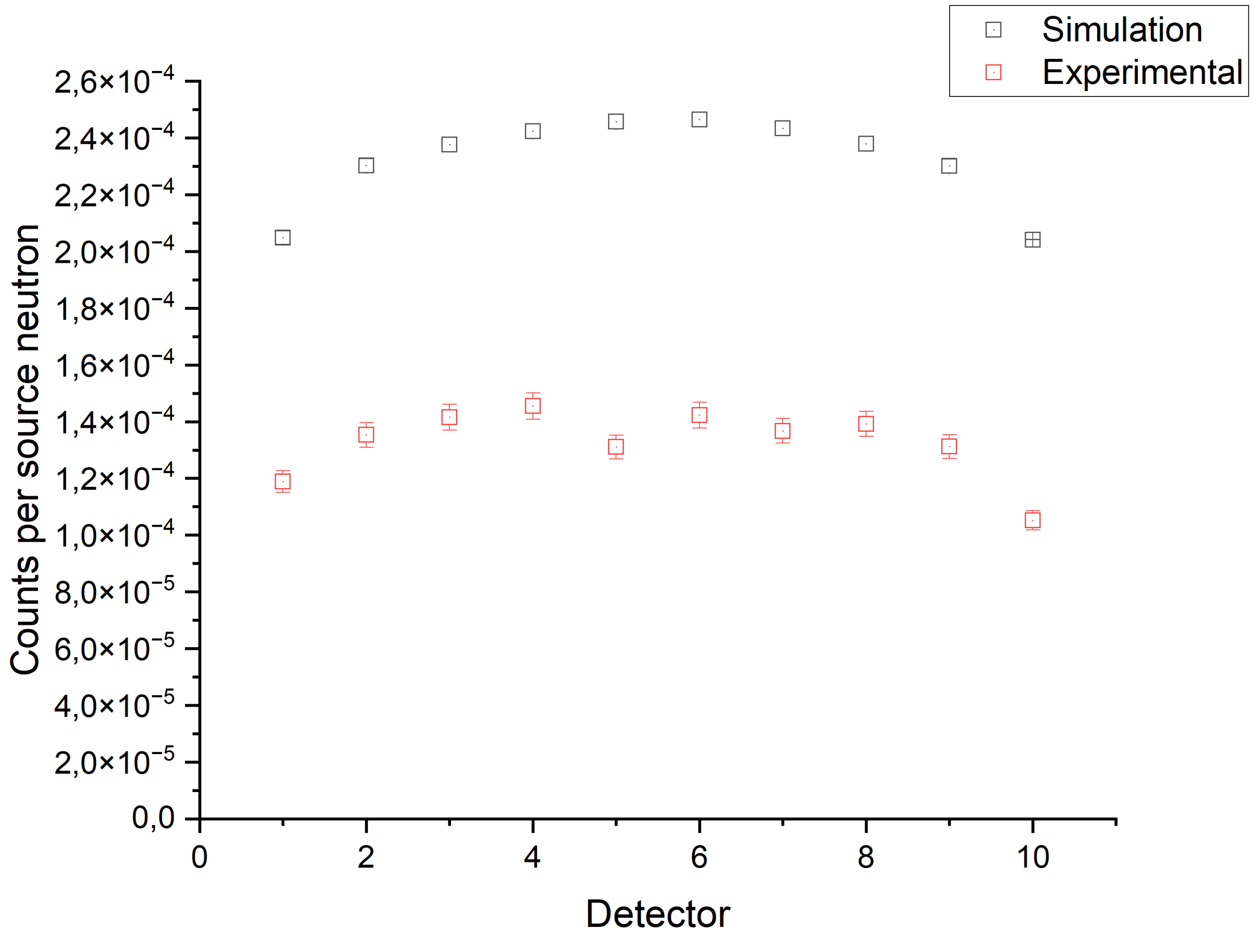


Figure 7: Average population of protons detected for the simulated and experimental back shot for the 4 atm detectors.

Figure 8 shows a comparison of experimental and simulated data for back firing. The simulated values were above the experimental value of 6 atm, although they are closer to the results shown in figure 7. The values were in the order of 10^{-5}.

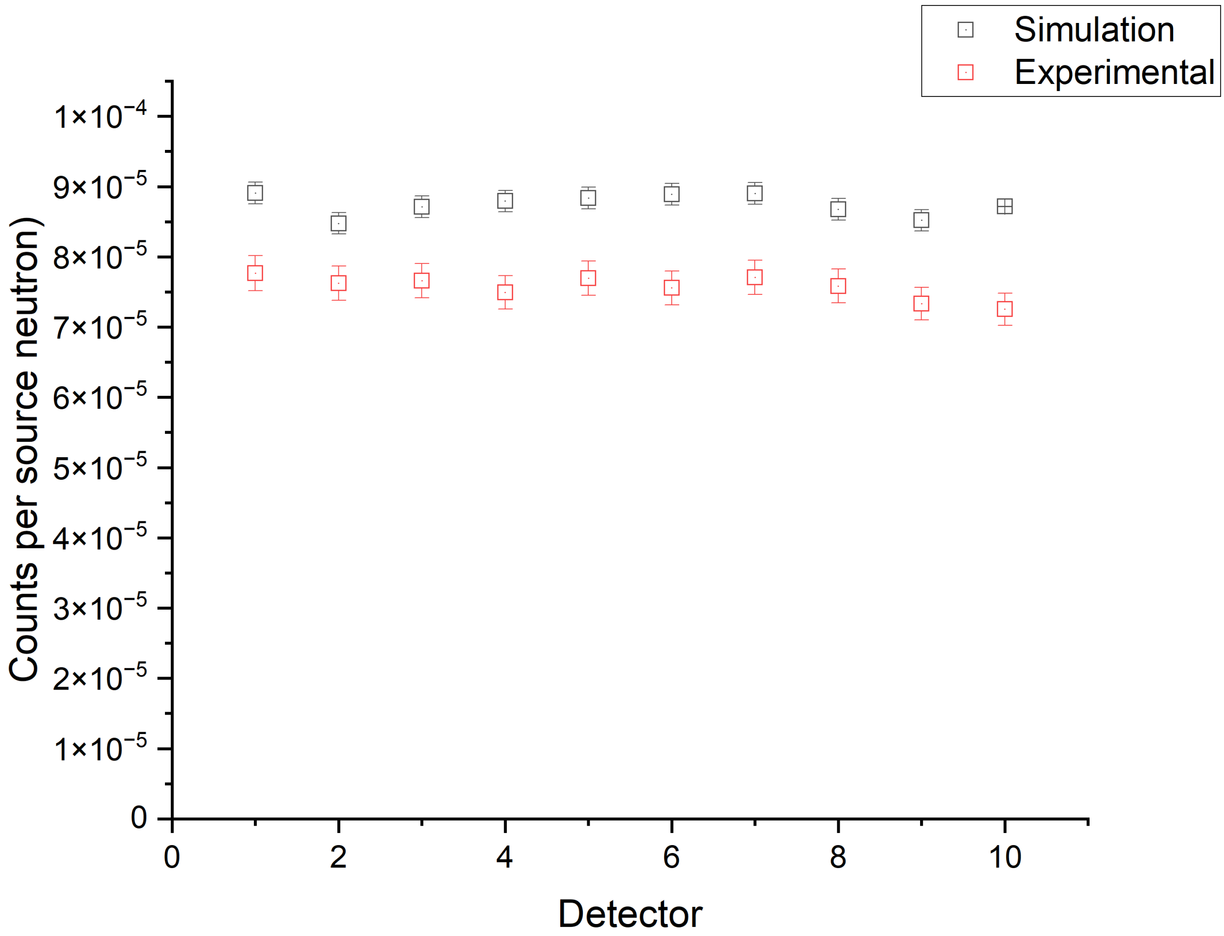


Figure 8: Average population of protons detected for the simulated and experimental back shot for the 6 atm detectors.

Figure 9 shows a comparison of experimental and simulated data for the shot on the side of the detector with the source next to detector 10. The simulated values were higher than the experimental values for the 4 atm detectors, although both curves maintain the same trend. The experimental results showed the detectors close to the source having values in the order of 10^{-4} and the three most distant ones in the order of 10^{-5}. The simulation follows this characteristic, but only the first detector remained at 10^{-5}.

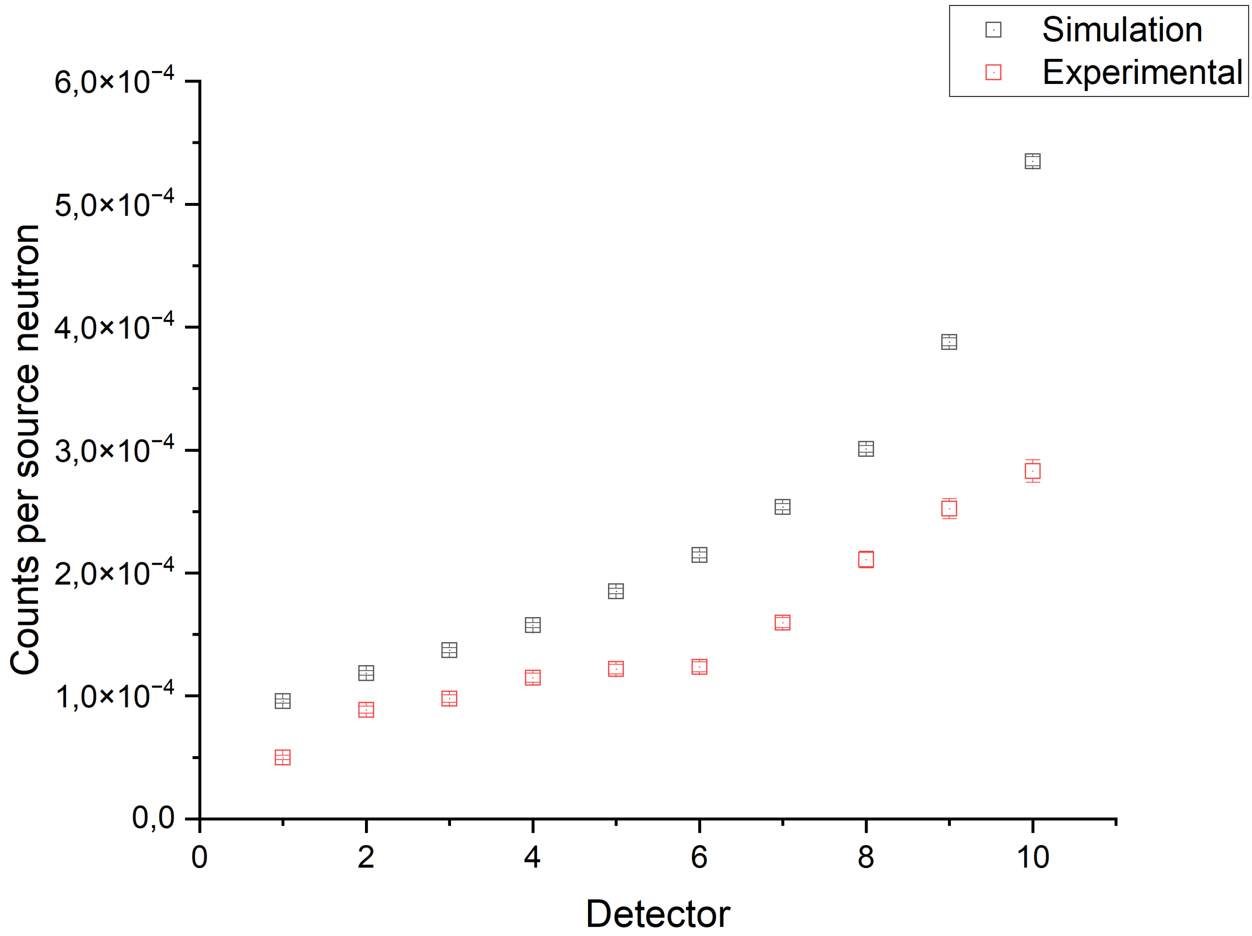


Figure 9: Average population of protons detected for the simulated and experimental side shot for the 4 atm detectors.

Figure 10 shows a comparison of experimental and simulated data for the side firing of the 6 atm detectors. The experimental data were above the simulated 6 atm data, except for detector 10, which was closest to the source. The simulated values were in the order of 10^{-5}, except for detectors 9 and 10, which were in the order of 10^{-4}. The experimental results followed this trend, but with detector 8 registering in the order of 10^{-4}.

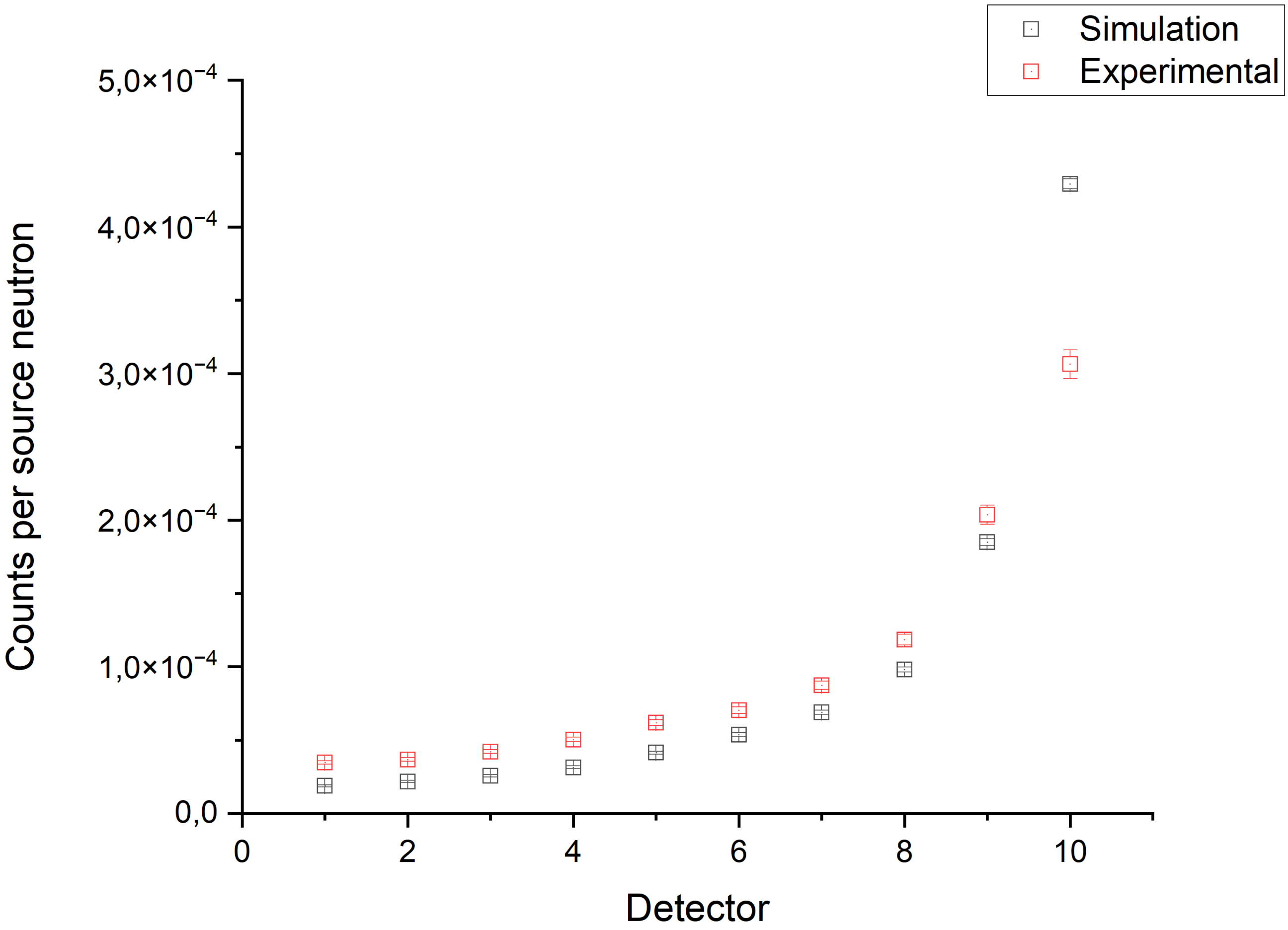


Figure 10: Average population of protons detected for the simulated and experimental side shot for the 6 atm detectors.

Despite the discrepancies, there is good agreement in the behavior between the detectors. In both data sets (simulated and experimental), the detectors with the highest response appear in similar positions. This demonstrates that the computational model is effective in reproducing the spatial distribution of neutrons and the relative response of the detectors. The simulation is suitable for predicting the performance of detectors in different positions, making it a useful tool for optimizing the geometric arrangement and evaluating the overall response of the system. However, to obtain reliable estimates of absolute detection efficiency, further refinement of the model is necessary. In addition, there may be inconsistencies between the material properties derived from the NIST library and those of the detection system. In addition to further investigating the geometry in Geant4 in terms of its positioning and dimensions. The uncertainties associated with the simulations in Geant4, estimated from the RMS (Root mean square) divided by the root of the number of events ($N = 10^8$), varied in the order of 10^{-6}. However, even with small uncertainties, there is a difference between the simulated and experimental values, suggesting that the discrepancy is not associated with the Monte Carlo statistical error, but rather with aspects of physical modeling or other aspects that will be investigated.

4 Conclusions

The present study is under development. At this stage, it has experimentally and through simulation presented the application of ^{3}He detectors at 4 and 6 atm to monitor cosmic radiation neutrons in the ground. The tests showed good uniformity among the detectors, with variation between 3% and 5%. The simulations performed in Geant4 reproduced the behavior observed in the experimental data, although with a discrepancy, which is being investigated. Even so, the model proved to be reliable, serving as a basis for future adjustments, especially in relation to the density of the materials used. The next steps include adjustments to the model and detailing of the internal components, seeking greater precision in the measurements, when the overall response of the system to the cosmic radiation spectrum incident at the intended operating site will also be estimated through computer simulation. Finally, the system will be installed at the Pico dos Dias Observatory, providing unprecedented data for the SAMA region and contributing to studies of radiation in flights and space weather.

Acknowledgements

The authors thank partial funding from the ERISA-D project from the Institute for Advanced Studies (IEAv), as well as support from the Advanced Scientific Computing and Modeling Laboratory (Lab-CCAM) at the Aeronautics Technological Institute (ITA) for providing the computational infrastructure used in the simulations and CNPQ (process 420156/2023-2), (processes n. 420156/2023-2) and Grants PQ (310745/2025-0), (304636/2023-1 and 409490/2023-7) for the partial financial support.

References

Almeida Filho D., Federico C., da Silva F., and Souza-Santos D. (2023) Evaluation of neutron doses in brazilian aircrew by the monte carlo method. *Radiation Measurements* 166:106959

Boudard A., Cugnon J., David J.-C., Leray S., and Mancusi D. (2013) New potentialities of the liège intranuclear cascade model for reactions induced by nucleons and light charged particles. *Physical Review C—Nuclear Physics* 87:014606

CERN (2025) Appendix – material names. geant4 user documentation for application developers. `https://geant4-userdoc.web.cern.ch/UsersGuides/ForApplicationDeveloper/html/Appendix/materialNames.html`. acesso em: 24 jul. 2025

EMBRAPA (2025) 5ª edição do sistema brasileiro de classificação de solos (sibcs). `https://www.agroapi.cnptia.embrapa.br/portal/assets/docs/SiBCS-2018-ISBN-9788570358004.pdf`. acesso em: 9 jul. 2025

Fares M., Messai A., Begaa S., Salem M., Negara K., Debili M. Y., and Messaoudi M. (2021) 3he proportional counter development for thermal neutron detection. *Radiation Detection Technology and Methods* 5:264–272

Federico C. A., Gonçalez O. L., Fonseca E. S. d., Patrão K. C. d. S., Pereira M. A., and Caldas L. V. E. (2013) Verification of response of neutron monitor for in-flight neutron dosimetry. *Journal of Aerospace Technology and Management* 5:315–322

Federico C. A., Gonçalez O. L., Fonseca E. S., Martin I. M., and Caldas L. V. E. (2010) Neutron spectra measurements in the south atlantic anomaly region. *Radiation Measurements* 45:1526–1528

Federico C. A., Oliveira W. A. d., Pereira M. A., and Gonçalez O. L. (2009) Avaliação da resposta de um contador do tipo long-counter para nêutrons do ^{241}am/be. *Journal of Aerospace Technology and Management* 01:243–246

Folger G., Ivanchenko V., and Wellisch J. P. (2004) The binary cascade: nucleon nuclear reactions. *The European Physical Journal A-Hadrons and Nuclei* 21:407–417

Heinrich W., Roesler S., and Schraube H. (1999) Physics of cosmic radiation fields. *Radiation protection dosimetry* 86:253–258

ISO S. (2001) Reference neutron radiations–part 1: Characteristics and methods of production. *Int Stand Organ* pages 8521–9

Knoll G. F. (2010) *Radiation Detection and Measurement*. John Wiley & Sons

Pazianotto M. T., Federico C. A., Gonçalez O. L., and Carlson B. V. (2011) Evaluation of the neutron detector response for cosmic ray energy spectrum by monte carlo transport simulation. in *Proc of XIII WONP '2011 and VII NURT '2011*. InSTEC-CEADEN, Havana

PNNL P. (2021) Data mining analysis and modeling cell compendium of material composition data for radiation transport modeling

Wright D. and Kelsey M. (2015) The geant4 bertini cascade. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 804:175–188

System for Visual Detection of Individual Dosimeters Using Convolutional Neural Networks

A. Vaz[1*], F. Mamed[1], P. Silveira[1], W. Vallory[1]

[1] Instituto Militar de Engenharia, Nuclear Engineering Department, 22290-270, Rio de Janeiro, Brazil.

[*] alessandra.vaz@ime.eb.br

Abstract

Monitoring occupational exposure to ionizing radiation in controlled environments requires the strict use of individual dosimeters. However, the effectiveness of radiological protection systems can be compromised by human errors, such as improper use or absence of these devices. This research proposes an artificial intelligence-based approach for the automatic detection of individual dosimeters using computer vision. A YOLOv8-type architecture trained with real images was used to develop a model capable of identifying and classifying different types of dosimeters worn over operational clothing. The system was evaluated using specific object detection metrics, achieving satisfactory results in terms of precision (0.89), recall (0.96), and mAP@50 (0.97). These results indicate high reliability in visual identification across various operational scenarios. The implementation of this technology in nuclear and hospital facilities has the potential to provide additional support for personal dosimetry, enhancing radiological control and safety through automated real-time verification systems.

Keywords: Personal dosimetry; Computer vision; YOLOv8; Object detection; Radiological safety

1 Introduction

Radiological safety is a critical component in environments that operate with sources of ionizing radiation, such as nuclear facilities, research centers, and healthcare units. In these settings, the protection of individuals occupationally exposed to radiation is a priority, given the need to mitigate health risks associated with high doses. Among the practices adopted, individual exposure monitoring plays a central role, with dosimeters being the primary tools for quantifying the radiation dose absorbed by workers. The use of these devices allows continuous tracking of exposure and the implementation of corrective measures when necessary [6].

In Brazil, the National Nuclear Energy Commission (CNEN), through regulation NN-3.01: Basic Guidelines for Radiological Protection, establishes the basic requirements for radiological safety. This regulation covers various exposure situations and requires that employers and license holders implement an Occupational Radiological Monitoring Program (PMRO). For external exposure assessment, individual monitoring of Occupationally Exposed Individuals (OEIs) must be carried out using personal dosimeters. These devices should be positioned at the most exposed point of the chest and calibrated according to the personal dose equivalent quantity (H_P(10)), allowing for the assessment of the OEI's effective dose [3].

Despite the fundamental importance of dosimeters, the integrity of radiological protection systems can be compromised by operational issues, such as improper use, incorrect positioning, or the absence of the device during exposure. These failures can result in inaccurate records of the effective dose, underestimating actual exposure and hindering the implementation of appropriate preventive measures [4]. Therefore, confidence in individual monitoring depends on both the accuracy of the dosimeter and the consistency of its use.

With recent advances in artificial intelligence, particularly in the field of computer vision, it has become feasible to develop systems capable of interpreting images in real time to detect specific objects in complex environments. Convolutional Neural Networks (CNNs) stand out as one of the most effective architectures for visual recognition tasks and have been widely applied in fields such as public safety, industrial manufacturing, autonomous transportation, and more recently, the nuclear sector [7] [8].

In this context, this study presents an automated system for the visual detection of individual dosimeters using CNNs. The goal is to develop a tool that complements personal dosimetry methods, contributing to the continuous verification of the presence and proper use of dosimeters in controlled areas. To this end, the YOLOv8 (You Only Look Once) object detection architecture is employed, which has proven particularly effective for real-time recognition.

The implementation of this technology has the potential to provide an additional support mechanism for radiological safety. By automating the visual verification of dosimeter usage, the system can act as a control component, reinforcing protection practices and contributing to the maintenance of a safer work environment for professionals exposed to radiation.

2 Methodology

The development of the detection system was carried out using the Python programming language, executed on the Google Colab platform. The architecture adopted was YOLOv8, known for its efficiency in real-time object detection tasks. Images were labeled using the Roboflow tool, and the dataset was split into 70% for training, 20% for validation, and 10% for testing.

The dataset consisted of 723 original images taken at the nuclear detection and instrumentation laboratory of IME – Rio de Janeiro, containing three distinct types of individual dosimeters:

a. TLD Dosimeter (Thermoluminescent) - a passive type, which uses crystals for thermoluminescence reading;

b. Film Dosimeter - also passive, based on photographic film, although currently less used;

c. Pen Dosimeter - a direct-reading device, operating based on ionization chambers.

Figure 1 presents an image with the three types of dosimeters used in this study, properly identified by name and class.

Figure 1 - Dosimeters Used with Their Respective Classes

Source: The Authors (Own).

The images were annotated with the following detection classes: TLD, Film, and Pen, respectively. Figure 2 shows an example of how the images were labeled during the annotation process.

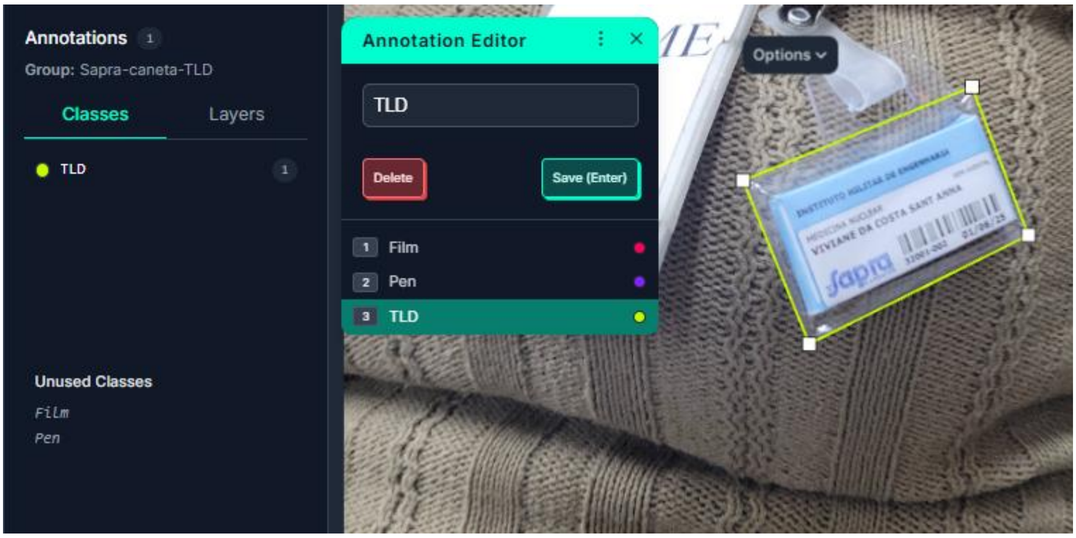


Figure 2- Example of Image Annotation in the Roboflow Tool

Source: The Authors (Own).

The main configurations used during the model training are presented in Table 1:

Table 1 – YOLOv8 Model Training Settings

Parameter	Description	YOLOv8
Number of epochs	Number of training iterations	50
Image size	Image dimensions	640 x 640
Batch size	Number of images per batch	16
Number of classes	Detection classes	3 (TLD, Film, Pen)
Architecture	Model type	YOLOv8
Optimization method	Optimization algorithm	AdamW

To evaluate the model's performance, the results were analyzed using standard computer vision metrics. Indicators such as Precision, Recall, and loss values across epochs were assessed. Overall performance was measured using mAP (mean Average Precision) at different Intersection over Union (IoU) thresholds, including mAP@0.5 and mAP@0.5:0.95. Additionally, the Precision-Recall Curve (PR-Curve) was generated to demonstrate the trade-off between the model's precision and sensitivity in various scenarios. The Confusion Matrix was used to analyze the network's ability to distinguish between classes and to identify false positives and false negatives in relation to the background of the images.

3 Results and Discussion

The automatic detection system developed demonstrated satisfactory performance in the visual recognition of different types of individual dosimeters, establishing itself as a promising tool for application in radiological control environments, such as nuclear instrumentation laboratories, medical areas, and industrial facilities that operate with sources of ionizing radiation.

The model's overall results are summarized in Table 2, which presents the main evaluation metrics.

Table 2 - General Results of the YOLOv8 Model for Dosimeter Detection

Metric	Value
Precision	0.889
Recall	0.969
mAP@0.5	0.977
mAP@0.5:0.95	0.715
Fitness	0.742
Inference time (ms)	0.900
Post-processing time (ms)	2.720

The overall precision of 0.889, meaning that approximately 89% of the detections were correct, indicates a low incidence of false positives. The imprecision rate of approximately 11% (1 - 0.889) is directly related to the network's difficulty in distinguishing some instances of the Film and Pen classes from the image background. This difficulty is detailed in the confusion matrix analysis and can be attributed to factors such as the low contrast between the dosimeter and the environment, as well as possible variations in lighting and capture angles.

The per-class results, with mAP@0.5 values of 0.970 for Pen, 0.967 for Film, and 0.994 for TLD, and an overall average of 0.977, reinforce the model's high performance. The superior performance of the TLD class shows that the model has learned its visual features with high fidelity, which is essential in radiological protection contexts, where the absence or misidentification of dosimeters may compromise occupational safety.

These results are complemented by the graphical representations generated during training. Figure 3 presents the Precision-Recall Curve (BoxPR Curve), which shows the balance between the model's precision and sensitivity at different confidence thresholds.

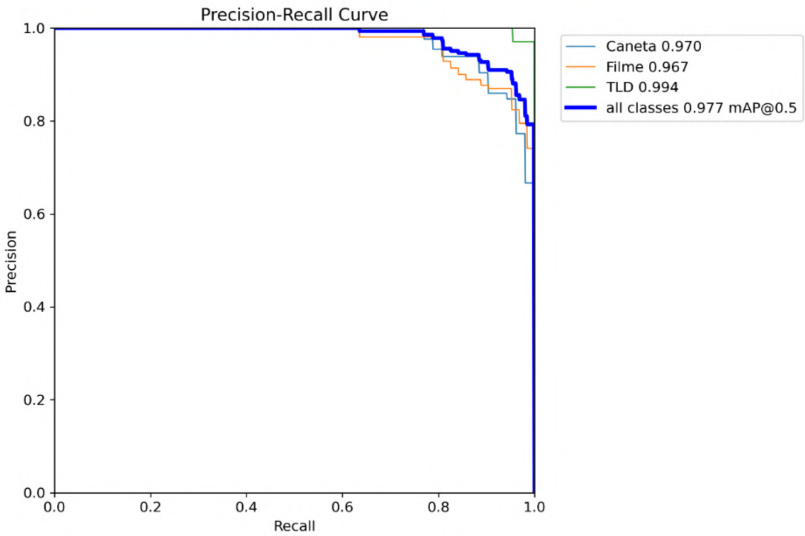


Figure 3 - Precision-Recall (BoxPR Curve)

Source: The Authors (Own).

In order to evaluate the performance of the YOLO-based object detection model training, a set of parameters can be assessed. Table 3 illustrates the parameters analyzed in this work, while Figure 4 provides an analysis of their respective evolution in terms of losses and metrics.

Table 3 - Training and Evaluation Parameters in YOLOv8

Parameter	**Practical Purpose**
train/box_loss	Evaluates whether the model is learning to draw the bounding boxes in the correct locations during training.
train/cls_loss	Evaluates whether the model is learning to correctly classify the objects during training.
train/dfl_loss	Evaluates whether the model is learning to accurately refine the bounding box edges (fine adjustment of detection shape).
val/box_loss	Same as train/box_loss, but on new data (validation) - checks whether the model generalizes well.
val/cls_loss	Same as train/cls_loss, but on validation data - shows whether the model classifies correctly on unseen samples.
val/dfl_loss	Same as train/dfl_loss, but on validation data - ensures that bounding box accuracy is not limited to training only.
metrics/precision(B)	Measures detection reliability: out of all detections made, how many were correct? (avoids false positives).
metrics/recall(B)	Measures detection coverage: out of all existing objects, how many were detected? (avoids false negatives).
metrics/mAP50(B)	The main quality indicator: summarizes model performance at a minimum overlap threshold (IoU $\geq$ 50%).
metrics/mAP50-95(B)	The most rigorous metric: evaluates model accuracy across multiple overlap thresholds (IoU 0.5 to 0.95).

Figure 4 shows the evolution of losses and evaluation metrics over the 50 training and validation epochs. The loss curves display a smooth and steady decline, while the precision, recall, and mAP metrics stabilize at high values after the 20th epoch.

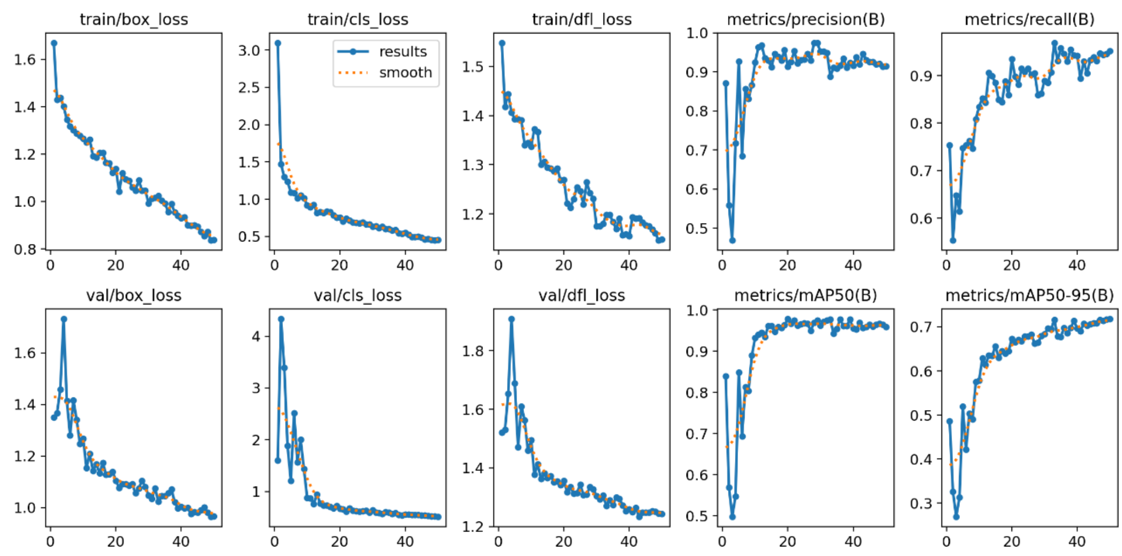


Figure 4 - Evolution of Losses and Metrics

Source: The Authors (Own).

Figure 5 presents the normalized confusion matrix, which details the network's performance in distinguishing between the different classes during validation.

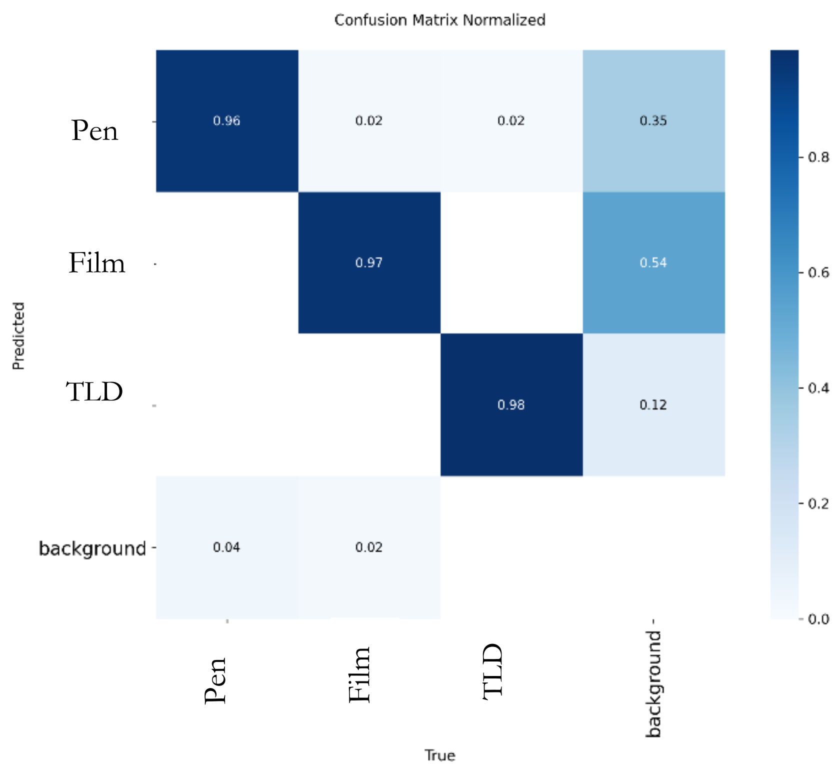


Figure 5 - Normalized Confusion Matrix

Source: The Authors (Own).

The analysis of the confusion matrix reveals that the TLD class was the most accurately classified, with 98% accuracy and only 12% misclassified as background. The Pen class achieved 96% accuracy, but was confused with the background in 35% of the cases. This suggests that, in some images, the model had difficulty distinguishing the dosimeter from the environment, which may be attributed to lighting conditions or occlusion.

The Film class reached 97% accuracy, but also showed 54% misclassification with the background. This result can be explained by the lower visual contrast between the Film dosimeter and the operational environment, making its detection more challenging for the model.

To demonstrate the model's visual performance in real-world situations, figure 6 presents a panel of examples from the validation set, showing detections made by the network with bounding boxes around the dosimeters.

Figure 6 - Panel of Real Images with Detections

Source: The Authors (Own).

In addition to its technical performance, the project proposes a practical application with potential impact in the field of radiation dosimetry monitoring. The idea is to integrate visual detection into an automated real-time verification system, using an IP camera installed at the entrance of controlled laboratories. The images would be transmitted to a local notebook running the YOLOv8 model, capable of identifying the presence and correct placement of dosimeters on clothing.

If the system detects the absence of a dosimeter, two automatic actions are foreseen: the emission of a local audible alert and the sending of notification messages (via WhatsApp or SMS) to the responsible personnel or the radiological safety team. This solution would act as an additional control barrier, minimizing human errors.

It is a low-cost, easy-to-implement, and scalable proposal, with potential application in both small laboratories and larger areas with multiple access points. As a future development, the goal is to build and test this prototype.

4 Conclusions

The proposal of this work demonstrated the technical feasibility of applying convolutional neural networks to the automated visual detection of individual dosimeters. The results obtained using the YOLOv8 architecture showed outstanding performance in the main evaluation metrics, with highlights including a mAP@0.5 of 0.977 and a recall of 0.969—indicators that validate the model's accuracy in real operational contexts. The analysis of the results by class revealed a high level of accuracy in identifying the different types of dosimeters, especially the TLD, which achieved nearly 100% accuracy.

Despite some limitations observed in distinguishing Film and Pen dosimeters against low-contrast backgrounds, the stability of the loss curves and the reduced inference time (under 1 ms) reinforce the system's potential for embedded and real-time applications.

From a practical standpoint, the proposed model constitutes an innovative solution to strengthen existing radiological protection systems. By enabling continuous and automated verification of correct dosimeter use through IP cameras and computer vision algorithms, the system acts as an additional control layer, capable of mitigating human errors and increasing the reliability of occupational exposure records.

Acknowledgements

The authors would like to express their gratitude to the Nuclear Detection and Instrumentation Laboratory of the Military Institute of Engineering (IME) for the technical support, infrastructure, and contributions that made this research possible.

References

[1] Biblioteca Ultralytics. Ultralytics YOLOv8. Disponível em: https://github.com/ultralytics/ultralytics. Acesso em 20 agost. 2025.

[2] Bin, Y. et al. Computer vision technology for monitoring of indoor and outdoor environments and HVAC equipment: a review. Sensors, v. 23, n. 13, p. 6186, 2023.

[3] Comissão Nacional De Energia Nuclear. NN-3.01: Diretrizes Básicas de Radioproteção. Rio de Janeiro: CNEN, 2024.

[4] European Commission. (2018). Practical aspects of personal dosimetry. Radiation Protection No. 195. Luxembourg: Publications Office of the European Union.

[5] ICRP Publication 103. The 2007 Recommendations of the International Commission on Radiological Protection. Annals of the ICRP, 37(2-4). Oxford: Pergamon Press, 2007.

[6] International Atomic Energy Agency (IAEA). Occupational Radiation Protection: Safety Guide. IAEA Safety Standards Series No. GSG-7. Vienna: IAEA, 2018.

[7] Redmon, J. et al. You Only Look Once: Unified, Real-Time Object Detection. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016.

[8] Vaz, A. et al. Computer Vision Applied to Nuclear Safety: A Comparison Between YOLOv5 and YOLOv8 in the Identification of Warning Signs in Nuclear Contexts. The Journal of Engineering and Exact Sciences, v. 11, n. 1, p. 21998-21998, 2025.

XXV ISSSD
International Symposium on Solid State Dosimetry
September 22 – 26, 2025
Belo Horizonte, Minas Gerais - Brazil
CDTN UFMG

Bayesian-Optimized Monte Carlo Modeling of CyberKnife Collimators and Source Parameters for Small-Field Dosimetry

T. M. da Cunha e Silva[1*], **H. Hyoriyaz**[1]

[1] Instituto de Pesquisas Energéticas e Nucleares, Centro de Engenharia Nuclear, 05508-000, São Paulo, Brasil

[*] tiago.c.silva@usp.br

Abstract

This work applies Bayesian optimization to infer physical and geometric parameters of the CyberKnife system, aiming to enhance dosimetric accuracy in small-field radiotherapy. The main objectives were to determine optimal source parameters in an open field and geometric parameters of conical collimators for 10 mm and 60 mm fields. The Bayesian approach was selected due to the stochastic nature and high computational cost of Monte Carlo simulations performed with the TOPAS code. This is particularly relevant as the objective function lacks known derivatives and an analytical structure. A Gaussian Process Regression (GPR) with a Matern kernel (v = 3/2) was used as the probabilistic model. It was coupled with the Upper Confidence Bound (UCB) acquisition function, which balanced exploration and exploitation during optimization. Model I was used to infer the primary source parameters (spatial distribution σ_{spt} and angular distribution σ_{ang}) in an open field , obtaining values corresponding to $\Gamma = 0.5$ mm and $\varphi = 8.59^{\circ}$. Model II was used to adjust conical collimator geometry for 10 mm and 60 mm fields based on beam profiles (lateral dose profiles), resulting in $\boldsymbol{\theta}_{II}^{*10} = \left[r_{low}^{*10}, r_{sup}^{*10}, t^{*10}\right] = [2.21, 3.07, 50.80]$ mm and $\boldsymbol{\theta}_{II}^{*60} = \left[r_{low}^{*60}, r_{sup}^{*60}, t^{*60}\right] = [13.99, 25.78, 59.96]$ mm. Validation via percentage depth dose (PDD) revealed deviations that are attributable to the absence of PDD data in the objective function, indicating that geometric parameters such as collimator thickness (t) have limited influence in the current model. In conclusion, Bayesian optimization proved effective for parameter inference in small-field dosimetry, though future models should incorporate PDD data to improve depth-dose accuracy.

Keywords: CyberKnife; Monte Carlo Simulation; Small-field; Bayesian Optimization; Dosimetry.

1 Introduction

Techniques such as Stereotactic Radiosurgery (SRS), Stereotactic Body Radiotherapy (SBRT), Intensity-Modulated Radiotherapy (IMRT), and Volumetric Modulated Arc Therapy (VMAT) have emerged as promising procedures capable of precisely depositing dose in solid tumors. All these techniques share the use of multiple highly collimated beams from different directions (Bagheri et al. (2017)). These techniques rely on accurate small-field dosimetry, which presents unique challenges due to electronic disequilibrium and detector size effects (Palmans et al. (2018)). These small photon fields are generated by linear accelerators and shaped using different types of collimators. The definitions provided in the TRS-483 report, published in 2017 by the IAEA, establish that for an external photon field to be defined as small, at least one of the following physical conditions must be satisfied: (i) Absence of Lateral Charged Particle Equilibrium (LCPE) on the beam axis; (ii) Partial occlusion of the primary photon source by the collimation system on the beam axis; (iii) The detector size being similar to or large compared to the field dimensions. The CyberKnife system (Accuray Inc., Sunnyvale, CA), which is designed to perform radiosurgery procedures and is the focus of this work, is a compact linear accelerator mounted on a robotic arm capable of producing small photon fields. The system has 12 interchangeable conical collimators that shape the beam into circular dimensions, ranging from 5 to 60 mm in diameter (beam size at isocenter) (Wang et al. (2021)). The accurate modeling of such systems is essential for treatment planning and dose delivery (Kilby et al. (2020)). To computationally study the transport and interaction of radiation produced by the CyberKnife system, simulations were performed using Monte Carlo (MC) methods, since beams simulated by MC methods can be accurately calibrated and dosimetric calculations can be accurately determined for any radiation field geometry – including situations of electronic disequilibrium, such as small fields (Bagheri et al. (2017)).

Monte Carlo methods can be described as a set of statistical methods that use sequences of random numbers to simulate a stochastic process. Any physical system can be computationally simulated using Monte Carlo methods, provided its behavior can be described by probability density functions (PDF). In terms of radiation transport, the stochastic process can be seen as a family of particles whose individual coordinates change randomly with each interaction (Yoriyaz (2009)). Thus, the essence of MC methods applied to radiation transport consists of estimating macroscopic quantities, such as fluence or dose, from individual observations of a large number of events. Among the codes available for radiation transport simulations, the TOPAS (Tool for Particle Simulation) code was chosen, a freely available, easy-to-learn, and easy-to-use interface for GEANT4. This choice is due to the fact that TOPAS's architecture allows the parameterization of any element of the simulation, a characteristic particularly useful in situations where some parameters of the system under study are unknown and one wishes to infer their values through mathematical optimization. Historically, in small-field radiotherapy simulations, the number of free parameters – both physical and geometric – has been sufficiently small to allow inference through manual optimization (trial and error). However, a more general optimization approach could enable a more sophisticated physical and geometric model, or even offer a greater degree of freedom in the choice of these parameters.

Optimization processes consist of systematically searching the domain A for a point, or set of points, $\mathbf{x}^* = [x_1, x_2, \ldots, x_n]$ that minimize (or maximize) a function $f(\mathbf{x})$. Clearly, the objective function is merely a mathematical abstraction, an attempt to convert, for example, a spatial dose distribution into a single dimensionless value. Due to their stochastic nature, optimizations based on radiation transport simulations using MC methods require that a large number of events be simulated; consequently, evaluating the objective function f is computationally expensive. Furthermore, f lacks previously known structures, and when evaluating it, we cannot obtain its derivatives, a fact that precludes the use of popular gradient-based search methods. To circumvent these difficulties, we seek a Bayesian optimization approach. Bayesian optimization has been widely adopted in problems involving expensive black-box functions, such as those arising from Monte Carlo simulations (Frazier (2018); Garnett (2023)). This optimization methodology does not require f to have a known form; Bayesian algorithms treat f as a "black box" and only require some mechanism to reveal information about the objective function.

The term Bayesian optimization refers not only to a specific algorithm but rather to an optimization philosophy, underpinned by the inference properties consequent to Bayes' Theorem, capable of reasoning about uncertain information during the optimization process. In a Bayesian inference approach, all unknown quantities are treated as random variables. This allows the representation of beliefs about the variables as probability density functions. Bayesian optimization policies are defined indirectly by optimizing the so-called acquisition function: this quantifies potential observation locations according to their potential to benefit the optimization process. Acquisition functions tend to be inexpensive to evaluate. In summary, Bayesian optimization is a class of methods for optimizing objective functions with unknown morphology, costly evaluation, and stochastic noise, in which the sampling process is guided by optimizing an acquisition function. In Bayesian logic, $f(\mathbf{x})$ is treated as a "black-box" function, such that its value for a parameter set $\mathbf{x}$ is only known after evaluation. Thus, the central idea of Bayesian optimization is that each piece of information collected about f can improve our knowledge of its behavior. The problem of optimizing an objective function $f(\mathbf{x}) : A \to \Re$ is defined as: $f^* = \max_{\mathbf{x}^* \in A} f(\mathbf{x}) = f(\mathbf{x}^*)$; where $A \subseteq \Re^D$ is typically a D-dimensional hyper-rectangle $A = \left\{\mathbf{x} \in \Re^D : a_i \leq x_i \leq b_i\right\}$ (Frazier (2018); Garnett (2023)). Nominally, we can assume that a given value $y = \phi + \varepsilon$, resulting from the observation of $\mathbf{x}$, is distributed according to an observational model with additive Gaussian noise dependent on the objective function value $\boldsymbol{\phi} = f(\mathbf{x})$. Through the Bayes' theorem, y can be related to ϕ via the observational model defined by the conditional probability: $p(y|\mathbf{x}, \boldsymbol{\phi})$.

There are two main components in Bayesian optimization: a Bayesian probabilistic observational model that models the objective function, usually based on a Gaussian Process Regression (GPR), and an acquisition function to be optimized in order to determine which points in A will be sampled at each iteration. This function must jointly consider exploitation – focusing searches on well-known regions of A where high values are already expected, through the model's predictive mean $\mu(\mathbf{x})$ – and exploration – betting on unknown regions, through the model's predictive uncertainty $\sigma(\mathbf{x})$. The mechanism responsible for balancing the choice between exploiting or exploring is called the acquisition function (or auxiliary function), denoted by $\alpha(\mathbf{x})$. More precisely, deciding the next observation consists of solving an auxiliary optimization problem (with negligible cost compared to evaluating $f(\mathbf{x})$) (Frazier (2018)). To perform the optimization *loop*, the Pyhton library TopasOpt (Whelan et al. (2023)) was used. This library is responsible for the dynamics between performing the simulations with Topas, collecting the computational results, and training the observational model, to then build and optimize the acquisition function in order to determine the free parameters that will be used in the next evaluation of the objective function.

The objective of this work is, therefore, to determine, through modeling and simulation of the CyberKnife system via the Monte Carlo Method, the values closest to the global optimum for the physical parameters of the source and the geometric parameters of the collimation system. It is sought that these parameters generate 10 mm and 60 mm fields that present the best match with experimental dosimetric data. The optimization will follow a Bayesian approach, with the observational model built using Gaussian Process Regression and an acquisition function responsible for defining the next points to be sampled.

2 Methods

Regression processes aim to formulate a function that adequately represents a distribution of points belonging to a dataset. Regarding GPR, this process is carried out by defining a distribution over a family of possible functions that represent the observed data. A Gaussian process is defined as (Garnett (2023)):

$$p(f) = GP(f|\mu, K). \tag{1}$$

The mean function $\mu : A \to \Re$ determines the expected value of the function $\boldsymbol{\phi} = f(\mathbf{x})$ for a given point $\mathbf{x}$: $\mu(\mathbf{x}) = \mathbb{E}[\boldsymbol{\phi}|\mathbf{x}] \in \Re^D$, representing the central tendency of f. The covariance function $K : A \times A \to \Re$ is the positive semi-definite covariance function (or kernel), responsible for determining how deviations from the mean are structured, encoding the expected behavior and properties of the function. Defining $\boldsymbol{\phi}' = f(\mathbf{x}')$, the covariance function is given by: $K(\mathbf{x}, \mathbf{x}') = \Sigma_{D\times D} = \text{cov}[\boldsymbol{\phi}, \boldsymbol{\phi}'|\mathbf{x}, \mathbf{x}']$. The covariance matrix encapsulates prior knowledge about the functions we intend to model. Let $\mathbf{x} \subset A$ be finite and $\boldsymbol{\phi} = f(\mathbf{x})$ the corresponding function values. For the equation 1, the distribution of $\boldsymbol{\phi}$ is multivariate normal:

$$p(\boldsymbol{\phi}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\phi}|\boldsymbol{\mu}, \Sigma), \tag{2}$$

where $\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{\phi}|\mathbf{x}] = \mu(\mathbf{x})$ and $\Sigma = \text{cov}[\boldsymbol{\phi}|\mathbf{x}] = K(\mathbf{x},\mathbf{x})$. We can condition a GP on the observation of any vector $\mathbf{y}$ through the marginal distribution $p(\mathbf{y})$, which shares a joint Gaussian distribution with f:

$$p(f,\mathbf{y}) = GP\left(\left[\begin{array}{c} f \\ \mathbf{y} \end{array}\right] \middle| \left[\begin{array}{c} \mu \\ \mathbf{m} \end{array}\right], \left[\begin{array}{cc} K & \kappa^T \\ \kappa & \mathbf{C} \end{array}\right]\right), \tag{3}$$

$$p(\mathbf{y}) = \mathcal{N}(\mathbf{y}|\mathbf{m},\mathbf{C}), \tag{4}$$

where $\mathbf{C} = \text{cov}[\mathbf{y}]$, $\mathbf{m} = \mathbb{E}[\mathbf{y}]$ and $\kappa(\mathbf{x}) = \text{cov}[\mathbf{y},\phi|\mathbf{x}]$ are, respectively, the prior covariance matrix, the prior expected value, and the cross-covariance matrix between $\mathbf{y}$ and f. Thus, we extend the Gaussian process in f to include the entries of $\mathbf{y}$. Noting $\mathcal{D} = \mathbf{y}$ for the observed data, we have:

$$\begin{aligned} &p(f|\mathcal{D}) = GP(f|\mu_{\mathcal{D}}, K_{\mathcal{D}}); \\ &\mu_{\mathcal{D}} = \mu(\mathbf{x}) + \kappa(\mathbf{x})^T \mathbf{C}^{-1}(\mathbf{y}-\mathbf{m}); \\ &K_{\mathcal{D}}(\mathbf{x},\mathbf{x'}) = K(\mathbf{x},\mathbf{x'}) - \kappa(\mathbf{x})^T \mathbf{C}^{-1}\kappa(\mathbf{x}'). \end{aligned} \tag{5}$$

For the GPR, the Matérn Kernel with $\nu = 3/2$ was chosen, expressed by:

$$K(\mathbf{x},\mathbf{x'}) = \exp\left(\frac{-d(\mathbf{x},\mathbf{x'})\sqrt{3}}{l}\right)\left[\frac{d(\mathbf{x},\mathbf{x'})\sqrt{3}}{l} + 1\right], \text{ where } l = 0.1. \tag{6}$$

A simple procedure can be executed to infer the posterior distribution of the *GP* from any observation vector satisfying 3: (i) compute the marginal distribution of $\mathbf{y}$ via 4; (ii) derive the cross-covariance function κ and, finally, (iii) find the posterior distribution of f via 5. Commonly, we adopt an optimistic stance towards the uncertainty of our observational model, such that the upper confidence bounds delimit the acquisition function through the Upper Confidence Bound (UCB) acquisition function:

$$\alpha_{UCB}(\mathbf{x}) = \mu(\mathbf{x}) + \varkappa\sigma(\mathbf{x}) \tag{7}$$

The scalar $\varkappa$ controls the exploration and exploitation process. Higher values of $\varkappa$ will cause the acquisition function to select sampling points in uncertain yet promising regions of the Gaussian model. To reduce the exploration level as iterations progress and model uncertainty decreases, we apply a decay rate to $\varkappa$. This ensures that α_{UCB} adopts a more exploitative behavior in the final iterations of the optimization loop.

The CyberKnife system uses a set of three collimators to shape the treatment beam: primary, intermediate, and secondary collimators, as shown in Figure 1. We divided the CyberKnife head into two main sections. An upper fixed section containing the linear accelerator (LINAC) and the primary tungsten collimator with a fixed aperture; A lower section containing the intermediate collimator, also made of tungsten, with a fixed size to reduce the beam to the maximum field size that the conical collimators can produce, and immediately below, the secondary collimator responsible for defining the treatment beam geometry. The CyberKnife has three different collimation systems: the Iris collimation system; the InCise multileaf system and, the focus of this work, a set of 12 conical collimators capable of producing field sizes between 5 mm and 60 mm.

Two computational models of the CyberKnife were built for parameter optimization. Model I of the CyberKnife head was designed to consider open-field conditions to infer the physical parameters of the source without dosimetric interference from secondary collimation. The primary collimator has a thickness of 100 mm and a trapezoidal aperture, with upper and lower apertures of 0.492 mm $\times$ 0.420 mm and 16.936 mm $\times$ 14.404 mm, respectively. The intermediate collimator simulates the CyberKnife MLC configured for an open field of 120 mm $\times$ 120 mm with 23 mm thickness. The secondary collimator was disregarded for Model I. We developed Model II to infer the geometric parameters of the collimators that define the beam's geometry; the primary collimator remains the same as used in Model I, but the intermediate collimator was adapted and the secondary included.

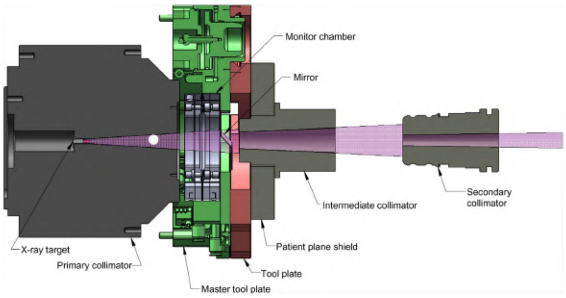


Figure 1: CyberKnife® head diagram out of scale (Kilby et al. (2020)).

Model I seeks to infer optimal parameters for the distributions governing the behavior of the CyberKnife source. Three probabilistic distributions govern the physical behavior of the primary photon source: energy, spatial, and angular distributions. To simulate *bremsstrahlung* photon production by a monoenergetic 7 MeV electron beam, we used the beam energy spectrum provided by IAEA, obtained from the phase space of a CyberKnife® IRIS™ 60 mm accelerator. The spatial distribution determines the initial coordinates of primary photons and is described by a symmetric bivariate normal distribution for the x and y axes, with parameters μ_{spt} and σ_{spt}. Let $\Gamma = 2\sigma_{spt}\sqrt{2\ln 2}$ be the Euclidean distance between points representing half the maximum distribution value (formally this distance is called full width at half maximum, but to avoid overloading notation, we denote it Γ); let the geometric center of the source be $\mu_{spt} = 0$ and the bounds $\pm 2\sigma_{spt} = \pm L$ ensure $\approx 95\%$ confidence. Thus, although the source is modeled as a circle of radius L centered at μ_{spt} (containing $\approx 95\%$ of photons), its effective size for dosimetric purposes is determined by Γ. Literature suggests good values of Γ lie in the interval 1 mm $\leq \Gamma \leq$ 3 mm. We take σ_{spt} as a free parameter subject to the constraints:

$$\frac{1}{2\sqrt{2\ln 2}}\ \text{mm} \leq \sigma_{spt} \leq \frac{3}{2\sqrt{2\ln 2}}\ \text{mm}. \tag{8}$$

The angular distribution is also bivariate normal, with parameters μ_{ang} and σ_{ang}, and determines the emission angle φ of primary photons relative to the normal vector of the z-axis. Analogously, $\mu_{ang} = 0$ rad is a fixed parameter and $\sigma_{ang} = \frac{\varphi\pi}{180^\circ}$ will be the free parameter. The parameter φ was constrained to the interval $1^\circ \leq \varphi \leq 10^\circ$, with the corresponding confidence bounds defined as $\pm L = \pm 2\sigma_{ang}$, such that σ_{ang} obeys:

$$\frac{1^\circ\pi}{180^\circ}\ \text{rad} \leq \sigma_{ang} \leq \frac{10^\circ\pi}{180^\circ}\ \text{rad}. \tag{9}$$

We define for Model I the vector storing the free parameters to be optimized as: $\boldsymbol{\theta}_I = [\sigma_{spt}, \sigma_{ang}]$.

We constructed Model II to infer geometric parameters of secondary collimation capable of generating 10 mm and 60 mm field sizes. TRS-483 establishes as good practice determining field size as the irradiation field specified at 50% of the relative dose level, also called Full Width at Half Maximum (FWHM) (Palmans et al. (2018)). Therefore, in this work ”field size” is used synonymously with ”irradiation field size at 50% of relative dose”. The primary collimator is identical to Model I, while the intermediate is a cone restricting the beam to a maximum size of 60 mm. The secondary collimator has a truncated cone aperture with free parameters: the collimator thickness t and the radii of the upper (r_{sup}) and lower (r_{low}) apertures. The geometric parameters were arbitrarily constrained to the intervals:

$$\begin{aligned} &1\ \text{mm} \leq t^{10,60} \leq 93.5\ \text{mm} \\ &1\ \text{mm} \leq r_{sup}^{10,60}, r_{low}^{10,60} \leq 30\ \text{mm}. \end{aligned} \tag{10}$$

Model II was optimized twice: once to infer parameters capable of producing a 10 mm field (t^{10}, r_{low}^{10} and r_{sup}^{10}) and once for a 60 mm field (t^{60}, r_{low}^{60} and r_{sup}^{60}). Thus the vector of free parameters is $\boldsymbol{\theta}_{II}^{10,60} = \left[r_{low}^{10,60}, r_{sup}^{10,60}, t^{10,60}\right]$.

For both Models, a water tank ($240 \times 240 \times 360$ mm) was centered on the z-axis, such that: (i) for PDD calculation, the distance between the primary source and the box surface is $SSD = 800$ mm; (ii) for PDD calculation, the distance between the primary source and the geometric center of the detection volume is $SAD = 800$ mm at a reference depth z_{ref}, such that $SSD = SAD - z_{ref}$. To achieve sufficient statistical precision in both Models, we used the *importance sampling* variance reduction technique in detection volumes. Different initial numbers of particles and importance values were determined according to the needs of each simulation.

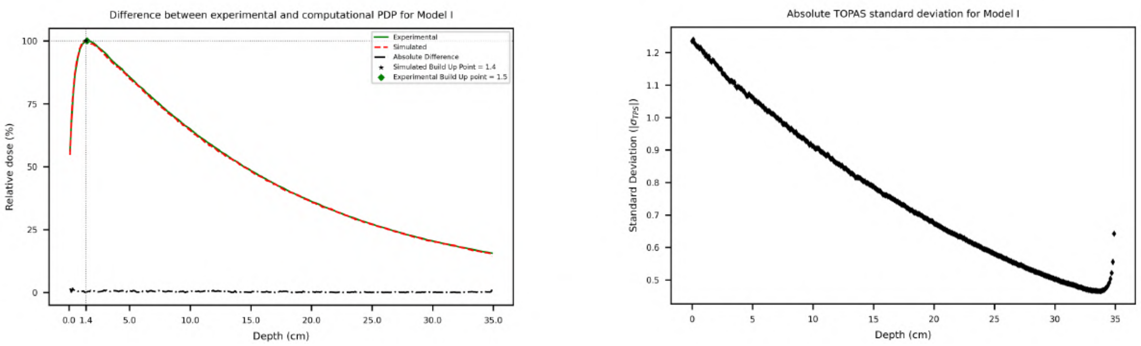
The objective function, which will be indirectly optimized through the optimization of $\alpha_{UCB}(x)$, was chosen to find the values of θ that minimize the mean absolute difference between experimental $\mathbf{D}^{exp} = [d_1^{exp}, \ldots, d_i^{exp}]$ and computational $\mathbf{D} = [d_1, \ldots, d_i]$ doses point-by-point in the detection grid:

$$f(\mathbf{D}|\boldsymbol{\theta}) = \frac{1}{n}\sum_{i}^{n} |d_i^{exp} - d_i| \,. \tag{11}$$

The values $\mathbf{D}^{exp}$ were obtained from dosimetric measurements of a CyberKnife beam using an EDGE detector (Sun Nuclear) in an internal study. Beam profiles for 10 mm and 60 mm fields were obtained at depths of 20 cm and 10 cm, respectively. PDDs were performed for 10 mm, 60 mm, and open fields.

3 Results and Discussions

First, we performed the optimization of Model I using the experimental PDD obtained with an open field to determine the physical parameters of the source without influence from secondary collimation. The optimal parameters found by the optimizer are: $\boldsymbol{\theta}_I^* = \left[\sigma_{spt}^*, \sigma_{ang}^*\right] = [0.25, 0.15] \rightarrow \Gamma = 0.5$ mm and $\varphi = 8.59°$; with the objective function value for the optimal parameters $f(\mathbf{D}|\boldsymbol{\theta}_I^*) = 0.31$. The PDD resulting from the optimization of Model I (shown in Figure 2) shows that the inferred parameters are indeed capable of computationally reproducing the experimental values.



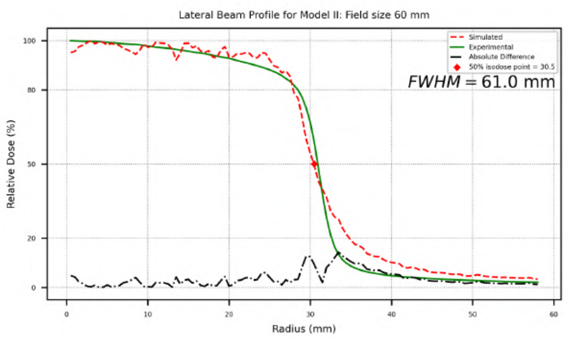
(a) PDD for open field.

(b) Uncertainty analysis of PDD.

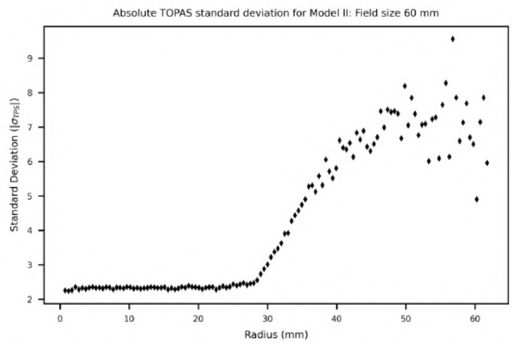
Figure 2: On the left the comparison of computed (red line) and experimental (green line) results with absolute difference (black line) for Model I with open size. On the right the scatter plot of the standard deviation. (a) Percentage Depth Dose curve; (b) Uncertainty of TOPAS-simulated PDD. All results are based on optimized physical parameters of the primary sources distributions.

Once the source parameters were determined, we performed the two optimizations of Model II for the 10 mm and 60 mm fields based on their respective experimental beam profile curves. Despite the high uncertainties at the beam edges, the profile curves respect the experimental values, the umbra regions Γ correspond to the expected field sizes, and the penumbras are small and non-overlapping. The parameters found, along with their associated f values, were: $\boldsymbol{\theta}_{II}^{*10} = \left[r_{inf}^{*10}, r_{sup}^{*10}, t^{*10}\right] = [2.21, 3.07, 50.80] \rightarrow f(\mathbf{D}|\boldsymbol{\theta}_{II}^{*10}) = 3.18$ and $\boldsymbol{\theta}_{II}^{*60} = \left[r_{inf}^{*60}, r_{sup}^{*60}, t^{*60}\right] = [13.99, 25.78, 59.96] \rightarrow f(\mathbf{D}|\boldsymbol{\theta}_{II}^{*60}) = 5.45$.

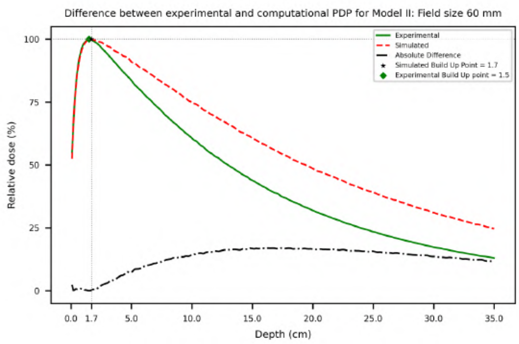
To validate the inferred parameters for Model II, we calculated the PDDs and compared them to reference results. It is notable that the computed PDD curves are shifted toward higher values on the dose axis, resulting in a more gradual dose fall-off at depth, as shown in Figures 3 and 4. We suspect this discrepancy between computational and experimental results is due to the objective function considering only the lateral beam profile to infer Model II parameters. When evaluating the objective function predicted by the Gaussian process as a function of each parameter - assuming the other parameters are fixed at their current optimal values - it is evident that the parameter $t^{10,60}$ has little influence on the objective function (as can be seen in Figure 5). Consequently, its role in defining the field size is minor compared to the collimator aperture radii. However, if an objective function were constructed using both the beam profile and PDD as reference, the collimator thickness could assume a more prominent role in the model.



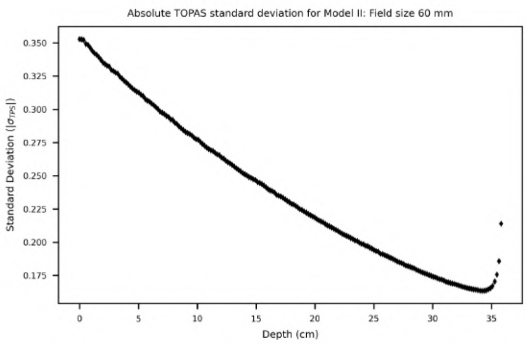
(a) Beam profile for 60 mm field.



(b) Uncertainty analysis of beam profile.

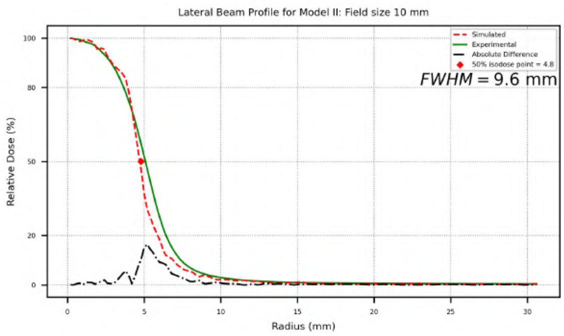


(c) PDD for 60 mm field.

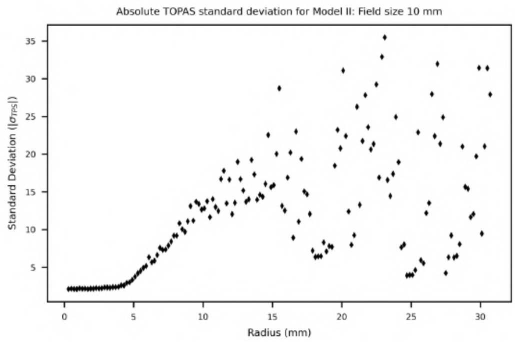


(d) Uncertainty analysis of PDD.

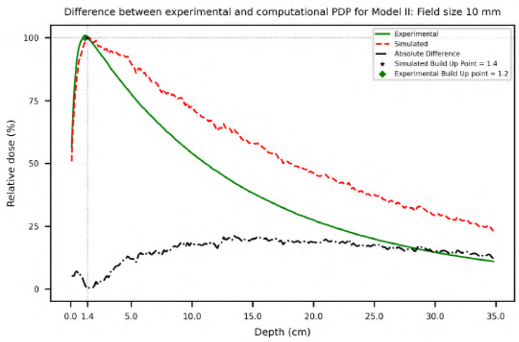
Figure 3: On the left the comparison of computed (red line) and experimental (green line) results with absolute difference (black line) for Model II with 60 mm field size. On the right the scatter plot of the standard deviation. (a) Lateral beam profile; (b) Uncertainty (one standard deviation) of TOPAS-simulated beam profile; (c) Percentage Depth Dose curve; (d) Uncertainty of TOPAS-simulated PDD. All results are based on optimized geometric parameters.



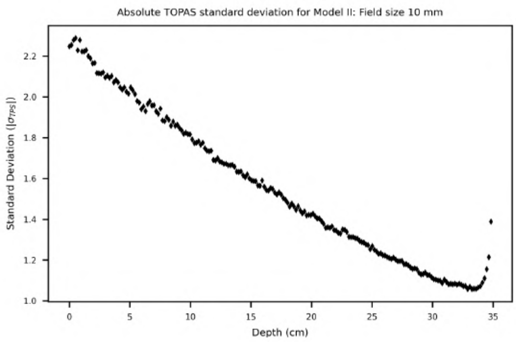
(a) Beam profile for 10 mm field.



(b) Uncertainty analysis of beam profile.

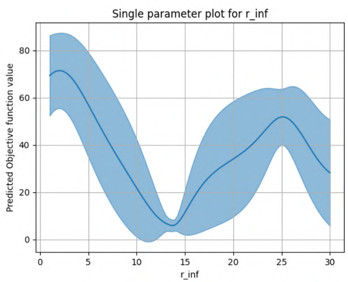


(c) PDD for 10 mm field.

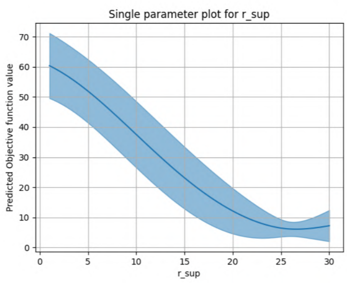


(d) Uncertainty analysis of PDD.

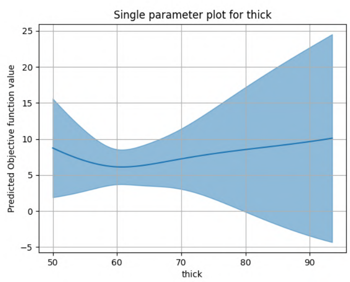
Figure 4: On the left the comparison of computed (red line) and experimental (green line) results with absolute difference (black line) for Model II with 10 mm field size. On the right the scatter plot of the standard deviation. (a) Lateral beam profile; (b) Uncertainty (one standard deviation) of TOPAS-simulated beam profile; (c) Percentage Depth Dose curve; (d) Uncertainty of TOPAS-simulated PDD. All results are based on optimized geometric parameters.



(a)



(b)



(c)

Figure 5: Univariate sensitivity analysis of the objective function for Model II (60 mm field size). The curves show the variation of the predicted objective function value when a single parameter is changed, while the others are held at their optimum values. (a) Variation for the lower radii (r_{low}); (b) Variation for the upper radii (r_{sup}); (c) Variation for the collimator thickness (t).

4 Conclusions

This study demonstrated the feasibility of Bayesian optimization coupled with Monte Carlo simulations (using TOPAS) to infer physical and geometric parameters of the CyberKnife system. For Model I (open field), the optimal source parameters ($\sigma_{spt} = 0.25$, $\sigma_{ang} = 0.15$) accurately reproduced the experimental PDD ($f = 0.31$), validating the spatial ($\Gamma = 0.5$ mm) and angular ($\varphi = 8.59^{\circ}$) photon distributions. In Model II, optimization of conical collimators for 10 mm and 60 mm fields yielded geometric parameters that adequately adjusted the beam profiles ($f = 3.18$ and $f = 5.45$). Nevertheless, simulated PDDs exhibited deviations from experimental data, indicating that an objective function based solely on lateral profiles is insufficient to capture depth-dependent effects. Collimator thickness (t) showed limited influence in the current objective function, suggesting that incorporating PDD data in future models would improve accuracy. The Bayesian optimization approach demonstrated notable efficiency, significantly reducing computational costs, providing a robust framework for calibrating complex radiotherapy systems.

Acknowledgements

This study was financed in part by the Conselho Nacional de Desenvolvimento Científico e Tecnológico - Brasil (CNPq) - Process 131354/2025-6. The author(s) also acknowledge the support of the Graduate Program in Universidade de São Paulo (USP) at Instituto de Pesquisas Energéticas e Nucleares (IPEN).

References

Bagheri H., Soleimani A., Gharehaghaji N., Mesbahi A., Manouchehri F., Shekarchi B., Dormanesh B., and Dadgar H. A. (2017) An overview on small-field dosimetry in photon beam radiotherapy: Developments and challenges. *Journal of cancer research and therapeutics* 13:175–185

Frazier P. I. (2018) A tutorial on bayesian optimization. *arXiv preprint arXiv:180702811*

Garnett R. (2023) *Bayesian Optimization*. Cambridge University Press

Kilby W., Naylor M., Dooley J. R., Maurer Jr C. R., and Sayeh S. (2020) A technical overview of the cyberknife system. *Handbook of robotic and image-guided surgery* pages 15–38

Palmans H., Andreo P., Huq M. S., Seuntjens J., Christaki K. E., and Meghzifene A. (2018) Dosimetry of small static fields used in external photon beam radiotherapy: summary of trs-483, the iaea–aapm international code of practice for reference and relative dose determination. *Medical physics* 45:e1123–e1145

Wang J., Wang L., Maxim P. G., and Loo B. W. (2021) An automated optimization strategy to design collimator geometry for small field radiation therapy systems. *Physics in Medicine & Biology* 66:075016

Whelan B., Loo Jr B. W., Wang J., and Keall P. (2023) Topasopt: An open-source library for optimization with topas monte carlo. *Medical Physics* 50:1121–1131

Yoriyaz H. (2009) Método de monte carlo: princípios e aplicações em física médica. *Revista Brasileira de Física Médica* 3:141–149

Characterization of the Ubatuba Formation: Integration of Gamma-Ray Spectrometry, Total Organic Carbon, and Well Logs from the Upper Cretaceous to the Lower Paleogene Intervals in the Campos Basin

Rubens C. Florenzano[1,2], Daniel V. Lamas[1,2], Bruno S. N. Rocha[1], Jardel L. Thalhofer[1,2,3], Antonio F. M. Freire[1,4]

[1] Group of Exploratory Interpretation and Reservoir Characterization/Center for Research and Development in Geological Well Operations (UFF/GIECAR/COPEGEO).

[2] Laboratory of Radiobiology and Radiometry (LARARA/UFF).

[3] Nuclear Engineering Program (PEN/COPPE/UFRJ).

[4] National Institute of Science and Technology/Petroleum Geophysics (INCT-GP).

Abstract

In the Campos Basin, the Turonian, Campanian, and the transition between the Lower Cretaceous and Paleocene intervals are characterized by the presence of rollover anticlines and listric faults, indicating intense tectonic activity during deposition and sedimentary reorganization. This tectonic process resulted in complex depositional patterns and significant environmental changes, observable through various geophysical methods, ranging from neritic to abyssal and bathyal settings. The objective of this study was to analyze the main characteristics of these stratigraphic units and to establish a quantitative relationship between the radioelements ^{40}K, ^{238}eU, and ^{232}eTh, measured through gamma-ray spectrometry, and the total organic carbon (TOC) data obtained from laboratory analyses of drilling cuttings samples. The methodology involved TOC laboratory analysis combined with radiometric data acquisition using a high-purity germanium (HPGe) detector. Based on these data, a predictive modeling approach was implemented to estimate TOC values from the measured concentrations of radioelements. It was observed that the relationship between TOC content and the radioelements is nonlinear, which required the application of regression techniques capable of capturing these complexities. This approach contributed to evaluating the potential of gamma spectrometry as an indirect tool for estimating organic matter in specific sedimentary contexts. The radiometric data allowed the identification of six stratigraphic surfaces in well 1-DEV-16-RJS, associated with ^{238}eU enrichment at maximum flooding surfaces. Furthermore, the interpretation of well log responses at these surfaces reinforced the effectiveness of the methodologies employed. As a result, it was possible to distinguish parasequence boundaries, identify intervals with anoxic potential and enhanced water column expansion, and demonstrate the usefulness of chemostratigraphic data derived from drilling cuttings in paleoenvironmental and paleoclimatic reconstructions.

Keywords: Gamma Spectrometry, Drilling Cuttings, TOC, Well Logs, Campos Basin.

1 Introduction

No intervalo do Turoniano Inferior ao topo do Cretáceo, destaca-se o “Ubatuba Sísmico”, caracterizado por fortes reflexões sísmicas, compactação diferenciada, variações deposicionais e erosão pronunciada relacionada a mudanças na coluna d’água e na distribuição de elementos químicos e radioativos [Chang et al. 2006]. Essa superfície pode ser identificada por métodos geofísicos como sísmica de alta resolução e espectrometria gama, essenciais para delimitar seus limites. Inserido em um contexto de tectônica salina e extensiva ligada à abertura do Atlântico, esses processos controlam o espaço de acomodação e modificam o arranjo deposicional. A espectrometria gama aplicada em perfis de poços é ferramenta robusta para avaliar essas influências e aprimorar a caracterização estratigráfica.

2 Materials and methods

This study focuses on the analysis of well 1-DEV-16-RJS, drilled in the southern portion of the Campos Basin, in an area characterized by more widely spaced bathymetric contours, indicating a deep and distal marine environment.

Approximately 50 cuttings samples were analyzed, collected during the drilling of the rock formations. These samples play a key role as geological markers for this study and serve as the basis for laboratory tests such as Total Organic Carbon (TOC) analysis and gamma spectrometry. These data provide the primary lithological reference used in the investigation.

2.1 Study Area

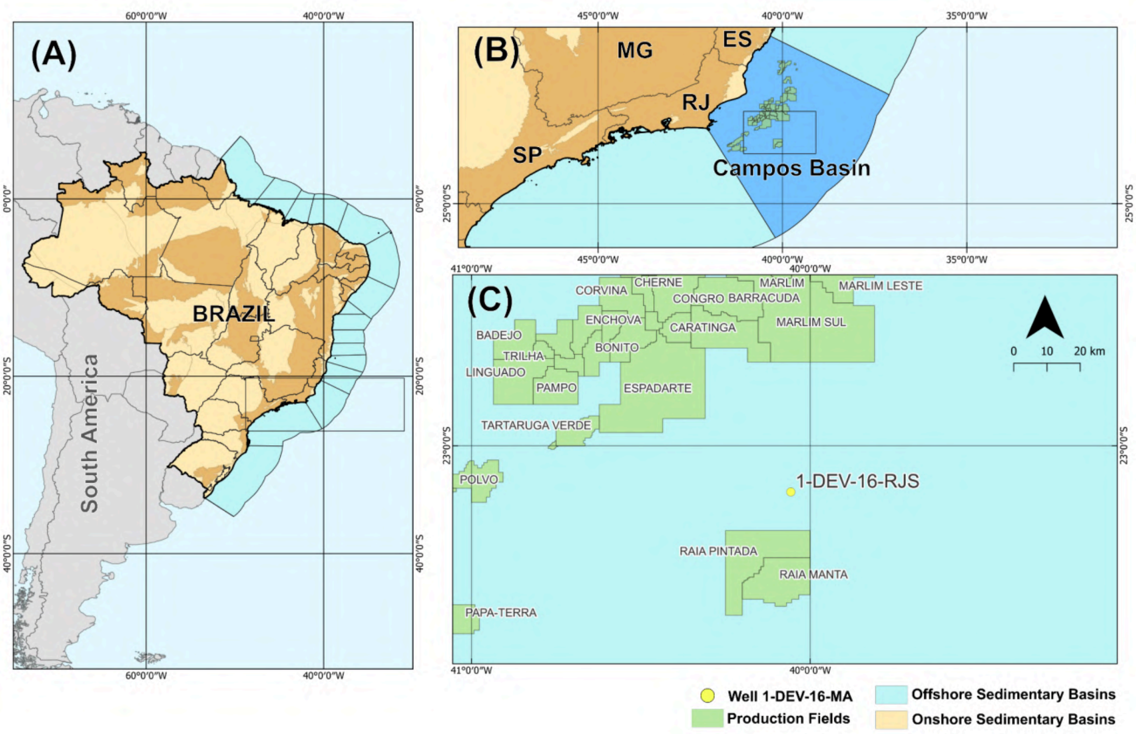


Figure1: Map showing the location of well 1-DEV-16-RJS in the Campos Basin. (A) Light blue highlights Brazil's coastal region bordering the marginal basins; (B) Region adjacent to the Campos Basin; and (C) Location of well 1-DEV-16-RJS.

3 Drilling Cuttings Samples

Cuttings samples are rock fragments obtained during geological monitoring of drilling operations, representing the material penetrated along the wellbore. These fragments are transported to the surface by the circulation of the drilling fluid, a process driven by pressure differentials and the buoyant force generated by the fluid’s movement. The return time of the material to the surface is carefully controlled, allowing for depth corrections to be made. This enables each rock fragment to be accurately associated with its original depth in the drilling profile, ensuring greater precision in stratigraphic interpretation and lithological characterization of the studied section.

4 Drill Fluid

The drilling fluid interacts with the rock and can affect the response of certain logging tools. Therefore, its presence must be taken into account when interpreting lithology from geophysical well logs. The distance between the receiver and the borehole wall is a critical factor, as many tools have readings that are influenced by this spacing [Ellis & Singer, 2008]. Drilling fluids are mixtures composed of both liquid and solid components.

5 Sample Preparation for TOC Analysis

The samples were washed with deionized water and detergent to remove contaminants from drilling fluids. They then underwent three ultrasonic cleaning cycles with n-hexane to separate carbon associated with the fluids. After drying in an oven at temperatures below 60 °C, the samples were ground and sieved (<106 µm) to optimize decarbonation, which was performed using diluted hydrochloric acid (20%) for 48 hours at 50 °C. The efficiency of decarbonation was visually verified by observing effervescence, using the sample with the highest calcium content (determined by XRF) as a reference. Following additional washing until neutral pH and complete detergent removal, the samples were analyzed at LaGen NAB by relating combustion smoke to TOC content.

6 Mobility of Radioactive Elements

A mobilidade relativa dos elementos potássio (K), tório (Th) e urânio (U) é influenciada por suas propriedades químicas e pelo ambiente geológico em que se encontram. Cada elemento apresenta comportamentos distintos de mobilidade em diferentes tipos de rochas e condições ambientais.

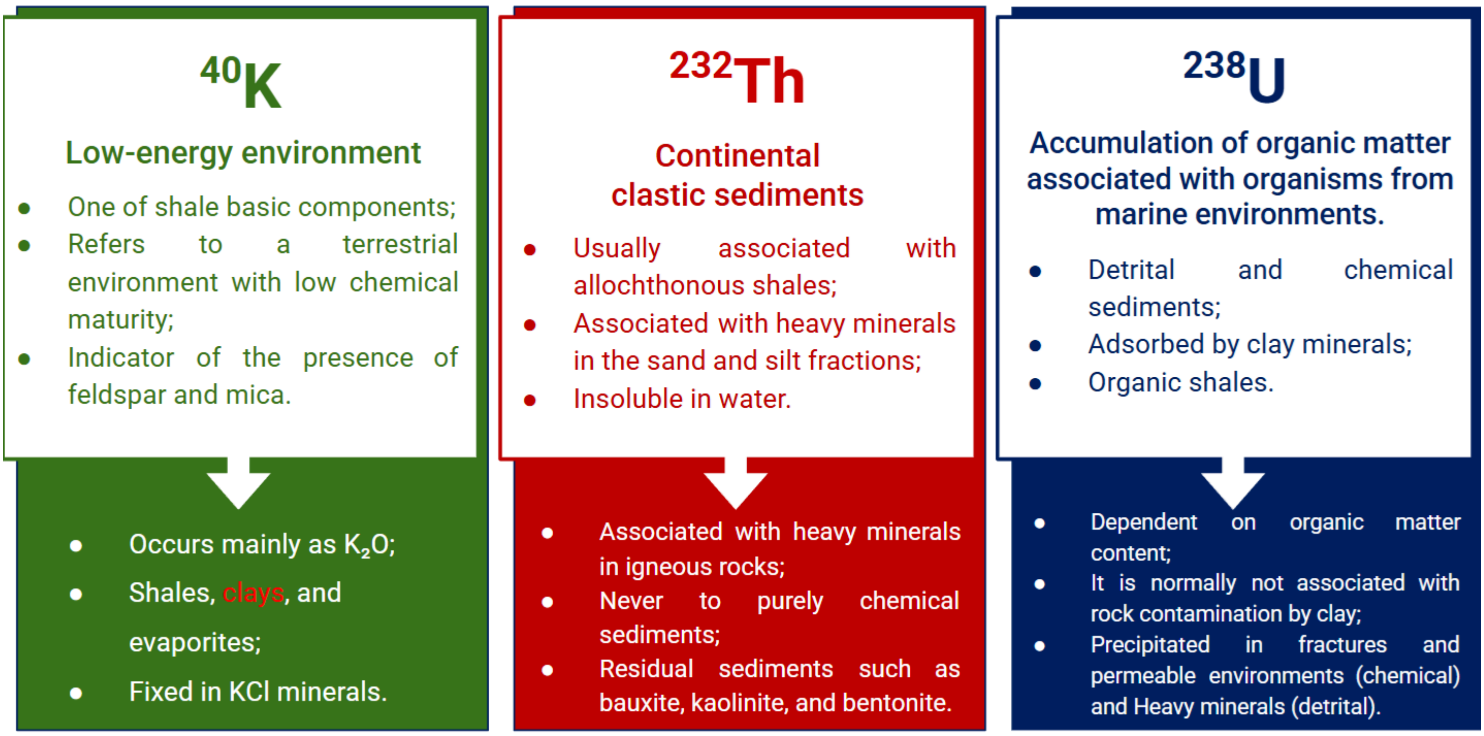


Figure 5: Association of K40 U238, Th232 with paleodepositional (Klaja & Dudek, 2016)

7 High-Purity Germanium (HPGe)

To obtain the gamma radiation spectrum of natural radionuclides, a coaxial high-purity germanium (HPGe) semiconductor detector with a relative efficiency of 30% (model GC3020 from Canberra

Industries) was used. A low-noise preamplifier (model 2002 CSL) coupled to a Digital Spectrum Analyzer (DSA 1000) with 8192 channels was connected to the detector along with the Genie 2000 software. The energy range scanned was from 50 keV to 2 MeV. The activity concentration is calculated using the following equation (IAEA 1989).

8 Well Logs

Well logging tools measure the physical properties of rocks during drilling, providing essential data to characterize formations and ensure operational safety. While effective, their accuracy can be affected by maintenance issues, calibration, and drilling fluid interference. Laboratory data, free from these influences, are used to validate and adjust field results, improving quality control and interpretation.

9 Electrical resistivity (RT)

Electrical resistivity is essential for characterizing formations and identifying fluids in wells, providing a complementary analysis to gamma ray detectors. Tools such as Laterolog and Induction logs measure resistivity at various depths. Conductive fluids, like saline water, lower resistivity, while oil and gas increase it. Conductive minerals can also influence the readings. Variations in resistivity can indicate drilling fluid invasion, which is useful for evaluating permeability and porosity (Ellis & Singer, 2008).

10 Sonic (DTCO)

Sonic logs (DTCO) measure the transit time of seismic waves through rocks, providing data on porosity, fractures, and correlation with seismic surveys. P- and S-waves, whose transit time is inversely related to seismic velocity, propagate consistently due to the physical properties of the medium. Tools equipped with sensitive transducers and proper corrections ensure accuracy even at great depths. The influence of fluids and multiple wave paths is minimal, making sonic data reliable and stable in many contexts (Ellis & Singer, 2008).

11 K-Nearest Neighbors Regressor (KNN - Regressor)

The K-Nearest Neighbors regression technique (KNN Regressor) was employed with the objective of estimating petrophysical properties in regions lacking direct well data.

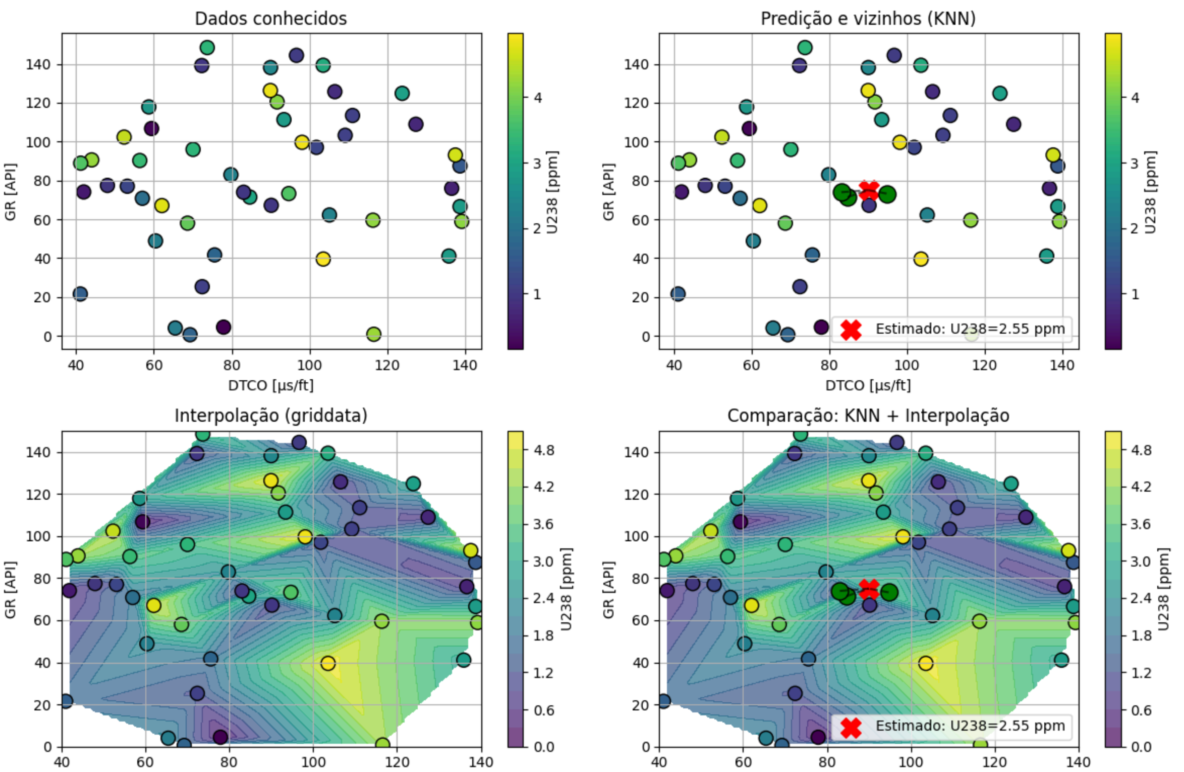


Figure 6: Interpolation using K-Nearest Neighbors (KNN Regressor)

The KNN method predicts values based on the similarity between samples, considering the k nearest neighbors in a multidimensional attribute space. This approach is flexible and effective at capturing local patterns, especially in geologically heterogeneous contexts (Hastie, Tibshirani & Friedman, 2009). It functions as an interpolation technique, using variables such as coordinates to estimate unknown values. In this study, cuttings samples collected at 9 × 9 meter intervals were interpolated using KNN to align their lithogeochemical profiles with the centimeter-scale resolution of continuous well log data.

12 Results

Well log interpretation identified two main stratigraphic sequences of shale deposits. The first, composed of greenish shales, indicates a deepening of the water column and anoxic conditions, reflected by an increase in U-238 concentrations. The second, located at the base of the unit, consists of light gray shales and shows a rise in Th-232, suggesting a greater influx of terrigenous material. K-40 and U-238 are strongly correlated, indicating an authigenic origin likely associated with transgressive events. Reddish shales, associated with the Emborê Formation, point to shallower, oxygenated environments, with organic matter preservation evidenced by TOC peaks, especially in greenish to darker-toned shales. However, U-238 does not consistently correlate with TOC in some zones, indicating a mixed influence of redox and terrigenous factors. Resistivity and sonic logs revealed compaction surfaces, but TOC estimation via the Delta Log R method proved unsatisfactory, possibly due to variations in temperature, depth, and organic matter maturation. Peaks in U-238 and DTCO values marked transitions in flooding and compaction stages.

The study applied machine learning, particularly surface regression and KNN, to relate Total Organic Carbon (TOC) to radioisotopes (U, Th, K), revealing that higher TOC values are associated with elevated uranium and thorium levels and lower potassium, indicating anoxic environments favorable to organic preservation. The model performed well in predicting low TOC, but showed greater error in intervals with high organic content. The greenish shales displayed sharp TOC peaks with strong terrigenous influence, while the basal light gray shale sequence showed more broadly distributed TOC. In contrast, the reddish shales reflected more oxygenated environments with low TOC content

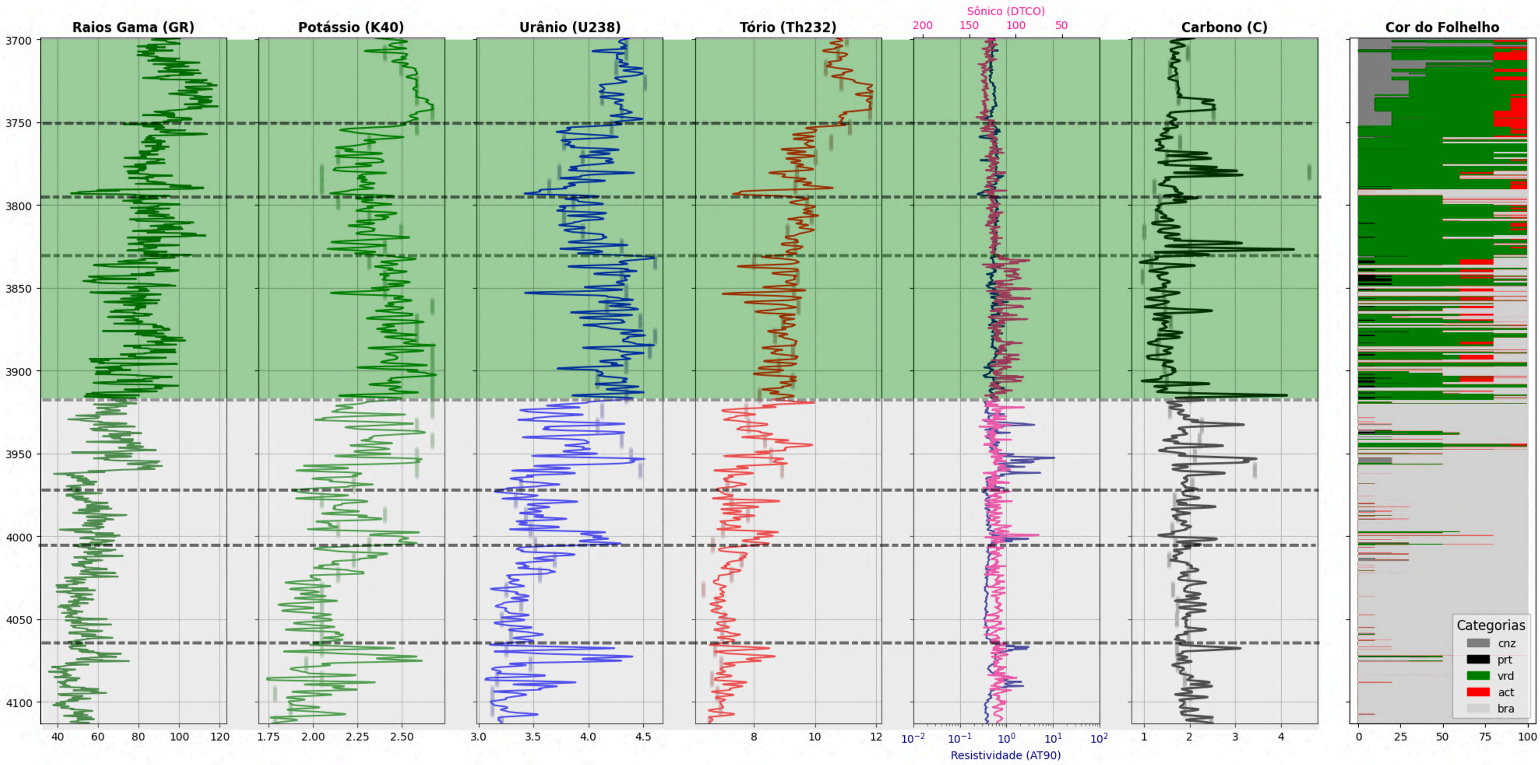


Figure 7: Gamma Ray (Track 1), K40 (Track 2), eU238 (Track 3), eTh232 (Track 4), DTCO and RT90 (track 5), COT (Track 6) and shale colors display the descriptions of drilling cuttings and shale colors, respectively, shown as percentages. Track 7 represents the lithological log of this sedimentary succession. (Track 7).

References

AGÊNCIA NACIONAL DO PETRÓLEO, GÁS NATURAL E BIOCOMBUSTÍVEIS. Resolução ANP Nº 71, de 31.12.2014. *Diário Oficial da União*, Brasília, 2 jan. 2015. Seção 1, p. 2.

CHANG, H. K. et al. Interpretação e Mapeamento dos Sistemas Petrolíferos da Bacia de Campos. Tomo I. São Paulo: ANP/LEBAC/UNESP, 2006.

EISENBUD, M.; GESELL, T. Environmental Radioactivity: from natural, industrial and military sources. New York: Academic Press, 1998. 656 p.

ELLIS, D. V.; SINGER, J. M. Well logging for earth scientists. 2. ed. Dordrecht: Springer, 2008.

GILL, R. Chemical Fundamentals of Geology. 2. ed. London: Chapman & Hall, 1996. 299 p.

GROWCOCK, F.; HARVEY, T. Drilling fluids. In: *Drilling fluids processing handbook*. Burlington: Elsevier, 2005.

HASTIE, T.; TIBSHIRANI, R.; FRIEDMAN, J. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. 2. ed. New York: Springer, 2009.

INTERNATIONAL ATOMIC ENERGY AGENCY (IAEA). Guidelines for Radioelement Mapping Using Gamma Ray Spectrometry Data. Vienna: IAEA, 2003. (Technical Report TECDOC 1363). 179 p.

KLAJA, J.; DUDEK, L. Geological interpretation of spectral gamma ray (SGR) logging in selected boreholes. Nafta-Gaz, v. 72, n. 1, p. 3-14,2016.

RIDER, M. H. The geological interpretation of well logs. Malta: Interprint Ltd, 2002.

ROCHA, L. A. S. et al. Perfuração direcional. In: ROCHA, L. A. S. (Org.). *Perfuração direcional*. 2. ed. Rio de Janeiro: Editora Interciência, 2006. Cap. 2, p. 47–98.

SOUZA, F. Determinação das Concentrações de atividade de ^{234}U e ^{238}U em águas subterrâneas de três poços perfurados em rochas da Suíte Intrusiva de Itu (SP). 2006. 106 p. Dissertação (Mestrado em Ciências – Área de Concentração: Geofísica) – Instituto de Astronomia, Geofísica e Ciências Atmosféricas, Universidade de São Paulo, São Paulo, 2006.

WINTER, W. R.; JAHNERT, R. J.; FRANÇA, A. B. Geologia da Bacia de Campos – aspectos estruturais, estratigráficos e petrolíferos. *Geociências (Petrobras)*, Rio de Janeiro, v. 15, n. 2, p. 511–529, maio/nov. 2007.

Comparison of Methods for Determining the Half-Value Layer (HVL) in Computed Tomography Equipment

H. M. C. Rhodes[1]*, P. C. Santana[2], D. S. Calheiro[1], P. M. C. Oliveira[2], P. L. Squair[1], O. G. M. Silva[2], M. P. A. Potiens[3], C. R. Soares[1], M. S. Nogueira[1]

[1] Nuclear Technology Development Center, 31270-901, Belo Horizonte, Brazil

[2] Federal University of Minas Gerais, 31270-901, Belo Horizonte, Brazil

[3] Nuclear and Energy Research Institute, 05508-000, São Paulo, Brazil

[*] heloisarhodes@gmail.com

Abstract

Evaluation of Half-Value Layer (HVL) in Computed Tomography (CT) equipment is a key parameter to ensure image quality, patient safety, and radiation dose optimization. Assessing the HVL in CT is a challenging task due to the equipment's geometry, and therefore it is not usually included in quality control programs. However, HVL monitoring can be useful as an indicator of potential irregularities in equipment performance. The aim of this work is to evaluate HVL values in a CT scanner using two different methods. A methodology based on the Tandem System will be tested, considering practical aspects to estimate the HVL of the clinical CT beam and thus infer characteristics of the X-ray spectrum. The detection system for the Tandem method consists of five cylindrical aluminum absorbers of 1 mm, 3 mm, 5 mm, 7 mm, and 10 mm thickness, and three PMMA absorbers of 15 mm, 25 mm, and 35 mm, manufactured at IPEN's workshop. A Radcal Corporation ionization chamber, model 10X6-3CT, coupled to a Radcal Accu Gold electrometer, was inserted into the absorbers to determine detection ratios for each component. The semiconductor detector Xi CT, coupled to a Platinum Plus unit from Unfors, was also used for HVL evaluation. Based on the different ionization chamber responses for various absorber materials and thicknesses, a Tandem curve was constructed to assess HVL. The results were compared and showed discrepancies between HVL values obtained with the two methods. For instance, at 120 kV, the difference between the values was 2.2 mmAl. This difference highlights the complexity of evaluating the X-ray beam quality in CT scanners and confirms that the combined use of different techniques can provide a broader and more critical view of beam quality.

Keywords: Computed Tomography, Half-Value Layer, Tandem System

1 Introduction

The Half-Value Layer (HVL) is a fundamental parameter for characterizing the quality of the X-ray beam, playing a key role in quality assurance and radiation protection programs in computed tomography (CT) (McCollough et al., 2015; Seibert et al., 2018). According to the International Electrotechnical Commission (IEC 60601-2-44:2012), the periodic verification of the HVL allows monitoring changes in the X-ray tube, such as anode wear and the effective thickness of the filter—factors that can directly impact image quality and the dose absorbed by the patient (McNitt-Gray, 2016).

Measuring the HVL in CT presents specific challenges due to the complex beam geometry, the rotational movement of the tube, and the use of automatic current modulation techniques (Bowen et al., 2014; Kalender, 2011). Therefore, several methods have been studied: among them, direct methods—which use semiconductor detectors or ionization chambers along with standardized absorbers—stand out, as well as indirect methods, which infer the HVL from calibration curves or Monte Carlo simulations (Mayo et al., 2022; Boone et al., 2019).

Fontes (2016) proposed in Brazil the use of a Tandem System composed of aluminum and PMMA layers, combined with an ionization chamber, as a practical alternative for measuring the HVL in clinical CT scanners. This approach makes it possible to estimate beam attenuation using different combinations of materials and thicknesses, which helps deduce characteristics of the emitted X-ray spectrum.

In this context, the present study aims to compare HVL values obtained using two different methods: (i) the Tandem System described by Fontes (2016) and (ii) an Xi CT semiconductor detector (Unfors), widely used in quality control services. The analysis seeks to assess the agreement between the methods, discuss their limitations, and suggest strategies to improve the accuracy of HVL measurements in CT.

2 Methodology

The proposed Tandem System consists of five cylindrical aluminum absorber sleeves with thicknesses of 1 mm, 3 mm, 5 mm, 7 mm, and 10 mm, as well as three PMMA absorber sleeves of 15 mm, 25 mm, and 35 mm. An ionization chamber coupled to an electrometer is inserted into these sleeves and used to determine the ratios between the responses of each set. This makes it possible to identify which configuration performs better in detecting the HVL through the Tandem curve. The second method employs a Xi CT semiconductor detector, manufactured by Unfors, to directly determine the HVL value.

Initially, measurements were performed on the computed tomography scanner using a Radcal Model 10X6-3CT ionization chamber, coupled to a Radcal Accu Gold electrometer, in air for different energy ranges of the CT scanner. The energy levels proposed by Fontes (2016) were used: 80 kV, 100 kV, and 120 kV.

The measurements were taken using the skull protocol from a facility equipped with a 64-slice General Electric (GE) Lightspeed VCT computed tomography scanner. The parameters used are shown in Table 1 below:

Table 1: Parameters used in the CT scanner

Nominal Slice Thickness	Detector Coverage	Pitch	Rotation Time	mA
0.625mm	20mm	0.969:1	1s	100

Figure 1 shows the setup arrangement used in the experiment.

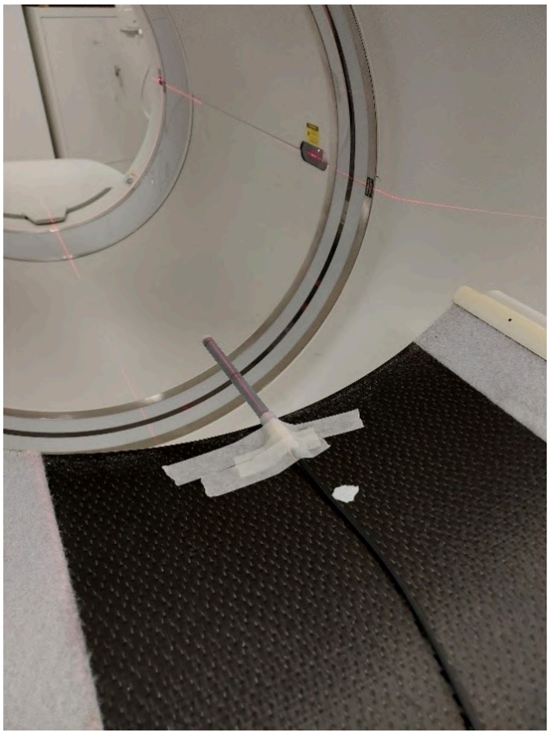
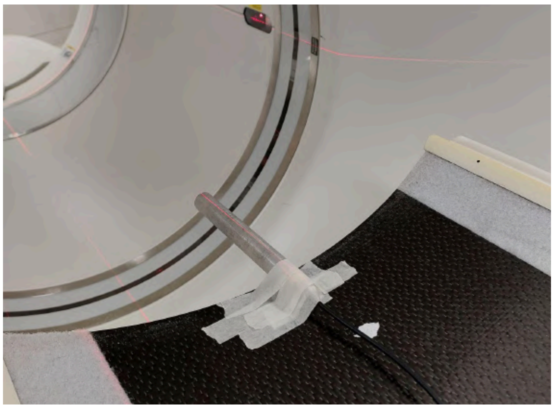


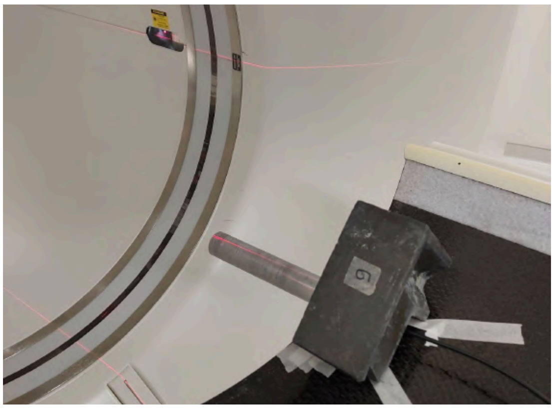
Figure 1: Measurement performed with the ionization chamber in air.

After performing the measurement with the ionization chamber in air, measurements were taken by attaching the aluminum cylinders and conducting the measurement for each set, as shown in Figure 2 (a), (b), (c), (d), (e).

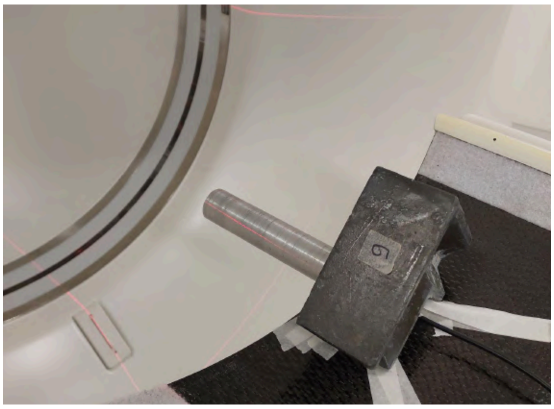
(a)



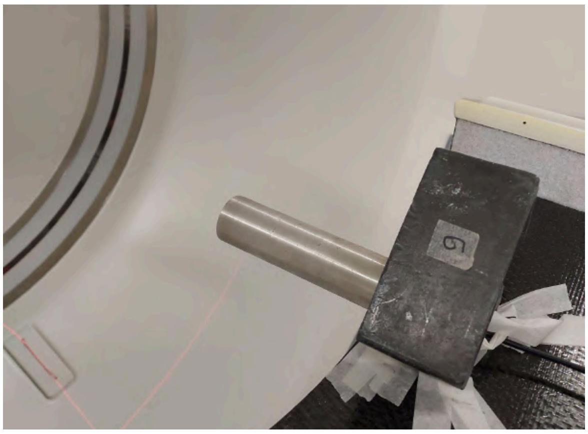
(b)



(c)



(d)



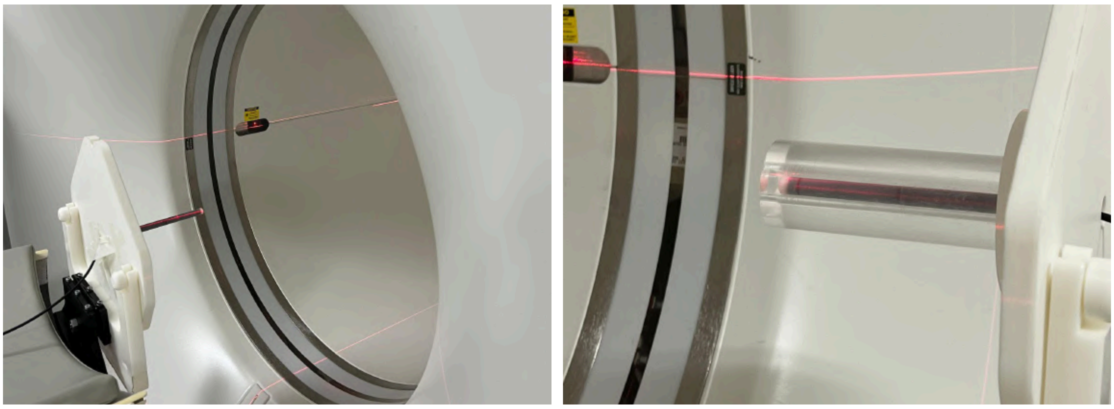
(e)

Figure 2: (a) Measurement performed with one aluminum layer of 1 mmAl thickness. (b) Measurement with two aluminum layers, resulting in a thickness of 3 mmAl. (c) Measurement with three aluminum layers, resulting in a thickness of 5 mmAl. (d) Measurement with four aluminum layers, resulting in a thickness of 7 mmAl. (e) Measurement with five aluminum layers, resulting in a thickness of 10 mmAl.

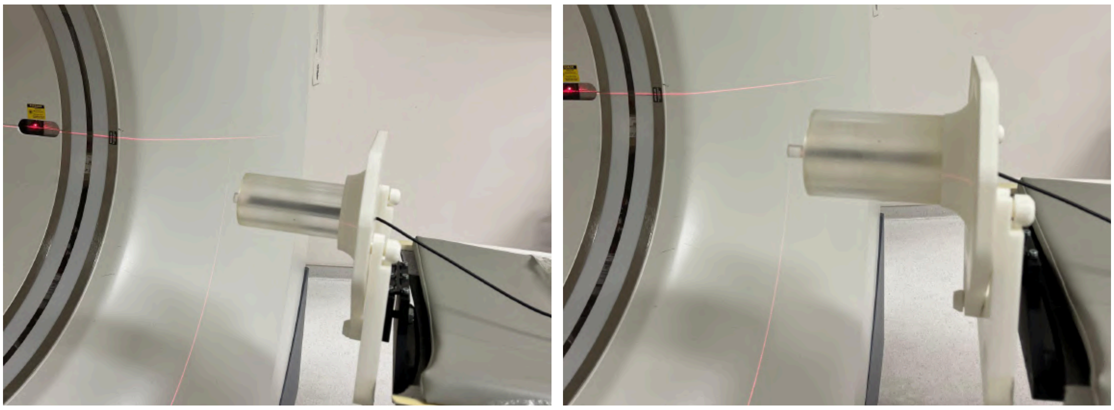
After completing the measurements with all the aluminum layers, measurements were also carried out using PMMA layers.

The PMMA absorber layers are heavier compared to the aluminum layers. Therefore, it was necessary to design a support to hold the PMMA layers in place. The software used to design the support was MicroStation V8i (Select Series 10), version 08.11.09.919, and the 3D printer used to print the part was the Creality Print V4.3.8.7062. The material used for printing was PLA (polylactic acid).

With the support designed to be attached to the CT scanner table, it was possible to position the PMMA layers and carry out the measurements by attaching the ionization chamber, as shown in Figure 3 (a), (b), (c), (d).



(a) (b)



(c) (d)

Figure 3: (a) Measurement with the ionization chamber in air. (b) Measurement performed with one PMMA layer and a thickness of 15 mmPMMA. (c) Measurement performed with two PMMA layers, resulting in a thickness of 25 mmPMMA. (d) Measurement performed with three PMMA layers, resulting in a thickness of 35 mmPMMA.

For the measurements performed with the Xi CT semiconductor detector (Unfors), the device was positioned at the isocenter of the CT scanner gantry. The exposures were carried out using the same parameters described in Table 1, varying only the nominal tube voltages (80 kV, 100 kV, and 120 kV).

3 Results

For each measurement performed with the aluminum filters, within a specific energy range, a table was generated containing the collected data. Table 2 presents the measurements taken with the aluminum filters, while Table 3 shows the results obtained using the PMMA filters.

Table 2: Measurements performed with aluminum filters at different energy ranges.

Attenuator Thickness	**Measurements at 120 kV**	**Measurements at 100 kV**	**Measurements at 80 kV**
--	6.83 mGy	4.64 mGy	2.83 mGy
1mmAl	6.35 mGy	4.31 mGy	2.53 mGy
3mmAl	5.60 mGy	3.67 mGy	2.03 mGy
5mmAl	4.95 mGy	3.13 mGy	1.65 mGy
7mmAl	4.37 mGy	2.69 mGy	1.35 mGy
10mmAl	3.79 mGy	2.25 mGy	1.07 mGy

Table 3: Measurements performed with PMMA filters at different energy ranges.

Attenuator Thickness	Measurements at 120 kV	Measurements at 100 kV	Measurements at 80 kV
--	6.80 mGy	4.67 mGy	2.83 mGy
15mmPMMA	7.13 mGy	4.85 mGy	2.88 mGy
25mmPMMA	6.99 mGy	4.72 mGy	2.76 mGy
35mmPMMA	6.55 mGy	4.38 mGy	2.52 mGy

According to Fontes (2016), the combination that shows the greatest variation—and therefore the best response is the ratio between the measurements with 10 mm Al and those with 35 mm PMMA. Table 4 presents the results of the 10 mm Al/35 mm PMMA response ratio for the different energy ranges used in this study.

Table 4: Result of the ratio between the responses 10 mmAl/35 mmPMMA for different energy ranges.

kV	10mmAl/35mmPMMA
80	0.42
100	0.51
120	0.57

Using the values of the 10 mm Al/35 mm PMMA response ratio, a graph was generated, shown in Figure 4, depicting the response ratio as a function of the applied kV.

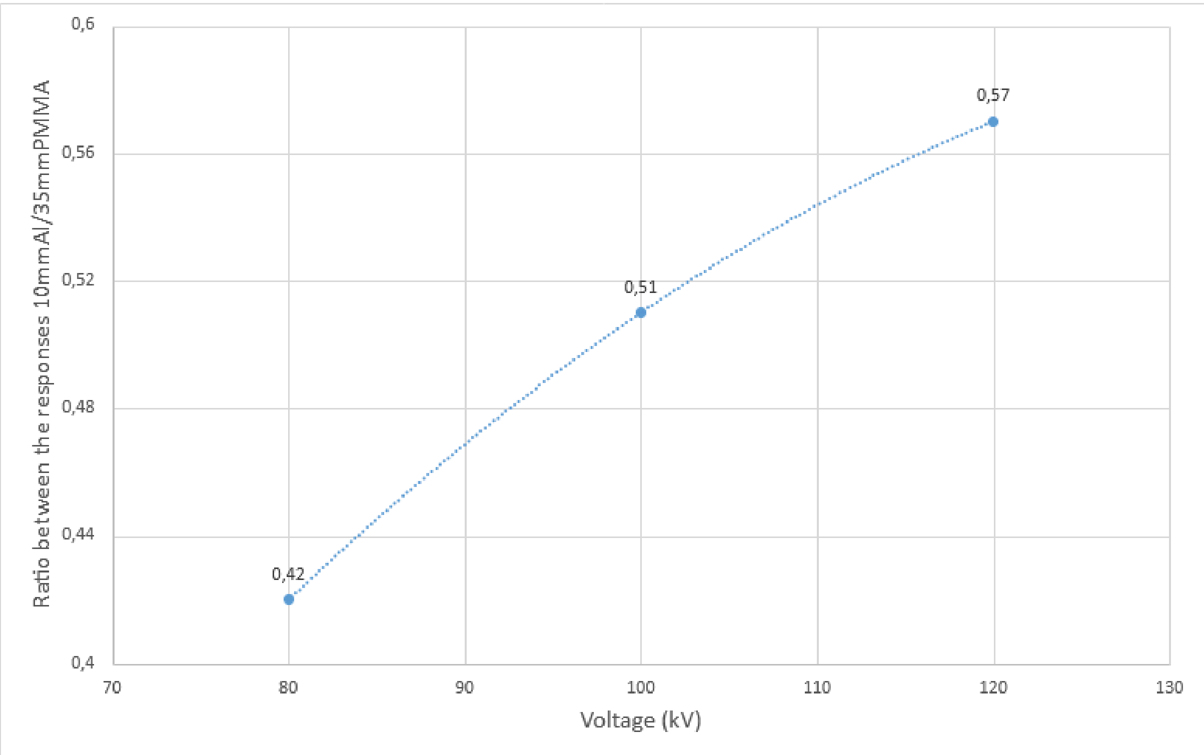


Figure 4: Graph of the response ratio as a function of the applied kV.

Using the Tandem calibration curve provided by Fontes (2016), expressed by Equation 1, it is possible to obtain HVL values for each applied voltage.

$$y = -\ 0,00251 \cdot x^2 + 0,08222 \cdot x + 0,14014 \quad (1)$$

In the given equation, the values of the ratio between the responses are substituted into y, and the HVL value is obtained through algebraic manipulation. In this way, for each response ratio of 10 mmAl/35 mmPMMA, we obtain a corresponding HVL value. Therefore, we can directly associate the HVL values with the kV values.

Table 5 shows the HVL values for the three voltage points, obtained by applying Equation 1.

Table 5: HVL results for different kV values using the Tandem curve equation.

kV	**HVL**
80	3.93 mmAl
100	5.44 mmAl
120	6.69 mmAl

Figure 5 shows the curve that relates the HVL values to the tube voltages used in CT imaging.

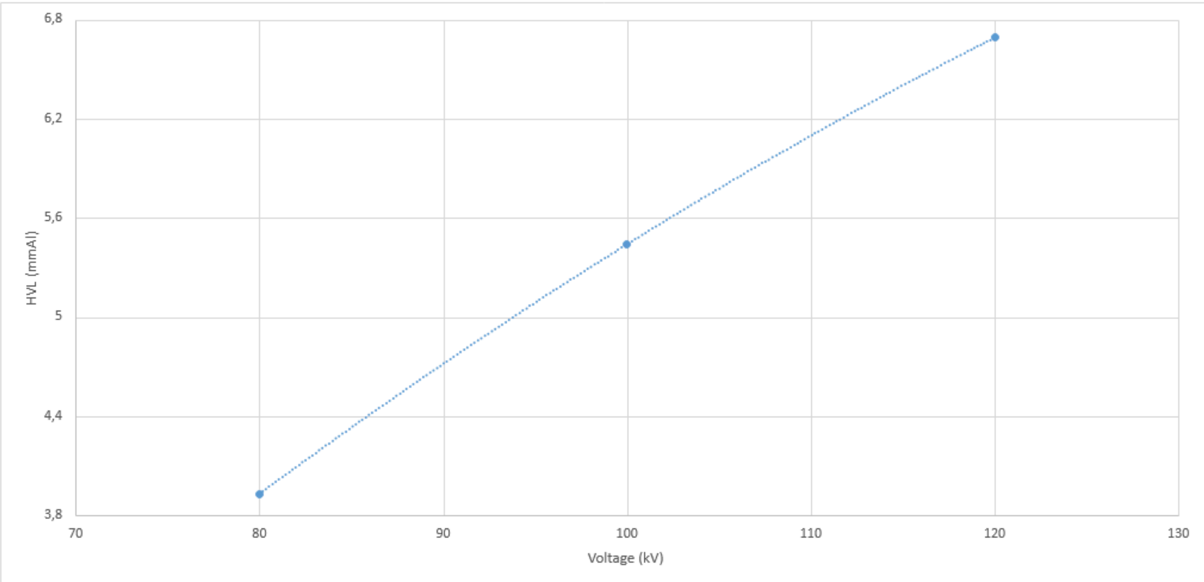


Figure 5: HVL values calculated using the Tandem curve equation for different CT voltage ranges.

By using the Xi CT semiconductor detector coupled to a Platinum Plus unit from the manufacturer Unfors, the HVL values shown in Table 6 were obtained directly.

Table 6: HVL results for different kV values using the semiconductor detector.

kV	**HVL**
80	3.30 mmAl
100	4.05 mmAl
120	4.77 mmAl

The results obtained in this study revealed considerable discrepancies between the Half-Value Layer (HVL) values determined using the method based on the Tandem System (Fontes, 2016) and those measured directly with the Xi CT semiconductor detector. This divergence highlights the complexity of assessing X-ray beam quality in clinical CT scanners, as described by authors such as Boone et al. (2019) and Mayo et al. (2022), who emphasize the challenges posed by beam geometry, energy variations, and the presence of automatic current modulation (Bowen et al., 2014).

Despite these limitations, the study confirms that the combined use of different techniques can provide a broader and more critical view of X-ray beam quality and its stability over time (McCollough et al., 2015; Seibert et al., 2018). It was observed that the Tandem method shows potential as a complementary tool for HVL monitoring, especially due to its adaptability and cost-effectiveness. However, the data demonstrated the need for rigorous and continuous calibration of the ionization chamber, as recommended by McNitt-Gray (2016), in addition to further studies to validate the applicability of the calibration curve used.

On the other hand, the technique based on a semiconductor detector, although it provides results more quickly and directly, also requires periodic calibration checks and a critical analysis of the detector's suitability for the specific type of beam used, in accordance with recommendations found in the literature (Kalender, 2011; Seibert et al., 2018).

4 Conclusion

This study highlighted the importance of adopting complementary methodologies for HVL assessment in computed tomography and pointed out directions for future improvements: expanding the sample size of measurements, developing more stable supports for absorbers, and increasing the integration of indirect methods such as Monte Carlo simulations. These advancements are essential to ensure greater accuracy, reproducibility, and, above all, to contribute to dose optimization and patient safety in CT examinations.

Acknowledgements

The authors acknowledge the financial support from CNPq and the institutions that made this research possible: the Nuclear Technology Development Center (CDTN), the Center for Technology in Molecular Medicine of UFMG, and the Federal University of Minas Gerais (UFMG).

References

Boone, J. M., et al. (2019). *Measurement of half-value layer in CT: A review*. Medical Physics, 46(9), e787–e803.

Bowen, S. R., et al. (2014). *Challenges of measuring HVL in modern CT scanners*. Journal of Applied Clinical Medical Physics, 15(6).

Fontes, L. P. (2016). *Development of a guidance document for dosimetry in computed tomography*. Master’s thesis, Nuclear and Energy Research Institute (IPEN) – University of São Paulo, São Paulo.

International Electrotechnical Commission. (2012). *IEC 60601-2-44:2012 – Medical electrical equipment – Part 2-44: Particular requirements for the basic safety and essential performance of X-ray equipment for computed tomography* (3rd ed.). Geneva: IEC.

Kalender, W. A. (2011). *Computed tomography: Fundamentals, system technology, image quality, applications*. Weinheim: Wiley.

Maia, A. F. (2005). *Standardization of X-ray beams and dosimetric methodology in computed tomography*. Doctoral thesis, Nuclear and Energy Research Institute (IPEN) – University of São Paulo, São Paulo.

McCollough, C. H., et al. (2015). *Quality assurance and radiation dose optimization in CT: The role of physics and engineering*. Radiology, 276(1), 12–23.

McNitt-Gray, M. F. (2016). *AAPM/RSNA physics tutorial for residents: Topics in CT: Radiation dose in CT*. Radiographics, 36(6), 1840–1850.

Mayo, J. R., et al. (2022). *CT dose and quality assurance: State of the art and future directions*. European Radiology, 32, 4643–4656.

Seibert, J. A., et al. (2018). *X-ray quality control testing in computed tomography*. Radiographics, 38(7), 2141–2156.

Comparison of the Dosimetric Properties of the EBT3 and EBT4 radiochromic Films for Application in Hemocomponent Dosimetry

N. M. C. Ignacio[1], M. E. F. F. Milo[1], T. C. Alonso[2], P. C. Santana[1, 2], P. M. C. Oliveira[1], L. C. Meira-Belo[2], D. S. Calheiro[1, 2]*

[1] Medical School, Federal University of Minas Gerais, UFMG, Av. Prof. Alfredo Balena, 190, 30130-100, Belo Horizonte, Brazil

[2] Nuclear Technology Development Center, CDTN, 31270-901, Belo Horizonte, Brazil

[*] danielcalheiro@gmail.com

Abstract

Gamma radiation is highly relevant due to its beneficial applications in several fields, particularly in medicine. In this context, dosimetry is employed alongside gamma radiation applications as it represents the scientific discipline responsible for measuring the absorbed radiation dose received by a given material, tissue, or organism. The objective of this study is to compare the different dosimetric properties of EBT3 and EBT4 radiochromic films for application in hemocomponent dosimetry, under the operational framework of the Gamma Irradiation Laboratory (LIG) at CDTN (Center for the Development of Nuclear Technology), aiming to assess the quality and safety of the irradiation process in compliance with ANVISA (Brazilian Health Regulatory Agency) and ISO/ASTM 51939:2017 (International Organization for Standardization/ASTM International) standards. Ultimately, this also aims to prevent transfusion-associated graft-versus-host disease (TA-GVHD). For this purpose, briefcases containing simulated blood bags (filled with water) and the EBT3 and EBT4 films were irradiated using the Cobalt-60 irradiator facility at CDTN/CNEN (National Nuclear Energy Commission). Twenty-four hours of post-irradiation, the radiochromic films were scanned using an HP Scanjet G4050 scanner, and image analysis was performed using ImageJ software. Dose-response data were processed and analyzed using OriginLab software. Parameters such as sensitivity, reproducibility, repeatability, self-shielding, and angular dependence were evaluated. The results demonstrated that the blue color channel of the visible spectrum exhibited the highest sensitivity within the irradiation dose range relevant to hemocomponents. EBT4 showed greater sensitivity to gamma radiation, consistently registering higher absorbed doses than EBT3, with a maximum difference of 19% in the scenario involving 20 blood bags. In terms of reproducibility, EBT4 showed less variation among repeated measurements (1.75%) compared to EBT3 (4.5%), indicating a more stable and precise response. However, regarding repeatability, EBT3 performed better, with an approximate variation of 2.0%, whereas EBT4 exhibited an 8.0% variation. The analysis of angular dependence indicated that EBT4 is less sensitive to rotational variations, with an average variation

of 3%, compared to the 7% observed in EBT3. Overall, although the manufacturer's documentation does not indicate significant compositional differences between the two film types, the study concludes that EBT4 exhibits superior sensitivity and reproducibility, making it more suitable for hemocomponent dosimetry applications.

Keywords: Gamma Radiation; Dosimetry; Blood Irradiation; EBT3; EBT4; TA-GVHD.

1 Introduction

Gamma radiation is a high-energy electromagnetic wave widely employed for sterilization purposes due to its significant penetration capacity, allowing interaction with sealed materials without compromising their structural integrity [3]. Gamma irradiation induces the formation of highly excited electrons and reactive free radicals through energy transfer from photons, initiating chemical reactions in any phase (gaseous, liquid, or solid) without the need for catalysts [3]. In the irradiation of blood components, Cobalt-60 and Cesium-137 sources are utilized, with a preference for Cobalt-60 due to its relatively short half-life, high production feasibility, and suitable energy levels, emitting photons of 1.17 MeV and 1.33 MeV. Cobalt-60 is an artificially produced radionuclide, obtained through the neutron bombardment of stable Cobalt-59 in a nuclear reactor [3].

Blood transfusion is a critical therapeutic intervention for restoring hematological homeostasis. However, it poses a recognized risk, the transfusion-associated graft-versus-host disease (TA-GVHD), a rare but potentially fatal complication [6,7], caused by the proliferation of donor T lymphocytes and the subsequent immune response in the recipient. To achieve inactivation of these cells and thus prevent TA-GVHD, blood components are irradiated [1, 2, 4, 6, 7] with a minimum absorbed dose of 25 Gy, potentially reaching up to 50 Gy [5, 6], effectively inhibiting T-cell proliferation due to their high radiosensitivity [1,6,7], while preserving the functional integrity of the remaining blood components.

Given the critical importance of blood component irradiation, rigorous monitoring is essential to ensure accurate and effective results [6]. In this context, dosimetry serves as a fundamental tool in the irradiation process [6], being applicable across various sectors, including medicine, the nuclear industry, and scientific research. It constitutes the scientific field responsible for quantifying the absorbed dose by a material or biological system during irradiation [2].

To this end, several methodologies are employed to measure absorbed doses, such as thermoluminescent dosimeters (TLDs), polymethyl methacrylate (PMMAs), and radiochromic films, among others. At the Gamma Irradiation Laboratory (LIG) of the Nuclear Technology Development Center (CDTN), which is affiliated with the National Nuclear Energy Commission (CNEN), radiochromic dosimeters are utilized. These are materials composed of thin polymeric layers that undergo a color change in response to gamma radiation exposure [2], enabling the assessment of the absorbed dose received by blood components and its adequacy for the inactivation of T lymphocytes and pathogens.

Therefore, the aim of this study is to compare the dosimetric properties of EBT3 and EBT4 radiochromic films for application in the irradiation of blood components, from the perspective of the LIG.

2 Methods and materials

2.1 Irradiation of blood components

The Gamma Irradiation Laboratory (LIG) is equipped with a GammaBeam-127, a category 2 multipurpose irradiator featuring a rotating table designed to ensure a homogeneous distribution of the absorbed dose. It is loaded with a dry-stored cobalt-60 source, which can be replaced every five years due to the natural radioactive decay (half-life) of the source. The equipment is used for scientific, technological, and service-related purposes, among others. In this study, blood component bags were irradiated with a reference absorbed dose of 25 Gy, by international standards. This dose level is estimated to be sufficient to inhibit T lymphocytes, as defined by regulatory guidelines concerning dose rates for blood products. Initially, the cobalt-60 source had an activity of 60 kCi; at the time of the experiments, the activity had decreased to approximately 48 kCi. Figure 1 shows an image of the GammaBeam-127 irradiator model used at LIG/CDTN.

Figure 1: Model of the Irradiator at the Gamma Irradiation Laboratory (LIG) of the Nuclear Technology Development Center (CDTN/CNEN).

2.2 Calibration of EBT3 and EBT4 Films

The calibration of the EBT3 and EBT4 radiochromic films was performed simultaneously to ensure that all dosimeters were exposed to identical irradiation conditions, thereby minimizing external experimental variables. Calibrations were conducted for the following absorbed dose values: 5 Gy, 10 Gy, 15 Gy, 20 Gy, 25 Gy, and 30 Gy.The selected values were intended to establish highly accurate calibration curves for each film type, thereby enabling the precise quantification of absorbed doses in the various analyses performed throughout the present study. Figure 2 shows the image of the dosimetry holder and the EBT3 and EBT4 films positioned for irradiation with a known absorbed dose (calibration).

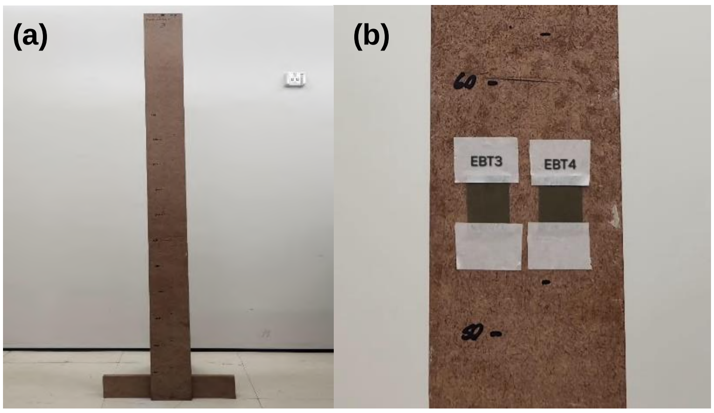


Figure 2: Positioning for the Calibration of EBT3 and EBT4 Films in the Irradiation Chamber at LIG/CDTN. (a) Dosimetry holder; (b) Dosimetry holder containing non-irradiated dosimeters;

2.3 Positioning of radiochromic films in the blood components case

In this study, EBT3 and EBT4 radiochromic films were employed as dosimetric tools to assess the absorbed dose delivered to blood component bags following exposure to gamma radiation. The purpose of using both film types was to perform a comparative analysis to evaluate their dosimetric properties and potential applications in the irradiation of blood products. A total of 20 dosimeters of each type were prepared for distinct irradiation conditions, positioned around an isothermal container filled with water bags, which were used to simulate real blood component bags due to their similar density. To ensure comprehensive dose mapping, six dosimeters were placed on the front and rear surfaces of the cooler (three on the upper region and three on the lower), corresponding to the largest spatial dimensions, while four dosimeters were positioned on the lateral surfaces (two upper and two lower) to detect radiation from all angles and heights. This configuration aimed to assess both dose uniformity and the angular dependence (rotational sensitivity) of the films. The containers were irradiated using different quantities of water bags in each experiment specifically, 5, 10, 15, and 25 bags to evaluate gamma radiation attenuation among the contents and to examine the sensitivity of the radiochromic films. Figure 3 shows the arrangement of water bags simulating blood components, while Figure 4 illustrates the positioning of the radiochromic films within the thermal container used in this study.

Figure 3: Bags filled with water simulating blood components.

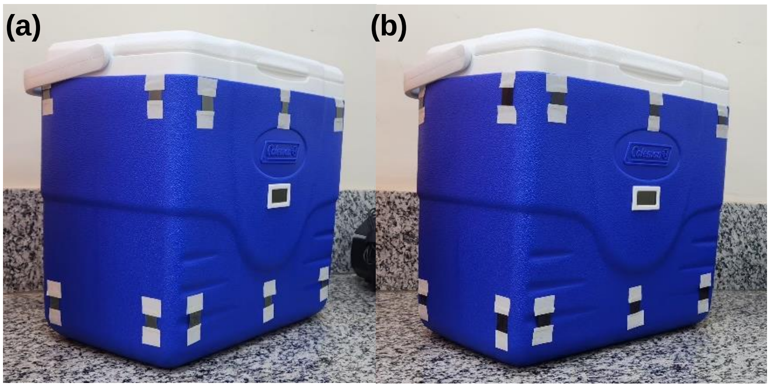


Figure 4: Positioning of the radiochromic films in the blood component container. (a) Container with non-irradiated dosimeters; (b) Container with irradiated dosimeters.

2.4 Analysis of EBT3 and EBT4 Radiochromic Films

To evaluate the sensitivity, reproducibility, and influence of radiation orientation on dosimetric response, three distinct experimental analyses were conducted using EBT3 and EBT4 radiochromic films. The sensitivity analysis involved irradiating sets of water bags (simulating blood components) in varying quantities, ranging from 1 to 20 units, to compare the variation in normalized absorbed dose recorded by each film type.

For the reproducibility analysis, five independent irradiations were performed, each with twenty dosimeters, totaling one hundred measurements per film type. The data were normalized based on the highest dose value recorded in each set, allowing for an assessment of result consistency across repetitions.

Lastly, the influence of radiation orientation was investigated by exposing the films to irradiation (five dosimeters of each model) in two distinct orientations: vertical and horizontal (rotated 90°) and analyzing the percentage variation in dosimetric response under each condition. All irradiation was carried out under identical operational conditions using a Category II multipurpose irradiator (Nordion model), and the data were comparatively analyzed between the two film types to determine the stability, accuracy, and reliability of the materials under different experimental variables.

2.5 Reading radiochromic films

For reading the radiochromic films, an HP Scanjet G4050 transmission scanner, available at the Dosimetry Laboratory of CDTN, was used to scan the irradiated dosimeters 24 hours after irradiation. Using the generated image, the open-source software ImageJ was employed to convert the image to the blue color channel. The blue channel was selected based on an analysis of the other color channels red and green which indicated greater effectiveness of the blue channel for the irradiation of blood components.

The absorbed dose calculation method consisted of evaluating the optical density of the EBT3 and EBT4 films using OriginLab software version 2025a. Following the calibration method previously described in Section 2.2 (Calibration of EBT3 and EBT4 Films) and the optical density values of the calibration films, a graph was generated in OriginLab using a polynomial fit and the R^2 (Coefficient of Determination) to derive an equation corresponding to the calibration values. Consequently, the absorbed dose values were determined from this equation, taking into account the optical density values obtained using ImageJ software for both dosimeters in each irradiation. Below is an example equation (Equation 1) used for the calculation of the absorbed dose of the dosimeters, where 'y' represents the absorbed dose and 'x' the measured optical density value.

$$y = -0.0017x^2 + 0.4261x + 4.9632 \tag{1}$$

3 Results and discussion

3.1 Sensitivity

Sensitivity is defined as the ability of a dosimeter to detect and accurately register variations in absorbed dose. The values presented in Table 1 were obtained through the experimental evaluation of EBT3 and EBT4 films, aiming to compare their performance by analyzing the absorbed dose under different quantities of water bags used in each irradiation, which simulate blood component bags.

Table 1: Table of Normalized Absorbed Dose Detection by Radiochromic Films as a Function of the Number of Bags.

Number of bags	Normalized Absorbed Dose EBT3	Normalized Absorbed Dose EBT4
1-5	0.88	0.98
6-10	0.89	0.97
11-15	0.91	0.99
16-20	0.84	1.00

The normalization of the results was performed considering the highest absorbed dose value obtained in the irradiations using EBT3 and EBT4 radiochromic films. As shown in the table, the irradiations with the EBT4 film under the condition of 16 to 20 simulated blood bags reached the highest value (EBT3: 17.50 and EBT4: 20.83), considering the reference absorbed dose of 25 Gy established by ISO/ASTM 51939:2017 [5]. Taking into account that the manufacturer of the EBT3 and EBT4 films confirmed that no changes were made to the dosimeter properties only an increase in the commercial sheet size (from 20.32cm×25.4cm to 32.512cm×43.18) and given the highest value recorded in the irradiations using the reference absorbed dose for blood components, the data were normalized by applying the value obtained in the irradiation with the EBT4 film under the 16 to 20 bag condition.

The data in Table 1 highlights the variation in normalized absorbed dose as a function of the number of bags irradiated, using EBT3 and EBT4 radiochromic films. For the range of 1 to 5 bags, the normalized absorbed doses were 0.88 for EBT3 and 0.98 for EBT4. As the number of bags increased, the value remained relatively stable: between 0.89 (EBT3) and 0.97 (EBT4) for 6 to 10 bags, and 0.91 (EBT3) and 0.99 (EBT4) for 11 to 15 bags. However, a slight decrease in normalized absorbed doses was observed for 16 to 20 bags, reaching a minimum of 0.84 for EBT3, while EBT4 remained at 1.00. These results indicate that EBT4 film exhibited only a 3% variation from the expected absorbed dose value, demonstrating greater stability and sensitivity for applications involving blood components under the irradiation conditions of LIG/CDTN. In contrast, EBT3 film showed variations of up to 16%, suggesting lower stability and greater sensitivity to positioning and quantity changes during the irradiation process.

3.2 Reproducibility

Reproducibility is related to the consistency of measurement results obtained under varying conditions, such as different operators, equipment, and time intervals. The evaluation method employed consisted of comparing normalized data and their means, recorded by the radiochromic films, based on four independent repetitions with 20 dosimeters per irradiation for each film type, as shown in Table 2. The data were normalized using the highest absorbed dose value obtained among the five irradiation sessions: for EBT3, the maximum value (1.0) was observed in the fifth session; for EBT4, in the fourth session (1.0).

Table 2: Table of Normalized Data from the Absorbed Dose Detection Analysis According to the Number of Bags.

Repetition	**EBT3**	**EBT4**	**Variation EBT3 (%)**	**Variation EBT4 (%)**
1	0.97	0.98	3	2
2	0.93	0.97	7	3
3	0.94	0.99	6	1
4	0.98	1.0	2	0
5	1.0	0.99	0	1

The results presented in Table 2 show that EBT3 exhibited greater variability in its values, with a decrease of up to 7%, indicating lower reproducibility. In contrast, EBT4 showed less variability, maintaining values between 0.97 and 1.00 throughout the tests, with a maximum variation of only 3%.

Conclusions regarding reproducibility were drawn by comparing the mean values obtained in the experiments. EBT4 (1.75%) demonstrated higher reproducibility due to the lower dispersion in its responses, in contrast to EBT3 (4.5%), which exhibited greater susceptibility to external factors. Therefore, EBT4 proved to be more suitable for high-precision and stable measurements in multiple repetitions, providing greater reliability in terms of reproducibility.

3.3 Radiation orientation

The impact of irradiation orientation on the absorbed dose was assessed by analyzing the response variation of EBT3 and EBT4 films under two experimental orientations: vertical and horizontal (rotated 90°). Five dosimeters of each type were employed for this dependency analysis. Table 4 presents the response variation values for the radiochromic films as a function of irradiation orientation.

Table 4: Table of Response Variation Values for EBT3 and EBT4 Radiochromic Films.

Mean	**Response Variation EBT3 (%)**	**Response Variation EBT4 (%)**
Values	7	3

The results indicate that EBT3 exhibited a 7% variation in response, while EBT4 showed only a 3% variation.Nevertheless, it cannot be conclusively stated that the evaluated dosimeters exhibit irradiation orientation dependence, as the values obtained are within or close to the range observed in the reproducibility analysis of the tested radiochromic films.

4 Conclusion

Based on the results obtained in this study, it is possible to conclude that EBT4 radiochromic film demonstrates superior performance compared to EBT3 in terms of sensitivity, reproducibility, and stability under varying irradiation orientations. The sensitivity assessment showed that EBT4 exhibits a more uniform response with only a 3% deviation across different quantities of irradiated water bags, while EBT3 showed up to 16% variation, indicating lower accuracy in absorbed dose detection.

Regarding reproducibility, the experimental data demonstrated that EBT4 provides more consistent results across repeated experiments, with a lower dispersion of values (mean deviation of 1.75%) when compared to EBT3 (4.5%). This stability is crucial to ensuring the reliability of measurements in applications involving blood component irradiation.

The analysis of irradiation orientation influence revealed that EBT4 had a response variation of approximately 3%, whereas EBT3 showed a variation of about 7%. However, it cannot be conclusively stated that both dosimeters present irradiation orientation dependence, given the proximity of the obtained results to the reproducibility test values for both films.

In conclusion, based on all analyses, the EBT4 film is better suited for use in dosimetric procedures involving the irradiation of blood components, particularly under the operational conditions of the irradiation chamber at LIG/CDTN. Furthermore, it meets international technical standards for precision, stability, and quality more reliably, as established by ISO/ASTM 51939:2017

Acknowledgments

The authors are grateful to the National Council for Scientific and Technological. Development (CNPq), the Coordination for the Improvement of Higher Education Personnel. (CAPES), The Minas Gerais State Research Support Foundation (FAPEMIG) and The National Nuclear Energy Commission (CNEN) for supporting the study.

References

[1] Alan, H. Y., Ünal, G., Atakul, T., Akyürek, S., & Yücel, H. (2024). A study on quality control check of dose distribution in blood bags irradiated by 137Cs source blood irradiator by using EBT3 radiochromic film dosimetry. *Radiation Physics and Chemistry*, *214*, 111303.

[2] Calheiro, D. S., Meira-Belo, L. C., & Alonso, T. C. (2024). Mapping absorbed dose distribution in containers of blood and blood components using radiochromic dosimeters irradiated in an industrial irradiator. *Radiation Physics and Chemistry*, *222*, 111797.

[3] da Silva, R. C., da Silva, R. M., & Aquino, K. A. S. (2014). A interação da radiação gama com a matéria no processo de esterilização. *Revista Virtual de Química*, *6*(6), 1624-1641.

[4] Farajzadeh, E., & Sina, S. (2021). Developing a radiochromic dosimeter for dosimetry in blood irradiation chambers. *Radiation Physics and Chemistry*, *188*, 109637.

[5] ISO/ASTM. (2017). ISO/ASTM 51939:2017 - Standard Practice for Blood Irradiation Dosimetry.

[6] Fonseca, L., Calheiro, D. S., Mesquita, A., Alonso, T., & Squair, P. (2024). Calibration and evaluation of the MTS-N dosimeter for absorbed dose measurement in blood components. *Brazilian Journal of Radiation Sciences*, *12*(4A (Suppl.)), e2502-e2502.

[7] Francis, R. O. (2025). Irradiation of blood products. In *Transfusion medicine and hemostasis* (pp. 229-232). Elsevier.

[8] Lopes, F., Rocha, R. C. G., Munhoz, E., Trigo, F. C., Feijó, V. B. E. R., Fernandes, M., ... & Araujo, G. S. (2023). Quality control in blood irradiation. *Brazilian Journal of Radiation Sciences*, *11*(1A (Suppl.)), 01-17.

XXV ISSSD
International Symposium on Solid State Dosimetry
September 22 – 26, 2025
Belo Horizonte, Minas Gerais - Brazil

Computational Tools for Automatic Conversion of Microscopy Images into Voxelized Cellular Phantoms

L. B. Silva[1*], T. C. F. Fonseca[2,3,4], L. F. Araujo[2]

[1] Universidade Federal de Minas Gerais, Departamento de Física/Instituto de Ciências Exatas, Av. Antônio Carlos, 6627, Pampulha, 31270-901, Belo Horizonte, Minas Gerais, Brazil

[2] Universidade Federal de Minas Gerais, Departamento de Engenharia Nuclear, Av. Antônio Carlos, 6627, Pampulha, 31270-901, Belo Horizonte, Minas Gerais, Brazil

[3] Universidade Federal de Minas Gerais, Programa de Pós-Graduação em Física, Av. Antônio Carlos, 6627, Pampulha, 31270-901, Belo Horizonte, Minas Gerais, Brazil

[4] Centro de Desenvolvimento da Tecnologia Nuclear, CDTN, Av. Antônio Carlos, 6627, Campus UFMG – Pampulha, 31270-901, Belo Horizonte, Minas Gerais, Brazil

*belotilucas4@gmail.com

Abstract

Computer simulations of particle transport using the Monte Carlo method are extensively employed in dosimetry, primarily at the macroscopic scale. However, there is an increasing necessity for high-fidelity representations of cellular environments at micro- and nanometric scales. At these dimensions, such simulations are essential for accurately quantifying energy deposition and dose distribution. The use of realistic, image-based target models provides two key advantages: it enhances dosimetric accuracy and enables a more detailed analysis of dose heterogeneity at the subcellular level. When integrated with radiobiological data, this approach can elucidate the mechanisms of radiation interaction within cells. This study introduces two distinct image processing methods for generating three-dimensional microscopic models designed to investigate the radiosensitizing effects of nanoparticles. The methodology utilizes freely available computational tools, including custom scripts and open-source software, to establish an automated workflow for converting microscopy images into voxelized cellular phantoms suitable for Monte Carlo simulations. The first method converts binary microscopy images into a voxelized format, processes the resulting cellular geometry, and simulates the spatial distribution of nanoparticles in the medium. The second method is designed for multichannel fluorescence microscopy images. It segments distinct, fluorescently-labelled cellular structures, reconstructs them as individual 3D mesh objects, and subsequently converts them into a unified voxelized model. Validation for the first method was performed quantitatively using object analysis software. For the second method, the generated models exhibit high morphological fidelity to the source microscopy images. While Monte Carlo simulations are well-established for macroscopic applications, their accuracy and efficiency at microscopic scales remain an area of active investigation. The methods presented here offer new possibilities for high-resolution modelling, though they could be enhanced by incorporating additional features. A critical factor is that the quality of the input images significantly influences the fidelity of the final model. Future work should therefore focus on developing adaptive image processing algorithms capable of preserving high-detail regions while mitigating noise and artifacts in lower-quality areas.

Keywords: Monte Carlo; Computational Dosimetry; Voxel; Simulation

1 Introduction

The use of radiation in medicine starting in the 19th century was a significant milestone in the development of clinical practice, but exposure to increased levels of radiation does carry significant risks. Thus, the primary objective of radiation therapy is to achieve the most effective dose deposition to tumour tissue while delivering the least possible dose to adjacent healthy tissues (Donya et al., 2015). As research in radiation therapy advanced, radiosensitization became more prevalent as a strategy to improve tumour control and optimize treatment efficiency. Radiosensitizers increase the sensitivity of tumour tissue to radiation and, to this day, many types of nanoparticles have been used to fulfil this purpose, such as gold, iron, carbon, bismuth, and titanium NPs (Abdollahi et al., 2020). Gold nanoparticles are proven effective in augmenting the performance of radiation therapy in preclinical studies in mice (Hainfeld et al., 2010) and their role as a radiosensitizer has also been the subject of investigation through Monte Carlo simulations (Chow, 2018).

1.1 Human Phantoms in Dosimetry Simulations

Good phantom representations of the human body and its organs allow for a better estimation of the dose delivered to each organ, which leads to increased patient safety in radiation treatments in radiotherapy (Paixão et al., 2019b; Fonseca et al., 2020). In structures like microscopic organisms and the human body, realistic models can better represent the complex heterogeneity of these structures and thus radiation dose can be measured in locations of interest Xu (2014). Advances in image processing software have enabled the replacement of basic representations with more accurate and realistic models Ferreira Fonseca et al. (2014).

The first human phantoms date back to the 1960s and 1970s and represented the human body using simple geometrical forms. A phantom of this type was developed by Kerr et al. (1976) at Oak Ridge National Laboratory, Figure 1.1(a) and 1.1(b), and portrayed an adult consisting of three sections built from cylinders, ellipsoids and cones. By that time they were already able to represent inner structures like organs and bones, Figure 1.1(a) and 1.1(b). Further progress was made in the 1980s with the use of CT data, a technique which allowed any physical phantom to be converted into computer files to be used for several applications. Zankl et al. (1988) used this to convert patient picture data into computer files, made out of volume elements (voxels), which could then be attached to a Monte Carlo code for the calculation of absorbed dose in organs and tissues, Figure 1.1(c). More advanced models began to appear in the last few decades that are based on boundary representation (BREP), as described by Xu (2014); Ferreira Fonseca et al. (2014). This technique yields very smooth surfaces that allow the inclusion of complex anatomical features. Figure 1.1(d) shows a male and a female phantom developed using BREP.

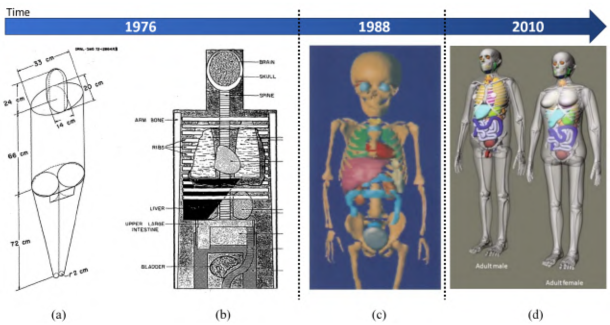


Figure 1: (a) Simple human phantom from 1976 and (b) Representation of internal organs in a computational phantom (Kerr et al., 1976) (c) Human phantom reconstructed from CT images (Zankl et al., 1988) (d) UF family phantoms developed from BREP methods Bolch et al. (2010).

The current models used to simulate the effects of nanoparticles as radiosensitizers remain relatively simplified, often lacking the level of detail and complexity found in concurrent macroscopic human phantoms (Retif et al., 2016; Cai et al., 2017; Paro et al., 2016; Seniwal et al., 2020). Our research group, the Monte Carlo Modelling and Expert Group ((MCMEG), 2025), specializes in Monte Carlo radiation transport modelling and simulation, with a focus on applications in radiation protection and dosimetry. The group has contributed to various studies involving radiotherapy dose calculations (Fonseca et al., 2020; Paixão et al., 2019a; Fonseca et al., 2017). In this context, (Seniwal et al., 2021) present a mathematical dosimetric study on low-dose-rate brachytherapy using radioactive nanoparticles. As another example (Araujo and Fonseca, 2025) used a three-dimensional cylinder to represent a homogeneous culture medium, consisting solely of water and randomly distributed gold NPs, to assess gold radiosensitization in a brachytherapy scenario. To investigate the effect of gold nanoparticles' distribution on cellular dose enhancement, Rasouli and Masoudi (2019) used ellipsoids to model the cytoplasm and nucleus of a single cell. While simpler models offer certain advantages, more complex models offer the possibility of a more detailed and accurate analysis of the problem at hand.

The aim of this study was to present two distinct image processing methods for creating three-dimensional microscopic models using freely available software tools commonly applied in 3D analysis and the construction of macroscopic human phantoms. The proposed idea is to develop cell models to investigate the radiosensitization effects of nanoparticles in biological tissues. This project's scripts integrate multiple tools and automate their operation, streamlining the process for the user.

2 Materials and Methods

Here two distinct image processing methods for creating three-dimensional microscopic models are presented with the goal of investigating the radiosensitization effects of nanoparticles in biological tissues. The key difference between the two approaches lies in the type of image data used and how cellular structures and nanoparticles are represented and processed, however, both images used here in Case Study 1 and 2 are in .tif format and were treated using imageJ (Schneider et al., 2012; ImageJ Development Team, 2025).

2.1 Method 1:

2.1.1 Development of the process

The first step of the developed workflow involves the use of specific tools to convert microscopic cellular images into a voxelized format. This process, known as voxelization, transforms the input image data into a structured 3D grid composed of cubic elements (voxels), each corresponding to a specific segment of the original image. The output is a text file (.txt) containing a three-dimensional matrix, where each voxel is assigned a numerical value representing distinct regions or structures within the image, such as the cytoplasm, nucleus, or extracellular medium. This voxel matrix serves as the basis for subsequent steps in defining the distribution of nanoparticles, taking into account the specific regions where the nanoparticles will be randomly fixed, say inside the nucleus, on the surface, near the nucleus, or in the surrounding medium. These spatial distributions are critical for the accuracy of the Monte Carlo simulations and dose calculations that follow. An overview of the developed process is illustrated in Figure 2.

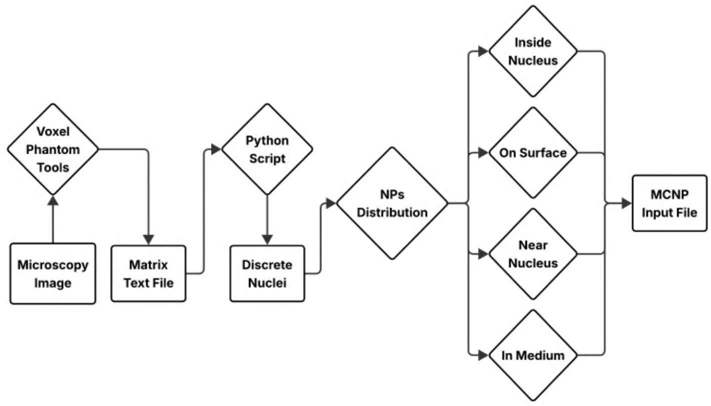


Figure 2: Flowchart of Method 1: Steps in the process of creating the voxel cell model.

2.1.2 3D Cellular Model

In vitro cell models are commonly used to study diseases and predict treatment effectiveness. However, conventional 2D models have limitations and fail to adequately reproduce *in vivo* tissue dynamics (Achilli et al., 2012; Lazzari et al., 2017). The 3D cell model, developed and made available by Maury et al., differs from 2D monolayers by incorporating the extracellular matrix, a structure that provides biochemical and structural support to the surrounding cells. This type of 3D *in vitro* model enables intercellular communication through the transmission of biochemical signals, playing a role in cell proliferation, differentiation, and gene expression.

2.1.3 Conversion into Voxelized Matrix: Voxel Phantom Tools (VPT)

The voxelization of this 3D cell image file was performed using the Voxel Phantom Tools (VPT). This tool is an ImageJ plug-in developed by Ros et al. (2007). It was also used by Paixão et al. (2019a) to develop a voxel phantom into EGSnrc code. The plug-in has the capacity to convert sliced images into a three-dimensional voxelized matrix, as well as to convert them into a format that the Monte Carlo MCNPx code (Werner et al., 2017) can read.

2.1.4 Random Nanoparticle Cluster Distribution

A custom Python script was developed to process the voxelized matrix in two main steps. In the first step, the script discretizes the cellular nuclei based on the voxelized .txt file and, optionally, calculates the centroid (centre of mass) and volume of each nucleus. This information can be used to support further analysis or guide nanoparticle placement. In the second step, the script performs the distribution of nanoparticles, with four possible spatial configurations: (i) inside the cell nucleus, (ii) on the nuclear surface, (iii) within a defined radius from the cell's centre of mass (perinuclear region), and (iv) randomly distributed in the extracellular medium. When nanoparticles are internalized by cells, via mechanisms such as phagocytosis or pinocytosis, they are typically encapsulated within vesicles (endosomes), preventing direct interaction with the cytosol and intracellular organelles (Radaic et al., 2016). This flexible framework allows for simulations that reflect different biological scenarios, enhancing the relevance of subsequent Monte Carlo dose calculations.

In all four possible spatial strategies, selected voxels within the 3D model are replaced with nanoparticles following spatial criteria specific to each case. For the intranuclear distribution, nanoparticles are placed randomly across all voxels labelled as nucleus. A voxel is classified as nuclear surface if it belongs to the nucleus and has at least one of its six immediate Neighbors (in the 3D grid) identified as background; nanoparticles are also randomly assigned within this subset. In the perinuclear region, nanoparticle placement depends on the voxel's distance from the cell's centre of mass: voxels within a specified radius are grouped into small spherical clusters to simulate localized endosomal accumulations, while those beyond this region are filled randomly. For the extracellular distribution, nanoparticles are randomly placed in voxels corresponding to the surrounding medium. The total number of nanoparticles placed in the model can be defined by the user, and a specific value may be assigned based on the desired nanoparticle concentration to be used in the dosimetric calculations. Once the distribution process is complete, the modified voxel matrix is compressed and formatted as an input file compatible with MCNP simulations.

To validate the efficiency of the voxelization process, the volume of each nucleus as well as their centre of mass (CM) was calculated and compared with the results obtained from 3D Objects Counter (Bolte and Cordelières, 2006), an ImageJ plug-in that analyses object data directly from the images.

2.2 Method 2:

2.2.1 Conversion into Voxelized Matrix: Binvox and Binvox2MCNP

Here the conversion of image files into 3D mesh objects was performed using Vedo, a Python module for the analysis and visualization of three-dimensional structures (Musy et al., 2021). To transform the mesh objects into voxelized 3D files, the tools Binvox and Binvox Merge were employed (Nooruddin and Turk, 2003; Min, 2004 - 2019). These freely available utilities convert object data into 3D voxel grid format and allow the merging of multiple structures into a single voxelized model, facilitating the creation of phantoms with distinct structural features.

The final step in the pipeline involved Binvox2MCNP, a C++ program developed to convert the Binvox output into an MCNPx input file containing the 3D voxel lattice of the phantom (Fonseca et al., 2014).

To generate the mesh models, the RGB colour channels from the original images were first separated into three individual stacks. A fourth stack, consisting of dark images with bright corner markers, was created to define a bounding box so as to avoid file dimension conflicts on Binvox. After applying thresholding to reduce noise, each stack was individually converted into a 3D mesh using Vedo and then voxelized with Binvox. The resulting voxel files were merged using Binvox Merge before being converted into the MCNP-compatible format by Binvox2MCNP. A flowchart illustrating this process is shown in Figure 3.

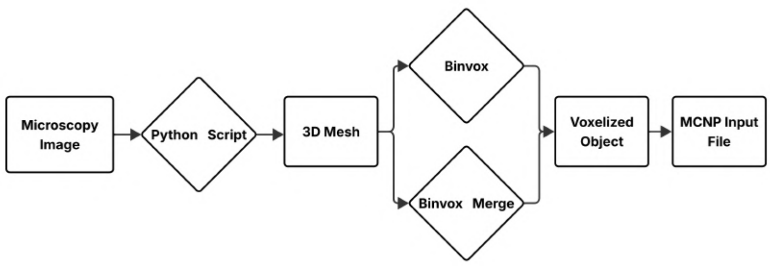


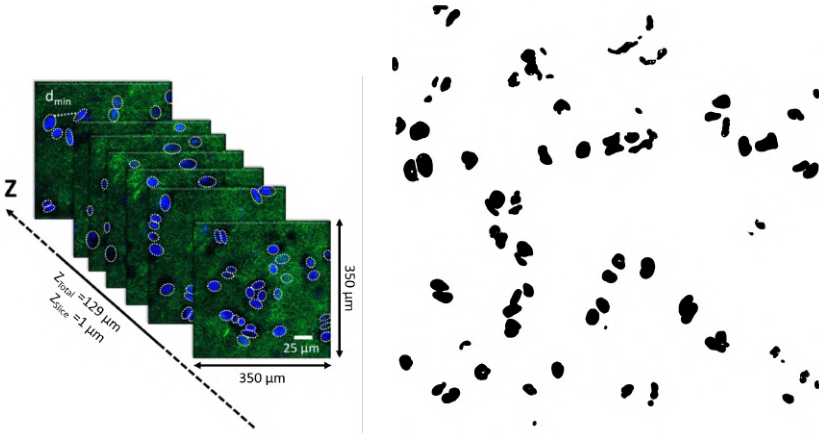
Figure 3: Flowchart of Method 2: Steps in the process of creating the voxel cell model.

3 Results and Discussion

3.1 Method 1

3.1.1 Case Study 1: 3D *In Vitro* HeLa Cell

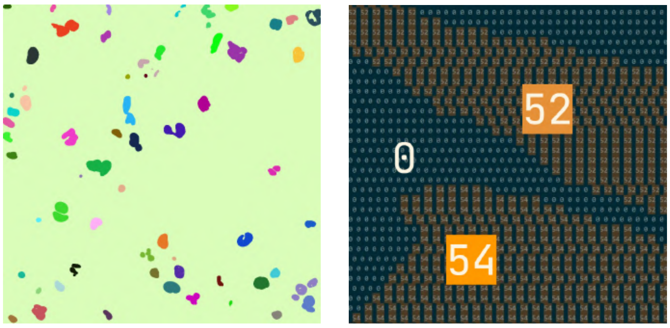
In the first method, a 3D model of *in vitro* cells, developed by Maury et al. (2021) was used. The authors provided image data acquired using complementary microscopy techniques: confocal and multi photon microscopy, the latter offering greater penetration depth and enabling high-resolution 3D imaging throughout the entire sample volume. The voxel model was generated from downsized versions of the original images with 80 binary (black and white) image layers of HeLa cells (human epithelial cells from cervical adenocarcinoma) with a resolution of 1000 x 1000 pixels. These images contain only a single channel and depict cellular morphology without the presence of nanoparticles. A voxel matrix was created with a single labelled structure (the nucleus), and nanoparticles were artificially added afterwards using a custom Python script, allowing for various spatial distributions (e.g., inside the nucleus, on its surface, or in the perinuclear region). Figures 4(a) and 4(b) illustrate the 3D arrangement of the cells and an example of one of the image layers made available in their supplementary material. The original HeLa Cell image files comprised 129 layers of 1822x1822 pixel resolution and their dimensions are shown in Figure 4(a), measuring 350 μm $\times$ 350 μm in the x and y directions, and 129 μm in the z direction.



(a) (b)

Figure 4: (a) HeLa cellular *in vitro* cells. Each image represents a single layer of the well in the vertical, Z, direction (Maury et al., 2021). (b) HeLa cellular nuclei after processing by (Maury et al., 2021): converted to a binary format and segmented.

The images were converted by VPT into an input file with a voxel matrix of 1000 x 1000 x 80 elements, formatted as a text file, where the pixels are represented by numbers that range from 0 to 256 according to their brightness. The selected sample was in binary format, with the background labelled as 0 and the nuclei labelled as 1. To enable analysis of nanoparticle radiosensitization effects at the level of individual nuclei, unique numerical indices were assigned to each nucleus using the Python script. Nuclei were identified using a connected component labelling algorithm, which detects connected regions by examining each voxel's 26 neighbours for matching indexes. The Z direction is defined relative to the sample's layered structure, with each layer comprising 1,000 consecutive lines while the X and Y direction are defined as the lines and rows of the matrix respectively. Once identified, the nuclei indexes are replaced by ever increasing numbers as shown in Figure 5(a) and 5(b). As a result of the discretization, a cell count can also be obtained allowing the extraction of additional information for each nucleus, such as its location, centre of mass, volume, and surface area. These parameters can be used to validate the voxelization by comparing our results with those obtained through other object counting and analysis methods.



(a) (b)

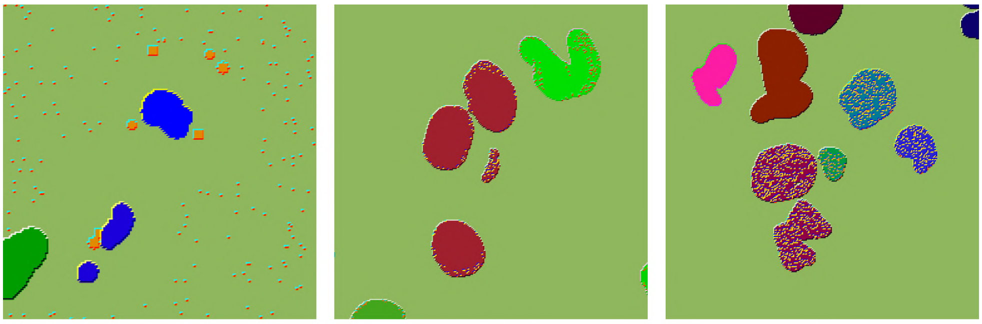
Figure 5: (a) HeLa Cells with different indexes. The image was reconstructed from the text file, each number is assigned a different colour. (b) Cells indexed 52 and 54 and background differentiated in the text file.

It was found that the data produced by the Python script matched that of 3D Objects Counter plug-in almost perfectly. Table 1 presents the average percentage difference between the mean values of the nuclei centre of mass coordinates and volumes.

Table 1: Average percentage differences between the results obtained using our script and those from the 3D Objects Counter.

Metric	Average Difference (%)
Volume (number of voxels)	0.000
X_{CM}	0.000
Y_{CM}	0.000
Z_{CM}	0.001

Figure 6 illustrates the distinct nanoparticle distribution strategies applied to the cell model, in which four types of distributions can be observed. Each voxel in the Figure represents a volume of $0.192\,\mu m \times 0.192 \mu m \times 1.000 \mu m$, meaning that each nanoparticle voxel corresponds to an agglomerate of particles rather than a single nanoparticle. In Figure 6a, nanoparticles are distributed in the medium surrounding the cells. Particles located beyond a defined radius from the cell's centre of mass are randomly distributed, while those within this radius, representing the cytoplasm, form clumps to simulate endocytosis. Nanoparticles that reach the nucleus are also randomly distributed and can either remain on the nuclear surface, as shown in Figure 6b, or penetrate into the nuclear region, as shown in Figure 6c. Although the distributions are presented separately in the Figure, they can be applied simultaneously or in any desired combination. Additionally, the number of nanoparticles to be distributed in each scenario can be defined by the user.



(a) (b) (c)

Figure 6: Four types of nanoparticle distribution: (a) Nanoparticles distributed outside the nuclei, in clumps within a certain radius and randomly in the medium. (b) Nanoparticles distributed on the surface. (c) Nanoparticles distributed inside the nuclei.

3.2 Method 2

3.2.1 Case Study 2: Osteosarcoma Cells

The second method was applied to osteosarcoma cell images provided by Rodrigues et al. (2024). Osteosarcoma (OS) is the second most common type of primary malignant bone tumour and is highly aggressive. This cancer has a poor prognosis, as approximately 30% of patients develop lung metastases. Hydroxyapatite (HA), a natural component of bone, has been extensively used as a biomaterial and is a popular choice for bone repair. Additionally, recent reports suggest that nanoparticles of this substance can suppress the proliferation of various cancer cell types. With the goal of investigating a form of osteosarcoma treatment by incorporating cytotoxic chemicals in nanostructures, the authors Rodrigues et al. (2024) assessed nanoparticle uptake by cells using immunofluorescence microscopy, in which α-tubulin proteins were visualized in red and cell nuclei in blue in Figure 7. The structural and morphological characterization of the nanoparticles, represented in green, was conducted using transmission electron microscopy (TEM).

A set of 400 microscopy images, 100 layers for each colour channel (red, green, and blue) and 100 corresponding images for the bounding box, were transformed into 3D objects, each with a resolution of 1200 × 1200 pixels, representing successive layers of the cell (Figure 7). In this case, each channel corresponds to a different cellular component and the nanoparticles are already visible in the images. Here, the 3D surface meshes are generated using Vedo, voxelized with Binvox, and then merged using Binvox Merge. This separation preserves the structural identity of each component and enables accurate conversion to MCNP input using Binvox2MCNP, making it possible to perform more realistic dose calculations based on the actual spatial arrangement of the nanoparticles and cell structures.

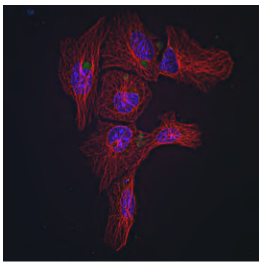
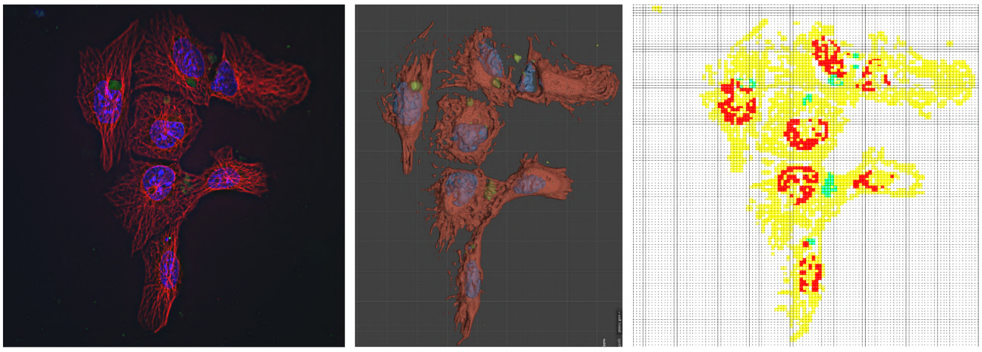


Figure 7: Immunofluorescence image of osteosarcoma cells: Nanoparticles in green, a-tubulin proteins in red, and nuclei in blue Rodrigues et al. (2024).

Figure 8 (c) shows the result of the conversion of the original image to a voxelized MCNP input in comparison to the original image Figure 8 (a). In the conversion of the mesh to a voxelized object through Binvox a resolution of 200 x 200 x 200 voxels was used, hence the lower amount of detail in Figure 8 (c), however Binvox supports resolutions of up to 512 cubed voxels in its free version and up to 4096 cubed voxels in its paid version. By voxelizing each colour channel individually and merging them using Binvox Merge, it was possible to generate a final input with distinctly indexed features. In Figure 8(c), the α-tubulin protein is shown in yellow, the nanoparticles in green, and the nuclei in red.

Careful manual adjustment of the threshold for each colour channel is crucial, since unfocused noise can otherwise be misclassified as part of the cell. This problem can be mitigated if better image sources are used, especially if the object imaged is in focus throughout the Z direction. This issue has been primarily observed in the first and last few images of the stack, suggesting that handling these layers differently during processing could improve the overall quality of the final model.



(a) (b) (c)

Figure 8: Voxelized model obtained from Method 2: (a) Original image, (b) 3D mesh created using Vedo and (c) slice of the voxelized MCNP input file.

4 Conclusions

This study introduced two image processing methods for creating three-dimensional microscopic models to support investigations into the radiosensitization effects of nanoparticles in biological tissues. This work is part of a doctoral thesis and is being applied in its final stage. In the first method, a voxel phantom of human epithelial cells (HeLa cell line) was generated from binary microscopy images. Nanoparticles were computationally distributed in various regions, randomly inside the nucleus, on its surface, in the surrounding medium, and clustered near the nucleus—to mimic endocytosis observed in real cells. The second method focused on osteosarcoma cells, processed through multichannel fluorescence microscopy. Individual cellular structures were segmented and reconstructed into 3D meshes using Vedo, voxelized with Binvox, and then merged with Binvox Merge. The final model was converted into an MCNPx compatible input file using Binvox2MCNP. While Monte Carlo simulations are well established for macroscopic applications, their accuracy and efficiency at microscopic scales remain underexplored. The first method presents new possibilities for high-resolution modelling but can be improved by incorporating additional features, such as nanoparticle agglomeration on the cell membrane, an aspect not addressed due to the absence of membrane data in the used images. It was found that, in the second method, the quality of the input images significantly affects the final model. Future work should focus on developing adaptive processing approaches that preserve high-quality regions while mitigating noise in low-quality areas. The scripts presented in this article are currently being improved and will be made publicly available as image processing tools in the near future.

Acknowledgements

The following Brazilian institutions support this research project: Research Support Foundation of the State of Minas Gerais (FAPEMIG), Brazilian Council for Scientific and Technological Development (CNPq), and Coordination for the Capacitation of Graduated Personnel (CAPES). Prof. T. C. F. Fonseca is grateful for the financial support provided by CNPq (101845/2024-3 and 302990/2021-6). We also thank the FAPEMIG Project: Rede Mineira de Pesquisa Translational em Oncologia (RMPTO) RED00059-23. The authors would like to thank Patrick Min, developer of the Binvox software, for his collaboration and clarification of information. We also express our thanks to the authors cited in this work for providing and sharing the microscopic images that served as the focus of this study.

References

Abdollahi B. B., Malekzadeh R., Azar F. P., Salehnia F., Naseri A. R., Ghorbani M., Hamishehkar H., and Farajollahi A. R. (2020) Main approaches to enhance radiosensitization in cancer cells by nanoparticles: A systematic review. *Advanced pharmaceutical bulletin* 11:212

Achilli T.-M., Meyer J., and Morgan J. R. (2012) Advances in the formation, use and understanding of multi-cellular spheroids. *Expert opinion on biological therapy* 12:1347–1360

Araujo L. and Fonseca T. (2025) In silico monte carlo with novel particle tagging: Assessing gold radiosensitivity in voxelized scenario of brachytherapy. *Applied Radiation and Isotopes* 220:111745

Bolch W., Lee C., Wayson M., and Johnson P. (2010) Hybrid computational phantoms for medical dose reconstruction. *Radiation and environmental biophysics* 49:155–168

Bolte S. and Cordelières F. P. (2006) A guided tour into subcellular colocalization analysis in light microscopy. *Journal of microscopy* 224:213–232

Cai Z., Kwon Y. L., and Reilly R. M. (2017) Monte carlo n-particle (mcnp) modeling of the cellular dosimetry of 64cu: comparison with mirdcell s values and implications for studies of its cytotoxic effects. *Journal of Nuclear Medicine* 58:339–345

Chow J. C. (2018) Recent progress in monte carlo simulation on gold nanoparticle radiosensitization.. *AIMS Biophysics* 5

Donya M., Radford M., ElGuindy A., Firmin D., and Yacoub M. H. (2015) Radiation in medicine: Origins, risks and aspirations. *Global Cardiology Science and Practice* 2014:57

Ferreira Fonseca T. C., Bogaerts R., Lebacq A. L., Ribeiro R. M., and Vanhavere F. (2014) Mamp and femp: computational mesh phantoms applied for studying the variation of wbc efficiency using a nai(tl) detector. *Journal of Radiological Protection* 34:529

Fonseca T., Antunes P., Belo M., Bastos F., Campos T., Geraldo J., Mendes A., Mendes B., Paixão L., Santana P., Seniwal B., Squair P., and Yoriyaz H. (2020) MCMEG: Intercomparison exercise on prostate radiotherapy dose assessment. *Radiation Physics and Chemistry* 167:108295

Fonseca T., Mendes B., Lacerda M., Silva L., Paixão L., Bastos F., Ramirez J., and Junior J. (2017) MCMEG: Simulations of both PDD and TPR for 6MV LINAC photon beam using different MC codes. *Radiation Physics and Chemistry* 140:386–391

Fonseca T. F., Bogaerts R., Hunt J., and Vanhavere F. (2014) A methodology to develop computational phantoms with adjustable posture for wbc calibration. *Physics in Medicine & Biology* 59:6811

Hainfeld J. F., Dilmanian F. A., Zhong Z., Slatkin D. N., Kalef-Ezra J. A., and Smilowitz H. M. (2010) Gold nanoparticles enhance the radiation therapy of a murine squamous cell carcinoma. *Physics in Medicine & Biology* 55:3045

ImageJ Development Team (2025) Imagej: Image processing and analysis in java. `https://imagej.net/ij/`. accessed: July 14, 2025

Kerr G., Hwang J., and Jones R. (1976) A mathematical model of a phantom developed for use in calculations of radiation dose to the body and major internal organs of a japanese adult. *Journal of Radiation Research* 17:211–229

Lazzari G., Couvreur P., and Mura S. (2017) Multicellular tumor spheroids: a relevant 3d model for the in vitro preclinical investigation of polymer nanomedicines. *Polymer Chemistry* 8:4947–4969

Maury P., Porcel E., Mau A., Lux F., Tillement O., Mahou P., Schanne-Klein M.-C., and Lacombe S. (2021) Rapid evaluation of novel therapeutic strategies using a 3d collagen-based tissue-like model. *Frontiers in Bioengineering and Biotechnology* 9:574035

(MCMEG) (2025) Monte Carlo Modelling and Expert Group. `https://sites.google.com/view/mcmeg-ufmg/`. accessed: July 14, 2025

Min P. (2004 - 2019) binvox. http://www.patrickmin.com/binvox

Musy M., Jacquenot G., Dalmasso G., de Bruin R., Pollack A., Claudi F., Badger C., Sullivan B., Hrisca D., Volpatto D. et al. (2021) vedo, a python module for scientific analysis and visualization of 3d objects and point clouds. *Zenodo* 1

Nooruddin F. S. and Turk G. (2003) Simplification and repair of polygonal models using volumetric techniques. *IEEE Transactions on Visualization and Computer Graphics* 9:191–205

Paixão L., Mendes B., and Fonseca T. (2019a) Validation study of voxel phantom Monte Carlo simulations with EGSnrc C++ class library. *Brazilian Journal of Radiation Sciences* 7

Paixão L., Mendes B. M., and Fonseca T. C. F. (2019b) Validation study of voxel phantom monte carlo simulations with EGSnrc C++ class library. *Brazilian Journal of Radiation Sciences* v.7

Paro A. D., Hossain M., Webster T. J., and Su M. (2016) Monte carlo and analytic simulations in nanoparticle-enhanced radiation therapy. *International Journal of Nanomedicine* pages 4735–4741

Radaic A., Pugliese G. O., Campese G. C., Pessine F. B. T., and Jesus M. B. d. (2016) Como estudar interaÇÕes entre nanopartÍculas e sistemas biolÓgicos. *Química Nova* 39:1236–1244

Rasouli F. S. and Masoudi S. F. (2019) Monte carlo investigation of the effect of gold nanoparticles' distribution on cellular dose enhancement. *Radiation Physics and Chemistry* 158:6–12

Retif P., Reinhard A., Paquot H., Jouan-Hureaux V., Chateau A., Sancey L., Barberi-Heyob M., Pinel S., and Bastogne T. (2016) Monte carlo simulations guided by imaging to predict the in vitro ranking of radiosensitizing nanoparticles. *International journal of nanomedicine* pages 6169–6179

Rodrigues M. A., Marinho J. P. N., Arantes Fernandes Vieira L., Gomes D. A., and Sousa E. M. B. d. (2024) Internalized hydroxyapatite nanoparticles conjugated with a modified form of curcumin functionalized with folic acid promote bone tumor cell necrosis. *ACS Applied Nano Materials* 7:21304–21318

Ros J. G., Moraleda M., López M., Navarro T., and Navarro J. (2007) Monte carlo based voxel phantoms for in vivo internal dosimetry. *Radiation protection dosimetry* 125:161–165

Schneider C. A., Rasband W. S., and Eliceiri K. W. (2012) Nih image to imagej: 25 years of image analysis. *Nature Methods* 9:671–675

Seniwal B., Bernal M. A., and Fonseca T. C. (2020) Microdosimetric calculations for radionuclides emitting and particles and auger electrons. *Applied Radiation and Isotopes* 166:109302

Seniwal B., Freitas F., Mendes B., Lugão A., Katti K., and Fonseca T. (2021) In silico dosimetry of low-dose rate brachytherapy using radioactive nanoparticles. *Physics in Medicine and Biology* 66

Werner C. J., Armstrong J. C., Brown F. B., Bull J. S., Casswell L., Cox L. J., Dixon D. A., Forster R. A. III., Goorley J. T., Hughes H. G. III., Favorite J. A., Martz R. L., Mashnik S. G., Rising M. E., Solomon C. J. Jr.., Sood A., Sweezy J. E., Zukaitis A. J., Anderson C. A., Elson J. S., Durkee J. W. Jr.., Johns R. C., McKinney G. W., McMath G. E., Hendricks J. S., Pelowitz D. B., Prael R. E., Booth T. E., James M. R., Fensin M. L., Wilcox T. A., and Kiedrowski B. C. (2017) MCNP User's Manual Code Version 6.2. Technical Report LA-UR-17-29981. Los Alamos National Laboratory. Los Alamos, NM, USA

Xu X. G. (2014) An exponential growth of computational phantom research in radiation protection, imaging, and radiotherapy: a review of the fifty-year history. *Physics in Medicine & Biology* 59:R233

Zankl M., Veit R., Williams G., Schneider K., Fendel H., Petoussi N., and Drexler G. (1988) The construction of computer tomographic phantoms and their application in radiology and radiation protection. *Radiation and environmental biophysics* 27:153–164

Construction of the Glioblastoma scenario in the right optic nerve region using DIP techniques for simulation in the MCNP code with conversion to lattice structure by SCAN2MCNP

L. S. Boia[1]*, A. F. Menezes[2], A. X. Silva[3], M. C. Martins[1], A. A. S. A. Nascimento[1], C. A. S. A. Nascimento[1], J. L. A. Rosa[1]

[1] Universidade do Estado do Rio de Janeiro (UERJ/ZO), Department of Computer Science, Zip code: 23070-200, Rio de Janeiro, Brazil

[2] Instituto de Medicina Integral Professor Fernando Figueira (IMIP), Zip code: 50070-902, Recife, Brazil

[3] Universidade Federal do Rio de Janeiro, COPPE/PEN – Nuclear Engineering Program, Zip code: 21945-970, Rio de Janeiro, Brazil

* leonardo.boia@uerj.br

Abstract

The aim of this work is to model a computational scenario for simulating the radiotherapy treatment of a glioblastoma near the right optic nerve in the MCNP code. To do this, the head images of the male phantom are captured in ICRP 110 voxels, visualized in the Moritz Geometry Tool, and edited via digital image processing (DIP) using the GIMP software for the insertion/construction of the right optic nerve, and a Glioblastoma in its vicinity, in order to make the editing of the final images a single composition that is saved in the standard DICOM file format. With this, the set of edited images of the phantom's head can be uploaded to treatment planning systems for scientific studies, and their conversion into a lattice structure by the Scan2MCNP program so that they can be saved in an input file for preparing a computer simulation of radiotherapy treatment for the MCNP code. The possibility of constructing various scenarios, with different exposures to ionizing radiation fields for scientific or academic studies, broadens the analysis of the secondary and late effects on the structures surrounding the tumour in the treatment through the results of the computer simulation due to the high complexity involved in the task of delivering the dose prescribed for the treatment and limiting the dose in the organs adjacent to the treatment target.

Keywords: Glioblastoma; DIP; DICOM; lattice structure; Radiotherapy; MCNP.

1 Introduction

The latest estimates presented by the National Cancer Institute (INCA/RJ - BRAZIL) in its official document for 2023, point to a growing incidence of tumors in the CNS (Central Nervous System) that can affect and disrupt different vital functions, leading to the most varied symptoms, requiring extreme observation and analysis by the doctor in charge (INCA, 2023). The increase in risk factors in recent decades has raised the level of attention of the scientific community in the search for improvements in planning and treatment in order to provide patients with a better quality of life.

One of the most aggressive tumors that can appear in the brain is called Glioblastoma, also known as Glioblastoma Multiforme, it is the fastest growing type of Astrocytoma, which spreads rapidly, invading normal brain tissue and contains areas of dead cells in the center of the tumor. Glioblastoma Multiforme (GBM) is the most common type of malignant brain tumor in adults and accounts for around 50% of cases. Although GBM can occur in any age group, it is most commonly seen in adults between 50 and 70 years of age (Vainboim, 2011). Less than 10% of childhood brain tumors are Glioblastomas. Glioblastoma Multiforme tends to occur more frequently in men than women, in a ratio of approximately 3:2.

Radiotherapy is recognized as the most effective adjuvant treatment for patients with Glioblastoma Multiforme, often including different approaches that can be through Conventional Radiotherapy (RT2D) which uses two-dimensional radiation beams to reach the tumor being a simpler (but less precise) technique, 3D Conformal Radiotherapy which allows the radiation beams to be shaped to the exact shape of the tumour, minimizing damage to the surrounding healthy tissue, or Intensity Modulated Radiotherapy (IMRT) which is an advanced technique that adjusts the intensity of the radiation beams, providing greater precision and efficacy in the treatment.

One of these approaches is indicated when after complete surgical removal of the Glioblastoma is performed to eliminate residual cells or delay recurrence of the disease, after partial surgical resection of the Glioblastoma, in cases where surgery cannot be performed safely due to the location of the tumor, or in cases where, due to advanced age or general condition, the patient is not considered a candidate for brain surgery.

This work models a computational scenario, from the capture of the head images of the male phantom in voxels (ICRP 110, 2009) visualized in the Moritz Geometry Tool program (Riper, 2008), and edited via digital image processing for the insertion/construction of the right optic nerve, and of a Glioblastoma in its vicinity, for the preparation of a radiotherapy treatment simulation using the Monte Carlo MCNP code (Rising, 2023). The single-tone part of the segmentation process for these images was reviewed in preparation for generating an input file for the MCNP code with these two components using the Scan2MCNP program (Riper, 2009).

2 Techniques and Materials

2.1 ICRP adult male voxel phantom (voxel model, ICRP 110)

The male voxel model recommended by the ICRP in publication no. 110 (ICRP 110, 2009), the anthropomorphic REX phantom, represents a standard reference adult.

For the construction of the reference male phantom, we used whole-body computed tomography images of a 38-year-old individual with a height of 176 centimeters and a mass of slightly less than 70 kg (Standard Male: 176 cm and 73 kg). The individual, who suffered from leukemia and underwent whole-body irradiation, had no obvious signs of disease that could appear on the image. The person was lying supine, with their arms parallel to their body.

The data set consisted of 220 slices of 256 x 256 pixels resolution. The original voxel size was 8 mm high with an in-plane resolution of 2.08 mm, resulting in a voxel size of 36.5 mm^3. In total, 122 structures were segmented (67 of these being bones or groups of bones), including many (but not all) of the organs and tissues later identified in the ICRP characterization of the reference anatomical data (Cordeiro, 2013).

This ICRP phantom has several validations in the literature and, according to a study from 2014 on the simulation of the calculation of dose conversion coefficients using Monte Carlo Geant4 compared to John Hunt's VMC software (Hunt, J.G. et al., 2004), the methodology has been validated.

The main characteristics of the reference adult male phantom are shown in Table 1.

Table 1 - Technical data of the ICRP male phantom in voxels.
(* Additional skin slices from the head and sole of the foot.)

Properties	REX
Height (m)	1.76
Mass (Kg)	73.0
Total voxel number	1,946,375
Slice Thickness (mm)	8.0
In-plane Voxel Resolution (mm)	2.137
Voxel volume (mm^3)	36.54
Number of columns	254
Number of lines	127
Number of slices	220 (+2)*

❶ Image of the ICRP male voxel phantom visualized in the Moritz graphic encoder and the main characteristics of the ICRP anthropomorphic voxel model.

2.2 Optic nerve

The optic nerve is the second of the twelve cranial nerves in our brain responsible for transporting visual information, through which the electrical signal formed in the retina reaches the brain for the image to be formed. This nerve is made up of a million small individual segments, in which it captures information through the cones and rods present in the retina, which are stimulated by light projected onto objects (Portal da Visão, 2018). The optic nerve of each eye is divided and half of the nerve fibers on each side cross to the opposite side in the optic chiasm. Because of this arrangement, the brain receives information from both the left and right visual fields via both optic nerves, so that the focused image can be identified and seen.

The total length of the optic nerve is 50 mm, of which 0.7 mm is intraocular, 33 mm intraorbital, 6 mm intracanalicular and 10 mm intracranial on average. The diameter of the intraorbital portion, with its meninges, is 3 to 4 mm, tapering to 1.5 mm in the scleral canal, when the meninges end and the axons lose their myelin coatings (Guimarães, 2016).

The intraorbital segment of the optic nerve is longer than the distance between the posterior pole of the globe and the apex of the orbit, allowing normal eye movements and occasional marked proptosis without damaging the nerve fibers. Figure 1 shows a representation of the human eye and the location of the optic nerve.

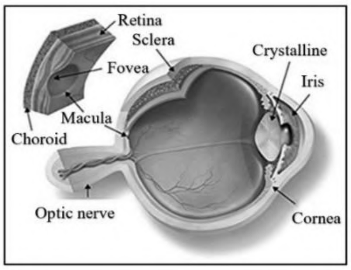


Figure 1: Representation of the human eye and the location of the optic nerve.
Source: https://www.institutoderetina.com.br/home/anatomia/

2.3 Glioblastoma

Glioblastoma, also known as Glioblastoma Multiforme, is the most common and aggressive type of primary brain tumor in humans. Consisting of glial cells, this type of cancer accounts for 52% of all parenchymal brain tumors and, after brainstem gliomas, is the one with the worst prognosis among intracranial malignancies. It is common in adults between 35 and 70 years of age, but it is not uncommon for it to occur at other ages (Oncoguia, 2008).

Glioblastoma arises from the brain itself, more specifically from astrocytes, which are the cells responsible for some of the functions of this noble area of the human being. When a tumor has its origin defined by astrocytes, it is said to be an astrocytoma.

The major problem with this type of tumor is its rapid growth, and even after surgery, further growth is expected. In addition, there is infiltration of isolated tumor cells into the apparent normal brain tissue 7 cm deep beyond the periphery of the lesion. Even if 99.99% of the neoplastic tissue is removed by surgery, the remaining infiltrated tissue or minimal local debris is capable of multiplying and, depending on the case, returning to its original size within 30 days. The remaining 0.01% represents something in the millions of cells, and a single tumor cell is enough for the lesion to multiply and recur.

There are various substances that inhibit or “hinder” tumor growth in some way, and it is on this basis that these alternative methods are applied. They are usually administered shortly after surgery, starting in conjunction with the standard treatment protocol (Rodrigues-Junior, D. M., 2022).

However, it is formally indicated for recurrent GBM, i.e. for all after surgery (which promotes cytoreduction, i.e. a reduction in the number of tumor cells) together with radiotherapy and/or chemotherapy treatments. Aberrant cells have special characteristics, including behavior very similar to that of fetal cells, which grow extremely quickly.

3 Computational Process

3.1 Editing the head images of the ICRP male phantom

In order to model a computational scenario for the radiotherapy treatment of a Glioblastoma near the right optic nerve region, the first step was to capture the images of the head of the ICRP male phantom in lattice structure from position Z=165.7 cm to Z=167.1 cm in which they were visualized using the Moritz Geometry Tool program (Riper, 2008), and then taken one by one to the GIMP program (GIMP, 2025) for editing (Figure 2). GIMP is an acronym for GNU Image Manipulation Program, which is used for various image manipulation tasks. Each of these images has a resolution of 256 pixels x 256 pixels and has been saved in .PNG format, as it is a format with lossless compression, which means that the quality of the image is not compromised.

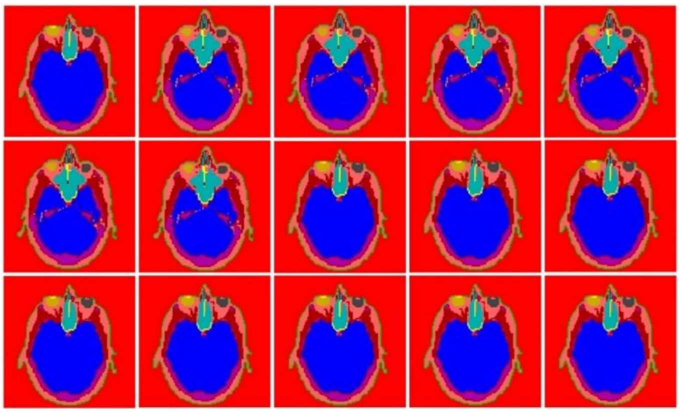


Figure 2: Images of the ICRP male phantom in lattice structure captured by the Moritz Geometry Tool program.

Based on the reference images (Figure 3a), the images of the ICRP 110 male phantom (Figure 3b) were correlated to construct the right optic nerve and glioblastoma multiforme (GBM) tumor. The tumor volume was modeled to have a morphology compatible with a typical GBM, with dimensions of approximately 3.2 cm x 2.6 cm x 2.4 cm. This morphology is characterized by an irregular shape, poorly defined margins, heterogeneous internal structure, and a possible mass effect on adjacent structures. To compose the dimensions, the minimum image element was used, which is 0.2139600 cm on the x and y axes (as shown in Figure 4a and Figure 4b), replicating it for the construction of the optic nerve and tumor based on the model image (Figure 5). To compose the dimension on the Z axis, three image slices were used, each 0.8 cm thick, according to the technical data of the male voxel phantom from ICRP 110 (2009). The sum of these slices results in the 2.4 cm required in the tumor volume, reflecting a highly complex clinical scenario, considering both the critical location of the tumor and its characteristic infiltrative behavior.

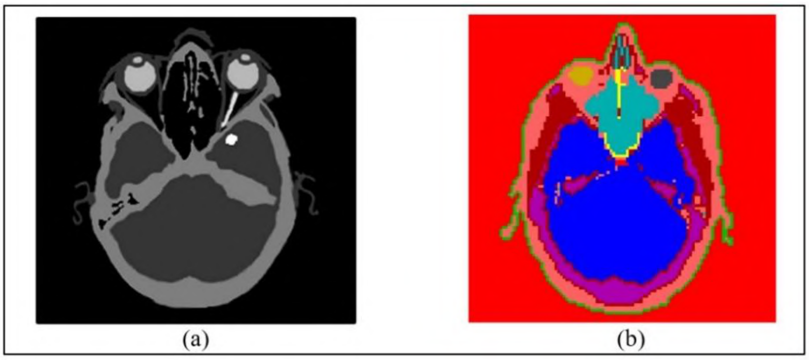


Figure 3: (a) Reference image. (b) Image of the lattice structure of the ICRP male phantom.

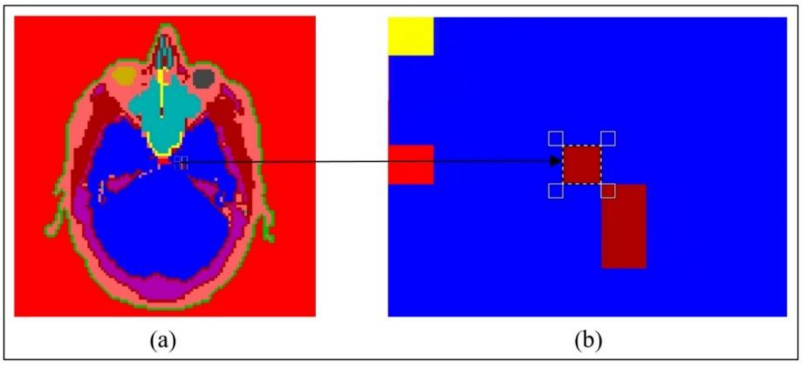


Figure 4: (a) Voxel geometry of the ICRP male phantom. (b) Minimum image element for the composition of ICRP male tissues and organs captured in the Moritz Geometry Tool software visualization.

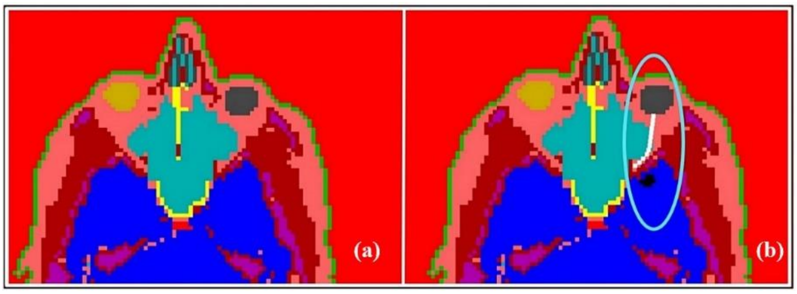


Figure 5: (a) Original image before alteration. (b) Construction of the right optic nerve and GBM on the image of the ICRP male phantom slice using the GIMP image editor (GIMP, 2025).

3.2 DICOM standard

The DICOM (Digital Imaging Communications in Medicine) standard is an image format that was developed with the aim of standardizing diagnostic images, such as CT scans, MRI scans, X-rays and ultrasounds (DICOM, 2025). The DICOM standard is a series of rules that allows medical images and associated information to be exchanged between imaging equipment, computers and hospitals, which allows for the standardization of a common language between equipment of different brands, which are generally not compatible, and between imaging equipment and computers, whether these are in hospitals, clinics or laboratories. Therefore, this is the standard for reading/uploading images into the planning system present in hospitals, as well as the standard for loading the Scan2MCNP program.

As the edited images of the ICRP male phantom are in .PNG, they need to be converted into DICOM, but first the RGB color scale needs to be changed to the Grayscale color scale, as can be seen in Figure 6a and Figure 6b. In a Grayscale image, the color scale ranges from 0 to 255, where 0 represents absolute black and 255 represents absolute white. Intermediate values represent the different shades of gray, with 128 being a medium gray tone. This scale of 256 (2^8) gray levels is common because it is based on 8 bits per pixel, which is sufficient for most applications. This means that each pixel in the image can have one of 256 possible values, providing a smooth transition between black and white.

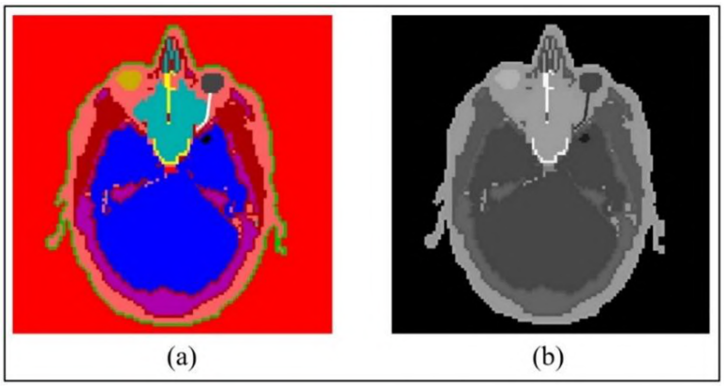
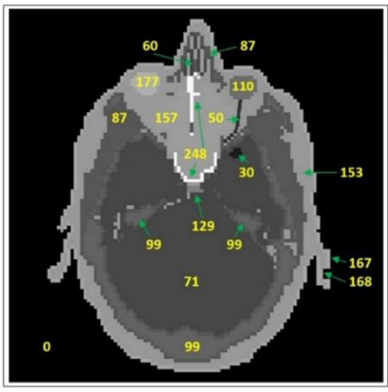


Figure 6: (a) Image of the ICRP male phantom slice in RGB. (b) Image of the ICRP male phantom slice in Grayscale.

In order for the Scan2MCNP program to generate the lattice structure correctly, it was essential to ensure that each area of the image received a unique gray tone through a segmentation process (Ambrósio, P. E. (Ed.), 2022). Each gray tone would then be correlated to a specific material in the MCNP input file (Rising, 2023). Figure 7 illustrates this identification process, based on technical data from the ICRP 110 male voxel phantom.



Grayscale	Material	ID INPUT
0	Air	1
30	GBM *	201
50	Right optic nerve *	200
60	Cortical skull	2
71	Brain	25
87	Muscular tissue of the head	22
99	Spongy skull	7
110	Bulb of the right eye	27
129	Pituitary gland	37
153	Residual head tissue	41
157	Residual head tissue	41
167	Head skin	20
168	Residual head tissue	41
177	Bulb of the right eye	27
248	Posterior nasal passage	4

Figure 7: Image of the head slice of the altered ICRP male phantom and description of the shades of gray present. (* Inserted during the image editing process.)

With segmentation complete, the grayscale images were converted to DICOM format, a crucial standard for loading into hospital planning systems and the Scan2MCNP software itself. This conversion was performed using the open source software GIMP with a specific plugin. The plugin was installed by adding the "file-dicom" folder to the GIMP plugin directory (Figure 8), followed by a restart of the software. For each image, the conversion process was performed individually, following the export path: File > Export as > DICOM. With the images now in the DICOM standard, it was possible to load them into a planning system, as shown in Figure 9, for use in scientific studies.

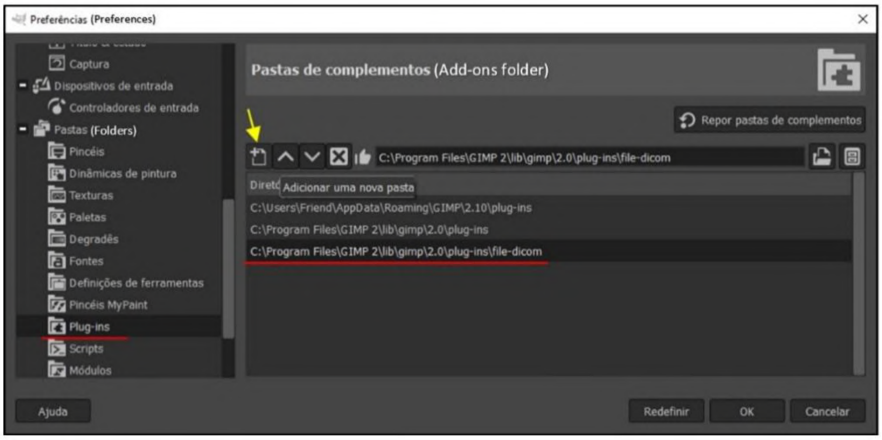


Figure 8: Working with the "Preferences" window of the GIMP software to insert the "file-dicom" plugin.

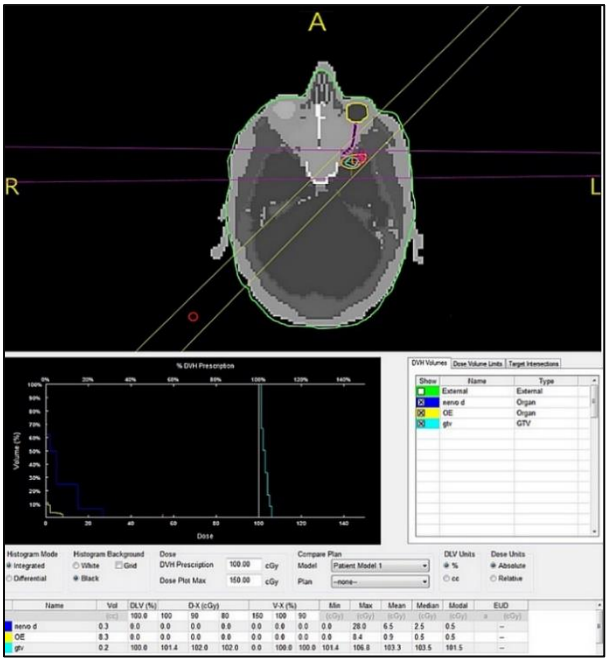


Figure 9: Images edited and converted into DICOM can be used in planning systems for scientific studies.

3.3 Conversion of DICOM images to input file

The process of converting DICOM images into an input file for the MCNP code is performed by the Scan2MCNP program (Figure 10). This software, with its various features, allows the selection of specific areas of the image, the assignment of identifiers (IDs) to delimit different regions, and the parameterization of the input file format. In addition, it exports data directly to the Sabrina (Riper, 1997) and Moritz (Riper, 2008) graphics encoders, which enables three-dimensional visualization of the constructed model.

In the images edited and converted to DICOM format, specific IDs were assigned to identify brain structures based on the documentation of the male voxel phantom from ICRP 110 (2009). The right optic nerve and glioblastoma multiforme (GBM) were identified by their characteristic gray level bands and correlated with the materials corresponding to their composition.

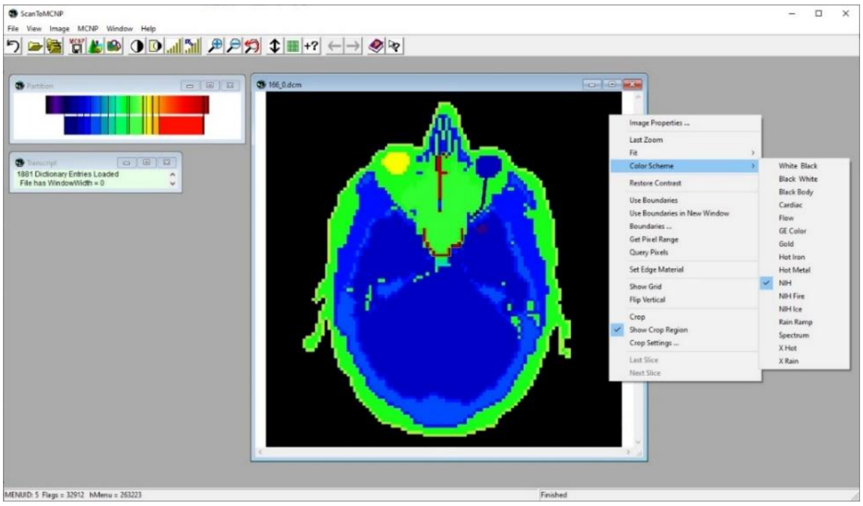


Figure 10: Determining the ID identifiers on the DICOM images in preparation for generating the input file. A color scheme (mask) is being applied to the uploaded image for better visualization.

For dosimetry simulations, the elemental composition and density of the optic nerve and GBM are often modeled with the same as the surrounding brain tissue. Adopting the composition of gray matter is a common practice to simplify the model, since the difference between tumor tissue and healthy brain tissue is considered insignificant for dose calculation on a macroscopic scale.

The program offers several configuration options, such as contrast equalization, use of a grid to analyze edge effects between materials, choice between lattice or cell input formats, definition of edge material to differentiate the air outside the phantom, delimitation of the image area to be considered, and optional generation of coincident surface planes.

After defining all relevant parameters, the DICOM image set is converted into an input file in lattice format, compatible with the MCNP code (Figure 11).

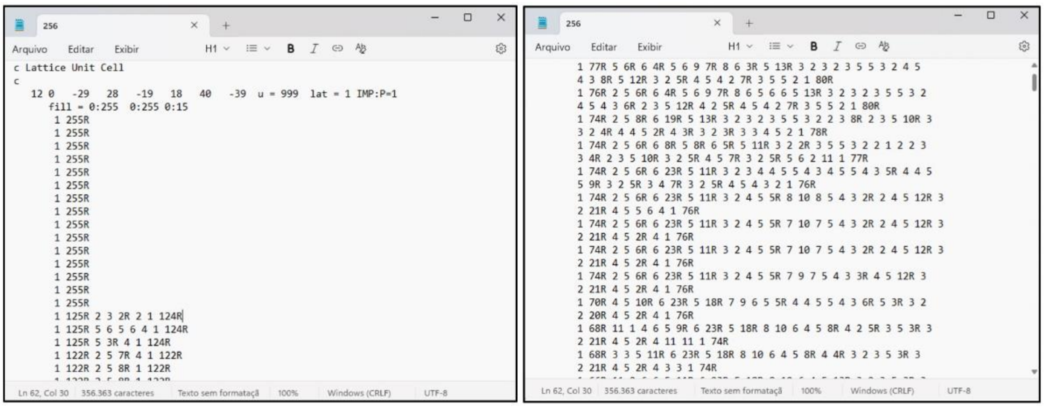
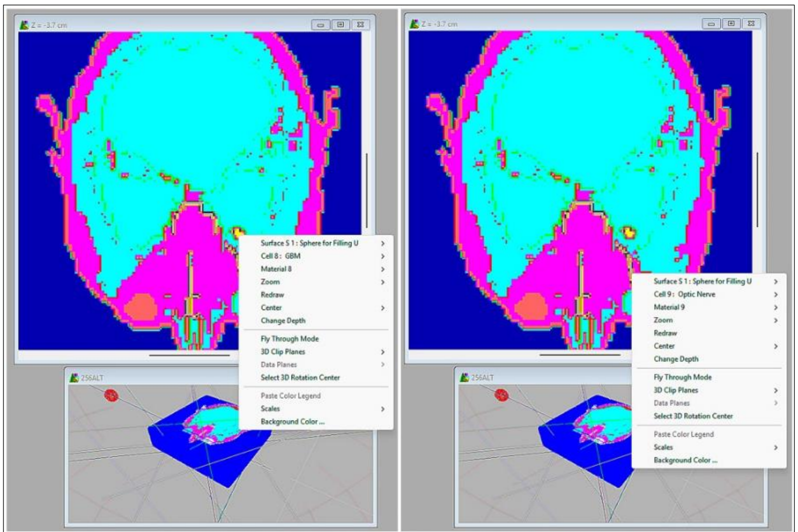


Figure 11: Excerpt from the generation of the input file with the lattice structure of the REX phantom altered with the insertion of the optic nerve and glioma.

4 Results

After completing the methodological process, the generated input file was viewed for verification purposes using the Moritz Geometry Tool graphic encoder. Figure 12 shows a detailed view of the simulation model geometry. In the Z plane, it is possible to observe the optic nerve and GBM inserted into one of the slices of the phantom head modified using DIP (Digital Image Processing), which validates the accuracy of the segmentation and insertion of both structures. This visual validation step is essential, since any error in the mesh structure, material cards, or other parameters in the input file can prevent Moritz from loading the model, serving as an indication that some step in the methodology was not performed as expected.

In addition to the cross-sectional view, the figure also displays a three-dimensional graphical representation of the model. This 3D perspective is crucial for confirming the integrity and correct orientation of the phantom, optic nerve, and GBM in space. Visual validation of the model in the Moritz Geometry Tool ensures that the geometry used for dose calculation in the MCNP code is accurate and faithfully reflects the clinical scenario of interest.



Identification of the GBM — Identification of the optic nerve

Figure 12. Images generated by the Moritz Geometry Tool showing the identification of the optic nerve and glioblastoma multiforme (GBM) in a slice of the modified phantom in the Z plane, together with the general 3D graphical representation.

5 Conclusions

The use of Digital Image Processing (DIP) techniques in the preparation of images for the construction of medical scenarios enables in-depth studies by teams using planning systems, as well as the conversion of these images into input files for the MCNP code, expanding the range of analyses in radiotherapy treatments and allowing for refinements in the studies to be carried out. The possibilities of image manipulation stand out, allowing regions of interest to be highlighted, noise to be removed and images to be edited for specific purposes, as well as segmentation and data extraction, providing the research team with numerous options for approaching the study to be carried out.

Converting images into lattice structures is a fundamental technique in the context of computer simulation, as it defines an accurate representation of complex geometries, and manages to reliably portray reality. The computational efficiency of work involving lattice structures allows for a more efficient distribution of computational calculations, improving the overall performance of simulations, especially in simulations that require a lot of processing power. Therefore, the insertion of DIP techniques within the computational process contributes to more detailed simulation results, allowing the identification of specific patterns and behaviors within the model.

References

Ambrósio, P. E. (Ed.) (2022). Digital Image Processing Applications. IntechOpen. DOI: 10.5772/intechopen.95685. Accessed: August 17, 2025.

Cordeiro, T. P. V. (2013) Coeficientes de conversão para a dose efetiva e equivalente de dose ambiente para feixes de raios X utilizados em radioterapia. PhD Thesis, COPPE, Universidade Federal do Rio de Janeiro, Rio de Janeiro, Brazil.

DICOM Standard (n.d.) Digital Imaging and Communications in Medicine. National Electrical Manufacturers Association, Rosslyn, VA, USA. Available at: https://www.dicomstandard.org/. Accessed: January 18, 2025.

GIMP Development Team (n.d.) GIMP (GNU Image Manipulation Program). [S.l.: s.n.]. Available at: https://www.gimp.org/. Accessed: January 15, 2025.

Guimarães, M. (2016) Curso de Histologia Ocular. Hospital de Olhos de Minas Gerais, Belo Horizonte, Brazil.

Hunt, J. G., et al. (2004) Voxel phantoms and Monte Carlo methods applied to internal and external dose calculations. In: Proceedings of the 11th International Congress of the International Radiation Protection Association – IRPA11, Madrid, Spain.

ICRP (2009) Publication 110: Adult Reference Computational Phantoms. Ann. ICRP 39(2).

INCA (2023) Estimativa 2023: Incidência de Câncer no Brasil. Instituto Nacional de Câncer. Available at: https://www.inca.gov.br/sites/ufu.sti.inca.local/files/media/document/estimativa-2023.pdf. Accessed: January 10, 2025.

Nunes, P. R., Silva, M. O., and Costa, R. H. (2025) Glioblastoma multiforme: avanços terapêuticos recentes e o papel atual da cirurgia no manejo integrado. Research, Society and Development, 14(3), e48572438142. DOI: 10.33448/rsd-v14i3.48572. Available at: https://rsdjournal.org/index.php/rsd/article/view/48572. Accessed: August 17, 2025.

Oncoguia (2008) Tumores Cerebrais / Sistema Nervoso Central – Material de apresentação. Available at: https://pt.slideshare.net/slideshow/tumores-de-crebro-e-sistema-nervoso-central/8537042. Accessed: January 27, 2025.

Portal da Visão (2018) O que é o nervo óptico. Available at: https://portaldavisaocuritiba.com.br/o-que-e-o-nervo-optico/#:~:text=Localizado%20na%20parte%20de%20tr%C3%A1s,parte%20do%20sistema%20nervoso%20central. Accessed: January 12, 2025.

Riper, K. A. V. (2008) Moritz Geometry Tool User's Guide – Windows Version (Manual). White Rock Science, USA.

Riper, K. A. V. (2009) Scan2MCNP User Manual (CT & MRI Scan Data to MCNP Input Format Conversion Software). White Rock Science, USA.

Rising, M. E., et al. (2023) MCNP® Code Version 6.3.0 Release Notes. Los Alamos National Laboratory Tech. Rep. LA-UR-22-33103, Rev. 1. Los Alamos, NM, USA.

Rodrigues-Junior, D. M., Raminelli, C., Hassanie, H., et al. (2022) Derivados de aporfina e isoquinolina bloqueiam células-tronco de glioblastoma e aumentam a citotoxicidade da temozolomida. Scientific Reports, 12, 21113. DOI: https://doi.org/10.1038/s41598-022-25534-2.

Vainboim, T. B. (2011) Qualidade de vida e intervenção psicoeducativa com cuidadores não-profissionais de pacientes portadores de glioblastoma multiforme. MSc Thesis, Faculdade de Medicina, Universidade de São Paulo, São Paulo, Brazil.

XXV ISSSD International Symposium on Solid State Dosimetry
September 22 – 26, 2025
Belo Horizonte, Minas Gerais - Brazil
CDTN UFMG

Development of a Dedicated System for Generating Input Files for the MCNP and Adaptation of a Phantom with Edema for Simulation of Prostate Brachytherapy with ^{125}I

L. S. Boia[1]*, A. F. Menezes[2], A. X. Silva[3]

[1] Universidade do Estado do Rio de Janeiro (UERJ/ZO), Department of Computer Science, Zip code: 23070-200, Rio de Janeiro, Brazil

[2] Instituto de Medicina Integral Professor Fernando Figueira (IMIP), Zip code: 50070-902, Recife, Brazil

[3] Universidade Federal do Rio de Janeiro, COPPE/PEN – Nuclear Engineering Program, Zip code: 21945-970, Rio de Janeiro, Brazil

* leonardo.boia@uerj.br

Abstract

The manual generation of MCNP code input files, especially the cell and surface cards for each ^{125}I seed, is unfeasible. This is because the large number of lines and the complex Monte Carlo parameterizations require excessive manual effort and time. For practical Monte Carlo simulation applications in the medical field, a fast and dynamic method for creating these input files is essential. This allows researchers to focus on running simulations and analyzing results. This paper presents a dedicated computer system, recently developed and improved. It automates the creation of MCNP input files based on real ^{125}I seed prostate brachytherapy planning data. In addition, the system modifies the ICRP 110 male voxel phantom to simulate edema. This allows studies on the effects of edema during low dose rate (LDR) brachytherapy treatment, including repositioning the seeds and evaluating dosimetric changes. The system significantly reduces file generation time and offers greater flexibility. It reproduces the planning data in the MCNP simulation environment and generates input files with an edema predictability factor. This capability supports more realistic studies, offering various possibilities for research teams to improve the brachytherapy technique. Ultimately, this system optimizes complex modeling in MCNP, paving the way for more efficient and accurate investigations into brachytherapy treatments affected by edema.

Keywords: LDR brachytherapy; Iodine-125 seeds; Voxel phantom; MCNP.

1 Introduction

According to data from the National Cancer Institute (INCA/RJ), based in Brazil, prostate cancer is responsible for the death of 28.6% of the male population who develop neoplasms. In 2023, around 17,000 men lost their lives to this disease, resulting in an alarming average of 47 deaths per day (CAPES, 2023). This malignancy is the second most common among men (second only to non-melanoma skin cancer). Age is an important risk factor, since both incidence and mortality increase significantly after the age of 60. It is estimated that between 2023 and 2025, Brazil will register 71,730 new cases of prostate cancer per year (INCA, 2023).

Depending on the type, prostate tumors may grow rapidly, spreading to other organs and consequently leading to death. However, the majority grow slowly (approximately 15 years to reach 1 cm^3) and do not show signs during their lifetime or even pose a threat to men's health (INCA, 2023). The prostate is a gland located in the lower abdomen. It is shaped like an apple and is situated below the bladder and in front of the rectum (the final part of the large intestine). The prostate surrounds the initial portion of the urethra and produces semen.

Among the treatment options for this disease, low dose rate (LDR) brachytherapy, which uses the insertion of multiple ^{125}I or ^{103}Pd seeds into the prostate, has been commonly used. However, this insertion causes an inflammatory response in the body, expressed as prostatic edema, the effects of which are not taken into account during treatment planning. The increased volume of the prostate, a consequence of this edema, results in the displacement of the sources, which can affect the dose administered. This deviation from the prescribed dose can impact both the target volume and the surrounding organs at risk (OARs). The actual dose administered and the deviations resulting from these displacements are, in fact, difficult to quantify and evaluate in LDR brachytherapy treatments (Almanza, 2018).

MCNP (Rising et al, 2023) is a computer simulation tool capable of evaluating the dosimetric effects resulting from seed migration after implantation in low dose rate (LDR) brachytherapy procedures, as a consequence of edema. It is important to note that these effects are not taken into account by the systems currently used in treatment planning.

To overcome this limitation, the use of a dedicated computer system in conjunction with the MCNP code makes it possible to carry out more reliable studies and analyses. This is because the generation of the input file can be correlated with the actual planning. Consequently, the generation of other files that take into account the predictability of edema makes it possible to compare the output data, which leads to a more accurate assessment.

2 Techniques and Materials

2.1 The specific features of brachytherapy

Brachytherapy is a treatment modality in which the source is inserted in or near the treatment site. For low dose rate (LDR) brachytherapy, the implant is usually permanent, and in most cases ^{125}I sources are used. In contrast, teletherapy is a technique in which the radiation source is located externally to the patient, and the dose is delivered by linear accelerators or Cobalt-60 equipment.

The ^{125}I seeds are inserted using needles guided by transrectal ultrasound, with the patient in the lithotomic position. This procedure is considered minimally invasive (Lim, S., and Kim, Y., 2021) and is performed by a multidisciplinary team, in which the physician and medical physicist play a prominent role in treatment planning.

2.2 ^{125}I seeds

The Amersham model 6711 ^{125}I seed (Figure 1), used as a reference in this work, was modeled as follows: an internal silver cylinder containing a ^{125}I emitting film with a thickness of 1 μm. This assembly is encapsulated externally by a titanium cylinder, which is 0.81 mm in diameter and 4.5 mm long, and is closed by two ellipsoids (Duggan, 2004).

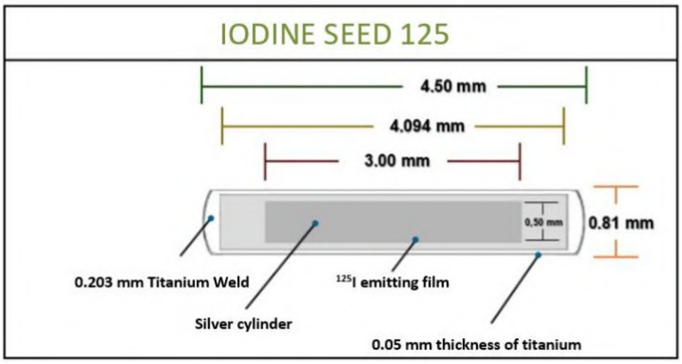


Figure 1: Geometric illustration of the ^{125}I Amershan Model 6711 seed.

2.3 Java and NetBeans IDE

The selection of the Java programming language in conjunction with the NetBeans integrated development environment (IDE) provided a robust, cross-platform environment, an essential attribute for the complexity of the project. The software was structured into 24 separate Java files, the organization of which optimized the modularity and maintainability of the code.

To ensure the core functionality of the system, in relation to the generation of input files for the MCNP (Monte Carlo N-Particle) and the subsequent recording in a report, specific Java modules were implemented for critical operations:

Calculation Modules: Responsible for executing the complex algorithms needed to determine the physical and geometric parameters of the simulation.

MCNP Input File Creation Modules: Dedicated to formatting and generating input files in the specific format of the MCNP code.

Reporting and Storage Modules: Developed to record essential metadata, such as the identification of the user who created the file, the basic specifications of the file generated, and other relevant information. This data is stored locally in an SQL database. SQL (Structured Query Language) is a standard language used to manage and retrieve data from relational databases, which organize data into structured tables with rows and columns.

In addition, other modules are present in the system:

Phantom Modification Module: Aimed at automatically adapting the phantom to the proposed scenario, which involves altering the voxel planes, lattice planes and rectangular parallelepiped.

Results Extraction Module: Responsible for extracting relevant data from the output file generated by the MCNP for post-simulation manipulation and analysis.

System Registration and Use Modules: Developed to manage access and use of the system by undergraduate students and researchers.

Figure 2 below illustrates the basic block diagram of the system modules.

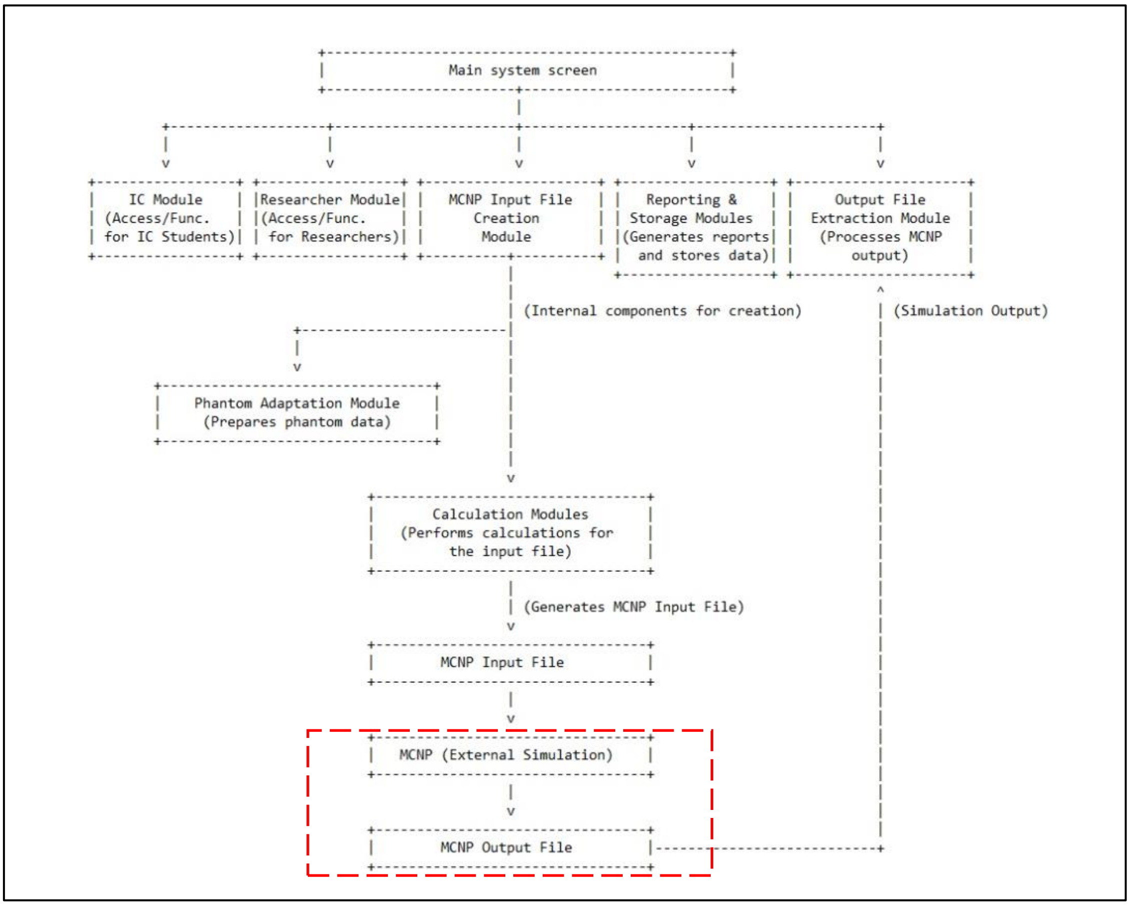


Figure 2: Basic block diagram of the system modules.

3 System

3.1 Workflow of the Developed System

To automate the entire process described in this article, from clinical planning itself to the generation of input files for the MCNP code, a dedicated system was used, developed and recently improved specifically for this research. This system was programmed in the Java programming language (high-level and object-oriented), in which all the calculations pertaining to the positioning of the seeds are generated and described in the input files, depending on whether edema is predictable or not. The seeds are placed in the prostate of a voxelized phantom (lattice structure), individually adapted for each case with edema.

The system has the following options on its home screen (Figure 3):

- **IC Student Module**: Aimed at future work with undergraduate students, allowing them to register their data in the system.
- **Researcher Module**: Records the date, time and person responsible for each input file generated.
- **Calculation/Generation of INPUT MCNP**: Module for entering data and selecting parameters to create the input file. It also includes the feature of modifying voxel phantom planes for edema predictability studies.
- **Reports**: Allows access to the complete history of the system's use in generating input files.
- **OUTPUT Export**: Extracts the results data from the MCNP output file.

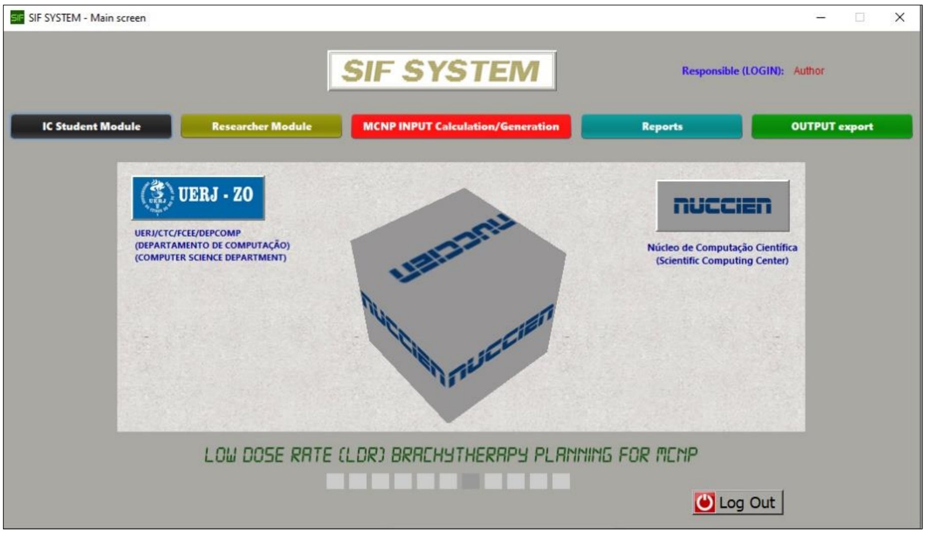


Figure 3: Overview of the system developed to generate input files.

Within the software interface, the system reproduces the clinical reference template (Figure 4a) used in the clinical process of inserting the ^{125}I seeds. Once the seeds have been inserted, it is possible to generate the input file with the "edema predictability" feature. To do this, simply activate this feature and select the desired percentage. In this configuration, the system repositions the seeds and performs a recalculation based on the new prostate volume, according to the conditions illustrated in Figure 4b.

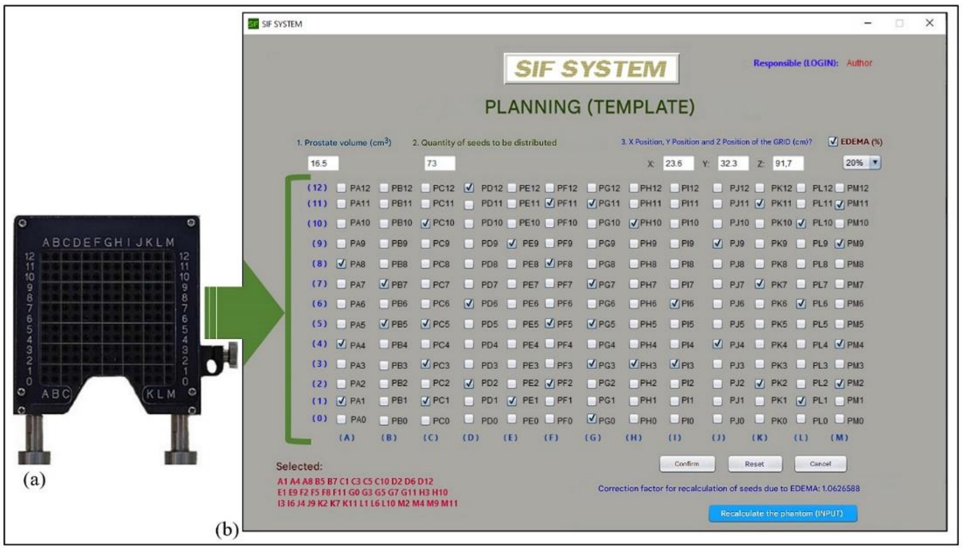


Figure 4: (a) The template used in the clinical procedure. (b) The computer system and the correlation with the template.

3.2 MCNP Input File Generated by the System

The "MCNP INPUT Calculation/Generation" option on the system interface is responsible for creating the input file for the MCNP code. The initial stage of this process asks for the prostate volume in cm³. After entering and confirming this information in the system, the total number of seeds for the treatment is determined (Figure 5). This total is calculated based on an Equation 1 implemented in the system's programming, which was published in the article by K. A. Mountris et al. (2017).

$$N_{seeds} = 4 + \frac{4.674 \times V^{0.562}}{A} \quad (1)$$

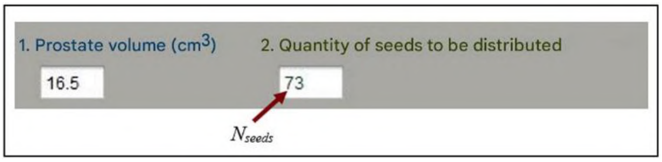


Figure 5: The initial step in the process is to request the volume of the prostate in cm^3, which, when filled in and confirmed in the system, returns the total number of seeds based on the technical equation of K. A. Mountris et al, 2017.

If the value returned by the system changes, a notification will be issued stating that the proposed value differs from the total seeds for the defined prostate volume. This indicates a deviation from the technical equation, which will result in a final dose delivery to the organ that is not as expected, leading to underdoses or overdoses in the treatment.

After the system returns the total number of seeds, the template location data must be entered for the spatial positioning of the seeds in the lattice structure (prostate geometry) of the phantom (Figure 6a). The exact position is obtained using the Moritz Geometry Tool program (Riper, 2008), which allows 3D visualization of the geometry in voxel structure.

The system then asks the user if they want to enable edema predictability. If enabled, the user must select the desired percentage (10%, 20%, 30%, 40% or 50%) for the recalculation to be carried out, impacting the total volume. After selecting the percentage of prostate inflammation, the system returns the correction factor for recalculating the seed distribution (Figure 6b).

In addition, the "Recalculate phantom (INPUT)" button is enabled. This allows the user to load the desired voxel phantom (Figure 6c).

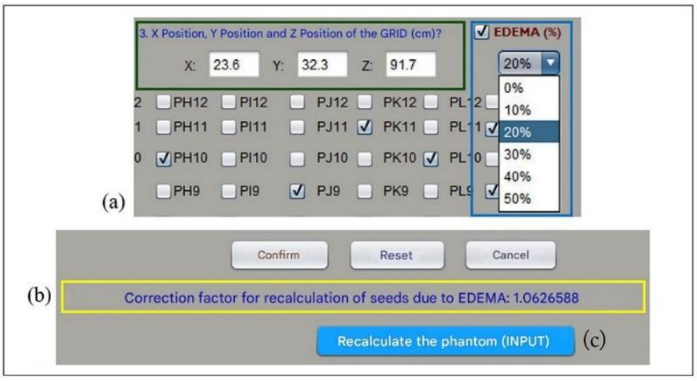


Figure 6: (a) Definition of the grid location (template) for the spatial distribution of seeds in the lattice structure. (b) Correction factor for recalculating the distribution of seeds. (c) enabling the "Recalculate phantom (INPUT)" button.

To study the impact of edema on prostate brachytherapy planning, it is essential that the working phantom is adapted to each proposed scenario. To simulate prostate inflammation, changes to the voxel planes, lattice planes and the rectangular parallelepiped that encompasses the phantom are necessary.

It is important to note that, in the computer simulation, the analysis will focus only on the prostate, using the definitions described in tally F8 of the MCNP input file. These changes to the plans are carried out automatically by the system: it tracks specific data extracted from the phantom input file and carries out all the necessary recalculations based on the percentage of edema selected. Figure 7 illustrates the sequence of steps and the functioning of this feature implemented in the system.

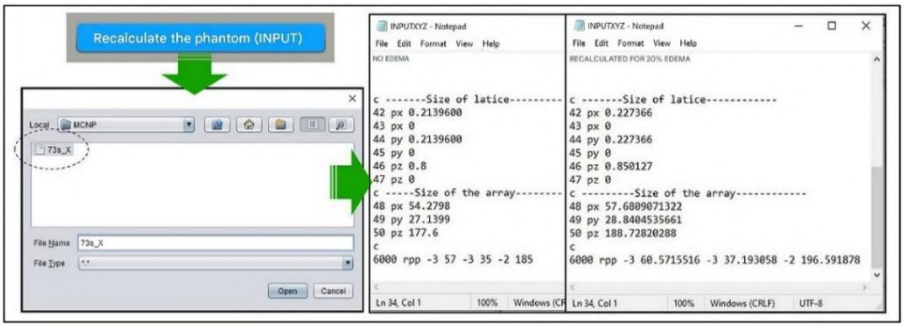


Figure 7: Illustration of the sequence of steps for the “Recalculate phantom (INPUT)” button function in the system.

The system interface visually mirrors the clinical procedure template. In it, the columns are represented by the letters A to M and the rows by the numbers 0 to 12.

For example, if you select position ‘A3’ for inserting seeds, the system will first request the number of seeds for that position (Figure 8a) and then the depths of each one (Figure 8b). At the end of the process, it will generate the description of the cards (instructions) relating to the positioned seeds. The same process is repeated for the other seeds.

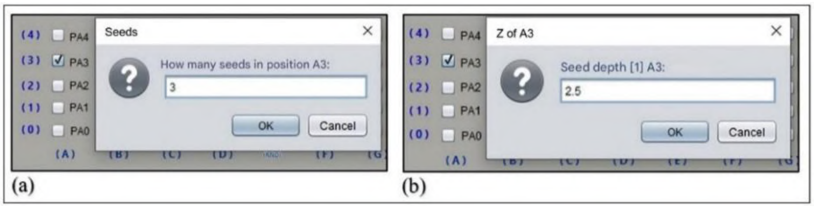


Figure 8: (a) Request for the number of seeds for a given position. (b) Depth of each seed in that position.

After the initial steps, a new window will open for the final specifications. The user can select the format of the instruction they want for the MCNP input file. The options available are: I-125 without TR and I-125 with TR (Figure 9).

The TR Card in MCNP is essential for simulating ^{125}I seeds, as it allows geometric transformations (such as rotations and translations) of surfaces and cells in the model (Rising et al, 2023). Using the TR Card, the system is able to replicate the seeds dynamically, translating them from a reference position (the origin of the coordinate system) to their previously planned positions and quantities. This process creates a link between the surfaces of the seeds, originally defined at the origin, and the transformation applied.

Finally, it is necessary to enter the initial block numbers for cells, materials and surfaces. These numbers will determine the starting point for all calculations related to these elements (Figure 9).

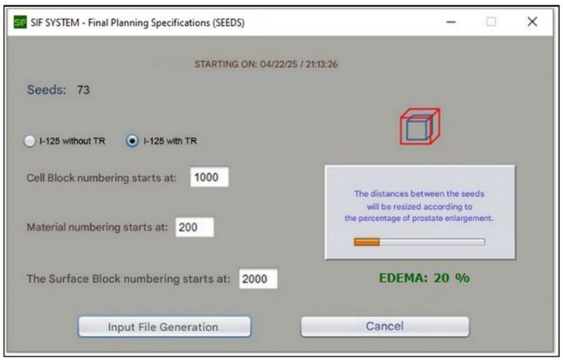


Figure 9: Final specifications for generating the input file.

To finish the process, click on the 'Generate Input File' button. The system will then ask for the 'File name (8 characters)' where the instructions will be saved.

The system will open the generated input file in Windows NotePad. This will allow the ^{125}I seed coding to be checked and then inserted into the altered phantom. Figure 10 shows an example of the MCNP code generated by the system.

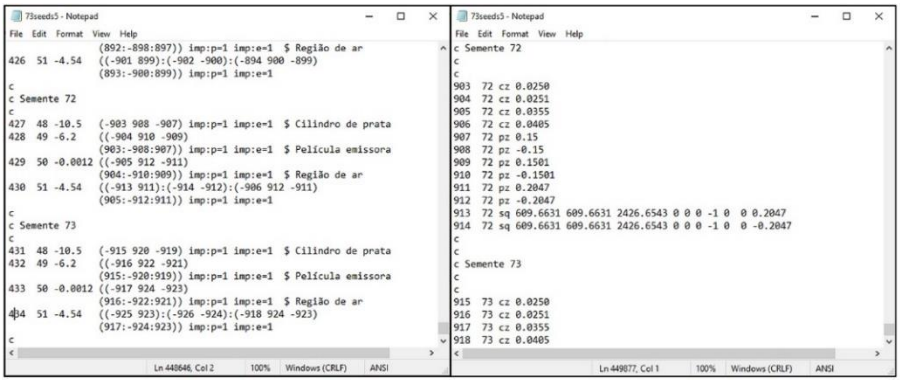


Figure 10: Example of the input file generated by the system with the "I-125 with TR" option.

3.3 System report

On the system's main screen, the "Reports" option makes it possible to monitor the files generated, whose data is stored and managed in a local SQL database. SQL (Structured Query Language) is a standard language for managing and manipulating relational databases, guaranteeing the organization and integrity of information. This data structure allows statistical analysis of the planning carried out, with a view to improving and adjusting the system's performance.

3.4 Exporting Results (OUTPUT)

To optimize the extraction and analysis of the results generated by the MCNP code, a specific algorithm was developed and implemented in the system. This algorithm analyzes the MCNP output file, identifying and collecting values relevant to energy deposition. This data, which details the energy deposited in different materials or regions of the simulation model, is then sequentially saved in a new file, facilitating later analysis.

Figure 11a illustrates the original format of the MCNP output file, containing a varied distribution of data. In contrast, Figure 11b demonstrates the organization and sequencing of the energy deposition data after processing by the system.

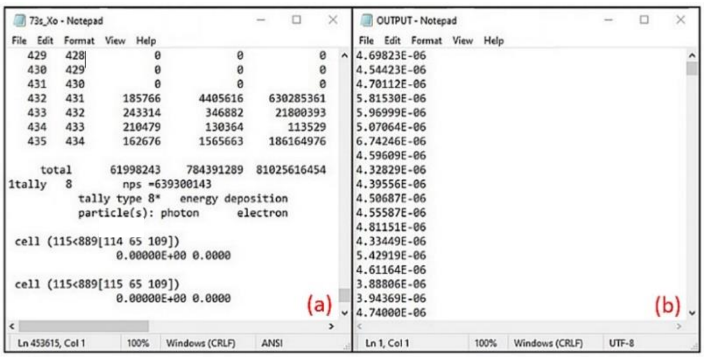


Figure 11: (a) Output file delivered by the MCNP code. (b) Data extracted and organized in another file created by the system.

4 Results

For system validation, the ICRP 110 (2009) male voxel phantom was used, which has a prostate gland with a volume of 36.5 cm³. Based on technical planning criteria for low dose rate (LDR) brachytherapy in the treatment of prostate cancer with ^{125}I seeds, 73 seeds should be inserted into the phantom's prostate, aiming to deliver a total dose of 144 Gy.

A simulation scenario was then proposed with a predictable prostate edema of 50%, considered after seed insertion. In this condition, the system automatically recalculated the spatial distribution of the seeds and adjusted the prostate volume to reflect the new anatomical configuration. This process was subsequently verified using the Moritz Geometry Tool graphic encoder.

The simulation, performed with the MCNPX code, considered two scenarios: without edema and with predictable prostate edema of 50%. After processing more than 7×10^{8} simulated histories, the results indicated that, in the scenario without edema, the approximate dose reached 144 Gy, as predicted in the planning. On the other hand, the scenario with 50% edema showed a significant loss in dose in the target region, resulting in approximately 87% of the total planned dose (about 125.28 Gy).

Figure 12 shows the dose-volume histogram (DVH), which illustrates the impact of edema on the simulated dose distribution.

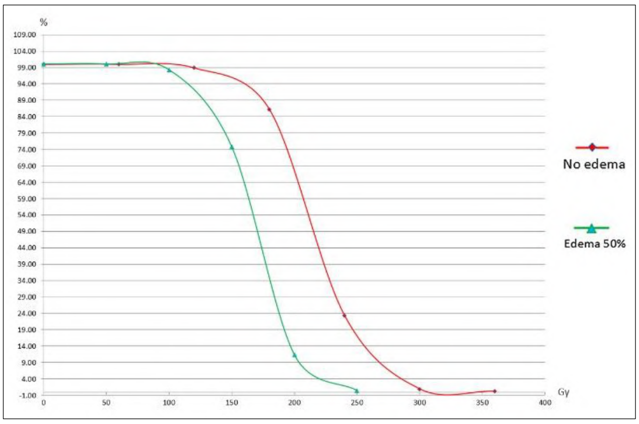


Figure 12: Dose-volume histogram (DVH) generated from the MCNPX simulation. The graph compares scenarios without edema and with 50% prostate edema, illustrating the impact on dose distribution.

Generated at the end of the simulation, the data contained in the ptrack file enabled detailed visualization of the trajectory of low-energy photons. It was possible to identify the occurrence of secondary events in adjacent organs at risk, such as the rectum and bladder, highlighting the importance of considering the protection of these structures in treatment planning.

Three-dimensional visualization and animation of particle behavior were performed using Moritz Geometry Tool software (Figure 13).

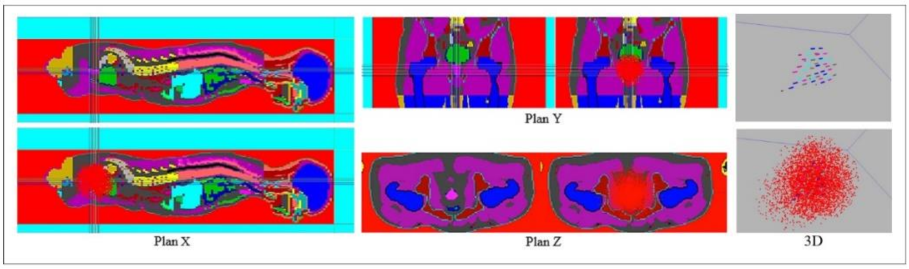


Figure 13: Capture of the animation in the X, Y, Z, and 3D planes in the Moritz Geometry Tool software, from the ptrack file generated by the simulation in the MCNPX code.

5 Conclusions

This work makes a significant contribution to the simulation of LDR brachytherapy for the treatment of prostate cancer by developing a system that generates input files for the MCNP code, taking into account the predictability of edema. This approach aligns computer simulations more closely with clinical reality, addressing a critical gap in current planning systems that ignore edema and source displacement. The introduction of these computational techniques is vital for more realistic scenarios.

The drastic reduction in the time needed to build input files for ^{125}I seeds and the automatic modification of planes and surfaces of the working phantom free the research team to focus on the physics of the study and in-depth analysis of the MCNP results. Accuracy in the spatial distribution of the seeds in the phantom's prostate is crucial to the quality of the simulations, which generate results with good statistics in 5 to 10 days.

The system developed in the Java programming language lays the foundations for the future application of artificial intelligence as a core technology. This integration will make it possible to significantly refine the prostate edema profile, correlating it precisely to the number and distribution of seeds. For this to happen, it will be crucial to develop a robust database of brachytherapy plans.

References

Almanza, V. L. D. (2018) Monte Carlo N-particle (MCNP) simulation of prostate cancer brachytherapy with edema and I-125 seed displacement. Master's Thesis. Master of Science in Physics. College of Science/Physics.

Apache Software Foundation. (s.d.). Apache NetBeans. Disponível em: https://netbeans.apache.org/. Acesso em: 05/05/2025.

CAPES (2023) Câncer de próstata: o segundo mais comum entre homens. Publicado em 19/10/2023. Disponível em: https://intranet.capes.gov.br/noticias/10505-cancer-de-prostata-o-segundo-mais-comum-entre-homens#:~:text=Este%20tipo%20de%20c%C3%A2ncer%20%C3%A9,de%2047%20mortes%20por%20dia. Acesso em: 06/05/2025.

Duggan, M. D. (2004) Improved radial dose function estimation using current version MCNP Monte Carlo Simulation: Model 6711 and ISC3500 ^{125}I brachytherapy sources. Appl. Radiat. Isot., v. 61, pp. 1443-1450.

ICRP 110 (2009) Adult Reference Computational Phantoms. International Commission on Radiological Protection, Pergamon Press, Oxford.

INCA (2023) Câncer de Próstata. Disponível em: https://www.gov.br/inca/pt-br/assuntos/cancer/tipos/prostata. Acesso em: 01/05/2024.

Mountris, K. A., Bert, J., Boussion, N., Valeri, A., Schick, U., & Visvikis, D. (2017) DVH-based optimization of LDR prostate brachytherapy using GPU-accelerated MC dosimetry. Apresentado na conferência Surgetica 2017, Strasbourg, França, de 20 a 22 de novembro.

Lim, S., and Kim, Y. (2021) Brachytherapy: A Comprehensive Review. Progress in Medical Physics, 32(2), 25–40. Disponível em: https://www.progmedphys.org/journal/view.html?doi=10.14316/pmp.2021.32.2.25. Acesso em 03/07/2025.

Oracle. (s.d.). Java. Disponível em: https://www.oracle.com/br/java/. Acesso em: 05/05/2025.

Rising, M. E. et al. (2023) MCNP® Code V.6.3.0 Release Notes. Los Alamos: Los Alamos National Laboratory. (LA-UR-22-33103, Rev. 1). Disponível em: https://doi.org/10.2172/1909545. Acesso em: 13 jul. 2025.

Riper, V. A. K. (2008) Moritz Geometry Tool User's Guide - Windows Version (Manual). White Rock Science.

Development of an Integrated National Database for Effective Dose Calculation from Environmental Radiation Exposure in Brazil

L. C. Takahashi[1]*, C.S. Lamounier[1], J.B. Severo[1], I.N.S Ferreira[1], T.O. Santos[2], Z. Rocha[1] and R. G. Passos[1]**

[1] Center for the Development of Nuclear Technology (CDTN), 31270-901, Belo Horizonte, Brazil

[2] Federal University of Minas Gerais (UFMG), 31270-901, Belo Horizonte, Brazil

* laura.takahashi@cdtn.br
** *In memoriam*

Abstract

The population's exposure to environmental radioactivity arises from both natural and anthropogenic sources. Natural sources include radionuclides such as uranium (U), thorium (Th), potassium-40 (K-40), and radon (Rn), as well as cosmic radiation (both primary and secondary). Anthropogenic sources encompass nuclear reactors, industrial sources, and medical procedures. This study aimed to construct an integrated database to characterize exposure and estimate the effective dose absorbed by the Brazilian population, considering multiple exposure sources and pathways (natural and artificial). To achieve this, secondary data were compiled from scientific literature and institutional databases, both national and international, as well as radiometric data provided by the Natural Radioactivity Laboratory of the Nuclear Technology Development Center (LRN/CDTN). The database also incorporates data from the Radon Gas Monitoring Campaign conducted in the metropolitan region of Belo Horizonte, which is essential for assessing internal exposure via radon inhalation. Its structure includes physical, environmental, geographic, and occupational parameters, systematically organized to allow targeted queries and integrated analyses. Considering the radon exposure data, a fundamental part in calculating the dose, which refers to Belo Horizonte, the pilot application and validation of the tool will take place in that city. The resulting database represents a methodological advancement within the national context by consolidating, standardizing, and making accessible previously dispersed information on radiological exposure. Its future implementation in computational tools will enable personalized dose estimation and may inform public policies related to environmental monitoring and population radiological protection. Additionally, it supports broader access to scientific data in a transparent and socially beneficial manner, fostering the responsible application of technical and scientific knowledge.

Keywords: Radon; Database; Effective Dose; Calculator.

1. Introduction

Human exposure to ionizing radiation is a continuous phenomenon, originating from both natural and artificial sources. The average annual effective dose received by the world population is estimated to be approximately 2.4 mSv, varying significantly according to geographical, occupational, and behavioral factors (UNSCEAR, 2020). Of this total, about 80% is of natural origin, while the remaining 20% derives from anthropogenic sources, mainly diagnostic imaging and therapeutic tests, which vary according to the type of test and the amount performed (ICRP, 2022).

Among the natural sources, radon gas (^{222}Rn) stands out as the main contributor to the global average effective dose, accounting for approximately 50% of the natural exposure, about 1.2 mSv/year. The inhalation of its decay products, especially indoors, represents a significant risk to public health, being associated with cases of lung cancer, in smokers and non-smokers (WHO, 2009). Other relevant contributions include cosmic radiation (0.4 mSv/year), radionuclides present in the human body and ingested (potassium-40, with 0.3 mSv/year), and terrestrial radiation from materials containing uranium and thorium (0.5 mSv/year) (UNSCEAR, 2020).

On the other hand, artificial sources of exposure have increased in importance, especially in the medical sector. It is estimated that diagnostic procedures with ionizing radiation (X-rays, computed tomography, scintigraphy, among others) are responsible for more than 80% of the effective anthropogenic dose, with a worldwide average of 0.5 mSv/year per person. In countries with more developed health systems, this value can exceed 3 mSv/year (UNSCEAR, 2020; IAEA, 2021).

In this scenario, it is increasingly necessary to develop integrated monitoring systems, capable of estimating, with greater precision, the population exposure by different routes (inhalation, ingestion, external exposure), considering regional, occupational and environmental aspects. In Brazil, despite the recognized geological and occupational diversity, data on radiological exposure are still fragmented and, in many cases, underutilized (Martinelli, 2023; Silva, 2020)

The present study proposes the construction of an integrated database, to characterize the exposure of the Brazilian population to natural and artificial radioactivity and estimate the effective dose by multiple routes. The database gathers secondary data from the scientific literature and national and international institutions, as well as radiometric information from the Natural Radioactivity Laboratory of the CDTN (LRN/CDTN). The framework also incorporates data from the Radon Monitoring Campaign carried out in the Metropolitan Region of Belo Horizonte, allowing the refinement of the assessment of internal exposure by inhalation.

Organized into physical, geographical, environmental and occupational parameters, this database enables targeted spatial and temporal analyses, and the city of Belo Horizonte was chosen as a pilot for the application and validation of the methodology, given the availability of regional data. This is a methodological advance in the national context, by consolidating, standardizing and making accessible a set of information that was previously dispersed, supporting the formulation of public policies aimed at radiological protection and environmental and health surveillance.

Finally, by enabling the future implementation of computational tools for personalized dose estimates and fostering the transparency and democratization of scientific data, the project reinforces the role of science as an instrument of social transformation and protection of public health.

2. Methodology

2.1 Selection and processing of data

After analyzing the literature and based on other international calculators, it was determined the variables that the calculation of the annual effective dose should consider, at least, four main categories of sources of ionizing radiation, both natural and artificial: exposure to radon gas, terrestrial radiation from natural radionuclides present in soil and building materials, cosmic radiation and exposure to artificial sources, especially medical diagnostic tests. In the case of exposure to radon gas (^{222}Rn), it was decided not to use regional data interpolation, since Brazil still does not have systematic campaigns for monitoring the gas in indoor environments with wide territorial coverage. In addition, radon concentrations in closed environments are strongly influenced by factors specific to each residence, such as the habits of the residents and the constructive characteristics, which makes it impossible to accurately estimate them using generalist models. Thus, for the database proposed in this work, the actual values measured in the homes of the participants of the Radon Gas Monitoring Campaign in Indoor Environments in the Metropolitan Region of Belo Horizonte, as well as other residents who have had their homes previously monitored through Solid-State Nuclear Track Detectors (SSNTDs) will be considered. In cases where measurement is not available, the absence of this variable will not compromise the estimate of the annual effective dose, which can be performed based on other sources of exposure.

The characterization of terrestrial radiation was performed from data obtained by in situ gamma spectrometry, using a portable spectrometer model RS-230 (Radiation Solutions Inc., Canada), sensitive to gamma radiation and coupled to a global positioning system (GPS) for the precise recording of the geographic coordinates of the sampled points. The information obtained was processed in a geoprocessing environment using the ArcGIS® software, allowing the quantitative and qualitative analysis of the spatial distribution of the natural radionuclides uranium-238, thorium-232 and potassium-40. The values collected, expressed in counts per second (cps), indicate the number of gamma decay events detected per second, which makes it possible to estimate the intensity of the natural radioactivity of the monitored surface. The data were subsequently interpolated using the Inverse Distance Weighting (IDW) method, which estimates the radiation values for non-sampled areas from the values recorded at the nearest points, attributing greater weight to neighboring samples. The study area was segmented according to the official delimitation of the neighborhoods of the city of Belo Horizonte, thus generating average values of natural radiation for each geographic sector. The information used was generated over the last few years by the Natural Radioactivity Laboratory (LRN) of the Center for the Development of Nuclear Technology (CDTN). The platform under development was structured to allow the incorporation of future data from other Brazilian cities, which will ensure its national applicability. In situations where specific radiometric data for the selected location are unavailable, standardized values of terrestrial radionuclide concentrations (uranium, thorium, and potassium) may be used, based on regional or national averages previously established in the scientific literature, in order to enable a technically consistent estimation of external gamma dose.

The estimate of the cosmic radiation dose was based on data from the scientific literature, which indicates an average annual dose at sea level ranging between 0.29 and 0.39 mSv, a value that can double every 2,000 meters of elevation. This exposure also varies according to latitude, due to the influence of the Earth's magnetic field. To account for these geographic variations, the average altitude of Brazilian capitals was collected, in order to adjust the estimate of the cosmic dose according to the location of the individual (CNSC, 2023; UNSCEAR 2000, 2008).

Regarding artificial sources of exposure, diagnostic imaging tests, such as radiographs and CT scans, were mainly included. The dose assigned to these procedures was estimated based on regional data on the frequency of these tests, according to institutional sources and technical literature. The platform was designed in a modular way, allowing the future inclusion of other types of medical tests and procedures that involve ionizing radiation, such as nuclear medicine and radiotherapy.

In addition to these main categories, complementary parameters based on international calculators of human radiation exposure, such as those developed by the U.S. Nuclear Regulatory Commission and the Oak Ridge Institute for Science and Education, were incorporated. These parameters include aspects of lifestyle and home environment that influence the annual effective dose, such as the type of construction of the residence (considering an increase of 0.07 mSv/year for stone, brick or concrete houses), the intake of food and water containing natural radionuclides such as potassium-40, carbon-14, uranium and thorium, the flight time in commercial airplanes (with an estimated increase of 0.005 mSv per flight hour), the use of porcelain dental prostheses (0.0007 mSv/year), exposure to X-ray machines for baggage screening at airports (0.00002 mSv per event), the use of televisions or CRT monitors (0.01 mSv/year), the habit of smoking half a pack of cigarettes per day (0.18 mSv/year), and the presence of smoke detectors containing americium-241 (0.00008 mSv/year). The consideration of these variables allows for a more personalized and accurate estimation of individual exposure to ionizing radiation, making the proposed database and the future digital effective dose calculator robust (EPA U.S., 2025; ENERGY EDUCATION, 2025).

2.2. Calculation of the effective dose

An integrated approach was adopted to calculate the annual effective dose, using conversion coefficients established by international organizations and methods consolidated in the literature. In the case of radon gas, the conversion of the concentration measured in Bq/m^3 to the annual effective dose in mSv was based on the methodology recommended by UNSCEAR (2000, 2019). Considering an equilibrium factor of 0.4 between radon and its decay products, an average indoor residence time of 7,000 hours per year, and a dosimetric conversion factor of 9×10^{-6} mSv/(Bq·h/m³). It is emphasized that, applying these parameters, for the effective dose to be 1 mSv, the minimum radon concentration is 40 Bq/m^3. In the model adopted by UNSCEAR, the annual effective dose from radon gas is estimated based on the following equation 1:

$$E = C_{Rn} \cdot F \cdot t \cdot k \tag{1}$$

Where:

E = annual effective dose (Sv)
C_{Rn} = radon activity concentration ($Bq \cdot m^{-3}$) (to be entered by the user)
F = equilibrium factor (dimensionless)
t = exposure time (h/year) (to be entered by the user)
k = dosimetric conversion factor

For the dose from terrestrial radiation, the count-per-second (cps) values obtained for the radionuclides uranium-238 (^{238}U), thorium-232 (^{232}Th) and potassium-40 (^{40}K) in CPS per ppm were converted into activity concentration (Bq/kg), and later into dose rate (nGy/h). Table 1 shows this conversion.

Table 1: CPS conversion of natural radionuclides to nGy.

Radionuclide	**CPS conversion factor → Bq/kg***	**Conversion factor Bq/kg → nGy/h****
^{238}U	3.29 CPS per ppm → 1 ppm = 12.35 Bq/kg → therefore factor ≈ 0.30 CPS/(Bq/kg)	0.462 nGy/h per Bq/kg
^{232}Th	0.13 CPS per ppm → 1 ppm = 4.06 Bq/kg → therefore factor ≈ 0.13 CPS/(Bq/kg)	0.604 nGy/h per Bq/kg
^{40}K	3.29 CPS per 1 % K → 1 % = 313 Bq/kg → therefore factor ≈ 0.0105 CPS/(Bq/kg)	0.0417 nGy/h per Bq/kg

*Values taken from the literature and can also be estimated in the laboratory through own calibration.
**Values according to UNSCEAR (2008).

Equation 2 for calculating the rate of absorbed dose in air from each radionuclide is given by:

$$D_i = C_i \times F_i \tag{2}$$

Where:

- D_i is the airborne absorbed dose rate (nGy/h) due to radionuclide i;
- C_i is the specific concentration of the radionuclide (Bq/kg);
- F_i is the conversion factor from Bq/kg to nGy/h according to UNSCEAR (2008).

The total terrestrial dose is then obtained by adding the individual contributions of the three radionuclides, as per Equation 3:

$$Terrestrial_D = (D_U + D_{Th} + D_K) \times T \times C_f \tag{3}$$

Where:

- $D_{terrestrial}$ = gamma dose rate (nGy/h)
- T= 8760 h/year
- $C_f = 0.7\times10^{-6}$ mSv/nGy (conversion factor from Gy to Sv)

For the dose from cosmic radiation, an estimate based on the altitude of the city of residence was adopted. The average dose at sea level is approximately 0.30 mSv/year, and it increases with elevation, and can double every 2,000 meters. Equation 4 used was adapted from the proposal by Shahbazi-Gahrouei et al. (2013):

$$Cosmic_D = 0.30 \times 2^{(h/2000)} \tag{4}$$

Where:

- Cosmic = cosmic dose (mSv/year)
- h = altitude of the city in meters

After calculating the partial doses related to radon, terrestrial radiation, and cosmic radiation, in addition to contributions from artificial sources and complementary parameters (such as lifestyle and occupational exposure), the sum of all plots was performed to estimate the total annual effective dose. The general equation 5 was expressed as:

$$Total_D = D_{Rn} + D_{Terrestrial} + D_{Cosmic} + D_{Artificial} + D_{Complementary} \qquad (5)$$

The final value obtained represents the estimate of individual exposure to ionizing radiation in a year, considering natural and artificial sources. The unit adopted to express the dose is the Sievert (Sv), which is the international unit for measuring the biological effects of ionizing radiation. If the estimated value exceeds 20 mSv/year, it is likely that the main contribution will come from medical examinations.

2.3. Selection of the platform for the development of the virtual calculator

The final stage of the methodology consisted of defining the most appropriate digital platform for the development of the virtual calculator for estimating the dose by exposure to natural and artificial radiation. The choice was guided by technical and functional criteria, such as multiplatform accessibility (web, desktop and mobile), ease of use and interaction with the end user, the ability to incorporate formulas and complex calculation logic, integration with databases and the possibility of exporting results, in addition to the maintainability and future scalability of the tool. After defining the logical structure for calculating the dose, the chosen programming was in Python, using the JetBrains PyCharm development environment. The calculator interface is developed with the use of the Streamlit framework, which allows the creation of interactive web applications in a simple and efficient way. This stage represents the transition between the theoretical foundation established in this work and the practical implementation of the tool, which will be structured based on the data and equations previously described.

The logic for calculating the dose was structured based on the equations presented in the previous sections, being parameterized to accept different sources of exposure, such as radon, terrestrial, cosmic and medical radiation, to allow the insertion of data according to the individual or regional profile. The choice for a web platform also aims to expand the reach and usefulness of the tool, favoring actions of education, health surveillance and support for the management of environmental and occupational risks related to radiation exposure.

3. Results

The first stage of this work consisted of the construction and organization of a database with the main parameters involved in the estimation of the annual effective dose by exposure to natural radiation (radon, terrestrial radiation and cosmic radiation) and artificial radiation (medical examinations). The database was structured in such a way as to allow both the individual analysis of the parameters and their integration into a digital tool.

The equations used to calculate the dose were based on recommendations from international organizations, such as UNSCEAR (2000, 2008, 2019), ICRP (1993, 2007) and recent scientific publications, and were adequately parameterized according to the units used in the national databases.

For each component of the exhibition, the appropriate conversion factor was applied and the annual exposure workload was considered, as detailed in the methodology section.

The preliminary tests of application of the conversion formulas and dose calculation were carried out and the results obtained demonstrated consistency with reference values for natural exposure in residential environments, as reported in the scientific literature and in international technical documents. The application of specific conversion factors for the measuring instruments used, such as the RS-230 portable gamma spectrometer, was also tested. In the case of cosmic radiation, the tests based on the altitude of different Brazilian capitals demonstrated an expected variation that was consistent with the altimetric models used.

The logical structure of the database allowed the simulation of exposure scenarios for different individual and population profiles and regions, anticipating the operation of the future calculator. Thus, it was possible to validate not only the coherence of the calculations but also the applicability of the model in real contexts through small simulations.

The next step of this work consists of the implementation of the virtual calculator, which will be developed based on the established data and methodology. Initially, a test version of the tool will be created, with a simple and functional interface, allowing the insertion of individual exposure parameters and providing as output the estimate of the annual effective dose. The platform adopted for the initial development is the Python environment using the Streamlit framework, as it offers easy integration with scientific libraries and real-time visualization.

It should be noted that this first version of the calculator will be experimental and will serve as a basis for tests with users. Based on the evaluation of demand and the experience of use, adaptations can be made, both in the calculation logic and in the technological infrastructure. Among the possible future evolutions are the migration to more robust platforms, with greater capacity for personalization, storage and cross-referencing of data, in addition to the expansion of the input database with national and regional sources of radiation that are obtained through research.

The creation of this tool seeks not only to facilitate access to information on radiation exposures, but also to promote awareness among the population and subsidize environmental and occupational health surveillance actions, contributing to risk management and the strengthening of public policies based on scientific evidence.

4. Conclusions

This work presented the construction of an integrated database and the formulation of a set of equations for estimating the annual effective dose by exposure to natural and artificial radiation, with the objective of subsidizing the development of an easy-to-use virtual calculator. The theoretical foundation adopted is in line with the recommendations of international reference organizations and the conversions and calculations applied were previously tested, resulting in estimates consistent with expected values in the scientific literature and environmental assessments carried out in the national territory.

By gathering data on radon concentration, natural radioactivity in the soil and cosmic radiation, in addition to medical exposure, the proposed model enables a more comprehensive estimate of the effective dose to which individuals may be exposed. The use of validated conversion factors adapted to the Brazilian reality gives greater precision to the system, especially in the transposition of measurements obtained by gamma spectrometry and other field tools.

The methodology developed proved to be efficient in organizing, processing and applying the data in an automated way, which will allow, in the next step, the creation of a digital calculator accessible to professionals from different areas, such as health, environment, engineering and education. This tool has the potential to expand access to information on exposure to ionizing radiation and contribute to prevention, monitoring, and environmental and health education actions.

The present work was fundamental for establishing the technical-scientific foundations necessary for the development of a reliable calculation tool. The creation of a structured database, based on specialized literature and validated through previous tests, is an indispensable step, as the virtual calculator depends directly on the quality and accuracy of this information to generate correct dose estimates. Without this well-designed and scientifically based database, the calculator would have no practical applicability or technical credibility. Thus, this study represents an essential step to ensure the robustness and reliability of the tool that will be implemented in the future.

It is concluded, therefore, that the structure developed in this work offers a solid basis for the construction of the proposed calculator, which can become an important tool to support decision making and awareness of the population about the effects of radiation exposure in indoor environments. The continuity of the project will include the development of the calculator interface, its validation in real scenarios and the eventual expansion of functionalities, according to the demands identified by users and institutions involved with the theme.

Acknowledgements

The author thanks to CDTN/CNEN.

References

da Silva J. B., Yoshimura E. M., and Fernandes N. N. (2022) Radiological mapping and characterization of the Poços de Caldas Plateau uranium province, Brazil. *Minerals* 12(10):1295
https://www.mdpi.com/2075-163X/12/10/1295

Energy Education (n.d.) *Radiation Dose Calculator Simulation*. University of Calgary https://energyeducation.ca/simulations/radiation_dose_calculator/index.html

EPA – United States Environmental Protection Agency (n.d.) *Calculate Your Radiation Dose*. https://www.epa.gov/radiation/calculate-your-radiation-dose

IAEA – International Atomic Energy Agency (2003) *Guidelines for radioelement mapping using gamma ray spectrometry data*. IAEA-TECDOC-1363, Vienna https://www-pub.iaea.org/MTCD/publications/PDF/te_1363_web.pdf

IAEA – International Atomic Energy Agency (n.d.) *Radiology Process in Medical Physics*. Human Health Campus https://www.iaea.org/resources/hhc/medical-physics/radiology/process

ICRP – International Commission on Radiological Protection (1993) *1990 Recommendations of the International Commission on Radiological Protection.* ICRP Publication 60. Annals of the ICRP 21(1–3). Elsevier, Oxford

ICRP – International Commission on Radiological Protection (2007) *The 2007 Recommendations of the International Commission on Radiological Protection.* ICRP Publication 103. Annals of the ICRP 37(2–4). Elsevier, Oxford

Martinelli S, Medeiros AN, de Souza RF, Marconi AM, Navoni JA. Analysis of the cancer occurrence related to natural radioactivity in the Rio Grande do Norte State, Brazil. Environ Sci Pollut Res Int. 2023 Aug; 30(38):89140-89152. DOI: 10.1007/S11356-023-28708-1. Epub 2023 Jul 14. PMID: 37442937. https://pubmed.ncbi.nlm.nih.gov/37442937/

Shahbazi-Gahrouei D., Gholami M., Setayandeh S., and Bahadorzadeh M. (2023) Indoor and outdoor gamma radiation exposure levels in selected cities of Iran: A comparative analysis and implications for public health. *Radiation and Environmental Biophysics* 62:329–336 https://pubmed.ncbi.nlm.nih.gov/37442937/

UNSCEAR – United Nations Scientific Committee on the Effects of Atomic Radiation (2020) *Sources, effects and risks of ionizing radiation: UNSCEAR 2020/2021 Report, Volume I*. United Nations, New York https://www.unscear.org/unscear/en/publications/2020_2021_1.html

UNSCEAR – United Nations Scientific Committee on the Effects of Atomic Radiation (2008) *Sources and effects of ionizing radiation: UNSCEAR 2008 Report, Volume I, Annex B*. United Nations, New York https://www.unscear.org/unscear/uploads/documents/unscear-reports/UNSCEAR_2008_Report_Vol.I-CORR.pdf

World Health Organization (2009) *WHO Handbook on Indoor Radon: A Public Health Perspective*. WHO Press, GenevaAvailable at: https://www.who.int/publications/i/item/9789241547673

Development of an optical sensor prototype for doses readings in Fricke gel dosimeters

Silva L. S. M.[1], Santos A. M. M.[1], Mesquita, A. Z[2], Fonseca T. C. F. [1,2]

[1] Federal University of Minas Gerais - UFMG, Department of Nuclear Engineering, Campus of UFMG, Pampulha, Zip: 31.270-901. Belo Horizonte, MG - Brazil.

[2] Nuclear Technology Development Center – CDTN, Brazilian National Nuclear Energy Commission, Campus of UFMG, Pampulha, Zip: 31.270-901, Belo Horizonte, MG - Brazil.

[*] leonardosantiago.lsms@gmail.com

Abstract

Fricke gel dosimeters, also known as hydrogels, are chemical solutions capable of recording absorbed radiation doses within their sensitive volume. These solutions are prepared in an acidified medium and saturated with Fe^{2+} ions, which are dispersed in a tissue-equivalent gel matrix. The radiation-induced oxidation of Fe^{2+} species allows for the correlation between the absorbed radiation dose and the amount of Fe^{3+} formed. However, despite the unique advantage of enabling three-dimensional (3D) dosimetry, this system remains limited, as measurements are typically performed in laboratory settings. Therefore, the aim of this study is to develop an optical sensor prototype composed of red, green, and blue (RGB) channels, capable of performing quantitative readings on colored samples of Fricke gel dosimeters. The RGB sensor was rigorously evaluated to identify which spectral bands, with use of filters, are capable of providing quantitative responses in both irradiated and non-irradiated samples.The effective response of the sensor—defined as its efficiency in illuminating and detecting electromagnetic radiation reflected by the samples—was observed in the range of 415–564 nm for the blue channel, 440–600 nm for the green channel, and 510–750 nm for the red channel. These data contribute to a better understanding of the interaction between the light emitted by the sensor and the signals captured by the RGB channels for dose quantification in the corresponding dosimeters. The analytical performance of the RGB sensor was assessed by comparing its readings with those obtained using a commercial spectrophotometer. This comparison yielded a linearity range between 0.005 and 0.040 mmol L^{-1}, as demonstrated by the colorimetric method for determining Fe^{3+} species. Additionally, the sensor's performance in response to both irradiated and non-irradiated Fricke dosimeters was evaluated through individual and combined RGB channel readings. Two equations were derived for these distinct reading parameters, allowing for the evaluation of randomly selected irradiated samples.The prototype demonstrated the ability to measure radiation dose in Fricke gel dosimeters, showing a proportional relationship between color intensity and the applied radiation dose. These analyses confirm that the RGB system can be optimized for reading Fricke gel samples in various formats and open up new perspectives for the use of Fricke dosimeters, with readings limited to specialized laboratory environments.

Keywords: Dosimetry; Fricke gel; optical sensor prototype

1.0 Introduction

Due to the increasing demand for dosimetric systems capable of efficiently validating the radiation dose distribution delivered to patients undergoing high-precision radiotherapy, gel dosimetry gained significant scientific attention in the 1990s. Fricke gel dosimeters, also known as hydrogels, compete with other dosimeters such as ionization chambers, thermoluminescent detectors, and alanine/electron paramagnetic resonance (EPR) dosimeters, owing to their ability to quantify three-dimensional (3D) dose distributions with high spatial resolution, whereas traditional dosimeters are limited to point measurements (De Deene, 2022; Moussous, 2021; Nezhad and Geraily, 2022). According to Sagsoz et al. (2025), radiotherapy continues to evolve as a discipline in which precise and personalized treatment planning is increasingly critical. In this context, the accurate and reliable application of radiation dosimetry techniques is essential to enhance treatment efficacy and safeguard surrounding tissues from unintended radiation exposure (Sagsoz et al., 2025) Fricke gel dosimetry is based on the radiation-induced oxidation of ferrous ions (Fe^{2+}) into ferric ions (Fe^{3+}). The absorbed radiation dose is linearly proportional to the concentration of Fe^{3+}. As an indirect dose measurement technique, these dosimeters can be evaluated using magnetic resonance imaging (MRI) and optical computed tomography (OCT), in addition to point-based measurements such as ultraviolet-visible (UV-Vis) spectrophotometry (Macchione, 2022; Baldock et al.,2001; Gore and Kang, 1984).

In order to optimize and keep pace with advancing technologies in the field of radiotherapy, the readout techniques for Fricke dosimeters must be improved. According to De Deene and Jirasek (2024), as treatment planning software has become more reliable and various treatment modalities have been increasingly integrated and standardized within the quality assurance workflow alongside electronic dosimetry, the need for gel dosimetry has declined for these well-established radiotherapy techniques. Therefore, gel dosimeters must be adapted to these specific challenges in order to align with emerging technologies, including new readout methods such as dose recording in anthropomorphic geometries with physiological motion, image guidance and tracking, among other parameters (De Deene and Jirasek, 2024). In this context, an optical sensor prototype based on red, green, and blue (RGB) channels was developed for the quantitative analysis of Fricke gel solutions. Analytical techniques based on colorimetric methods are frequently employed in quantitative chemical analyses (Roy Choudhury, 2015). The colorimetry-based analytical approach has broad applicability due to its precision and simplicity. This is attributed to several advantages, such as its easy integration with portable and widely available imaging devices, resulting in user-friendly analytical procedures in many fields that require out-of-laboratory applications for in situ and real-time monitoring (Capitán-Vallvey et al., 2015).

In the RGB system, each sensor operates by capturing the intensity of light in the red (R), green (G), or blue (B) spectral range, respectively. The application of this system for reading Fricke gel dosimeters involves the digital capture of images of irradiated samples, correlating color intensity with the absorbed radiation dose. This analytical procedure comprises two main processes. First, a point-based analysis is performed on a small group of pixels to detect fine features within the sample. Second, a global analysis is conducted—such as a color histogram—to evaluate the overall homogeneity of the sample under investigation (Brosnan and Sun, 2004; Du and Sun, 2004). In this approach, color information can be described through the object's color space, represented by the three primary light colors defined by the chromaticities of the standard red, green, and blue components (RGB) (León et al., 2006). This study aims to develop a portable spectrophotometric device based on colorimetry, employing an RGB color sensor, for the effective readout of Fricke gel dosimeters.

2.0 Materials and Methods

2.1 Preparation of Fricke gel and chemical reagents

The Fricke gel dosimeters analyzed in this study were prepared in a gelling medium using only a sodium carboxymethylcellulose (CMC) matrix. The fabrication protocol for this dosimeter was based on the work of Santos et al. (2024), which had already been radiologically characterized through UV-Vis spectroscopy. To validate the RGB detection system, a standard ferric chloride sample was prepared in varying concentrations ranging from 0.005 to 0.150 $mmol \cdot L^{-1}$. Using these same samples, measurements were performed with a commercial spectrophotometer (Hitachi U-2900 UV-Vis, Hitachi High-Technologies, Tokyo, Japan) for comparison purposes. Sample irradiation for optical sensor readings was carried out using a cobalt-60 (^{60}Co) source at the Gamma Irradiation Laboratory (LIG), located at the

Nuclear Technology Development Center (CDTN). During irradiation, a dose rate of approximately 100 Gy/hour was used, with a source-to-sample distance of 1.5 meters. The irradiation time varied depending on the source activity. The analytical-grade (PA) chemical reagents used in the preparation of the dosimeters were: ammonium ferrous sulfate or Mohr's salt [FeSO4(NH4)2SO4·6H2O], purity [95–97%]; sulfuric acid (H2SO4), density = 1.84 kg/cm³, purity [95–97%]; tetrasodium salt of xylenol orange ACS (C31H28N2Na4O13S); and sodium carboxymethyl cellulose – CMC (C28H30Na8O27), all supplied by Sigma-Aldrich-Merck KGaA®, Germany. Sodium chloride (NaCl), purity >99%, Fmaia®, Cotia, Brazil; ferric chloride hexhydrate ($FeCl_3 \cdot 6H_2O$), purity ≥99.0% and glycerol (C3H8O3), purity >99.5%, Synth®, Diadema, Brazil.

2.2 Optical sensor prototype assembly

The optical sensor developed in this work consists of three main components: a computer running a control and data acquisition program written in Python, responsible for instruction, data collection, and analysis; a light chamber composed of white LED lights with known wavelengths, corresponding to the sample compartment. This chamber serves to enhance the light reflected from the samples, which is then measured by the color sensor; and a console, which integrates the electronic components, including a microcontroller (Arduino Uno) and a TCS230 color sensor with an external filter (Hoya cmc 500). These devices were connected using wires and connectors, forming the electrical interface between the microcontroller and the TCS230 sensor. All electronic components were assembled within an MDF enclosure measuring 10 × 10 × 0.65 cm, which comprises the hardware of the prototype. The assembly of the optical sensor prototype is illustrated in Figure 1.

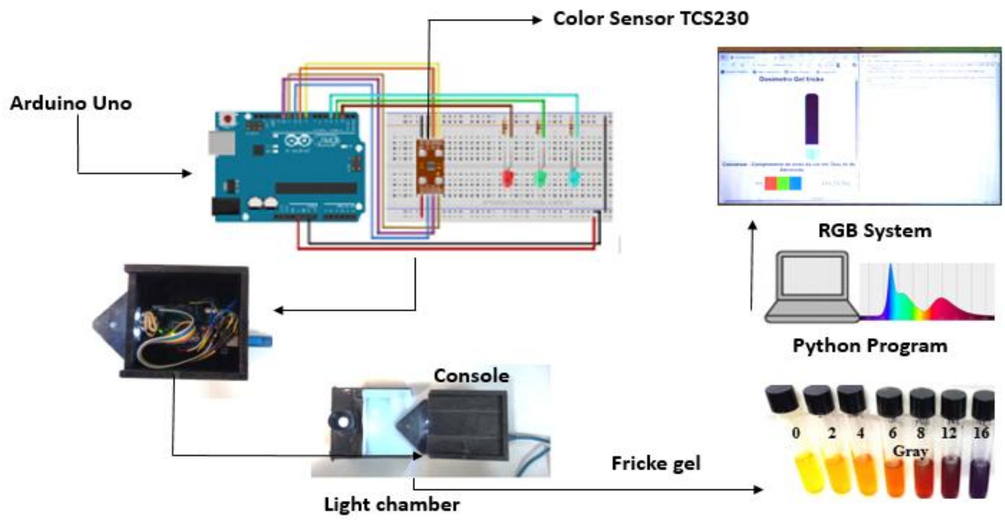


Figure 2.1: Diagram of the RGB color sensor for quantitative doses in Fricke gel dosimeters.

2.2 System Calibration

Before beginning the readout of the dosimeters, the prototype undergoes a checklist to ensure it responds correctly across a color range from white to black. Typically, calibration cards with these shades are placed in front of the sensor, positioned within the light chamber (sample holder), following protocols defined by the Python program. Subsequently, Fricke gel dosimeter samples irradiated with different doses—exhibiting colors ranging from golden yellow to deep blue—are individually placed in the same compartment. The color shade of each sample is associated with the concentration of Fe^{3+} ions formed during irradiation, which is proportional to the absorbed radiation dose. The sensor captures the different color intensities of the samples and outputs them as numerical values. Based on this data, the Python program displays the corresponding colors in RGB format, translating the values on the computer screen. Initially, the optical sensor prototype was developed to read Fricke gel samples contained in sealed tubes with ground-glass screw caps. This section of the light chamber can be adapted for future studies, potentially being reconfigured to accommodate containers of different shapes filled with hydrogel.

Results and discussion

3.1 Fundamental aspects:

The TCS230 color sensor provides a digital output corresponding to three distinct channels: red, green, and blue (RGB). In addition to the color sensors, the module includes a set of four white LEDs that serve as a constant light source directed at the sample being measured. The sensor has a detection system based on a set of color filters composed of 64 photodiodes. These are subdivided into 16 for red, 16 for green, 16 for blue, and 16 for white light (no filter). The photodiodes are uniformly distributed to capture light, filter the received colors, and subsequently generate a square-wave output signal with a well-defined frequency. This signal provides information about the intensity of the primary RGB color components (Taos, 2009).

3.2 RGB sensor analysis capability

The sensor's ability to respond to different colored solutions was critically evaluated to better understand which spectral bands are filtered and processed by each sensor channel, according to the manufacturer's data (Taos, 2009). The effective response of the RGB sensor is defined as its ability to emit and detect the electromagnetic radiation reflected by the samples. Figure 3.1A shows the response spectra for the red, green, and blue channels, along with white light, all without filters. With the application of filters to this same system, a narrowing of the absorption bands can be observed in the ranges of 625–750 nm, 500–565 nm, and 450–485 nm for the red, green, and blue channels, respectively, as shown in Figure 3.1B.

To evaluate the signals processed by the detector and their correlations with the different measured colors, readings were performed using the developed prototype. The effective capability of the RGB sensor was determined by combining the results (multiplication of responses) of LED illumination emission and the detector response for each RGB channel (Figure 3.2A). The color information can be described by the sample's color spaces, represented by the chromaticities of the standard components: red, green, and blue, which correspond to the primary colors (Leó, 2006). In this analysis, with the prototype already assembled, the readings identified the same bands for the red, green, and blue channels. Overlaps between colors were observed, especially for the green channel. The spectrum of the blue channel was less affected

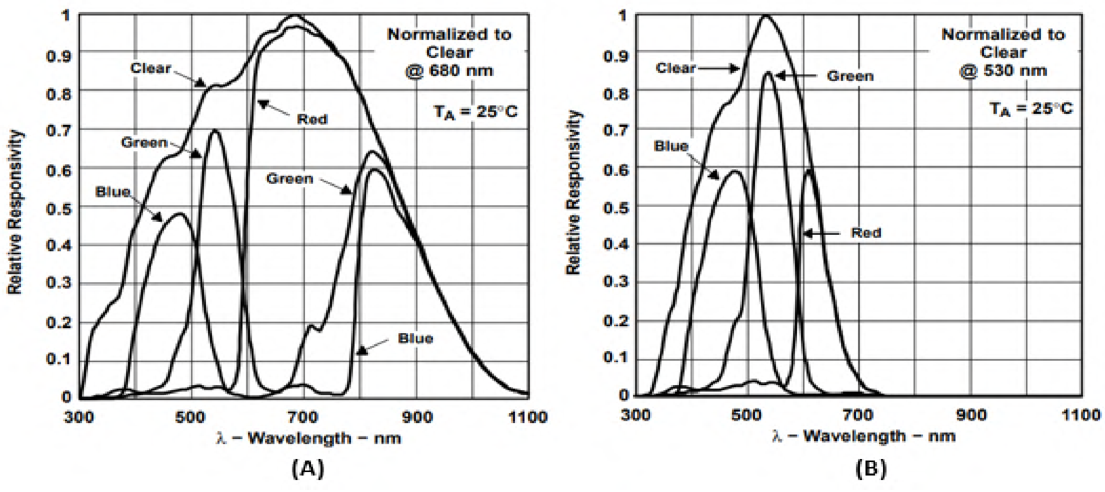


Figure 3.1: Photodiode spectral responsivity: No filter (A) with external Hoya cmc 500 filter (B).

by the red channel at wavelengths up to 500 *nm*, and by the green channel up to 485 *nm*. However, it was observed that the influence of blue and green colors on the red channel was less significant.
Regarding the composition of the samples, the colored iron complexes (main analytes) exhibit maximum absorption bands in the regions of 435 nm (Fe^{2+}) and 585 nm (Fe^{3+}). Based on the scan performed in the visible region, it was possible to identify that the maximum absorption bands of the iron species overlap the sensor signal that spans all three channels. However, it is expected that the analytical signal behavior will decrease as the concentrations of the complexes increase, due to the light absorption by the colored complex.

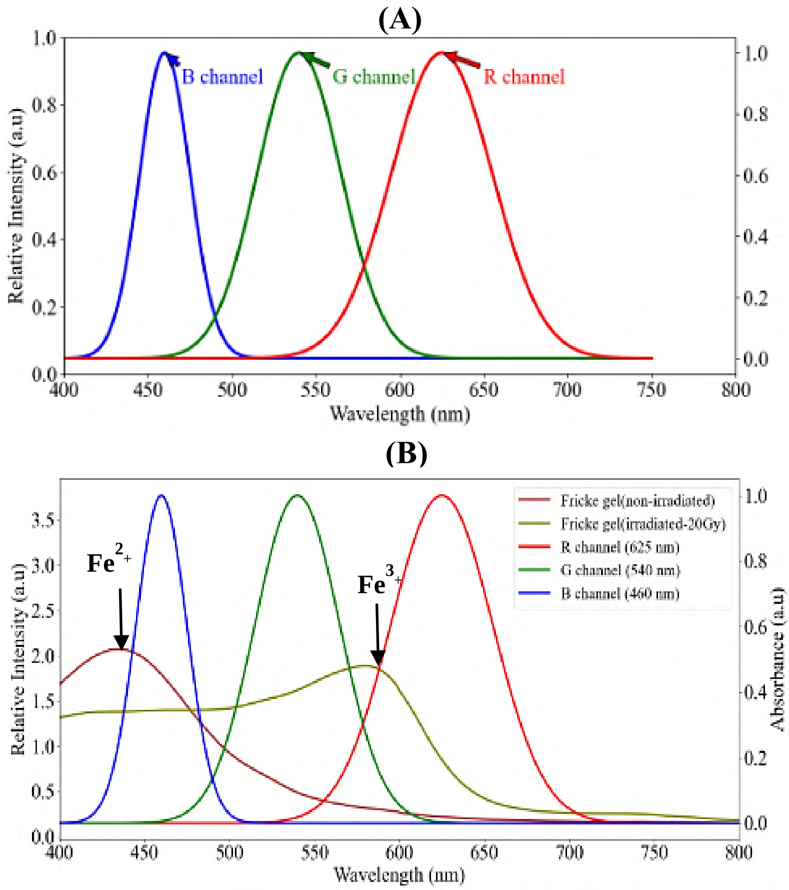


Figure 3.2: Effective capability of the RGB sensor (A) and overlap of **Fe^{2+}/Fe^{3+}**complex absorption spectrum and effective sensor response (B).

3.3 Performance and validation of the RGB system for quantitative analysis

To evaluate the colorimetric quantification using the RGB sensor, a calibration curve of ferric chloride standards at varying concentrations (0.000 to 150 mMol·L^{-1}) was analyzed and compared with readings from a commercial spectrophotometer**.** The linear RGB color model utilized intensity data from the three channels (red, green, and blue) to form a single color for each image pixel (Figure 3.3(A)). Numerically, the color is represented by three values and can be described by the analytical characteristics of the reading system, according to the equation below:

$$\mathbf{A} = -\log \frac{I_{(RGB)}}{255} \quad (1)$$

Where the absorbance (A) of the image is given by the ratio of the actual intensity value (IRGB) obtained from the RGB channels to the logarithm of the maximum intensity (255), as shown in Figure 3.3(B).

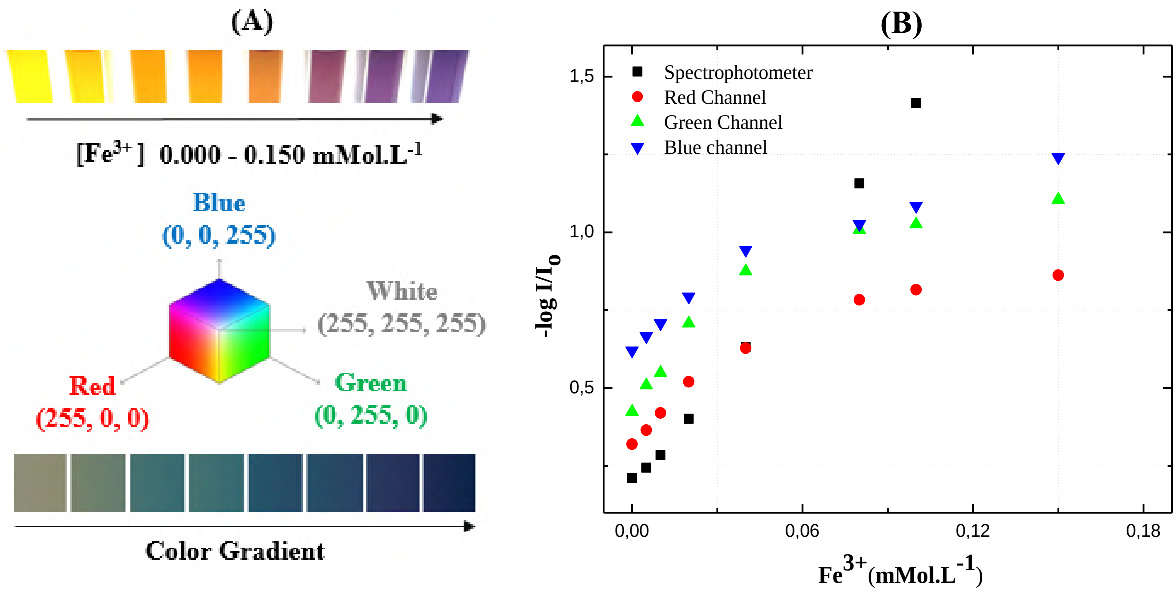


Figure 3.3 - Standard curves for iron determination (A) Scheme of reading of iron standards using the RGB sensor and result of the color gradiente. (B) Sensitivity of reading the curve of ferric standards (RGB sensor and commercial spectrophotometer).

Regarding the composition of the Fricke gel standard, the concentration of Fe^{3+} species complexed with xylenol orange produced a color gradient in the calibration curve, demonstrating the sensor's ability to detect differences in hue. For the lightest solution (white), maximum intensity values were recorded due to the absence of absorbing species. For the other solutions, an increase in color intensity was observed across a range from golden yellow to violet, corresponding to higher iron concentrations. Based on the data presented in Table 3.1, a linear response was observed between the values obtained by the RGB sensor and Fe(III) concentrations up to 0.040 mMol·L^{-1}. In comparison, the spectrophotometer showed linearity in the concentration range between 0.005 and 0.150 mMol·L^{-1} for the same analyte.
However, RGB color system readings can still be optimized to extend linearity to higher analyte concentration ranges. This would require adjustment of integration time and gain, which are tunable parameters used to enhance sensitivity. According to the manufacturer's recommendations, gain can be set to one of the following values: 1x, 4x, 16x, or 60x, while

Parameters	RGB sensor	Spectrophotometer
Curve equation	R - A = 3.7256 + 0.4007[Fe^{+3}] G - A = 4.4921 + 0.5483[Fe^{+3}] B - A = 3.4926 + 0.7012[Fe^{+3}]	A = 12.1946 + 0.1754[Fe^{+3}]
Linear range (mMol. L-1)	R 0.005 – 0.040 G 0.005 – 0.040 B 0.005 – 0.040	0.005 – 0.150
R-square	R 0.88697 G 0.85151 B 0.90596	0.99896

integration time can be adjusted to: 2.4, 24, 50, 101, 154, or 700 ms (De Carvalho Oliveira, 2022).

3.4 Colorimetric analytical technique for dose determination

To evaluate the use of the RGB sensor for quantitative measurements on irradiated and non-irradiated Fricke gel samples, readings were performed using the senso óptico with Python programming. Based on the numerical values obtained from the RGB channel readings, a nonlinear equation relating intensity as a function of radiation dose was determined, with their respective deviations (error bars).

3.4.1 Dose determination - Individual colorimetric readings per channel and average readings from the RGB system.

Dose determination in Fricke gel using the red, green, and blue channels was obtained through individually defined polynomial equations (Figure 3.4 (A)). The red channel (R) exhibited the highest sensitivity to dose variation (greater curve slope). The green (G) and blue (B) channels followed the same general decreasing trend but with a narrower dynamic range. The intensity of the color centers used to determine the polynomial equations is presented in Table 3.2. (by channel). Each color intensity is represented by 256 levels (from 0 to 255). Within this color scheme, a total of 16.777.216 distinct colors can be generated. The value R = 0, G = 0, B = 0 corresponds to pure black, while R = 255, G = 255, B = 255 corresponds to pure white. With this system, unique combinations of R, G, and B values are possible, providing millions of different hues, saturations, and brightness levels (Levkowitz, 1993).

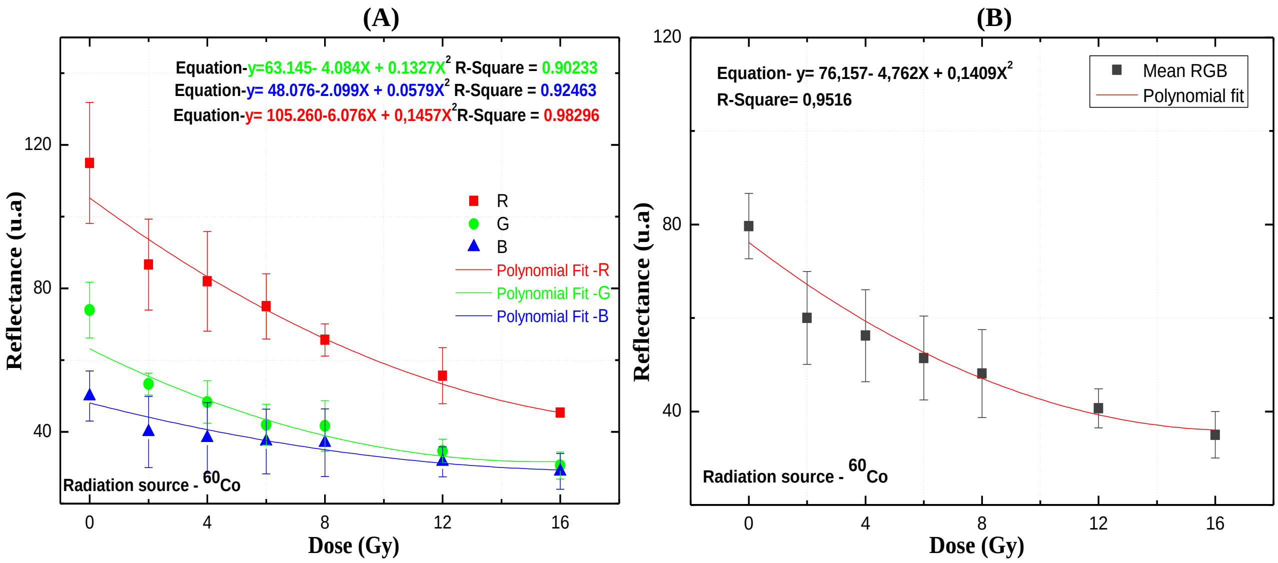


Figure 3.4 - Intensity versus dose curve: irradiated and non-irradiated Fricke gel samples with readings per channel (A) and readings using the system average (B).

For the colorimetric measurements using all RGB channels, the arithmetic mean of the intensities from the complete color system was used, given by RGB. The color intensity was defined as $I=(R+G+B)/3$, as shown in Table 3.2. Based on these data, a curve was plotted, resulting in a polynomial equation representing the average of the RGB channels (Figure 3.4(B)). Using the equations presented in Figures3.4, it is possible to estimate the radiation dose in the Fricke gel dosimeter by applying the corresponding parameters. The colorimetric analytical technique using RGB systems enables the application of individual channel models or aggregated metrics (such as average RGB) for enhanced calibration. The Python program automatically utilizes the open-source (Scikit-Image) library for image processing. The original images are converted into NumPy arrays via C-based programming. Once these data are compiled, a multiple linear regression model is applied, resulting in Equation 2. This linear equation allows for the estimation of the absorbed dose using the RGB values as independent variables.

In this model, higher G values increase the estimated dose, while higher R and B values decrease it—consistent with the typical darkening (loss of intensity) of the gel as the dose increases.

$$\textbf{Dose (Gy)} = 30.93 - 0.345*R + 0.493*G - 0.566*B \quad (2)$$

Samples Fricke gel Doses (Gray)	R	G Mean /SD (by channel)	B	RGB Mean/SD (System)
0	115.00 ± 26.87	74.00 ± 7.79	50.00 ± 6.98	79.67 ± 6,98
2	86.67 ± 12.66	53.33 ± 3.09	40.00 ±9.933	60.01 ± 9,33
4	82.00 ± 13.88	48.33 ± 5.91	38.33 ± 9.84	56.22 ± 9,84
6	75.00 ± 9.09	42.00 ± 5.72	37.33 ± 8.99	51.44 ±8,99
8	65.67 ± 4.50	41.67 ± 7.04	37.00 ± 9.42	48.11 ± 9,42
12	55.67 ± 7.85	34.67 ± 3.30	31.67 ± 4.19	40.67 ± 4,19
16	45.33 ± 0.47	30.67 ± 3.77	29.00 ± 4.98	35.00 ± 4,97

Table 3.2 - Arithmetic mean of individual colorimetric readings per channel and average readings of the RGB system

Another approach to colorimetric dose assessment using the RGB variables independently involves analyzing the pixel values of the images generated in the Python program. This analysis can be performed using a heatmap, as shown in Figure 3.5.

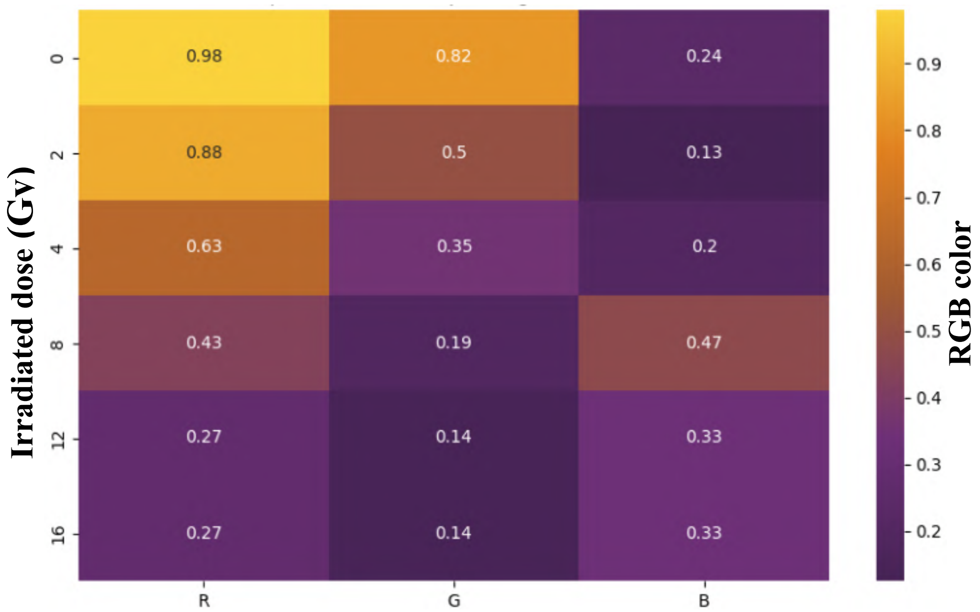


Figure 3.5 - Heat map, using measurements of the number of pixels as a function of radiation dose.

As presented, the heatmap is divided into cells. Each cell contains a specific number of pixels, which corresponds to the color intensity readings obtained from the image generated by the RGB system. For each irradiated sample, a set of cells was created for the R, G, and B variables, respectively. It is possible to correlate these color intensities with their corresponding absorbed doses. This process optimizes the reading of Fricke dosimeters, enabling timely dose assessments. However, this color scale must be standardized in advance using reference standards and subsequently integrated into the Python program's database.

3.0 Conclusions

The results obtained in this study revealed that the optical sensor prototype based on RGB channels has effective capability for dose quantification in Fricke gel dosimeters. The various observations, using color increments reflected by the samples, helped to understand the interaction between the light emitted by the sensor and the signals detected by the RGB channels in colorimetric quantifications. Regarding the analytical performance of the optical sensor, based on color center readings, it exhibited a linear relationship with the values obtained by conventional spectrophotometry, especially for the lower concentration ranges of the analyte (0.005 and 0.040 $mmol \cdot L^{-1}$). However, this response requires further system optimization to allow the expansion of this linear range. Two equations were derived as distinct parameters for sample reading (individualized by channel or combined), which enabled the evaluation of randomly selected irradiated samples. The channel-based analysis demonstrated that the red channel is the most sensitive to dose variations, making it the most promising for more precise calibrations. Therefore, the developed prototype represents a viable, low-cost, and portable alternative for reading gel dosimeters, with potential for application outside laboratory environments, contributing to the broader adoption of three-dimensional dosimetry in both clinical and experimental settings. Future optimizations will allow its integration into complex geometries and embedded systems for real-time monitoring.

Acknowledgements

The following Brazilian institutions support this research project: Research Support Foundation of the State of Minas Gerais (FAPEMIG), Brazilian Council for Scientific and Technological Development (CNPq), and Coordination for the Capacitation of Graduated Personnel (Capes). We also thank the Fapemig Project: Minas Gerais Translational Research Network in Oncology (RMPTO) RED00059-23.

References

Baldock, C., Harris, P. J., Piercy, A. R., & Healy, B. (2001). Experimental determination of the diffusion coefficient in two-dimensions in ferrous sulphate gels using the finite element method. Australasian *Physics & Engineering Sciences in Medicine*, 24(1), 19-30.

Brosnan, T., & Sun, D. W. (2004). Improving quality inspection of food products by computer vision a review. *Journal of food engineering*, 61(1), 3-16.

De Carvalho Oliveira, G., Machado, C. C. S., Inácio, D. K., da Silveira Petruci, J. F., & Silva, S. G. (2022). RGB color sensor for colorimetric determinations: Evaluation and quantitative analysis of colored liquid samples. *Talanta*, *241*, 123244.

De Deene, Y. (2022). Radiation Dosimetry by Use of Radiosensitive Hydrogels and Polymers: Mechanisms, State-of-the-Art and Perspective from 3D to 4D. *Gels*. 19;8 (9):599.

De Deene, Y. and Jirasek, A. (2024). Gel dosimetry: An overview of dosimetry systems and readout methods *Radiation Measurements*, 179, 2024,107321, https://doi.org/10.1016/j.radmeas.2024.107321.

Dos Santos, M. M., Mesquita, A. Z., Sebastião, R. D. C. O., Fonseca, T. C. F. (2024). Feasibility of new polymeric matrices in the production of ferrous sulphate dosimeters. *Applied Radiation and Isotopes* 214, 111526.

Du, C. J., & Sun, D. W. (2004). Recent developments in the applications of image processing techniques for food quality evaluation. *Trends in food science & technology*, 15(5), 230-249.

Gore, J. C., & Kang, Y. S. (1984). Measurement of radiation dose distributions by nuclear magnetic resonance (NMR) imaging. *Physics in Medicine & Biology*, 29(10), 1189.

Choudhury, A. K.R (2015). Visual measures of colour. Principles of colour and appearance measurement, 2, 1-25.

León, K., Mery, D., Pedreschi, F., León, J. (2006). Color measurement in L∗a∗b∗ units from RGB digital images. Food Research International, 39(10), 1084-1091, https://doi.org/10.1016/j.foodres.2006.03.006.

Levkowitz, H. and Gabor T. H. (1993). GLHS: a generalized lightness, hue, and saturation color model. *CVGIP: Graphical Models and Image Processing*. 55: 271-285

Macchione, M. A., Lechón Páez, S., Strumia, M. C., Valente, M., & Mattea, F. (2022). Chemical Overview of Gel Dosimetry Systems: A Comprehensive Review. *Gels*, *8*(10), 663. https://doi.org/10.3390/gels8100663

Moussous, O. Characterization of ferrous-agarose-xylenol gel dosimeter at 60Co γ-rays beam therapy unit. *Radiological Physics and Technology*, 14, 105–112 (2021). https://doi.org/10.1007/s12194-020-00600-4.

Nezhad, Z. A., Geraily, G. (2022) A review study on application of gel dosimeters in low energy radiation dosimetry. *Applied Radiation and Isotopes*, 179,110015.

Sagsoz, M. E., Korkut, O., Gallo, S. (2025). Advancements in Tissue-Equivalent Gel Dosimeters. *Gels*, 21:11 (2):81. doi: 10.3390/gels11020081.

Taos (2009). TCS230: Programable light-to-grequency converter. Datasheet Taos tcs3200, tcs3210. Plano: Jupter Road 205.

Dose Rate Estimation by Extracting and Analyzing Colors from an Image Generated by a CMOS Image Sensor

W. Viana[1], D. A Campolina[2], A. Z. Mesquita[3], A. C. O. Alves[4]

[1] Centro de Desenvolvimento da Tecnologia Nuclear - CDTN, 31270-901, Belo Horizonte, Brazil

[2] Centro de Desenvolvimento da Tecnologia Nuclear - CDTN, 31270-901, Belo Horizonte, Brazil

[3] Centro de Desenvolvimento da Tecnologia Nuclear - CDTN, 31270-901, Belo Horizonte, Brazil

[4] Centro Federal de Educação Tecnológica de Minas Gerais, 37022-560, Varginha, Brazil

[1] wallacy.viana@gmail.com

[2] campolina@cdtn.br

[3] amir@cdtn.br

[4] annaclara98alves@gmail.com

Abstract

This work explores the potential of using images taken by a CMOS-based camera to estimate gamma radiation dose rates. The device, originally integrated into a remotely operated vehicle (ROV) for inspection tasks, was subjected to controlled exposure at the Gamma Irradiation Laboratory (LIG) at CDTN. A series of tests were conducted under varying dose rates to examine the behavior of the image sensor. Specific pixel regions in each image were selected, and a color extraction tool was employed to determine the most prevalent colors and their respective proportions. These color patterns were compared across samples captured at different dose levels, including one image acquired during an uncontrolled phase when the radiation source was being shielded. The color data obtained were then used to generate trend lines, which served as the basis for estimating the dose rate of the unknown sample. Findings revealed that certain colors became increasingly prominent as the dose rate rose. Using mathematical models derived from these observations, it was possible to predict the dose rate for the sample taken under unknown conditions. The results were consistent with the experimental context, and the analysis confirmed the impact of radiation on the camera’s components, which eventually led to its permanent failure beyond a certain threshold. This study demonstrates that a low-cost, widely available CMOS camera can offer valuable insights into radiation levels through color-based analysis. While not a replacement for conventional dosimetry, this method suggests a novel auxiliary tool that could be refined further.

Keywords: CMOS sensors; Gamma-rays; Cameras; Dosimetry;

1 Introduction

In recent years, the use of image sensors utilizing complementary metal oxide semiconductors (CMOS) have become a new trend in radiation detection due to factors such as their low cost, general availability, and their direct response to X-rays and gamma rays. [1]

This paper presents the estimation of dose rate using a process of extracting the predominant colors of an image taken from a CMOS image sensor while being irradiated, and analyzing their percentage in relation to the image as a whole.
In a project in which the development of a remotely controlled submersible robot (ROV) was carried out, a generic camera commonly used in surveillance devices or car maneuvering assistance systems was used as means of visual inspection. Said camera had a CMOS sensor, and it was subjected to tests under different dose rates of gamma radiation at the CDTN gamma irradiation laboratory.
The Gamma Irradiation Laboratory – LIG, located on the CDTN premises is classified as a Category II Multipurpose Panoramic Irradiator, manufactured by MDS Nordion in Canada, Model/serial number IR-214 and type GB-127, equipped with a Cobalt source -60 dry stoke with maximum activity of 2,200 TBq or 60,000 Ci. [2]
It is worth mentioning that there are also applications that can be installed on smartphones in order to carry out measurements of low dose rates using the image sensors installed on them. However, the process used by these apps consists of the contrast between a static background and a count of the number of CMOS pixels that light up during irradiation. [3]

2 Methodology

Several tests were carried out under different dose rates, with the aim of carrying out resistance tests on the ROV's internal circuit which was equipped with the before mentioned camera. It is worth highlighting here the possible variation of 10% in the recorded dose.
Samples were taken from the images, specifically from a black square of 100 by 100 pixels on their corners. Image 1 shows the samples taken from the images, at three different dose rates.

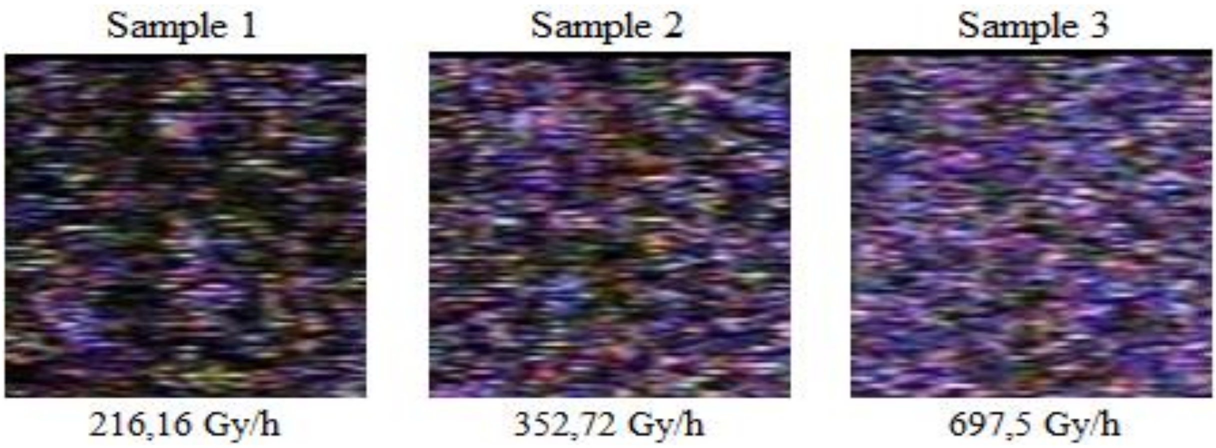


Figure 1: Three samples taken from three different images at different dose rates.

In all tests, the camera exhibited artifacts due to the passage of electrons in the circuit during irradiation, returning to the normal state as soon as irradiation stopped. A critical camera failure occurred at a total dose of approximately 250 Gy, 45 minutes after the tests began. This critical failure was repeated in all subsequent tests a few seconds after the start of exposure, indicating permanent damage to the camera's semiconductors, which prevented further data collection.
It is also worth highlighting the capture of an image in which the dose rate was unknown. Such an image was captured during the cover-up of the Cobalt-60 source, which resulted in such an unknown dose rate. This image generated sample number four.

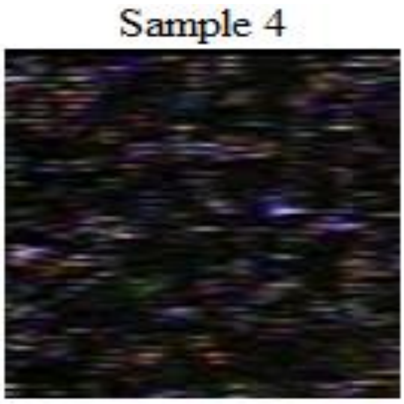


Figure 2: Sample number 4, taken from an image of unknown dose rate.

Subsequently, an online tool was used to extract the predominant colors in the image. The PHP Image Color Extract tool extracts the most common colors from an image file. Color values are displayed in hexadecimal, along with their percentage in relation to the image. It is worth noting that such color extraction can be performed with most graphic design programs. [4]
The collected data were then displayed in table I, and the control group was added in order to establish a standard.

Table 1: Percentage of each predominant color extracted from the samples.

Color (Code)	Control (%)	Sample 1 (%)	Sample 2 (%)	Sample 3 (%)	Sample 4 (%)
Black (#000000)	100	51,6	24,51	8,03	84,9
Gray (#646464)	0	25,25	38,78	47,36	9
Blue (#000064)	0	6,4	10,36	10,07	1,79
Purple (#640064)	0	6,18	11,8	15,45	1,23
Red (#640000)	0	4,14	3,4	1,17	1,54
Light gray (#c8c8c8)	0	1,71	3	3,24	0,2
Yellow (#646400)	0	1,51	1,03	0	0,7
Light blue (#6464c8)	0	1,2	3,48	9,09	0,09

Using the MiniTab® program, it was possible to create graphs and generate trend lines which in turn generated equations for each of the colors, making it possible to carry out calculations to estimate the dose rate using the percentage values of each color in sample four.

3 Results

Considering the physical integrity of the camera, the radiation darkened the glass lens in front of the camera, making it significantly difficult to view the image at the end of the tests. It is possible to compare the data regarding the radiation resistance of the camera collected through the tests with other data collected by other authors. Tests were performed using a similar camera, which showed SNR-related damage at different irradiation rates, but even at 100 Gy/h for 1 h, totaling a dose of 100 Gy, the damage was still small. [5]
This was confirmed by the tests carried out, as even with a dose of around 200 Gy, the camera still worked normally.
Carrying out the calculations using the equations generated from the trend lines mentioned above, and considering the percentages of colors in sample four, it is possible to construct a graph which demonstrates the predicted dose rate for each of the colors.

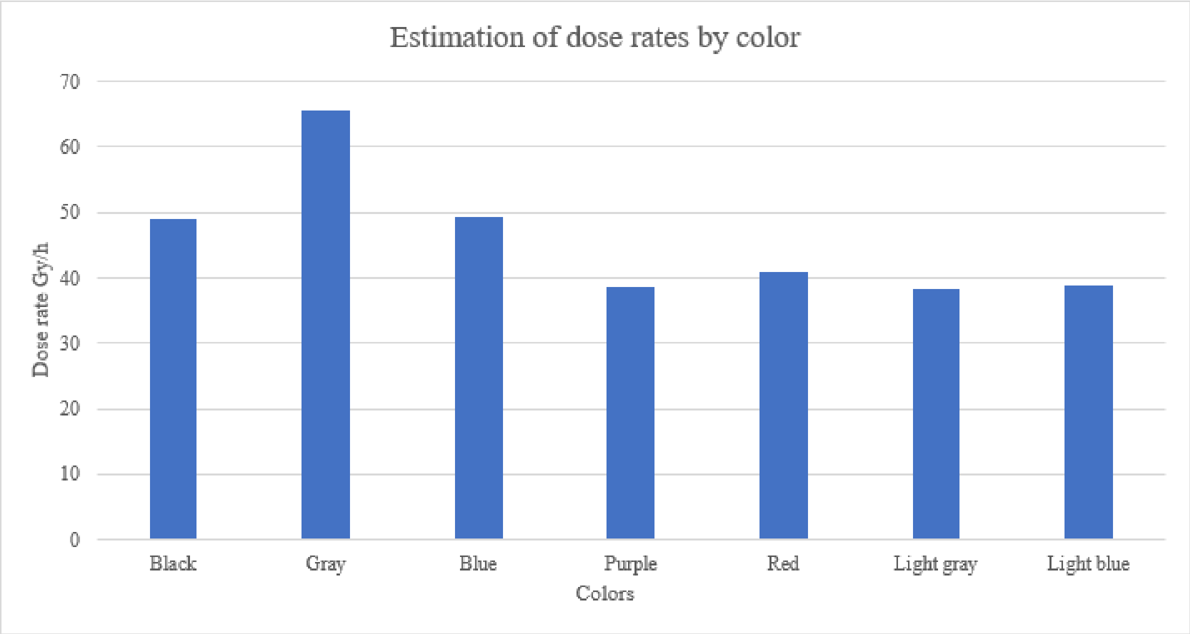


Figure 3: Graph demonstrating the predicted dose rate by each color.

Analyzing the results, it is possible that there were small discrepancies in relation to some colors, however, they all appeared between the range of 38 Gy/h and 65 Gy/h. Taking the arithmetic mean of the previously mentioned values, a value of approximately 45.84 Gy/h is obtained, adjusted for the possible variation of 10% previously mentioned.

4 Conclusions

Nuclear technology has been promoted and applied throughout the world, but as it poses a certain threat to public safety, methods of detecting nuclear radiation are necessary. Image sensors using complementary metal oxide semiconductors (CMOS) present a promising addition to radiation detection methods.

This paper analyzed a method for estimating high dose rates using a camera. To this end, image color extraction and quantization methods were used, as well as mathematical models in order to recognize patterns in color variations and estimate the dose rate in an image whose dose rate was previously unknown.

The results of the dose rate estimate were also consistent with the appearance of the image presented in sample four, as it follows mathematical models that were generated from other images, and the fact that the radiation source was being covered at the time.

The camera's irradiation tests also obtained results consistent with previously carried out tests in terms of radiation resistance.

Subsequently, more tests should be carried out with better cameras, in tests with varying dose rates and of shorter duration, in order to preserve the integrity of the camera for more comprehensive data collection and creation of more accurate mathematical models. An algorithm can also be developed in order to process and analyze this data automatically.

Acknowledgements

This research project was supported by the following institutions: Nuclear Technology Development Center (CDTN), Brazilian Nuclear Energy Commission (Cnen), Research Support Foundation of the State of Minas Gerais (Fapemig), The Gamma Irradiation Laboratory – LIG, and Brazilian Council for Scientific and Technological Development (CNPq).

References

[1] Kang, H.G. et al. (2016). Nucl. Instrum. Methods Phys. Res. A 823: 126-134.

[2] CDTN. (2024). Centro de Desenvolvimento da Tecnologia Nuclear – CDTN. Available at: https://www.gov.br/cdtn/pt-br/laboratorios/laboratorio-de-irradiacao-gama

[3] Van Hoey, O. et al. (2015). Radiat. Prot. Dosimetry 168: 352.

[4] Wang, C. et al. (2018). Sensors 18: 514.

Dosimetric comparison of methods for obtaining dose profiles in head CT scans using 120 kV X-ray beams

G. C. A. Souza[1,2]*, F. S. Santos[2], C. F. T. Portela[2], A. P. Mourão[2]

[1]Centro de Desenvolvimento da Tecnologia Nuclear, CEP: 31270-901, Belo Horizonte, MG, Brazil

[2]Universidade Federal de Minas Gerais, Departamento de Engenharia Nuclear, CEP:31270-901, Belo Horizonte, MG, Brazil.

*guilhermecavalcante45@gmail.com

Computed tomography (CT) is a radiodiagnostic modality used in various clinical and research sectors. The increased use of this equipment is explained by the technological advances they have made in improving image quality and reducing image acquisition times. The use of computed tomography results in an increase in the dose absorbed by the population. The quantification of the dose absorbed by the patient is defined as CTDI, computed tomography dose index. The standard dosimetric measurement is carried out using a pencil-type ionization chamber and a standard cylindrical head simulator made of polymethylmethacrylate (PMMA). Alternatively, radiochromic film can be used to replace the ionization chamber. This dose measurement method uses a radiochromic film, a standard headpiece simulator, and a complementary PMMA cylinder. The aim of this work is to compare the volumetric CTDI values ($CTDI_{vol}$) obtained using the two forms of measurement. In this study, a GE LightSpeed VCT 64-channel CT scanner was used for the irradiations. For the standard methodology, a RADCAL ACCU-GOLD pencil-type ionization chamber and a standard cylindrical head simulator 16 cm in diameter and 15 cm long, with five openings for positioning the chamber, four peripheral and one central, defined as C, were used. The irradiations were carried out with the head simulator positioned at the isocenter of the CT gantry, and the ionization chamber was positioned alternately in each of the openings. A central 10 mm slice of the phantom was irradiated. Five irradiations were carried out for each positioning of the chamber, for a total of twenty-five, and the $CTDI_{vol}$ values were calculated from these measurements. To irradiate the GAFCHROMIC® XR-QA2 radiochromic film, a 16 cm circular sheet was used, which was positioned between the head phantom and the PMMA cylinder with a 16 cm diameter and 7.5 cm in length. Digital images of the film were generated before and after irradiation, using an HP Photosmart C4480 scanner, for subsequent evaluation of the dose values. The IMAGE J software was used to obtain and evaluate the dose profile. The values obtained in grayscale intensity were converted into absorbed dose (mGy) using a calibration curve. A 10 cm scan, with pitch equal to 1, was carried out with the film positioned in the central slice. As well as being able to observe the dose variation profile in the central slice, it was possible to obtain the dose values recorded in the five aperture positions. The $CTDI_{vol}$ value was calculated from the $CTDI_{100}$ values recorded on the films. Two experiments were carried out, with a voltage of 120 kV and loads of 150 mA.s. The $CTDI_{vol}$ values obtained were equivalent for both protocols, considering the two dose measurement methods. The use of radiochromic film enabled rapid data acquisition through a single irradiation, whereas the ionization chamber required at least twenty-five exposures. However, unlike the ionization chamber, the radiochromic film cannot be reused for new measurements.

Keywords: Computed Tomography; Dosimetry; Human phantom; Radiochromic Film.

1 Introduction

Computed tomography (CT) is a radiodiagnostic method widely used in various clinical specialties, such as cardiology, oncology, neurology, traumatology and medical research (Mourão, 2008; Mourão, 2018). CT images are generated from the difference in absorption of the X-ray beam depending on the characteristics of the tissues (Mourão, 2018**;** Goo, 2012). CT equipment has made significant advances, such as improving reconstruction algorithms and software, reducing image acquisition times, among others. As a result, CT machines have proved to be essential for radiodiagnostic and have led to an increase in the number of examinations, leading to a greater radiation dose deposition in the population (Mourão, 2018; Dance, 2014). CT dose measurement is defined as the Computed Tomography Dose Index (CTDI) and can be obtained as a point value ($CDTI_{100}$)**,** weighted from $CTDI_{100}$ measurements ($CTDI_w$) or as a volumetric value ($CTDI_{vol}$) (AAPM, 2008; AAPM, 2011; Tack, 2007).
The CTDI calculation can be used as a tool to carry out protocol optimization tests with the aim of reducing dose deposition in patients. The $CTDI_w$ and $CTDI_{vol}$ values must be indicated in order to comply with Normative Instruction 93 (IN93) established by the Collegiate Board of the National Health Surveillance Agency. The CTDI can be calculated in a standard way using a pencil-type ionization chamber and a standard head phantom (Tack, 2007; IAEA, 2007; Devic, 2011). Alternatively, dosimetric measurements can be carried out using radiochromic film and thermoluminescent dosimeters (Devic, 2016; Brady, 2010; Oliveira, 2013; Dixon, 2010). The aim of this study was to measure the point and volumetric dose index using the two dosimetric measurement methods, with the pencil-type ionization chamber and the radiochromic film, as well as obtaining the dose profile for the X and Y axes, for a 120 kV and 150 mA.s X-ray beam.

2 Methodology

In this work was used tomograph GE LightSpeed VCT model 64 channels. Dosimetric measurements were carried out using a standard head phantom coupled to a pencil-type ionization chamber and radiochromic film. The standard head phantom is made of polymethylmethacrylate (PMMA), cylindrical shape with a diameter of 16 cm and length of 15 cm, contains five openings, four peripheral and one central, for positioning dosimeters, such as the pencil-type ionization chamber, radiochromic films or TLD. These peripheral oppenings are identified with a clock, 3, 6, 9 and 12. Figure 1 shows an axial section of a computed tomography image of the standard head phantom, with the numbers arranged analogously to a clock.

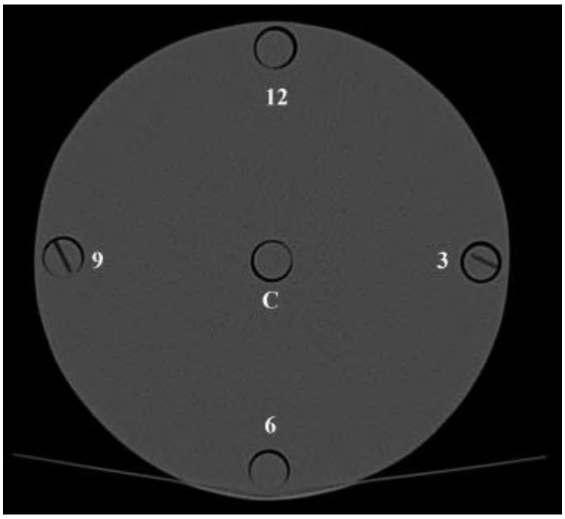


Figure 1: Illustration of head phantom and complementary cylinder.
Source: Santos, 2023.

This phantom is considered the standard for dosimetric reference in head CT scan (Dixon, 2010). A complementary cylinder with a length of 7.5 cm and a diameter of 16 cm was used to enlarge the head phantom. Figure 2 shows an illustration with the dimensions of the standard head phantom, the filling billets and the complementary cylinder.

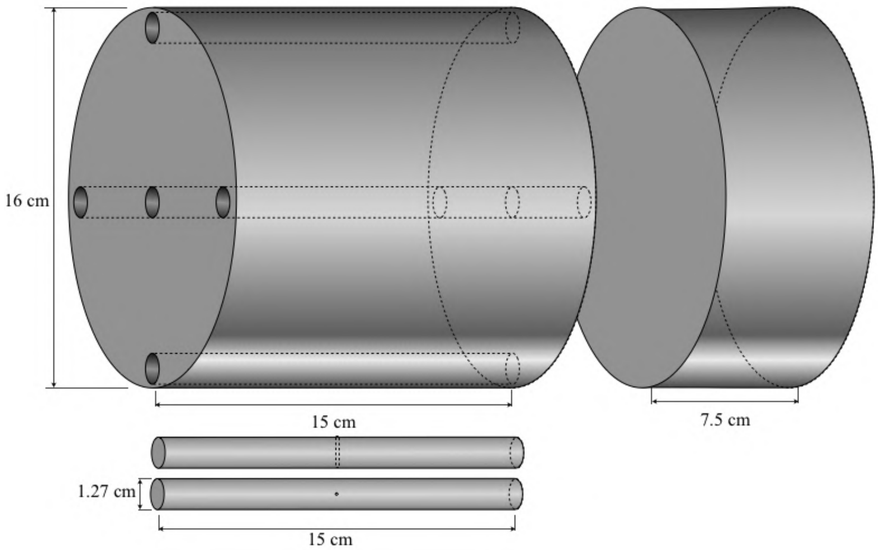


Figure 2: Illustration of head phantom and complementary cylinder.

A pencil-type ionization chamber, a RADCAL ACCU-GOLD model 10X6-3CT, was used to fill the openings of the head phantom for measuring the point dose index. The ionization chamber was designed to measure the X-ray beam in CT, in open air or inside a body or head phantom. The pencil-type ionization chamber is 15 cm long, with a useful length of 10 cm, cylindrical and has a dose detection range between 20 nGy and 1 kGy with a calibration accuracy of ±4% for X-rays generated at up to 150 kV. Figure 3 shows pencil-type ionization chamber (Radical, 2022).

Figure 3: Ionization chamber
Source: Santos, 2023.

The standard head phantom was placed at the isocenter of the gantry and the ionization chamber was positioned inside the openings, making it unnecessary to use the complementary phantom. Five scans, with a 1 cm longitudinal scan in helical mode, were carried out for each opening, totaling twenty-five scans. The ionization chamber scan records values for the Kerma quantity in air, but when positioned inside the phantom, they are Kerma values in PMMA (Ck_{100}). Using the average beam energy value and the mass absorption coefficient tables, the conversion factors from Ck_{100} to $CTDI_{100}$ were obtained. $CTDI_{vol}$ values depend on the weighted dose index ($CTDI_w$), as shown in Equation 1. $CTDI_{vol}$ values were obtained using $CTDI_w$, as shown in Equation 2.

$$CTDI_w = \frac{1}{3} CDTI100, c + \frac{2}{3} CTDI100, p \quad (1)$$

where:

$CTDI_{100}$, c: $CTDI_{100}$ measured in the central position of the head phantom;
$CTDI_{100}$, p: CTD_{I100} measured in the peripheral position of the head phantom;

$$CTDIvol = \frac{CTDIw}{pitch} \quad (2)$$

Radiochromic film XR-QA2 GAFCHROMIC® model, manufactured by ASHLAND, is specifically for diagnostic radiology with applications in dosimetry and control and can be used as a substitute for the ionization chamber to obtain dose values in computed tomography. This film has high sensitivity to ionizing radiation with doses in the range of 1.0 to 200 mGy and can be used in X-ray beams generated by voltages of 20 to 200 kV (Ashland, 2023). In addition, the films are easy to handle and make it possible to carry out qualitative analyses before and after irradiation processes, as they undergo a color change directly proportional to the dose of radiation deposited (Gotanda, 2007; Goo, 2012; Fernandes, 2018). Figure 4 show two radiochromic film strips, before irradiation (a) and after irradiation (b).

Figure 4: Scanned images of the radiochromic film before (a) and after irradiation (b).

The radiochromic film was cut into a circular shape with a diameter of 16 cm and placed between the standard head phantom and the complementary cylinder (Figure 5). The set was then positioned in the center of the equipment gantry for a subsequent 10 cm helical scan (Figure 6).

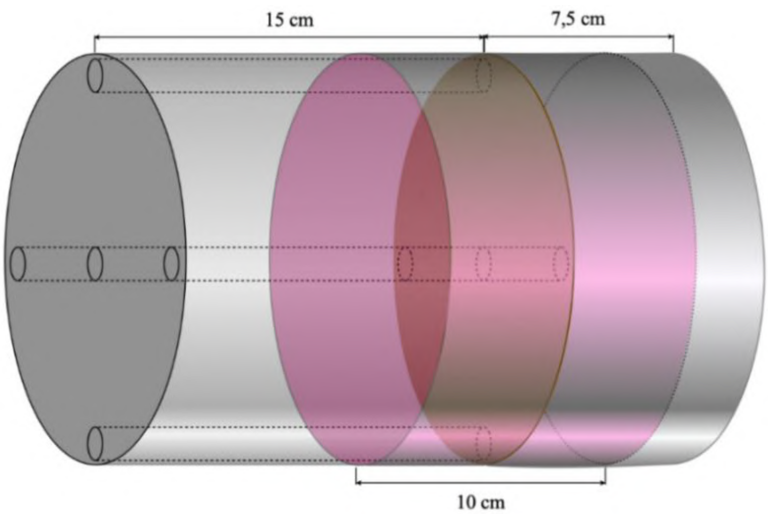


Figure 5: Radiographic film placed between the phantoms.

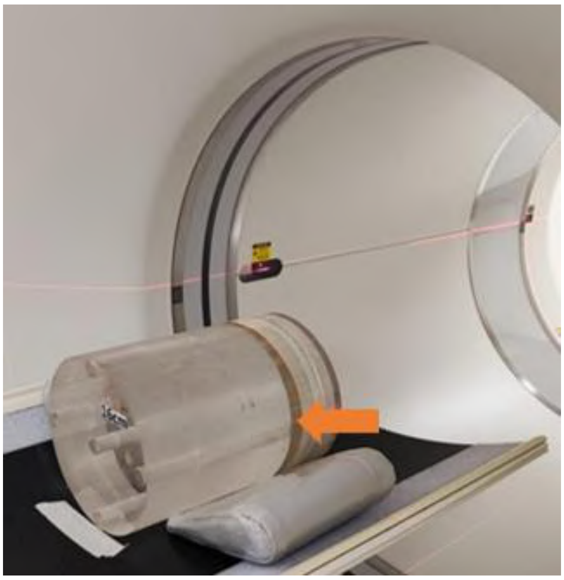


Figure 6: Circular sheet of radiochromic film, indicated by the arrow, positioned between head phantom and complementary cylinder.

Metrological reliability of the radiochromic films was demonstrated through homogeneity and repeatability tests and by calibrating it in a reference radiation for CT that were reproduced in the Calibration Laboratory of the Development Center of Nuclear Technology (CDTN). Digital images in extension .tiff and 200 dpi of the film sheet was obtained in a HP Photosmart C4480 reflective type scanner, before and after its irradiation. These images were worked using the software Image J. The red channel of the color image has a main absorption peak in the red region of the visible spectrum (636nm) and was used to observe the variations in grey scale recorded in the film (Figure 6A) (Stevens, 1996; Jackson, 2013). Dose variation profile were obtained for horizontal and vertical axis (X and Y) of the central slice (Figure 6B).

The X-axis starts at openings 3, central and 9 of the head phantom, whose centers of the openings are located at positions 1, 8 and 15 cm, respectively. The Y axis starts at openings 6, central and 12 of the head phantom, whose centers of the openings are located at positions 1, 8 and 15 cm, respectively.

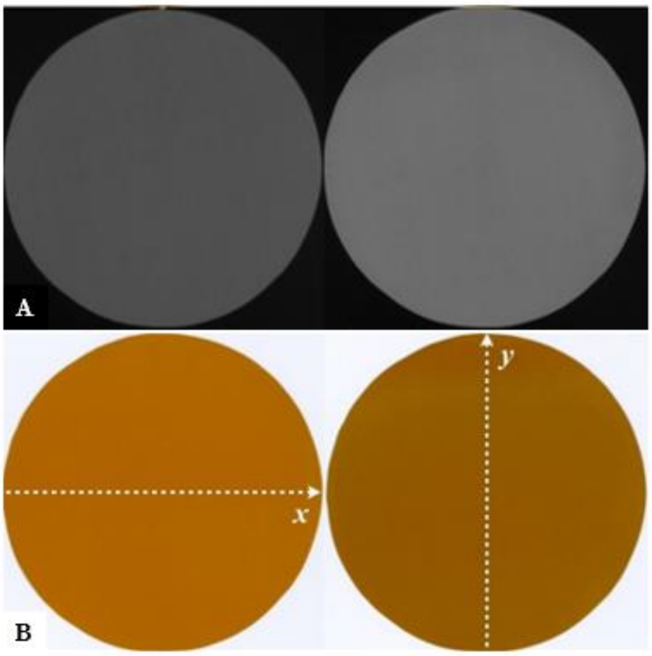


Figure 6: Radiochromic film images: (a) red channel and (b) before and after CT scanning.

The radiochromic film records the absorbed dose in MSAD (multiple scan average dose). MSAD can be correlated with $CTDI_{vol}$ according to Equation 3. MSAD and $CTDI_{vol}$ are equivalent because the pitch used was 0.985.

$$MSAD = \frac{CTDIvol}{pitch} \tag{3}$$

The film sheet records the MSAD (multiple scan average dose), which is correlated with the $CTDI_{vol}$ according to Equation 3. The MSAD and $CTDI_{vol}$ values are equivalent because the pitch used in the protocol was 0,985. Intensity is a parameter used to measure the darkening of the radiochromic film. The color intensity of the radiochromic film was obtained after irradiation and converted into absorbed dose using the conversion factor obtained from the absorbed dose values measured by the ionization chamber at each of the five points of the standard phantom. Table 1 shows the parameters used in the experiment protocol.

Table 1: Protocol of the CT scanner.

Voltage (kV)	**Charge (mA.s)**	**Tube time (s)**	**Pitch**
120	150	1	0,985

3 Results and discussion

3.1 $CTDI_{100}$ and $CTDI_{vol}$

Tables 2 shows the average values and standard deviation of the absorbed dose in $CTDI_{100}$ and CTDIvol for measurements with the pencil-type ionization chamber and radiochromic film.

Table 2: Index of absorbed dose $CTDI_{100}$ and $CTDI_{vol}$ using chamber ionization and radiochromic film.

Ionization Chamber			**Radiochromic Film**	
Position	**Absorbed Dose (mGy)**	**SD**	**Absorbed Dose (mGy**	**SD**
3	30.92	0.03	32.88	0.29
6	28.84	0.14	21.00	0.40
9	30.22	0.06	31.21	0.50
12	33.96	0.60	35.87	0.33
C	27.81	0.05	29.22	0.20
$CTDI_{vol}$	29.93	0.15	29.90	0.32

Irradiation of the standard head phantom with a pencil-type ionization chamber showed the highest record at position 12, with 33.96±0.60 mGy. The lowest record was in the central position, with 27.81±0.05 mGy, due to the interaction of the X-rays with the PMMA head phantom, followed by position 6, with 28.84±0.14 mGy, due to the interaction of the X-rays with the examination bed. Radiochromic film irradiation showed the highest record at position 12, with 35.87 ± 0.33 mGy. The lowest record was at position 6, with 21.00 ± 0.40 mGy, due to the interaction of the X-rays with the examination bed. Analysis of the $CTDI_{vol}$ between the two dose measurement methods showed equivalent absorbed dose values with a variation of 0.03 mGy.

3.2 Axial dose profile

Figures 7 and 8 show the dose line profile for the X-axis and Y-axis, respectively.

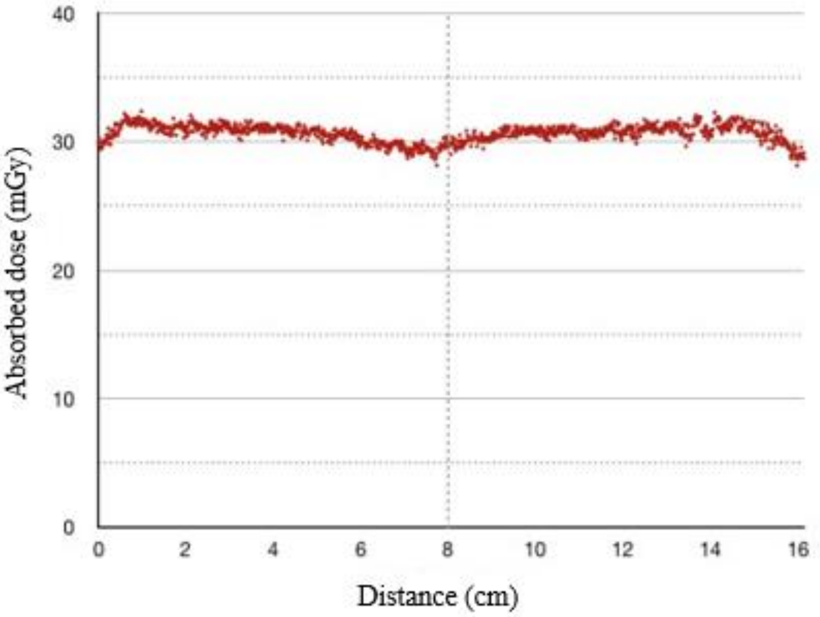


Figure 7: Dose line profile for the X axis.

The X-axis passes through openings 3, central, and 9 of the simulator object, whose opening centers are located at positions 1, 8, and 15 cm, respectively. The maximum absorbed dose recorded on this curve was 32.36 mGy, the minimum dose was 28.18 mGy, and the average dose was 30.65±0.68 mGy. The curve showed a decrease in values in the central region. The X-axis recorded approximate absorbed dose values at positions 3 and 9. This dose behavior can be explained by the correct positioning of the simulator set in relation to the gantry isocenter. The lowest dose value recorded in the central position can be explained by the fact that at this point the X-ray beam is always filtered by 8 cm of PMMA, regardless of the direction of propagation of the X-ray beam.

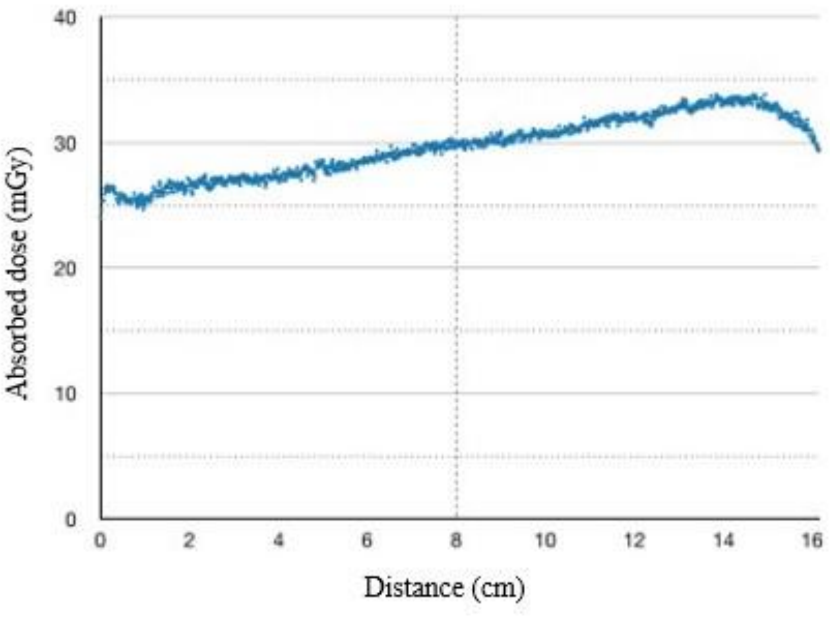


Figure 8: Dose line profile for the X axis

The Y axis passes through openings 6, central, and 12 of the simulator object, whose opening centers are located at positions 1, 8, and 15 cm, respectively. The maximum absorbed dose recorded on this curve was 33.81 mGy, the minimum absorbed dose was 24.04 mGy, and the average dose was 29.63±2.43 mGy. The curve showed an increase in dose values from its beginning. The Y-axis recorded a lower dose value near point 6, due to the fact that at this point, the portion that contributes most to the absorbed dose is from the beam coming from the floor to the ceiling, and this beam is filtered by the table, causing its intensity to be lower. The highest absorbed dose value occurs in the region of opening 12. This is the position in which the beam contribution has the shortest distance between the radiation source and the region of opening 12, in addition to not undergoing external filtration such as the examination table.

4 Conclusions

The absorbed dose index, $CTDI_{100}$ and $CTDI_{vol}$, was measured using two different methods, pencil type ionization chamber and radiochromic film. The absorbed dose values obtained were equivalent. However, the radiochromic film allows measurements to be made more quickly with just one scan, while the ionization chamber uses twenty-five irradiations. In addition, radiochromic films make it possible to obtain variations in the absorbed dose on the X and Y axes. Radiochromic films have proven to be efficient dosimetric tools and can be used to test protocols for reducing doses to patients.

Acknowledgements

The authors would like to acknowledge the financial support of the Minas Gerais State Research and Development Agency (FAPEMIG) and the Coordination for the Improvement of Higher Education Personnel - Brazil (CAPES).

References

AAPM. Report no. 96: The Measurement, Reporting, and Management of Radiation Dose in CT. Colege Park: *American Association of Physicists in Medicine*, 2008. 28 p.

AAPM. Size-specific dose estimates (ssde) in pediatric and adult body ct examinations. *American Association of Physicists in Medice*, v. 204, p. 1–30, 2011.

Ashland. GAFCHROMIC XR-QA2. Wilmington, Delaware: [s.n.], 2023. Disponível em: Acesso em: 27 apr. 2023.

Brady, S. et al. Implementation of radiochromic film dosimetry protocol for volumetric dose assessments to various organs during diagnostic ct procedures. Medical physics, Wiley Online Library, v. 37, n. 9, p. 4782–4792, 2010.

Dance, D. et al. Diagnostic radiology physics: A handbook for teachers and students. endorsed by: American association of physicists in medicine, asia-oceania federation of organizations for medical physics, european federation of organisations for medical physics. 2014.

Devic, S. Radiochromic film dosimetry: past, present, and future. Physica medica, Elsevier, v. 27, n. 3, p. 122–134, 2011.

Devic, S.; Tomic, N.; Lewis, D. Reference radiochromic film dosimetry: review of technical aspects. *Physica Medica*, Elsevier, v. 32, n. 4, p. 541–556, 2016.

Dixon, Robert L. et al. Comprehensive methodology for the evaluation of radiation dose in x-ray computed tomography. 2010.

Fernandes, L. C. Estudos dosimétricos das varreduras de cabeça em tomografia computadorizada: avaliação da qualidade de imagem visando à otimização de protocolos. Universidade Federal de Minas Gerais, 2018.

Gomez, M.; Mourão, A. P. Radiochromic film calibration for dosimetry in chest ct scans. *Brazilian Journal of Radiation Sciences*, v. 9, n. 1A, 2021.

Goo, H. W. Ct radiation dose optimization and estimation: an update for radiologists. Korean journal of radiology, *The Korean Society of Radiology*, v. 13, n. 1, p. 1–11, 2012.

Gotanda, R. et al. Computed tomography phantom for radiochromic film dosimetry. *Australasian Physics & Engineering Sciences in Medicine*, Springer, v. 30, p. 194–199, 2007.

IAEA. Dosimetry in Diagnostic Radiology: An International Code of Practice Publication 457. [S.l.]: *International Atomic Energy Agency*, 2007.

Jackson, S. R., Ahmad, S., Hu, Y., & Ruan, C. (2013). Evaluation of different techniques for CT radiation profile width measurement. *Journal of Applied Clinical Medical Physics*, 14(4), 227-237.

Mourão, A. P.; Oliveira, F. A. de. Fundamentos de radiologia e imagem. [S.l.]: Difusão Editora, 2009.

Mourão, A. P. Tomografia computadorizada: tecnologias e aplicações. 2. ed. São Caetano do Sul: Difusão Editora, 2018.

Oliveira, B. B.; Mourão, A. P.; Silva, T. A. D. Estimativa dos valores de msad em procedimentos de tomografia computadorizada utilizando filmes radiocrômicos. *Radiologia Brasileira*, SciELO Brasil, v. 46, p. 96–100, 2013.

Radical. The Chamber for Computed Tomography Dose Index (CTDI). California, USA: [s.n.], 2023. Disponível em: Acesso em: 09 maio 2022.

Stevens, M. A., Turner, J. R., Hugtenburg, R. P., & Butler, P. H. (1996). High-resolution dosimetry using radiochromic film and a document scanner. *Physics in Medicine & Biology*, 41(11), 2357.

Tack, D.; Gevenois, P. A.; Abada, H. Radiation dose from adult and pediatric multidetector computed tomography. New York: Springer, 2007.

Estimation of the soil-to-plant transfer factor of natural radionuclides in organic crop from Pedra Branca State Park

N. R. dos Santos[1]*, A. M. Domingues[1], I. S. de Souza[1], L. A. Neves[1], C. O. P. Orejuela[1], A. X. da Silva[1], E. M. de Souza[2], I. C. B. Lima[1]

[1] Federal University of Rio de Janeiro, Nuclear Engineering Program, 21941-914, Rio de Janeiro, Brazil

[2] State University of Rio de Janeiro, Laboratory of Reduced Models, 23070-200, Rio de Janeiro, Brazil

* nadia.rodrigues@coppe.ufrj.br

Abstract

Humans incorporate the natural radionuclides present in the Earth's crust through the consumption of food and water. This contributes the largest share of the internal radiation dose. The transfer of natural radionuclides from soil to plants, and subsequently to the food chain, is important for estimating ingestion doses. The transfer factor (TF) is a parameter used to predict the presence of contaminants in agricultural systems Therefore, given the global increase in the production and consumption of organic food samples of soil and vegetables were collected from a location within the Pedra Branca State Park (PEPB, Brazil), a Conservation Unit in the municipality of Rio de Janeiro, and were analyzed using gamma spectrometry. The activity concentrations of the radionuclides ^{40}K, ^{228}Ra, ^{226}Ra, and ^{228}Th were determined using a High Purity Germanium (HPGe) detector to estimate the soil-plant transfer factor. The range of activity concentration values of ^{40}K, ^{228}Ra, ^{226}Ra, and ^{228}Th identified in vegetables was 134.08 ± 19.38 to 1984.75 ± 271.72 (Bq/kg); 3.15 ± 1.28 to 32.75 ± 13.60 (Bq/kg), 1.67 ± 0.78 to 5.47 ± 1.44 (Bq/kg), and 1.59 ± 0.92 to 9.86 ± 5.07 (Bq/kg), respectively. The range of TFs for ^{40}K, ^{228}Ra, ^{226}Ra, and ^{228}Th were (0.16 ± 0.045 – 2.36 ±0.46); 0.01 ± 0.021 – 0.33 ± 0.28; 0.03 ± 0.02 – 0.40 ± 0.13, and 0.0047 ± 0,005 – 0.072 ± 0.073, respectively. Despite the transfer factor varying greatly, it can still be used as comparative data for future investigations in the same area.

Keywords: Gamma spectrometry; Soil-to-plant transfer factor; Activity concentration.

1 Introduction

There are two main sources of exposure to natural ionizing radiation: high-energy cosmic rays interacting with the Earth's atmosphere, and radionuclides from the Earth's crust, which are distributed throughout the environment. Various routes are taken by these natural radionuclides as they pass through water sources, plants, and animals until they are ingested by humans (Cinelli *et al.*, 2019). The main contributors to ingestion doses are ^{40}K and the decay products of ^{238}U and ^{232}Th present in food (UNSCEAR 2000). The study of natural radioactivity generally aims to gather information about current levels of harmful pollutants being released into the environment (Al-Kharouf *et al.*, 2008). The migration and accumulation of contaminants in arable soils is a complex process involving leaching, capillary rise, surface runoff, sorption, root absorption and resuspension in the atmosphere. Other factors that influence the migration of radionuclides in soil and their absorption by plants include the radionuclides' physicochemical properties, soil characteristics, species, clay content, cation exchange capacity, and soil pH (Barnett *et al.*, 2009; Tome *et al.*,2003). The transfer factor is a parameter used to determine the ability of various plant species to absorb radionuclides from the soil or another substrate. It corresponds to the ratio of the activity concentration of a given radionuclide in plant tissue to the concentration in the soil or another substrate. TF is used to estimate the transport of radionuclides and other elements through the food chain, and also serves as an indicator of potential contamination in agricultural systems (Oliveira, 2012; Makki, 2024; Mortvedt, 1994). Even though there are many variables involved in the migration of radionuclides from soil to plant, it is important and necessary to study and obtain data showing the concentrations of naturally radioactive elements that will be ingested. This is especially important for monitoring levels of natural radioactivity, since the soil and plant materials analyzed in this research are from an environmentally protected area.

In addition to the above, this study considered that global consumption of organic food has intensified, and that there will be a substantial expansion of organic farmland in 2022, with an increase seen on all continents. (Willer *et al*., 2024; Imani *et al*., 2025) In Brazil, the production and consumption of organic products has increased and is spreading to different regions of the country (Tandon *et al*, 2021; Farias et al, 2022). Given this, in this work, the transfer factor from soil to organic vegetables for ^{40}K, ^{228}Ra, ^{226}Ra, and ^{228}Th was estimated.

2 Materials and Methods

2.1 Sample collection site

Organic food from the PEPB, located in the municipality of Rio de Janeiro, was analyzed. The Park is a Full Protection Conservation Unit, created by State Law (Rio de Janeiro, 1974), and also a place for ecological tourism (Dias, 2018). Despite it is a conservation unit, it is home to communities of family farmers who cultivate on sites belonging to a huge area of 12,495 hectares. Due to its size and accessibility, one of these sites was selected for plant and soil collection.

2.2 Vegetables collection and preparation

Organic vegetables from different food groups were collected: pineapple, lady's finger, banana, beet, broccoli, persimmon, carrot, kale, bahia orange, Persian lime orange, Galician lemon, Tahiti lemon, papaya, turnip, and radicchio.

The vegetables were hand-picked and washed to remove any traces of soil, insects, or other impurities. Depending on the characteristics of each food item, the edible parts were then either sliced or juiced. In order to check the rate of radionuclide transfer from soil to vegetables, the samples were dried in a conventional microwave oven (voltage 100V-120V, power 1300W). The drying process was stopped once the mass had stabilized. After drying, using a food processor and a mixer, the solid samples were ground, sieved, and stored in acrylic jars for a period of at least 45 days (Lopes, 2018) in order to establish the secular equilibrium.

Food is typically dried in an oven (Coelho, 2022; Hassan, 2021; Oladele, 2023; Eliud, 2023; Raj et al., 2024). In this study, we chose to reduce the moisture content using a conventional microwave oven. Pastorini et al., De Souza, Nogueira and Rassini (2002) have shown the advantages of using microwaves to dry food in terms of practicality and time optimization.

2.3 Soil collection and preparation

According to the International Union of Radioecology (IUR, 2023), the depth of the soil layer considered in the calculation is crucial for radionuclides deposited in the soil with an inhomogeneous depth distribution. To minimize the impact of this variable, it is recommended that the location of the roots in the soil be standardized, assuming that all radionuclides in the rooting zone are within 0–10 cm for grass and 0–20 cm for other crops, including trees (Guillén, 2018). For statistical analysis, it is assumed that the radionuclides are homogeneously mixed in a layer extending from the surface to a certain depth. For all crops except grass, this depth is assumed to be 20 cm (Frisssel, 1988).

Based on these observations, three random soil samples were taken from the same location as the plant at a depth of 20 cm (Al-Oudat et al., 2021), totaling approximately 1 kg, to prepare a representative sample. Any roots, stones, gravel or other unwanted materials were removed. The soil was then dried in an oven at 105 °C until it reached a constant weight. After grinding and sieving where necessary, the samples were stored in polypropylene containers for 45 days of analysis.

2.4 Gamma-ray spectrometry

The gamma-ray spectra were obtained using a Canberra vertical HPGe detector, model GC3020, a germanium cylinder with an n-type contact on the outer surface and a p-type contact on the axial surface, with a relative efficiency of 30% and an energy resolution of 1816 keV for the 1332.5 keV peak of 60Co. The coaxial hyperpure germanium crystal is 62 mm in diameter and 40 mm in height. This data acquisition system also has a Canberra Model 747 E low background radiation lead shield with a thickness of 10.16 cm, internally plated with 1 mm of tin covered with 1.6 mm of copper. This internal part has a cavity with a diameter of 27.9 cm and a depth of 27.5 cm. Externally, the shield has a 9.5 mm coating of low carbon steel. An RC type preamplifier, model 2002 C, coupled to the detector in the cryostat, model 7500 SL, and a 30 L dewar. The multichannel system used was a DSA 1000 (Digital Spectrum Analyzer) with 8192 channels.

The energy calibration of the detection system was carried out with sources with gamma emissions that coincide with or are close to the areas of interest in the spectrum. The certified standards used were ^{137}Cs, ^{60}Co, ^{22}Na, and ^{40}K, totaling 5 experimental points of energy 661.6 keV, 1173 keV, 1332.5 keV, 1274 keV, and 1460 keV. The emissions tracked in this study are from the nuclides ^{212}Pb, ^{214}Bi, ^{228}Ac, and 40K, with gamma emissions at energies of 238.6 keV, 609 keV, 911 keV, and 1460 keV, respectively. The Genie 2000 software was used to simulate the efficiencies according to the energies and to make corrections related to self-attenuation. The time to obtain the area under the photopic interest of samples and background was 28,000s (Lopes, 2018).

2.5 Activity concentration

The radionuclide activity concentration was calculated according to Equation 1 (IAEA, 2003).

$$\mathrm{AC} = \frac{\mathrm{N_L}}{\varepsilon.\mathrm{P}_\gamma.\mathrm{t.m}} \tag{1}$$

Where AC is the specific activity concentration of the radionuclide (Bq/Kg); NL is the net area under the photopeak of interest; ε is the detection efficiency of the system at the energy of interest; Pγ is the transition probability of the measured gamma ray (γ); t is the counting time (s); and m is the mass of the sample (kg).

Equation 2 (Curie, 1968) was used to calculate the minimum detectable activity (MDA) per unit mass (Bq/Kg).

$$\mathrm{MDA} = \frac{k^2 \pm \sqrt{8}\mathrm{k}\sigma}{\varepsilon.\mathrm{P}_\gamma.\mathrm{t.m}} \tag{2}$$

Where k is the percentage of measurements outside a defined interval from the mean value [defined as 5%, k= 1.96]; σ is the standard deviation of the background radiation from the shielding plus the sample holder measured over a time, ε is the energy detection efficiency; $P\gamma$ is the probability of emitting a certain energy; t is the measurement time (s), and m is the sample mass(kg).

2.6 Soil to plant crop TF

The soil to plant crop TF was calculated using the following Equation 3 (Rout, Yadav, Pulhani, 2021).

$$\mathrm{TF} = \frac{A_R^P}{A_R^S} \tag{3}$$

Where A_R^P is the activity concentration (Bq/Kg dry weight) of “radionuclide R” in the plant and A_R^S is the activity concentration (Bq/Kg dry weight) of “radionuclide R” in the soil.

3 Results and discussion

The activity concentrations of ^{40}K, ^{228}Ra, ^{226}Ra, and ^{228}Th in organic plants and soils collected at a particular site in PEPB, Rio de Janeiro, were calculated by gamma spectrometry to estimate the transfer factor of these radionuclides from soil to plant.

Tables 1, 2, 3, and 4 present the activity concentrations of ^{40}K, ^{228}Ra, ^{226}Ra, and ^{228}Th in the analyzed vegetables and soils, along with the corresponding soil-to-plant transfer factors.

Table 1: Activity concentration of ^{40}K in the samples and the soil-to-plant transfer factor.

Sample	**Activity concentration in plant (Bq/Kg)**	**Activity concentration in Soil (Bq/Kg)**	**Soil to plant TF**
Pineapple	265.54 ±38.64	863.91 ±117.86	0.31 ±0.06
Banana	441.71 ±61.74	797.41 ±108.83	0.55 ±0.11
Beet	628.62 ±87.26	664.28 ±90.78	0.95 ±0.18
Broccoli	765.79 ±106.09	815.01 ±111.23	0.94 ±0.18
Persimmon	257.92 ±36.84	663.42 ±90.70	0.39 ±0.08
Carrot	546.82 ±76.15	638.55 ±87.10	0.86 ±0.17
Kale	792.97 ±109.59	854.08 ±116.52	0.93 ±0.18
Bahia orange	319.28 ±44.39	838.53 ±114.44	0.38 ±0.07
Persian lime	261.18 ±36.44	736.46 ±100.58	0.35±0.07
Galician lemon	282.31 ±39.41	1112.98 ±151.82	0.25 ±0.05
Tahiti lemon	134.08 ±19.38	823.29 ±112.39	0.16 ±0.03
Papaya	462.86 ±64.67	846.74 ±115.53	0.55 ±0.11

Turnip	664.08 ±92.07	585.93 ±80.10	1.13 ±0.22
Radicchio	1984.75 ±271.72	839.55 ±114.57	2.36 ±0.46

The lowest ^{40}K activity concentration value for vegetables was found in Tahiti lemon (134.08 ± 19.38) Bq/kg, and the highest was found in Radicchio (1984.75 ± 271.72) Bq/kg. The lowest ^{40}K activity concentration value for soils was found for Turnip cultivation (585.93 ± 80.10) Bq/kg, and the highest was found for Galician lemon cultivation (1112.98 ± 151.82) Bq/kg. For TF, the lowest and highest values were (0.16 ±0.03) in Tahiti lemon cultivation and (2.36 ±0.46) in radicchio, respectively. The TFs corresponding to the turnip and radicchio plantations indicate a higher activity concentration of ^{40}K in the plant than in the soil. In the case of turnips, one possible explanation is that they grow underground, putting them in direct contact with radionuclides and enabling them to absorb them from the soil.

Table 2: Activity concentration of ^{228}Ra in the samples and the soil-to-plant transfer factor.

Sample	Activity concentration in plant (Bq/Kg)	Activity concentration in Soil (Bq/Kg)	Soil to plant TF
Pineapple	<MDA	114.43 ±46.17	-
Banana	<MDA	69.38 ±28.02	-
Beet	16.79 ±7.37	162.71 ±65.63	0.10 ±0.06
Broccoli	19.72 ±8.27	74.37 ±30.02	0.26 ±0.15
Persimmon	5.51 ±3.37	82.90 ±33.50	0.07 ±0.05
Carrot	32.75 ±13.60	164.54 ±66.49	0.20 ±0.12
Kale	12.21 ±6.12	74.09 ±29.91	0.16 ±0.11
Bahia orange	6.34 ±2.85	84.21 ±34.02	0.08 ±0.05
Persian lime	3.15 ±1.28	78.98 ±31.89	0.04±0.02
Galician lemon	4.59 ±2.58	40.30 ±16.73	0.11±0.08
Tahiti lemon	5.77 ±2.70	74.18 ±29.99	0.078 ±0.05
Papaya	3.96 ±2.31	72.99 ±29.47	0.05 ±0.04
Turnip	12.95 ±5.86	39.21 ±15.90	0.33 ±0.20
Radicchio	14.32 ±6.30	76.05 ±30.72	0.19 ±0.11

The lowest ^{228}Ra activity concentration value for vegetables was found in Persian lime orange (3.15 ±1.28) Bq/kg, and the highest was found in Carrot (32.75 ±13.60) Bq/kg. The lowest ^{228}Ra activity concentration value for soils was found for Galician lemon cultivation (40.30 ±16.73) Bq/kg, and the highest was found for Beet cultivation (162.71 ± 65.63) Bq/kg. For TF, the lowest and highest values were (0.04 ±0.02) in Persian lime cultivation and (0.33 ±0.20) in turnip, respectively.

Table 3: Activity concentration of ^{226}Ra in the samples and the soil-to-plant transfer factor.

Sample	Activity concentration in plant (Bq/Kg)	Activity concentration in Soil (Bq/Kg)	Soil to plant TF
Pineapple	<MDA	16.39 ±2.96	-
Banana	<MDA	16.51 ±2.88	-
Beet	4.47 ±1.53	21.32 ±3.71	0.21 ±0.08
Broccoli	3.95 ±1.12	19.66 ±3.41	0.20 ±0.07
Persimmon	4.74 ±1.30	16.75 ±2.97	0.28 ±0.09
Carrot	<MDA	21.14 ±3.68	-
Kale	<MDA	15.69 ±2.78	-
Bahia orange	<MDA	21.10 ±3.70	-
Persian lime	4.13 ±0.95	14.42 ±2.59	0.28 ±0.08
Galician lemon	1.67 ±0.78	52.61 ±9.10	0.03 ±0.02
Tahiti lemon	2.13 ±0.77	14.46 ±2.62	0.15 ±0.06
Papaya	4.67 ±1.62	18.86 ±3.27	0.25 ±0.10
Turnip	5.47 ±1.44	13.59 ±2.39	0.40 ±0.13
Radicchio	4.24 ±1.01	20.03 ±3.49	0.21 ±0.06

The lowest ^{226}Ra activity concentration value for vegetables was found in Galician lemon (1.67 ±0.78) Bq/kg, and the highest was found in turnip (5.47 ±1.44) Bq/kg. The lowest ^{226}Ra activity concentration value for soils was found for turnip cultivation (13.59 ±2.39) Bq/kg, and the highest was found for Galician lemon cultivation (52.61 ±9.10) Bq/kg. For TF, the lowest and highest values were (0.03 ±0.02) in Galician lemon cultivation and (0.40 ±0.13) in turnip, respectively.

The ^{226}Ra and ^{228}Ra commonly occur in water and soil in a 1:1 ratio and the geochemistry of these two radionuclides is practically the same (National Research Council, 1999). The sorption of radium is highly dependent on the ionic strength and concentration of the competing ions (Powell et al., 2015). Clay minerals and organic matter can absorb radium, and the amount of radium in soil can vary considerably both locally and regionally. This can be the result of factors such as soil formation and the amount of time that weathering has occurred (National Research Council, 1999).

Table 4: Activity concentration of ^{228}Th in the samples and the soil-to-plant transfer factor.

Sample	Activity concentration in plant (Bq/Kg)	Activity concentration in Soil (Bq/Kg)	Soil to plant TF
Pineapple	3.10 ± 1.79	97.16 ±50.13	0.03 ±0.02
Banana	3.65 ± 1.97	76.87 ±39.02	0.05 ±0.04
Beet	7.35 ± 3.83	183.39 ±93.67	0.04 ±0.03

Broccoli	5.35 ± 2.83	94.95 ±48.97	0.06 ±0.04
Persimmon	3.26 ± 1.76	89.58 ±45.37	0.04 ±0.03
Carrot	9.86 ± 5.07	137.10 ±70.44	0.07 ±0.05
Kale	4.29 ± 2.34	59.79 ±30.29	0.07 ±0.05
Bahia orange	1.96 ± 1.11	96.98 ±49.96	0.02±0.02
Persian lime	1.59 ± 0.92	86.66 ±43.89	0.018 ±0.01
Galician lemon	2.01 ± 1.13	431.24 ± 219.11	0.0047 ± 0.005
Tahiti lemon	2.00 ± 1.13	90.51 ± 45.84	0.022 ± 0.02
Papaya	3.30 ± 1.85	82.90 ± 41.99	0.040 ± 0.03
Turnip	4.41 ± 2.41	120.21 ± 61.80	0.037 ± 0.03
Radicchio	3.86 ± 2.17	62.12 ± 31.47	0.062 ± 0.05

The lowest ^{228}Th activity concentration value for vegetables was found in Persian lime orange (1.59 ±0.92) Bq/kg, and the highest was found in carrot (9.86 ±5.07) Bq/kg. The lowest ^{228}Th activity concentration value for soils was found for Kale cultivation (59.79 ±30.29) Bq/kg, and the highest was found for Galician lemon cultivation (431.24 ± 219.11) Bq/kg. For TF, the lowest and highest values were (0.0047 ±0.005) in Galician lemon cultivation and (0.07 ±0.05) in turnip and carrot, respectively.

According to Viana (2007), the existence of different types of soil, vegetation and climate means that the same radionuclide can exhibit a wide range of transfer values. In his work, Viana (2007) presents a variation in FT of ^{226}Ra for carrot samples ranging from 3.0 x 10^{-2} to 1.5 x 10^{-1} (in tropical climate soils) and from 2.5 x 10^{-3} to 5.7 x 10^{-3} (in temperate climate soils).

The low TF values can be explained by the mobility of thorium, which is restricted in plants due to its adsorption by cell wall materials (Azeez *et al*., 2019). In general, the concentration of radionuclides in soil reflects their total concentration rather than the amount available to plants. Consequently, most TF values are quite low (Mortvedt, 1994).

4 Conclusions

This study analyzed the levels of natural radionuclides in organic vegetables and soil samples collected from a site within the PEPB in the municipality of Rio de Janeiro. The range of activity concentration values of ^{40}K, ^{228}Ra, ^{226}Ra, and ^{228}Th identified in vegetables was 134.08 ± 19.38 to 1984.75 ± 271.72 (Bq/kg); 3.15 ± 1.28 to 32.75 ± 13.60 (Bq/kg), 1.67 ± 0.78 to 5.47 ± 1.44 (Bq/kg), and 1.59 ± 0.92 to 9.86 ± 5.07 (Bq/kg), respectively. The range of activity concentration values of ^{40}K, ^{228}Ra, ^{226}Ra, and ^{228}Th identified in soil was 585.93 ± 80.10 to 1112.98 ± 151.82 (Bq/kg); 40.30 ±16.73 to 162.71 ± 65.63 (Bq/kg); 13.59 ±2.39 to 52.61 ±9.10 (Bq/kg), and 59.79 ±30.29 to 431.24 ± 219.11 (Bq/kg). The range of TFs for ^{40}K, ^{228}Ra, ^{226}Ra, and ^{228}Th were (0.16 ± 0.045 – 2.36 ±0.46); 0.01 ± 0.021 – 0.33 ± 0.28; 0.03 ± 0.02 – 0.40 ± 0.13, and 0.0047 ± 0,005 – 0.072 ± 0.073, respectively. Soil and plant samples analyzed showed a higher activity concentration of ^{40}K than that of the other radionuclides. Despite the transfer factor varying greatly, it can still be used as comparative data for future investigations in the same area.

Acknowledgements

The authors would like to thank INEA (Instituto Estadual do Ambiente) for authorizing the research (INEA authorization nº. 27/2021) and FAPERJ (Fundação Carlos Chagas Filho de Amparo à Pesquisa do Estado do Rio de Janeiro).

References

Al-Kharouf, S. J., Al-Hamarneh, I. F., & Dababneh, M. (2008). Natural radioactivity, dose assessment and uranium uptake by agricultural crops at Khan Al-Zabeeb, Jordan. Journal of Environmental Radioactivity, 99(7), 1192-1199.

Al-Oudat, M., Al Attar, L., & Othman, I. (2021). Transfer factor of 137Cs and 90Sr to various crops in semi-arid environment. *Journal of Environmental Radioactivity*, *228*, 106525.

Azeez, H. H., Mansour, H. H., & Ahmad, S. T. (2019). Transfer of natural radioactive nuclides from soil to plant crops. *Applied Radiation and Isotopes*, *147*, 152-158.

Barnett, C. L., Belli, M., Beresford, N. A., Bossew, P., Boyer, P., Brittain, J. E., ... & Zeiller, L. (2009). Quantification of radionuclide transfer in terrestrial and freshwater environments for radiological assessments. iaea-tecdoc-1616. IAEA.

Cinelli Giorgia et al. European Atlas of Natural Radiation, Publication Office of the European Union, Luxembourg, 2019. Disponível em: https://op.europa.eu/en/publication-detail/-/publication/a8c08fd3-56b6-11ea-aece-01aa75ed71a1/language-en Acesso em:09 out.2022

Coelho, Kelly de Souza. *Determinação da matéria seca de alimentos volumosos úmidos em estufa e em micro-ondas.* 2022. Trabalho de Conclusão de Curso (Graduação em Zootecnia) – Universidade Federal de Uberlândia, Faculdade de Medicina Veterinária, Uberlândia, MG.

Currie, L. A. Limits for qualitative detection and quantitative determination. Application to radiochemistry. Analytical chemistry, v. 40, n. 3, p. 586-593, 1968.

Dias, M. C. O. (2018). *Agricultura urbana na cidade maravilhosa: os agricultores familiares do Maciço da Pedra Branca*. Revista Idealogando, 2(1), 98-110.

Farias, L. F., Soares, J. P. G., & de Sousa, R. L. (2022). O mercado de orgânicos e os preços praticados nos principais canais de comercialização na cidade de Goiânia, GO. Iheringia, Série Botânica., 77.

Hassan, Yarima Mudassir et al. Radioactivity in staple foodstuffs and concomitant dose to the population of Jigawa state, Nigeria. Radiation Physics and Chemistry, v. 178, p. 108945, 2021.

Imami, D., Ahmadi Kaliji, S., Miftari, I., Canavari, M., & Gjokaj, E. (2025). Exploring consumer segments: integrating food-related lifestyles and perceptions of organic food in Kosovo. Organic Agriculture, 1–17.

INTERNATIONAL ATOMIC ENERGY AGENCY, Guidelines for Radioelement Mapping Using Gamma Ray Spectrometry Data, IAEA-TECDOC-1363, IAEA, Vienna (2003).

Lopes, J.M. Dose efetiva comprometida devido aos radionuclídeos K-40, Ra-226, Ra-228 E Th228 contidos nos alimentos da dieta da população do estado do Rio de Janeiro. Tese (Doutorado), Programa de Engenharia Nuclear/COPPE/Universidade Federal do Rio de Janeiro, 2018.

Makki, A.A., Hajjar, D., Bagais, O., & Soliman, M. (2024). *Soil-to-plant transfer factor of natural radionuclides and associated health risk in some vegetable species in arid area*. Journal of Radioanalytical and Nuclear Chemistry, 333(8), 3947–3954.

Mortvedt, J. J. (1994). Plant and soil relationships of uranium and thorium decay series radionuclides—a review. Journal of Environmental Quality, 23(4), 643-650.

National Research Council, Division on Earth, Life Studies, Commission on Life Sciences, & Committee on Evaluation of EPA Guidelines for Exposure to Naturally Occurring Radioactive Materials. (1999). Evaluation of guidelines for exposures to technologically enhanced naturally occurring radioactive materials. National Academies Press.

Oliveira, Kerley Alberto Pereira de. Fator de transferência de elementos terras raras em solos tropicais tratados com fosfogesso. 2012. Tese (Doutorado em Saneamento, Meio Ambiente e Recursos Hídricos) – Universidade Federal de Minas Gerais, Escola de Engenharia, Belo Horizonte.

Powell, B. A., Miller, T., & Kaplan, D. I. (2015). On the influence of ionic strength on radium and strontium sorption to sandy loam soils. Journal of the South Carolina Academy of Science, 13(1), 4.

Rout, S.; Yadav, S.; Pulhani, V. Transfer of radionuclides from soil to selected tropical plants of Indian Subcontinent: A review. Journal of Environmental Radioactivity, 2021.

Tandon, A., Jabeen, F., Talwar, S., Sakashita, M., & Dhir, A. (2021). Facilitators and inhibitors of organic food buying behavior. *Food Quality and Preference*, *88*, 104077.

Tome, F. V., Rodríguez, M. B., & Lozano, J. C. (2003). Soil-to-plant transfer factors for natural radionuclides and stable elements in a Mediterranean area. Journal of Environmental Radioactivity, 65(2), 161-175.

United Nations Scientific Committee on the Effects of Atomic Radiation. (2000). *Sources and Effects of Ionizing Radiation, United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR) 2000 report, volume I: report to the general assembly, with scientific annexes-sources*. United Nations.

Viana, A. G. (2007). Avaliação do potencial de mobilidade do 90Sr em solos brasileiros. Dissertação (Mestrado em Ciências) – Instituto de Radioproteção e Dosimetria, Comissão Nacional de Energia Nuclear, Rio de Janeiro, 2007.

Willer, H., Trávníček, J., & Schlatter, B. (2024). The world of organic agriculture. Statistics and emerging trends 2024

Evaluating the Performance of Bayesian Optimization in High-Dose-Rate Brachytherapy Planning

C. N. C. Souza[1*], P. C.G. Antunes [1,2], H. Yoriyaz[1]

[1] Instituto de Pesquisas Energéticas e Nucleares, *IPEN/CNEN*, 05508-000, São Paulo, SP, Brasil

[2] Faculdade Israelita de Ciência da Saúde Albert Eistein - Hospital Albert Einstein, 05653-120, São Paulo, SP, Brasil

[*] camila.nc.souza@usp.br

Abstract

This study explores Bayesian Optimization (BO) in automating high-dose-rate (HDR) brachytherapy planning to optimize source dwell times. The methodology compared two hyperparameter approaches, using cases from the TROTS database. Results show that automatic optimization produces more stable plans with a higher success rate than fixed-parameter methods, which proved highly sensitive to manual selection. A case study, validated using the stringent Gamma Index criterion, demonstrates the method's efficacy and safety for clinical application.

Keywords: Bayesian Optimization; Brachytherapy; HDR; Hyperparameters tuning;

1 Introduction

High-dose-rate brachytherapy (HDR-BT) stands out as an effective technique in radiotherapy by enabling the precise delivery of high radiation doses over short exposure times, there by minimizing damage to healthy tissues. However, the complexity of treatment planning in this modality demands optimized and automated approaches to ensure treatment quality and efficiency Aluwini et al. (2016); Breedveld et al. (2019).

In this context, computational methods have been developed to optimize variables such as the dwell times of radioactive sources . A promising possibility is the use of Bayesian Optimization (BO), which is particularly well-suited for problems involving computationally expensive evaluations. This technique relies on probabilistic models to guide the search for optimal solutions, requiring the careful tuning of hyperparameters—elements that directly influence model performance and convergence Morén et al. (2021); Breedveld and Heijmen (2017); Breedveld et al. (2019).

This study focuses on evaluating the impact of hyperparameters on the behavior of Bayesian models applied to brachytherapy, exploring how the choice of these parameters—a decision analogous to training models in *machine learning*—directly affects the efficiency and robustness of treatment planning, highlighting its potential as an automated planning tool within a framework aligned with machine learning principles Gelbert (2015); Shahriari et al. (2016).

1.1 Bayesian Optimization

Process optimization, a branch of operations research, aims to find the best solution to real-world problems through mathematical modeling. The process involves: problem definition, mathematical modeling (variables, objectives, and constraints), selection of the solution method, validation of the results, and implementation of the solution Arenales et al. (2007).

In general, an optimization model consists of an objective function (OF) subject to constraints, as follows:

$$\begin{aligned} \text{Minimize/Maximize} \quad & f(x) = c^t x, && (1) \\ \text{subject to} \quad & Ax = b, && (2) \\ & x \geq 0, && (3) \end{aligned}$$

where c^t is the vector of coefficients of the objective function f, x is the vector of decision variables, A is the constraint coefficient matrix, and b is the demand vector.

The goal is to find an optimal solution that minimizes (or maximizes) the objective function within the feasible solution set. When it is not possible to achieve the global minimum, a feasible solution is sought instead.

The choice of the solution method depends on factors such as the nature of the model (linear, quadratic, etc.), number of variables, computational resources, and efficiency criteria. In this context, Bayesian Optimization Algorithms (BOA) stand out for their ability to deal with expensive and analytically intractable functions. Based on Bayes' Theorem, these algorithms iteratively update their estimates of the objective function as new observations are made, guiding the search for effective solutions Crocomo and Delbem (2011); Florêncio et al. (2014); Shahriari et al. (2016).

BOA relies on two fundamental components: (i) a probabilistic model that initially represents prior beliefs about the behavior of the objective function, based on the available data, and is continuously updated as new observations are acquired; (ii) the acquisition function, responsible for evaluating the current model and selecting the next points to be sampled based on exploration and exploitation criteria Shahriari et al. (2016).

The application strategy of BOA may vary depending on the type of information provided by the data. When the form of the data distribution is known and there is a finite number of parameters, a fixed structure can be applied to formulate the model—this is the approach known as **parametric estimation**. On the other hand, when the data do not follow a fixed functional form and may increase in complexity as new observations are incorporated, the model must be formulated using **non-parametric estimation**. In the latter, a distribution over entire functions is defined by assigning probability distributions to each point in the input space based on correlation (via kernel) with the observed data Gelman et al. (2015); Theodoridis and Koutroubas (2008).

A viable alternative for non-parametric modeling is the use of Multidimensional Gaussian Processes (GPs) for the construction of the Bayesian surrogate model.

Considering $f_i = f(x_i)$ as the values of the unknown function f evaluated at n points x_i, and y as the noisy observations of the unknown function f, corrupted by additive Gaussian noise $\mathcal{V} \sim \mathcal{N}(0, \sigma^2_{\mathcal{V}})$, we have, for an arbitrary x Ruela and Bessani (2021); Shahriari et al. (2016):

$$f(x) \sim \mathcal{GP}(m, K) \tag{4}$$

$$y \mid f, \sigma^2 \sim \mathcal{N}(f, \sigma^2 I) \tag{5}$$

where $f(x)$ follows a multivariate normal distribution with mean function m and covariance matrix K, and the noisy observations y are obtained from f corrupted by additive Gaussian noise $\nu \sim \mathcal{N}(0, \sigma^2_\nu)$, where σ^2_ν is the hyperparameter referred to in this work as *noise*.

The analytical form of the GP, also known as the Marginal Likelihood Function, is:

$$p(y \mid x, \theta) = \frac{1}{(2\pi)^{n/2}\|\Sigma\|^{1/2}} exp(-\frac{1}{2}(y-m)^T \Sigma^{-1}(y-m)) \tag{6}$$

where θ is the set of model hyperparameters implicit in the total covariance matrix; $\Sigma := K_{ij} + \sigma_\nu^2 I$ is the total covariance matrix, which accounts for both the covariance matrix and the variance of the additive Gaussian noise, σ_ν^2.

The covariance matrix is constructed based on a kernel function, whose choice is essential to ensure that the model accurately reflects the behavior of the data, which directly impacts the quality of the evaluation performed by the acquisition function. Among the classes of kernel functions most widely explored in the literature, the Matérn family (Eq. (6)) stands out due to its flexibility and differentiability properties, which are controlled by tunable hyperparameters. These characteristics provide the model with adaptability, making it physically realistic and suitable for a wide range of applications.

The class of Matérn covariance functions is given by Rasmussen and Williams (2006):

$$k_\nu(r) = \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu}r}{\ell} \right)^\nu K_\nu \left(\frac{\sqrt{2\nu}r}{\ell} \right), \tag{7}$$

where r is the Euclidean distance between the sampled points, $r = \|x_i - x_j\|$; ν is the hyperparameter that controls the differentiability of the function; $\Gamma(.)$ é a função gamma; ℓ is the correlation scale hyperparameter, also known as the lengthscale (ls); and K_ν is the modified Bessel function of the second kind.

It is common to set ν as a half-integer: $\nu = p + \frac{1}{2}$, where p is a non-negative integer. This strategy makes the function particularly simple and computationally efficient.

In this work, we consider $\nu = \frac{5}{2}$, whose analytical form is given by Rasmussen and Williams (2006):

$$k_{\nu=\frac{5}{2}}(r) = \left(1 + \frac{\sqrt{5}r}{\ell} + \frac{5r^2}{3\ell^2} \right) \exp\left(\frac{-\sqrt{5}r}{\ell} \right) \tag{8}$$

The *lengthscale* hyperparameter can be specified in two ways: as a **scalar**, when the same pattern of variation is assumed across all input space dimensions; or as a **vector**, which allows modeling different variations along each dimension.

Another hyperparameter that can be incorporated for a more complete characterization of the model is the *outputscale* (ots), denoted by (σ_f^2). While the *lengthscale* controls the correlation in the input space (horizontal dimension), the *outputscale* regulates the scale of variation in the output space (vertical dimension). Essentially, this hyperparameter adjusts the amplitude of the function, determining the magnitude of the covariance and, consequently, the range of values the objective function can attain.

An analytical formulation incorporating the *outputscale* is given by Gardner et al. (2018):

$$k_{final}(x_i, x_j) = \sigma_f^2 k_{base}(x_i, x_j), \tag{9}$$

where k_{base} is the kernel function previously chosen, in our case, Eq. (8).

The computational approach adopted for optimizing these hyperparameters is the Maximization of the

Log-Marginal Likelihood:

$$log\ p(y \mid x, \theta) = -\frac{1}{2}(y-m)^T \Sigma^{-1}(y-m) - \frac{1}{2}\log \|\Sigma\| - \frac{n}{2} log(2\pi) \tag{10}$$

$$\theta^* = \arg\max_{\theta} \log p(y \mid x, \theta) \tag{11}$$

Gaussian Processes (GPs) are used to model the objective function during the optimization process. However, the selection of the points to be evaluated at each iteration—that is, the generation of candidates for the next evaluation—is carried out through the **acquisition functions (α)**.

One of the best-known acquisition functions in the literature is the *Expected Improvement (EI)*. This function estimates the expected gain when observing f relative to a reference value η, which generally represents the best solution observed so far. Its standard formulation is given by Gelbert (2015); Ruela and Bessani (2021):

$$\alpha_{EI}(x) = \mathbb{E}[max(0, \eta - y)] = \int_{-\infty}^{\infty} max(0, \eta - y) p(y|x) dy, \tag{12}$$

in which $p(y|x)$ is the prior predictive distribution of f at x. Since we are using a Gaussian model, the EI has a closed-form expression Gelbert (2015):

$$\alpha_{EI}(x) = \sigma(x)(z(x)\Phi) + \rho(z(x)) \tag{13}$$

where $\sigma(x)$ is the standard deviation and $\mu(x)$ is the predicted mean of f, $z(x) = \frac{\eta - \mu(x)}{\sigma(x)}$ is the standardized normal variable, Φ denotes the cumulative distribution function (CDF), and ρ is the probability density function (PDF), both corresponding to the standard normal distribution.

However, this EI formulation assumes unconstrained problems, which does not apply to the case addressed in this work. For constrained problems, it is necessary to adjust the acquisition function by incorporating the probability that the constraints are satisfied. When the constraints are considered independent, we have the **Expected Improvement with Constraints (EIC)**, given by Gelbert (2015):

$$\alpha_{EIC}(x) = \alpha_{EI_f}(x) \prod_{k=1}^{n_g} \mathbb{P}(g_k(x) \leq 0) \tag{14}$$

where α_{EI_f} is the acquisition function for the objective function, $g_k(x)$ represents the k-th constraint, and n_g is the total number of constraints.

If each constraint is modeled as a Gaussian process, the probability of feasibility can be expressed using the CDF of a standard normal distribution. Thus, the EIC takes the following form:

$$\alpha_{EIC}(x) = \sigma_f(x)(z_f(x)\Phi(z_f(x)) + \rho(z_f(x))) \prod_{k=1}^{n_f} \Phi(\frac{\mu_k(x)}{\sigma_k(x)}) \tag{15}$$

where $\mu_k(x)$ and $\sigma_k(x)$ are, respectively, the mean and standard deviation of the prediction for the k-th constraint. These parameters directly influence the exploration–exploitation dynamics of the function. When the uncertainties associated with the prediction are high (i.e., large $\sigma(x)$), the EI value tends to be lower, favoring exploration of less-known regions of the search space. On the other hand, when uncertainties are low and the predictive mean $\mu(x)$ is close to f^* (the best known objective value), EI tends to assume higher values, promoting exploitation of promising regions.

The iterative optimization process, known as the Bayesian optimization loop, consists of an initial sampling of points in the search space, followed by Gaussian process (GP) modeling based on this data, using a previously selected kernel. Next, the model's hyperparameters are optimized and incorporated into the GP. New candidate points are then selected based on exploration–exploitation criteria defined by the chosen acquisition function. This new dataset is added to the initial one, contributing to the refinement of the predictive model. The cycle is repeated until the predefined number of iterations is reached.

2 Methodology

In this work, Bayesian optimization routines were developed using the *Python* programming language and the **BoTorch** library (Bayesian Optimization with Torch). In one of the scripts, the hyperparameter optimization method provided by the library itself was implemented, using automatic adjustment based on marginal likelihood. In another, the model hyperparameters were manually fixed, based on the values that yielded the most promising results (i.e., greater dose coverage of the target or a higher number of optimal solutions) in the runs with automatic optimization. This approach aims to simulate a scenario in which an expert uses prior knowledge to configure the model. Simulations were performed on a Lenovo computer equipped with an **AMD Ryzen 5 3500U with Radeon Vega Mobile Gfx** processor and **12 GB of RAM**.

The input data for the Gaussian process model were derived from dose distributions extracted using the Dose–Volume Histogram (DVH), employing the V_{d^c} metric, which represents the fraction of the volume that receives at least a dose d^c. This metric is defined by the following equation:

$$V_{d^c}^{Exact} = \frac{1}{m}\sum_{i=1}^{m} I_{d_i > d^c}(d_i) \tag{16}$$

where m is the total number of voxels in the analyzed tissue, d_i is the dose in voxel i, and I is an indicator function that assumes the value 1 when $d_i > d^c$, and 0 otherwise.

The optimization problem was formulated as multi-objective, with the primary goal of maximizing dose coverage in the prostate target volume, represented by the $V_{100\%}$ metric, with a minimum target of 95%. Simultaneously, the objective was to minimize the dose received by organs-at-risk (OARs), such as the rectum and bladder, through the minimization of the $V_{70\%}$ metrics. The urethra, in turn, was treated differently, with the goal of minimizing a risk metric based on a variation of the logarithmic tumor control probability (LTCP) function, expressed as:

$$LTCP = \frac{1}{m}\sum_{i=1}^{m} e^{-\alpha(d_i - d^c)} \tag{17}$$

where d^c was fixed at 90% of the reference dose and $\alpha = -0.5$.

The clinical constraints were modeled based on maximum acceptable dose limits for organs at risk. The urethra was not allowed to exceed 120% of the reference dose in 1% of its volume ($D_{1\%}$); the rectum and bladder could not surpass 100% of the maximum dose and had to comply with the limit of 80% for a volume of 1 cm^3 (D_{1cc}). The mathematical equivalence between these constraints and the V_{d^c} metric was established through the following relation:

$$D_x < d^c \Longleftrightarrow V_{d^c} < x \tag{18}$$

For the purpose of comparative analysis, only those solutions that fully satisfied all imposed constraints were considered viable. The subsequent performance evaluation, presented in the Results section, focuses on the relative effectiveness of the methods in optimizing prostate dose coverage while preserving the safety of adjacent tissues.

The data used in this study—including dose matrices, source positions, treatment reference values, as well as evaluation metrics and protocol-defined thresholds—were obtained from the public database, The Radiotherapy Optimisation Test Set Breedveld and Heijmen (2017); Breedveld et al. (2019).

In this work, we adopt the **isotropic parameterization** of the kernel, that is, a scalar *lengthscale*, which implies that the resulting covariance surface assumes the shape of a **hypersphere** in the input space. This simplifies the model, as it assumes that correlation decays equally in all directions.

3 Results

Ten clinical cases were analyzed, in each of which the script was executed four times with automatic adjustment of the model's hyperparameters. From the results obtained in these runs, the timing configurations that provided the best dose delivery—considering optimality or clinical feasibility criteria—were selected. Based on these ideal configurations, the sets of hyperparameters responsible for achieving such performance were identified. Once selected, these hyperparameters were then fixed and used in a new round of simulations for the respective patients, without allowing further optimization. In this way, it was possible to compare the performance between models with automatically adjusted hyperparameters and those with fixed hyperparameters. All values used and results obtained are summarized in Table 1.

3.1 Dosimetric Validation Using the Gamma Index: A Case Study

The database also provides a set of routines in *MATLAB* that allow an approximation of the final dose matrix. This enabled a detailed comparison of dose distributions in a representative case (Patient 4), using the Gamma Index method, as proposed by Low et al. (1998). The reference dose distribution and the resulting distributions in three distinct scenarios were analyzed: the plan generated by the method with hyperparameter optimization, the plan with the best-performing fixed hyperparameters, and the plan with the worst-performing fixed hyperparameters.

For evaluation, three tolerance criteria were selected, combining dose difference (DD) and distance-to-agreement (DTA): 3%/3mm, 2%/2mm, and 1%/1mm.

Quantitative results of this analysis are summarized in Table 2. A clear performance hierarchy can be observed among the approaches. The method with optimized hyperparameters stood out, achieving 100% passing rates under all three stringent evaluation criteria. The plan generated with the best fixed configuration also achieved a high level of agreement. In contrast, the plan with the worst fixed configuration, while acceptable under the least strict 3%/3mm criterion, showed a significant drop in performance under the more demanding criteria, highlighting its lower dosimetric quality.

Visually, the Gamma Index distribution maps in Figure 1 corroborate these findings. The maps for the "Optimized" and "Fixed (Best)" plans exhibit very high agreement, with few or no failing points (γ 1). In contrast, the map for the "Fixed (Worst)" plan reveals a concentration of failing points in the central region, indicating a more systematic and clinically relevant dosimetric disagreement.

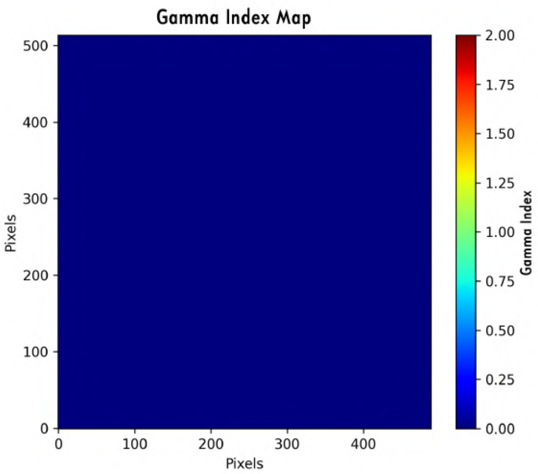
This case study illustrates and reinforces the conclusions of the main analysis: while the optimized approach consistently generates high-quality and clinically acceptable plans, the fixed-parameter approach is highly sensitive to the expert's choices. The high passing rate of the optimized plan in this practical example demonstrates the feasibility of the method as a robust and safe tool for automated brachytherapy planning.

Table 1: Comparative summary of the performance between the Optimized and Fixed approaches, where ”Nº Viables” indicates the number of feasible answers and ”Nº Optimal” the number of optimal solutions found for each clinical case.

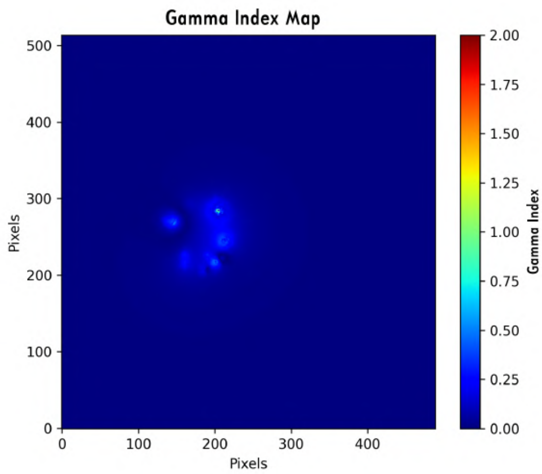
Case	**Method**	**Time (min)**	**Nº. Viables**	**Nº. Optimal**	**Prostate Dose (V_{100}%)**	**Hyperparameters (ots, ls, noise)**
1	Optimized (Mean)	51.1 ± 10.9	42.8 ± 7.5	8	96.71%	*Optimized*
	Fixed (Best Result)	71.13	49	1	96.63%	[2.95; 0.74; 0.0061]
	Fixed (Worst Result)	45.98	54	0	-	[2.88; 0.56; 0.0067]
2	Optimized (Mean)	60.9 ± 7.7	35.5 ± 0.6	2	95.02%	*Optimized*
	Fixed (Only Result)	47.94	54	6	97.40%	[2.10; 0.87; 0.0066]
3	Optimized (Mean)	56.2 ± 10.6	41.0 ± 7.2	10	98.92%	*Optimized*
	Fixed (Best Result)	28.34	31	1	95.85%	[1.54; 1.12; 0.0053]
	Fixed (Worst Result)	59.22	48	0	-	[3.64; 0.80; 0.0073]
4	Optimized (Mean)	49.1 ± 10.9	43.8 ± 4.5	20	99.00%	*Optimized*
	Fixed (Best Result)	61.70	55	2	98.28%	[3.00; 0.94; 0.0067]
	Fixed (Worst Result)	57.89	51	1	95.00%	[11.77; 1.80; 0.0066]
5	Optimized (Mean)	53.1 ± 18.0	40.8 ± 3.8	6	97.76%	*Optimized*
	Fixed (Best Result)	52.63	51	1	95.00%	[2.17; 0.47; 0.0066]
	Fixed (Worst Result)	46.58	54	0	-	[2.54; 0.50; 0.0066]
6	Optimized (Mean)	46.5 ± 14.8	48.3 ± 7.6	19	98.19%	*Optimized*
	Fixed (Best Result)	52.25	49	1	98.08%	[4.94; 1.33; 0.0067]
	Fixed (Worst Result)	69.08	51	0	-	[1.44; 1.30; 0.0054]
7	Optimized (Mean)	46.8 ± 11.2	40.8 ± 1.5	0	-	*Optimized*
	Fixed (Best Result)	70.70	53	0	-	[1.43; 1.46; 0.0026]
	Fixed (Worst Result)	31.85	33	0	-	[1.44; 2.47; 0.0014]
8	Optimized (Mean)	57.0 ± 6.8	38.5 ± 3.3	3	95.29%	*Optimized*
	Fixed (Best Result)	64.80	37	1	95.00%	[4.59; 3.08; 0.0055]
	Fixed (Worst Result)	69.04	47	0	-	[2.68; 0.65; 0.0070]
9	Optimized (Mean)	50.1 ± 9.1	39.3 ± 3.7	0	-	*Optimized*
	Fixed (Best Result)	64.39	52	1	97.07%	[5.34; 1.23; 0.0065]
	Fixed (Worst Result)	65.81	42	0	-	[19.86; 3.10; 0.0060]
10	Optimized (Mean)	53.6 ± 12.1	39.8 ± 5.7	3	95.41%	*Optimized*
	Fixed (Best Result)	62.38	49	2	97.35%	[6.76; 1.81; 0.0063]
	Fixed (Worst Result)	56.86	50	1	95.00%	[1.90; 2.35; 0.0040]

Table 2: Gamma Index passing rates for Patient 4 under different tolerance criteria.

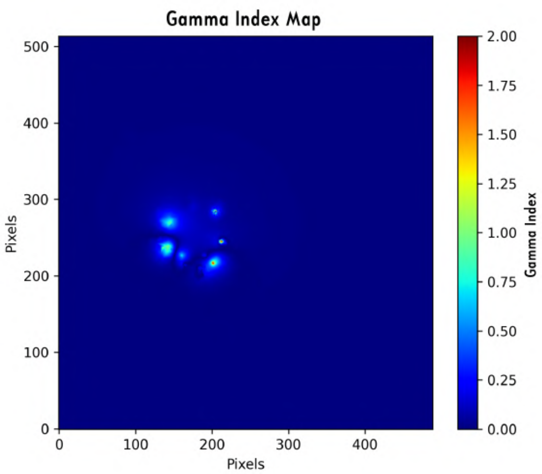
Comparison Scenario	**Passing Rate (%)**		
	3%/3mm	**2%/2mm**	**1%/1mm**
Reference vs. Optimized	100%	100%	100%
Reference vs. Fixed (Best)	100%	99.99%	99.90%
Reference vs. Fixed (Worst)	99.99%	99.90%	99.56%



(a) Optimized vs. Reference



(b) Fixed (Best) vs. Reference



(c) Fixed (Worst) vs. Reference

Figure 1: Gamma Index distribution maps (3%/3mm criterion) for Patient 4. Warm colors (yellow to red) indicate regions of disagreement where $\gamma > 1$. Agreement is visibly higher in scenarios (a) and (b) compared to (c).

4 Discussion and Conclusion

The comparative results between the approach with optimized hyperparameters and the configurations with fixed parameters are summarized in Table 1. A general analysis of the data reveals a distinct and complementary performance profile between the two methods. The optimized approach proved to be a more robust and consistent strategy, while the fixed-parameter approach exhibited greater variability in results.

The hyperparameter optimization method served as a solid benchmark, consistently finding optimal solutions in all analyzed cases. On average, this approach achieved consistently high dose coverage in the prostate, as observed in Case 4 (99.0%) and Case 3 (98.92%). Although it did not always reach the absolute performance peak, its ability to adapt to different scenarios and ensure a high-quality outcome highlights its robustness as an autonomous optimization strategy.

In contrast, the fixed-hyperparameter approach demonstrated a pronounced sensitivity to predefined choices. On one hand, a well-chosen manual configuration was able to outperform the average performance of the optimized method, as seen in Case 6, where the configuration $[4.94; 1.33; 0.0067]$ achieved a dose coverage of 98.08%. On the other hand, poor hyperparameter choices led to a complete failure in finding optimal solutions, as exemplified in Cases 5, 7, and 9. Notably, excessively high *outputscale* values, such as 19.86 in Case 9, proved detrimental, preventing model convergence.

This duality of outcomes suggests a clear *trade-off* between exploration and exploitation. The optimized method performs a broader search, adapting its *surrogate* model (Gaussian Process) to the complexity of each case, which ensures robustness. In contrast, fixing hyperparameters imposes a strong *a priori* bias on the model, forcing intensive "exploitation" in a specific region of the search space. When that region is appropriate, the results are excellent; otherwise, the risk of failure becomes imminent. This dynamic is analogous to the concepts of *transfer learning* in deep learning models, where prior knowledge is used to specialize and accelerate the solution of a problem.

This study illustrates that hyperparameter tuning in Gaussian Processes is not a mere technical adjustment, but a strategic decision that defines the model's behavior between exploration and specialization. The ability of a Bayesian model to optimize its own assumptions represents a powerful form of autonomous learning, whereas fixing parameters based on prior knowledge reflects the essence of *transfer learning*, one of the pillars of modern Deep Learning success.

Acknowledgements

This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior – Brasil (CAPES) – Process 88887.984027/2024-00. The author(s) also acknowledge the support of the Postgraduate Program at Instituto de Pesquisas Energéticas e Nucleares.

References

Aluwini S., Busser W., Baartman L., Bhawanie A., Alemayehu W., Boormans J., and Kolkman-Deurloo I. K. (2016) Fractionated high-dose-rate brachytherapy as monotherapy in prostate cancer: does implant displacement and its correction influence acute and late toxicity?. *Brachytherapy* 15:707–713

Arenales M. N., Armentano V. A., Neto R. M., and Yanasse H. H. (2007) *Pesquisa operacional*. Elsevier Editora Ltda

Breedveld S., Bennan A. B. A., Aluwini S., Schaart D. R., Kolkman-Deurloo I.-K. K., and Heijmen B. J. M. (2019) Fast automated multi-criteria planning for hdr brachytherapy explored for prostate cancer. *Physics in Medicine & Biology* 64:205002

Breedveld S. and Heijmen B. (2017) Data for trots – the radiotherapy optimisation test set. *Data in Brief* 12:143–149

Crocomo M. K. and Delbem A. C. B. (2011) Otimização bayesiana com detecção de comunidades: relatório técnico. Technical report. Instituto de Ciências Matemáticas e de Computação, Universidade de São Carlos

Florêncio P. H. B., Neto A. S. S., and Dantas M. J. P. (2014) Análise do problema de monty hall: um enfoque bayesiano. IX SAEPRO, Universidade Federal de Viçosa

Gardner J. R., Pleiss G., Bindel D., Weinberger K. Q., and Wilson A. G. (2018) Gpytorch: blackbox matrix-matrix gaussian process inference with gpu acceleration. *Advances in Neural Information Processing Systems* 31:7576–7586

Gelbert M. A. (2015) *Constrained bayesian optimization and applications*. Ph.D. thesis. Harvard University, Graduate School of Arts & Sciences

Gelman A., Carlin J. B., Stern H. S., Dunson D. B., Vehtari A., and Rubin D. B. (2015) *Bayesian data analysis*. Taylor & Francis Group. 3rd edition

Low D. A., Harms W. B., Mutic S., and Purdy J. A. (1998) A technique for the quantitative evaluation of dose distributions. *Medical Physics* 25:656–661

Morén B., Larsson T., and Tedgren C. (2021) Optimization in treatment planning of high dose-rate brachytherapy — review and analysis of mathematical models. *Medical Physics* 48

Rasmussen C. E. and Williams C. K. I. (2006) *Gaussian processes for machine learning*. MIT Press, Cambridge, MA

Ruela V. S. P. and Bessani M. (2021) Investigação de abordagens evolutivas e otimização bayesiana restrita no problema da otimização em tempo-real

Shahriari B., Swersky K., Wang Z., Adams R. P., and de Freitas N. (2016) Taking the human out of the loop: a review of bayesian optimization. *Proceedings of the IEEE* 104

Theodoridis S. and Koutroubas K. (2008) *Pattern recognition*. Academic Press. 4th edition

Evaluation of Absorbed Dose Uniformity in Irradiated Foods with Different Packaging Geometries Using PMMA Dosimetry

M. R. P. S. Barros[1]*, T. C. Alonso[2], A. Z. Mesquita, P. C. Santana[1,2], P. M. C. Oliveira[1], D. S. Calheiro[1].

[1]Universidade Federal de Minas Gerais, Faculdade de Medicina, 30130-100, Belo Horizonte, Brasil.
[2] Centro de Desenvolvimento da Tecnologia Nuclear, Laboratório de Irradiação Gama, 31270-901, Belo Horizonte, Brasil.

* mrpsbarros@gmail.com

Abstract

Dosimetry assumes a pivotal role in a wide range of irradiation applications by ensuring that materials receive the precise absorbed dose required to fulfill specific processing goals, such as sterilization of medical products, polymer modification, and microbial reduction in food products (Rodrigues & Amaral, 2014; ISO/ASTM, 2012). In the present study, the spatial distribution of the absorbed dose was evaluated in food-containing boxes arranged in different geometric configurations within an irradiation chamber. The objective was to determine the most suitable arrangement to ensure dose uniformity throughout the box volume, using a reference dose of 10 kGy. Absorbed dose measurements were performed using polymethylmethacrylate (PMMA) dosimeters—specifically, the Harwell Red Perspex 4034 model (Olejnik, 1979; Fairand, 1998; Santos & Marques, 2008). Irradiations were carried out at the Laboratório de Irradiação Gama (LIG), part of the Centro de Desenvolvimento da Tecnologia Nuclear (CDTN), utilizing a Category II Panoramic Multipurpose Irradiator, model IR-214 (type GB-127), manufactured by MDS Nordion (Calvo et al., 2009; Macedo et al., 2024). A calibration curve was first established across a range of absorbed doses following ISO/ASTM 51276:2012(E) (ISO/ASTM, 2012). Two distinct geometric arrangements were then selected to investigate dose distribution using a total of 72 Perspex 4034 dosimeters. Dosimeter readings were taken 24 hours post-irradiation using spectrophotometry, as recommended by the manufacturer and consistent with routine high-dose dosimetry practices (Rodrigues & Amaral, 2014; Vargas-Segura & Rojas-Rojas, 2024). All boxes were subjected to rotation during irradiation. The laterally stacked rotating configuration yielded an improved distribution, with a maximum average dose of 10.2 kGy and a minimum of 9.4 kGy. The most uniform distribution was observed in the vertically stacked rotating configuration, which achieved a maximum average of 10.1 kGy and a minimum of 9.9 kGy. These findings demonstrate that rotational movement during irradiation significantly enhances dose uniformity, in agreement with operational insights reported for panoramic irradiators (Calvo et al., 2009; Macedo et al., 2024). Among the configurations analyzed, the arrangement of four boxes positioned vertically in a row, with dosimeters distributed across their surfaces, proved to be the most effective in achieving homogeneity in absorbed dose distribution. Complementary studies further underscore the importance of routine calibration and verification of dose fields in different irradiation contexts (Ferreira et al., 2025; Vargas-Segura & Rojas-Rojas, 2024).

Keywords: PMMA; Absorbed dose; Dosimetry; Red Perspex.

1 Introduction

Since the discovery of ionizing radiation in the late 19th century, its use and significance have played an essential role in various fields and applications, such as medicine and industry. In the context of this study, food irradiation was explored extensively throughout the 20th century as a promising alternative for preservation, extending shelf life and reducing microbial load without chemical additives (Rodrigues & Amaral, 2014). Dosimetry—i.e., the measurement of absorbed dose—is crucial in industrial irradiation to ensure that products receive the appropriate amount of radiation to meet specific goals without compromising quality (ISO/ASTM, 2012; Rodrigues & Amaral, 2014). Among dosimetric systems, polymethylmethacrylate (PMMA) films such as Harwell Red Perspex 4034 provide reliable measurements via radiation-induced changes in optical absorbance (Olejnik, 1979; Fairand, 1998; Santos & Marques, 2008). The reliability of these systems is fundamental for validating and optimizing large-scale irradiation processes, and routine high-dose dosimetry implementations have been reported across different irradiator types (Vargas-Segura & Rojas-Rojas, 2024). Additionally, verification and calibration efforts in related health-care irradiation scenarios highlight good practice in maintaining traceability (Ferreira et al., 2025). Given that geometry and positioning within the irradiation chamber influence radiation distribution, understanding spatial arrangements is essential to avoid under- and over-dose regions (Calvo et al., 2009; Macedo et al., 2024).

2 Materials and Methods

2. 1 Irradiator

The GammaBeam-127 is a Category II Panoramic Multipurpose Irradiator, model/serial number IR-214, with a dry-storage design that uses the F-127 transport container from Nordion for storage and operation of the source. It contains a Cobalt-60 source with a maximum activity of 60,000 Ci. The products to be irradiated are placed on rotating platforms or directly on the floor. After activation of safety interlocks, the source is raised by a pneumatic hoist for the prescribed time and then returned to its secure position inside the container. The distance between the source and the product can be varied to improve dose uniformity across a wide range of densities (Calvo et al., 2009). Correlation between height (position) and absorbed dose in GammaBeam-127 installations has been characterized in recent work (Macedo et al., 2024).

Figure 1: Category II Panoramic Multipurpose Irradiator currently installed at LIG/CDTN.

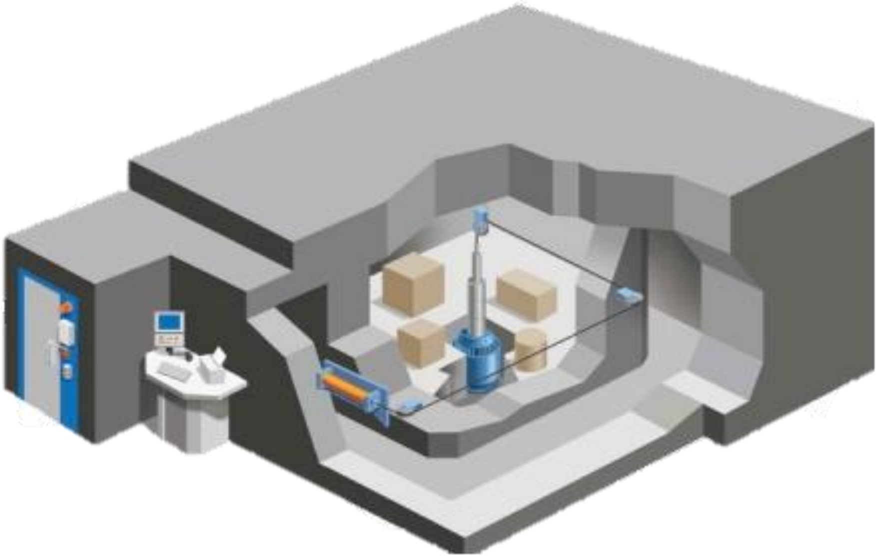


Figure 2: Irradiation Chamber.

2.2 The Dosimeters

Red Perspex 4034 dosimeters are devices used to measure high doses of radiation, especially in industrial irradiation processes. Initially developed in the 1960s, they continue to be widely used due to their effectiveness and reliability. These dosimeters are made of polymethylmethacrylate (PMMA) dyed with a special colorant that darkens when exposed to radiation. They have a functional measurement range from 5 to 50 kGy and are supplied in sealed pouches. The dosimeters are irradiated while still sealed, and after exposure, the pouches are opened so that the dose reading can be taken 24 hours later using a spectrophotometer, as recommended by the manufacturer (Olejnik, 1979; Fairand, 1998; Santos & Marques, 2008).

2.3 Spectrophotometry

In this study, a K37-UV-VIS spectrophotometer, manufactured by Kasvi, was used for dosimeter readings. The absorbance values were obtained using the UV-Professional software. A spectrophotometer is a laboratory instrument used to measure the amount of light absorbed by a substance as a function of the wavelength of electromagnetic radiation. Its operating principle is based on the Beer-Lambert law, which states that the amount of light absorbed by a solution is proportional to the path length (optical path) and the concentration of the absorbing substance - in other words, the higher the concentration of the substance or, in the case of this study, the greater the thickness of the sample, the more light will be absorbed. In the case of Red Perspex dosimeters, which are solid (PMMA), a practical version of the Beer-Lambert equation is used where, Aesp corresponds to the specific absorbance, A to the absorbance, and l to the thickness of the material, as shown in Equation 1.

$$Aesp = \frac{A}{l} \quad (1)$$

Where the specific absorbance is the absorbance divided by the thickness of the piece (in cm). This specific absorbance was compared to a calibration curve to determine the absorbed radiation dose (Rodrigues & Amaral, 2014).

2. 4 Dosimeter Calibration

The calibration of the Red Perspex 4034 dosimeters was carried out using absorbed dose values ranging from 5 kGy to 45 kGy, with 5 kGy increments between each calibration point. This procedure strictly followed the guidelines established by the ISO/ASTM 51276:2012(E) standard, ensuring traceability and reliability of the obtained results (ISO/ASTM, 2012).. After irradiating the dosimeters at their respective absorbed dose values, spectrophotometric readings of the specific absorbance for each sample were performed. Based on these data, the experimental points were fitted to construct the dosimetric curve. This curve was specifically obtained for the batch designated “PY” of Red Perspex dosimeters, ensuring that all readings in this study were conducted using the same dosimetric curve (Vargas-Segura & Rojas-Rojas, 2024).

2 5 Dosimeter Distribution And Box Positioning

The boxes are made of cardboard - a thicker, more robust type of paper - and filled with a material that simulates an apparent density of 0.65 g/cm³. For the analysis, three distinct geometric configurations of boxes were selected, each with dimensions of 49.0 cm in length, 34.0 cm in width, and 18.5 cm in height and apparent density of approximately 0.65 g/cm³, to evaluate dose distribution. To obtain the absorbed dose corresponding to the expected value, all boxes were irradiated at a distance of 84 cm from the surface of the chimney for a period of 24 hours.

In the first arrangement, four boxes were positioned vertically in a row, with dosimeters distributed across all their sides, as shown in Figure 3, totaling 36 dosimeters, nine on each side, arranged in three tiers: upper, middle, and lower. The boxes were subjected to irradiation on a rotating table.

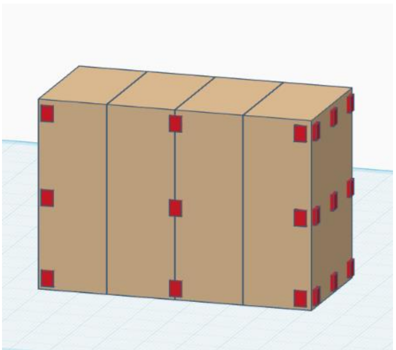


Figure 3: First arrangement

In the second arrangement, three boxes were positioned vertically in a row, with one additional box placed horizontally on top of them, as shown in Figure 4. Dosimeters were distributed across their sides, nine per side, distributed across three tiers (upper, middle, and lower), totaling 36 dosimeters. The boxes were subjected to irradiation on a rotating table.

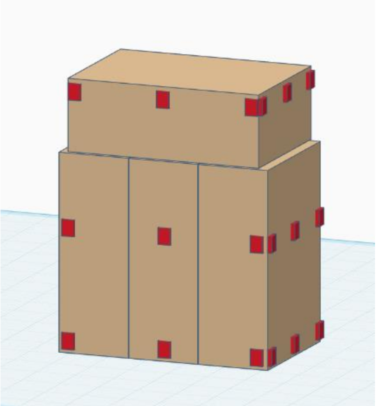


Figure 4: Second arrangement

3 Results

The results showed that among the two geometric arrangements, only one achieved a satisfactory absorbed dose distribution by the criteria established by the ISO/ASTM 51276:2012(E) standard.

The objective was to determine the most suitable arrangement to ensure dose uniformity throughout the box volume, using a reference dose of 10 kGy.

The first arrangement - four boxes positioned vertically side by side and subjected to rotation during irradiation - demonstrated an average absorbed dose of 9.9 kGy, with a standard deviation of ± 1.0 kGy. This configuration exhibited the best dose uniformity among the evaluated sides, remaining within the acceptable limits established by the standard, which allows variations of up to ±10% around the target dose, as presented in Table 1.

Table 1: First arrangement - dose distribution

Measurement	Absorbed Dose (kGy)	Average Absorbed Dose (kGy)
	Side 1	
Measurement 1	9.7 ± 1.0	
Measurement 2	10.0 ± 1.0	9.9 ± 1.0
Measurement 3	10.1 ± 1.0	
	Side 2	
Measurement 1	10.2 ± 1.0	
Measurement 2	10.1 ± 1.0	9.9 ± 1.0
Measurement 3	9.4 ± 1.0	
	Side 3	
Measurement 1	9.6 ± 1.0	
Measurement 2	9.5 ± 1.0	9.8 ± 1.0
Measurement 3	10.2 ± 1.0	
	Side 4	
Measurement 1	10.1 ± 1.0	
Measurement 2	9.8 ± 1.0	9.9 ± 1.0
Measurement 3	9.7 ± 1.0	
Mean absorbed dose distribution:		**9.9 ± 1.0 kGy**

In the second arrangement - three boxes vertically aligned with one box placed horizontally on top, also subjected to rotation during irradiation - the average absorbed dose was 8.9 kGy, with a standard deviation of ± 1.0 kGy. This configuration presented a lower average compared to the reference absorbed dose of 10 kGy. Furthermore, greater heterogeneity was observed among the sides of the boxes, with individual values ranging from 8.2 kGy to 9.5 kGy, as shown in Table 2, compromising the quality of the process.

Table 2: Second arrangement - dose distribution

Measurement	**Absorbed Dose (kGy)**	**Average Absorbed Dose (kGy)**
	Side 1	
Measurement	9,3 ± 1,0	
Measurement 2	9,1 ± 1,0	9,3 ± 1,0
Measurement 3	9,5 ± 1,0	
	Side 2	
Measurement 1	8,6 ± 1,0	
Measurement 2	8,5 ± 1,0	8,7 ± 1,0
Measurement 3	9,1 ± 1,0	
	Side 3	
Measurement 1	9,3 ± 1,0	
Measurement 2	9,0 ± 1,0	8,8 ± 1,0
Measurement 3	8,2 ± 1,0	
	Side 4	
Measurement 1	8,6 ± 1,0	
Measurement 2	8,8 ± 1,0	8,6 ± 1,0
Measurement 3	8,4 ± 1,0	
Mean absorbed dose distribution:		**8,9 ± 1,0 kGy**

4 Conclusion

This study enabled the evaluation of absorbed dose uniformity in boxes surfaces subjected to gamma irradiation, using Harwell Red Perspex 4034-type PMMA dosimeters, with a focus on different geometric positioning configurations. Among the three arrangements analyzed, only the configuration composed of four vertically stacked boxes with continuous rotation yielded satisfactory results in terms of average absorbed dose (9.9 ± 1.0 kGy), meeting the criteria established by the ISO/ASTM 51276:2012(E) standard.

The other configurations - the arrangement with three vertical boxes - showed average doses below the reference (8.9 ± 1.0 kGy), as well as greater heterogeneity in radiation distribution. These results highlight the box geometry and positioning on the ineffectiveness of irradiation.

Therefore, it is concluded that, to ensure uniformity and effectiveness in the irradiation process of products with dimensions similar to those used in this study, the configuration with vertically stacked boxes subjected to continuous rotation is the most recommended. This arrangement proved capable of ensuring homogeneity in absorbed dose distribution within regulatory limits, making it the most suitable for industrial applications that require strict control of dosimetric parameters, such as sterilization and decontamination processes using gamma radiation.

Acknowledgements

The authors are grateful to the National Council for Scientific and Technological. Development (CNPq), the Coordination for the Improvement of Higher Education Personnel. (CAPES), The Minas Gerais State

Research Support Foundation (FAPEMIG), The National Nuclear Energy Commission (CNEN) for supporting the study.

References

Calvo, W. A. P., Rela, P. R., Napolitano, C. M., Kodama, Y., Omi, N. M., da Costa, F. E., and de Andrade e Silva, L. G. (2009) Development of an irradiation system for a small size continuous run multipurpose gamma irradiator. Nukleonika 54(2), 93–97.

Fairand, B. (1998) Calibration of a polymethylmethacrylate routine dosimetry system. Radiation Physics and Chemistry 52(1–6), 523–526.

Ferreira, I. L., Barros, M. R. P. S., Fonseca, L. S., Mesquita, A. Z., Alonso, T. C., Santana, P. C., Oliveira, P. M. C., and Calheiro, D. S. (2025) Verification of absorbed dose in blood component cases using radiochromic film EBT3. Brazilian Journal of Radiation Sciences.

ISO/ASTM (2012). Standard Practice for Use of a Polymethylmethacrylate Dosimetry System (ISO/ASTM 51276:2012(E)). West Conshohocken, PA: ASTM International.

Macedo, L. G. S., Calheiro, D. S., Guimarães, M. C., Mesquita, A. Z., and Alonso, T. C. (2024) Análise da correlação entre altura e dose absorvida em um irradiador GammaBeam-127 por meio de dosímetro acrílico. VII Semana Nacional de Engenharia Nuclear e da Energia e Ciências das Radiações – SENCIR, Belo Horizonte, Brasil, 2024.

Olejnik, T. A. (1979) PERSPEX dosimeters in industrial radiation sterilization process. Radiation Physics and Chemistry 14, 431–447. Pergamon Press Ltd.

Rodrigues, L. B., and Amaral, J. L. do (2014) Radioproteção e dosimetria: fundamentos. Rio de Janeiro: IRD/CNEN, 95 p.

Santos, J. P., and Marques, J. G. (2008) Dosimetria de fótons usando dosímetros Harwell Red 4034 em um reator de pesquisa do tipo piscina. Radiation Measurements 43(2–6), 1166–1169.

Vargas-Segura, W., and Rojas-Rojas, L. (2024) Implementation of a high dose routine dosimetry in a self-shielded irradiator. UNED Research Journal 16, e5229.